

**Bangladesh University of Engineering and Technology**

Department of Computer Science & Engineering

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## **CHEM 113 Question Bank With Answers**

Chemistry

CHEM 113 / CHEM 101 (L-1/T-2)

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Year Coverage: 2006–07 to 2023–24

Topics: Quantum Mechanics, Bonding, Solutions, Colloids, Phase Diagrams, Electrochemistry & more

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# Chemistry

Section S1

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Quantum Concept & Atomic Structure

## S1 — Quantum Concept & Atomic Structure

**Q1.** Explain why each element having a characteristic emission and absorption spectra is significant in understanding dual nature of electron? (5–10 marks) [CHEM 113 | 2023–24; 2020–21]

### Answer

#### Characteristic Spectra & the Dual Nature of Electron

(8 marks)

##### (i) Characteristic Emission & Absorption Spectra:

Each element emits or absorbs photons at fixed, discrete wavelengths — its unique spectral “fingerprint.” This occurs because electrons in an atom occupy *quantized energy levels*. When an electron drops from  $E_2 \rightarrow E_1$ , it emits a photon:

$$\Delta E = E_2 - E_1 = h\nu = \frac{hc}{\lambda}$$

Since each element has a unique set of energy levels, each produces a unique spectrum. Similarly, electrons *absorb* only those photons whose energy exactly matches an allowed transition.

##### (ii) Link to the Dual Nature of the Electron:

The existence of discrete spectra *cannot* be explained if electrons behave purely as particles moving in classical orbits. It was de Broglie who proposed that electrons have a **wave nature**, with wavelength:

$$\lambda = \frac{h}{mv}$$

The key connection is this: only those electron orbits are *allowed* in which the electron wave fits as a **standing wave** — i.e., the circumference of the orbit is an integer multiple of the wavelength:

$$2\pi r = n\lambda, \quad n = 1, 2, 3, \dots$$

This restricts electrons to discrete orbits  $\Rightarrow$  discrete energy levels  $\Rightarrow$  discrete spectral lines.

##### (iii) Significance:

- Discrete spectra **prove** that electron energies are quantized — a purely wave-like property.
- They provide direct experimental **evidence** for de Broglie’s hypothesis: matter (electrons) has wave properties.
- The fact that only certain wavelengths appear confirms that electrons do *not* follow arbitrary classical paths — their wave nature imposes restrictions.
- This laid the foundation for the **quantum mechanical model** of the atom, replacing the classical particle picture entirely.

**Conclusion:** Characteristic spectra are not merely an identification tool — they are direct physical evidence that electrons possess both particle nature (they carry momentum  $p = mv$ ) and wave nature (they interfere, form standing waves, and produce discrete energy levels). Without this wave–particle duality, the very existence of sharp spectral lines would remain unexplained.

**Q2.** Describe the relationship between electron shielding and  $Z_{\text{eff}}$  on the outermost electrons of an atom. Predict how chemical reactivity is affected by a decreased effective nuclear charge. (5–10 marks) [CHEM 113 | 2020–21; 2022–23; 2023–24]

### Answer

#### 1. Electron Shielding and Effective Nuclear Charge ( $Z_{\text{eff}}$ ):

In a multi-electron atom, inner (core) electrons repel outer electrons, partially cancelling the attraction of the nucleus. This is called **shielding** or **screening**. The net charge felt by the outermost electron is the **effective nuclear charge**:

$$Z_{\text{eff}} = Z - \sigma$$

where  $Z$  is the atomic number and  $\sigma$  is the **shielding constant**. The relationship is **inverse**:

$$\uparrow \sigma \implies \downarrow Z_{\text{eff}} \quad \text{and} \quad \downarrow \sigma \implies \uparrow Z_{\text{eff}}$$

#### 2. Factors that Determine Shielding:

- Inner shell electrons** shield most effectively — they lie entirely between the valence electron and the nucleus.
- Same-shell electrons** shield poorly — they do not fully block the nuclear charge from each other (Slater weight  $\approx 0.35$  vs. 0.85 for inner shells).
- Penetrating orbitals** (e.g.  $s > p > d > f$ ) experience less shielding because they have a higher probability of being close to the nucleus, thus seeing a higher  $Z_{\text{eff}}$ .

#### 3. Trend across the Periodic Table:

**Across a period** (left  $\rightarrow$  right):  $Z$  increases by 1 each step but  $\sigma$  increases by only  $\sim 0.35$  (same-shell shielding), so  $Z_{\text{eff}}$  *increases*:

$$\text{e.g., Na: } Z_{\text{eff}} \approx 2.2 \quad \text{Cl: } Z_{\text{eff}} \approx 6.1$$

**Down a group** (top  $\rightarrow$  bottom): each new period adds a full inner shell, greatly increasing  $\sigma$ , so  $Z_{\text{eff}}$  increases only slowly while the valence electron moves to a larger, more shielded orbital — the *net* attraction on the outermost electron **decreases**.

| Element | $Z$ | $\sigma$ | $Z_{\text{eff}}$ | Valence shell |
|---------|-----|----------|------------------|---------------|
| Li      | 3   | 1.70     | 1.30             | 2s            |
| Na      | 11  | 8.85     | 2.15             | 3s            |
| K       | 19  | 16.80    | 2.20             | 4s            |
| Cl      | 17  | 10.90    | 6.10             | 3p            |
| F       | 9   | 2.85     | 6.15             | 2p            |

4. Effect of Decreased  $Z_{\text{eff}}$  on Chemical Reactivity:

A **decreased**  $Z_{\text{eff}}$  means the outermost electrons are held more loosely. This affects reactivity in several interconnected ways:

- (a) **Lower ionization energy (IE):**  
Loosely held electrons are easier to remove. Elements with low  $Z_{\text{eff}}$  (e.g. alkali metals) have very low IE and readily lose electrons to form cations:  
$$\text{Na} \rightarrow \text{Na}^+ + e^- \quad \text{IE}_1 = 496 \text{ kJ/mol}$$
- (b) **Increased metallic/reducing character:**  
Low  $Z_{\text{eff}}$  elements are strong **reducing agents** — they easily donate electrons in redox reactions. This is why alkali and alkaline earth metals are the most reactive metals.
- (c) **Larger atomic radius:**  
Reduced nuclear attraction allows the electron cloud to expand, increasing atomic size. Larger atoms have more diffuse, reactive valence electrons:  
$$r \propto \frac{1}{Z_{\text{eff}}}$$
- (d) **Weaker electronegativity:**  
With low  $Z_{\text{eff}}$ , the atom has less ability to attract bonding electrons toward itself, resulting in low electronegativity and a tendency to form **ionic** rather than covalent bonds.
- (e) **Lower electron affinity:**  
Less nuclear pull means the atom gains little stabilisation by accepting an extra electron — low  $Z_{\text{eff}}$  atoms have low or even negative electron affinity.

5. Illustrative Comparison:

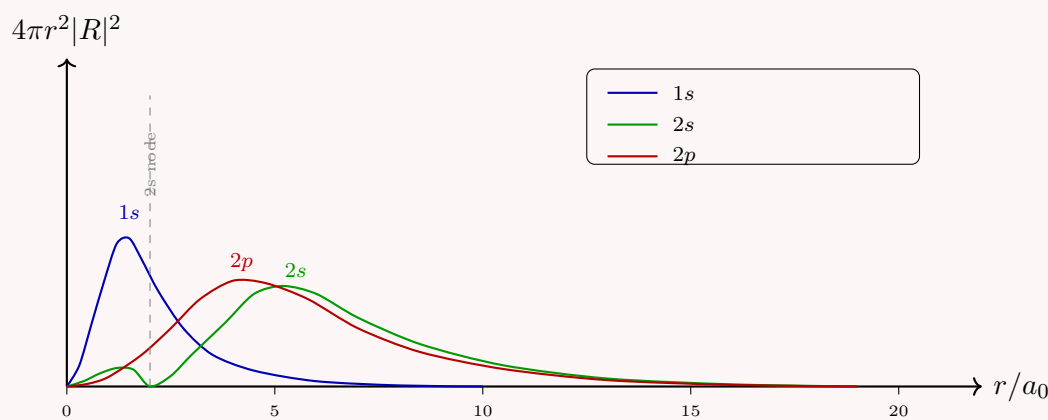
| Property           | Low $Z_{\text{eff}}$ | High $Z_{\text{eff}}$ | Example pair |
|--------------------|----------------------|-----------------------|--------------|
| Ionization energy  | Low                  | High                  | Na vs. Cl    |
| Atomic radius      | Large                | Small                 | K vs. F      |
| Electronegativity  | Low                  | High                  | Cs vs. F     |
| Metallic character | High                 | Low                   | Na vs. Si    |
| Reducing power     | Strong               | Weak                  | Li vs. Ne    |

**Conclusion:** Electron shielding and  $Z_{\text{eff}}$  are **inversely related** — greater shielding reduces  $Z_{\text{eff}}$ , weakening the nucleus’s grip on valence electrons. A decreased  $Z_{\text{eff}}$  leads to **lower ionization energy, larger atomic radius, weaker electronegativity, and higher metallic and reducing reactivity** — making such elements far more chemically reactive toward electron donation and ionic bond formation.

**Q3.** Analyze the radial distribution functions for the hydrogenic 1s, 2s, and 2p orbitals, clearly illustrating their nodal structures. Based on your analysis, justify why the electron in the 2s orbital exhibits a greater probability of being found closer to the nucleus compared to the 2p orbital. **(10 marks)** [CHEM 113 | 2023–24]

Answer

The **radial distribution function (RDF)** gives the probability of finding an electron in a spherical shell of radius  $r$  and thickness  $dr$  from the nucleus:  
$$P(r) dr = 4\pi r^2 |\psi|^2 dr = 4\pi r^2 [R_{nl}(r)]^2 dr$$



### 1. Analysis of Each Orbital:

(i) **1s orbital** ( $n = 1, l = 0$ ):

$$R_{10}(r) = 2 \left( \frac{1}{a_0} \right)^{3/2} e^{-r/a_0}$$

- RDF rises steeply from zero, peaks sharply at  $r = a_0$  (Bohr radius = 0.529 Å), then decays exponentially.
- **No radial nodes** —  $(n - l - 1) = 0$  nodes.
- Electron is tightly confined closest to the nucleus.

(ii) **2s orbital** ( $n = 2, l = 0$ ):

$$R_{20}(r) = \frac{1}{2\sqrt{2}} \left( \frac{1}{a_0} \right)^{3/2} \left( 2 - \frac{r}{a_0} \right) e^{-r/2a_0}$$

- RDF has **one radial node** at  $r = 2a_0$  (where  $2 - r/a_0 = 0$ ), giving *two peaks*: a small inner peak near the nucleus and a larger outer peak around  $r \approx 5a_0$ .
- Number of radial nodes =  $(n - l - 1) = 1$ .
- The **inner peak** (penetration) gives the 2s electron significant probability density near the nucleus.

(iii) **2p orbital** ( $n = 2, l = 1$ ):

$$R_{21}(r) = \frac{1}{2\sqrt{6}} \left( \frac{1}{a_0} \right)^{3/2} \frac{r}{a_0} e^{-r/2a_0}$$

- RDF starts at zero (due to  $r/a_0$  factor), rises to a **single broad peak** around  $r \approx 4a_0$ , then decays.
- **No radial nodes** —  $(n - l - 1) = 0$  nodes.
- Has **one angular node** (nodal plane,  $l = 1$ ), but no inner peak.
- The electron has *no significant probability* of being found close to the nucleus.

### 2. Nodal Structure Summary:

| Orbital | Radial nodes       | Angular nodes   | Total nodes | Inner peak?     |
|---------|--------------------|-----------------|-------------|-----------------|
| 1s      | 0                  | 0               | 0           | Yes (only peak) |
| 2s      | 1 (at $r = 2a_0$ ) | 0               | 1           | Yes             |
| 2p      | 0                  | 1 (nodal plane) | 1           | No              |

### 3. Why 2s Electron is Found Closer to Nucleus than 2p:

Although both 2s and 2p have  $n = 2$  and similar average distances from the nucleus, the 2s orbital exhibits a phenomenon called **penetration**:

- The **inner lobe** of the 2s RDF (before the radial node at  $r = 2a_0$ ) gives the 2s electron a non-negligible probability of being found very close to the nucleus — even inside the 1s shell.
- The 2p RDF, by contrast, goes to zero at  $r = 0$  (due to the centrifugal barrier from angular momentum,  $l = 1$ ) and has *no* inner peak — it cannot penetrate to the nucleus.
- This is quantified by the **penetration effect**: 2s electrons experience a higher effective nuclear charge  $Z_{\text{eff}}$  because they spend more time near the unshielded nucleus:

$$Z_{\text{eff}}(2s) > Z_{\text{eff}}(2p)$$

- As a consequence, 2s electrons are **lower in energy** and more tightly bound than 2p electrons — which is why in multi-electron atoms, 2s fills before 2p.

**Conclusion:** The presence of a **radial node** in the 2s orbital creates an inner probability lobe that allows 2s electrons to *penetrate* closer to the nucleus compared to 2p electrons, which are excluded from the nuclear region by their angular momentum. This penetration difference is the quantum mechanical basis for the 2s–2p energy splitting in multi-electron atoms.



- Q4.** Evaluate the case of a  $\text{Ca}^{19+}$  ion with its electron in the 6th excited state: (i) Calculate the longest wavelength of light emitted when the electron transitions to a lower energy level. (ii) Suppose the same transition as in part (i) took place in a hydrogen atom. Would the wavelength of emission be longer than, shorter than, or the same as your answer? **(10 marks)** [CHEM 113 | 2023–24]

### Answer

#### Given:

- $\text{Ca}^{19+}$  is a **hydrogenic ion** (one electron) with atomic number  $Z = 20$
- 6th excited state means  $n = 7$  (ground state  $n = 1$ , so excited state  $n = 1 + 6 = 7$ )
- Energy of a hydrogenic ion:

$$E_n = -\frac{Z^2 \times 13.6}{n^2} \text{ eV}$$

#### (i) Longest wavelength of emitted light:

Longest wavelength  $\Rightarrow$  lowest energy transition  $\Rightarrow$  smallest  $\Delta E \Rightarrow$  electron drops to the **nearest lower level**, i.e.  $n = 7 \rightarrow n = 6$ .

Calculate  $E_7$  and  $E_6$ :

$$E_7 = -\frac{(20)^2 \times 13.6}{7^2} = -\frac{400 \times 13.6}{49} = -\frac{5440}{49} = -111.02 \text{ eV}$$

$$E_6 = -\frac{(20)^2 \times 13.6}{6^2} = -\frac{400 \times 13.6}{36} = -\frac{5440}{36} = -151.11 \text{ eV}$$

#### Energy of emitted photon:

$$\Delta E = E_7 - E_6 = -111.02 - (-151.11) = 40.09 \text{ eV}$$

Converting to Joules:

$$\Delta E = 40.09 \times 1.602 \times 10^{-19} = 6.422 \times 10^{-18} \text{ J}$$

#### Wavelength:

$$\lambda = \frac{hc}{\Delta E} = \frac{(6.626 \times 10^{-34})(3.0 \times 10^8)}{6.422 \times 10^{-18}}$$

$$\boxed{\lambda = 3.09 \times 10^{-8} \text{ m} = 30.9 \text{ nm}}$$

This falls in the **UV region** of the electromagnetic spectrum.

#### (ii) Same transition in hydrogen ( $Z = 1$ ), $n = 7 \rightarrow n = 6$ :

For hydrogen:

$$\Delta E_H = 13.6 \left( \frac{1}{6^2} - \frac{1}{7^2} \right) = 13.6 \left( \frac{1}{36} - \frac{1}{49} \right) = 13.6 \times \frac{49 - 36}{36 \times 49} = 13.6 \times \frac{13}{1764} = 0.1003 \text{ eV}$$

$$\lambda_H = \frac{hc}{\Delta E_H} = \frac{(6.626 \times 10^{-34})(3.0 \times 10^8)}{0.1003 \times 1.602 \times 10^{-19}} = 1.235 \times 10^{-5} \text{ m} = 12,350 \text{ nm}$$

This falls in the **infrared region**.

#### Comparison:

| System                         | $\Delta E$ | $\lambda$ | Region   |
|--------------------------------|------------|-----------|----------|
| $\text{Ca}^{19+}$ ( $Z = 20$ ) | 40.09 eV   | 30.9 nm   | UV       |
| H ( $Z = 1$ )                  | 0.100 eV   | 12,350 nm | Infrared |

The wavelength for hydrogen is **much longer** than for  $\text{Ca}^{19+}$ .

**Reason:** The energy of a hydrogenic system scales as  $Z^2$ :

$$\Delta E \propto Z^2 \quad \Rightarrow \quad \lambda \propto \frac{1}{Z^2}$$

For  $\text{Ca}^{19+}$ ,  $Z = 20$ , so  $\Delta E$  is  $20^2 = 400$  times larger than for hydrogen for the same transition. Since  $\lambda = hc/\Delta E$ , the wavelength of  $\text{Ca}^{19+}$  is 400 times *shorter* than that of hydrogen:

$$\frac{\lambda_H}{\lambda_{\text{Ca}^{19+}}} = Z^2 = 400 \quad \checkmark \quad \left( \frac{12350}{30.9} \approx 400 \right)$$

**Conclusion:** The hydrogen emission wavelength (12,350 nm, infrared) is **far longer** than that of  $\text{Ca}^{19+}$  (30.9 nm, UV) for the same  $n = 7 \rightarrow 6$  transition, because the much higher nuclear charge of calcium pulls the electron to much lower (more negative) energy levels, making transitions far more energetic.

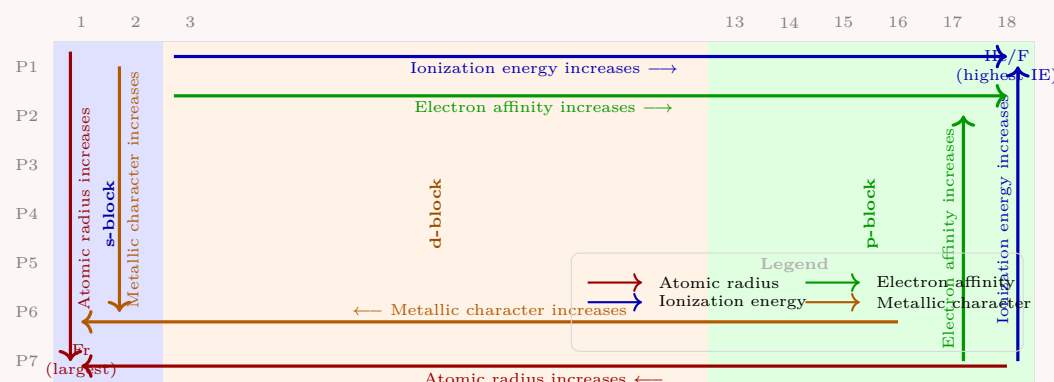
- Q5.** Discuss the periodic trends in ionization energy, electron affinity, atomic size, and metallic properties of elements, illustrating your explanation with the help of a labeled outline of the periodic table. **(10 marks)** [CHEM 113 | 2023–24]

## Answer

### Periodic Trends — Overview:

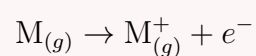
Four key periodic properties all depend on **effective nuclear charge** ( $Z_{\text{eff}}$ ) and **atomic radius**, and show consistent trends across periods and down groups.

### Labeled Periodic Table Showing All Four Trends:



#### 1. Ionization Energy (IE):

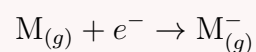
IE is the energy required to remove the outermost electron from a gaseous atom:



- **Across a period** ( $\rightarrow$ ):  $Z_{\text{eff}}$  increases, electrons are held more tightly  $\Rightarrow$  **IE increases**. Exception: slight dip at group 3 (e.g. Al < Mg) due to  $2p$  being higher than  $2s$ , and at group 6 (e.g. S < P) due to pairing repulsion.
- **Down a group** ( $\downarrow$ ): atomic radius increases, outer electron farther from nucleus, more shielded  $\Rightarrow$  **IE decreases**.
- Highest IE: He (2372 kJ/mol); Lowest: Cs (376 kJ/mol).

#### 2. Electron Affinity (EA):

EA is the energy change when a gaseous atom gains one electron:



- **Across a period** ( $\rightarrow$ ): increasing  $Z_{\text{eff}}$  makes atoms more attractive to electrons  $\Rightarrow$  **EA becomes more negative** (more favourable). Exception: group 2 (filled  $s$ ) and group 5 (half-filled  $p$ ) show anomalously low EA.
- **Down a group** ( $\downarrow$ ): larger radius, increased shielding  $\Rightarrow$  **EA decreases** (less favourable). Exception: F < Cl due to electron-electron repulsion in the compact  $2p$  orbital.
- Most negative EA: Cl (−349 kJ/mol).

#### 3. Atomic Size (Atomic Radius):

- **Across a period** ( $\rightarrow$ ): increasing  $Z$  pulls electrons closer with little extra shielding  $\Rightarrow$  **radius decreases**.
- **Down a group** ( $\downarrow$ ): new electron shells are added, greatly increasing shielding  $\Rightarrow$  **radius increases**.
- Largest atom: Fr ( $\sim 270$  pm); Smallest: He ( $\sim 31$  pm).

#### 4. Metallic Character:

Metallic character reflects the tendency to **lose electrons** and form cations — linked directly to low IE and large atomic radius.

- **Across a period** ( $\rightarrow$ ): IE increases, electron loss becomes harder  $\Rightarrow$  **metallic character decreases** (metals on left, non-metals on right).
- **Down a group** ( $\downarrow$ ): IE decreases, electrons lost more easily  $\Rightarrow$  **metallic character increases**.
- Most metallic: Fr (bottom-left); Least metallic: F/He (top-right).

#### Summary Table:

| Property           | Across period<br>( $\rightarrow$ ) | Down group<br>( $\downarrow$ ) |
|--------------------|------------------------------------|--------------------------------|
| Ionization energy  | Increases                          | Decreases                      |
| Electron affinity  | More negative                      | Less negative                  |
| Atomic radius      | Decreases                          | Increases                      |
| Metallic character | Decreases                          | Increases                      |

**Conclusion:** All four trends are governed by the interplay of **nuclear charge**, **shielding**, and **atomic radius**. Increasing  $Z_{\text{eff}}$  (across a period) tightens electron hold, raising IE and EA while shrinking the atom and reducing metallic character. Increasing shielding (down a group) has the opposite effect, making valence electrons easier to remove and atoms more metallic.



- Q6.** Consider a photoelectric experiment in which light of variable frequency strikes a metal surface. Plot the kinetic energy of emitted electrons as a function of the frequency of the incident light. From this graph, derive the work function of the metal, and discuss how the graph would differ if the light source were replaced with a source of lower frequency which is below the threshold frequency. (10 marks) [CHEM 113 | 2022–23]

### Answer

#### Photoelectric Effect: KE vs. Frequency Graph & Work Function

(10 marks)

##### Part 1: The KE vs. Frequency Graph

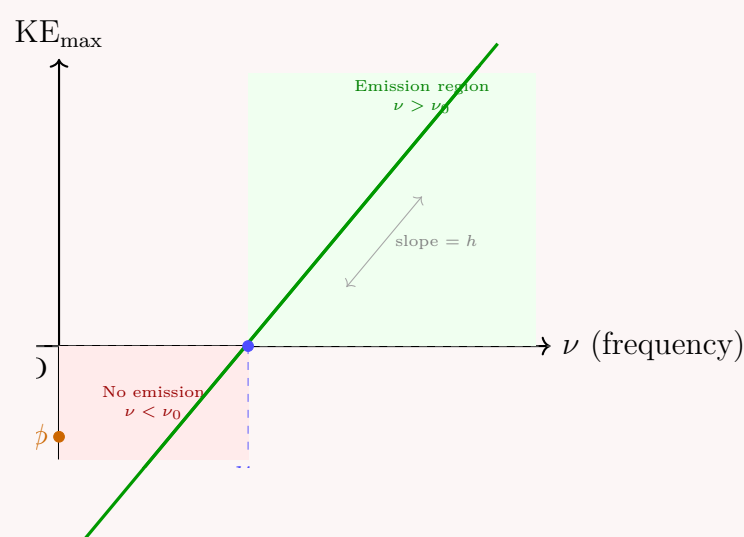
(3 marks)

From Einstein's photoelectric equation:

$$\frac{1}{2}mv_{\max}^2 = h\nu - \phi = h\nu - h\nu_0 = h(\nu - \nu_0)$$

Plotting  $KE_{\max}$  on the  $y$ -axis against frequency  $\nu$  on the  $x$ -axis gives a **straight line** with:

$$\begin{array}{ccc} \text{slope} = h & \text{x-intercept} = \nu_0 & \text{y-intercept} = -\phi \\ \text{Planck's constant} & \text{threshold frequency} & \text{(negative work function)} \end{array}$$



##### Part 2: Deriving the Work Function from the Graph

(4 marks)

The Einstein photoelectric equation is a straight line of the form  $y = mx + c$ :

$$KE_{\max} = h\nu - \phi \quad \longleftrightarrow \quad y = \underbrace{h}_{\text{slope}} \cdot \nu + \underbrace{(-\phi)}_{\text{y-intercept}}$$

##### Method 1 — From the $x$ -intercept:

At the threshold frequency  $\nu_0$ ,  $KE_{\max} = 0$ . Setting  $KE = 0$ :

$$0 = h\nu_0 - \phi \quad \Rightarrow \quad \boxed{\phi = h\nu_0}$$

So the work function equals Planck's constant multiplied by the  $x$ -intercept read directly off the graph.

##### Method 2 — From the $y$ -intercept:

Extrapolating the straight line back to the  $y$ -axis gives a negative intercept. The magnitude of this intercept is directly equal to  $\phi$ :

$$\phi = |\text{y-intercept}|$$

##### Method 3 — From the slope:

The slope of the line equals  $h$  (Planck's constant). Once the slope is measured experimentally, and two points  $(\nu_1, KE_1)$  and  $(\nu_2, KE_2)$  are known:

$$h = \frac{\Delta KE}{\Delta \nu} = \frac{KE_2 - KE_1}{\nu_2 - \nu_1} \quad \Rightarrow \quad \phi = h\nu - KE \text{ at any known point}$$

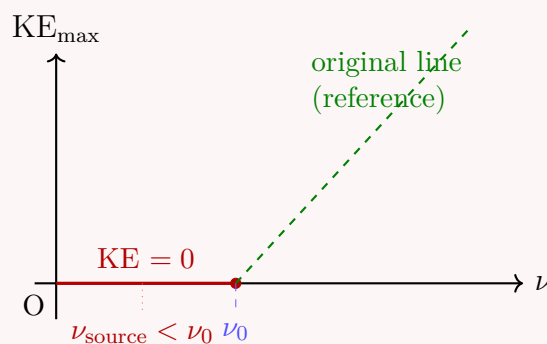
##### Part 3: Effect of Replacing with a Sub-threshold Light Source

(3 marks)

If the incident light has frequency  $\nu < \nu_0$  (below the threshold):

- **No electrons are emitted** — regardless of how intense the light source is or how long it illuminates the surface.
- Each photon carries energy  $h\nu < \phi$ , so no single photon has sufficient energy to overcome the work function and free an electron.
- **The graph does not extend** into the positive KE region. The entire operating range lies to the *left* of  $\nu_0$ , where  $KE = 0$  for all frequencies.

- **No straight line is observed** — instead, the KE axis remains flat at zero for all  $\nu < \nu_0$ , as illustrated below:



The red flat line shows that the sub-threshold source produces *zero* kinetic energy for all frequencies up to  $\nu_0$ . The dashed green line is the original graph shown for reference. This stark contrast demonstrates that the photoelectric effect is **not** a gradual phenomenon — it has a sharp, well-defined cutoff at  $\nu_0$ .

**Conclusion:** The KE vs.  $\nu$  graph is a powerful tool — its slope gives Planck's constant  $h$ , its  $x$ -intercept gives  $\nu_0$ , and its  $y$ -intercept gives  $-\phi$ . A sub-threshold source produces no graph at all in the positive KE region, confirming that photon energy must exceed  $\phi$  for any emission to occur.

**Q7.** Inside an atom, the uncertainty in the position of an electron is 0.1 nm. Calculate the minimum uncertainty in the momentum of the particle. If the electron has a de Broglie wavelength  $\lambda = 2.0 \times 10^{-10}$  m, calculate its velocity and discuss how the uncertainty in the particle's position influences its momentum and velocity. **(10 marks)** [CHEM 113 | 2022–23]

### Answer

#### Heisenberg Uncertainty & de Broglie Wavelength of an Electron

**(10 marks)**

#### Part 1: Minimum Uncertainty in Momentum

**(3 marks)**

Heisenberg's Uncertainty Principle states that the position and momentum of a particle cannot both be known precisely at the same time:

$$\Delta x \cdot \Delta p \geq \frac{h}{4\pi}$$

For *minimum* uncertainty, we use the equality:

$$\Delta p_{\min} = \frac{h}{4\pi \Delta x}$$

**Given:**  $\Delta x = 0.1 \text{ nm} = 0.1 \times 10^{-9} \text{ m} = 1.0 \times 10^{-10} \text{ m}$

Substituting:

$$\Delta p_{\min} = \frac{6.626 \times 10^{-34}}{4\pi \times 1.0 \times 10^{-10}} = \frac{6.626 \times 10^{-34}}{1.2566 \times 10^{-9}}$$

$$\Delta p_{\min} = 5.27 \times 10^{-25} \text{ kg}\cdot\text{m/s}$$

#### Part 2: Velocity from de Broglie Wavelength

**(3 marks)**

The de Broglie relation connects wavelength to momentum:

$$\lambda = \frac{h}{mv} \quad \Rightarrow \quad v = \frac{h}{m\lambda}$$

**Given:**  $\lambda = 2.0 \times 10^{-10} \text{ m}$ ,  $m_e = 9.109 \times 10^{-31} \text{ kg}$

Substituting:

$$v = \frac{6.626 \times 10^{-34}}{9.109 \times 10^{-31} \times 2.0 \times 10^{-10}} = \frac{6.626 \times 10^{-34}}{1.822 \times 10^{-40}}$$

$$v = 3.64 \times 10^6 \text{ m/s}$$

This is approximately 1.2% of the speed of light — confirming the electron is moving at a highly energetic but still non-relativistic speed.

**Part 3: How Uncertainty in Position Influences Momentum & Velocity**
**(4 marks)**
**(i) The Inverse Relationship:**

 From the uncertainty principle  $\Delta x \cdot \Delta p \geq h/4\pi$ , position uncertainty  $\Delta x$  and momentum uncertainty  $\Delta p$  are *inversely* related:

$$\Delta p \propto \frac{1}{\Delta x}$$

| Situation                | $\Delta x$        | $\Delta p$             |
|--------------------------|-------------------|------------------------|
| Position known precisely | Very small        | Very large             |
| Position poorly known    | Very large        | Very small             |
| Balanced (atomic scale)  | $\sim 10^{-10}$ m | $\sim 10^{-25}$ kg·m/s |

**(ii) Influence on Momentum:**

 Since  $p = mv$  and  $m$  is fixed for an electron, the uncertainty in momentum directly translates to uncertainty in velocity:

$$\Delta p = m_e \Delta v \quad \Rightarrow \quad \Delta v = \frac{\Delta p}{m_e} = \frac{5.27 \times 10^{-25}}{9.109 \times 10^{-31}} \approx 5.79 \times 10^5 \text{ m/s}$$

 This means the velocity is uncertain by about  $5.79 \times 10^5$  m/s — roughly **16%** of the calculated velocity  $v = 3.64 \times 10^6$  m/s. This is a significant uncertainty, not a negligible one.

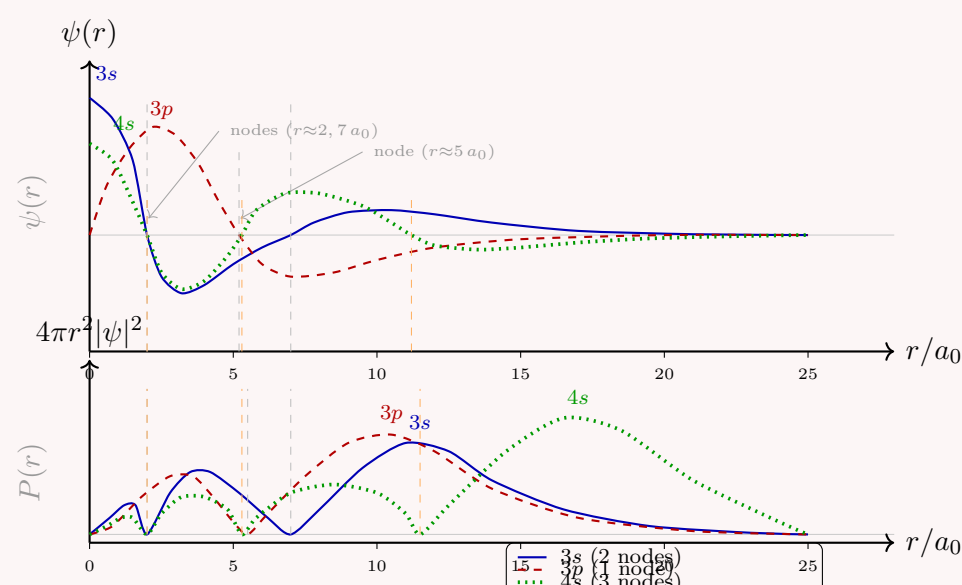
**(iii) Physical Interpretation:**

- Confining an electron to a small region (small  $\Delta x$ , as inside an atom) *forces* a large spread in its momentum  $\Delta p$  — the electron cannot be at rest; it must be in vigorous motion.
- This is why electrons do *not* collapse into the nucleus despite electrostatic attraction — confinement to the nucleus ( $\Delta x \sim 10^{-15}$  m) would require an impossibly large momentum uncertainty.
- Unlike macroscopic objects, for an electron at the atomic scale ( $\Delta x \sim 10^{-10}$  m), the uncertainty  $\Delta v \sim 10^5$ – $10^6$  m/s is comparable to the actual velocity — making a precise classical trajectory *physically meaningless*.
- This is precisely why quantum mechanics replaces the concept of a definite orbit (Bohr) with a **probability cloud** — the electron's position and momentum are fundamentally uncertain simultaneously.

**Conclusion:** The minimum momentum uncertainty is  $5.27 \times 10^{-25}$  kg·m/s and the electron velocity from the de Broglie wavelength is  $3.64 \times 10^6$  m/s. The resulting velocity uncertainty of  $\sim 16\%$  confirms that at the atomic scale, the Heisenberg principle is not a measurement limitation — it is a fundamental property of nature that makes classical trajectory descriptions impossible.

**Q8. Sketch the wave functions and the probabilities for 3s, 3p, and 4s orbitals. (5 marks)**

[CHEM 113 | 2022–23]

**Answer**
**Wave Functions  $\psi(r)$  and Probability Distributions  $4\pi r^2 |\psi|^2$  for 3s, 3p, and 4s orbitals:**

**Key Features Summary:**

| Orbital | Radial nodes<br>( $n-l-1$ ) | Angular nodes ( $l$ ) | RDF peaks | Sign changes in $\psi$           |
|---------|-----------------------------|-----------------------|-----------|----------------------------------|
| 3s      | 2                           | 0                     | 3         | 2 (at $r \approx 2, 7 a_0$ )     |
| 3p      | 1                           | 1                     | 2         | 1 (at $r \approx 5 a_0$ )        |
| 4s      | 3                           | 0                     | 4         | 3 (at $r \approx 2, 5, 11 a_0$ ) |

**Note:** The  $\psi(r)$  plots show sign changes at each radial node. The RDF  $4\pi r^2|\psi|^2$  is always non-negative and touches zero at each radial node. The 4s orbital, having more nodes than 3s and 3p, has greater penetration toward the nucleus — explaining why  $E_{4s} < E_{3d}$  in multi-electron atoms.

**Q9.** Demonstrate the relationship between blackbody radiation and Planck's quantum theory. (10 marks) [CHEM 113 | 2021–22]

### Answer

#### Blackbody Radiation & Planck's Quantum Theory

(10 marks)

#### Part 1: What is Blackbody Radiation?

(2 marks)

A **blackbody** is an ideal object that:

- **Absorbs** all electromagnetic radiation falling on it, regardless of frequency or angle — no reflection, no transmission.
- **Emits** radiation purely as a function of its temperature  $T$ , not its composition or shape.

The radiation emitted by such a body is called **blackbody radiation**. Key experimental observations about blackbody radiation:

- The emitted radiation spans a *continuous* range of frequencies, forming a smooth bell-shaped curve of intensity vs. frequency.
- As temperature increases, the peak of the curve shifts to **higher frequencies** (shorter wavelengths) — Wien's Displacement Law:

$$\lambda_{\max} T = b, \quad b = 2.898 \times 10^{-3} \text{ m} \cdot \text{K}$$

- The total power radiated increases rapidly with temperature — Stefan-Boltzmann Law:

$$E = \sigma T^4, \quad \sigma = 5.67 \times 10^{-8} \text{ W} \cdot \text{m}^{-2} \cdot \text{K}^{-4}$$

#### Part 2: Failure of Classical Theory — The Ultraviolet Catastrophe

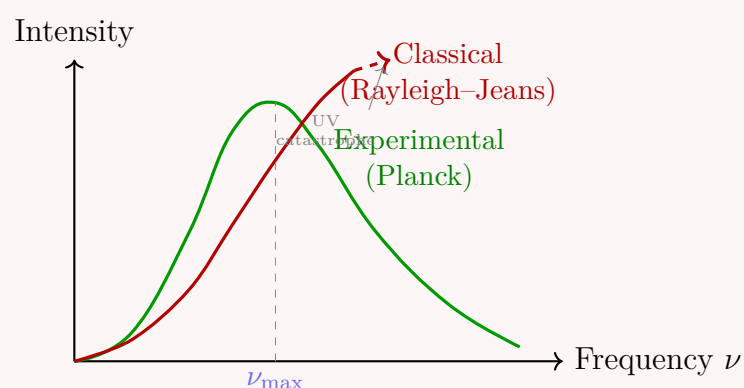
(2 marks)

Classical physics attempted to explain blackbody radiation using the **Rayleigh-Jeans Law**, which treated radiation as continuous waves and applied the equipartition theorem of thermodynamics:

$$u(\nu, T) = \frac{8\pi\nu^2}{c^3} k_B T$$

where  $u(\nu, T)$  is the energy density per unit frequency,  $k_B$  is Boltzmann's constant, and  $c$  is the speed of light.

**The Problem:** This equation predicts that energy density increases without limit as frequency increases ( $u \rightarrow \infty$  as  $\nu \rightarrow \infty$ ). Integrating over all frequencies gives *infinite* total energy — a physically absurd result known as the **ultraviolet catastrophe**.



#### Part 3: Planck's Quantum Theory as the Solution

(4 marks)

(i) Planck's Revolutionary Hypothesis (1900):

Max Planck resolved the ultraviolet catastrophe by abandoning the classical assumption of continuous energy. He proposed that oscillating atoms in the blackbody walls can only emit or absorb energy in discrete packets called **quanta**:

$$E = nh\nu, \quad n = 0, 1, 2, 3, \dots$$

where  $h = 6.626 \times 10^{-34}$  J·s is **Planck's constant** and  $\nu$  is the frequency of oscillation. Energy is therefore *quantized* — not continuous.

**(ii) Planck's Radiation Law:**

Using this quantization assumption and Boltzmann statistics, Planck derived the correct energy density formula:

$$u(\nu, T) = \frac{8\pi h\nu^3}{c^3} \cdot \frac{1}{e^{h\nu/k_B T} - 1}$$

This single equation perfectly matched *all* experimental blackbody curves at every temperature — a complete triumph over classical theory.

**(iii) Why Quantization Eliminates the Catastrophe:**

The key lies in the exponential term  $e^{h\nu/k_B T}$ . At high frequencies:

$$h\nu \gg k_B T \quad \Rightarrow \quad e^{h\nu/k_B T} \rightarrow \infty \quad \Rightarrow \quad u(\nu, T) \rightarrow 0$$

Physically: at high frequencies, each quantum  $h\nu$  is so large that thermal energy  $k_B T$  cannot excite the oscillators — they are *frozen out*. This naturally suppresses emission at high frequencies and prevents the divergence that plagued classical theory.

At low frequencies ( $h\nu \ll k_B T$ ), the exponential reduces to  $e^{h\nu/k_B T} \approx 1 + h\nu/k_B T$ , and Planck's law reduces exactly to the classical Rayleigh–Jeans law — confirming it is correct at low frequencies:

$$u(\nu, T) \xrightarrow{h\nu \ll k_B T} \frac{8\pi\nu^2}{c^3} k_B T \quad \checkmark$$

**Part 4: Significance of the Relationship**

**(2 marks)**

| Aspect                  | Classical Theory                     | Planck's Quantum Theory        |
|-------------------------|--------------------------------------|--------------------------------|
| Nature of energy        | Continuous                           | Discrete quanta $E = h\nu$     |
| High-frequency behavior | $u \rightarrow \infty$ (catastrophe) | $u \rightarrow 0$ (frozen out) |
| Low-frequency behavior  | Matches experiment                   | Reduces to Rayleigh–Jeans      |
| Overall spectral curve  | Fails completely                     | Perfect agreement              |
| Physical basis          | Equipartition theorem                | Boltzmann + quantization       |

The blackbody problem was the **birthplace of quantum theory**. Planck himself called his quantization assumption “an act of desperation” — yet it launched the entire field of modern physics. It directly inspired Einstein's photon concept (1905), Bohr's atomic model (1913), and ultimately the full development of quantum mechanics.

**Conclusion:** Blackbody radiation was the phenomenon that classical physics *could not* explain. Planck's revolutionary insight — that energy is quantized in units of  $h\nu$  — not only resolved the ultraviolet catastrophe perfectly but gave birth to quantum theory itself. The relationship between blackbody radiation and Planck's quantum theory is therefore not merely historical: it is the *origin* of our entire modern understanding of the subatomic world.

**Q10.** Concisely describe the physical significance of  $\Psi^2$  for a hydrogen atom. **(5 marks)**

[CHEM 113 | 2021–22]

**Answer**

**Physical Significance of  $\Psi^2$  for a Hydrogen Atom**

**(5 marks)**

In quantum mechanics, the wave function  $\Psi$  itself has no direct physical meaning. However, its square,  $\Psi^2$  (or more precisely  $|\Psi|^2$  for complex wave functions), carries deep physical significance:

**(i) Probability Density:**

$|\Psi|^2$  gives the **probability density** — the probability of finding the electron per unit volume at a given point in space. Specifically, the probability of finding the electron in a small volume element  $dV$  at position  $\mathbf{r}$  is:

$$P = |\Psi|^2 dV$$

A large value of  $|\Psi|^2$  at a point means the electron is *likely* to be found there; a small value means it is *unlikely*.

**(ii) Electron Density Cloud:**

For a hydrogen atom,  $|\Psi|^2$  describes the **electron density cloud** (or probability cloud) around the nucleus. This replaces the classical idea of a fixed circular orbit (Bohr model) with a three-dimensional probability distribution. The shape of this cloud defines what we call an **atomic orbital**.

**(iii) Normalization Condition:**

Since the electron must exist *somewhere* in space,  $|\Psi|^2$  integrated over all space must equal 1:

$$\int_{-\infty}^{\infty} |\Psi|^2 dV = 1$$

This ensures  $|\Psi|^2$  is a valid probability distribution.

**(iv) Radial Distribution:**

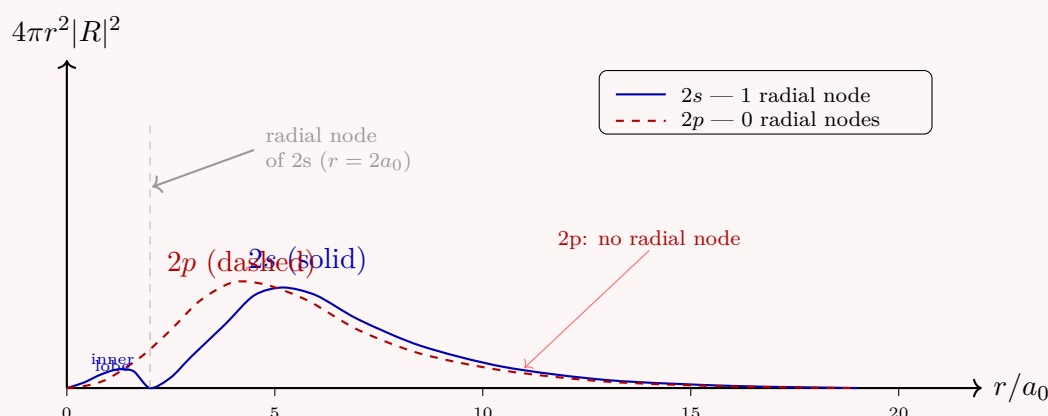
For hydrogen,  $|\Psi|^2$  depends on quantum numbers  $n$ ,  $l$ ,  $m_l$ . The most probable distance of the electron from the nucleus for the ground state ( $n = 1$ ) corresponds to the Bohr radius:

$$a_0 = 0.529 \text{ \AA}$$

confirming that  $|\Psi|^2$  correctly reproduces known atomic properties.

**Conclusion:**  $|\Psi|^2$  is the **probability density** of finding the electron at a given location. It transforms the abstract wave function into a physically measurable quantity — the spatial distribution of the electron around the nucleus — and forms the foundation of the orbital model of the atom.

- Q11.** Plot the radial probability distribution for a 2s orbital (as a solid line) and a 2p orbital (as a dashed line). For each orbital indicate any radial nodes with an arrow. Compare the shielding experienced by a 2s electron to that of a 2p electron and provide a brief explanation. **(10 marks)** [CHEM 113 | 2021–22]

**Answer**
**Radial Probability Distribution — 2s vs 2p:**

**Comparison of Shielding — 2s vs 2p:**

**Shielding** (or screening) refers to the reduction in the effective nuclear charge  $Z_{\text{eff}}$  experienced by an electron, caused by the repulsion from inner-shell electrons that partially cancel the nuclear attraction:

$$Z_{\text{eff}} = Z - \sigma$$

where  $Z$  is the atomic number and  $\sigma$  is the shielding constant.

**(a) 2s electron — less shielded (higher  $Z_{\text{eff}}$ ):**

The 2s radial wavefunction:

$$R_{20}(r) \propto \left(2 - \frac{r}{a_0}\right) e^{-r/2a_0}$$

has a **small inner lobe** (before the radial node at  $r = 2a_0$ ) that gives the 2s electron significant probability of **penetrating inside** the 1s core electrons. In this inner region, the electron is nearly unshielded and experiences the full nuclear charge  $Z$ . This penetration **reduces** the effective shielding:

$$\sigma_{2s} < \sigma_{2p} \implies Z_{\text{eff}}(2s) > Z_{\text{eff}}(2p)$$

**(b) 2p electron — more shielded (lower  $Z_{\text{eff}}$ ):**

The 2p radial wavefunction:

$$R_{21}(r) \propto \frac{r}{a_0} e^{-r/2a_0}$$

vanishes at  $r = 0$  due to the angular momentum barrier ( $l = 1$ ). The 2p electron has **no inner lobe** and **cannot penetrate** the 1s core. It therefore spends all its time *outside* the core electrons, which fully shield it from the nucleus, resulting in a lower  $Z_{\text{eff}}$ .



(c) **Consequence on energy:**

Since the 2s electron experiences a higher  $Z_{\text{eff}}$ , it is more strongly attracted to the nucleus and has a **lower (more negative) energy** than the 2p electron:

$$E_{2s} < E_{2p}$$

This is why, in multi-electron atoms, the 2s subshell **fills before** the 2p subshell (aufbau principle), and the 2s–2p energy gap increases across Period 2.

**Summary:**

| Orbital | Penetration            | Shielding $\sigma$ | $Z_{\text{eff}}$             |
|---------|------------------------|--------------------|------------------------------|
| 2s      | High (inner lobe)      | Lower              | Higher — closer to nucleus   |
| 2p      | None (angular barrier) | Higher             | Lower — farther from nucleus |

**Conclusion:** The **penetration** of the 2s orbital through the inner electron core, arising from its inner probability lobe, causes it to experience **less shielding** and a **higher effective nuclear charge** compared to the 2p orbital. This makes the 2s electron more tightly bound and lower in energy.

**Q12.** Consider a  $\text{Ca}^{19+}$  ion with its electron in the 5th excited state. (i) Calculate the longest wavelength of light that could be emitted when the  $\text{Ca}^{19+}$  electron transitions to a lower energy state. (ii) Suppose the same transition as in part (i) took place in a hydrogen atom. Would the wavelength of emission be longer than, shorter than, or the same as your answer? **(10 marks)** [CHEM 113 | 2021–22]

**Answer**

**Given:**

- $\text{Ca}^{19+}$  is a **hydrogenic ion** (one electron), atomic number  $Z = 20$
- 5th excited state  $\Rightarrow n = 1 + 5 = 6$
- Energy formula for a hydrogenic ion:

$$E_n = -\frac{Z^2 \times 13.6}{n^2} \text{ eV}$$

**(i) Longest wavelength of emitted light:**

Longest wavelength  $\Rightarrow$  lowest energy  $\Rightarrow$  smallest  $\Delta E \Rightarrow$  transition to the **nearest lower level**:  $n = 6 \rightarrow n = 5$

Calculate  $E_6$  and  $E_5$ :

$$E_6 = -\frac{(20)^2 \times 13.6}{6^2} = -\frac{400 \times 13.6}{36} = -\frac{5440}{36} = -151.11 \text{ eV}$$

$$E_5 = -\frac{(20)^2 \times 13.6}{5^2} = -\frac{400 \times 13.6}{25} = -\frac{5440}{25} = -217.60 \text{ eV}$$

**Energy of emitted photon:**

$$\Delta E = E_6 - E_5 = -151.11 - (-217.60) = 66.49 \text{ eV}$$

Converting to Joules:

$$\Delta E = 66.49 \times 1.602 \times 10^{-19} = 1.065 \times 10^{-17} \text{ J}$$

**Wavelength:**

$$\lambda = \frac{hc}{\Delta E} = \frac{(6.626 \times 10^{-34})(3.0 \times 10^8)}{1.065 \times 10^{-17}}$$

$$\lambda = 1.867 \times 10^{-8} \text{ m} = 18.67 \text{ nm}$$

This falls in the **extreme UV region** of the electromagnetic spectrum.

**(ii) Same transition ( $n = 6 \rightarrow n = 5$ ) in hydrogen ( $Z = 1$ ):**

$$\Delta E_H = 13.6 \left( \frac{1}{5^2} - \frac{1}{6^2} \right) = 13.6 \left( \frac{1}{25} - \frac{1}{36} \right) = 13.6 \times \frac{36 - 25}{25 \times 36} = 13.6 \times \frac{11}{900} = 0.1662 \text{ eV}$$

$$\Delta E_H = 0.1662 \times 1.602 \times 10^{-19} = 2.663 \times 10^{-20} \text{ J}$$

$$\lambda_H = \frac{hc}{\Delta E_H} = \frac{(6.626 \times 10^{-34})(3.0 \times 10^8)}{2.663 \times 10^{-20}} = 7.468 \times 10^{-6} \text{ m}$$

$$\lambda_H = 7,468 \text{ nm} \approx 7.47 \text{ } \mu\text{m}$$

This falls in the **infrared region**.

**Comparison:**

| System                         | $\Delta E$ | $\lambda$ | Region     |
|--------------------------------|------------|-----------|------------|
| $\text{Ca}^{19+}$ ( $Z = 20$ ) | 66.49 eV   | 18.67 nm  | Extreme UV |
| H ( $Z = 1$ )                  | 0.166 eV   | 7,468 nm  | Infrared   |

The hydrogen wavelength is **much longer** than that of  $\text{Ca}^{19+}$ .

**Reason:** For the same transition, energy scales as  $Z^2$ :

$$\Delta E \propto Z^2 \quad \Rightarrow \quad \lambda \propto \frac{1}{Z^2}$$

Since  $Z = 20$  for  $\text{Ca}^{19+}$ , the transition energy is  $Z^2 = 400$  times larger than for hydrogen, making the wavelength 400 times *shorter*. Verification:

$$\frac{\lambda_H}{\lambda_{\text{Ca}^{19+}}} = \frac{7468}{18.67} \approx 400 = Z^2 \quad \checkmark$$

**Conclusion:** For the same  $n = 6 \rightarrow 5$  transition, hydrogen emits at 7,468 nm (infrared), which is **far longer** than the 18.67 nm (extreme UV) emitted by  $\text{Ca}^{19+}$ . The  $Z^2$  dependence of transition energies in hydrogenic systems means that higher nuclear charge produces far more energetic — and therefore far shorter wavelength — emissions.

**Q13.** Explain why elements produce their own characteristic colors when they emit photons. How is the concept of electron density used to describe the motions and energies of submicroscopic particles in the quantum mechanical treatment of an atom? **(15 marks)**  
[CHEM 113 | 2020–21 (Jan 21)]

### Answer

#### Part 1: Characteristic Colors from Photon Emission

**(7 marks)**

Each element emits photons of specific wavelengths (colors) because its electrons occupy *discrete, quantized energy levels* unique to that element.

When an electron transitions from a higher energy level  $E_2$  to a lower level  $E_1$ , it emits a photon whose energy equals the difference:

$$\Delta E = E_2 - E_1 = h\nu = \frac{hc}{\lambda}$$

where  $h$  is Planck's constant,  $\nu$  is the photon frequency, and  $\lambda$  is its wavelength. Since  $\lambda$  determines color, and each element has a *unique* set of energy levels, each element produces its own unique spectrum of colors — its spectral “fingerprint.”

**Example — Hydrogen:** The Balmer series gives visible lines (e.g.  $n = 3 \rightarrow 2$  gives red at 656 nm;  $n = 4 \rightarrow 2$  gives blue-green at 486 nm).

#### Part 2: Electron Density in Quantum Mechanical Treatment

**(8 marks)**

In classical mechanics, electron motion is described by a definite trajectory. Quantum mechanics replaces this with the **wave function**  $\psi$ , obtained by solving the Schrödinger equation:

$$\hat{H}\psi = E\psi$$

**(i) Physical meaning of  $|\psi|^2$ :** The square of the wave function,  $|\psi|^2$ , gives the *probability density* — i.e., the probability of finding the electron per unit volume at a given point in space. This is called the **electron density**.

**(ii) Describing Motion:** Instead of saying “the electron moves in a circular orbit” (Bohr), quantum mechanics states that the electron exists as a *probability cloud* or *electron density cloud* around the nucleus. Regions of high  $|\psi|^2$  are where the electron is most likely to be found.

**(iii) Describing Energy:** Each wave function  $\psi_{n,l,m_l}$  corresponds to a definite, quantized energy  $E_n$ . For hydrogen:

$$E_n = -\frac{13.6 \text{ eV}}{n^2}, \quad n = 1, 2, 3, \dots$$

Higher  $n$  means a more diffuse electron cloud (larger orbital) and higher (less negative) energy.

**(iv) Heisenberg Uncertainty & Electron Density:** Because we cannot simultaneously know the exact position and momentum of an electron (Heisenberg's Uncertainty Principle:  $\Delta x \cdot \Delta p \geq h/4\pi$ ), electron density is the most physically meaningful description possible — it tells us *where* the electron is likely to be, not a fixed path.

**Q14.** What role did Einstein's explanation of the photoelectric effect play in the development of the particle-wave interpretation of the nature of electromagnetic radiation? Does a baseball in motion possess wave properties? If so, why can we not determine its wave properties? **(20 marks)**  
[CHEM 113 | 2020–21 (Jan 21)]

### Answer

#### Part 1: Einstein's Photoelectric Effect & Wave–Particle Duality of Light

**(12 marks)**

##### (i) Background — The Classical Wave Picture & Its Failure:

Before Einstein, light was understood purely as an electromagnetic *wave* (Maxwell, 1865). The classical wave theory predicted that:

- Any frequency of light, given enough intensity, should eject electrons.



- Higher intensity  $\Rightarrow$  higher kinetic energy of ejected electrons.
- There should be a measurable time delay before electrons are emitted.

Experiments showed *none* of these predictions were correct — classical wave theory completely failed to explain the photoelectric effect.

### (ii) Einstein's Explanation (1905):

Einstein proposed that light is not a continuous wave but is made of discrete energy packets called **photons**. The energy of each photon depends only on its frequency:

$$E = h\nu$$

where  $h = 6.626 \times 10^{-34}$  J·s is Planck's constant and  $\nu$  is the frequency of light.

A single photon interacts with a single electron. The electron is ejected only if the photon energy  $h\nu$  exceeds the **work function**  $\phi$  (the minimum energy needed to remove the electron from the metal surface):

$$h\nu = \phi + \frac{1}{2}mv_{\max}^2$$

$$\Rightarrow \frac{1}{2}mv_{\max}^2 = h\nu - h\nu_0 = h(\nu - \nu_0)$$

where  $\nu_0 = \phi/h$  is the **threshold frequency**. Below  $\nu_0$ , no electrons are emitted regardless of intensity.

### (iii) Key Experimental Facts Explained:

| Observation                                                  | Einstein's Explanation                                                          |
|--------------------------------------------------------------|---------------------------------------------------------------------------------|
| No emission below threshold frequency $\nu_0$                | Photon energy $h\nu < \phi$ ; not enough to free electron                       |
| KE of electrons depends on $\nu$ , not intensity             | Each electron absorbs exactly one photon; $\text{KE} = h(\nu - \nu_0)$          |
| Emission is instantaneous                                    | One photon $\rightarrow$ one electron; no energy accumulation needed            |
| More intensity $\Rightarrow$ more electrons, not faster ones | More photons $\Rightarrow$ more collisions, but each photon carries same $h\nu$ |

### (iv) Role in Wave-Particle Duality of Electromagnetic Radiation:

Einstein's work created a profound dilemma:

- Young's double-slit experiment (1801) and Maxwell's equations *proved* light is a wave — it diffracts and interferes.
- Einstein's photoelectric effect *proved* light behaves as a particle (photon) — it collides with electrons one-to-one.

Neither picture alone is complete. Light exhibits **wave-particle duality**:

$$\text{Wave nature: } \lambda, \nu, \text{ diffraction, interference} \quad \text{Particle nature: } E = h\nu, p = \frac{h}{\lambda}$$

This became the cornerstone of quantum mechanics and directly inspired de Broglie (1924) to ask: *if waves can behave as particles, can particles behave as waves?*

## Part 2: Does a Baseball Possess Wave Properties?

(8 marks)

### (i) de Broglie's Hypothesis:

In 1924, de Broglie proposed that *all* matter — not just light — has an associated wavelength:

$$\lambda = \frac{h}{mv} = \frac{h}{p}$$

where  $m$  is the mass,  $v$  is the velocity, and  $p = mv$  is the momentum of the object. This applies to *every* moving object, including a baseball.

### (ii) Calculation for a Baseball:

Consider a baseball of mass  $m = 0.145$  kg moving at  $v = 40$  m/s:

$$\lambda = \frac{h}{mv} = \frac{6.626 \times 10^{-34}}{0.145 \times 40} = \frac{6.626 \times 10^{-34}}{5.8} \approx 1.14 \times 10^{-34} \text{ m}$$

### (iii) Why We Cannot Observe the Baseball's Wave Properties:

- The de Broglie wavelength of the baseball,  $\lambda \approx 10^{-34}$  m, is *incomprehensibly* smaller than anything measurable. For comparison:

$$\text{diameter of a proton} \approx 10^{-15} \text{ m} \quad \gg \quad \lambda_{\text{baseball}} \approx 10^{-34} \text{ m}$$

- Wave properties such as diffraction and interference are only observable when the wavelength is *comparable* to the size of the obstacle or slit. No physical apparatus can come close to resolving  $10^{-34}$  m.
- For *subatomic* particles like electrons ( $m \approx 9.1 \times 10^{-31}$  kg),  $\lambda$  falls in the Ångström range ( $\sim 10^{-10}$  m), comparable to atomic spacings — hence electrons *do* show measurable diffraction (Davisson-Germer experiment, 1927).

- The Heisenberg Uncertainty Principle also tells us that for a macroscopic object the uncertainty in momentum required to detect wave behavior would be physically meaningless:

$$\Delta x \cdot \Delta p \geq \frac{h}{4\pi}$$

For a baseball,  $\Delta p$  is negligibly small compared to  $mv$ , so quantum effects are entirely undetectable.

#### Conclusion:

Yes, a baseball *does* possess wave properties according to de Broglie's relation  $\lambda = h/mv$ . However, because its mass is enormous compared to subatomic particles, its wavelength is so vanishingly small ( $\sim 10^{-34}$  m) that no experiment can detect it. Wave-particle duality is a universal property of matter, but it is only *observable* at the quantum (subatomic) scale where  $\lambda$  becomes comparable to atomic dimensions.

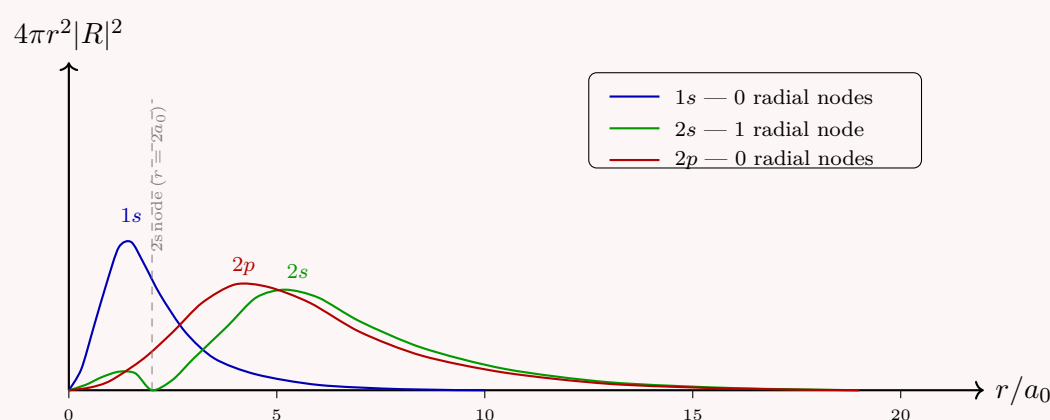
**Q15.** Show the radial distribution function for hydrogenic 1s, 2s and 2p orbitals with nodes. Why does 2s give the electron a greater probability of close approach to the nucleus compared to 2p orbital? **(10 marks)** [CHEM 113 | 2020–21]

#### Answer

The **radial distribution function (RDF)** gives the probability of finding an electron in a thin spherical shell of radius  $r$ :

$$P(r) = 4\pi r^2 [R_{nl}(r)]^2$$

#### RDF Plots for 1s, 2s, and 2p:



#### Nodal Structure:

| Orbital | Radial nodes<br>( $n - l - 1$ ) | Angular nodes ( $l$ ) | Peak position            |
|---------|---------------------------------|-----------------------|--------------------------|
| 1s      | 0                               | 0                     | $r = a_0$                |
| 2s      | 1 (at $r = 2a_0$ )              | 0                     | $r \approx 5a_0$ (major) |
| 2p      | 0                               | 1 (nodal plane)       | $r \approx 4a_0$         |

#### Why 2s gives greater probability of close approach than 2p:

- (a) **Penetration via inner lobe:** The radial wavefunction of the 2s orbital is:

$$R_{20}(r) = \frac{1}{2\sqrt{2}} \left( \frac{1}{a_0} \right)^{3/2} \left( 2 - \frac{r}{a_0} \right) e^{-r/2a_0}$$

The factor  $(2 - r/a_0)$  does not vanish at  $r = 0$ , so the 2s RDF has a **small but nonzero inner peak** between  $r = 0$  and the node at  $r = 2a_0$ . This inner lobe represents a real probability of finding the 2s electron very close to the nucleus — even inside the 1s shell.

- (b) **Angular momentum barrier for 2p:** The radial wavefunction of the 2p orbital contains the factor  $r/a_0$ :

$$R_{21}(r) = \frac{1}{2\sqrt{6}} \left( \frac{1}{a_0} \right)^{3/2} \frac{r}{a_0} e^{-r/2a_0}$$

This factor forces  $R_{21}(0) = 0$ , meaning the 2p RDF is **strictly zero at the nucleus**. Physically, the angular momentum ( $l = 1$ ) creates a centrifugal barrier that repels the electron away from the nucleus — there is no inner peak and no penetration.

- (c) **Consequence — effective nuclear charge:** Because the 2s electron penetrates closer to the nucleus, it experiences less shielding from inner electrons and sees a higher effective nuclear charge:

$$Z_{\text{eff}}(2s) > Z_{\text{eff}}(2p)$$

This makes the 2s electron **more tightly bound** and lower in energy than the 2p electron — which is why 2s fills before 2p in the aufbau principle.

**Conclusion:** The presence of a **radial node** in the 2s orbital creates an inner probability lobe (penetration), giving the 2s electron significant access to the nuclear region. The 2p electron, excluded by its angular momentum barrier ( $l = 1$ ), has no such inner peak — making 2s electrons closer to and more strongly bound by the nucleus than 2p electrons.

**Q16.** In making a transition from an orbital with a principal quantum number of 4 to an orbital with a principal quantum number of 7, does the electron of a hydrogen atom emit or absorb a photon of energy? What would be the energy of the photon? To what region of the electromagnetic spectrum does this energy correspond? **(10 marks)** [CHEM 113 | 2020–21]

### Answer

#### Given:

- Hydrogen atom:  $Z = 1$
- Initial state:  $n_i = 4$
- Final state:  $n_f = 7$
- Energy of hydrogen atom:

$$E_n = -\frac{13.6}{n^2} \text{ eV}$$

#### (i) Emission or Absorption?

The electron transitions from  $n = 4$  to  $n = 7$ , i.e. from a **lower** to a **higher** energy level. Moving away from the nucleus requires energy input. Therefore, the electron **absorbs** a photon.

$$n_i = 4 \longrightarrow n_f = 7 \quad (\text{upward transition} \Rightarrow \text{absorption})$$

#### (ii) Energy of the photon:

Calculate  $E_4$  and  $E_7$ :

$$E_4 = -\frac{13.6}{4^2} = -\frac{13.6}{16} = -0.850 \text{ eV}$$

$$E_7 = -\frac{13.6}{7^2} = -\frac{13.6}{49} = -0.2776 \text{ eV}$$

#### Energy of absorbed photon:

$$\Delta E = E_7 - E_4 = -0.2776 - (-0.850) = +0.5724 \text{ eV}$$

The positive sign confirms **absorption**.

Converting to Joules:

$$\Delta E = 0.5724 \times 1.602 \times 10^{-19} = 9.17 \times 10^{-20} \text{ J}$$

$$\Delta E = +0.5724 \text{ eV} = 9.17 \times 10^{-20} \text{ J}$$

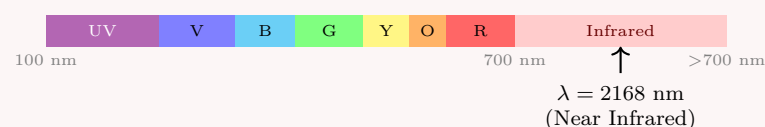
#### (iii) Wavelength and region of electromagnetic spectrum:

$$\lambda = \frac{hc}{\Delta E} = \frac{(6.626 \times 10^{-34})(3.0 \times 10^8)}{9.17 \times 10^{-20}}$$

$$\lambda = \frac{1.988 \times 10^{-25}}{9.17 \times 10^{-20}} = 2.168 \times 10^{-6} \text{ m}$$

$$\lambda = 2168 \text{ nm} \approx 2.17 \mu\text{m}$$

#### Electromagnetic Spectrum Region:



Since  $\lambda = 2168 \text{ nm}$  lies well beyond the visible red limit of  $\sim 700 \text{ nm}$ , this photon belongs to the **near-infrared (NIR)** region of the electromagnetic spectrum.

#### Summary:

| Quantity        | Value                                                 |
|-----------------|-------------------------------------------------------|
| Transition      | $n = 4 \rightarrow n = 7$ (absorption)                |
| $\Delta E$      | $+0.5724 \text{ eV} = 9.17 \times 10^{-20} \text{ J}$ |
| $\lambda$       | $2168 \text{ nm} = 2.17 \mu\text{m}$                  |
| Spectral region | Near-infrared (NIR)                                   |
| Spectral series | Brackett series ( $n_f = 4$ )                         |

**Conclusion:** Since the transition is from a lower ( $n = 4$ ) to a higher ( $n = 7$ ) energy level, the electron **absorbs** a photon of energy 0.5724 eV. The corresponding wavelength of 2168 nm places this absorption in the **near-infrared** region, specifically within the **Brackett series** of the hydrogen spectrum.

**Q17.** Account for the large decrease in electron affinity between Li and Be despite the increase in nuclear charge? **(5 marks)** [CHEM 113 | 2020–21]

### Answer

#### Electron Affinities:

$$\text{Li} : -59.6 \text{ kJ/mol} \quad \text{Be} : \approx +18 \text{ kJ/mol}$$

Be has a **positive** (unfavourable) electron affinity despite having a higher nuclear charge ( $Z = 4$ ) than Li ( $Z = 3$ ). This seems contradictory but is explained by their electron configurations:

#### Electron Configurations:

$$\text{Li} : 1s^2 2s^1 \quad \text{Be} : 1s^2 2s^2$$

**Li** adding an electron:



The incoming electron enters the **vacant**  $2s$  orbital — no extra repulsion, moderate  $Z_{\text{eff}}$ , stabilisation occurs.

**Be** adding an electron:



The  $2s$  subshell is **completely full** in Be. The incoming electron must enter the **higher energy**  $2p$  orbital.

#### Three reasons for the large decrease:

(a) **Subshell energy penalty:**

The  $2p$  orbital is **higher in energy** than  $2s$  due to poorer penetration and greater shielding:

$$E_{2p} > E_{2s}$$

Placing the extra electron in  $2p$  requires overcoming this energy gap — making the process *endothermic*.

(b) **Increased shielding in  $2p$ :**

The added electron in the  $2p$  orbital is **more shielded** by the filled  $2s^2$  electrons than Li's added electron was. This reduces  $Z_{\text{eff}}$  felt by the incoming electron in Be, despite the higher nuclear charge:

$$Z_{\text{eff}}(\text{Be}, 2p) < Z_{\text{eff}}(\text{Li}, 2s)$$

(c) **Stability of filled subshell:**

Be has a **completely filled**  $2s^2$  subshell, which confers extra stability (exchange energy). Disrupting this closed-subshell configuration by forcing an electron into  $2p$  is energetically unfavourable — Be *resists* gaining an electron.

#### Summary:

| Atom | Config.     | Added $e^-$ en-<br>ters        | EA                     |
|------|-------------|--------------------------------|------------------------|
| Li   | $1s^2 2s^1$ | $2s$ (half-filled, low energy) | $-59.6 \text{ kJ/mol}$ |
| Be   | $1s^2 2s^2$ | $2p$ (empty, higher energy)    | $+18 \text{ kJ/mol}$   |

**Conclusion:** Despite Be having a higher nuclear charge, its **completely filled  $2s^2$  subshell** forces the incoming electron into the higher-energy, more-shielded  $2p$  orbital — making electron gain energetically unfavourable and causing a large *decrease* in electron affinity compared to Li.

**Q18.** How is the concept of electron density used to describe the position of an electron in the quantum mechanical treatment of an atom? **(7–15 marks)** [CHEM 113 | 2017–18; 2019–20]

### Answer

#### Electron Density in the Quantum Mechanical Treatment of an Atom

**(15 marks)**

#### Part 1: Failure of the Classical Picture

**(2 marks)**

In Bohr's model, the electron moves in a fixed circular orbit around the nucleus — a precise, deterministic trajectory. This picture completely contradicts the **Heisenberg Uncertainty Principle**:

$$\Delta x \cdot \Delta p \geq \frac{h}{4\pi}$$

It is fundamentally impossible to know both the position and momentum of an electron simultaneously. Therefore, a fixed orbit is physically meaningless at the atomic scale. Quantum mechanics replaces the concept of a trajectory with the concept of **electron density**.

## Part 2: The Wave Function $\Psi$ and Electron Density

(3 marks)

In quantum mechanics, the state of an electron is described by the **wave function**  $\Psi$ , obtained by solving the Schrödinger equation:

$$\hat{H}\Psi = E\Psi$$

The wave function  $\Psi$  itself has no direct physical meaning. However, its square  $|\Psi|^2$  — called the **electron density** or **probability density** — has a precise physical interpretation:

$$\text{Probability of finding electron in volume } dV = |\Psi|^2 dV$$

- Where  $|\Psi|^2$  is **large** — the electron is *likely* to be found (high electron density).
- Where  $|\Psi|^2$  is **small** — the electron is *unlikely* to be found (low electron density).
- Where  $|\Psi|^2 = 0$  — the electron is *never* found (a **nodal surface**).

Since the electron must exist somewhere in space,  $|\Psi|^2$  is normalized:

$$\int_{-\infty}^{\infty} |\Psi|^2 dV = 1$$

## Part 3: Electron Density as a Probability Cloud

(3 marks)

The electron density  $|\Psi|^2$  is visualized as a **probability cloud** or **charge cloud** around the nucleus. The density of dots in the cloud at any point represents the value of  $|\Psi|^2$  there:

- **Dense region** (near nucleus for 1s)  $\Rightarrow$  electron spends most time here.
- **Sparse region** (far from nucleus)  $\Rightarrow$  electron rarely found here.
- The cloud has *no sharp boundary* — it fades gradually, which is why orbitals are drawn as 90% probability surfaces.

This cloud picture is fundamentally different from Bohr's orbit: the electron does not “travel” along a path — it simply has a *higher or lower probability* of being at each location.

## Part 4: Radial Distribution Function

(3 marks)

To describe *how far* the electron is from the nucleus, we use the **radial distribution function** (RDF), defined as:

$$P(r) = 4\pi r^2 |\Psi|^2$$

This gives the probability of finding the electron in a thin spherical shell of radius  $r$  and thickness  $dr$ .

For the hydrogen 1s orbital:

$$\Psi_{1s} = \frac{1}{\sqrt{\pi}} \left( \frac{1}{a_0} \right)^{3/2} e^{-r/a_0}$$

$$P(r) = 4\pi r^2 \cdot \frac{1}{\pi a_0^3} e^{-2r/a_0} = \frac{4r^2}{a_0^3} e^{-2r/a_0}$$

The maximum of  $P(r)$  occurs at  $r = a_0 = 0.529 \text{ \AA}$  — exactly the Bohr radius. This confirms that quantum mechanics *predicts* the same most-probable distance as Bohr, but as a *probability maximum* rather than a fixed orbit.

## Part 5: Quantum Numbers and Electron Density Shapes

(2 marks)

The shape of the electron density cloud depends on the quantum numbers ( $n, l, m_l$ ):

| Orbital | $n$ | $l$ | Electron Density Shape                        |
|---------|-----|-----|-----------------------------------------------|
| 1s      | 1   | 0   | Spherically symmetric; maximum at nucleus     |
| 2s      | 2   | 0   | Spherical with one nodal sphere at $r = 2a_0$ |
| 2p      | 2   | 1   | Dumbbell-shaped along $x$ , $y$ , or $z$ axis |
| 3d      | 3   | 2   | Complex lobed shapes; two nodal surfaces      |

**Nodes:** A **node** is a surface where  $|\Psi|^2 = 0$  — the electron has zero probability of being found there.

- Radial nodes =  $n - l - 1$
- Angular nodes =  $l$
- Total nodes =  $n - 1$

**Conclusion:** Electron density  $|\Psi|^2$  replaces the classical orbit with a *probability cloud* that tells us where the electron is *likely* to be found — never where it *will* be. The radial distribution function identifies the most probable distance, which correctly predicts the Bohr radius for hydrogen. The shape of the cloud, determined by quantum numbers, defines the geometry of atomic orbitals and ultimately governs chemical bonding.



- Q19.** Point out three characteristic features in the photoelectric effect that cannot be explained on the basis of wave theory of light. Provide a brief explanation of those features. **(15 marks)** [CHEM 113 | 2019–20]

### Answer

#### Three Characteristic Features of the Photoelectric Effect Inexplicable by Wave Theory

**(15 marks)**

##### Feature 1: Existence of a Threshold Frequency ( $\nu_0$ )

**(5 marks)**

**Observation:** Electrons are emitted from a metal surface only when the frequency of incident light exceeds a minimum value  $\nu_0$ , called the **threshold frequency**. Below  $\nu_0$ , *no* electrons are ejected, no matter how intense the light is or how long it is shone on the surface.

##### Why Wave Theory Fails:

According to the classical wave theory, light energy is spread continuously over the wavefront. A wave of *any* frequency, given sufficient intensity or time, should deliver enough energy to eject electrons. There is no reason for a threshold frequency to exist. Yet experiments clearly show one does — a direct contradiction of the wave picture.

##### Einstein's Explanation:

Light consists of photons, each carrying energy  $E = h\nu$ . An electron is ejected only if a single photon has energy at least equal to the work function  $\phi$  of the metal:

$$h\nu \geq \phi = h\nu_0 \quad \Rightarrow \quad \nu \geq \nu_0 = \frac{\phi}{h}$$

Below  $\nu_0$ , even an enormous number of low-energy photons cannot collectively free a single electron, because each electron interaction involves exactly *one* photon.

##### Feature 2: Kinetic Energy of Ejected Electrons Depends on Frequency, Not Intensity

**(5 marks)**

**Observation:** The maximum kinetic energy of the emitted electrons increases linearly with the *frequency* of light, and is completely *independent* of the intensity. Increasing the intensity only increases the *number* of ejected electrons, not their speed.

##### Why Wave Theory Fails:

Classical wave theory predicts that a more intense wave carries more energy. Therefore, higher intensity should deliver more energy to the electrons and eject them with greater kinetic energy. Experiments show the exact opposite — intensity has *no effect* on the kinetic energy of individual electrons. This is completely at odds with the wave model.

##### Einstein's Explanation:

Each electron absorbs exactly one photon. The photon's energy  $h\nu$  goes partly to overcoming the work function  $\phi$ , and the rest becomes kinetic energy:

$$\frac{1}{2}mv_{\max}^2 = h\nu - \phi = h(\nu - \nu_0)$$

Since  $h$  and  $\phi$  are constants, KE depends *only* on  $\nu$ . Higher intensity simply means more photons per second  $\Rightarrow$  more electrons ejected, but each photon still carries the same  $h\nu$ .

##### Feature 3: Instantaneous Emission — No Time Delay

**(5 marks)**

**Observation:** Electron emission begins **instantaneously** (within  $\sim 10^{-9}$  s) as soon as light of frequency  $\nu > \nu_0$  strikes the metal, even if the light is extremely dim.

##### Why Wave Theory Fails:

A continuous wave spreads its energy over a large area of the metal surface. Classical theory predicts that a faint beam of light would need to shine for a considerable time — sometimes minutes or even hours — before any single electron accumulates enough energy to escape. Experiments detect no such delay whatsoever, even with the faintest light sources.

##### Einstein's Explanation:

Because light energy is concentrated in discrete photons, a single photon delivers its entire energy  $h\nu$  to a single electron *all at once* — there is no gradual accumulation. As long as  $h\nu \geq \phi$ , emission is immediate, regardless of how few photons are present:

$$\Delta t_{\text{emission}} \approx 10^{-9} \text{ s} \quad \longleftrightarrow \quad \text{single photon-electron collision}$$

#### Summary Table:

| Feature             | Wave Theory Predicts                               | Experiment Shows                                      |
|---------------------|----------------------------------------------------|-------------------------------------------------------|
| Threshold frequency | Any frequency should work (given enough intensity) | Emission only above $\nu_0$ , regardless of intensity |
| KE of electrons     | KE increases with intensity                        | KE increases with $\nu$ only; intensity has no effect |
| Time delay          | Long delay for dim light                           | Instantaneous emission always                         |

**Q20.** The wave nature of matter is not noticeable in our daily observations — Why? **(5 marks)**

[CHEM 113 | 2017–18]

### Answer

The wave nature of matter is described by the **de Broglie wavelength**:

$$\lambda = \frac{h}{mv}$$

where  $h = 6.626 \times 10^{-34}$  J·s is Planck's constant,  $m$  is the mass, and  $v$  is the velocity of the object.

**Why it is undetectable in daily life:**

For a macroscopic object (e.g., a cricket ball,  $m \approx 0.15$  kg,  $v \approx 40$  m/s):

$$\lambda = \frac{6.626 \times 10^{-34}}{0.15 \times 40} = 1.1 \times 10^{-34} \text{ m}$$

This is *incredibly smaller* than any measurable length — far smaller than an atomic nucleus ( $\sim 10^{-15}$  m).

In contrast, for an **electron** ( $m = 9.11 \times 10^{-31}$  kg,  $v \approx 10^6$  m/s):

$$\lambda = \frac{6.626 \times 10^{-34}}{9.11 \times 10^{-31} \times 10^6} \approx 7.27 \times 10^{-10} \text{ m}$$

This is comparable to atomic/molecular dimensions, so wave behavior *is* observable.

**Conclusion:** Since  $h$  is extremely small, macroscopic objects have wavelengths far too tiny to detect with any instrument, making wave nature completely imperceptible in everyday observations. Only **subatomic particles** have significant de Broglie wavelengths.

**Q21.** Describe the experimental basis for believing that the electrons in an atom behave as tiny bar magnets. **(15 marks)** [CHEM 113 | 2017–18]

### Answer

The experimental evidence for electrons behaving as tiny bar magnets comes from three key observations:

#### 1. The Stern–Gerlach Experiment (1922)

(most direct evidence)

Stern and Gerlach passed a beam of silver atoms through a **non-uniform magnetic field**. Classical theory predicted the beam should spread into a *continuous* band. Instead, the beam split into **exactly two discrete spots**, indicating that the magnetic moment of the electron can only take two orientations:

$$m_s = +\frac{1}{2} \quad (\text{spin up, } \uparrow) \quad \text{and} \quad m_s = -\frac{1}{2} \quad (\text{spin down, } \downarrow)$$

This is analogous to a bar magnet having a north and south pole — it aligns either with or against an external field.

#### 2. Electron Spin & Magnetic Moment

To explain the Stern–Gerlach result, Uhlenbeck and Goudsmit (1925) proposed that every electron possesses an intrinsic angular momentum called **spin**, giving rise to a **spin magnetic moment**:

$$\mu_s = -g_s \frac{e}{2m_e} \mathbf{S}, \quad g_s \approx 2.002$$

where  $\mathbf{S}$  is the spin angular momentum,  $e$  is electron charge, and  $m_e$  is electron mass. The magnitude of the spin magnetic moment is:

$$|\mu_s| = g_s \sqrt{s(s+1)} \mu_B \approx \sqrt{3} \mu_B$$

where  $\mu_B = \frac{e\hbar}{2m_e} = 9.274 \times 10^{-24}$  J/T is the **Bohr magneton** — the fundamental unit of electron magnetism.

#### 3. The Anomalous Zeeman Effect

When atoms are placed in an external magnetic field, spectral lines split. The **normal Zeeman effect** (orbital motion only) predicts splitting into 3 lines. However, experiments showed splitting into **more than 3 lines** — the *anomalous* Zeeman effect. This could only be explained by including the **intrinsic spin magnetic moment** of the electron in addition to its orbital magnetic moment:

$$\mu_{\text{total}} = \mu_L + \mu_S$$

The additional splitting confirmed that electrons carry an independent magnetic dipole moment, just like a bar magnet.

#### 4. Fine Structure of Spectral Lines (Spin–Orbit Coupling)

High-resolution spectroscopy reveals that single spectral lines are actually **closely spaced doublets** (e.g., the sodium D-line at 589 nm splits into 589.0 and 589.6 nm). This **fine structure** arises from the interaction between the electron's *orbital* magnetic moment and its *spin* magnetic moment:

$$E_{SO} \propto \mathbf{L} \cdot \mathbf{S}$$

This spin–orbit coupling is only possible if the electron possesses a true magnetic moment, further confirming its bar-magnet behavior.

**Summary Table:**



| No. | Experiment / Observation  | Evidence Provided                   |
|-----|---------------------------|-------------------------------------|
| 1   | Stern–Gerlach experiment  | Two discrete magnetic orientations  |
| 2   | Electron spin & $\mu_B$   | Intrinsic magnetic dipole moment    |
| 3   | Anomalous Zeeman effect   | Spin magnetic moment beyond orbital |
| 4   | Fine structure (doublets) | Spin–orbit magnetic interaction     |

**Conclusion:** All four independent experimental observations consistently demonstrate that electrons possess an intrinsic, quantized magnetic moment — behaving exactly like tiny bar magnets with only two allowed orientations in a magnetic field.

**Q22.** Describe the role played in elucidating the atomic structures by each of the following: (i) Positive rays, (ii) Cathode rays, (iii) X-rays.  
(18 marks) [CHEM 113 | 2017–18]

### Answer

Each of these three discoveries played a foundational role in building our modern understanding of atomic structure:

#### (i) Positive Rays

(6 marks)

Positive rays (also called **canal rays**) were discovered by Goldstein (1886) using a discharge tube with a perforated cathode. Rays passed *through* the holes in the cathode, travelling opposite to cathode rays.

**Key contributions to atomic structure:**

- Proved the existence of **positively charged particles** inside atoms, later identified as **protons**.
- J.J. Thomson’s parabola method (1913) used positive rays to measure the **charge-to-mass ratio** ( $e/m$ ) of positive ions, showing it varied with the gas used — unlike cathode rays.
- Aston’s **mass spectrograph** (developed from positive ray analysis) revealed the existence of **isotopes** — atoms of the same element with different masses — proving that atomic nuclei are not all identical.
- Established that the proton has:

$$m_p = 1.673 \times 10^{-27} \text{ kg}, \quad \text{charge} = +1.6 \times 10^{-19} \text{ C}$$

- Confirmed **electrical neutrality** of atoms: the number of positive charges (protons) equals the number of negative charges (electrons).

#### (ii) Cathode Rays

(6 marks)

Cathode rays were discovered by Julius Plücker (1858) and studied extensively by J.J. Thomson (1897) using evacuated discharge tubes. They travel from the cathode to the anode regardless of the cathode material.

**Key contributions to atomic structure:**

- Thomson proved cathode rays consist of **negatively charged particles** — the **electron** — by deflecting them in electric and magnetic fields.
- Measured the charge-to-mass ratio of the electron:

$$\frac{e}{m_e} = 1.758 \times 10^{11} \text{ C/kg}$$

- Millikan’s oil-drop experiment (using knowledge from cathode rays) determined the **absolute charge** of the electron:

$$e = 1.602 \times 10^{-19} \text{ C}, \quad m_e = 9.109 \times 10^{-31} \text{ kg}$$

- The **same**  $e/m$  ratio regardless of cathode material proved electrons are **universal constituents** of all atoms.
- Led directly to Thomson’s **plum pudding model** (1904) — the first quantitative atomic model, proposing electrons embedded in a diffuse positive sphere.
- Paved the way for Rutherford’s nuclear model and Bohr’s quantized model.

#### (iii) X-rays

(6 marks)

X-rays were discovered by Wilhelm Röntgen (1895) when cathode rays struck a metal target. They are high-energy electromagnetic radiation with wavelengths  $\sim 10^{-10}$  m.

**Key contributions to atomic structure:**

- Moseley’s Law (1913):** Moseley measured the characteristic X-ray frequencies emitted by different elements and found:

$$\sqrt{\nu} = a(Z - b)$$

where  $\nu$  is X-ray frequency,  $Z$  is the **atomic number**, and  $a, b$  are constants. This established  $Z$  (number of protons) as the true identity of an element — more fundamental than atomic mass.

- Moseley’s work **corrected the order** of elements in the periodic table (e.g., Ar before K) and predicted missing elements.

- **X-ray diffraction** (von Laue, 1912; Bragg, 1913) used the Bragg equation:

$$n\lambda = 2d \sin \theta$$

to determine **crystal structures** and inter-atomic distances, providing direct evidence of atomic arrangement in matter.

- Characteristic X-ray spectra ( $K_\alpha$ ,  $K_\beta$  lines) arise from **electron transitions between inner shells**, confirming the **shell structure** of atoms predicted by Bohr.
- Compton scattering (1923) of X-rays by electrons confirmed the **particle nature of photons** and the wave-particle duality central to quantum atomic theory.

**Summary:**

| Ray           | Key Scientist(s)          | Atomic Structure Contribution                         |
|---------------|---------------------------|-------------------------------------------------------|
| Positive rays | Goldstein, Thomson, Aston | Discovered proton; identified isotopes                |
| Cathode rays  | Thomson, Millikan         | Discovered electron; $e/m$ ratio                      |
| X-rays        | Röntgen, Moseley, Bragg   | Atomic number $Z$ ; shell structure; crystal geometry |

**Conclusion:** Together, positive rays, cathode rays, and X-rays revealed the three fundamental building blocks of atomic structure — the **proton**, the **electron**, and the **quantized shell arrangement** — forming the experimental foundation of modern atomic theory.

- Q23.** Crystal diffraction can be studied by electron diffraction as well as by neutron diffraction method. What should be the ratio of the velocities of electron and neutron for the de Broglie wavelength to be the same? (You don't need to use any data to solve this problem.) **(5 marks)** [CHEM 113 | 2017–18]

**Answer**

For the de Broglie wavelength to be the same for both electron and neutron:

$$\lambda_e = \lambda_n$$

$$\frac{h}{m_e v_e} = \frac{h}{m_n v_n}$$

Cancelling  $h$  from both sides:

$$m_e v_e = m_n v_n$$

$$\boxed{\frac{v_e}{v_n} = \frac{m_n}{m_e} \approx \frac{1836 m_e}{m_e} = 1836}$$

**Conclusion:** The electron must travel **1836 times faster** than the neutron to have the same de Broglie wavelength, since the neutron is 1836 times heavier than the electron.

- Q24.** Schrödinger equation can be represented in a simpler form as  $H\Psi = E\Psi$ , where  $E$  is the energy and  $\Psi$  is called a wave function. What information can you get from the equation about an electron in an atom? **(6 marks)** [CHEM 113 | 2016–17]

**Answer**

The **Schrödinger equation** in its compact operator form is:

$$\hat{H}\Psi = E\Psi$$

where  $\hat{H}$  is the **Hamiltonian operator** (representing total energy = kinetic + potential),  $\Psi$  is the **wave function**, and  $E$  is the **total energy** of the electron.

**Information obtained from the equation:**

- (a) **Allowed Energy Levels ( $E$ ):** Solving the equation gives *discrete, quantized* energy values  $E_1, E_2, E_3, \dots$  for the electron — explaining why atomic spectra consist of sharp lines rather than a continuum. For hydrogen:

$$E_n = -\frac{13.6}{n^2} \text{ eV}, \quad n = 1, 2, 3, \dots$$

- (b) **The Wave Function ( $\Psi$ ) — Atomic Orbitals:** Each solution  $\Psi$  (called an **eigenfunction**) describes a specific **atomic orbital**, characterized by three quantum numbers ( $n, l, m_l$ ). The shape and orientation of orbitals (s, p, d, f) are directly obtained from  $\Psi$ .

- (c) **Probability Distribution ( $\Psi^2$ ):**  $|\Psi|^2$  at any point gives the **probability density** of finding the electron at that location:

$$P = |\Psi|^2 dV$$

This replaces the classical concept of fixed orbits with **electron clouds** or probability maps.

(d) **Quantum Numbers:** The boundary conditions imposed while solving  $\hat{H}\Psi = E\Psi$  naturally give rise to the three quantum numbers:

- $n$  — principal quantum number (energy level, size of orbital)
- $l$  — azimuthal quantum number (shape of orbital)
- $m_l$  — magnetic quantum number (orientation of orbital)

(e) **Total Energy of the Electron:** The eigenvalue  $E$  gives the **total mechanical energy** (kinetic + potential) of the electron in a given state, allowing calculation of ionization energies and spectral transition frequencies via:

$$\Delta E = E_{\text{final}} - E_{\text{initial}} = h\nu$$

(f) **Nodal Surfaces:** Regions where  $\Psi = 0$  (and hence  $|\Psi|^2 = 0$ ) are called **nodes** — surfaces where the probability of finding the electron is zero. The number of nodes =  $(n - 1)$ , giving insight into the spatial structure of orbitals.

**Conclusion:** The Schrödinger equation provides a **complete quantum mechanical description** of the electron in an atom — its energy, spatial distribution, orbital shape, and the quantum numbers governing its state — replacing the limited Bohr model with a fully probabilistic and mathematically rigorous framework.

**Q25.** Justify the statement: “The probability of finding two electrons with the same four quantum numbers in an atom is zero.” (10 marks) [CHEM 113 | 2016–17]

### Answer

The statement is a direct expression of the **Pauli Exclusion Principle**, and can be justified from multiple perspectives:

#### 1. Statement of the Pauli Exclusion Principle (1925):

Wolfgang Pauli proposed that:

*No two electrons in an atom can have the same set of all four quantum numbers ( $n, l, m_l, m_s$ ).*

Since the four quantum numbers completely specify the **quantum state** of an electron, two electrons sharing all four would occupy the *identical* quantum state — which is forbidden.

#### 2. The Four Quantum Numbers and Their Roles:

| Sym-<br>bol | Name          | Values                       | Describes              |
|-------------|---------------|------------------------------|------------------------|
| $n$         | Principal     | $1, 2, 3, \dots$             | Energy level & size    |
| $l$         | Azimuthal     | $0$ to $(n - 1)$             | Shape of orbital       |
| $m_l$       | Magnetic      | $-l$ to $+l$                 | Orientation of orbital |
| $m_s$       | Spin magnetic | $+\frac{1}{2}, -\frac{1}{2}$ | Spin direction         |

The first three ( $n, l, m_l$ ) define a unique **orbital**. The fourth,  $m_s$ , distinguishes the **two electrons** within that orbital.

#### 3. Quantum Mechanical Justification — Antisymmetry of Wave Functions:

Electrons are **fermions** (spin- $\frac{1}{2}$  particles). The total wave function of a multi-electron system must be **antisymmetric** with respect to exchange of any two electrons:

$$\Psi(1, 2) = -\Psi(2, 1)$$

If two electrons had identical quantum numbers, exchanging them would give:

$$\Psi(1, 2) = \Psi(2, 1)$$

But antisymmetry requires  $\Psi(1, 2) = -\Psi(2, 1)$ . Both conditions can only be satisfied simultaneously if:

$$\Psi(1, 2) = 0$$

A zero wave function means **zero probability density**  $|\Psi|^2 = 0$  — the state simply does not exist. This is the deepest quantum mechanical proof of the exclusion principle.

#### 4. The Slater Determinant:

For a two-electron system, the antisymmetric wave function is written as a **Slater determinant**:

$$\Psi(1, 2) = \frac{1}{\sqrt{2}} \begin{vmatrix} \psi_a(1) & \psi_b(1) \\ \psi_a(2) & \psi_b(2) \end{vmatrix}$$

If both electrons occupy the same state,  $\psi_a = \psi_b$ , then two rows of the determinant become identical, and:

$$\Psi(1, 2) = 0 \quad \Rightarrow \quad |\Psi|^2 = 0$$

This mathematically confirms that the **probability is exactly zero**.

#### 5. Physical Consequences & Justification:

- Each orbital (defined by  $n, l, m_l$ ) can hold **at most 2 electrons**, with opposite spins ( $m_s = +\frac{1}{2}$  and  $-\frac{1}{2}$ ).
- A subshell with  $l$  has  $(2l + 1)$  orbitals, so it holds at most  $2(2l + 1)$  electrons.
- The **electronic configuration** of all elements, the **periodic table structure**, and **chemical bonding** are all direct consequences of this principle.
- Without the exclusion principle, all electrons would collapse to the lowest  $n = 1$  state — atoms as we know them would not exist.

**Example — Helium (He,  $Z = 2$ ):**

Both electrons occupy the  $1s$  orbital:

$$n = 1, l = 0, m_l = 0, m_s = +\frac{1}{2} \quad (\text{electron 1})$$

$$n = 1, l = 0, m_l = 0, m_s = -\frac{1}{2} \quad (\text{electron 2})$$

The first three quantum numbers are the same, but  $m_s$  differs — Pauli's principle is satisfied. A third electron with  $m_s = +\frac{1}{2}$  or  $-\frac{1}{2}$  in  $1s$  is impossible, which is why He has a **completely filled**  $n = 1$  shell.

**Conclusion:** The probability of finding two electrons with identical quantum numbers is zero because: (i) the antisymmetry requirement of fermionic wave functions forces  $\Psi = 0$ , and (ii) the Slater determinant vanishes when two quantum states are identical. This is not merely a rule but a **fundamental law of quantum mechanics**.

**Q26.** The  $\text{He}^+$  ion contains only one electron and is therefore hydrogen-like. Calculate the wavelength, in increasing order, of the first four transitions in the Balmer series of the  $\text{He}^+$  ion. Compare these wavelengths with the same transitions in an H atom. Comment on the differences. (The Rydberg constant for  $\text{He}^+$  is  $8.72 \times 10^{-18}$  J and that for H is  $2.18 \times 10^{-18}$  J.) (9 marks) [CHEM 113 | 2016–17]

**Answer**
**Given:**

- $\text{He}^+$ : one electron,  $Z = 2$ , Rydberg constant  $R_{\text{He}^+} = 8.72 \times 10^{-18}$  J
- H:  $Z = 1$ , Rydberg constant  $R_H = 2.18 \times 10^{-18}$  J
- Balmer series: transitions from  $n_i = 3, 4, 5, 6$  down to  $n_f = 2$
- Energy of emitted photon:

$$\Delta E = R \left( \frac{1}{n_f^2} - \frac{1}{n_i^2} \right), \quad \lambda = \frac{hc}{\Delta E}$$

- $hc = 6.626 \times 10^{-34} \times 3.0 \times 10^8 = 1.988 \times 10^{-25}$  J·m

**Calculations for  $\text{He}^+$  Balmer series ( $n_f = 2$ ):**

**Transition 1:**  $n = 3 \rightarrow 2$

$$\Delta E = 8.72 \times 10^{-18} \left( \frac{1}{4} - \frac{1}{9} \right) = 8.72 \times 10^{-18} \times \frac{5}{36} = 1.211 \times 10^{-18} \text{ J}$$

$$\lambda = \frac{1.988 \times 10^{-25}}{1.211 \times 10^{-18}} = 164.2 \text{ nm}$$

**Transition 2:**  $n = 4 \rightarrow 2$

$$\Delta E = 8.72 \times 10^{-18} \left( \frac{1}{4} - \frac{1}{16} \right) = 8.72 \times 10^{-18} \times \frac{3}{16} = 1.635 \times 10^{-18} \text{ J}$$

$$\lambda = \frac{1.988 \times 10^{-25}}{1.635 \times 10^{-18}} = 121.6 \text{ nm}$$

**Transition 3:**  $n = 5 \rightarrow 2$

$$\Delta E = 8.72 \times 10^{-18} \left( \frac{1}{4} - \frac{1}{25} \right) = 8.72 \times 10^{-18} \times \frac{21}{100} = 1.831 \times 10^{-18} \text{ J}$$

$$\lambda = \frac{1.988 \times 10^{-25}}{1.831 \times 10^{-18}} = 108.6 \text{ nm}$$

**Transition 4:**  $n = 6 \rightarrow 2$

$$\Delta E = 8.72 \times 10^{-18} \left( \frac{1}{4} - \frac{1}{36} \right) = 8.72 \times 10^{-18} \times \frac{8}{36} = 1.938 \times 10^{-18} \text{ J}$$

$$\lambda = \frac{1.988 \times 10^{-25}}{1.938 \times 10^{-18}} = 102.6 \text{ nm}$$

**Calculations for H Balmer series ( $n_f = 2$ ):**

**Transition 1:**  $n = 3 \rightarrow 2$

$$\Delta E = 2.18 \times 10^{-18} \times \frac{5}{36} = 3.028 \times 10^{-19} \text{ J}, \quad \lambda = 656.7 \text{ nm}$$

**Transition 2:**  $n = 4 \rightarrow 2$

$$\Delta E = 2.18 \times 10^{-18} \times \frac{3}{16} = 4.088 \times 10^{-19} \text{ J}, \quad \lambda = 486.2 \text{ nm}$$

**Transition 3:**  $n = 5 \rightarrow 2$

$$\Delta E = 2.18 \times 10^{-18} \times \frac{21}{100} = 4.578 \times 10^{-19} \text{ J}, \quad \lambda = 434.3 \text{ nm}$$

**Transition 4:**  $n = 6 \rightarrow 2$

$$\Delta E = 2.18 \times 10^{-18} \times \frac{8}{36} = 4.844 \times 10^{-19} \text{ J}, \quad \lambda = 410.4 \text{ nm}$$

**Summary Table (wavelengths in increasing order):**

| Transition        | Factor<br>$\left(\frac{1}{n_f^2} - \frac{1}{n_i^2}\right)$ | He <sup>+</sup> $\lambda$ (nm) | H $\lambda$ (nm) | Ratio<br>$\lambda_H/\lambda_{\text{He}^+}$ |
|-------------------|------------------------------------------------------------|--------------------------------|------------------|--------------------------------------------|
| 6 $\rightarrow$ 2 | 8/36                                                       | 102.6                          | 410.4            | $\approx 4$                                |
| 5 $\rightarrow$ 2 | 21/100                                                     | 108.6                          | 434.3            | $\approx 4$                                |
| 4 $\rightarrow$ 2 | 3/16                                                       | 121.6                          | 486.2            | $\approx 4$                                |
| 3 $\rightarrow$ 2 | 5/36                                                       | 164.2                          | 656.7            | $\approx 4$                                |

**Comments on the Differences:**

(a) **He<sup>+</sup> wavelengths are exactly 4 times shorter than H:**

For hydrogenic systems,  $R \propto Z^2$ , so:

$$\frac{R_{\text{He}^+}}{R_H} = \frac{8.72 \times 10^{-18}}{2.18 \times 10^{-18}} = 4.00 = Z^2 \quad (Z = 2)$$

Since  $\lambda \propto 1/R$ , the He<sup>+</sup> wavelengths are exactly  $\frac{1}{4}$  of the corresponding H wavelengths for every transition.

(b) **Spectral region shifts entirely:**

The H Balmer series lies in the **visible region** (410–657 nm), producing the familiar coloured lines of hydrogen. The He<sup>+</sup> Balmer series is shifted entirely into the **UV region** (103–164 nm) — invisible to the naked eye.

(c) **Physical reason — higher nuclear charge:**

He<sup>+</sup> has  $Z = 2$ , so its electron is bound *four times* more strongly than in H. Transitions between the same quantum levels therefore release four times more energy, producing photons with four times shorter wavelengths:

$$\Delta E_{\text{He}^+} = Z^2 \Delta E_H = 4 \Delta E_H \quad \Rightarrow \quad \lambda_{\text{He}^+} = \frac{\lambda_H}{4}$$

(d) **Same series structure, different scale:**

Despite the wavelength difference, both series show the same *pattern* — the gap between successive lines decreases as  $n_i$  increases, converging to a series limit at  $n_i \rightarrow \infty$ .

**Conclusion:** The He<sup>+</sup> Balmer series is structurally identical to that of H but compressed by a factor of  $Z^2 = 4$  in wavelength due to the stronger nuclear attraction, shifting all four lines from the visible into the UV region.

**Q27.** What are photons? What role did Einstein's explanation of the photoelectric effect play in the development of the particle-wave interpretation of the nature of electromagnetic radiation? **(9–12 marks)** [CHEM 113 | 2015–16; 2011–12]

### Answer

#### Photons & Einstein's Role in Wave-Particle Duality

**(10 marks)**

#### Part 1: What are Photons?

**(4 marks)**

A **photon** is a discrete, massless packet (quantum) of electromagnetic energy. Einstein (1905) proposed that light is not a continuous wave but is instead composed of these individual energy packets. Key properties:

- **Energy:** Each photon carries energy proportional to its frequency:

$$E = h\nu = \frac{hc}{\lambda}$$

where  $h = 6.626 \times 10^{-34}$  J·s,  $\nu$  = frequency,  $\lambda$  = wavelength,  $c$  = speed of light.

- **Rest mass:** A photon has zero rest mass ( $m_0 = 0$ ), yet it carries momentum:

$$p = \frac{h}{\lambda} = \frac{h\nu}{c}$$



- **Speed:** All photons travel at  $c = 3.0 \times 10^8$  m/s in vacuum, regardless of frequency.
- **Indivisibility:** A photon is absorbed or emitted as a whole unit — it cannot be split. One photon interacts with exactly one electron.
- **No charge:** Photons are electrically neutral and are not deflected by electric or magnetic fields.

## Part 2: Einstein's Photoelectric Effect & the Wave-Particle Interpretation

(6 marks)

### (i) The Classical Wave Picture Before Einstein:

Maxwell's electromagnetic theory (1865) described light purely as a continuous wave characterized by frequency  $\nu$  and wavelength  $\lambda$ . This wave model successfully explained reflection, refraction, diffraction, and interference. However, it *completely failed* to explain the photoelectric effect — specifically, the existence of a threshold frequency and the instantaneous emission of electrons.

### (ii) Einstein's Particle Explanation (1905):

Einstein proposed that light energy is quantized into photons. A single photon ejects a single electron only if its energy exceeds the metal's work function  $\phi$ :

$$h\nu = \phi + \frac{1}{2}mv_{\max}^2 \quad \Rightarrow \quad \frac{1}{2}mv_{\max}^2 = h(\nu - \nu_0)$$

This perfectly explained all experimental observations — threshold frequency, instantaneous emission, and KE depending on  $\nu$  alone — none of which the wave theory could account for.

### (iii) The Dilemma — Two Irrefutable Bodies of Evidence:

Einstein's success created a fundamental conflict in physics:

| Evidence for Wave Nature       | Evidence for Particle Nature    |
|--------------------------------|---------------------------------|
| Young's double-slit experiment | Photoelectric effect            |
| Diffraction & interference     | Compton scattering              |
| Polarization of light          | Photon momentum $p = h/\lambda$ |
| Maxwell's equations            | Blackbody radiation             |

Neither picture alone was complete. Light had to be *both* — a conclusion that was deeply unsettling to classical physicists.

### (iv) Birth of Wave-Particle Duality:

The resolution was to accept that electromagnetic radiation exhibits **wave-particle duality** — behaving as a wave or a particle depending on the experimental context:

$$\underbrace{E = h\nu}_{\substack{\text{particle property} \\ \text{(energy of photon)}}} \longleftrightarrow \underbrace{\lambda = \frac{c}{\nu}}_{\substack{\text{wave property} \\ \text{(wavelength)}}$$

Einstein's photoelectric explanation was the **turning point** — it forced physicists to abandon the purely classical wave model and accept that light has a dual nature. This directly inspired de Broglie (1924) to extend the idea to matter itself:

$$\lambda_{\text{matter}} = \frac{h}{mv}$$

which ultimately led to the full development of **quantum mechanics**.

**Conclusion:** Photons are the quantum units of light. Einstein's photoelectric explanation shattered the purely wave-based view of electromagnetic radiation, established the particle nature of light, and — in combination with existing wave evidence — gave rise to the revolutionary concept of wave-particle duality that forms the bedrock of modern quantum theory.

Q28. What do you mean by Eigen values and Eigen functions? (6–8 marks)

[CHEM 113 | 2015–16; 2011–12]

## Answer

### Eigenvalues and Eigenfunctions

(7 marks)

### Part 1: The Schrödinger Equation — Starting Point

(2 marks)

In quantum mechanics, the state of a particle is described by a wave function  $\Psi$ . The total energy of the system is obtained by applying the **Hamiltonian operator**  $\hat{H}$  to  $\Psi$ , giving the time-independent Schrödinger equation:

$$\hat{H}\Psi = E\Psi$$

This is an **eigenvalue equation** — the most fundamental equation in quantum mechanics. Its solutions come in two parts:

- The function  $\Psi$  that satisfies the equation — called the **eigenfunction**



- The constant  $E$  that appears on the right side — called the **eigenvalue**

### Part 2: Eigenfunctions

(2 marks)

An **eigenfunction**  $\Psi_n$  is a well-behaved wave function that, when operated on by  $\hat{H}$ , returns the *same function* multiplied by a constant:

$$\hat{H} \Psi_n = E_n \Psi_n$$

For a valid eigenfunction,  $\Psi$  must satisfy the following conditions:

- Single-valued** — only one value of  $\Psi$  at each point in space
- Continuous** — no abrupt jumps or discontinuities
- Finite** —  $\Psi \rightarrow 0$  as  $r \rightarrow \infty$
- Normalizable** —  $\int |\Psi|^2 dV = 1$

Each eigenfunction  $\Psi_n$  corresponds to a specific *allowed state* of the system and is characterized by a set of quantum numbers  $(n, l, m_l)$ . For example, for the hydrogen atom ground state:

$$\Psi_{1s} = \frac{1}{\sqrt{\pi}} \left( \frac{1}{a_0} \right)^{3/2} e^{-r/a_0}$$

where  $a_0 = 0.529 \text{ \AA}$  is the Bohr radius.

### Part 3: Eigenvalues

(2 marks)

An **eigenvalue**  $E_n$  is the specific, *quantized* value of energy that the system possesses when it is in the state described by  $\Psi_n$ . For the hydrogen atom, the energy eigenvalues are:

$$E_n = -\frac{13.6 \text{ eV}}{n^2}, \quad n = 1, 2, 3, \dots$$

Key properties of eigenvalues:

- They are always **real** and **discrete** (quantized) — only certain values are allowed.
- Each eigenvalue  $E_n$  is the *only* possible result of an energy measurement when the system is in state  $\Psi_n$ .
- Different eigenfunctions may share the same eigenvalue — this is called **degeneracy** (e.g. the  $2s$  and  $2p$  orbitals of hydrogen both have  $E_2 = -3.4 \text{ eV}$ ).

### Part 4: Summary & Significance

(1 marks)

| Term                   | Definition                              | Physical Meaning                                                       |
|------------------------|-----------------------------------------|------------------------------------------------------------------------|
| Eigenfunction $\Psi_n$ | Solution to $\hat{H}\Psi = E\Psi$       | Describes the quantum state; $ \Psi_n ^2 = \text{probability density}$ |
| Eigenvalue $E_n$       | Constant in $\hat{H}\Psi_n = E_n\Psi_n$ | Allowed energy of the system in state $\Psi_n$                         |
| Degeneracy             | Multiple $\Psi_n$ share same $E_n$      | Different orbitals at same energy level                                |

The concept of eigenvalues and eigenfunctions is significant because it explains *why* atomic energies are quantized — only those wave functions satisfying the boundary conditions yield valid eigenfunctions, and each gives a specific discrete eigenvalue. This directly accounts for the **line spectra** of atoms.

**Conclusion:** An **eigenfunction** is the allowed wave function of a quantum system, and its corresponding **eigenvalue** is the precise, quantized energy of that state. Together they form the complete solution to the Schrödinger equation and explain the discrete nature of atomic energy levels.

**Q29.** The electron in the hydrogen atom makes a transition from an energy state of principal quantum number  $n_i$  to the  $n = 2$  state. If the photon emitted has a wavelength of 434 nm, what is the value of  $n_i$ ? **(7–8 marks)** [CHEM 113 | 2015–16; 2014–15]

**Answer**
**Given:**

- Hydrogen atom:  $Z = 1$
- Final state:  $n_f = 2$  (Balmer series)
- Wavelength of emitted photon:  $\lambda = 434 \text{ nm} = 434 \times 10^{-9} \text{ m}$
- $h = 6.626 \times 10^{-34} \text{ J}\cdot\text{s}$ ,  $c = 3.0 \times 10^8 \text{ m/s}$ ,  $R_H = 2.18 \times 10^{-18} \text{ J}$

**Step 1 — Energy of the emitted photon:**

$$\Delta E = \frac{hc}{\lambda} = \frac{(6.626 \times 10^{-34})(3.0 \times 10^8)}{434 \times 10^{-9}} = \frac{1.988 \times 10^{-25}}{4.34 \times 10^{-7}} = 4.580 \times 10^{-19} \text{ J}$$

**Step 2 — Apply the Rydberg energy equation:**

 For a transition from  $n_i$  to  $n_f = 2$ :

$$\Delta E = R_H \left( \frac{1}{n_f^2} - \frac{1}{n_i^2} \right)$$

$$4.580 \times 10^{-19} = 2.18 \times 10^{-18} \left( \frac{1}{4} - \frac{1}{n_i^2} \right)$$

**Step 3 — Solve for  $n_i$ :**

$$\frac{1}{4} - \frac{1}{n_i^2} = \frac{4.580 \times 10^{-19}}{2.18 \times 10^{-18}} = 0.2101$$

$$\frac{1}{n_i^2} = \frac{1}{4} - 0.2101 = 0.2500 - 0.2101 = 0.0399$$

$$n_i^2 = \frac{1}{0.0399} = 25.06 \approx 25$$

$$\boxed{n_i = 5}$$

**Verification:**

$$\Delta E = 2.18 \times 10^{-18} \left( \frac{1}{4} - \frac{1}{25} \right) = 2.18 \times 10^{-18} \times \frac{21}{100} = 4.578 \times 10^{-19} \text{ J}$$

$$\lambda = \frac{1.988 \times 10^{-25}}{4.578 \times 10^{-19}} = 434.2 \text{ nm} \approx 434 \text{ nm} \quad \checkmark$$

**Conclusion:** The electron transitions from  $n_i = 5$  to  $n_f = 2$ . This is the **third line** of the Balmer series ( $H_\gamma$  line), falling in the **violet region** of the visible spectrum.

**Q30.** Matter and radiation have a “dual nature” — Explain. (5 marks)

[CHEM 113 | 2015–16]

**Answer**

 The concept of **dual nature** means that both matter and radiation exhibit *both* wave-like and particle-like properties — not simultaneously, but depending on the type of experiment performed.

**1. Dual Nature of Radiation (Light):**

- Wave nature** — evidenced by *interference*, *diffraction*, and *polarization* of light, which can only be explained if light travels as a wave with wavelength  $\lambda$  and frequency  $\nu$ .
- Particle nature** — evidenced by the **photoelectric effect** (Einstein, 1905) and **Compton scattering** (1923), where light behaves as discrete packets of energy called **photons**, each carrying:

$$E = h\nu, \quad p = \frac{h}{\lambda}$$

**2. Dual Nature of Matter:**

- Particle nature** — matter has definite mass and occupies space, well established in classical mechanics.
- Wave nature** — proposed by de Broglie (1924): any moving particle has an associated wavelength:

$$\lambda = \frac{h}{mv}$$

 This was experimentally confirmed by **Davisson and Germer** (1927), who observed *diffraction* of electrons by a nickel crystal — a purely wave phenomenon.

**Summary Table:**

| Entity            | Wave Evidence             | Particle Evidence                        | Evidence |
|-------------------|---------------------------|------------------------------------------|----------|
| Radiation (light) | Diffraction, interference | Photoelectric effect, Compton scattering | ef-      |
| Matter (electron) | Electron diffraction      | Definite mass, momentum                  |          |

**Conclusion:** Wave and particle properties are **complementary** — described by Bohr’s *Complementarity Principle*. Neither description alone is complete; together they form the foundation of **quantum mechanics**.

**Q31.** What is meant by the term “Shielding of electrons” in an atom? Using the Li atom as an example, describe the effect of shielding on the energy of electrons in an atom. (7 marks) [CHEM 113 | 2015–16]

**Answer**
**1. Shielding of Electrons:**

In a multi-electron atom, electrons in inner shells repel the outer electrons, partially cancelling the attractive force of the nucleus. This reduction in the effective nuclear attraction experienced by an outer electron, caused by the presence of inner electrons, is called **electron shielding** (or screening).

The net nuclear charge felt by an outer electron is expressed as the **effective nuclear charge**:

$$Z_{\text{eff}} = Z - \sigma$$

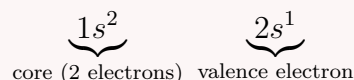
where  $Z$  is the true atomic number and  $\sigma$  is the **shielding constant**, representing the cumulative repulsive effect of all inner electrons on the electron of interest.

**2. Mechanism of Shielding:**

- Inner shell electrons ( $n < n_{\text{outer}}$ ) are most effective at shielding — they lie *between* the outer electron and the nucleus, directly blocking nuclear attraction.
- Same-shell electrons shield each other only partially (Slater weight  $\approx 0.35$ ) since they do not fully intercept the nuclear field.
- The effectiveness of shielding also depends on orbital penetration:  $s > p > d > f$ . An  $s$  electron penetrates closer to the nucleus and is therefore *less* shielded than a  $p$  or  $d$  electron in the same shell.

**3. Example — Lithium (Li,  $Z = 3$ ):**

The electron configuration of Li is:



The two  $1s$  core electrons lie between the  $2s$  valence electron and the nucleus. They shield the valence electron from the full  $+3$  nuclear charge. Using Slater’s rules:

$$\sigma = 2 \times 0.85 = 1.70$$

$$Z_{\text{eff}}(2s) = 3 - 1.70 = \mathbf{1.30}$$

So the  $2s$  valence electron feels only  $+1.30$  instead of the full  $+3$  — less than half the true nuclear charge.

**4. Effect of Shielding on Electron Energy:**

The energy of an electron in a hydrogenic-like environment scales as:

$$E \propto -\frac{Z_{\text{eff}}^2}{n^2}$$

Shielding lowers  $Z_{\text{eff}}$ , which directly raises (makes less negative) the energy of the outer electron:

**(a) Valence electron is less stable (higher energy):**

The  $2s$  electron of Li, shielded from the full  $+3$  charge, has a much higher (less negative) energy than it would without shielding. This makes it easier to remove — Li has a low first ionization energy:

$$\text{IE}_1(\text{Li}) = 520 \text{ kJ/mol}$$

**(b) Core electrons are more stable (lower energy):**

The two  $1s$  electrons experience almost the full  $+3$  nuclear charge (they shield each other only slightly,  $\sigma_{1s} \approx 0.30$ ):

$$Z_{\text{eff}}(1s) = 3 - 0.30 = 2.70$$

Their energy is therefore much lower (more negative) and they are tightly bound — a far higher energy is needed to remove them:

$$\text{IE}_2(\text{Li}) = 7298 \text{ kJ/mol} \gg \text{IE}_1$$

(c) **Energy difference between subshells:**

In hydrogen,  $2s$  and  $2p$  are degenerate. In Li, shielding breaks this degeneracy: the  $2s$  orbital penetrates closer to the nucleus than  $2p$ , experiencing less shielding and a higher  $Z_{\text{eff}}$ , making it *lower in energy* than  $2p$ :

$$E(2s) < E(2p) \quad (\text{in Li and all multi-electron atoms})$$

This is why Li fills  $2s$  before  $2p$ .

**5. Summary for Li:**

| Electron         | Shell   | $\sigma$ | $Z_{\text{eff}}$ | Energy/stability     |
|------------------|---------|----------|------------------|----------------------|
| $1s^2$ (core)    | $n = 1$ | 0.30     | 2.70             | Low (tightly bound)  |
| $2s^1$ (valence) | $n = 2$ | 1.70     | 1.30             | High (loosely bound) |

**Conclusion:** Shielding by inner electrons reduces  $Z_{\text{eff}}$  for outer electrons, raising their energy and making them easier to remove. In Li, the two  $1s$  core electrons shield the  $2s$  valence electron so effectively that it experiences less than half the true nuclear charge — directly explaining Li's low ionization energy and its high chemical reactivity as an alkali metal.

**Q32.** In your own words, explain the photoelectric effect. (4 marks)

[CHEM 113 | 2014–15]

**Answer**

**The Photoelectric Effect**

(4 marks)

When light of sufficiently high frequency strikes the surface of a metal, electrons are ejected from that surface. This phenomenon is called the **photoelectric effect**.

The key points are:

- Light must exceed a minimum **threshold frequency**  $\nu_0$  — below this, no electrons are emitted regardless of intensity.
- Each photon of energy  $E = h\nu$  collides with one electron and transfers its energy. If  $h\nu \geq \phi$  (the work function of the metal), the electron is freed:

$$\frac{1}{2}mv_{\text{max}}^2 = h\nu - \phi$$

- Higher frequency  $\Rightarrow$  faster electrons.
- Higher intensity  $\Rightarrow$  more electrons, not faster ones.

**Q33.** The binding energy of an electron in a metal surface is  $5.86 \times 10^{-19}$  J. (i) Calculate the minimum frequency of light required to release electrons from the metal. (ii) Calculate the kinetic energy of the ejected electron if light frequency  $2.00 \times 10^{15} \text{ s}^{-1}$  is used for irradiating the metal ( $h = 6.63 \times 10^{-34} \text{ J}\cdot\text{s}$ ). (6 marks)

[CHEM 113 | 2014–15]

**Answer**

**Given:**

- Binding energy (work function):  $\phi = 5.86 \times 10^{-19} \text{ J}$
- $h = 6.63 \times 10^{-34} \text{ J}\cdot\text{s}$
- Frequency used in part (ii):  $\nu = 2.00 \times 10^{15} \text{ s}^{-1}$

**(i) Minimum frequency required:**

At the minimum frequency (threshold frequency  $\nu_0$ ), all photon energy is used to overcome the binding energy, with no kinetic energy remaining:

$$\begin{aligned} h\nu_0 &= \phi \\ \nu_0 &= \frac{\phi}{h} = \frac{5.86 \times 10^{-19}}{6.63 \times 10^{-34}} \\ \nu_0 &= 8.84 \times 10^{14} \text{ s}^{-1} \end{aligned}$$

**(ii) Kinetic energy of the ejected electron:**

Using the **photoelectric equation** (Einstein, 1905):

$$\begin{aligned} KE &= h\nu - \phi \\ KE &= (6.63 \times 10^{-34})(2.00 \times 10^{15}) - 5.86 \times 10^{-19} \\ KE &= 13.26 \times 10^{-19} - 5.86 \times 10^{-19} \\ KE &= 7.40 \times 10^{-19} \text{ J} \end{aligned}$$

**Note:** Since  $\nu = 2.00 \times 10^{15} \text{ s}^{-1} > \nu_0 = 8.84 \times 10^{14} \text{ s}^{-1}$ , electron ejection is confirmed.

**Q34.** What is meant by the term “Effective nuclear charge ( $Z_{\text{eff}}$ )” in an atom? Using the Sodium ( $z = 11$ ) atom as an example, describe the effect of shielding on the 1st ionization energy. (4 marks) [CHEM 113 | 2014–15]

### Answer

#### Effective Nuclear Charge ( $Z_{\text{eff}}$ ):

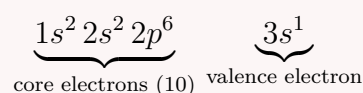
The **effective nuclear charge** is the net positive charge actually experienced by an electron in a multi-electron atom, after accounting for the **shielding** (screening) effect of inner electrons:

$$Z_{\text{eff}} = Z - \sigma$$

where  $Z$  is the actual atomic number and  $\sigma$  is the **shielding constant** — a measure of how much the inner electrons cancel the nuclear attraction felt by the outer electron.

#### Example — Sodium (Na, $Z = 11$ ):

The electron configuration of Na is:



The single  $3s$  valence electron is shielded by the 10 inner core electrons. Using Slater’s rules:

$$\sigma \approx 10 \times 0.85 + 0 \times 0.35 = 8.50 + 0 = 8.85$$

$$Z_{\text{eff}} = 11 - 8.85 = 2.15$$

So instead of feeling the full +11 nuclear charge, the  $3s$  electron experiences only  $\approx +2.15$  — greatly weakened by the shielding of the inner shells.

#### Effect on 1st Ionization Energy:

Because  $Z_{\text{eff}}$  is much smaller than  $Z$ , the  $3s$  electron is **loosely held** — it is far from the nucleus and experiences only a fraction of the nuclear attraction. This makes it relatively easy to remove, giving sodium a **low first ionization energy**:

$$\text{IE}_1(\text{Na}) = 496 \text{ kJ/mol}$$

In contrast, if there were no shielding ( $\sigma = 0$ ), the  $3s$  electron would experience the full  $Z = +11$  charge, and the ionization energy would be far higher. Shielding is therefore directly responsible for the low reactivity barrier of alkali metals.

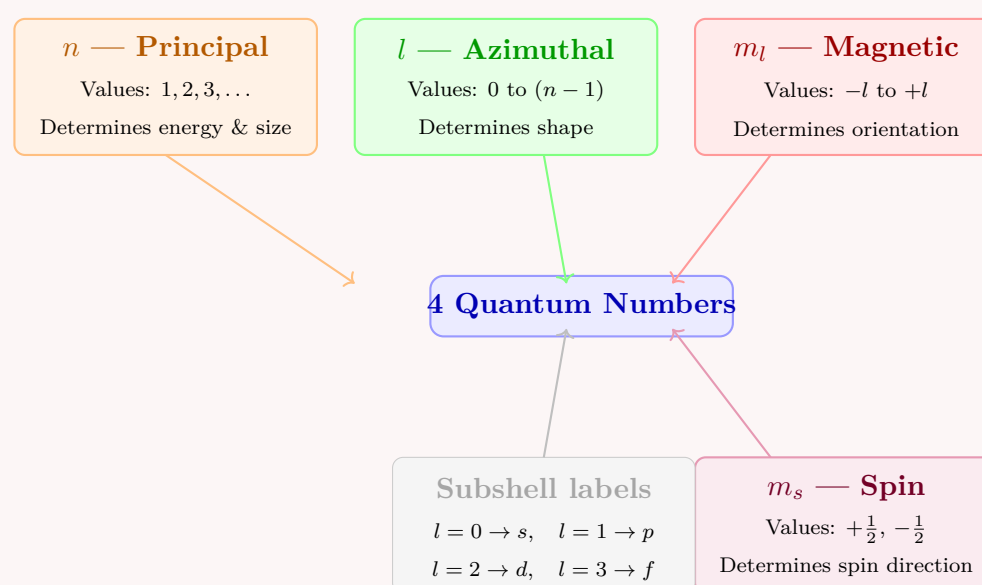
**Conclusion:**  $Z_{\text{eff}}$  determines how tightly an electron is bound. Greater shielding  $\Rightarrow$  lower  $Z_{\text{eff}}$   $\Rightarrow$  lower ionization energy — as clearly illustrated by sodium’s easily removed  $3s$  valence electron.

**Q35.** Describe how you build up the detailed electron configuration with the help of all four quantum numbers. (10 marks) [CHEM 113 | 2014–15]

### Answer

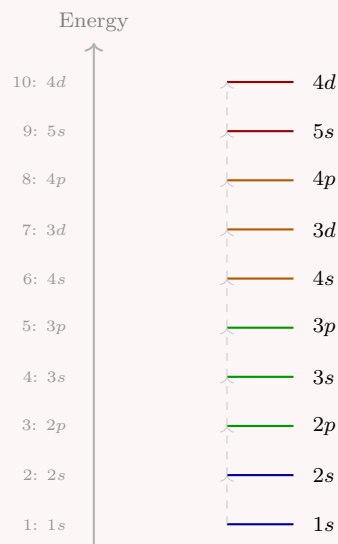
The detailed electron configuration of an atom is built up using three fundamental rules, each governed by the four quantum numbers ( $n, l, m_l, m_s$ ).

#### 1. The Four Quantum Numbers:



#### 2. Three Rules for Building Configurations:

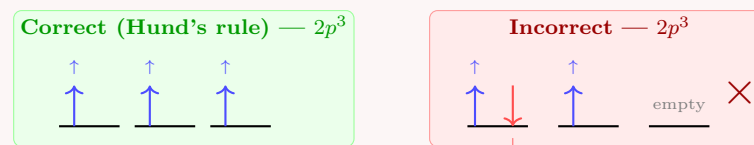
**Rule 1 — Aufbau Principle:** Electrons fill orbitals in order of **increasing energy**. The filling order follows the  $(n + l)$  rule — lower  $(n + l)$  fills first; if equal, lower  $n$  fills first:



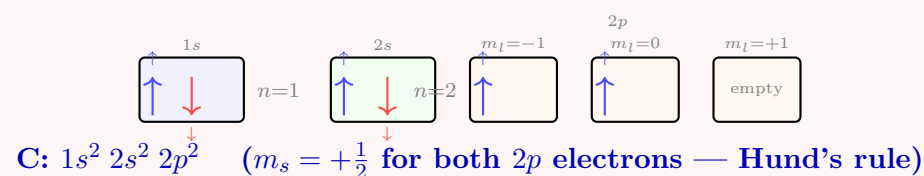
**Rule 2 — Pauli Exclusion Principle:** No two electrons in an atom can have the same four quantum numbers. Each orbital holds **at most 2 electrons** with opposite spins:



**Rule 3 — Hund's Rule of Maximum Multiplicity:** In degenerate orbitals (same  $n$  and  $l$ ), electrons occupy **separate orbitals** with **parallel spins** before pairing:



### 3. Worked Example — Carbon ( $Z = 6$ ):



### 4. Quantum Numbers for Each Electron in Carbon:

| Electron | $n$ | $l$ | $m_l$ | $m_s$          | Orbital     |
|----------|-----|-----|-------|----------------|-------------|
| 1        | 1   | 0   | 0     | $+\frac{1}{2}$ | $1s$        |
| 2        | 1   | 0   | 0     | $-\frac{1}{2}$ | $1s$        |
| 3        | 2   | 0   | 0     | $+\frac{1}{2}$ | $2s$        |
| 4        | 2   | 0   | 0     | $-\frac{1}{2}$ | $2s$        |
| 5        | 2   | 1   | -1    | $+\frac{1}{2}$ | $2p$ (Hund) |
| 6        | 2   | 1   | 0     | $+\frac{1}{2}$ | $2p$ (Hund) |

**Conclusion:** Electron configurations are built by applying the **Aufbau principle** (fill lowest energy first), **Pauli exclusion principle** (no two electrons share all four quantum numbers), and **Hund's rule** (maximise parallel spins in degenerate orbitals). Together, the four quantum numbers ( $n, l, m_l, m_s$ ) uniquely identify every electron in an atom and completely define its energy, orbital shape, orientation, and spin.

**Q36.** Draw a rough sketch of a periodic table (no details required). Indicate where alkali metals, chalcogens and halogens are located. Write the name and symbol for an element in each of these groups. (— marks) [CHEM 113 | 2014–15]



## Answer

|    | 1             | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 |
|----|---------------|---|---|---|---|---|---|---|---|----|----|----|----|----|----|----|----|----|
| P1 | H             |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |    |    |
| P2 | Li            |   |   |   |   |   |   |   |   |    |    |    |    |    |    | O  | F  |    |
| P3 | Na            |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    | Cl |    |
| P4 | Alkali metals |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |    |    |
| P5 | Alkali metals |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |    |    |
| P6 | Cs            |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |    |    |
| P7 | Fr            |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |    |    |

Legend

- Alkali metals (Gp. 1)
- Chalcogens (Gp. 16)
- Halogens (Gp. 17)

## Named Elements in Each Group:

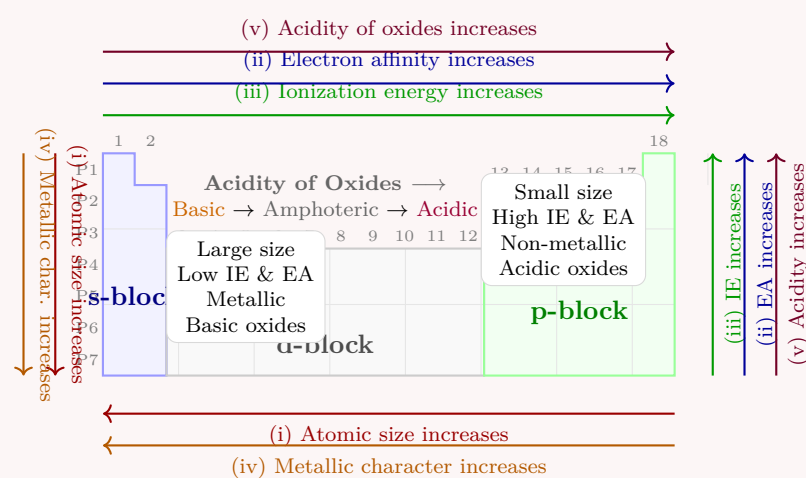
| Group         | Group No. | Element       | Properties                                               |
|---------------|-----------|---------------|----------------------------------------------------------|
| Alkali metals | 1         | Sodium (Na)   | Soft, silvery metal; reacts vigorously with water        |
| Chalcogens    | 16        | Oxygen (O)    | Colourless gas; essential for combustion and respiration |
| Halogens      | 17        | Chlorine (Cl) | Toxic yellow-green gas; strong oxidising agent           |

## Key observations:

- Alkali metals** (Group 1) occupy the far left column. They have one valence electron ( $ns^1$ ) and are the most reactive metals — reactivity *increases* down the group.
- Chalcogens** (Group 16) are two columns from the right. They have six valence electrons ( $ns^2np^4$ ) and commonly form  $-2$  anions or covalent bonds.
- Halogens** (Group 17) occupy the second-to-last column. They have seven valence electrons ( $ns^2np^5$ ), need only one more electron to complete their octet, and are the most reactive non-metals — reactivity *decreases* down the group.

**Q37.** Explain with a figure how each of these properties of the representative elements generally increases or decreases across the periodic table: (i) atomic size, (ii) electron affinity, (iii) ionization energy, (iv) metallic character, (v) acidity of oxides. **(12 marks)** [CHEM 113 | 2014–15]

## Answer



## (i) Atomic Size:

Atomic size is the distance from the nucleus to the outermost electron.

- Across a period ( $\rightarrow$ ):** nuclear charge  $Z$  increases with minimal extra shielding, pulling electrons inward  $\Rightarrow$  **atomic size decreases**.
- Down a group ( $\downarrow$ ):** new electron shells are added, increasing shielding  $\Rightarrow$  **atomic size increases**.
- Largest: Fr ( $\sim 270$  pm); Smallest: He ( $\sim 31$  pm).

## (ii) Electron Affinity (EA):

EA is the energy released when a gaseous atom gains one electron:  $M_{(g)} + e^- \rightarrow M_{(g)}^-$

- Across a period ( $\rightarrow$ ):** increasing  $Z_{\text{eff}}$  attracts electrons more strongly  $\Rightarrow$  **EA increases** (more negative). Exceptions: group 2 (filled  $2s^2$ ) and group 5 (half-filled  $p^3$ ) show anomalously low EA.

- **Down a group** ( $\downarrow$ ): larger radius and greater shielding reduce nuclear attraction  $\Rightarrow$  **EA decreases**.
- Most negative EA: Cl ( $-349$  kJ/mol).

### (iii) Ionization Energy (IE):

IE is the energy required to remove the outermost electron:  $M_{(g)} \rightarrow M_{(g)}^+ + e^-$

- **Across a period** ( $\rightarrow$ ): increasing  $Z_{\text{eff}}$  holds electrons more tightly  $\Rightarrow$  **IE increases**. Exceptions: slight dip at group 13 (Al  $<$  Mg, due to  $2p > 2s$ ) and group 16 (S  $<$  P, due to pairing repulsion).
- **Down a group** ( $\downarrow$ ): outer electrons are farther away and more shielded  $\Rightarrow$  **IE decreases**.
- Highest: He (2372 kJ/mol); Lowest: Cs (376 kJ/mol).

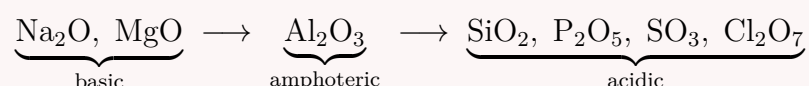
### (iv) Metallic Character:

Metallic character reflects the tendency to lose electrons and form cations — directly linked to low IE and large atomic radius.

- **Across a period** ( $\rightarrow$ ): IE increases, electron loss becomes harder  $\Rightarrow$  **metallic character decreases** (metals on left, non-metals on right).
- **Down a group** ( $\downarrow$ ): IE decreases, electrons lost more easily  $\Rightarrow$  **metallic character increases**.
- Most metallic: Fr; Least metallic: F.

### (v) Acidity of Oxides:

The nature of element oxides shifts systematically across the table:



- **Across a period** ( $\rightarrow$ ): as metallic character decreases, oxides change from **basic** (ionic, e.g.  $\text{Na}_2\text{O}$ ) through **amphoteric** (e.g.  $\text{Al}_2\text{O}_3$ ) to **acidic** (covalent, e.g.  $\text{Cl}_2\text{O}_7$ ).
- **Down a group** ( $\downarrow$ ): metallic character increases  $\Rightarrow$  oxides become **more basic**. E.g. in group 14:  $\text{CO}_2$  (acidic)  $\rightarrow$   $\text{SnO}_2$  (amphoteric)  $\rightarrow$   $\text{PbO}$  (basic).
- The shift from basic to acidic oxide reflects the increasing  $Z_{\text{eff}}$  and covalent bonding character of the element–oxygen bond across a period.

### Summary Table:

| Property                | Across<br>riod ( $\rightarrow$ ) | pe-<br>Down<br>group<br>( $\downarrow$ ) |
|-------------------------|----------------------------------|------------------------------------------|
| (i) Atomic size         | Decreases                        | Increases                                |
| (ii) Electron affinity  | Increases                        | Decreases                                |
| (iii) Ionization energy | Increases                        | Decreases                                |
| (iv) Metallic character | Decreases                        | Increases                                |
| (v) Acidity of oxides   | Increases                        | Decreases                                |

**Conclusion:** All five properties are governed by the interplay of **nuclear charge**, **electron shielding**, and **atomic radius**. Increasing  $Z_{\text{eff}}$  across a period shrinks atoms, raises IE and EA, reduces metallic character, and shifts oxide character from basic to acidic. The reverse holds going down a group as shielding dominates.

**Q38.** Give a brief description of the wave nature of light. What are the two main characteristics of light wave? (4 marks) [CHEM 113 | 2014-15]

### Answer

#### Wave Nature of Light:

Light is an **electromagnetic wave** consisting of oscillating electric and magnetic fields propagating perpendicular to each other and to the direction of travel. This wave nature was established by Maxwell (1865) and confirmed by Hertz (1887). It travels through vacuum at:

$$c = 3.0 \times 10^8 \text{ m/s}$$

#### Two Main Characteristics:

- (a) **Wavelength ( $\lambda$ ):** The distance between two consecutive crests (or troughs) of the wave, measured in metres (m) or nanometres

(nm). It determines the *colour* of light — visible light spans  $\lambda \approx 400\text{--}700$  nm.

- (b) **Frequency ( $\nu$ ):** The number of wave cycles passing a point per second, measured in Hertz (Hz). It is related to wavelength by:

$$c = \nu\lambda$$

Higher frequency means higher energy, given by Planck's relation  $E = h\nu$ .

**Q39.** Give the equation that relates wave-particle properties of light. Explain the meaning of each symbol in the equation. **(4 marks)**  
[CHEM 113 | 2014-15]

#### Answer

The equation that relates the wave and particle properties of light is the **Planck–Einstein relation**:

$$E = h\nu = \frac{hc}{\lambda}$$

**Meaning of each symbol:**

- $E$  — Energy of a single photon (particle property), measured in Joules (J)
- $h$  — Planck's constant =  $6.626 \times 10^{-34}$  J·s; the proportionality constant linking energy to frequency
- $\nu$  — Frequency of the light wave (wave property), measured in Hz ( $\text{s}^{-1}$ )
- $c$  — Speed of light in vacuum =  $3.0 \times 10^8$  m/s
- $\lambda$  — Wavelength of the light wave (wave property), measured in metres (m)

**Significance:** The left side ( $E$ ) represents the *particle* nature of light (photon energy), while the right side ( $\nu$  and  $\lambda$ ) represents the *wave* nature — thus the equation **bridges both natures** of light in a single expression.

**Q40.** What is Heisenberg's Uncertainty principle? Prove that the product of uncertainty in position of an electron with uncertainty in momentum of the same electron is of the order of Planck's constant. **(11 marks)**  
[CHEM 113 | 2013–14]

#### Answer

#### Heisenberg's Uncertainty Principle & Proof

**(11 marks)**

#### Part 1: Heisenberg's Uncertainty Principle

**(3 marks)**

**Statement:** It is fundamentally impossible to determine *simultaneously* and *precisely* both the position and the momentum of a subatomic particle. The more precisely one is known, the less precisely the other can be known. Mathematically:

$$\Delta x \cdot \Delta p \geq \frac{h}{4\pi}$$

where  $\Delta x$  is the uncertainty in position,  $\Delta p$  is the uncertainty in momentum, and  $h$  is Planck's constant.

**Other conjugate pairs** governed by the same principle:

$$\Delta E \cdot \Delta t \geq \frac{h}{4\pi}$$

where  $\Delta E$  is uncertainty in energy and  $\Delta t$  is uncertainty in time.

**Key Points:**

- This is *not* a limitation of instruments or measurement technique — it is a **fundamental law of nature**.
- It arises because any attempt to measure the position of an electron (e.g. by shining light on it) disturbs its momentum, and vice versa.
- For macroscopic objects (baseball, car),  $h$  is so tiny relative to their mass and momentum that the uncertainty is completely negligible. It is only significant at the **subatomic scale**.
- It fundamentally rules out the concept of a definite electron orbit (as in Bohr's model) and necessitates the **quantum mechanical probability cloud** picture.

#### Part 2: Proof Using the Gamma-Ray Microscope (Thought Experiment)

**(8 marks)**

**(i) Setup:**

To locate an electron, we must illuminate it with a photon (of wavelength  $\lambda$ ) and observe the scattered photon through a microscope. Let the microscope have a lens of half-angle  $\theta$  (the semi-angle of the cone of light entering the lens).

**(ii) Uncertainty in Position ( $\Delta x$ ):**

The resolving power of the microscope limits how precisely the electron's position can be determined. By Rayleigh's criterion, the minimum resolvable distance (uncertainty in position) is:

$$\Delta x \approx \frac{\lambda}{\sin \theta}$$

To reduce  $\Delta x$ , we need smaller  $\lambda$  (higher frequency, higher energy photons).

**(iii) Uncertainty in Momentum ( $\Delta p$ ):**

The incident photon has momentum  $p = h/\lambda$ . When the photon collides with the electron and scatters into the lens (anywhere within the cone of half-angle  $\theta$ ), it transfers momentum to the electron. The  $x$ -component of the scattered photon's momentum lies anywhere in the range:

$$-\frac{h \sin \theta}{\lambda} \leq p_x \leq +\frac{h \sin \theta}{\lambda}$$

Therefore the uncertainty in the electron's  $x$ -momentum is:

$$\Delta p \approx \frac{h \sin \theta}{\lambda}$$

**(iv) Taking the Product  $\Delta x \cdot \Delta p$ :**

Multiplying the two uncertainties:

$$\begin{aligned} \Delta x \cdot \Delta p &\approx \frac{\lambda}{\sin \theta} \times \frac{h \sin \theta}{\lambda} \\ \Delta x \cdot \Delta p &\approx h \end{aligned}$$

A more rigorous quantum mechanical derivation (using wave packets and Fourier analysis) gives the exact lower bound:

$$\boxed{\Delta x \cdot \Delta p \geq \frac{h}{4\pi}}$$

**(v) Physical Interpretation of the Trade-off:**

- To improve position accuracy, use **shorter**  $\lambda$  (e.g. gamma rays):

$$\lambda \downarrow \Rightarrow \Delta x \downarrow \Rightarrow \frac{h}{\lambda} \uparrow \Rightarrow \Delta p \uparrow$$

- To improve momentum accuracy, use **longer**  $\lambda$  (e.g. radio waves):

$$\lambda \uparrow \Rightarrow \Delta p \downarrow \Rightarrow \frac{\lambda}{\sin \theta} \uparrow \Rightarrow \Delta x \uparrow$$

- In both cases, the *product*  $\Delta x \cdot \Delta p$  remains of the order of  $h$  — you gain precision in one only at the cost of the other.

**(vi) Numerical Verification:**

For an electron confined to an atom,  $\Delta x \sim 10^{-10}$  m:

$$\Delta p \geq \frac{h}{4\pi \Delta x} = \frac{6.626 \times 10^{-34}}{4\pi \times 10^{-10}} \approx 5.27 \times 10^{-25} \text{ kg}\cdot\text{m/s}$$

$$\Delta x \cdot \Delta p = 10^{-10} \times 5.27 \times 10^{-25} = 5.27 \times 10^{-35} \approx \frac{h}{4\pi} \sim \mathcal{O}(h) \quad \checkmark$$

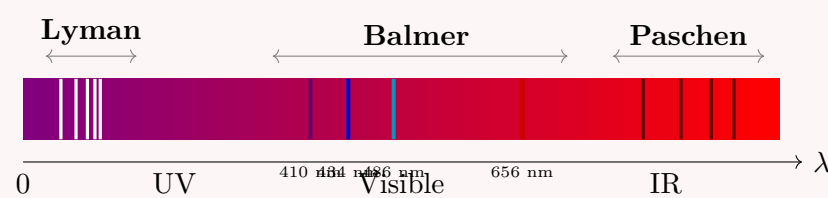
This confirms that the product is indeed of the order of Planck's constant  $h = 6.626 \times 10^{-34}$  J·s.

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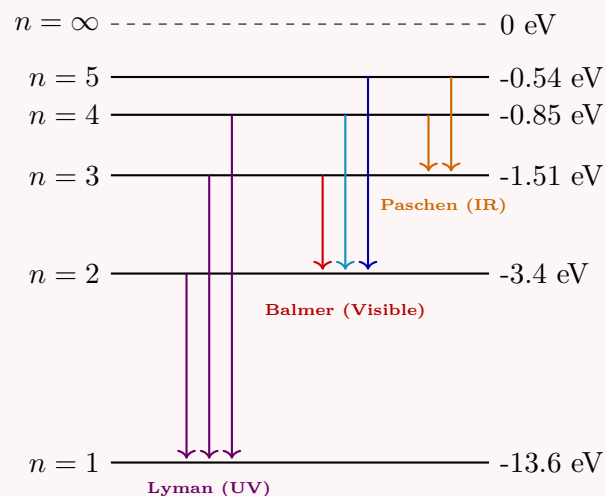
**Conclusion:** Heisenberg's Uncertainty Principle is not a measurement artifact but a fundamental quantum law. The gamma-ray microscope thought experiment elegantly proves that  $\Delta x \cdot \Delta p \approx h$  — any attempt to reduce one uncertainty inevitably amplifies the other, and their product is always of the order of Planck's constant. This irreducible uncertainty at the quantum level makes the classical concept of a precise electron trajectory physically impossible.

**Q41.** Draw emission spectra of hydrogen atom. Why does a hydrogen atom give so many lines in its spectrum although there is only one electron in it? (8 marks) [CHEM 113 | 2012–13]

**Answer**
**Emission Spectrum of Hydrogen Atom**
**Part 1: Diagram of Emission Spectrum**



### Part 2: Energy Level Transitions



### Part 3: Why Does One Electron Give Many Spectral Lines?

Although hydrogen has only **one electron**, it has **infinitely many allowed energy levels** ( $n = 1, 2, 3, \dots$ ). The large number of spectral lines arises because:

- In a hydrogen gas sample, a very large number of atoms are present simultaneously. Each atom may have its electron excited to a *different* energy level depending on the energy it absorbed.
- When electrons transition from a higher level  $n_2$  to a lower level  $n_1$ , a photon of a specific wavelength is emitted. The number of possible transitions between  $N$  energy levels is:

$$\text{Number of lines} = \frac{N(N-1)}{2}$$

- Each unique pair ( $n_2 \rightarrow n_1$ ) produces a *distinct* spectral line. For example, with 5 levels:  $\frac{5 \times 4}{2} = 10$  possible lines.
- These transitions are grouped into **spectral series**:

| Series   | Region      | Transition                |
|----------|-------------|---------------------------|
| Lyman    | Ultraviolet | $n_2 \rightarrow n_1 = 1$ |
| Balmer   | Visible     | $n_2 \rightarrow n_1 = 2$ |
| Paschen  | Infrared    | $n_2 \rightarrow n_1 = 3$ |
| Brackett | Far IR      | $n_2 \rightarrow n_1 = 4$ |

**Conclusion:** It is not a single electron in one atom producing all lines at once. Rather, *different atoms* in the sample emit *different wavelengths* as their electrons fall from various excited states — collectively producing the rich emission spectrum of hydrogen.

**Q42.** Derive the equation by which you can calculate the wavelength of a line in the hydrogen spectrum. Calculate the wavelength of the line in the Brackett series of hydrogen spectrum that is associated with a drop of the electron from the 7th orbit. (Rydberg constant =  $1097 \text{ m}^{-1}$ ) **(17 marks)** [CHEM 113 | 2012–13]

### Answer

#### Part 1: Derivation of Wavelength Equation (Rydberg Equation)

##### Step 1: Energy of Electron in the $n$ -th Bohr Orbit

From Bohr's model, balancing centripetal and Coulombic forces:

$$\frac{m_e v^2}{r} = \frac{e^2}{4\pi\epsilon_0 r^2}$$

Using the quantization of angular momentum  $m_e v r = \frac{nh}{2\pi}$ , the total energy of the electron in the  $n$ -th orbit is:

$$E_n = -\frac{m_e e^4}{8\epsilon_0^2 h^2} \cdot \frac{1}{n^2}$$

The negative sign indicates that the electron is bound to the nucleus.

##### Step 2: Energy of the Emitted Photon

When an electron drops from a higher orbit  $n_2$  to a lower orbit  $n_1$  ( $n_2 > n_1$ ), the energy of the emitted photon is:

$$\Delta E = E_{n_1} - E_{n_2} = -\frac{m_e e^4}{8\varepsilon_0^2 h^2} \cdot \frac{1}{n_1^2} - \left( -\frac{m_e e^4}{8\varepsilon_0^2 h^2} \cdot \frac{1}{n_2^2} \right)$$

$$\Delta E = \frac{m_e e^4}{8\varepsilon_0^2 h^2} \left( \frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

### Step 3: Relating Energy to Wavelength

Since the energy of a photon is  $\Delta E = h\nu = \frac{hc}{\lambda}$ :

$$\frac{hc}{\lambda} = \frac{m_e e^4}{8\varepsilon_0^2 h^2} \left( \frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

Dividing both sides by  $hc$ :

$$\frac{1}{\lambda} = \frac{m_e e^4}{8\varepsilon_0^2 h^3 c} \left( \frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

### Step 4: The Rydberg Equation

The constant  $\frac{m_e e^4}{8\varepsilon_0^2 h^3 c}$  is the **Rydberg constant**  $R_H$ :

$$R_H = \frac{m_e e^4}{8\varepsilon_0^2 h^3 c} = 1.097 \times 10^7 \text{ m}^{-1}$$

Therefore, the **Rydberg equation** is:

$$\boxed{\frac{1}{\lambda} = R_H \left( \frac{1}{n_1^2} - \frac{1}{n_2^2} \right)}$$

where  $n_1 < n_2$  are positive integers representing the lower and upper energy levels respectively.

## Part 2: Calculation of Wavelength — Brackett Series ( $n_2 = 7$ )

Given:

- Series: Brackett  $\Rightarrow n_1 = 4$
- Upper orbit:  $n_2 = 7$
- Rydberg constant:  $R_H = 1.097 \times 10^7 \text{ m}^{-1}$

Applying the Rydberg Equation:

$$\frac{1}{\lambda} = R_H \left( \frac{1}{n_1^2} - \frac{1}{n_2^2} \right) = 1.097 \times 10^7 \left( \frac{1}{4^2} - \frac{1}{7^2} \right)$$

$$\frac{1}{\lambda} = 1.097 \times 10^7 \left( \frac{1}{16} - \frac{1}{49} \right)$$

$$\frac{1}{16} - \frac{1}{49} = \frac{49 - 16}{16 \times 49} = \frac{33}{784}$$

$$\frac{1}{\lambda} = 1.097 \times 10^7 \times \frac{33}{784} = 1.097 \times 10^7 \times 0.042092$$

$$\frac{1}{\lambda} = 4.6175 \times 10^5 \text{ m}^{-1}$$

$$\boxed{\lambda = \frac{1}{4.6175 \times 10^5} = 2.166 \times 10^{-6} \text{ m} = 2166 \text{ nm}}$$

This wavelength lies in the **infrared region** of the electromagnetic spectrum, which is consistent with the Brackett series.

**Q43.** Give a brief account of the wave-particle duality nature of the electron. **(10 marks)**

[CHEM 113 | 2012–13]

### Answer

#### Wave-Particle Duality of the Electron

##### 1. Background: Dual Nature of Light

Before the concept was applied to electrons, it was established that **light exhibits dual nature**:

- **Wave nature:** shown by interference, diffraction, and polarization.
- **Particle nature:** shown by the photoelectric effect (Einstein, 1905), where light behaves as discrete packets of energy called **photons**:

$$E = h\nu$$



## 2. de Broglie's Hypothesis (1924)

Louis de Broglie proposed that **matter, like light, also has dual nature** — every moving particle has an associated wave. For a particle of mass  $m$  moving with velocity  $v$ , the **de Broglie wavelength** is:

$$\lambda = \frac{h}{mv} = \frac{h}{p}$$

where  $h$  is Planck's constant and  $p = mv$  is the linear momentum of the particle.

This relation connects the **wave property** ( $\lambda$ ) with the **particle property** ( $p$ ), elegantly unifying both natures.

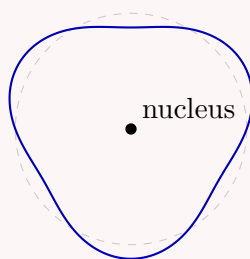
## 3. Justification via Bohr's Quantization Condition

de Broglie provided a physical basis for Bohr's postulate. If an electron of wavelength  $\lambda$  revolves in a circular orbit of radius  $r$ , a stable (non-destructively interfering) orbit requires the circumference to be an integer multiple of the wavelength:

$$2\pi r = n\lambda = \frac{nh}{mv}$$

$$\therefore m_e v r = \frac{nh}{2\pi}$$

This is exactly **Bohr's quantization condition**, now derived naturally from wave theory.



Standing wave for  $n = 3$ :  $2\pi r = 3\lambda$

## 4. Experimental Verification

- (i) **Davisson–Germer Experiment (1927)**: Electrons were diffracted by a nickel crystal, producing a diffraction pattern identical to that of X-rays. This conclusively proved the **wave nature of electrons**.
- (ii) **G.P. Thomson's Experiment (1927)**: A beam of electrons passed through a thin gold foil produced concentric diffraction rings on a photographic plate, further confirming wave behaviour.

## 5. Heisenberg's Uncertainty Principle — A Consequence

The wave–particle duality directly leads to Heisenberg's Uncertainty Principle: it is impossible to simultaneously determine the exact position and momentum of a particle:

$$\Delta x \cdot \Delta p \geq \frac{h}{4\pi}$$

This is not a limitation of instruments, but a fundamental property of matter arising from its wave nature.

## 6. Significance

- Wave–particle duality is the **cornerstone of quantum mechanics**.
- It led to the development of the **Schrödinger wave equation**, replacing Bohr's classical orbits with quantum mechanical **orbitals** (probability distributions).
- For macroscopic objects,  $\lambda = h/mv$  is negligibly small, so wave nature is unobservable. For subatomic particles (electrons, protons), the wavelength is comparable to atomic dimensions, making wave behaviour significant.

### Summary Table:

| Property     | Particle Nature      | Wave Nature                |
|--------------|----------------------|----------------------------|
| Evidence     | Photoelectric effect | Electron diffraction       |
| Described by | Momentum $p = mv$    | Wavelength $\lambda = h/p$ |
| Equation     | $E = mc^2$           | $\lambda = h/mv$           |
| Principle    | Newton's laws        | Schrödinger equation       |

**Q44.** Group the species that are isoelectronic:  $\text{Be}^{2+}$ ,  $\text{F}^-$ ,  $\text{Fe}^{2+}$ ,  $\text{N}^{3-}$ ,  $\text{He}$ ,  $\text{S}^{2-}$ ,  $\text{Co}^{3+}$ ,  $\text{Ar}$ . (8 marks)

[CHEM 113 | 2011–12]

### Answer

#### Isoelectronic Species

Isoelectronic species are atoms or ions that have the **same number of electrons**.

**Step 1: Count the electrons in each species**

| Species          | Atomic No. / Configuration | Charge | Electrons |
|------------------|----------------------------|--------|-----------|
| He               | $Z = 2$                    | 0      | 2         |
| Be <sup>2+</sup> | $Z = 4$                    | +2     | 2         |
| N <sup>3-</sup>  | $Z = 7$                    | -3     | 10        |
| F <sup>-</sup>   | $Z = 9$                    | -1     | 10        |
| S <sup>2-</sup>  | $Z = 16$                   | -2     | 18        |
| Ar               | $Z = 18$                   | 0      | 18        |
| Fe <sup>2+</sup> | $Z = 26$                   | +2     | 24        |
| Co <sup>3+</sup> | $Z = 27$                   | +3     | 24        |

Step 2: Group the Isoelectronic Species

- (i) 2 electrons:

He, Be<sup>2+</sup>

Both have the electron configuration 1s<sup>2</sup>.
- (ii) 10 electrons:

N<sup>3-</sup>, F<sup>-</sup>

Both have the electron configuration 1s<sup>2</sup> 2s<sup>2</sup> 2p<sup>6</sup>.
- (iii) 18 electrons:

S<sup>2-</sup>, Ar

Both have the electron configuration 1s<sup>2</sup> 2s<sup>2</sup> 2p<sup>6</sup> 3s<sup>2</sup> 3p<sup>6</sup>.
- (iv) 24 electrons:

Fe<sup>2+</sup>, Co<sup>3+</sup>

Both have the electron configuration [Ar] 3d<sup>6</sup>.

Q45. Ionization energy is usually measured in the gaseous state. Why? (7 marks)

[CHEM 113 | 2011–12]

Answer

**Why is Ionization Energy Measured in the Gaseous State?**

**Definition:** Ionization energy (IE) is the minimum energy required to remove the most loosely bound electron from a neutral isolated atom:

$$A(g) \longrightarrow A^+(g) + e^- \quad \Delta E = IE$$

**Reasons for Using the Gaseous State:**

(i) **Atoms are isolated and independent:** In the gaseous state, atoms are far apart from one another with negligible inter-molecular forces. This ensures that the measured energy corresponds only to the **removal of an electron from a single, isolated atom**, without any interference from neighbouring atoms or molecules.

(ii) **Eliminates lattice energy contributions:** In the solid state, atoms are held together by strong **lattice forces** (in ionic/metallic solids) or intermolecular forces. Removing an electron from a solid would require overcoming both the ionization energy *and* these lattice interactions, making it impossible to isolate the true IE.

(iii) **Eliminates solvation energy contributions:** In the liquid or aqueous state, atoms/ions are surrounded by solvent molecules, contributing **solvation energy**. This would distort the measured value of IE since extra energy is involved in breaking solute–solvent interactions.

(iv) **Gives a pure, intrinsic atomic property:** The gaseous state provides a **reference state** where the atom is in its standard, unperturbed condition. The IE measured is therefore a true **intrinsic property** of the atom, reflecting only the nuclear attraction on the outermost electron.

(v) **Allows meaningful comparison between elements:** Since all disturbing effects (lattice, solvation, intermolecular forces) are absent in the gas phase, ionization energies measured in this state can be **directly and fairly compared** across different elements in the periodic table.

(vi) **Consistent with theoretical calculations:** Quantum mechanical models (Bohr model, Schrödinger equation) calculate ionization energies for **isolated gaseous atoms**. Measuring IE in the gas phase ensures that experimental values match theoretical predictions without correction factors.

**Summary:**

| State  | Problem                  | Effect on IE                      |
|--------|--------------------------|-----------------------------------|
| Solid  | Lattice energy present   | IE appears larger than true value |
| Liquid | Solvation energy present | IE value gets distorted           |
| Gas    | No external interactions | True, intrinsic IE is measured    |

**Conclusion:** The gaseous state is chosen because it eliminates all external energetic contributions, ensuring that the measured ionization energy reflects **only the electrostatic attraction between the nucleus and the outermost electron** of an isolated atom.

Q46. What are the limitations of Bohr's atomic theory? (4 marks)

[CHEM 113 | 2011–12]

### Answer

#### Limitations of Bohr's Atomic Theory

- Fails for Multi-electron Atoms:** Bohr's model works only for **hydrogen-like species** (H, He<sup>+</sup>, Li<sup>2+</sup>, etc.). It fails completely for atoms with two or more electrons because it does not account for **electron–electron repulsions**.
- Cannot Explain Fine Structure of Spectral Lines:** When hydrogen spectral lines are observed under high resolution or in the presence of a magnetic field, each line splits into several closely spaced lines (**Zeeman effect** and **Stark effect**). Bohr's theory cannot explain this fine and hyperfine splitting.
- Violates Heisenberg's Uncertainty Principle:** Bohr assumed that electrons move in **fixed circular orbits** with precisely defined radius and velocity. This contradicts Heisenberg's Uncertainty Principle, which states that the position and momentum of an electron cannot be simultaneously determined with exactness:

$$\Delta x \cdot \Delta p \geq \frac{h}{4\pi}$$

- Does Not Account for Wave Nature of Electron:** Bohr treated the electron purely as a **classical particle**. It ignores the wave–particle duality of the electron established by de Broglie, which is fundamental to modern quantum mechanics.

Q47. State the general periodic trends in size of atomic radii. Arrange the following atoms in order of decreasing atomic radius: Na, Al, P, Cl, Mg. (9 marks)

[CHEM 113 | 2011–12]

### Answer

#### Part 1: Periodic Trends in Atomic Radii

**Definition:** Atomic radius is defined as **half the distance between the nuclei of two identical adjacent atoms** in a molecule or crystal.

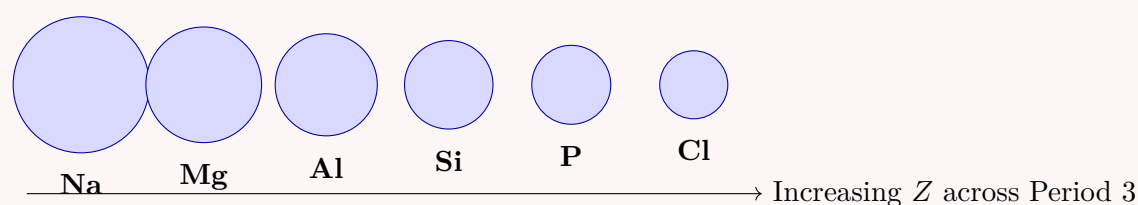
#### Trend 1: Across a Period (Left → Right)

Atomic radius **decreases** across a period from left to right.

- Moving across a period, the **number of protons increases** (higher nuclear charge  $Z$ ), but electrons are added to the **same principal shell**.
- The increased nuclear charge pulls the electron cloud **closer to the nucleus**, reducing the atomic radius.
- Shielding effect remains approximately constant across a period.

Atomic radius: Na > Mg > Al > Si > P > Cl > Ar

Radius decreases →



#### Trend 2: Down a Group (Top → Bottom)

Atomic radius **increases** down a group.

- Moving down a group, each successive element has electrons in a **higher principal quantum shell** ( $n$  increases).
- The outermost electrons are **farther from the nucleus**.
- Increased **shielding by inner electrons** reduces effective nuclear charge felt by the outermost electrons.
- Both effects cause the atomic radius to **increase**.

Atomic radius: Li < Na < K < Rb < Cs

#### Summary of Trends:

| Direction                   | Trend     | Reason                          |
|-----------------------------|-----------|---------------------------------|
| Across a period (L → R)     | Decreases | Increasing $Z$ , same shell     |
| Down a group (Top → Bottom) | Increases | New shell added, more shielding |

**Part 2: Arrange Na, Al, P, Cl, Mg in Decreasing Atomic Radius**

All five elements belong to **Period 3** of the periodic table:

| Element | Atomic No. | Period 3 Position | Atomic Radius (pm) |
|---------|------------|-------------------|--------------------|
| Na      | 11         | 1st               | 186                |
| Mg      | 12         | 2nd               | 160                |
| Al      | 13         | 3rd               | 143                |
| P       | 15         | 5th               | 110                |
| Cl      | 17         | 7th               | 99                 |

Since all are in the same period, atomic radius **decreases with increasing atomic number**. Therefore, the order of **decreasing atomic radius** is:

$$\text{Na} > \text{Mg} > \text{Al} > \text{P} > \text{Cl}$$

**Q48.** Derive an expression applying Bohr atom model for the calculation of wavelength of radiation obtained in the emission spectrum of hydrogen. **(10 marks)** [CHEM 101 | 2010–11]

**Answer**
**Derivation of Wavelength Expression Using Bohr's Atomic Model**
**Bohr's Postulates (relevant):**

- (i) Electrons revolve in fixed circular orbits without radiating energy.
- (ii) Energy is emitted or absorbed only when an electron transitions between orbits.
- (iii) The angular momentum of the electron is quantized:

$$m_e v r = \frac{n h}{2 \pi}$$

**Step 1: Energy of an Electron in the  $n$ -th Orbit**

Balancing centripetal and electrostatic forces:

$$\frac{m_e v^2}{r} = \frac{e^2}{4 \pi \epsilon_0 r^2} \implies r = \frac{e^2}{4 \pi \epsilon_0 m_e v^2}$$

From the quantization condition:

$$v = \frac{n h}{2 \pi m_e r}$$

Solving simultaneously, the radius of the  $n$ -th orbit:

$$r_n = \frac{n^2 h^2 \epsilon_0}{\pi m_e e^2}$$

The total energy (KE + PE) of the electron in the  $n$ -th orbit:

$$E_n = -\frac{m_e e^4}{8 \epsilon_0^2 h^2} \cdot \frac{1}{n^2}$$

**Step 2: Energy of Emitted Photon**

When an electron transitions from a higher orbit  $n_2$  to a lower orbit  $n_1$  ( $n_2 > n_1$ ), the energy of the emitted photon is:

$$\Delta E = E_{n_2} - E_{n_1} = \frac{m_e e^4}{8 \epsilon_0^2 h^2} \left( \frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

**Step 3: Expression for Wavelength**

Since  $\Delta E = h \nu = \frac{h c}{\lambda}$ :

$$\frac{h c}{\lambda} = \frac{m_e e^4}{8 \epsilon_0^2 h^2} \left( \frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

$$\frac{1}{\lambda} = \frac{m_e e^4}{8 \epsilon_0^2 h^3 c} \left( \frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

**Step 4: The Rydberg Constant**

The term  $\frac{m_e e^4}{8 \epsilon_0^2 h^3 c}$  is the **Rydberg constant**  $R_H$ :

$$R_H = \frac{m_e e^4}{8 \epsilon_0^2 h^3 c} = 1.097 \times 10^7 \text{ m}^{-1}$$

Thus the final expression becomes:

$$\frac{1}{\lambda} = R_H \left( \frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

where  $n_1 = 1, 2, 3, \dots$  and  $n_2 = n_1 + 1, n_1 + 2, \dots$

**Spectral Series of Hydrogen:**

| Series   | Region       | $n_1$ | $n_2$        |
|----------|--------------|-------|--------------|
| Lyman    | Ultraviolet  | 1     | 2, 3, 4, ... |
| Balmer   | Visible      | 2     | 3, 4, 5, ... |
| Paschen  | Infrared     | 3     | 4, 5, 6, ... |
| Brackett | Infrared     | 4     | 5, 6, 7, ... |
| Pfund    | Far Infrared | 5     | 6, 7, 8, ... |

**Q49.** What does Schrödinger wave equation describe? Deduce the equation and comment on its solution. **(10 marks)** [CHEM 101 | 2010–11]

### Answer

#### The Schrödinger Wave Equation

##### Part 1: What Does It Describe?

The Schrödinger wave equation is the **fundamental equation of quantum mechanics**. It describes:

- The **wave-like behaviour** of a particle (e.g. an electron) in a given potential field.
- The **allowed energy states** (energy quantization) of a quantum mechanical system.
- The **wavefunction**  $\psi$ , which contains all information about the state of the particle.
- The **probability** of finding an electron at a given point in space, given by  $|\psi|^2$  (Born interpretation).

It replaces Bohr's fixed circular orbits with **orbitals** — three-dimensional regions of space where the probability of finding an electron is highest.

##### Part 2: Deduction of the Schrödinger Wave Equation

##### Step 1: Classical Wave Equation

A classical wave travelling in the  $x$ -direction is described by:

$$\psi(x) = A \sin \left( \frac{2\pi x}{\lambda} \right)$$

Differentiating twice with respect to  $x$ :

$$\begin{aligned} \frac{d\psi}{dx} &= A \cdot \frac{2\pi}{\lambda} \cos \left( \frac{2\pi x}{\lambda} \right) \\ \frac{d^2\psi}{dx^2} &= -A \left( \frac{2\pi}{\lambda} \right)^2 \sin \left( \frac{2\pi x}{\lambda} \right) = -\frac{4\pi^2}{\lambda^2} \psi \end{aligned}$$

Rearranging:

$$\frac{d^2\psi}{dx^2} + \frac{4\pi^2}{\lambda^2} \psi = 0 \quad (1)$$

##### Step 2: Substitute de Broglie Wavelength

From de Broglie's relation:

$$\lambda = \frac{h}{mv} \implies \frac{1}{\lambda} = \frac{mv}{h} \implies \frac{4\pi^2}{\lambda^2} = \frac{4\pi^2 m^2 v^2}{h^2}$$

Substituting into equation (1):

$$\frac{d^2\psi}{dx^2} + \frac{4\pi^2 m^2 v^2}{h^2} \psi = 0 \quad (2)$$

##### Step 3: Express Kinetic Energy in Terms of Total Energy

The total energy  $E$  of the particle is the sum of kinetic energy (KE) and potential energy (V):

$$E = \text{KE} + V = \frac{1}{2}mv^2 + V$$

$$\text{KE} = \frac{1}{2}mv^2 = E - V \implies m^2v^2 = 2m(E - V)$$

##### Step 4: Substituting into Equation (2)

Replacing  $m^2v^2$  with  $2m(E - V)$ :

$$\begin{aligned} \frac{d^2\psi}{dx^2} + \frac{4\pi^2}{h^2} \cdot 2m(E - V) \psi &= 0 \\ \frac{d^2\psi}{dx^2} + \frac{8\pi^2 m}{h^2} (E - V) \psi &= 0 \end{aligned}$$

**Step 5: Three-Dimensional Form**

 Extending to three dimensions  $(x, y, z)$ :

$$\frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} + \frac{\partial^2 \psi}{\partial z^2} + \frac{8\pi^2 m}{h^2} (E - V) \psi = 0$$

 Using the Laplacian operator  $\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$ :

$$\nabla^2 \psi + \frac{8\pi^2 m}{h^2} (E - V) \psi = 0$$

 This is the **time-independent Schrödinger wave equation**.

 Using  $\hbar = \frac{h}{2\pi}$ , it is equivalently written as:

$$-\frac{\hbar^2}{2m} \nabla^2 \psi + V\psi = E\psi \quad \text{or} \quad \hat{H}\psi = E\psi$$

 where  $\hat{H}$  is the **Hamiltonian operator**.

**Part 3: Comments on the Solution**

- (i) **The Wavefunction  $\psi$ :** The solution of the Schrödinger equation gives the wavefunction  $\psi$ , which is a mathematical function of the coordinates of the electron  $(x, y, z)$ . By itself,  $\psi$  has no direct physical meaning.
- (ii) **Physical Interpretation — Born's Interpretation:** The square of the wavefunction,  $|\psi|^2$ , gives the **probability density** of finding the electron at a point in space. A large  $|\psi|^2$  at a point means a high probability of finding the electron there.
- (iii) **Quantization of Energy:** Acceptable solutions exist only for **specific values of energy  $E$** , called **eigenvalues**. This naturally explains the quantization of energy levels without any arbitrary assumption (unlike Bohr's model).
- (iv) **Quantum Numbers:** The solutions introduce three quantum numbers naturally:
  - $n$  — principal quantum number
  - $l$  — azimuthal quantum number
  - $m_l$  — magnetic quantum number
- (v) **Acceptable (Well-behaved) Wavefunctions:** Not all mathematical solutions are physically acceptable. A valid  $\psi$  must be:
  - **Single-valued** at every point
  - **Continuous** and **finite** everywhere
  - **Normalizable:**  $\int_{-\infty}^{+\infty} |\psi|^2 d\tau = 1$
- (vi) **Orbitals:** Each acceptable solution  $\psi_{n,l,m_l}$  describes an **atomic orbital** — a three-dimensional probability distribution for the electron, replacing Bohr's classical circular orbits.

**Q50.** Mention the variations of the following properties of the elements within periods and groups in the periodic table and give reasons for these variations: (i) Oxidation power, (ii) Metallic character, (iii) Ionic radii, (iv) Reducing power, (v) Atomic size. **(7 marks)**  
[CHEM 101 | 2010–11]

**Answer**
**Periodic Variations of Properties in the Periodic Table**
**(i) Oxidation Power**

- **Across a period (L→R):** Oxidation power **increases**. As nuclear charge increases across a period, the tendency to attract electrons (electronegativity) increases, so elements become stronger oxidizing agents.
- **Down a group (Top→Bottom):** Oxidation power **decreases**. Atomic radius increases and shielding increases down the group, reducing the tendency to attract electrons.

$$\text{e.g., } F_2 > Cl_2 > Br_2 > I_2 \quad (\text{oxidizing power})$$

**(ii) Metallic Character**

- **Across a period (L→R):** Metallic character **decreases**. Increasing nuclear charge makes it harder to lose electrons, so elements become less metallic and more non-metallic.
- **Down a group (Top→Bottom):** Metallic character **increases**. Increased atomic radius and shielding reduce the effective nuclear charge on valence electrons, making them easier to lose.

$$\text{e.g., } Li < Na < K < Rb \quad (\text{metallic character})$$

**(iii) Ionic Radii**



- **Cations** are always **smaller** than their parent atoms (loss of electrons reduces electron–electron repulsion, and nuclear charge pulls remaining electrons inward).
- **Anions** are always **larger** than their parent atoms (gain of electrons increases repulsion, expanding the electron cloud).
- **Across a period (L→R):** For isoelectronic ions, ionic radius **decreases** with increasing nuclear charge.

$$\text{e.g., } \text{N}^{3-} > \text{O}^{2-} > \text{F}^- > \text{Na}^+ > \text{Mg}^{2+}$$

- **Down a group (Top→Bottom):** Ionic radius **increases** as electrons occupy higher principal shells.

$$\text{e.g., } \text{F}^- < \text{Cl}^- < \text{Br}^- < \text{I}^-$$

#### (iv) Reducing Power

- **Across a period (L→R):** Reducing power **decreases**. Ionization energy increases across a period, making it harder for atoms to donate electrons and act as reducing agents.
- **Down a group (Top→Bottom):** Reducing power **increases**. Ionization energy decreases down a group (larger atomic size, more shielding), making electron donation easier.

$$\text{e.g., } \text{Li} < \text{Na} < \text{K} < \text{Rb} < \text{Cs} \quad (\text{reducing power})$$

#### (v) Atomic Size

- **Across a period (L→R):** Atomic size **decreases**. Nuclear charge  $Z$  increases while electrons are added to the **same shell**, so the nucleus pulls the electron cloud inward more strongly.
- **Down a group (Top→Bottom):** Atomic size **increases**. Each new period adds a new principal shell, placing valence electrons farther from the nucleus. Increased shielding by inner electrons further reduces the effective nuclear charge.

$$\text{e.g., } \text{Li} < \text{Na} < \text{K} < \text{Rb} < \text{Cs} \quad (\text{atomic size})$$

#### Summary Table:

| Property           | Across Period (L→R) | Down Group (Top→Bottom) |
|--------------------|---------------------|-------------------------|
| Oxidation power    | Increases           | Decreases               |
| Metallic character | Decreases           | Increases               |
| Ionic radii        | Decreases           | Increases               |
| Reducing power     | Decreases           | Increases               |
| Atomic size        | Decreases           | Increases               |

**Q51.** Calculate the magnitude of energy in erg of a quantum associated with light of wavelength 60.0 Å. **(8 marks)**

[CHEM 101 | 2010–11]

#### Answer

##### Energy of a Quantum of Light

##### Given:

- Wavelength:  $\lambda = 60.0 \text{ Å} = 60.0 \times 10^{-8} \text{ cm}$
- Planck's constant:  $h = 6.626 \times 10^{-27} \text{ erg} \cdot \text{s}$
- Speed of light:  $c = 3.0 \times 10^{10} \text{ cm} \cdot \text{s}^{-1}$

##### Formula:

The energy of a photon (quantum) is given by:

$$E = h\nu = \frac{hc}{\lambda}$$

##### Calculation:

$$E = \frac{hc}{\lambda} = \frac{6.626 \times 10^{-27} \text{ erg} \cdot \text{s} \times 3.0 \times 10^{10} \text{ cm} \cdot \text{s}^{-1}}{60.0 \times 10^{-8} \text{ cm}}$$

$$E = \frac{6.626 \times 3.0}{60.0} \times 10^{-27+10+8} \text{ erg}$$

$$E = \frac{19.878}{60.0} \times 10^{-9} \text{ erg}$$

$$E = 0.3313 \times 10^{-9} \text{ erg}$$

$$E = 3.313 \times 10^{-10} \text{ erg}$$

**Note:** This wavelength ( $60.0 \text{ \AA} = 6.0 \text{ nm}$ ) lies in the **extreme ultraviolet (EUV)** region of the electromagnetic spectrum. The high energy of this quantum ( $3.313 \times 10^{-10} \text{ erg}$ ) is consistent with the short wavelength, as energy and wavelength are inversely related:

$$E \propto \frac{1}{\lambda}$$

**Q52.** What is quantum theory of radiation? Discuss photoelectric effect in relation to quantization of energy. **(10 marks)** [CHEM 101 | 2009–10]

### Answer

#### Quantum Theory of Radiation and Photoelectric Effect

##### Part 1: Quantum Theory of Radiation (Planck, 1900)

Classical physics assumed that energy is emitted or absorbed **continuously** in any amount. This failed to explain blackbody radiation (the “ultraviolet catastrophe”).

Max Planck proposed the **Quantum Theory of Radiation** based on the following postulates:

- (i) Energy is not emitted or absorbed continuously, but in **discrete packets** called **quanta** (singular: quantum).
- (ii) The energy of each quantum is directly proportional to the frequency  $\nu$  of the radiation:

$$E = h\nu$$

where  $h = 6.626 \times 10^{-34} \text{ J}\cdot\text{s}$  is **Planck’s constant**.

- (iii) A body can emit or absorb energy only in **whole number multiples** of  $h\nu$ :

$$E = nh\nu \quad (n = 1, 2, 3, \dots)$$

- (iv) Energy is **quantized** — it cannot take arbitrary values, only discrete allowed values.

This was a revolutionary departure from classical physics and laid the foundation of **modern quantum mechanics**.

##### Part 2: Photoelectric Effect

**Definition:** The **photoelectric effect** is the phenomenon in which **electrons are ejected from the surface of a metal** when light of sufficient frequency is incident upon it.

##### Experimental Observations:

- (i) Electrons are emitted **only if the frequency** of incident light exceeds a minimum value called the **threshold frequency**  $\nu_0$ , regardless of the intensity of light.
- (ii) **Below**  $\nu_0$ , no electrons are emitted even if the light is very intense.
- (iii) The **kinetic energy** of emitted electrons increases with increasing frequency, but is **independent of intensity**.
- (iv) Increasing the **intensity** of light increases the **number** of emitted electrons, not their kinetic energy.
- (v) Electron emission is **instantaneous** — there is no time delay.

##### Failure of Classical Theory:

Classical wave theory predicted that:

- Any frequency of light should eventually eject electrons if intensity is high enough.
- There should be a time delay before emission.

Both predictions **contradicted** experimental results.

##### Einstein’s Quantum Explanation (1905):

Einstein extended Planck’s quantum theory and proposed that light consists of **photons** — discrete packets of energy. Each photon carries energy:

$$E = h\nu$$

When a photon strikes a metal surface:

- A part of the photon energy is used to overcome the **work function**  $\phi = h\nu_0$  (the minimum energy needed to eject an electron).
- The remaining energy appears as the **kinetic energy** of the ejected electron.

This gives **Einstein’s photoelectric equation**:

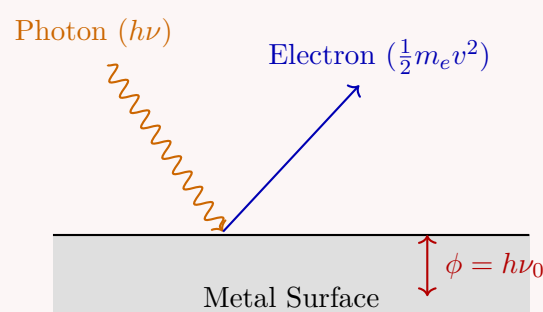
$$h\nu = h\nu_0 + \frac{1}{2}m_e v^2$$

$$\frac{1}{2}m_e v^2 = h\nu - h\nu_0 = h(\nu - \nu_0)$$

where:

- $h\nu$  = energy of incident photon
- $h\nu_0 = \phi$  = work function of the metal
- $\frac{1}{2}m_e v^2$  = kinetic energy of emitted electron

Diagram:



Relation to Quantization of Energy:

- The photoelectric effect provides **direct experimental proof** that light energy is quantized into photons of energy  $h\nu$ .
- A single photon interacts with a single electron in an **all-or-nothing** fashion — the electron is either ejected or not, depending on whether  $h\nu \geq h\nu_0$ .
- The **instantaneous** nature of emission confirms that energy is delivered in discrete packets, not spread continuously over time as a wave.
- The **threshold frequency**  $\nu_0$  is direct evidence of energy quantization: only photons with  $E = h\nu \geq \phi$  can eject electrons.

Summary Table:

| Observation                    | Classical Prediction | Quantum Explanation            |
|--------------------------------|----------------------|--------------------------------|
| Threshold frequency            | Not explained        | $E = h\nu \geq \phi$           |
| KE depends on $\nu$            | Not explained        | $\text{KE} = h(\nu - \nu_0)$   |
| Intensity $\rightarrow$ number | Not explained        | More photons = more electrons  |
| Instantaneous emission         | Time delay expected  | One photon ejects one electron |

Q53. What are the limitations of Bohr's atomic theory and how are they overcome? (10 marks)

[CHEM 101 | 2009–10]

Answer

Limitations of Bohr's Atomic Theory and How They Are Overcome

Part 1: Limitations of Bohr's Atomic Theory

- Fails for Multi-electron Atoms:** Bohr's model works only for **hydrogen-like species** (H,  $\text{He}^+$ ,  $\text{Li}^{2+}$ , etc.). It completely fails for atoms with two or more electrons because it does not account for **electron–electron repulsion**, which significantly affects energy levels.
- Cannot Explain Fine Structure of Spectral Lines:** Under high resolution, each spectral line of hydrogen splits into several closely spaced lines. Bohr's theory predicts only single lines and cannot account for this **fine and hyperfine splitting**.
- Cannot Explain Zeeman and Stark Effects:** When atoms are placed in an external **magnetic field** (Zeeman effect) or **electric field** (Stark effect), spectral lines split further. Bohr's model has no mechanism to explain this splitting.
- Violates Heisenberg's Uncertainty Principle:** Bohr assumed electrons move in **fixed circular orbits** with precisely defined positions and velocities simultaneously. This directly contradicts Heisenberg's Uncertainty Principle:
 
$$\Delta x \cdot \Delta p \geq \frac{h}{4\pi}$$
- Does Not Account for Wave Nature of Electron:** Bohr treated the electron purely as a **classical particle**, ignoring de Broglie's wave–particle duality. It does not explain why only certain orbits are stable.
- Cannot Explain Relative Intensities of Spectral Lines:** Bohr's model predicts the **positions** of spectral lines but cannot predict why some lines are more intense than others.
- Restricted to Circular Orbits:** Bohr assumed only **circular orbits**. Sommerfeld extended this to elliptical orbits, but even then the model remained fundamentally inadequate for multi-electron systems.
- Cannot Explain Chemical Bonding:** Bohr's model provides no satisfactory explanation for **how atoms form bonds** to make molecules, since it treats each atom in isolation.

Part 2: How the Limitations Are Overcome

- (i) **Quantum Mechanical Model (Schrödinger, 1926):** Schrödinger replaced Bohr's classical circular orbits with **orbitals** by solving the wave equation:

$$\nabla^2\psi + \frac{8\pi^2m}{h^2}(E - V)\psi = 0$$

This model works for **all atoms**, including multi-electron systems, by incorporating electron–electron repulsions through approximate methods (e.g. Hartree–Fock).

- (ii) **Introduction of Four Quantum Numbers:** The quantum mechanical model introduces **four quantum numbers** to fully describe the state of an electron:

| Quantum No. | Symbol | Describes                                          |
|-------------|--------|----------------------------------------------------|
| Principal   | $n$    | Energy level / shell size                          |
| Azimuthal   | $l$    | Shape of orbital (subshell)                        |
| Magnetic    | $m_l$  | Orientation in space                               |
| Spin        | $m_s$  | Electron spin ( $+\frac{1}{2}$ or $-\frac{1}{2}$ ) |

The **spin quantum number**  $m_s$  explains the fine structure of spectral lines and the Zeeman effect.

- (iii) **Heisenberg's Uncertainty Principle Incorporated:** The quantum mechanical model abandons the concept of fixed orbits. Instead,  $|\psi|^2$  gives the **probability density** of finding an electron, fully consistent with the uncertainty principle.
- (iv) **de Broglie's Wave Nature Incorporated:** The Schrödinger equation is itself a **wave equation**, naturally incorporating the wave nature of the electron. Stable orbits correspond to **standing waves**:

$$2\pi r = n\lambda \implies m_e v r = \frac{nh}{2\pi}$$

- (v) **Zeeman and Stark Effects Explained:** By including the magnetic quantum number  $m_l$  and spin quantum number  $m_s$ , quantum mechanics successfully explains the splitting of spectral lines in external magnetic and electric fields.
- (vi) **Relative Intensities Explained:** Quantum mechanics introduces **selection rules** and **transition probabilities** that determine which transitions are allowed and how intense the corresponding spectral lines are.
- (vii) **Chemical Bonding Explained:** The quantum mechanical model forms the basis of **molecular orbital theory** and **valence bond theory**, which successfully explain how atoms combine to form molecules through overlap of orbitals.

Summary Table:

| Limitation of Bohr's Model      | How Overcome                          |
|---------------------------------|---------------------------------------|
| Fails for multi-electron atoms  | Schrödinger equation + approximations |
| No fine structure explanation   | Spin quantum number $m_s$             |
| Violates uncertainty principle  | Probability-based orbital model       |
| Ignores wave nature of electron | de Broglie + wave equation            |
| No Zeeman/Stark effect          | Magnetic & spin quantum numbers       |
| Cannot explain bonding          | MO theory & VB theory                 |

Q54. Justify the stability of a nucleus of an atom. (8 marks)

[CHEM 101 | 2009–10]

### Answer

#### Stability of a Nucleus of an Atom

##### 1. Composition of the Nucleus

The nucleus of an atom consists of **protons** (positively charged) and **neutrons** (electrically neutral), collectively called **nucleons**. Since protons are positively charged and confined in an extremely small volume ( $\sim 10^{-15}$  m), they experience enormous **electrostatic repulsion**. Yet the nucleus remains stable — this requires explanation.

##### 2. Nuclear Force (Strong Force)

The primary reason for nuclear stability is the **strong nuclear force**, which acts between all nucleons (proton–proton, neutron–neutron, proton–neutron). It is much stronger than electrostatic repulsion (about 100× stronger at short range), is **short-range** (effective only up to  $\sim 10^{-15}$  m), and is **charge independent**. As long as the strong nuclear force overcomes electrostatic repulsion, the nucleus remains stable.

##### 3. Neutron-to-Proton Ratio ( $n/p$ Ratio)

The ratio of neutrons to protons is a key factor in nuclear stability. For light nuclei ( $Z \leq 20$ ), stable nuclei have  $n/p \approx 1$  (e.g.  ${}^{12}_6\text{C}$  has 6 protons and 6 neutrons). For heavier nuclei ( $Z > 20$ ), more neutrons are needed to dilute proton–proton repulsion, so  $n/p > 1$  (e.g.  ${}^{208}_{82}\text{Pb}$  has  $n/p \approx 1.54$ ). If  $n/p$  is too high or too low, the nucleus becomes unstable and undergoes radioactive decay.

##### 4. Nuclear Binding Energy

**Binding energy** is the energy required to completely separate all nucleons in a nucleus. It arises from the **mass defect**  $\Delta m$ , the difference between the mass of the nucleus and the sum of masses of its individual nucleons:

$$\Delta m = Z m_p + N m_n - m_{\text{nucleus}}$$

By Einstein’s mass–energy relation:

$$E_b = \Delta m \cdot c^2$$

The **binding energy per nucleon** ( $E_b/A$ ) measures nuclear stability. Higher  $E_b/A$  means a more stable nucleus. The maximum  $E_b/A \approx 8.8$  MeV occurs near  $^{56}_{26}\text{Fe}$  (iron), making it the most stable nucleus. Nuclei lighter or heavier than iron undergo **fusion** or **fission** respectively to reach greater stability.

5. Magic Numbers

Certain nuclei are exceptionally stable when the number of protons or neutrons equals a **magic number**: 2, 8, 20, 28, 50, 82, 126. These correspond to completely filled nuclear shells, providing extra stability. Examples include  $^4_2\text{He}$  (doubly magic),  $^{16}_8\text{O}$  (doubly magic), and  $^{208}_{82}\text{Pb}$  (doubly magic).

6. Even–Odd Rule

The stability of a nucleus also depends on whether the number of protons and neutrons is even or odd:

| Protons | Neutrons | Stable nuclides | Stability         |
|---------|----------|-----------------|-------------------|
| Even    | Even     | ~157            | Most stable       |
| Even    | Odd      | ~53             | Moderately stable |
| Odd     | Even     | ~50             | Moderately stable |
| Odd     | Odd      | ~5              | Least stable      |

Even numbers of both protons and neutrons allow nucleons to pair up with opposite spins, lowering energy and enhancing stability.

**Summary:** Nuclear stability is determined by a combination of strong nuclear force, optimal  $n/p$  ratio, high binding energy per nucleon, magic numbers, and even–even nucleon pairing.

Q55. What is periodic law? How many periods and groups are there in a periodic table? Why are the periods not equal? (8 marks)  
[CHEM 101 | 2009–10]

Answer

**Periodic Law, Periods, Groups and Their Inequality**

**Part 1: Periodic Law**  
The **Modern Periodic Law** (Moseley, 1913) states:

*“The physical and chemical properties of elements are periodic functions of their **atomic numbers**.”*

This was a refinement of **Mendeleev’s Periodic Law** (1869), which stated that properties are periodic functions of **atomic masses**. Moseley corrected this by showing that **atomic number** (number of protons), not atomic mass, is the fundamental property that determines the position and properties of an element.

**Part 2: Periods in the Periodic Table**  
A **period** is a horizontal row in the periodic table. There are **7 periods** in total:

| Period | Elements | No. of Elements | Highest Shell Filled     |
|--------|----------|-----------------|--------------------------|
| 1      | H to He  | 2               | $n = 1$ (1s)             |
| 2      | Li to Ne | 8               | $n = 2$ (2s, 2p)         |
| 3      | Na to Ar | 8               | $n = 3$ (3s, 3p)         |
| 4      | K to Kr  | 18              | $n = 4$ (4s, 3d, 4p)     |
| 5      | Rb to Xe | 18              | $n = 5$ (5s, 4d, 5p)     |
| 6      | Cs to Rn | 32              | $n = 6$ (6s, 4f, 5d, 6p) |
| 7      | Fr to Og | 32              | $n = 7$ (7s, 5f, 6d, 7p) |

**Part 3: Groups in the Periodic Table**  
A **group** is a vertical column in the periodic table. There are **18 groups** in the modern periodic table (IUPAC), numbered 1 to 18. Elements in the same group have the same **valence electron configuration** and hence similar chemical properties. For example, Group 1 (alkali metals) all have one valence electron and are highly reactive.

**Part 4: Why Are the Periods Not Equal in Length?**  
The periods are unequal in length because the **number of elements in each period depends on the number of electrons that can be accommodated in the subshells** of the corresponding principal quantum shell.  
The maximum number of electrons in a shell is given by  $2n^2$ , and the subshells available determine the number of elements per period. The key reasoning is:

(i) **Period 1 (2 elements):** Only the 1s subshell is filled. It holds a maximum of 2 electrons:

$$1s^2 \Rightarrow 2 \text{ elements}$$

(ii) **Periods 2 and 3 (8 elements each):** The s and p subshells of  $n = 2$  and  $n = 3$  are filled. The d subshell of  $n = 3$  is not filled until period 4:

$$ns^2 np^6 \Rightarrow 2 + 6 = 8 \text{ elements}$$



- (iii) **Periods 4 and 5 (18 elements each):** The  $s$ ,  $d$ , and  $p$  subshells are filled. The  $3d$  subshell (10 electrons) is included in period 4:

$$ns^2(n-1)d^{10}np^6 \Rightarrow 2 + 10 + 6 = 18 \text{ elements}$$

- (iv) **Periods 6 and 7 (32 elements each):** The  $s$ ,  $f$ ,  $d$ , and  $p$  subshells are all filled. The  $4f$  subshell (14 electrons) is included:

$$ns^2(n-2)f^{14}(n-1)d^{10}np^6 \Rightarrow 2 + 14 + 10 + 6 = 32 \text{ elements}$$

**Summary:**

| Subshells Filled | Electrons       | Period Length |
|------------------|-----------------|---------------|
| $s$              | 2               | 2             |
| $s + p$          | 2 + 6           | 8             |
| $s + d + p$      | 2 + 10 + 6      | 18            |
| $s + f + d + p$  | 2 + 14 + 10 + 6 | 32            |

The periods are not equal because different principal shells contain different numbers of subshells, and each subshell has a different capacity for electrons ( $s$ : 2,  $p$ : 6,  $d$ : 10,  $f$ : 14).

- Q56.** How would you conceive the idea of having an electron cloud around the nucleus of an atom? How is it experimentally proved that the electron has wave nature? (11 marks) [CHEM 101 | 2009–10]

### Answer

#### Electron Cloud and Experimental Proof of Wave Nature of Electron

##### Part 1: The Concept of Electron Cloud

##### Background — Failure of Bohr's Orbit:

Bohr's model assumed electrons move in **fixed circular orbits** with a definite position and velocity at every instant. However, **Heisenberg's Uncertainty Principle** states:

$$\Delta x \cdot \Delta p \geq \frac{h}{4\pi}$$

It is fundamentally impossible to know the exact position and momentum of an electron simultaneously. Therefore, the concept of a fixed orbit is physically meaningless.

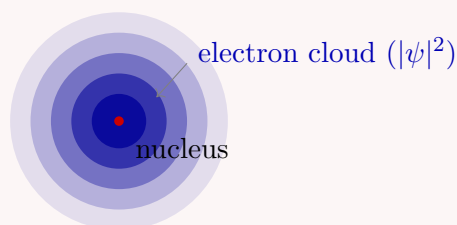
##### Quantum Mechanical Picture — The Electron Cloud:

In the quantum mechanical model, the electron is not confined to a fixed orbit. Instead, its behaviour is described by the **wave-function**  $\psi$ , obtained by solving the Schrödinger equation:

$$\nabla^2\psi + \frac{8\pi^2m}{h^2}(E - V)\psi = 0$$

The physical meaning of  $\psi$  is given by **Born's interpretation**: the quantity  $|\psi|^2$  at any point in space represents the **probability density** of finding the electron at that point. This leads to the concept of the **electron cloud**:

- The electron does not follow a fixed path. Instead, it occupies a **three-dimensional region** around the nucleus called an **orbital**.
- The **density of the cloud** at any point is proportional to  $|\psi|^2$  — regions of high density indicate high probability of finding the electron, and regions of low density indicate low probability.
- The electron cloud is **not a physical rotating cloud**; it is a **probability map** showing where the electron is most likely to be found.
- For the hydrogen atom ground state ( $1s$  orbital), the electron cloud is **spherically symmetric** and most dense near the nucleus, becoming diffuse further away.



Electron cloud for  $1s$  orbital of hydrogen

##### Part 2: Experimental Proof of Wave Nature of Electron

##### Theoretical Basis — de Broglie's Hypothesis (1924):

de Broglie proposed that every moving particle has an associated wavelength:

$$\lambda = \frac{h}{mv} = \frac{h}{p}$$

This predicted that electrons, being particles, should also exhibit **wave behaviour** such as diffraction and interference.

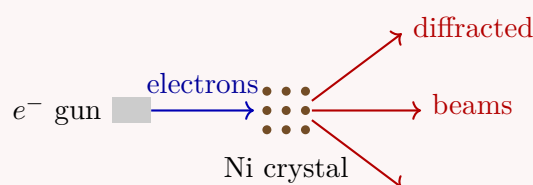
##### Experiment 1: Davisson–Germer Experiment (1927)



- A beam of electrons was directed at a **nickel crystal** surface at various angles.
- The crystal acted as a **diffraction grating** (the inter-atomic spacing  $\sim 10^{-10}$  m is comparable to the de Broglie wavelength of electrons).
- A **diffraction pattern** was observed — electrons were scattered at specific angles with maximum and minimum intensities, exactly like X-rays (which are known waves).
- The measured wavelength matched the de Broglie prediction:

$$\lambda = \frac{h}{mv}$$

- This was **direct experimental proof** that electrons have wave nature.



### Experiment 2: G.P. Thomson's Experiment (1927)

- A beam of high-energy electrons was passed through a **thin gold foil**.
- On a photographic plate placed behind the foil, **concentric circular diffraction rings** were observed — identical to those produced by X-rays passing through a crystalline material.
- This pattern is only possible if electrons behave as **waves** and undergo diffraction through the regular atomic planes of the gold crystal.
- G.P. Thomson was awarded the **Nobel Prize** in 1937 for this discovery (shared with Davisson).

#### Significance:

- Both experiments confirmed **de Broglie's hypothesis** that matter has wave nature.
- The wave nature of the electron is the physical basis for the **electron cloud model** and the Schrödinger equation.
- It established **wave-particle duality** as a fundamental property of all matter.

**Q57.** Deduce an equation by which the propagation of electron wave in space can be described. How can the solutions of the equation be used to describe the different shapes of orbitals? **(12 marks)** [CHEM 101 | 2009–10]

#### Answer

#### Schrödinger Wave Equation and Shapes of Orbitals

##### Part 1: Deduction of the Schrödinger Wave Equation

##### Step 1: Classical Wave Equation

A wave travelling in the  $x$ -direction is mathematically described by:

$$\psi(x) = A \sin\left(\frac{2\pi x}{\lambda}\right)$$

Differentiating twice with respect to  $x$ :

$$\begin{aligned} \frac{d\psi}{dx} &= A \cdot \frac{2\pi}{\lambda} \cos\left(\frac{2\pi x}{\lambda}\right) \\ \frac{d^2\psi}{dx^2} &= -A \left(\frac{2\pi}{\lambda}\right)^2 \sin\left(\frac{2\pi x}{\lambda}\right) = -\frac{4\pi^2}{\lambda^2} \psi \end{aligned}$$

Rearranging:

$$\frac{d^2\psi}{dx^2} + \frac{4\pi^2}{\lambda^2} \psi = 0 \quad (1)$$

##### Step 2: Introduce de Broglie Wavelength

From de Broglie's relation  $\lambda = \frac{h}{mv}$ :

$$\frac{1}{\lambda^2} = \frac{m^2 v^2}{h^2} \Rightarrow \frac{4\pi^2}{\lambda^2} = \frac{4\pi^2 m^2 v^2}{h^2}$$

Substituting into equation (1):

$$\frac{d^2\psi}{dx^2} + \frac{4\pi^2 m^2 v^2}{h^2} \psi = 0 \quad (2)$$

##### Step 3: Express Using Total Energy

Total energy  $E = \text{Kinetic energy} + \text{Potential energy } V$ :

$$E = \frac{1}{2}mv^2 + V \Rightarrow \frac{1}{2}mv^2 = E - V \Rightarrow m^2v^2 = 2m(E - V)$$

Substituting into equation (2):

$$\frac{d^2\psi}{dx^2} + \frac{8\pi^2m}{h^2}(E - V)\psi = 0$$

#### Step 4: Extend to Three Dimensions

Extending to three dimensions  $(x, y, z)$ :

$$\frac{\partial^2\psi}{\partial x^2} + \frac{\partial^2\psi}{\partial y^2} + \frac{\partial^2\psi}{\partial z^2} + \frac{8\pi^2m}{h^2}(E - V)\psi = 0$$

Using the Laplacian operator  $\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$ , the **Schrödinger wave equation** is:

$$\nabla^2\psi + \frac{8\pi^2m}{h^2}(E - V)\psi = 0$$

Using  $\hbar = \frac{h}{2\pi}$ , this is equivalently written as:

$$\hat{H}\psi = E\psi \quad \text{where} \quad \hat{H} = -\frac{\hbar^2}{2m}\nabla^2 + V$$

is the **Hamiltonian operator**.

## Part 2: Solutions and Shapes of Orbitals

### Nature of Solutions:

Solving the Schrödinger equation in spherical polar coordinates  $(r, \theta, \phi)$  gives solutions of the form:

$$\psi_{n,l,m_l}(r, \theta, \phi) = R_{n,l}(r) \cdot Y_{l,m_l}(\theta, \phi)$$

where:

- $R_{n,l}(r)$  is the **radial wavefunction** — describes how  $\psi$  varies with distance  $r$  from the nucleus.
- $Y_{l,m_l}(\theta, \phi)$  is the **angular wavefunction** — describes the **shape and orientation** of the orbital.

Each solution is characterized by three quantum numbers:

| Quantum Number | Symbol                          | Determines                 |
|----------------|---------------------------------|----------------------------|
| Principal      | $n = 1, 2, 3 \dots$             | Size and energy of orbital |
| Azimuthal      | $l = 0, 1, \dots, n - 1$        | Shape of orbital           |
| Magnetic       | $m_l = -l, \dots, 0, \dots, +l$ | Orientation in space       |

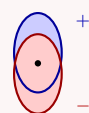
### Shapes of Orbitals from Solutions:

- (i) **s Orbitals ( $l = 0$ ):** The angular wavefunction  $Y_{0,0}$  is a constant, independent of  $\theta$  and  $\phi$ . Therefore,  $|\psi|^2$  is the same in all directions — the orbital is **perfectly spherical**. For each value of  $n$ , there is one  $s$  orbital. As  $n$  increases, the sphere becomes larger and contains  $(n - 1)$  radial nodes.



1s (spherical)

- (ii) **p Orbitals ( $l = 1$ ):** There are three  $p$  orbitals ( $m_l = -1, 0, +1$ ), namely  $p_x$ ,  $p_y$ , and  $p_z$ . The angular wavefunction gives a **dumbbell (figure-eight) shape** with two lobes on opposite sides of the nucleus along the  $x$ ,  $y$ , or  $z$  axis. There is a **nodal plane** passing through the nucleus where  $\psi = 0$ .



$p_z$  (dumbbell)

- (iii) **d Orbitals ( $l = 2$ ):** There are five  $d$  orbitals ( $m_l = -2, -1, 0, +1, +2$ ). Four of them ( $d_{xy}$ ,  $d_{xz}$ ,  $d_{yz}$ ,  $d_{x^2-y^2}$ ) have a **four-leaf clover (double dumbbell) shape** with four lobes. The fifth ( $d_{z^2}$ ) has two lobes along the  $z$ -axis with a **doughnut-shaped ring** in the  $xy$ -plane.
- (iv) **f Orbitals ( $l = 3$ ):** There are seven  $f$  orbitals ( $m_l = -3$  to  $+3$ ) with complex **multi-lobed shapes**. They are involved in the chemistry of lanthanides and actinides.

**Nodes in Orbitals:**

The number of nodes (regions where  $\psi = 0$ ) also follows from the solutions:

| Type          | Number      | Depends on               |
|---------------|-------------|--------------------------|
| Radial nodes  | $n - l - 1$ | Principal & azimuthal QN |
| Angular nodes | $l$         | Azimuthal QN             |
| Total nodes   | $n - 1$     | Principal QN only        |

**Summary:** The solutions  $\psi_{n,l,m_l}$  of the Schrödinger equation directly yield the **shapes, sizes, orientations, and energies** of atomic orbitals through the three quantum numbers  $n$ ,  $l$ , and  $m_l$ . The probability density  $|\psi|^2$  gives the **electron cloud distribution**, which defines the physical shape of each orbital.

**Q58.** What are the transition elements and which periods of the periodic table do they belong to? Discuss the important properties of transition elements. (7 marks) [CHEM 101 | 2008–09]

**Answer**
**Transition Elements: Definition, Periods and Properties**
**Part 1: Definition of Transition Elements**

Transition elements are defined as those elements in which the  **$d$  subshell is partially filled** either in the ground state of the atom or in any of their commonly occurring ions. They are also called  **$d$ -block elements**.

Formally, they are elements with the general electronic configuration:

$$(n-1)d^{1-10} ns^{1-2}$$

**Note:** Zn, Cd, and Hg are sometimes excluded from the strict definition because their  $d$  subshell is completely filled ( $d^{10}$ ) in both ground state and common ions.

**Part 2: Periods of the Periodic Table**

Transition elements occupy **Groups 3–12** and are found in **Periods 4, 5, 6, and 7** of the periodic table, forming four transition series:

| Series   | Period   | Elements                                 | Configuration                     |
|----------|----------|------------------------------------------|-----------------------------------|
| 1st (3d) | Period 4 | Sc ( $Z = 21$ ) to Zn ( $Z = 30$ )       | $[Ar] 3d^{1-10} 4s^{1-2}$         |
| 2nd (4d) | Period 5 | Y ( $Z = 39$ ) to Cd ( $Z = 48$ )        | $[Kr] 4d^{1-10} 5s^{1-2}$         |
| 3rd (5d) | Period 6 | La, Hf ( $Z = 72$ ) to Hg ( $Z = 80$ )   | $[Xe] 4f^{14} 5d^{1-10} 6s^{1-2}$ |
| 4th (6d) | Period 7 | Ac, Rf ( $Z = 104$ ) to Cn ( $Z = 112$ ) | $[Rn] 5f^{14} 6d^{1-10} 7s^{1-2}$ |

**Part 3: Important Properties of Transition Elements**

- (i) **Variable Oxidation States:** Transition metals exhibit **multiple oxidation states** because both  $(n-1)d$  and  $ns$  electrons can participate in bonding. For example:



- (ii) **Formation of Coloured Compounds:** Most transition metal compounds are **coloured** because electrons in lower  $d$  orbitals absorb visible light and are excited to higher  $d$  orbitals ( **$d-d$  transitions**). The complementary colour is observed. For example,  $\text{CuSO}_4$  is blue,  $\text{KMnO}_4$  is purple, and  $\text{K}_2\text{Cr}_2\text{O}_7$  is orange. Ions with  $d^0$  ( $\text{Sc}^{3+}$ ) or  $d^{10}$  ( $\text{Zn}^{2+}$ ) are colourless as  $d-d$  transitions are not possible.

- (iii) **Magnetic Properties:** Transition metal compounds are often **paramagnetic** due to the presence of unpaired  $d$  electrons. The magnetic moment is:

$$\mu = \sqrt{n(n+2)} \text{ B.M.}$$

where  $n$  is the number of unpaired electrons. Compounds with no unpaired electrons are **diamagnetic**.

- (iv) **Formation of Complex Compounds:** Transition metals readily form **coordination complexes** with ligands due to their small ionic size, high charge density, and available empty  $d$  orbitals to accept lone pairs. Examples:  $[\text{Fe}(\text{CN})_6]^{3-}$ ,  $[\text{Cu}(\text{NH}_3)_4]^{2+}$ .
- (v) **Catalytic Properties:** Transition metals are excellent **catalysts** due to their variable oxidation states and ability to adsorb reactants on their surfaces. Examples: Fe in the Haber process,  $\text{V}_2\text{O}_5$  in the Contact process, Pt in catalytic converters.
- (vi) **High Melting and Boiling Points:** Transition metals have **high melting points** because both  $s$  and  $d$  electrons participate in strong metallic bonding. Tungsten (W) has the highest melting point of all metals ( $3422^\circ\text{C}$ ).
- (vii) **Formation of Alloys:** Transition metals readily form **alloys** with each other due to similar atomic radii. Alloys are harder and more corrosion resistant than pure metals. Examples: steel (Fe + C), brass (Cu + Zn), nichrome (Ni + Cr).

**Summary Table:**

| Property                  | Reason                                           |
|---------------------------|--------------------------------------------------|
| Variable oxidation states | <i>d</i> and <i>s</i> electrons both participate |
| Coloured compounds        | <i>d</i> – <i>d</i> electronic transitions       |
| Paramagnetism             | Unpaired <i>d</i> electrons                      |
| Complex formation         | Empty <i>d</i> orbitals, high charge density     |
| Catalytic activity        | Variable valency, surface adsorption             |
| High melting points       | Strong metallic bonding via <i>d</i> electrons   |

Q59. What are noble gases? Discuss the important physical and chemical properties of noble gases. (7 marks) [CHEM 101 | 2008–09]

Answer

**Noble Gases: Definition and Properties**

**Part 1: What are Noble Gases?**

Noble gases are the elements of **Group 18** (Group 0) of the periodic table. They are also called **inert gases** or **rare gases**. The six noble gases are:

| Symbol | Name    | Atomic No. | Configuration                                                          |
|--------|---------|------------|------------------------------------------------------------------------|
| He     | Helium  | 2          | 1s <sup>2</sup>                                                        |
| Ne     | Neon    | 10         | [He] 2s <sup>2</sup> 2p <sup>6</sup>                                   |
| Ar     | Argon   | 18         | [Ne] 3s <sup>2</sup> 3p <sup>6</sup>                                   |
| Kr     | Krypton | 36         | [Ar] 3d <sup>10</sup> 4s <sup>2</sup> 4p <sup>6</sup>                  |
| Xe     | Xenon   | 54         | [Kr] 4d <sup>10</sup> 5s <sup>2</sup> 5p <sup>6</sup>                  |
| Rn     | Radon   | 86         | [Xe] 4f <sup>14</sup> 5d <sup>10</sup> 6s <sup>2</sup> 6p <sup>6</sup> |

Their outermost shell is **completely filled** (*ns<sup>2</sup>np<sup>6</sup>*, except He which has 1s<sup>2</sup>), giving them exceptional stability and very low chemical reactivity. This stable configuration is called the **octet configuration** (or duplet for He).

**Part 2: Physical Properties of Noble Gases**

(i) **State and Appearance:** All noble gases are **colourless, odourless, and tasteless** monoatomic gases under normal conditions. They exist as **single atoms** (monoatomic) because they have no tendency to form diatomic molecules.

(ii) **Low Boiling and Melting Points:** Noble gases have **very low boiling and melting points** because the only intermolecular forces present are weak **London dispersion (van der Waals) forces**. These increase down the group as atomic size and polarizability increase:

B.P.: He < Ne < Ar < Kr < Xe < Rn

Helium has the **lowest boiling point** of any substance (−269°C).

(iii) **Low Density:** Noble gases have **very low densities** compared to other gases due to their low atomic masses (except Rn). Helium is the second lightest gas after hydrogen.

(iv) **Low Solubility in Water:** Noble gases have **low but measurable solubility** in water, which increases down the group with increasing atomic size and polarizability.

(v) **Non-conductors:** In their normal state, noble gases are **non-conductors** of electricity. However, under high voltage, they **glow with characteristic colours** (used in neon signs): Ne (red-orange), Ar (blue-violet), Kr (pale violet), Xe (blue).

(vi) **Monoatomic and Ideal Gas Behaviour:** Noble gases behave nearly as **ideal gases** because of negligible intermolecular interactions. Their  $C_p/C_v = 5/3$ , which is the theoretical value for a monoatomic ideal gas.

**Part 3: Chemical Properties of Noble Gases**

(i) **General Inertness:** Noble gases are **chemically very inert** due to their completely filled valence shells. They have:

- **Zero electron affinity** — no tendency to gain electrons.
- **Very high ionization energies** — very difficult to remove electrons.
- **No unpaired electrons** — cannot form covalent bonds easily.

(ii) **Ionization Energy:** Noble gases have the **highest ionization energies** in their respective periods, decreasing down the group:

$IE:$  He > Ne > Ar > Kr > Xe > Rn

(iii) **Compounds of Noble Gases:** Although generally inert, heavier noble gases (Kr, Xe, Rn) can form compounds under special conditions because their outer electrons are more loosely held (larger atomic radius, lower ionization energy). Important examples include:

- **Xenon compounds:** XeF<sub>2</sub>, XeF<sub>4</sub>, XeF<sub>6</sub>, XeO<sub>3</sub>, XeOF<sub>4</sub> (first synthesized by Bartlett, 1962).
- **Krypton compounds:** KrF<sub>2</sub> (unstable at room temperature).
- **Radon compounds:** RnF<sub>2</sub> (radioactive, difficult to study).

He, Ne, and Ar form **no stable chemical compounds** under any known conditions.

- (iv) **Clathrate Compounds:** Noble gases can be **physically trapped** inside cavities of other molecules (e.g. ice, hydroquinone) to form **clathrates** or cage compounds. These are not true chemical compounds as no covalent bonds are formed. Example: Ar, Kr, Xe clathrates with water.
- (v) **Reactivity Trend Down the Group:** Chemical reactivity **increases down the group:**

$$\text{He} \approx \text{Ne} \approx \text{Ar} < \text{Kr} < \text{Xe} < \text{Rn}$$

This is because larger atoms have more polarizable electron clouds and lower ionization energies, making electron donation easier.

**Summary Table:**

| Property                | Reason                          |
|-------------------------|---------------------------------|
| Chemical inertness      | Completely filled valence shell |
| Low boiling points      | Only weak van der Waals forces  |
| High ionization energy  | Stable $ns^2np^6$ configuration |
| Xe forms compounds      | Large size, low IE, polarizable |
| He/Ne form no compounds | Very high IE, small size        |

**Q60.** Discuss the quantum theory of radiation. How is Planck's hypothesis for radiation of energy related to the spectral properties of an atom described by Bohr? **(13 marks)** [CHEM 101 | 2008–09]

### Answer

#### Quantum Theory of Radiation and Its Relation to Bohr's Spectral Theory

##### Part 1: Quantum Theory of Radiation (Planck, 1900)

##### Background — Failure of Classical Theory:

Classical physics (Rayleigh–Jeans law) assumed that a **blackbody** emits radiation continuously at all frequencies with energy proportional to temperature. This predicted that intensity should increase indefinitely at high frequencies (ultraviolet region) — a catastrophic failure known as the “**ultraviolet catastrophe**”, which completely contradicted experimental observations.

##### Planck's Postulates:

To resolve this, Max Planck (1900) proposed the **Quantum Theory of Radiation** based on the following postulates:

- A heated body does not emit or absorb energy **continuously**, but in **discrete packets** called **quanta** (singular: quantum). Each quantum is an indivisible unit of energy.
- The energy of each quantum is directly proportional to the **frequency**  $\nu$  of the radiation:

$$E = h\nu$$

where  $h = 6.626 \times 10^{-34}$  J·s is **Planck's constant**. Since  $\nu = c/\lambda$ :

$$E = \frac{hc}{\lambda}$$

- A body can emit or absorb energy only in **whole number multiples** of  $h\nu$ :

$$E = nh\nu \quad (n = 1, 2, 3, \dots)$$

Energy is therefore **quantized** — it cannot take arbitrary values, only specific discrete values.

- At **low frequencies**, the quantized description merges with classical theory (**Bohr's correspondence principle**).

##### Significance of Planck's Theory:

- It successfully explained the observed **blackbody radiation spectrum**.
- It introduced the revolutionary idea that energy is **not continuous** but comes in discrete packets.
- It laid the foundation of **modern quantum mechanics**.
- Einstein extended it to explain the **photoelectric effect** (1905), proposing that light itself consists of photons of energy  $h\nu$ .

##### Part 2: Bohr's Atomic Model and Spectral Properties

Bohr (1913) applied Planck's quantum hypothesis directly to the hydrogen atom. His model rests on the following key ideas:

- Electrons revolve in **fixed quantized orbits** without emitting energy (stationary states).
- Angular momentum is quantized:

$$m_e v r = \frac{nh}{2\pi}$$



- (iii) Energy is emitted or absorbed as a photon **only during transitions** between orbits:

$$\Delta E = E_{n_2} - E_{n_1} = h\nu$$

The total energy of the electron in the  $n$ -th orbit is:

$$E_n = -\frac{m_e e^4}{8\varepsilon_0^2 h^2} \cdot \frac{1}{n^2}$$

### Part 3: Relation Between Planck's Hypothesis and Bohr's Spectral Theory

The connection between Planck's quantum hypothesis and Bohr's description of atomic spectra is **deep and direct**. The key links are:

- (i) **Quantization of Energy Levels:** Bohr adopted Planck's idea that energy is quantized and applied it to electron orbits. Just as Planck quantized the energy of oscillators as  $E = nh\nu$ , Bohr quantized the **angular momentum** of electrons:

$$m_e v r = \frac{nh}{2\pi}$$

This leads naturally to **discrete energy levels**  $E_n \propto -1/n^2$ , meaning only specific energies are allowed for the electron.

- (ii) **Energy of Emitted Photon:** When an electron transitions from level  $n_2$  to  $n_1$  ( $n_2 > n_1$ ), the energy difference is released as a **photon** — directly using Planck's relation  $E = h\nu$ :

$$\Delta E = E_{n_2} - E_{n_1} = \frac{m_e e^4}{8\varepsilon_0^2 h^2} \left( \frac{1}{n_1^2} - \frac{1}{n_2^2} \right) = h\nu$$

- (iii) **Derivation of the Rydberg Equation:** Since  $h\nu = hc/\lambda$ , dividing both sides by  $hc$ :

$$\frac{1}{\lambda} = \frac{m_e e^4}{8\varepsilon_0^2 h^3 c} \left( \frac{1}{n_1^2} - \frac{1}{n_2^2} \right) = R_H \left( \frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

where  $R_H = 1.097 \times 10^7 \text{ m}^{-1}$  is the **Rydberg constant**. This equation predicts the exact positions of all spectral lines of hydrogen.

- (iv) **Explanation of Line Spectra:** Classical theory predicted a **continuous spectrum** from atoms. Planck's quantization, as incorporated by Bohr, explains why atoms emit only **discrete spectral lines** — because only specific energy transitions  $\Delta E = h\nu$  are allowed between quantized levels.
- (v) **Spectral Series of Hydrogen:** Different values of  $n_1$  and  $n_2$  in Bohr's equation give rise to different spectral series, all derivable using Planck's  $E = h\nu$ :

| Series   | Region       | $n_1$ | $n_2$      |
|----------|--------------|-------|------------|
| Lyman    | Ultraviolet  | 1     | 2, 3, 4... |
| Balmer   | Visible      | 2     | 3, 4, 5... |
| Paschen  | Infrared     | 3     | 4, 5, 6... |
| Brackett | Far Infrared | 4     | 5, 6, 7... |
| Pfund    | Far Infrared | 5     | 6, 7, 8... |

- (vi) **Ionization Energy from Planck's Relation:** The ionization energy of hydrogen is obtained by setting  $n_1 = 1$  and  $n_2 = \infty$ :

$$IE = h\nu_{\max} = \frac{m_e e^4}{8\varepsilon_0^2 h^2} = 13.6 \text{ eV}$$

This matches experimental values exactly, confirming that Planck's quantization is built into Bohr's model.

### Summary Table:

| Planck's Contribution         | Bohr's Application               |
|-------------------------------|----------------------------------|
| $E = h\nu$ (energy quantized) | Quantized electron energy levels |
| Energy in discrete packets    | Only specific orbits allowed     |
| Photon emitted: $E = h\nu$    | Spectral line at frequency $\nu$ |
| $\nu \propto \Delta E$        | Rydberg equation for $\lambda$   |
| No continuous energy emission | Discrete line spectrum explained |

**Conclusion:** Planck's quantum hypothesis is the **direct foundation** of Bohr's spectral theory. Bohr used  $E = h\nu$  to connect the energy difference between quantized atomic levels to the frequency of emitted radiation, thereby explaining the origin, position, and series of all spectral lines of hydrogen with remarkable accuracy.



2008–09]

**Answer**
**Limitations of Bohr's Model and Sommerfeld's Modification**
**Part 1: Limitations of Bohr's Atomic Model**

- (i) **Fails for Multi-electron Atoms:** Bohr's model works only for **hydrogen and hydrogen-like species** ( $\text{He}^+$ ,  $\text{Li}^{2+}$ , etc.). It completely fails for atoms with two or more electrons because it does not account for **electron–electron repulsions**, which significantly alter the energy levels.
- (ii) **Cannot Explain Fine Structure of Spectral Lines:** Under high-resolution spectroscopy, each spectral line of hydrogen is found to consist of **several closely spaced lines** (fine structure). Bohr's model predicts only single lines for each transition and cannot explain this splitting.
- (iii) **Cannot Explain Zeeman and Stark Effects:** When atoms are placed in an external **magnetic field** (Zeeman effect) or **electric field** (Stark effect), spectral lines split into multiple components. Bohr's model has no mechanism to account for this splitting.
- (iv) **Violates Heisenberg's Uncertainty Principle:** Bohr assigned both a **definite position** (radius  $r_n$ ) and a **definite velocity** to electrons simultaneously. This directly violates Heisenberg's Uncertainty Principle:

$$\Delta x \cdot \Delta p \geq \frac{h}{4\pi}$$

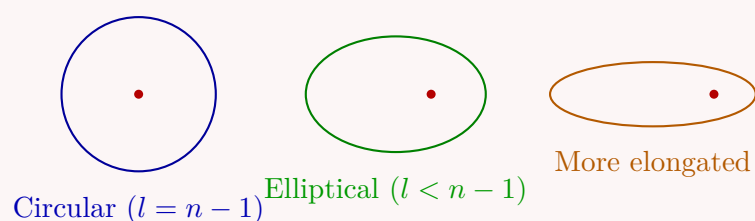
- (v) **Ignores Wave Nature of Electron:** Bohr treated the electron as a **classical point particle** moving in circular orbits. It completely ignores de Broglie's wave–particle duality of the electron, which is fundamental to modern quantum mechanics.
- (vi) **Cannot Explain Relative Intensities of Spectral Lines:** Bohr's model can predict the **positions** of spectral lines but gives no information about their **relative intensities** — why some transitions are stronger than others.
- (vii) **Restricted to Circular Orbits Only:** Bohr assumed that electrons move only in **circular orbits**. In reality, electrons can move in orbits of various shapes, which Bohr's model does not consider.
- (viii) **Cannot Explain Chemical Bonding:** Bohr's model treats atoms in complete isolation and provides no explanation for how atoms combine to form **molecules** through chemical bonding.

**Part 2: Sommerfeld's Modification of Bohr's Model (1916)**

Arnold Sommerfeld extended Bohr's model to address some of its limitations, particularly the **fine structure** of spectral lines. His key modifications are:

**Modification 1: Elliptical Orbits**

Sommerfeld proposed that electrons move not only in **circular orbits** (as Bohr assumed) but also in **elliptical orbits** with the nucleus at one focus. A circular orbit is a special case of an elliptical orbit where the two axes are equal.


**Modification 2: Introduction of Azimuthal Quantum Number  $l$** 

To describe elliptical orbits, Sommerfeld introduced a second quantum number called the **azimuthal (subsidiary) quantum number  $l$** , which defines the **shape** of the orbit (ratio of semi-minor to semi-major axis):

$$l = 1, 2, 3, \dots, n$$

(in Sommerfeld's original notation; in modern notation  $l = 0, 1, 2, \dots, n - 1$ ).

For a given principal quantum number  $n$ , there are  $n$  possible values of  $l$ , corresponding to  $n$  possible orbital shapes. The **eccentricity** of the ellipse is:

$$\frac{b}{a} = \frac{l}{n}$$

where  $a$  and  $b$  are the semi-major and semi-minor axes. When  $l = n$ , the orbit is **circular**; when  $l < n$ , it is **elliptical**.

**Modification 3: Relativistic Correction**

As electrons move in elongated elliptical orbits, their velocity **varies significantly** — fastest near the nucleus (perihelion) and slowest far away. Sommerfeld applied **Einstein's theory of relativity** to account for the variation of electron mass with velocity:

$$m = \frac{m_0}{\sqrt{1 - v^2/c^2}}$$

This relativistic variation of mass causes the elliptical orbits to **precess** (rotate slowly) around the nucleus, resulting in slightly different energy levels for orbits with the same  $n$  but different  $l$ .

**Modification 4: Explanation of Fine Structure**

The combination of elliptical orbits and relativistic correction means that for a given  $n$ , electrons in orbits with **different values of  $l$  have slightly different energies**. Transitions between these sub-levels produce **closely spaced spectral lines** — the fine structure. For example, for  $n = 2$ : two possible orbits ( $l = 1$  circular,  $l = 2$  elliptical) with slightly different energies, giving rise to two closely spaced lines instead of one.

#### Remaining Limitations of Sommerfeld's Model:

Despite its improvements, Sommerfeld's model still:

- Failed for **multi-electron atoms**.
- Could not explain the **Zeeman effect** completely.
- Did not account for the **wave nature** of the electron.
- Could not predict **relative intensities** of spectral lines.

These limitations were ultimately resolved only by the full **quantum mechanical model** (Schrödinger, 1926).

#### Summary Table:

| Aspect            | Bohr's Model   | Sommerfeld's Model      |
|-------------------|----------------|-------------------------|
| Orbit shape       | Circular only  | Circular and elliptical |
| Quantum numbers   | $n$ only       | $n$ and $l$             |
| Relativistic mass | Not considered | Included                |
| Fine structure    | Not explained  | Partially explained     |
| Multi-electron    | Fails          | Still fails             |
| Wave nature       | Ignored        | Ignored                 |

**Q62.** Deduce de Broglie's equation. Explain with suitable examples the physical significance of the equation. **(11 marks)** [CHEM 101 | 2008–09]

#### Answer

#### de Broglie's Equation: Deduction and Physical Significance

##### Part 1: Deduction of de Broglie's Equation

##### Step 1: Dual Nature of Light

It was well established that light exhibits **dual nature**:

- **Wave nature:** shown by diffraction and interference.
- **Particle nature:** shown by the photoelectric effect. Each photon carries energy:

$$E = h\nu \quad (1)$$

##### Step 2: Momentum of a Photon

From Einstein's mass-energy relation:

$$E = mc^2 \quad (2)$$

Equating (1) and (2):

$$mc^2 = h\nu \Rightarrow mc = \frac{h\nu}{c} = \frac{h}{\lambda} \quad (3)$$

since  $\nu/c = 1/\lambda$ . Here  $mc$  is the **momentum**  $p$  of the photon, so:

$$p = \frac{h}{\lambda} \Rightarrow \lambda = \frac{h}{p} \quad (4)$$

##### Step 3: de Broglie's Hypothesis (1924)

Louis de Broglie argued: if light (a wave) can behave as a particle, then **matter (a particle) should also behave as a wave**. By analogy with equation (4), for a material particle of mass  $m$  moving with velocity  $v$ , replacing the photon momentum  $mc$  with the particle momentum  $mv$ :

$$\lambda = \frac{h}{mv} = \frac{h}{p}$$

This is the **de Broglie equation**, where:

- $\lambda$  = de Broglie wavelength of the particle
- $h = 6.626 \times 10^{-34}$  J·s (Planck's constant)
- $m$  = mass of the particle (kg)
- $v$  = velocity of the particle ( $\text{m}\cdot\text{s}^{-1}$ )
- $p = mv$  = linear momentum ( $\text{kg}\cdot\text{m}\cdot\text{s}^{-1}$ )

#### Step 4: Alternative Forms

If a particle of charge  $q$  is accelerated through a potential difference  $V$ :

$$\frac{1}{2}mv^2 = qV \Rightarrow mv = \sqrt{2mqV}$$

$$\lambda = \frac{h}{\sqrt{2mqV}}$$

In terms of kinetic energy  $E_k = \frac{1}{2}mv^2$ :

$$\lambda = \frac{h}{\sqrt{2mE_k}}$$

### Part 2: Physical Significance of de Broglie's Equation

#### 1. Wave–Particle Duality of Matter

The most fundamental significance of de Broglie's equation is that it establishes **wave–particle duality** as a universal property of all matter — not just light. Every moving particle, whether an electron, proton, or cricket ball, has an associated **matter wave** (de Broglie wave) whose wavelength is given by  $\lambda = h/mv$ .

#### 2. Significance Depends on Mass — Comparison of Examples

| Object       | Mass (kg)              | Velocity   | $\lambda$         | Observable? |
|--------------|------------------------|------------|-------------------|-------------|
| Cricket ball | 0.15                   | 40 m/s     | $\sim 10^{-34}$ m | No          |
| Proton       | $1.67 \times 10^{-27}$ | $10^5$ m/s | $\sim 10^{-12}$ m | Barely      |
| Electron     | $9.1 \times 10^{-31}$  | $10^6$ m/s | $\sim 10^{-9}$ m  | Yes         |

#### Example 1: Macroscopic Object (Cricket Ball)

For a cricket ball of mass  $m = 0.15$  kg moving at  $v = 40$  m/s:

$$\lambda = \frac{6.626 \times 10^{-34}}{0.15 \times 40} = \frac{6.626 \times 10^{-34}}{6.0} \approx 1.1 \times 10^{-34} \text{ m}$$

This is **far smaller** than any known measurable length (nuclear diameter  $\sim 10^{-15}$  m). Therefore, the wave nature of macroscopic objects is **completely undetectable** — they behave purely as classical particles. This is consistent with our everyday experience.

#### Example 2: Electron

For an electron ( $m_e = 9.109 \times 10^{-31}$  kg) accelerated through  $V = 100$  V:

$$\lambda = \frac{h}{\sqrt{2m_e eV}} = \frac{6.626 \times 10^{-34}}{\sqrt{2 \times 9.109 \times 10^{-31} \times 1.602 \times 10^{-19} \times 100}}$$

$$\lambda = \frac{6.626 \times 10^{-34}}{\sqrt{2.919 \times 10^{-47}}} \approx 1.226 \times 10^{-10} \text{ m} = 1.226 \text{ Å}$$

This wavelength is **comparable to the interatomic spacing in crystals** ( $\sim 1\text{--}3$  Å). Therefore, electrons can be **diffracted by crystals**, exactly as X-rays are. This was confirmed by:

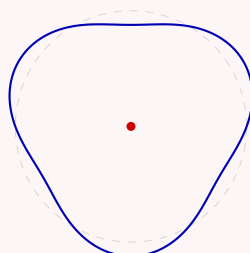
- **Davisson–Germer experiment (1927):** electrons diffracted by a Ni crystal produced diffraction maxima at angles predicted by  $\lambda = h/mv$ .
- **G.P. Thomson's experiment (1927):** electrons passed through a gold foil produced concentric diffraction rings identical to X-ray patterns.

#### 3. Justification of Bohr's Quantization Condition

de Broglie's equation provides a **physical basis** for Bohr's otherwise arbitrary quantization postulate. For a stable electron orbit of radius  $r$ , the electron wave must form a **standing wave** — the circumference must be an integral multiple of  $\lambda$ :

$$2\pi r = n\lambda = \frac{nh}{mv} \Rightarrow mvr = \frac{nh}{2\pi}$$

This is exactly **Bohr's quantization condition**, now derived naturally from wave theory rather than assumed.



Standing wave for  $n = 3$ :  $2\pi r = 3\lambda$

#### 4. Foundation of Quantum Mechanics

de Broglie's equation directly inspired **Schrödinger** to develop the wave equation for electrons (1926). The wavefunction  $\psi$  in the Schrödinger equation is the mathematical description of the de Broglie matter wave, and its square  $|\psi|^2$  gives the probability density of finding the electron.

#### 5. Electron Microscopy

The short de Broglie wavelength of high-energy electrons is exploited in the **electron microscope**, which achieves **much higher resolution** than optical microscopes (limited by the longer wavelength of visible light). This is a direct practical application of de Broglie's equation.

**Summary of Physical Significance:**

| Significance                        | Implication                                         |
|-------------------------------------|-----------------------------------------------------|
| Wave-particle duality of matter     | Universal property of all moving objects            |
| $\lambda$ negligible for large $m$  | Classical behaviour of macroscopic objects          |
| $\lambda$ significant for electrons | Electron diffraction observable                     |
| Justifies Bohr's quantization       | Standing wave condition $\Rightarrow mvr = nh/2\pi$ |
| Foundation of quantum mechanics     | Inspired Schrödinger wave equation                  |
| Electron microscopy                 | High-resolution imaging using electron waves        |

**Q63.** State Heisenberg's Uncertainty principle. Show that the uncertainty principle is applicable only for small particles like the electron.  
(9 marks) [CHEM 101 | 2007–08]

#### Answer

##### Heisenberg's Uncertainty Principle and Its Applicability to Small Particles

##### Part 1: Statement of Heisenberg's Uncertainty Principle

Heisenberg's Uncertainty Principle (1927) states:

*“It is fundamentally impossible to determine simultaneously and exactly both the **position** and the **momentum** (or velocity) of a moving particle.”*

Mathematically:

$$\Delta x \cdot \Delta p \geq \frac{h}{4\pi}$$

where:

- $\Delta x$  = uncertainty in position
- $\Delta p = m \Delta v$  = uncertainty in momentum
- $h = 6.626 \times 10^{-34}$  J·s = Planck's constant

Substituting  $\Delta p = m \Delta v$ :

$$\Delta x \cdot m \Delta v \geq \frac{h}{4\pi} \Rightarrow \Delta x \cdot \Delta v \geq \frac{h}{4\pi m}$$

This means: the more precisely the **position** is known ( $\Delta x$  small), the more uncertain the **momentum** ( $\Delta p$  large), and vice versa. This is not a limitation of instruments — it is a **fundamental property of nature**.

There is also an analogous relation between energy and time:

$$\Delta E \cdot \Delta t \geq \frac{h}{4\pi}$$

##### Part 2: Uncertainty Principle Applicable Only to Small Particles

The uncertainty principle is significant only when the uncertainties  $\Delta x$  and  $\Delta v$  are **comparable to the actual values** of position and velocity. We show this with two examples:

##### Example 1: Macroscopic Object (Cricket Ball)

Consider a cricket ball of mass  $m = 0.15$  kg. Suppose the uncertainty in its position is  $\Delta x = 10^{-6}$  m (already far smaller than measurable). Then the minimum uncertainty in velocity is:

$$\Delta v \geq \frac{h}{4\pi m \Delta x} = \frac{6.626 \times 10^{-34}}{4\pi \times 0.15 \times 10^{-6}} \approx 3.5 \times 10^{-28} \text{ m/s}$$

This uncertainty is **negligibly small** compared to the actual velocity of a cricket ball ( $\sim 40$  m/s). Therefore, the uncertainty principle has **no practical significance** for macroscopic objects.

##### Example 2: Electron (Small Particle)

Consider an electron of mass  $m_e = 9.109 \times 10^{-31}$  kg moving in a hydrogen atom. The diameter of a hydrogen atom is approximately  $10^{-10}$  m, so the maximum uncertainty in position is  $\Delta x \approx 10^{-10}$  m. Then:

$$\Delta v \geq \frac{h}{4\pi m_e \Delta x} = \frac{6.626 \times 10^{-34}}{4\pi \times 9.109 \times 10^{-31} \times 10^{-10}} \approx 5.79 \times 10^5 \text{ m/s}$$

The actual velocity of an electron in the first Bohr orbit is  $v = 2.18 \times 10^6$  m/s. The uncertainty  $\Delta v \approx 5.79 \times 10^5$  m/s is therefore **about 27% of the actual velocity** — enormously significant! It is impossible to track the electron's position and velocity simultaneously with any meaningful precision.

**Comparison Table:**

| Object       | Mass (kg)             | $\Delta v$ (m/s)      | Significance                         |
|--------------|-----------------------|-----------------------|--------------------------------------|
| Cricket ball | 0.15                  | $3.5 \times 10^{-28}$ | Negligible — classically predictable |
| Electron     | $9.1 \times 10^{-31}$ | $5.8 \times 10^5$     | Enormous — quantum behaviour         |

**Conclusion:** The uncertainty principle is **universally applicable** to all objects, but its effects are only significant for **subatomic particles** like electrons, protons, and neutrons, where the uncertainties are comparable to the actual physical quantities. For macroscopic objects, the product  $h/4\pi m$  is so tiny that uncertainties are completely negligible, and classical mechanics applies perfectly.

**Q64.** Discuss the classification of the elements present in the periodic table on the basis of electronic configuration. Mention the characteristics of two important classes of elements. **(15 marks)** [CHEM 101 | 2007–08]

### Answer

#### Classification of Elements Based on Electronic Configuration

##### Introduction

The modern periodic table classifies all known elements into **four blocks** based on which subshell receives the **last (differentiating) electron** in their electronic configuration. These are the **s, p, d, and f blocks**.

##### 1. s-Block Elements

Elements in which the **last electron enters the s subshell** of the outermost shell.

General configuration:  $ns^{1-2}$

- Includes **Group 1** (alkali metals: Li, Na, K, Rb, Cs, Fr) and **Group 2** (alkaline earth metals: Be, Mg, Ca, Sr, Ba, Ra).
- Also includes **H** and **He**.
- Located on the **far left** of the periodic table (Periods 1–7).

##### 2. p-Block Elements

Elements in which the **last electron enters the p subshell** of the outermost shell.

General configuration:  $ns^2 np^{1-6}$

- Includes **Groups 13–18**.
- Contains metals, metalloids, non-metals, and noble gases.
- Located on the **right side** of the periodic table (Periods 2–7).

##### 3. d-Block Elements (Transition Elements)

Elements in which the **last electron enters the d subshell** of the penultimate shell.

General configuration:  $(n-1)d^{1-10} ns^{1-2}$

- Includes **Groups 3–12**.
- Four series: 3d (Period 4), 4d (Period 5), 5d (Period 6), 6d (Period 7).
- All are **metals** with variable oxidation states, coloured compounds, and catalytic properties.

##### 4. f-Block Elements (Inner Transition Elements)

Elements in which the **last electron enters the f subshell** of the anti-penultimate shell.

General configuration:  $(n-2)f^{1-14} (n-1)d^{0-1} ns^2$

- Two series:
  - **Lanthanides** (4f series): Ce ( $Z = 58$ ) to Lu ( $Z = 71$ ), Period 6.
  - **Actinides** (5f series): Th ( $Z = 90$ ) to Lr ( $Z = 103$ ), Period 7.
- Placed separately at the **bottom** of the periodic table.
- All are **metals**; actinides are mostly radioactive.

**Summary of Block Classification:**

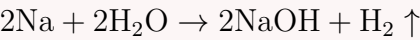


| Block    | Last electron in    | Groups  | Periods | Max electrons |
|----------|---------------------|---------|---------|---------------|
| <i>s</i> | $ns^{1-2}$          | 1–2     | 1–7     | 2             |
| <i>p</i> | $ns^2np^{1-6}$      | 13–18   | 2–7     | 6             |
| <i>d</i> | $(n-1)d^{1-10}ns^2$ | 3–12    | 4–7     | 10            |
| <i>f</i> | $(n-2)f^{1-14}$     | (below) | 6–7     | 14            |

Characteristics of Two Important Classes

Class 1: s-Block Elements

- (i) **Low ionization energies:** Only 1 or 2 valence electrons, easily lost to form  $M^+$  or  $M^{2+}$  ions. They are the **most electropositive** elements.
- (ii) **Highly reactive metals:** React vigorously with water, oxygen, and halogens. Group 1 metals react with water to release  $H_2$ :



- (iii) **Strong reducing agents:** Readily donate electrons; reducing power increases down the group.
- (iv) **Form ionic compounds:** Compounds are predominantly ionic (e.g.  $NaCl$ ,  $CaCO_3$ ), soluble in water, and good electrolytes.
- (v) **Flame colours:** Give characteristic flame colours (Li: red, Na: yellow, K: violet, Ca: brick-red, Sr: crimson, Ba: green).
- (vi) **Soft metals with low melting points:** Metallic bonding is weak due to only 1–2 valence electrons; melting points decrease down the group.

Class 2: d-Block Elements (Transition Metals)

- (i) **Variable oxidation states:** Both  $d$  and  $s$  electrons participate in bonding, giving multiple oxidation states (e.g. Mn: +2 to +7).
- (ii) **Coloured compounds:** Partially filled  $d$  orbitals allow  $d-d$  electronic transitions in the visible region (e.g.  $CuSO_4$ : blue,  $KMnO_4$ : purple).
- (iii) **Paramagnetic behaviour:** Unpaired  $d$  electrons give rise to paramagnetism. Magnetic moment:

$$\mu = \sqrt{n(n+2)} \text{ B.M.}$$

- (iv) **Catalytic properties:** Variable valency and surface adsorption make them excellent catalysts (Fe in Haber process,  $V_2O_5$  in Contact process).
- (v) **Complex formation:** Small ionic size, high charge, and empty  $d$  orbitals allow formation of stable complexes with ligands (e.g.  $[Fe(CN)_6]^{3-}$ ).
- (vi) **High melting and boiling points:** Both  $s$  and  $d$  electrons contribute to strong metallic bonding (W has the highest melting point,  $3422^\circ C$ ).

**Q65.** What are the postulates of Bohr’s atomic model? Based on the postulates, deduce an equation for the calculation of the difference in energies between two levels for an electron in an atom. How far is the Bohr model successful in explaining the spectral properties of an atom? **(17 marks)** [CHEM 101 | 2006–07]

Answer

**Bohr’s Atomic Model: Postulates, Energy Equation and Success**

**Part 1: Postulates of Bohr’s Atomic Model (1913)**

- (i) **Circular Orbits (Stationary States):** Electrons revolve around the nucleus in **fixed circular orbits** called **stationary states** or **allowed orbits**. While in these orbits, the electron does **not emit or absorb energy**.
- (ii) **Quantization of Angular Momentum:** An electron can revolve only in those orbits in which its **angular momentum** is an integral multiple of  $\frac{h}{2\pi}$ :
$$m_e v r = \frac{nh}{2\pi} \quad (n = 1, 2, 3, \dots)$$
where  $n$  is the **principal quantum number**,  $m_e$  is the electron mass,  $v$  is its velocity, and  $r$  is the orbital radius.
- (iii) **Energy Emission and Absorption:** Energy is emitted or absorbed **only when an electron transitions** from one stationary state to another. When an electron jumps from a higher energy level  $E_2$  to a lower energy level  $E_1$ , a photon of frequency  $\nu$  is emitted:
$$\Delta E = E_2 - E_1 = h\nu$$
- (iv) **Electrostatic Force Provides Centripetal Force:** The **Coulombic attraction** between the nucleus (charge  $+Ze$ ) and the



electron (charge  $-e$ ) provides the centripetal force necessary for circular motion:

$$\frac{m_e v^2}{r} = \frac{Z e^2}{4\pi\epsilon_0 r^2}$$

## Part 2: Derivation of Energy Difference Between Two Levels

### Step 1: Find the orbital radius $r_n$

Balancing centripetal and Coulombic forces:

$$\frac{m_e v^2}{r} = \frac{Z e^2}{4\pi\epsilon_0 r^2} \Rightarrow m_e v^2 r = \frac{Z e^2}{4\pi\epsilon_0} \quad (1)$$

From quantization of angular momentum:

$$m_e v r = \frac{nh}{2\pi} \Rightarrow v = \frac{nh}{2\pi m_e r} \quad (2)$$

Substituting (2) into (1):

$$\begin{aligned} m_e \left( \frac{nh}{2\pi m_e r} \right)^2 r &= \frac{Z e^2}{4\pi\epsilon_0} \\ \frac{n^2 h^2}{4\pi^2 m_e r} &= \frac{Z e^2}{4\pi\epsilon_0} \\ \boxed{r_n = \frac{n^2 h^2 \epsilon_0}{\pi m_e Z e^2}} \end{aligned} \quad (3)$$

For hydrogen ( $Z = 1$ ) and  $n = 1$ :  $r_1 = a_0 = 0.529 \text{ \AA}$  (Bohr radius).

### Step 2: Find the total energy $E_n$

**Kinetic energy** of the electron:

$$\text{KE} = \frac{1}{2} m_e v^2 = \frac{Z e^2}{8\pi\epsilon_0 r} \quad (4)$$

(using equation 1:  $m_e v^2 = \frac{Z e^2}{4\pi\epsilon_0 r}$ )

**Potential energy** of the electron:

$$\text{PE} = -\frac{Z e^2}{4\pi\epsilon_0 r} \quad (5)$$

**Total energy:**

$$E_n = \text{KE} + \text{PE} = \frac{Z e^2}{8\pi\epsilon_0 r} - \frac{Z e^2}{4\pi\epsilon_0 r} = -\frac{Z e^2}{8\pi\epsilon_0 r}$$

Substituting  $r_n$  from equation (3):

$$\begin{aligned} E_n &= -\frac{Z e^2}{8\pi\epsilon_0} \cdot \frac{\pi m_e Z e^2}{n^2 h^2 \epsilon_0} \\ \boxed{E_n = -\frac{m_e Z^2 e^4}{8\epsilon_0^2 h^2} \cdot \frac{1}{n^2}} \end{aligned} \quad (6)$$

The negative sign indicates the electron is **bound** to the nucleus.

### Step 3: Energy difference between two levels

When an electron transitions from level  $n_2$  (higher) to level  $n_1$  (lower), the energy difference is:

$$\begin{aligned} \Delta E = E_{n_2} - E_{n_1} &= -\frac{m_e Z^2 e^4}{8\epsilon_0^2 h^2 n_2^2} - \left( -\frac{m_e Z^2 e^4}{8\epsilon_0^2 h^2 n_1^2} \right) \\ \boxed{\Delta E = \frac{m_e Z^2 e^4}{8\epsilon_0^2 h^2} \left( \frac{1}{n_1^2} - \frac{1}{n_2^2} \right)} \end{aligned} \quad (7)$$

Since  $\Delta E = h\nu = \frac{hc}{\lambda}$ , dividing both sides by  $hc$ :

$$\frac{1}{\lambda} = \frac{m_e Z^2 e^4}{8\epsilon_0^2 h^3 c} \left( \frac{1}{n_1^2} - \frac{1}{n_2^2} \right) = R_H Z^2 \left( \frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

where  $R_H = \frac{m_e e^4}{8\epsilon_0^2 h^3 c} = 1.097 \times 10^7 \text{ m}^{-1}$  is the **Rydberg constant**.

## Part 3: Success of Bohr's Model in Explaining Spectral Properties

(i) **Explanation of Hydrogen Spectrum:** Bohr's model **successfully explained** the line spectrum of hydrogen. Each spectral line corresponds to a specific electron transition  $n_2 \rightarrow n_1$ , giving a photon of definite frequency. This perfectly matched the experimentally observed spectral lines.

(ii) **Prediction of Spectral Series:** The model correctly predicted all known spectral series of hydrogen:

| Series   | Region       | $n_1$ | $n_2$      |
|----------|--------------|-------|------------|
| Lyman    | Ultraviolet  | 1     | 2, 3, 4... |
| Balmer   | Visible      | 2     | 3, 4, 5... |
| Paschen  | Infrared     | 3     | 4, 5, 6... |
| Brackett | Infrared     | 4     | 5, 6, 7... |
| Pfund    | Far Infrared | 5     | 6, 7, 8... |

- (iii) **Calculation of Rydberg Constant:** The model derived the **Rydberg constant**  $R_H$  theoretically from fundamental constants, and its value matched the experimentally determined value perfectly:

$$R_H = 1.097 \times 10^7 \text{ m}^{-1}$$

- (iv) **Hydrogen-like Species:** By introducing  $Z$  (atomic number) in equation (7), Bohr's model successfully explained the spectra of **hydrogen-like ions** such as  $\text{He}^+$ ,  $\text{Li}^{2+}$ ,  $\text{Be}^{3+}$ , etc.

- (v) **Ionization Energy of Hydrogen:** Setting  $n_1 = 1$  and  $n_2 = \infty$  in equation (7):

$$IE = \frac{m_e e^4}{8\epsilon_0^2 h^2} = 13.6 \text{ eV}$$

This matches the **experimentally measured** ionization energy of hydrogen (13.6 eV) exactly.

- (vi) **Quantization of Energy:** The model explained why atoms emit **discrete line spectra** (not continuous spectra) — because only specific energy transitions are allowed between quantized levels.

**Limitations (Brief):** Despite its successes, Bohr's model fails for multi-electron atoms, cannot explain the fine structure of spectral lines, violates Heisenberg's Uncertainty Principle, and does not account for the wave nature of the electron. These were later resolved by the **quantum mechanical model**.

**Q66.** What is the general electronic configuration of transition elements? Discuss the general properties of transition elements. **(10 marks)**  
[CHEM 101 | 2006–07]

### Answer

#### Electronic Configuration and General Properties of Transition Elements

##### Part 1: General Electronic Configuration

Transition elements are those in which the  **$d$  subshell is partially filled** either in the ground state or in their common oxidation states. They occupy Groups 3–12 of the periodic table (3d, 4d, 5d, and 6d series).

The general electronic configuration is:

$$(n-1)d^{1-10} ns^{1-2}$$

where  $n$  is the outermost principal quantum number. The four transition series are:

| Series   | Elements     | Configuration                            | Period |
|----------|--------------|------------------------------------------|--------|
| 1st (3d) | Sc to Zn     | $[\text{Ar}] 3d^{1-10} 4s^{1-2}$         | 4      |
| 2nd (4d) | Y to Cd      | $[\text{Kr}] 4d^{1-10} 5s^{1-2}$         | 5      |
| 3rd (5d) | La, Hf to Hg | $[\text{Xe}] 4f^{14} 5d^{1-10} 6s^{1-2}$ | 6      |
| 4th (6d) | Ac, Rf to Cn | $[\text{Rn}] 5f^{14} 6d^{1-10} 7s^{1-2}$ | 7      |

**Note:** Some exceptions occur due to extra stability of half-filled ( $d^5$ ) and completely filled ( $d^{10}$ ) subshells. For example:

- Cr:  $[\text{Ar}] 3d^5 4s^1$  (not  $3d^4 4s^2$ )
- Cu:  $[\text{Ar}] 3d^{10} 4s^1$  (not  $3d^9 4s^2$ )

##### Part 2: General Properties of Transition Elements

- (i) **Metallic Character:** All transition elements are **metals**. They are hard, lustrous, malleable, ductile, and good conductors of heat and electricity. This is due to the presence of delocalized  $d$  electrons in addition to  $s$  electrons contributing to metallic bonding. They generally have **high melting and boiling points** (e.g. W has the highest melting point,  $3422^\circ\text{C}$ ).
- (ii) **Variable Oxidation States:** Transition metals exhibit **multiple oxidation states** because both  $(n-1)d$  and  $ns$  electrons can be used in bond formation. For example:



Lower oxidation states are found in covalent compounds while higher oxidation states are found with electronegative ligands.

- (iii) **Formation of Coloured Compounds:** Most transition metal compounds are **coloured** in the solid state or in solution. The colour arises from  **$d-d$  electronic transitions**: electrons in lower  $d$  orbitals absorb visible light and are promoted to higher  $d$  orbitals (crystal field splitting). The complementary colour is observed. For example:

- $\text{CuSO}_4$  solution: blue
- $\text{K}_2\text{Cr}_2\text{O}_7$ : orange
- $\text{KMnO}_4$ : purple
- $\text{FeCl}_3$ : yellow

$\text{Zn}^{2+}$  ( $3d^{10}$ ) and  $\text{Sc}^{3+}$  ( $3d^0$ ) are colourless because  $d-d$  transitions are not possible.

(iv) **Formation of Complex Compounds:** Transition metals have a strong tendency to form **coordination complexes** with ligands (Lewis bases). This is due to:

- Small ionic size and high charge density.
- Availability of empty  $d$  orbitals to accept lone pairs from ligands.

Examples:  $[\text{Fe}(\text{CN})_6]^{3-}$ ,  $[\text{Cu}(\text{NH}_3)_4]^{2+}$ ,  $[\text{Co}(\text{en})_3]^{3+}$ .

(v) **Magnetic Properties:** Transition metal ions are often **paramagnetic** due to the presence of **unpaired  $d$  electrons**. The magnetic moment  $\mu$  is given by:

$$\mu = \sqrt{n(n+2)} \text{ B.M.}$$

where  $n$  is the number of unpaired electrons and B.M. is the Bohr magneton. Compounds with no unpaired electrons are **diamagnetic** (e.g.  $\text{Zn}^{2+}$ ,  $3d^{10}$ ).

(vi) **Catalytic Properties:** Transition metals and their compounds are widely used as **catalysts** due to:

- Variable oxidation states allowing electron transfer.
- Ability to adsorb reactants on their surfaces.

Examples:

- Fe in Haber process ( $\text{N}_2 + \text{H}_2 \rightarrow \text{NH}_3$ )
- $\text{V}_2\text{O}_5$  in Contact process ( $\text{SO}_2 \rightarrow \text{SO}_3$ )
- Pt/Pd in catalytic converters
- Ni in hydrogenation of oils

(vii) **Formation of Alloys:** Transition metals readily form **alloys** with each other and with other metals because their atomic radii are similar. Alloys are generally harder and more resistant to corrosion than pure metals. Examples: steel (Fe + C), brass (Cu + Zn), nichrome (Ni + Cr).

(viii) **Atomic and Ionic Radii:** Across a transition series, atomic radii **decrease slowly** from left to right (due to increasing nuclear charge) but then **slightly increase** towards the end as  $d-d$  electron repulsion becomes significant. This is in contrast to the sharp decrease seen across main group elements.

(ix) **High Ionization Energies:** Transition elements have **higher ionization energies** than  $s$ -block elements but lower than  $p$ -block non-metals, reflecting their intermediate metallic character and the effective nuclear charge acting on  $d$  electrons.

(x) **Interstitial Compound Formation:** Small atoms such as H, C, N, and B can occupy **interstitial voids** in the transition metal crystal lattice, forming **interstitial compounds** (e.g.  $\text{TiC}$ ,  $\text{Fe}_3\text{C}$ ,  $\text{TiN}$ ). These compounds are harder and have higher melting points than the pure metals.

**Summary Table:**

| Property                  | Reason                                     |
|---------------------------|--------------------------------------------|
| Variable oxidation states | Participation of $d$ and $s$ electrons     |
| Coloured compounds        | $d-d$ transitions in visible range         |
| Complex formation         | Empty $d$ orbitals, high charge density    |
| Paramagnetism             | Unpaired $d$ electrons                     |
| Catalytic activity        | Variable valency, surface adsorption       |
| High melting points       | Strong metallic bonding with $d$ electrons |

**Q67.** Deduce de Broglie's equation. With a suitable example, discuss the applicability of the equation in the case of electron-like particles. (9 marks) [CHEM 101 | 2006–07]

**Answer**

**de Broglie's Equation: Deduction and Applicability**

**Part 1: Deduction of de Broglie's Equation**

**Step 1: Energy of a Photon (Particle Nature of Light)**

From Planck's quantum theory, the energy of a photon of frequency  $\nu$  is:

$$E = h\nu \quad (1)$$

From Einstein's mass-energy relation:

$$E = mc^2 \quad (2)$$

Equating (1) and (2):

$$mc^2 = h\nu \quad (3)$$

### Step 2: Introduce Wavelength

Since  $\nu = \frac{c}{\lambda}$ , substituting into (3):

$$mc^2 = h \cdot \frac{c}{\lambda} \Rightarrow mc = \frac{h}{\lambda} \Rightarrow \lambda = \frac{h}{mc} \quad (4)$$

Here  $mc$  is the **momentum**  $p$  of the photon.

### Step 3: Extend to Material Particles (de Broglie, 1924)

de Broglie proposed that **every moving material particle**, like a photon, also has an associated wave. By analogy with equation (4), for a particle of mass  $m$  moving with velocity  $v$ , the photon momentum  $mc$  is replaced by the particle momentum  $mv$ :

$$\lambda = \frac{h}{mv} = \frac{h}{p}$$

This is **de Broglie's equation**, where:

- $\lambda$  = de Broglie wavelength of the particle
- $h$  = Planck's constant ( $6.626 \times 10^{-34}$  J·s)
- $m$  = mass of the particle
- $v$  = velocity of the particle
- $p = mv$  = linear momentum of the particle

The equation beautifully connects the **wave property** ( $\lambda$ ) with the **particle property** ( $p$ ), establishing the dual nature of matter.

### Step 4: Relation to Kinetic Energy

If a particle is accelerated through a potential difference  $V$  (in volts), its kinetic energy is:

$$\frac{1}{2}mv^2 = eV \Rightarrow mv = \sqrt{2meV}$$

Substituting into de Broglie's equation:

$$\lambda = \frac{h}{\sqrt{2meV}}$$

## Part 2: Applicability to Electron-like Particles

### Example: de Broglie Wavelength of an Electron

Consider an electron (mass  $m_e = 9.109 \times 10^{-31}$  kg) accelerated through a potential difference of  $V = 100$  V.

**Calculating the wavelength:**

$$\begin{aligned} \lambda &= \frac{h}{\sqrt{2m_e eV}} \\ &= \frac{6.626 \times 10^{-34}}{\sqrt{2 \times 9.109 \times 10^{-31} \times 1.602 \times 10^{-19} \times 100}} \\ \lambda &= \frac{6.626 \times 10^{-34}}{\sqrt{2.919 \times 10^{-47}}} = \frac{6.626 \times 10^{-34}}{1.708 \times 10^{-23.5}} \\ \lambda &\approx 1.226 \times 10^{-10} \text{ m} = 1.226 \text{ \AA} \end{aligned}$$

This wavelength is **comparable to the spacing between atoms in a crystal** ( $\sim 1\text{--}3 \text{ \AA}$ ), which is why electrons can be **diffracted by crystals** — directly proving their wave nature.

### Comparison with Macroscopic Objects:

| Object       | Mass (kg)              | Velocity (m/s) | $\lambda$                        |
|--------------|------------------------|----------------|----------------------------------|
| Cricket ball | 0.15                   | 40             | $\sim 10^{-34}$ m (undetectable) |
| Proton       | $1.67 \times 10^{-27}$ | $10^5$         | $\sim 10^{-12}$ m                |
| Electron     | $9.1 \times 10^{-31}$  | $10^6$         | $\sim 10^{-9}$ m (observable)    |

For **macroscopic objects** (cricket ball, car, etc.), the de Broglie wavelength is **negligibly small** ( $\sim 10^{-34}$  m), far below any measurable scale. Hence their wave nature is **undetectable** in practice.

For **subatomic particles** like electrons, the wavelength is **comparable to atomic dimensions**, making wave behaviour **physically significant and observable**.

### Experimental Verification:

- Davisson-Germer Experiment (1927):** Electrons were diffracted by a nickel crystal, producing diffraction maxima at angles predicted exactly by de Broglie's equation. The measured  $\lambda$  matched  $h/mv$  perfectly.

- (ii) **G.P. Thomson's Experiment (1927):** Electrons passed through a thin gold foil produced concentric diffraction rings, identical to X-ray diffraction patterns, confirming the wave nature of electrons.

#### Application in Bohr's Model:

de Broglie's equation provides a **physical justification** for Bohr's quantization condition. For a stable circular orbit of radius  $r$ , the electron wave must form a **standing wave**:

$$2\pi r = n\lambda = \frac{nh}{mv} \Rightarrow mvr = \frac{nh}{2\pi}$$

This is exactly **Bohr's quantization postulate**, now derived naturally from wave theory.

**Conclusion:** de Broglie's equation  $\lambda = h/mv$  is highly applicable to electron-like particles because their small mass gives a wavelength large enough to be experimentally observed and utilized in electron microscopy, electron diffraction, and quantum mechanical modelling of atomic structure.

**Q68.** How would you conceive the idea of having an electron cloud around the nucleus of an atom? (6 marks) [CHEM 101 | 2006–07]

#### Answer

#### The Concept of Electron Cloud Around the Nucleus

##### 1. Failure of Bohr's Fixed Orbit Model

Bohr's model assumed that electrons move in **fixed circular orbits** with a precisely defined position and velocity at every instant. However, **Heisenberg's Uncertainty Principle** (1927) showed that it is **fundamentally impossible** to determine the exact position and momentum of an electron simultaneously:

$$\Delta x \cdot \Delta p \geq \frac{h}{4\pi}$$

Since position and momentum cannot both be known precisely, the concept of a fixed orbit is **physically meaningless** for an electron. This necessitated a new model.

##### 2. The Quantum Mechanical Description

In the quantum mechanical model, the behaviour of an electron is described by the **wavefunction**  $\psi$ , obtained by solving the Schrödinger wave equation:

$$\nabla^2\psi + \frac{8\pi^2m}{h^2}(E - V)\psi = 0$$

According to **Born's interpretation**, the wavefunction itself has no direct physical meaning, but its square  $|\psi|^2$  gives the **probability density** of finding the electron at a given point in space:

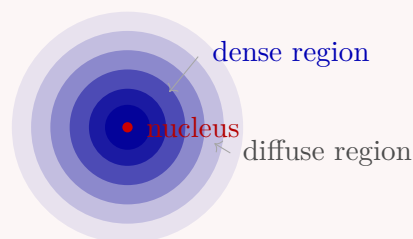
$$\text{Probability} \propto |\psi|^2 \cdot dV$$

where  $dV$  is a small volume element around the point.

##### 3. Conception of the Electron Cloud

Since the electron does not follow a fixed path, its position at any instant is **uncertain**. Over time, the electron can be found at various locations around the nucleus with different probabilities. This gives rise to the idea of an **electron cloud**:

- The electron cloud is a **three-dimensional probability map** around the nucleus, representing all the possible locations of the electron.
- Regions where  $|\psi|^2$  is **large** appear as **dense** parts of the cloud — the electron is most likely to be found here.
- Regions where  $|\psi|^2$  is **small** appear as **diffuse** parts of the cloud — the electron is unlikely to be found here.
- The cloud has **no sharp boundary** — there is always a small but non-zero probability of finding the electron far from the nucleus.



Electron cloud for 1s orbital ( $|\psi|^2$  map)

##### 4. Orbital vs. Cloud

The electron cloud defines the **atomic orbital** — the region of space where the probability of finding the electron is **95%** or more. It replaces Bohr's sharp circular orbit with a **fuzzy, probabilistic region** that correctly reflects the quantum mechanical nature of the electron.

**Conclusion:** The electron cloud arises naturally from the inability to assign a fixed path to the electron (Heisenberg's principle) and from the probabilistic interpretation of the wavefunction ( $|\psi|^2$ ). It represents the **spatial distribution of the probability** of finding the electron, giving the atom its characteristic "cloud-like" electron density around the nucleus.

BUET · CHEM 113

# Chemistry

Section S2

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Molecular Geometry



## S2 — Molecular Geometry

**Q1.** Predict the shapes of (i)  $\text{SO}_3^{2-}$  and (ii)  $\text{PCl}_4^+$  ions showing all the necessary steps. **(10 marks)**

[CHEM 113 | 2020–21]

### Answer

**Using VSEPR Theory:** The shape is determined by the number of bond pairs (BP) and lone pairs (LP) around the central atom.

**(i)  $\text{SO}_3^{2-}$  (Sulphite ion)**

**Step 1. Valence electrons:**

$$\text{S} = 6, \text{O} = 6 \times 3 = 18, \text{charge} = +2$$

$$\text{Total} = 6 + 18 + 2 = 26 \text{ electrons}$$

**Step 2. Lewis structure:** S is the central atom, bonded to 3 oxygen atoms.

After forming 3 S–O bonds (6 electrons used) and completing octets on O atoms (18 electrons), remaining =  $26 - 6 - 18 = 2$  electrons  $\rightarrow$  1 lone pair on S.

**Step 3. Electron geometry count:**

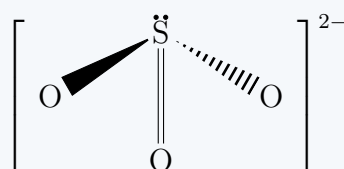
$$\text{BP} = 3, \text{LP} = 1 \Rightarrow \text{Total electron pairs} = 4$$

$\Rightarrow$  Electron geometry: **Tetrahedral**

**Step 4. Molecular shape:**

With 3 BP and 1 LP  $\Rightarrow$  **Trigonal Pyramidal**

Bond angle  $< 109.5^\circ$  (due to LP–BP repulsion)  $\approx 106^\circ$



**(ii)  $\text{PCl}_4^+$  (Tetrachlorophosphorus cation)**

**Step 1. Valence electrons:**

$$\text{P} = 5, \text{Cl} = 7 \times 4 = 28, \text{charge} = -1$$

$$\text{Total} = 5 + 28 - 1 = 32 \text{ electrons}$$

**Step 2. Lewis structure:** P is the central atom, bonded to 4 Cl atoms.

4 P–Cl bonds use 8 electrons; remaining 24 electrons fill lone pairs on Cl atoms (6 each). No lone pair remains on P.

**Step 3. Electron geometry count:**

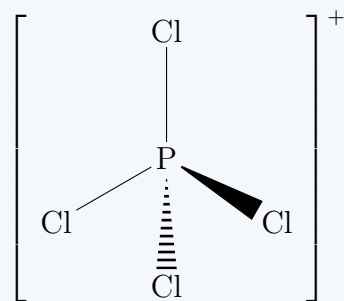
$$\text{BP} = 4, \text{LP} = 0 \Rightarrow \text{Total electron pairs} = 4$$

$\Rightarrow$  Electron geometry: **Tetrahedral**

**Step 4. Molecular shape:**

With 4 BP and 0 LP  $\Rightarrow$  **Tetrahedral**

Bond angle =  $109.5^\circ$



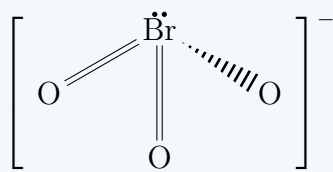
**Q2.** Sketch the shape or three-dimensional structure of each of the following:  $\text{BrO}_3^-$  and  $\text{AsF}_4^-$ . Name the shapes and state the status of hybridization in the central atoms. **(6 marks)**

[CHEM 113 | 2017–18]

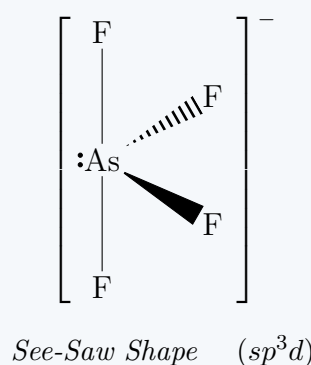
### Answer

**(i)  $\text{BrO}_3^-$  (Bromate ion)**

- **Valence electrons:**  $7 + 3 \times 6 + 1 = 26$  electrons
- Br is the central atom bonded to 3 O atoms; 1 LP on Br
- $\text{BP} = 3, \text{LP} = 1 \Rightarrow \text{Total electron pairs} = 4$
- **Hybridization of Br:**  $sp^3$
- Electron geometry: Tetrahedral
- **Molecular Shape:** Trigonal Pyramidal, bond angle  $< 109.5^\circ \approx 106^\circ$

(ii)  $\text{AsF}_4^-$  (Tetrafluoroarsenate ion)

- **Valence electrons:**  $5 + 4 \times 7 + 1 = 34$  electrons
- As is the central atom bonded to 4 F atoms; 1 LP on As
- BP = 4, LP = 1  $\Rightarrow$  Total electron pairs = 5
- **Hybridization of As:**  $sp^3d$
- Electron geometry: Trigonal Bipyramidal
- LP occupies equatorial position (less repulsion)
- **Molecular Shape: See-Saw (Sawhorse),**  
axial bond angle  $\approx 180^\circ$ , equatorial bond angle  $\approx 120^\circ$ ,  
axial-equatorial angle  $< 90^\circ$

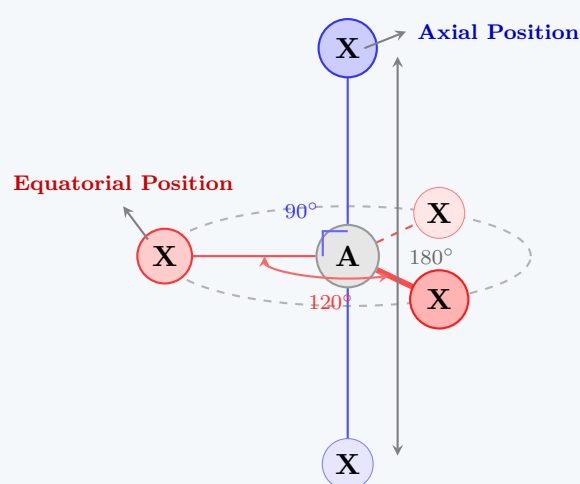


**Q3.** Draw the trigonal bipyramidal geometry and mark the axial and equatorial positions. Which position is suitable for the accommodation of a lone pair in the system? Why? **(6 marks)** [CHEM 113 | 2016–17]

**Answer****Trigonal Bipyramidal Geometry**

In trigonal bipyramidal geometry, the central atom has **5 electron pairs** arranged via  $sp^3d$  hybridization. There are two distinct positions:

- **Axial positions (2):** Above and below the equatorial plane, bond angle =  $180^\circ$  with each other,  $90^\circ$  with equatorial bonds.
- **Equatorial positions (3):** In the trigonal plane, bond angle =  $120^\circ$  with each other,  $90^\circ$  with axial bonds.



Trigonal Bipyramidal Geometry ( $sp^3d$ )

**Suitable Position for Lone Pair: Equatorial**

The **equatorial position** is preferred for lone pair accommodation. The reasons are:

- **Fewer repulsions:** A lone pair in the *equatorial* position experiences only **2 repulsions at  $90^\circ$**  (with 2 axial bond pairs). A lone pair in the *axial* position would experience **3 repulsions at  $90^\circ$**  (with 3 equatorial bond pairs).
- **LP–BP repulsion is stronger** than BP–BP repulsion, so minimizing the number of  $90^\circ$  LP–BP interactions lowers the overall energy of the molecule.
- **More space:** Equatorial positions have  $120^\circ$  angles between them, providing more room for the spatially diffuse lone pair cloud.

| LP Position | No. of 90° LP–BP interactions | Stability            |
|-------------|-------------------------------|----------------------|
| Axial       | 3                             | Less stable          |
| Equatorial  | 2                             | <b>More stable ✓</b> |

**Conclusion:** Lone pairs always occupy **equatorial positions** in trigonal bipyramidal systems to minimize repulsion and maximize stability. This is why molecules like  $\text{SF}_4$  adopt a *see-saw* shape and  $\text{ClF}_3$  adopts a *T-shape*.

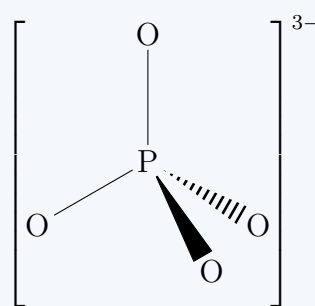
**Q4.** Draw the geometry of the following and predict the angles: (i)  $\text{PO}_4^{3-}$  (ii)  $\text{I}_3^-$ . (6 marks)

[CHEM 113 | 2016–17]

#### Answer

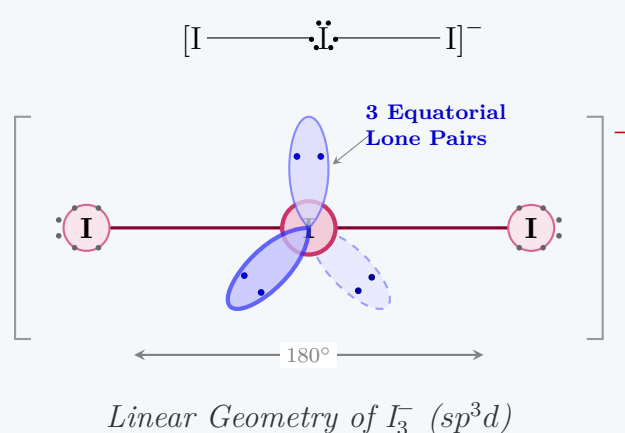
(i)  $\text{PO}_4^{3-}$  (Phosphate ion)

- **Valence electrons:**  $5 + 4 \times 6 + 3 = 32$  electrons
- P is the central atom bonded to 4 O atoms
- BP = 4, LP = 0  $\Rightarrow$  Total electron pairs = 4
- **Hybridization:**  $sp^3$
- Electron geometry = Molecular geometry
- **Shape:** Tetrahedral, bond angle =  $109.5^\circ$



(ii)  $\text{I}_3^-$  (Triiodide ion)

- **Valence electrons:**  $3 \times 7 + 1 = 22$  electrons
- Central I is bonded to 2 terminal I atoms; **3 LP** on central I
- BP = 2, LP = 3  $\Rightarrow$  Total electron pairs = 5
- **Hybridization:**  $sp^3d$
- Electron geometry: Trigonal Bipyramidal
- LP occupy all 3 equatorial positions (minimize repulsion)
- **Shape:** Linear, bond angle =  $180^\circ$



**Q5.** How does VBT apply the concept of hybridization? Illustrate the hybridization process of water. (9 marks)

[CHEM 113 | 2016–17]

#### Answer

##### Valence Bond Theory (VBT) and Hybridization

**Valence Bond Theory (VBT)** states that a covalent bond is formed by the overlap of two half-filled atomic orbitals from adjacent atoms. The key concepts are:

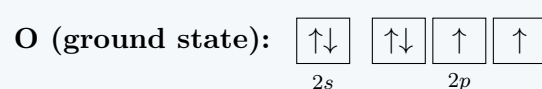
- **Orbital Overlap:** A bond forms when orbitals overlap and electrons pair up with opposite spins. Greater overlap  $\Rightarrow$  stronger bond.

- **Problem with pure VBT:** Pure atomic orbitals often fail to explain observed molecular geometries and bond angles. For example, pure  $p$  orbitals of O are at  $90^\circ$ , but the H–O–H angle in water is  $104.5^\circ$ .
- **Hybridization concept:** To resolve this, VBT introduces **hybridization** — the mathematical mixing of two or more atomic orbitals of similar energy on the *same atom* to form a new set of **hybrid orbitals** of equal energy and shape.
- **Key rules of hybridization:**
  - (i) Number of hybrid orbitals formed = number of atomic orbitals mixed.
  - (ii) Hybrid orbitals are equivalent in energy and shape.
  - (iii) Hybrid orbitals are directed in space to minimize repulsion.
  - (iv) Only orbitals of similar energy can hybridize.
  - (v) Hybridization explains both geometry *and* bond angles.
- **Types of hybridization and geometry:**

| Hybridization | Orbitals mixed          | No. of hybrids | Geometry             |
|---------------|-------------------------|----------------|----------------------|
| $sp$          | $s + p$                 | 2              | Linear               |
| $sp^2$        | $s + p + p$             | 3              | Trigonal Planar      |
| $sp^3$        | $s + p + p + p$         | 4              | Tetrahedral          |
| $sp^3d$       | $s + p + p + p + d$     | 5              | Trigonal Bipyramidal |
| $sp^3d^2$     | $s + p + p + p + d + d$ | 6              | Octahedral           |

### Hybridization of Water ( $\text{H}_2\text{O}$ )

#### Step 1: Ground state electronic configuration of O

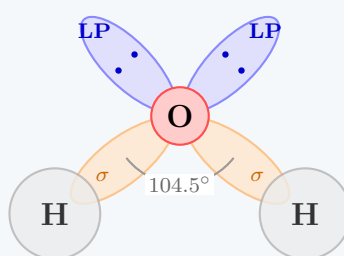


O has 2 half-filled  $2p$  orbitals  $\Rightarrow$  pure VBT predicts bond angle =  $90^\circ$ , but observed angle is  $104.5^\circ$ . So hybridization must occur.

#### Step 2: Hybridization — mixing of $2s$ and three $2p$ orbitals



- $2s + 2p_x + 2p_y + 2p_z$  mix to give **four  $sp^3$  hybrid orbitals**.
- **2 hybrid orbitals** contain lone pairs (LP).
- **2 hybrid orbitals** are half-filled  $\Rightarrow$  overlap with  $1s$  orbitals of two H atoms to form **2 O–H  $\sigma$  bonds**.



$sp^3$  hybridized O: 2 LP + 2 O–H  $\sigma$  bonds

#### Step 4: Why bond angle is $104.5^\circ$ and not $109.5^\circ$

- Perfect tetrahedral  $sp^3$  predicts  $109.5^\circ$ , but in  $\text{H}_2\text{O}$  there are **2 lone pairs** which exert stronger repulsion than bonding pairs.
- LP–LP > LP–BP > BP–BP repulsion compresses the H–O–H angle from  $109.5^\circ$  down to  **$104.5^\circ$** .

| Property          | Value                                      |
|-------------------|--------------------------------------------|
| Central atom      | O                                          |
| Hybridization     | $sp^3$                                     |
| Bonding pairs     | 2                                          |
| Lone pairs        | 2                                          |
| Electron geometry | Tetrahedral                                |
| Molecular shape   | Bent / V-shaped                            |
| Bond angle        | $104.5^\circ$                              |
| Bond type         | O–H $\sigma$ (overlap of $sp^3$ and $1s$ ) |

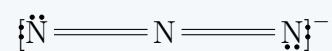
**Q6.** Use the VSEPR model to predict the geometry of the following ions: (i)  $\text{N}_3^-$  (ii)  $\text{BH}_4^-$  (iii)  $\text{SO}_3^{2-}$  (iv)  $\text{NO}_2^-$ . **(8 marks)** [CHEM 113 | 2015–16]

### Answer

**Using VSEPR Theory** — Total electron pairs (BP + LP) determine geometry.

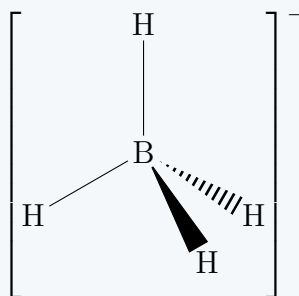
**(i)  $\text{N}_3^-$  (Azide ion)**

- Valence electrons:  $3 \times 5 + 1 = 16$  electrons
- Central N is double-bonded to both terminal N atoms: **linear arrangement**
- BP = 2, LP = 1 on central N (but linear due to resonance/triple bond character)
- Geometry: Linear**, bond angle =  $180^\circ$



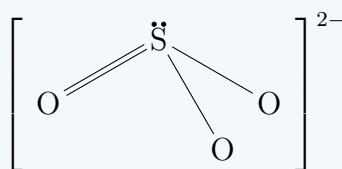
**(ii)  $\text{BH}_4^-$  (Tetrahydroborate ion)**

- Valence electrons:  $3 + 4 \times 1 + 1 = 8$  electrons
- B is the central atom bonded to 4 H atoms
- BP = 4, LP = 0
- Geometry: Tetrahedral**, bond angle =  $109.5^\circ$



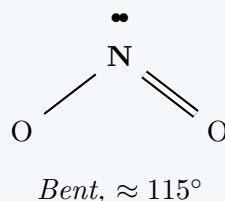
**(iii)  $\text{SO}_3^{2-}$  (Sulphite ion)**

- Valence electrons:  $6 + 3 \times 6 + 2 = 26$  electrons
- S is the central atom bonded to 3 O atoms; 1 LP on S
- BP = 3, LP = 1  $\Rightarrow$  electron geometry: Tetrahedral
- Geometry: Trigonal Pyramidal**, bond angle  $\approx 106^\circ$



**(iv)  $\text{NO}_2^-$  (Nitrite ion)**

- Valence electrons:  $5 + 2 \times 6 + 1 = 18$  electrons
- N is the central atom bonded to 2 O atoms; 1 LP on N
- BP = 2, LP = 1  $\Rightarrow$  electron geometry: Trigonal Planar
- Geometry: Bent (V-shaped)**, bond angle  $\approx 115^\circ$



**Q7.** In the trigonal bipyramidal arrangement for a system  $\text{AB}_4\text{E}$ , why does the lone pair occupy an equatorial position rather than an axial position? Sketch the bond moments and resultant dipole moments for the following molecules:  $\text{H}_2\text{O}$ ,  $\text{PCl}_3$ ,  $\text{XeF}_4$ ,  $\text{PCl}_5$ ,  $\text{SF}_6$ . **(12 marks)** [CHEM 113 | 2014–15]

## Answer

**Part 1: Why lone pair occupies equatorial position in  $AB_4E$** 

In a trigonal bipyramidal system, an **equatorial lone pair** experiences fewer repulsions than an axial lone pair:

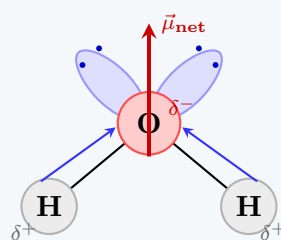
| LP Position | 90° LP–BP interactions | 120° LP–BP interactions | Stability            |
|-------------|------------------------|-------------------------|----------------------|
| Axial       | 3                      | 0                       | Less stable          |
| Equatorial  | 2                      | 2                       | <b>More stable ✓</b> |

- LP–BP repulsion at 90° is very strong; minimizing these interactions lowers energy.
- An **axial** LP has 3 interactions at 90° with equatorial BPs.
- An **equatorial** LP has only 2 interactions at 90° with axial BPs.
- Equatorial positions also have more angular space (120° separation) to accommodate the diffuse lone pair cloud.
- Therefore the lone pair always prefers the **equatorial position**  $\Rightarrow$  giving  $AB_4E$  a **see-saw shape**.

**Part 2: Bond moments and resultant dipole moments**
**(i)  $H_2O$** 

*Bent,  $sp^3$ , Net dipole  $\neq 0$*

- O is more electronegative than H  $\Rightarrow$  bond moments point **toward O**.
- Two O–H bond moments add up (do not cancel) due to bent geometry.
- **Resultant dipole moment is non-zero**, pointing from H–H midpoint toward O.

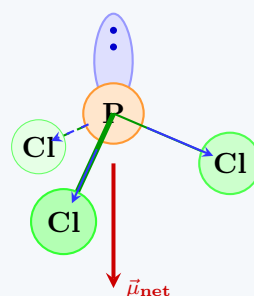


*Net dipole  $\neq 0$*

**(ii)  $PCl_3$** 

*Trigonal Pyramidal,  $sp^3$ , Net dipole  $\neq 0$*

- Cl is more electronegative than P  $\Rightarrow$  bond moments point **toward Cl**.
- 1 LP on P adds to the net moment in the same direction.
- Three P–Cl bond moments do **not** cancel  $\Rightarrow$  resultant points downward.
- **Net dipole moment  $\neq 0$ .**

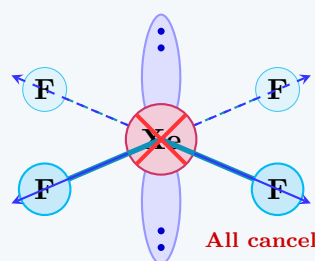


*Net dipole  $\neq 0$*

**(iii)  $XeF_4$** 

*Square Planar,  $sp^3d^2$ , Net dipole = 0*

- Xe has 2 LP occupying axial positions (opposite each other).
- 4 Xe–F bond moments point toward F (in the equatorial plane).
- Opposite bond moments **cancel in pairs**  $\Rightarrow$  net bond moment = 0.
- Two axial LP moments also cancel each other.
- **Net dipole moment = 0.**



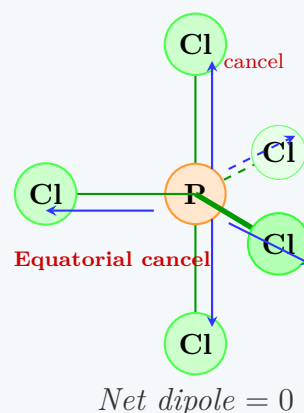
*Net dipole = 0*



(iv)  $\text{PCl}_5$ 

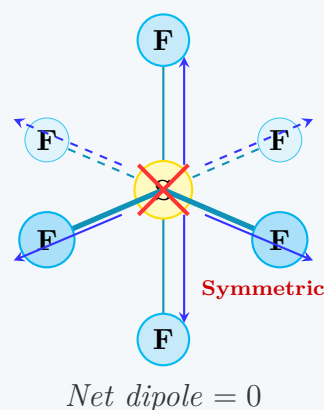
 Trigonal Bipyramidal,  $sp^3d$ , **Net dipole** = 0

- 2 axial P–Cl bond moments are equal and **opposite**  $\Rightarrow$  cancel.
- 3 equatorial P–Cl bond moments are symmetric at  $120^\circ \Rightarrow$  cancel.
- **Net dipole moment** = 0.


 (v)  $\text{SF}_6$ 

 Octahedral,  $sp^3d^2$ , **Net dipole** = 0

- 6 S–F bond moments arranged symmetrically along  $\pm x$ ,  $\pm y$ ,  $\pm z$  axes.
- Every bond moment is cancelled by the equal and opposite bond moment.
- **Net dipole moment** = 0.



| Molecule             | Shape                | Hybrid    | Dipole Moment    |
|----------------------|----------------------|-----------|------------------|
| $\text{H}_2\text{O}$ | Bent                 | $sp^3$    | $\neq 0$ (polar) |
| $\text{PCl}_3$       | Trigonal Pyramidal   | $sp^3$    | $\neq 0$ (polar) |
| $\text{XeF}_4$       | Square Planar        | $sp^3d^2$ | = 0 (non-polar)  |
| $\text{PCl}_5$       | Trigonal Bipyramidal | $sp^3d$   | = 0 (non-polar)  |
| $\text{SF}_6$        | Octahedral           | $sp^3d^2$ | = 0 (non-polar)  |

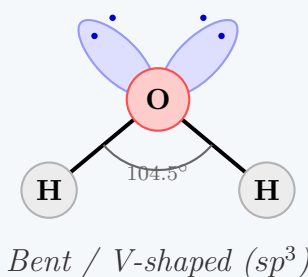
 Q8. Discuss the shapes of the following molecules:  $\text{H}_2\text{O}$ ,  $\text{NH}_3$ ,  $\text{BaCl}_2$ ,  $\text{CH}_4$ . (8 marks)

[CHEM 113 | 2013–14]

## Answer

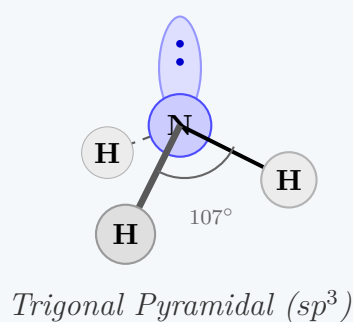
 (i)  $\text{H}_2\text{O}$  (Water)

- **Valence electrons:**  $6 + 2 \times 1 = 8$  electrons
- O is the central atom bonded to 2 H atoms; **2 LP** on O
- BP = 2, LP = 2  $\Rightarrow$  Total electron pairs = 4
- **Hybridization:**  $sp^3$
- Electron geometry: Tetrahedral
- **Shape: Bent (V-shaped)**, bond angle  $\approx 104.5^\circ$   
(reduced from  $109.5^\circ$  due to strong LP–LP and LP–BP repulsions)



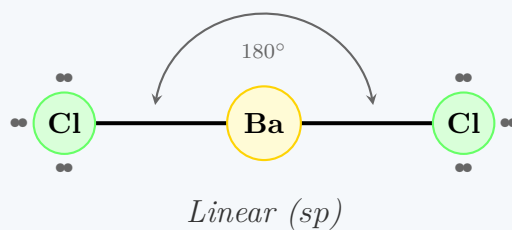
(ii)  $\text{NH}_3$  (Ammonia)

- **Valence electrons:**  $5 + 3 \times 1 = 8$  electrons
- N is the central atom bonded to 3 H atoms; **1 LP** on N
- BP = 3, LP = 1  $\Rightarrow$  Total electron pairs = 4
- **Hybridization:**  $sp^3$
- Electron geometry: Tetrahedral
- **Shape: Trigonal Pyramidal**, bond angle  $\approx 107^\circ$   
(reduced from  $109.5^\circ$  due to LP–BP repulsion)



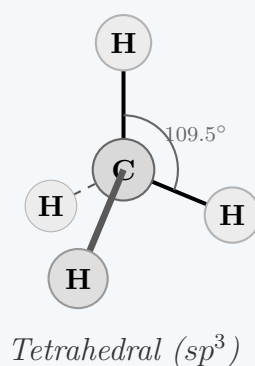
(iii)  $\text{BaCl}_2$  (Barium Chloride)

- **Valence electrons:**  $2 + 2 \times 7 = 16$  electrons
- Ba is the central atom bonded to 2 Cl atoms; **0 LP** on Ba
- BP = 2, LP = 0  $\Rightarrow$  Total electron pairs = 2
- **Hybridization:**  $sp$
- Electron geometry = Molecular geometry
- **Shape: Linear**, bond angle =  $180^\circ$



(iv)  $\text{CH}_4$  (Methane)

- **Valence electrons:**  $4 + 4 \times 1 = 8$  electrons
- C is the central atom bonded to 4 H atoms; **0 LP** on C
- BP = 4, LP = 0  $\Rightarrow$  Total electron pairs = 4
- **Hybridization:**  $sp^3$
- Electron geometry = Molecular geometry
- **Shape: Tetrahedral**, bond angle =  $109.5^\circ$   
(all BP–BP repulsions equal  $\Rightarrow$  perfect tetrahedral)



| Molecule          | BP | LP | Hybrid | Shape              | Bond Angle |
|-------------------|----|----|--------|--------------------|------------|
| H <sub>2</sub> O  | 2  | 2  | $sp^3$ | Bent               | 104.5°     |
| NH <sub>3</sub>   | 3  | 1  | $sp^3$ | Trigonal Pyramidal | 107°       |
| BaCl <sub>2</sub> | 2  | 0  | $sp$   | Linear             | 180°       |
| CH <sub>4</sub>   | 4  | 0  | $sp^3$ | Tetrahedral        | 109.5°     |

**Q9.** Discuss the basic features of the VSEPR model. Explain why the magnitude of repulsion decreases in this order: lone pair–lone pair > lone pair–bond pair > bond pair–bond pair. **(13 marks)** [CHEM 113 | 2011–12]

### Answer

#### Basic Features of VSEPR Model

- Electron pairs (both BP and LP) in the valence shell of the central atom **repel each other** and arrange to be as far apart as possible to minimize total repulsion energy.
- The **total number of electron pairs** (BP + LP) around the central atom determines the **electron pair geometry**.
- The **molecular shape** is determined only by the positions of the **bonding pairs**; lone pairs are invisible in the final shape but influence bond angles.
- Lone pairs occupy more space** than bonding pairs since they are attracted to only one nucleus (not shared), creating a larger, more diffuse electron cloud.
- Order of repulsion:**

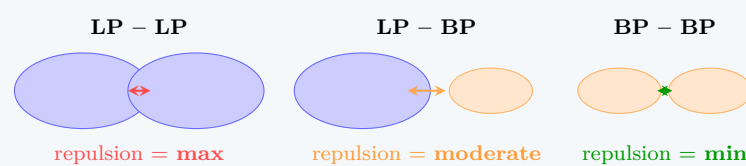
$$\text{LP-LP} > \text{LP-BP} > \text{BP-BP}$$

- Multiple bonds** (double, triple) repel more strongly than single bonds and are treated as a single “super pair” in VSEPR.
- Bond angles** decrease as lone pairs replace bonding pairs around the central atom.

| BP | LP | Total | Electron Geometry | Molecular Shape    |
|----|----|-------|-------------------|--------------------|
| 2  | 0  | 2     | Linear            | Linear             |
| 3  | 0  | 3     | Trigonal Planar   | Trigonal Planar    |
| 2  | 1  | 3     | Trigonal Planar   | Bent               |
| 4  | 0  | 4     | Tetrahedral       | Tetrahedral        |
| 3  | 1  | 4     | Tetrahedral       | Trigonal Pyramidal |
| 2  | 2  | 4     | Tetrahedral       | Bent               |
| 5  | 0  | 5     | Trig. Bipyramidal | Trig. Bipyramidal  |
| 4  | 1  | 5     | Trig. Bipyramidal | See-Saw            |
| 3  | 2  | 5     | Trig. Bipyramidal | T-Shape            |
| 6  | 0  | 6     | Octahedral        | Octahedral         |
| 5  | 1  | 6     | Octahedral        | Square Pyramidal   |
| 4  | 2  | 6     | Octahedral        | Square Planar      |

#### Why LP–LP > LP–BP > BP–BP repulsion

- Lone pair–Lone pair (LP–LP) repulsion is greatest:**  
A lone pair is attracted to **only one nucleus**. It is not confined between two nuclei like a bonding pair, so it spreads out and occupies a **larger region of space**. When two such diffuse, large electron clouds are close to each other, their electrostatic repulsion is very strong. Hence LP–LP repulsion is maximum.
- Lone pair–Bond pair (LP–BP) repulsion is intermediate:**  
A bonding pair is shared between two nuclei and is more **concentrated/localized** along the bond axis. When a lone pair (large cloud) interacts with a bonding pair (smaller, directed cloud), the repulsion is less than LP–LP but more than BP–BP since one partner is still a diffuse lone pair.
- Bond pair–Bond pair (BP–BP) repulsion is least:**  
Both electron pairs are bonding pairs, each pulled toward **two nuclei** and therefore confined to a narrow region between the atoms. Two such localized, compact clouds repel each other the least.



**Effect on bond angles:** Each lone pair added to the central atom compresses the remaining bond angles, e.g.,  $\text{CH}_4$  ( $109.5^\circ$ )  $\rightarrow$   $\text{NH}_3$  ( $107^\circ$ )  $\rightarrow$   $\text{H}_2\text{O}$  ( $104.5^\circ$ ) as LP count increases from 0 to 2.

**Q10.** Predict whether each of the following molecules has a dipole moment:  $\text{BrCl}$ ;  $\text{BF}_3$ ;  $\text{CH}_2\text{Cl}_2$ . (6 marks) [CHEM 113 | 2011–12]

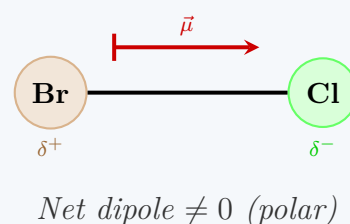
### Answer

#### Dipole Moment Predictions

A molecule has a net dipole moment if bond moments **do not cancel** due to asymmetry in geometry or electronegativity.

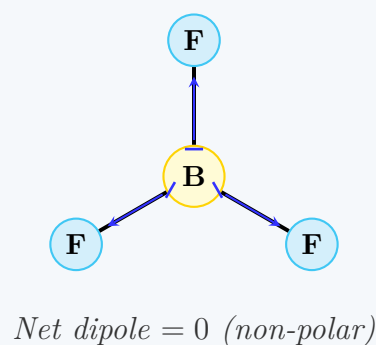
##### (i) $\text{BrCl}$ (Bromine monochloride)

- Diatomic molecule; Cl is more electronegative than Br.
- Bond moment points from Br  $\rightarrow$  Cl (toward Cl).
- Only one bond  $\Rightarrow$  **no cancellation possible**.
- **Dipole moment  $\neq 0$  (polar molecule),  $\mu > 0$**



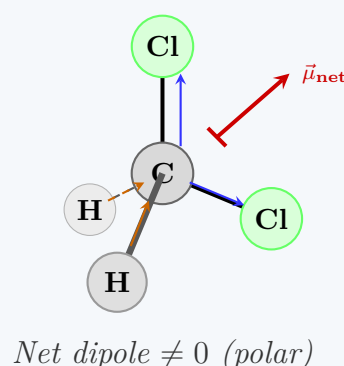
##### (ii) $\text{BF}_3$ (Boron trifluoride)

- B is central atom; BP = 3, LP = 0; **Hybridization:**  $sp^2$
- **Shape:** **Trigonal Planar**, bond angle =  $120^\circ$
- F is more electronegative  $\Rightarrow$  each B–F bond moment points toward F.
- Three bond moments at  $120^\circ$  are **symmetrically arranged**  $\Rightarrow$  they cancel completely.
- **Dipole moment = 0 (non-polar molecule)**



##### (iii) $\text{CH}_2\text{Cl}_2$ (Dichloromethane)

- C is central atom; BP = 4, LP = 0; **Hybridization:**  $sp^3$
- **Shape:** **Tetrahedral** (distorted), bond angle  $\approx 109.5^\circ$
- Cl is more electronegative than H  $\Rightarrow$  C–Cl bond moments point toward Cl; C–H bond moments point toward C (away from H).
- The molecule is **asymmetric**: 2 Cl on one side, 2 H on the other.
- Bond moments **do not cancel**  $\Rightarrow$  net resultant exists.
- **Dipole moment  $\neq 0$  (polar molecule)**



| Molecule                        | Shape             | Bond moments cancel? | Dipole moment     |
|---------------------------------|-------------------|----------------------|-------------------|
| BrCl                            | Linear (diatomic) | No                   | $\neq 0$ (polar)  |
| BF <sub>3</sub>                 | Trigonal Planar   | Yes                  | $= 0$ (non-polar) |
| CH <sub>2</sub> Cl <sub>2</sub> | Tetrahedral       | No                   | $\neq 0$ (polar)  |

**Q11.** Discuss the basis of determining the molecular structure by Valence Shell Electron Pair Repulsion (VSEPR) theory. Determine structure of H<sub>2</sub>O, BrF<sub>3</sub> and ICl<sub>4</sub><sup>-</sup> according to VSEPR theory. (10 marks) [CHEM 101 | 2009–10]

### Answer

#### Basis of VSEPR Theory

The **Valence Shell Electron Pair Repulsion (VSEPR)** theory predicts molecular geometry based on the following principles:

- Electron pairs (both bonding and lone pairs) in the valence shell of the central atom **repel each other** and arrange themselves to be as far apart as possible to minimize repulsion.
- The **order of repulsion** between electron pairs is:

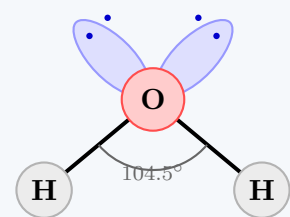
$$\text{LP-LP} > \text{LP-BP} > \text{BP-BP}$$

- Lone pairs occupy **more space** than bonding pairs (not shared between two nuclei), causing greater repulsion and **compressing bond angles**.
- The total number of electron pairs (BP + LP) around the central atom determines the **electron geometry**; the positions of bonding pairs alone determine the **molecular shape**.
- **Steps to determine structure:**
  - (i) Count total valence electrons.
  - (ii) Identify the central atom and form bonds with surrounding atoms.
  - (iii) Distribute remaining electrons as lone pairs.
  - (iv) Count BP and LP on central atom.
  - (v) Use the table below to predict geometry.

| BP | LP | Total | Electron Geometry | Molecular Shape    |
|----|----|-------|-------------------|--------------------|
| 2  | 0  | 2     | Linear            | Linear             |
| 3  | 0  | 3     | Trigonal Planar   | Trigonal Planar    |
| 2  | 1  | 3     | Trigonal Planar   | Bent               |
| 4  | 0  | 4     | Tetrahedral       | Tetrahedral        |
| 3  | 1  | 4     | Tetrahedral       | Trigonal Pyramidal |
| 2  | 2  | 4     | Tetrahedral       | Bent               |
| 5  | 0  | 5     | Trig. Bipyramidal | Trig. Bipyramidal  |
| 4  | 1  | 5     | Trig. Bipyramidal | See-Saw            |
| 3  | 2  | 5     | Trig. Bipyramidal | T-Shape            |
| 6  | 0  | 6     | Octahedral        | Octahedral         |
| 5  | 1  | 6     | Octahedral        | Square Pyramidal   |
| 4  | 2  | 6     | Octahedral        | Square Planar      |

#### (i) H<sub>2</sub>O (Water)

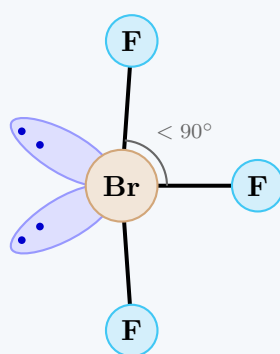
- **Valence electrons:**  $6 + 2 \times 1 = 8$  electrons
- O is central atom; 2 O–H bonds formed; remaining  $8 - 4 = 4$  electrons  $\Rightarrow$  2 LP on O
- BP = 2, LP = 2  $\Rightarrow$  Total = 4
- **Hybridization:**  $sp^3$
- Electron geometry: Tetrahedral
- **Molecular shape: Bent (V-shaped)**
- Bond angle  $\approx 104.5^\circ$  (compressed from  $109.5^\circ$  by strong LP–LP repulsion)



Bent / V-shaped ( $sp^3$ )

(ii)  $\text{BrF}_3$  (Bromine Trifluoride)

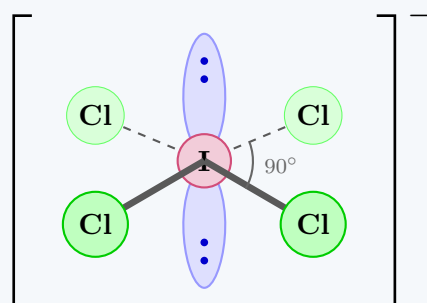
- **Valence electrons:**  $7 + 3 \times 7 = 28$  electrons
- Br is central atom; 3 Br–F bonds formed (6 electrons); remaining  $28 - 6 - 18 = 4$  electrons  $\Rightarrow$  2 LP on Br
- BP = 3, LP = 2  $\Rightarrow$  Total = 5
- **Hybridization:**  $sp^3d$
- Electron geometry: Trigonal Bipyramidal
- LPs occupy **equatorial** positions (minimize  $90^\circ$  repulsions)
- **Molecular shape: T-shaped**
- Bond angles  $< 90^\circ$  (compressed by LP–BP repulsions)



T-shaped ( $sp^3d$ )

(iii)  $\text{ICl}_4^-$  (Tetrachloroiodate ion)

- **Valence electrons:**  $7 + 4 \times 7 + 1 = 36$  electrons
- I is central atom; 4 I–Cl bonds formed (8 electrons); remaining  $36 - 8 - 24 = 4$  electrons  $\Rightarrow$  2 LP on I
- BP = 4, LP = 2  $\Rightarrow$  Total = 6
- **Hybridization:**  $sp^3d^2$
- Electron geometry: Octahedral
- LPs occupy **opposite axial positions** (trans to each other  $\Rightarrow$  LP–LP repulsion minimized at  $180^\circ$ )
- **Molecular shape: Square Planar**
- Bond angle =  $90^\circ$



Square Planar ( $sp^3d^2$ )

| Species              | BP | LP | Hybrid    | Shape         | Bond Angle    |
|----------------------|----|----|-----------|---------------|---------------|
| $\text{H}_2\text{O}$ | 2  | 2  | $sp^3$    | Bent          | $104.5^\circ$ |
| $\text{BrF}_3$       | 3  | 2  | $sp^3d$   | T-shaped      | $< 90^\circ$  |
| $\text{ICl}_4^-$     | 4  | 2  | $sp^3d^2$ | Square Planar | $90^\circ$    |



**Q12.** Discuss hybridization and the valence shell electron pair repulsion model in determining the structure of molecules. Cite suitable examples. (12 marks) [CHEM 101 | 2008–09]

### Answer

#### Hybridization and VSEPR in Determining Molecular Structure Hybridization (VBT):

- Hybridization is the **mixing of atomic orbitals** of similar energy on the same atom to produce a new set of equivalent hybrid orbitals.
- Number of hybrid orbitals = number of orbitals mixed.
- Hybrid orbitals are directed in space to **minimize repulsion** and form stronger bonds by better overlap.
- Explains molecular geometry and bond angles.

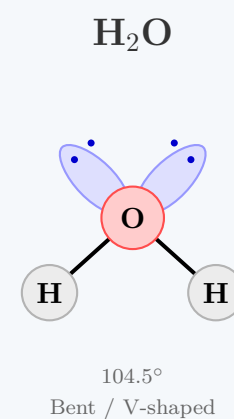
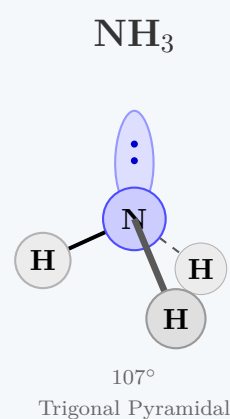
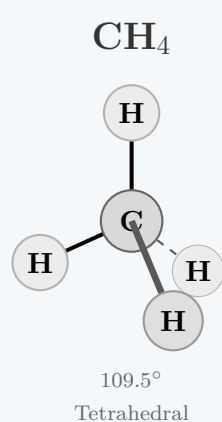
| Hybrid    | Orbitals                | No. | Geometry          | Example         |
|-----------|-------------------------|-----|-------------------|-----------------|
| $sp$      | $s + p$                 | 2   | Linear            | $\text{BeCl}_2$ |
| $sp^2$    | $s + p + p$             | 3   | Trigonal Planar   | $\text{BF}_3$   |
| $sp^3$    | $s + p + p + p$         | 4   | Tetrahedral       | $\text{CH}_4$   |
| $sp^3d$   | $s + p + p + p + d$     | 5   | Trig. Bipyramidal | $\text{PCl}_5$  |
| $sp^3d^2$ | $s + p + p + p + d + d$ | 6   | Octahedral        | $\text{SF}_6$   |

#### VSEPR Model:

- Electron pairs (BP + LP) repel each other and arrange to **minimize repulsion**.
- Repulsion order:  $\text{LP-LP} > \text{LP-BP} > \text{BP-BP}$ .
- Molecular shape depends on BP arrangement; LP invisible but compress angles.

#### Illustrative Examples ( $sp^3$ Family Comparison):

- $\text{CH}_4$  (Tetrahedral): BP = 4, LP = 0; pure  $sp^3$ , no LP compression ( $109.5^\circ$ ).
- $\text{NH}_3$  (Trigonal Pyramidal): BP = 3, LP = 1; LP-BP repulsion compresses angle to  $107^\circ$ .
- $\text{H}_2\text{O}$  (Bent): BP = 2, LP = 2; strong LP-LP repulsion compresses angle to  $104.5^\circ$ .



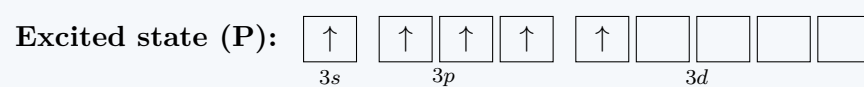
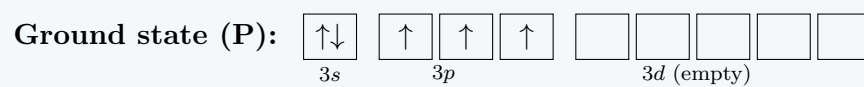
**Conclusion:** Both hybridization and VSEPR complement each other — hybridization explains the *type* of bonds formed, while VSEPR explains the *spatial arrangement* and bond angles.

**Q13.** What should be the possible geometrical structure of  $\text{PCl}_5$  molecule? Justify on the basis of hybridization and VSEPR model. (8 marks) [CHEM 101 | 2006–07]

### Answer

#### Geometrical Structure of $\text{PCl}_5$ Part 1: Hybridization Approach

- Valence electrons of P:**  $1s^2 2s^2 2p^6 3s^2 3p^3$  — ground state has 3 unpaired electrons, but P can expand its octet using empty  $3d$  orbitals.
- Excited state of P:** One electron from  $3s$  is promoted to empty  $3d$  orbital, giving 5 unpaired electrons available for bonding:



- The 5 half-filled orbitals ( $3s + 3p_x + 3p_y + 3p_z + 3d$ ) undergo  $sp^3d$  **hybridization** to form 5 equivalent hybrid orbitals.
- Each hybrid orbital overlaps with a half-filled  $3p$  orbital of Cl to form **5 P–Cl  $\sigma$  bonds**.
- BP = 5, LP = 0  $\Rightarrow$  **Trigonal Bipyramidal geometry**.

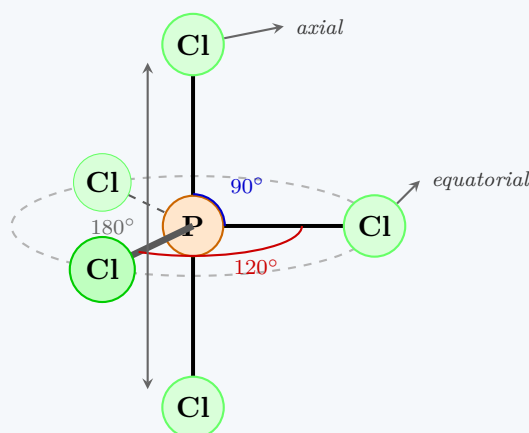
### Part 2: VSEPR Approach

- Valence electrons:**  $5 + 5 \times 7 = 40$  electrons
- P forms 5 bonds with Cl; no lone pairs remain on P.
- BP = 5, LP = 0  $\Rightarrow$  Total electron pairs = 5
- To minimize repulsion, 5 electron pairs arrange in a **Trigonal Bipyramidal** geometry.
- This gives **two distinct types** of P–Cl bonds:

| Position              | No. of bonds | Bond angle                                |
|-----------------------|--------------|-------------------------------------------|
| Equatorial (in plane) | 3            | 120° with each other                      |
| Axial (above & below) | 2            | 90° with equatorial, 180° with each other |

- Equatorial P–Cl bonds are slightly **longer** than axial bonds (more BP–BP repulsion from equatorial side).
- The molecule is **non-polar** as all bond moments cancel by symmetry.

### 3D Structure of $\text{PCl}_5$



Trigonal Bipyramidal ( $sp^3d$ ), BP = 5, LP = 0

### Summary

| Property                    | Value                |
|-----------------------------|----------------------|
| Central atom                | P                    |
| Hybridization               | $sp^3d$              |
| BP / LP                     | 5 / 0                |
| Electron geometry           | Trigonal Bipyramidal |
| Molecular shape             | Trigonal Bipyramidal |
| Equatorial bond angle       | 120°                 |
| Axial–equatorial bond angle | 90°                  |
| Axial–axial bond angle      | 180°                 |
| Dipole moment               | 0 (non-polar)        |

BUET · CHEM 113

# Chemistry

Section S3

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Chemical Bonding

### S3 — Chemical Bonding

Q1. Can you explain the existence of the transient ions like  $\text{H}_2^+$ ? How? (5 marks)

[CHEM 113 | 2023–24]

#### Answer

#### Existence of Transient Ion $\text{H}_2^+$

(5 marks)

##### (i) What is $\text{H}_2^+$ ?

$\text{H}_2^+$  is a **one-electron species** formed by removing one electron from  $\text{H}_2$ . It consists of two protons held together by a single electron. It is called *transient* because it exists only briefly under special conditions (e.g. electric discharge, cosmic rays, or mass spectrometry).

##### (ii) MO Explanation of its Existence:

Using LCAO, the two  $1s$  atomic orbitals combine to give:

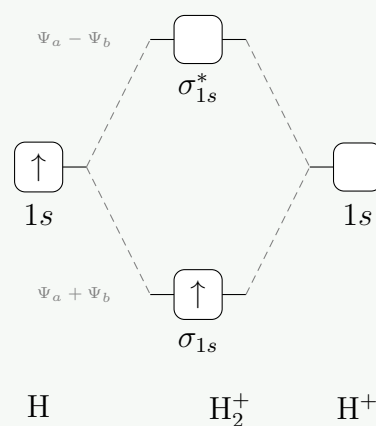
$$\Psi_b = \phi_{1s}^A + \phi_{1s}^B \quad (\sigma_{1s}, \text{ bonding MO})$$

$$\Psi^* = \phi_{1s}^A - \phi_{1s}^B \quad (\sigma_{1s}^*, \text{ antibonding MO})$$

$\text{H}_2^+$  has only **one electron**, which occupies the bonding MO  $\sigma_{1s}$ :

$$\text{Configuration: } (\sigma_{1s})^1$$

$$\text{Bond Order} = \frac{1}{2}(1 - 0) = \frac{1}{2}$$



Dihydrogen Cation (Bond order = 0.5)

##### (iii) Why it Can Exist:

- Bond order =  $\frac{1}{2} > 0$  — a net bonding interaction exists.
- The single electron in  $\sigma_{1s}$  creates **constructive interference** between the two nuclei, providing enough electron density to partially overcome nuclear repulsion.
- The system has a potential energy **minimum** at  $r_0 = 1.06 \text{ \AA}$  — confirming it is a bound state.
- Experimentally,  $\text{H}_2^+$  has been detected in **mass spectrometers** and exists in **interstellar space**.

##### (iv) Why it is Transient:

- Bond order of only  $\frac{1}{2}$  means the bond is very **weak** (bond energy = 255 kJ/mol vs 436 kJ/mol for  $\text{H}_2$ ).
- Easily destroyed by collision with other molecules or by further ionization.
- Cannot be isolated under normal conditions.

**Conclusion:**  $\text{H}_2^+$  exists because MO theory predicts a bond order of  $\frac{1}{2}$  — sufficient for a weak but real bond. It is transient because this half-bond is too weak to survive normal chemical conditions, yet strong enough to be detected spectroscopically and in mass spectrometry.

Q2. From the three possible ways to connect carbon, nitrogen, and oxygen to form a monoanion:  $\text{CNO}^-$ ,  $\text{CON}^-$  and  $\text{OCN}^-$ , one is the cyanate ion, a common and stable species; one is the fulminate ion, salts of which are used as explosive detonators; and one is so unstable that it has never been isolated. Use the concept of formal charge to determine which isomer is cyanate, which is the fulminate, and which is the least stable one. (10 marks)

[CHEM 113 | 2023–24; 2020–21]

Answer

Identifying  $\text{CNO}^-$ ,  $\text{CON}^-$ , and  $\text{OCN}^-$  Using Formal Charge

(10 marks)

Background: Formal Charge Formula

(1 marks)

The formal charge on an atom is calculated as:

$$\text{Formal Charge} = (\text{valence electrons}) - (\text{lone pair electrons}) - \frac{1}{2}(\text{bonding electrons})$$

Valence electrons: C = 4, N = 5, O = 6. Total valence electrons in the monoanion =  $4 + 5 + 6 + 1 = 16$ .

Isomer 1:  $\text{OCN}^-$  (Cyanate Ion)

(3 marks)

The dominant Lewis structure of  $\text{OCN}^-$  is:

$$[\ddot{\text{O}} = \text{C} = \text{N}]^- \longleftrightarrow [:\ddot{\text{O}} - \text{C} \equiv \text{N}:]^-$$

Calculating formal charges for the structure  $\text{O} = \text{C} = \text{N}^-$ :

| Atom | Valence $e^-$ | LP + $\frac{1}{2}\text{BP}$ | Formal Charge |
|------|---------------|-----------------------------|---------------|
| O    | 6             | $4 + \frac{1}{2}(4) = 6$    | $6 - 6 = 0$   |
| C    | 4             | $0 + \frac{1}{2}(8) = 4$    | $4 - 4 = 0$   |
| N    | 5             | $4 + \frac{1}{2}(4) = 6$    | $5 - 6 = -1$  |

Formal charges: O = 0, C = 0, N = -1. The negative charge resides on **nitrogen**, the most electronegative atom after oxygen — a chemically reasonable distribution. The overall formal charges are small and the negative charge is on a suitably electronegative atom.

$\therefore \text{OCN}^-$  is the **cyanate ion** — common and stable.

Isomer 2:  $\text{CNO}^-$  (Fulminate Ion)

(3 marks)

The Lewis structure of  $\text{CNO}^-$  places carbon at the terminal end:

$$[:\text{C} \equiv \text{N} - \ddot{\text{O}}:]^-$$

Calculating formal charges for  $\text{C} \equiv \text{N}-\text{O}$ :

| Atom | Valence $e^-$ | LP + $\frac{1}{2}\text{BP}$ | Formal Charge |
|------|---------------|-----------------------------|---------------|
| C    | 4             | $0 + \frac{1}{2}(6) = 3$    | $4 - 3 = +1$  |
| N    | 5             | $0 + \frac{1}{2}(8) = 4$    | $5 - 4 = +1$  |
| O    | 6             | $6 + \frac{1}{2}(2) = 7$    | $6 - 7 = -1$  |

Formal charges: C = -1, N = +1, O = -1. Large formal charges of opposite signs create a highly *polar*, *strained* structure with significant internal electrostatic tension. This stored energy is released explosively upon detonation.

$\therefore \text{CNO}^-$  is the **fulminate ion** — explosive, used in detonators.

Isomer 3:  $\text{CON}^-$  (Unstable, Never Isolated)

(2 marks)

The Lewis structure of  $\text{CON}^-$  places oxygen in the middle:

$$[:\text{C} = \text{O} = \text{N}:]^-$$

Calculating formal charges for  $\text{C}=\text{O}=\text{N}$ :

| Atom | Valence $e^-$ | LP + $\frac{1}{2}\text{BP}$ | Formal Charge |
|------|---------------|-----------------------------|---------------|
| C    | 4             | $2 + \frac{1}{2}(4) = 4$    | $4 - 4 = 0$   |
| O    | 6             | $0 + \frac{1}{2}(8) = 4$    | $6 - 4 = +2$  |
| N    | 5             | $4 + \frac{1}{2}(4) = 6$    | $5 - 6 = -1$  |

Formal charges: C = 0, O = +2, N = -1. A formal charge of +2 on oxygen — the most electronegative element — is extremely unfavorable. Placing positive charge on the most electronegative atom violates chemical intuition and creates a highly unstable electronic configuration. No resonance structure can relieve this charge distribution.

$\therefore$  **CON<sup>-</sup>** is the **least stable isomer** — never isolated.

### Summary

(1 marks)

| Ion              | Identity  | Formal Charges         | Stability                 |
|------------------|-----------|------------------------|---------------------------|
| OCN <sup>-</sup> | Cyanate   | O = 0, C = 0, N = -1   | Stable — small charges    |
| CNO <sup>-</sup> | Fulminate | C = -1, N = +1, O = -1 | Metastable — explosive    |
| CON <sup>-</sup> | Unknown   | C = 0, O = +2, N = -1  | Unstable — never isolated |

**Conclusion:** Formal charge analysis reveals that **OCN<sup>-</sup>** (cyanate) is most stable due to minimal charge separation; **CNO<sup>-</sup>** (fulminate) is metastable with large opposing charges that release energy explosively; and **CON<sup>-</sup>** is never isolated because a +2 formal charge on the most electronegative atom (O) is energetically intolerable.

**Q3.** Apply molecular orbital theory to explain the polar nature of HF molecule. (5–10 marks)

[CHEM 113 | 2023–24; 2022–23]

### Answer

#### Polar Nature of HF using Molecular Orbital Theory (MOT)

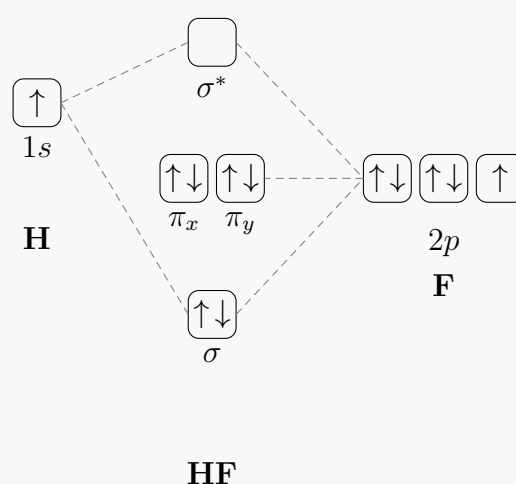
##### Atomic orbital energies:

- H: 1s orbital energy = -13.6 eV
- F: 2p orbital energy = -18.6 eV (significantly lower than H 1s)
- The large energy difference means the MOs are **unequal mixtures** of the two AOs.

##### MO formation:

- The bonding MO ( $\sigma$ ) is formed by constructive overlap of H(1s) and F(2p<sub>z</sub>).
- The antibonding MO ( $\sigma^*$ ) is formed by destructive overlap.
- Since F 2p is **lower in energy** than H 1s, the bonding MO has **more F character** (electron density closer to F).
- The antibonding MO has more H character.

##### MO energy diagram of HF:

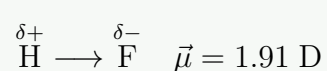


##### MO electronic configuration of HF:

$$(\sigma)^2 (n_{2p_x})^2 (n_{2p_y})^2$$

##### Explanation of polar nature:

- The bonding  $\sigma$  MO is **closer in energy to F(2p)** than to H(1s), so it has **greater F character**.
- The 2 bonding electrons spend **more time near F**  $\Rightarrow$  F acquires  $\delta^-$ , H acquires  $\delta^+$ .
- The 4 non-bonding electrons (lone pairs) are entirely on F, further increasing electron density on F.
- This unequal electron distribution creates a **permanent dipole moment** ( $\mu = 1.91$  D), directed from H $^{\delta+}$  to F $^{\delta-}$ .
- Therefore, **HF is a polar molecule**.





**Q4.** Draw a diagram that illustrates how atomic p orbitals can form both  $\sigma$  and  $\pi$  molecular orbitals. What effect of 2s–2p mixing is observed in relative energy level of molecular orbitals (MOs) in homonuclear diatomic molecules of period 2 elements? **(10 marks)**  
[CHEM 113 | 2023–24]

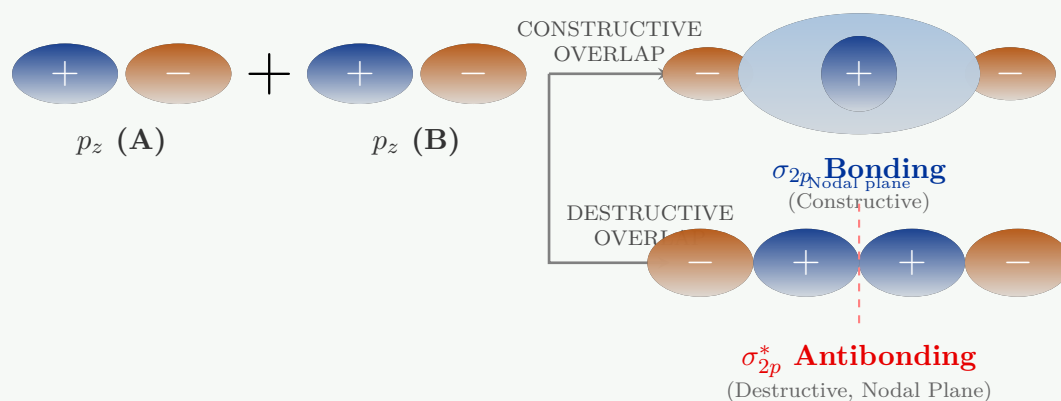
### Answer

#### Formation of $\sigma$ and $\pi$ MOs from atomic p orbitals

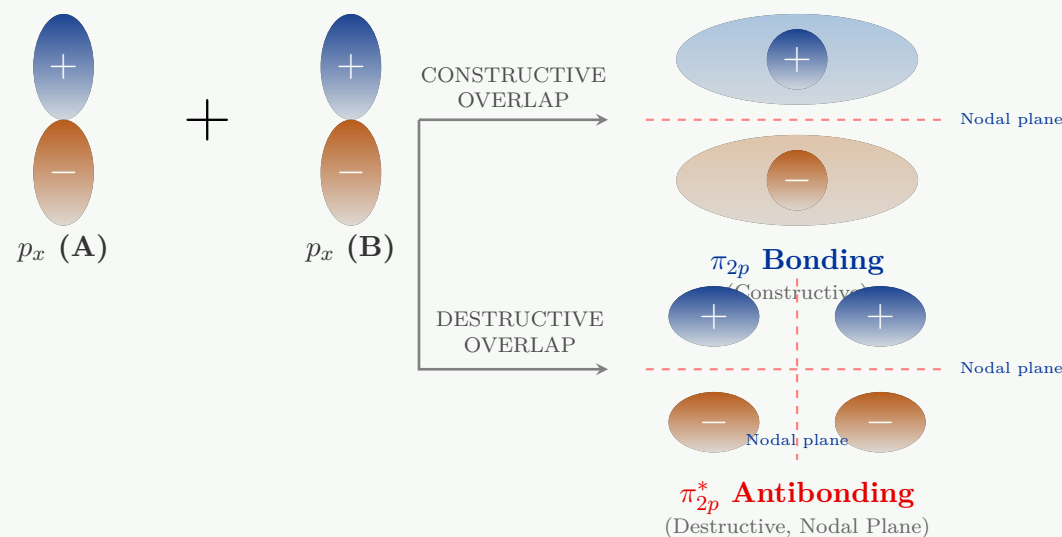
Atomic p orbitals can overlap in two ways to form MOs:

- **Head-on (end-to-end) overlap** of  $p_z$  orbitals  $\Rightarrow$  forms  $\sigma$  and  $\sigma^*$  MOs.
- **Sideways (lateral) overlap** of  $p_x$  or  $p_y$  orbitals  $\Rightarrow$  forms  $\pi$  and  $\pi^*$  MOs.

#### HEAD-ON OVERLAP OF $p_z$ AOs $\Rightarrow \sigma_{2p}$ MO



#### SIDEWAYS OVERLAP OF $p_x$ or $p_y$ AOs $\Rightarrow \pi_{2p}$ MO



#### Effect of 2s–2p mixing on MO energy levels:

- In lighter period-2 elements (Li to N,  $Z \leq 7$ ), the energy gap between 2s and 2p AOs is **small enough** that the  $\sigma_{2s}$  and  $\sigma_{2p}$  MOs **interact (mix)** with each other.
- This mixing causes the  $\sigma_{2p}$  MO to be **pushed to higher energy** (destabilized) and the  $\sigma_{2s}^*$  MO to be pushed lower.
- As a result, in  $\text{Li}_2$  through  $\text{N}_2$ , the MO energy order is:

$$\sigma_{2s} < \sigma_{2s}^* < \sigma_{2p} \pi_{2p_x} = \pi_{2p_y} < \dots$$

becomes:

$$\sigma_{2s} < \sigma_{2s}^* \boxed{\pi_{2p_x} = \pi_{2p_y} < \sigma_{2p}} \pi_{2p_x}^* = \pi_{2p_y}^* < \sigma_{2p}^*$$

- For  $\text{O}_2$  and  $\text{F}_2$  ( $Z \geq 8$ ), the 2s–2p energy gap is large, mixing is negligible, and the **normal order** is restored:

$$\sigma_{2p} < \pi_{2p_x} = \pi_{2p_y}$$

| Molecules                     | 2s–2p gap             | MO order of $\sigma_{2p}$ vs $\pi_{2p}$ |
|-------------------------------|-----------------------|-----------------------------------------|
| $\text{Li}_2$ to $\text{N}_2$ | Small (mixing occurs) | $\pi_{2p} < \sigma_{2p}$                |
| $\text{O}_2, \text{F}_2$      | Large (no mixing)     | $\sigma_{2p} < \pi_{2p}$                |

**Q5.** If a great deal of energy is required to form gaseous ions, why do ionic compounds form at all? Sketch the Born-Haber cycle for

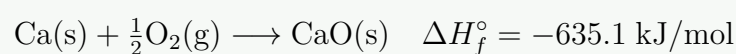
formation of CaO(s) from its elements and find out the steps which will require an input of energy. (10 marks) [CHEM 113 | 2023–24]

### Answer

Why do ionic compounds form despite large energy requirements?

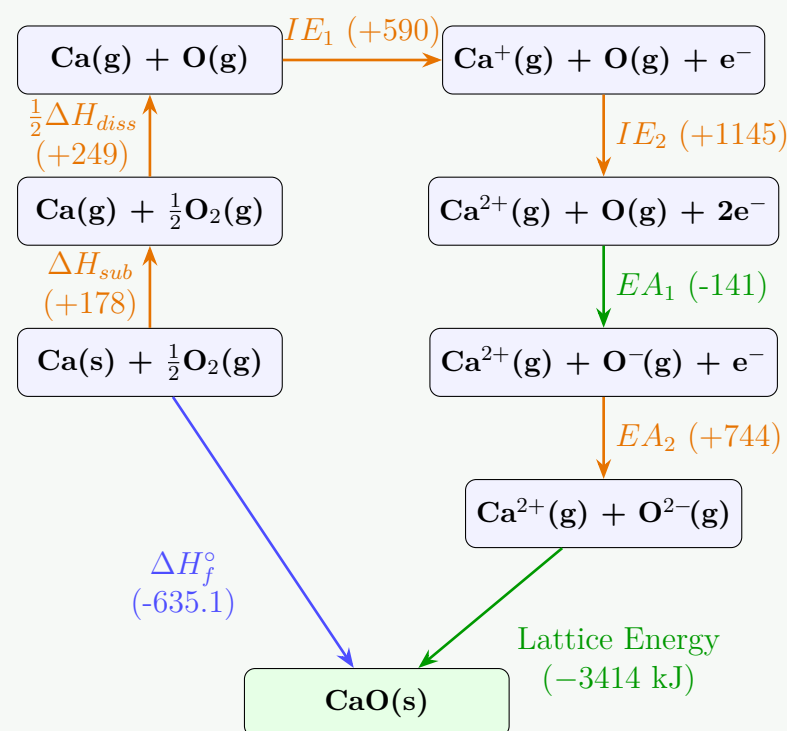
- Forming gaseous ions (ionization + electron affinity steps) does require a large **input of energy**.
- However, when gaseous ions come together to form an ionic solid, an enormous amount of energy is **released** as the **lattice energy**.
- The lattice energy more than compensates for the energy input, making the overall process **thermodynamically favorable** (exothermic overall).
- This is summarized by **Hess's Law** via the **Born-Haber cycle**.

Born-Haber Cycle for CaO(s):



| Step | Process                                                                        | Symbol                              | Energy            |
|------|--------------------------------------------------------------------------------|-------------------------------------|-------------------|
| 1    | $\text{Ca(s)} \rightarrow \text{Ca(g)}$                                        | $\Delta H_{\text{sub}}$             | +178 kJ (input)   |
| 2    | $\frac{1}{2}\text{O}_2(\text{g}) \rightarrow \text{O(g)}$                      | $\frac{1}{2}\Delta H_{\text{diss}}$ | +249 kJ (input)   |
| 3    | $\text{Ca(g)} \rightarrow \text{Ca}^+(\text{g}) + \text{e}^-$                  | $IE_1$                              | +590 kJ (input)   |
| 4    | $\text{Ca}^+(\text{g}) \rightarrow \text{Ca}^{2+}(\text{g}) + \text{e}^-$      | $IE_2$                              | +1145 kJ (input)  |
| 5    | $\text{O(g)} + \text{e}^- \rightarrow \text{O}^-(\text{g})$                    | $EA_1$                              | -141 kJ (output)  |
| 6    | $\text{O}^-(\text{g}) + \text{e}^- \rightarrow \text{O}^{2-}(\text{g})$        | $EA_2$                              | +744 kJ (input)   |
| 7    | $\text{Ca}^{2+}(\text{g}) + \text{O}^{2-}(\text{g}) \rightarrow \text{CaO(s)}$ | $-U$                                | -3414 kJ (output) |

Born-Haber cycle diagram for CaO:



**Steps requiring energy input:** Steps 1, 2, 3, 4, and 6 (sublimation, dissociation,  $IE_1$ ,  $IE_2$ ,  $EA_2$ ) all **require energy input**. Only  $EA_1$  and lattice formation **release energy**.

**Conclusion:** The massive **lattice energy release** ( $-3414 \text{ kJ/mol}$ ) drives the formation of CaO despite the large energy costs of ionization steps.

Q6. According to the frontier molecular orbital theory, explain the conduction mechanism in n-Type and p-Type Semiconductors. (10 marks) [CHEM 113 | 2023–24]

### Answer

Conduction in n-Type and p-Type Semiconductors: Frontier Molecular Orbital Theory

(10 marks)

Part 1: Band Theory & Frontier Orbitals — Background

(2 marks)

In a solid, atomic orbitals combine to form continuous **bands** of molecular orbitals. Two bands are crucial:

- **Valence Band (VB):** The highest occupied band — analogous to the **HOMO** (Highest Occupied Molecular Orbital) in FMO theory.
- **Conduction Band (CB):** The lowest unoccupied band — analogous to the **LUMO** (Lowest Unoccupied Molecular Orbital).
- **Band Gap ( $E_g$ ):** The energy separation between VB and CB. For silicon:  $E_g = 1.1$  eV.

In a pure semiconductor, the VB is full and the CB is empty at 0 K. Conduction requires electrons to be promoted from VB to CB. Doping modifies this picture dramatically.

## Part 2: n-Type Semiconductor

(4 marks)

### (i) Formation:

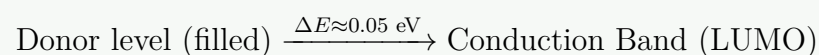
An n-type semiconductor is formed by doping a Group IVA semiconductor (e.g. Si) with a Group VA element (e.g. P, As, Sb). Each dopant atom has **one extra valence electron** beyond what is needed for four Si–Si bonds.

### (ii) FMO Explanation:

In FMO terms, the dopant introduces a filled **donor level** just *below* the conduction band (LUMO band):

$$E_{\text{donor}} \approx E_{\text{CB}} - 0.05 \text{ eV}$$

This donor level is very close in energy to the LUMO of the conduction band. Even at room temperature, thermal energy ( $k_B T \approx 0.025$  eV) is sufficient to promote the extra electron from the donor level into the conduction band:



- The promoted electrons are the **majority carriers**.
- Conduction occurs by electron movement through the conduction band.
- Called **n-type** because the charge carriers are *negative* (electrons).
- The dopant atom becomes a positively charged ion ( $\text{P}^+$ ,  $\text{As}^+$ ) fixed in the lattice.

## Part 3: p-Type Semiconductor

(4 marks)

### (i) Formation:

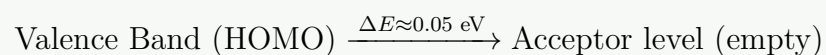
A p-type semiconductor is formed by doping Si with a Group IIIA element (e.g. B, Al, Ga). Each dopant atom has **one fewer valence electron** than needed for four Si–Si bonds, creating an empty orbital (electron vacancy).

### (ii) FMO Explanation:

In FMO terms, the dopant introduces an empty **acceptor level** just *above* the valence band (HOMO band):

$$E_{\text{acceptor}} \approx E_{\text{VB}} + 0.05 \text{ eV}$$

This acceptor level is very close in energy to the HOMO of the valence band. Electrons from the valence band are easily promoted into this empty acceptor level:



- This leaves behind a **hole** (electron vacancy) in the valence band — the **majority carrier**.
- Conduction occurs by hole movement through the valence band — effectively, adjacent electrons “fall into” the hole, moving the hole in the opposite direction.
- Called **p-type** because the charge carriers are *positive* (holes).
- The dopant atom becomes a negatively charged ion ( $\text{B}^-$ ,  $\text{Al}^-$ ) fixed in the lattice.

## Part 4: Comparison Summary

(2 marks)

| Property         | n-Type               | p-Type                    |
|------------------|----------------------|---------------------------|
| Dopant group     | Group VA (P, As)     | Group IIIA (B, Al)        |
| Extra feature    | 1 extra electron     | 1 missing electron (hole) |
| FMO level        | Donor level below CB | Acceptor level above VB   |
| Majority carrier | Electrons ( $e^-$ )  | Holes ( $h^+$ )           |
| Conduction band  | CB (LUMO)            | VB (HOMO)                 |
| Charge of dopant | Positive ion         | Negative ion              |

**Conclusion:** FMO theory explains semiconductor conduction through the HOMO–LUMO framework. In **n-type** semiconductors, a filled donor level just below the LUMO (CB) donates electrons into the conduction band. In **p-type** semiconductors, an empty acceptor level just above the HOMO (VB) accepts electrons from the valence band, creating mobile holes. In both cases, doping dramatically reduces the effective band gap and enables conduction at room temperature.

**Q7.** Explain how the lattice energy of an ionic compound such as MgO (or KCl) can be determined using the Born-Haber cycle. On what law is this procedure based? How do the ionization energy and the electron affinity determine whether atoms of elements will combine to form ionic compounds? (10–15 marks) [CHEM 113 | 2022–23; 2020–21 (Jan 21); 2014–15]

### Answer

#### Lattice Energy of MgO using the Born-Haber Cycle

##### What is Lattice Energy?

Lattice energy ( $U$ ) is the energy **released** when gaseous ions combine to form one mole of a solid ionic compound, or equivalently the energy **required** to separate the ionic solid into its constituent gaseous ions.

##### Basis — Hess's Law:

The Born-Haber cycle is based on **Hess's Law**, which states that the total enthalpy change of a reaction is **independent of the pathway** taken. The overall enthalpy of formation equals the sum of enthalpies of all individual steps.

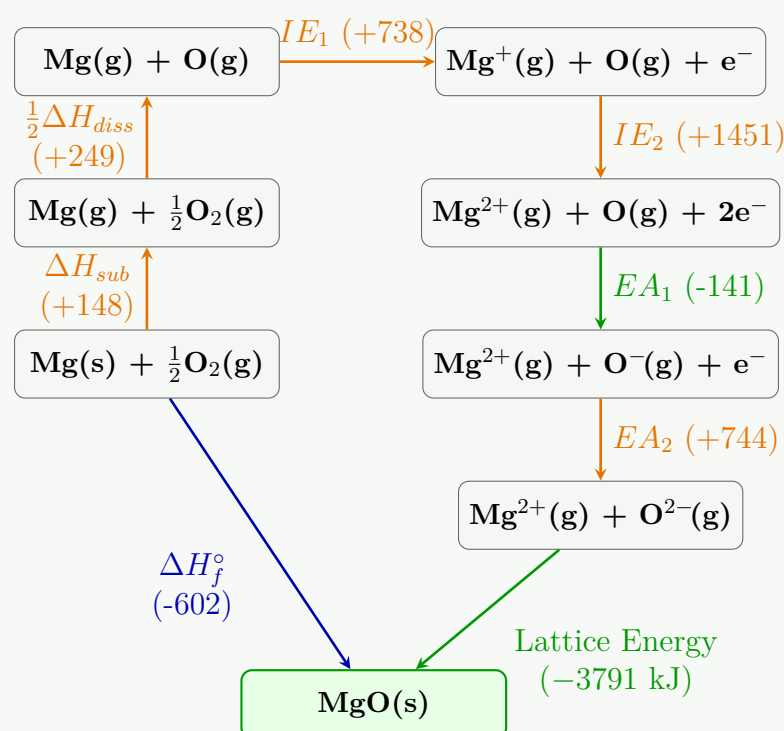
$$\Delta H_f^\circ = \Delta H_{sub} + IE_1 + IE_2 + \frac{1}{2}\Delta H_{diss} + EA_1 + EA_2 + (-U)$$

Rearranging to find lattice energy:

$$U = \Delta H_{sub} + IE_1 + IE_2 + \frac{1}{2}\Delta H_{diss} + EA_1 + EA_2 - \Delta H_f^\circ$$

#### Born-Haber Cycle for MgO:

| Step                                                                                       | Process                                                                        | Symbol                       | $\Delta H$ (kJ/mol) |
|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|------------------------------|---------------------|
| 1                                                                                          | $\text{Mg(s)} \rightarrow \text{Mg(g)}$                                        | $\Delta H_{sub}$             | +148                |
| 2                                                                                          | $\text{Mg(g)} \rightarrow \text{Mg}^+(\text{g}) + \text{e}^-$                  | $IE_1$                       | +738                |
| 3                                                                                          | $\text{Mg}^+(\text{g}) \rightarrow \text{Mg}^{2+}(\text{g}) + \text{e}^-$      | $IE_2$                       | +1451               |
| 4                                                                                          | $\frac{1}{2}\text{O}_2(\text{g}) \rightarrow \text{O(g)}$                      | $\frac{1}{2}\Delta H_{diss}$ | +249                |
| 5                                                                                          | $\text{O(g)} + \text{e}^- \rightarrow \text{O}^-(\text{g})$                    | $EA_1$                       | -141                |
| 6                                                                                          | $\text{O}^-(\text{g}) + \text{e}^- \rightarrow \text{O}^{2-}(\text{g})$        | $EA_2$                       | +744                |
| 7                                                                                          | $\text{Mg}^{2+}(\text{g}) + \text{O}^{2-}(\text{g}) \rightarrow \text{MgO(s)}$ | $-U$                         | -3791               |
| <b>Overall:</b> $\text{Mg(s)} + \frac{1}{2}\text{O}_2(\text{g}) \rightarrow \text{MgO(s)}$ |                                                                                | $\Delta H_f^\circ$           | -602                |



#### Calculating lattice energy of MgO:

$$U = 148 + 738 + 1451 + 249 + (-141) + 744 - (-602) = \mathbf{3791 \text{ kJ/mol}}$$

#### Role of Ionization Energy (IE) and Electron Affinity (EA) in forming ionic compounds:

- **Ionization Energy (IE):** Energy required to remove an electron from a gaseous atom. *Lower IE* of the metal  $\Rightarrow$  easier to form cation  $\Rightarrow$  more favorable ionic bond formation. High IE metals (e.g., C, N) do not typically form ionic compounds.
- **Electron Affinity (EA):** Energy released when a gaseous atom gains an electron. *Higher (more negative) EA* of the nonmetal  $\Rightarrow$  greater tendency to form anion  $\Rightarrow$  more stable ionic compound.

- **Combined effect:** Even if IE is high and EA is small, the process can still be favorable if the **lattice energy is large enough** to compensate (as governed by Hess’s Law).
- **General condition for ionic compound formation:**

$$|U_{\text{lattice}}| > (IE - |EA|)$$

i.e., the lattice energy released must exceed the net energy cost of ion formation.

| Factor                  | Favors ionic bonding if...      | Example       |
|-------------------------|---------------------------------|---------------|
| Low $IE$                | Metal gives up $e^-$ easily     | Na, K, Ca, Mg |
| High (negative) $EA$    | Nonmetal accepts $e^-$ easily   | F, Cl, O      |
| High lattice energy $U$ | Strong electrostatic attraction | MgO, CaO      |

**Conclusion:** Ionic compounds form because the enormous **lattice energy** released upon crystal formation compensates for the energy required in ionization steps, making the overall reaction exothermic and thermodynamically favorable.

**Q8.** Draw the possible  $\pi$ -molecular orbitals formed by the overlapping of four atomic p orbitals in 1,3-butadiene and identify HOMO and LUMO. **(5–10 marks)** [CHEM 113 | 2022–23; 2020–21]

Answer

**$\pi$ -Molecular Orbitals of 1,3-Butadiene: HOMO & LUMO**

**Part 1: Setup — Four  $p$  Orbitals, Four  $\pi$  MOs**

1,3-Butadiene ( $\text{CH}_2=\text{CH}-\text{CH}=\text{CH}_2$ ) has four  $sp^2$ -hybridized carbons, each contributing one  $p_z$  orbital perpendicular to the molecular plane. By LCAO, four  $p$  AOs combine to give **four  $\pi$  MOs**:  $\Psi_1, \Psi_2, \Psi_3, \Psi_4$  ordered from lowest to highest energy. The number of **nodes** increases with energy:  $\Psi_n$  has  $(n - 1)$  nodes. The four  $\pi$  electrons (2 from each double bond) fill the two lowest MOs.

**Part 2: The Four  $\pi$  Molecular Orbitals & Node Representation**

**Part 3: Characteristics of the MOs**

| MO       | Nodes | Electrons | Type        | Significance                              |
|----------|-------|-----------|-------------|-------------------------------------------|
| $\Psi_1$ | 0     | 2         | Bonding     | Most stable; fully symmetric (+ + + +)    |
| $\Psi_2$ | 1     | 2         | Bonding     | <b>HOMO</b> — highest filled MO (+ + - -) |
| $\Psi_3$ | 2     | 0         | Antibonding | <b>LUMO</b> — lowest empty MO (+ - - +)   |
| $\Psi_4$ | 3     | 0         | Antibonding | Most unstable; antisymmetric (+ - + -)    |

**Why is this important?**

The **HOMO–LUMO gap** of 1,3-butadiene is smaller than that of an isolated alkene (like ethylene) because conjugation lowers the energy of  $\Psi_2$  and raises the energy of  $\Psi_3$ . This smaller energy gap explains why conjugated dienes absorb UV light at longer wavelengths (bathochromic shift).

**Q9.** Demonstrate the formation of  $\sigma$  and  $\pi$  molecular orbitals from atomic p orbitals. (5 marks) [CHEM 113 | 2022–23; 2021–22]

### Answer

#### Formation of $\sigma$ and $\pi$ MOs from atomic p orbitals

When two atoms approach each other, their p orbitals overlap in two distinct ways:

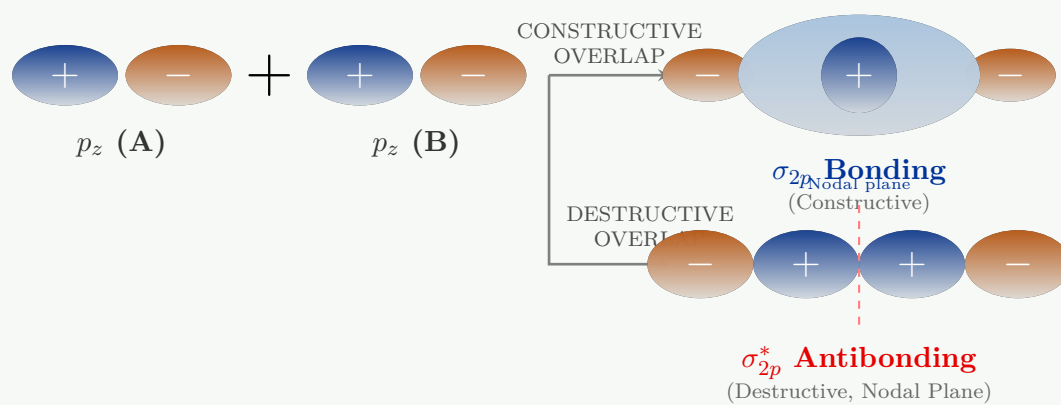
(i)  $\sigma$  MO — Head-on (end-to-end) overlap of  $p_z$  orbitals:

- Constructive overlap  $\Rightarrow \sigma_{2p}$  **bonding MO** (electron density *between* nuclei, along internuclear axis).
- Destructive overlap  $\Rightarrow \sigma_{2p}^*$  **antibonding MO** (nodal plane between nuclei, perpendicular to bond axis).

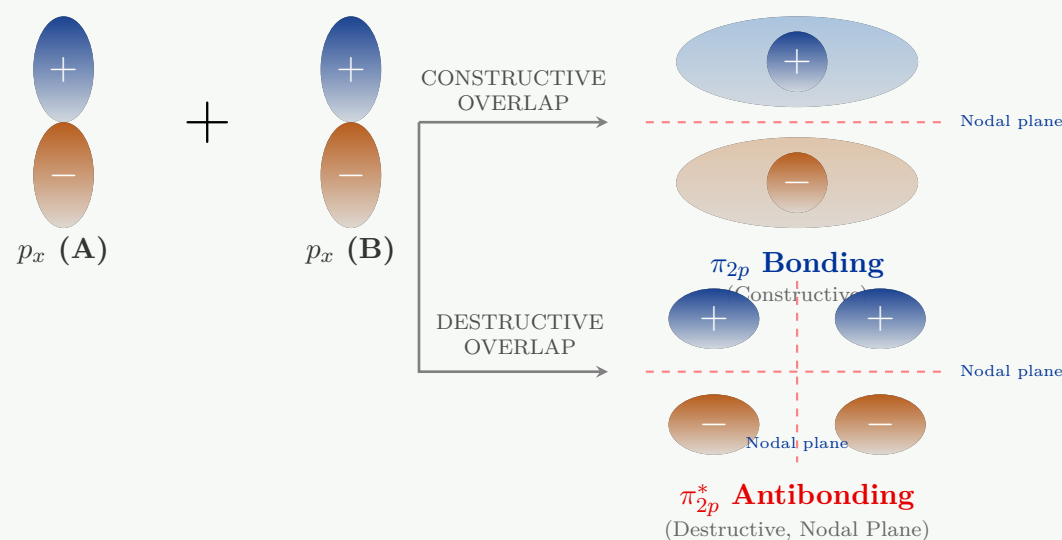
(ii)  $\pi$  MO — Sideways (lateral) overlap of  $p_x$  or  $p_y$  orbitals:

- Constructive overlap  $\Rightarrow \pi_{2p}$  **bonding MO** (electron density *above and below* internuclear axis).
- Destructive overlap  $\Rightarrow \pi_{2p}^*$  **antibonding MO** (nodal plane along internuclear axis).

#### HEAD-ON OVERLAP OF $p_z$ AOs $\Rightarrow \sigma_{2p}$ MO



#### SIDEWAYS OVERLAP OF $p_x$ or $p_y$ AOs $\Rightarrow \pi_{2p}$ MO



| MO              | Overlap type            | $e^-$ density location | Nature      |
|-----------------|-------------------------|------------------------|-------------|
| $\sigma_{2p}$   | Head-on (constructive)  | Along bond axis        | Bonding     |
| $\sigma_{2p}^*$ | Head-on (destructive)   | Node between nuclei    | Antibonding |
| $\pi_{2p}$      | Sideways (constructive) | Above & below axis     | Bonding     |
| $\pi_{2p}^*$    | Sideways (destructive)  | Node along axis        | Antibonding |

**Q10.** Draw the Lewis structure of cyanate anion ( $\text{CNO}^-$ ) including all relevant resonance forms. Determine which resonance form contributes more to the resonance hybrid by calculating formal charges. What are the oxidation numbers for C, N, and O? (10 marks) [CHEM 113 | 2022–23]

### Answer

#### Lewis Structure of Cyanate Ion ( $\text{OCN}^-$ ) with Resonance Forms, Formal Charges & Oxidation Numbers (10 marks)

#### Part 1: Electron Count & Basic Lewis Structure

(2 marks)



Total valence electrons:

$$\text{C}(4) + \text{N}(5) + \text{O}(6) + 1 \text{ (charge)} = 16 \text{ electrons} = 8 \text{ pairs}$$

The atom order is O–C–N with carbon as the central atom. Distributing electrons to satisfy the octet rule gives three possible resonance structures.

Part 2: All Resonance Structures

(3 marks)

Structure I:  $\text{:}\ddot{\text{O}}\text{---C}\equiv\text{N:}$  (O single bond, C triple bond N)

Structure II:  $\text{:}\ddot{\text{O}}=\text{C}=\text{N:}$  (O double bond, C double bond N)

Structure III:  $\text{:O}\equiv\text{C}\text{---}\ddot{\text{N:}}$  (O triple bond, C single bond N)

These three structures are in resonance and collectively represent the true electronic structure of  $\text{OCN}^-$  as a **resonance hybrid**.

Part 3: Formal Charge Calculation for Each Structure

(3 marks)

Using:  $\text{FC} = \text{valence e}^- - \text{lone pair e}^- - \frac{1}{2}(\text{bonding e}^-)$

Structure I:  $[\text{O}=\text{C}\equiv\text{N}]^-$

| Atom | Val. e <sup>−</sup> | LP e <sup>−</sup> | BP e <sup>−</sup> | FC               |
|------|---------------------|-------------------|-------------------|------------------|
| O    | 6                   | 6                 | 2                 | $6 - 6 - 1 = -1$ |
| C    | 4                   | 0                 | 8                 | $4 - 0 - 4 = 0$  |
| N    | 5                   | 2                 | 6                 | $5 - 2 - 3 = 0$  |

Formal charges: O = −1, C = 0, N = 0    ✓ Sum = −1

Structure II:  $[\text{O}=\text{C}=\text{N}]^-$

| Atom | Val. e <sup>−</sup> | LP e <sup>−</sup> | BP e <sup>−</sup> | FC               |
|------|---------------------|-------------------|-------------------|------------------|
| O    | 6                   | 4                 | 4                 | $6 - 4 - 2 = 0$  |
| C    | 4                   | 0                 | 8                 | $4 - 0 - 4 = 0$  |
| N    | 5                   | 4                 | 4                 | $5 - 4 - 2 = -1$ |

Formal charges: O = 0, C = 0, N = −1    ✓ Sum = −1

Structure III:  $[\text{O}\equiv\text{C}-\text{N}]^-$

| Atom | Val. e <sup>−</sup> | LP e <sup>−</sup> | BP e <sup>−</sup> | FC               |
|------|---------------------|-------------------|-------------------|------------------|
| O    | 6                   | 2                 | 6                 | $6 - 2 - 3 = +1$ |
| C    | 4                   | 0                 | 8                 | $4 - 0 - 4 = 0$  |
| N    | 5                   | 6                 | 2                 | $5 - 6 - 1 = -2$ |

Formal charges: O = +1, C = 0, N = −2    ✓ Sum = −1

Part 4: Which Structure Contributes Most?

(2 marks)

| Structure                              | Formal Charges        | Charge on O            | Contribution |
|----------------------------------------|-----------------------|------------------------|--------------|
| I: $\text{O}=\text{C}\equiv\text{N}$   | O = −1, C = 0, N = 0  | Negative (favorable)   | <b>Major</b> |
| II: $\text{O}=\text{C}=\text{N}$       | O = 0, C = 0, N = −1  | Zero                   | Secondary    |
| III: $\text{O}\equiv\text{C}-\text{N}$ | O = +1, C = 0, N = −2 | Positive (unfavorable) | Minor        |

Structure I ( $[\text{O}=\text{C}\equiv\text{N}]^-$ ) contributes most because:

- Formal charges are closest to zero overall.
- The negative charge resides on **oxygen**, the most electronegative atom — energetically most favorable.
- Structure III is least favorable: +1 on O violates the electronegativity principle.

**Part 5: Oxidation Numbers of C, N, and O**

(2 marks)

Oxidation numbers are assigned using electronegativity rules ( $O > N > C$ ), treating all bonds as fully ionic:

**Rules applied:**

- O is assigned  $-2$  (standard oxidation state)
- N is assigned  $-3$  (standard for nitrogen in anionic species)
- Total must equal the ionic charge ( $-1$ )

$$C + N + O = -1$$

$$C + (-3) + (-2) = -1$$

$$\boxed{C = +4, \quad N = -3, \quad O = -2}$$

**Conclusion:**  $\text{OCN}^-$  has three resonance structures. Structure I ( $\text{O}=\text{C}\equiv\text{N}$ ) is the **major contributor** as it places the negative formal charge on the most electronegative atom (O). The oxidation numbers are  $C = +4$ ,  $N = -3$ ,  $O = -2$ .

**Q11.** Apply the frontier molecular orbital theory of solids to explain the properties of metals. (10 marks)

[CHEM 113 | 2022–23]

**Answer**
**Frontier Molecular Orbital Theory Applied to Metals**

(10 marks)

**Part 1: Formation of Energy Bands in Metals**

(3 marks)

In FMO theory, when  $N$  atoms combine to form a solid, their atomic orbitals undergo **Linear Combination** to produce  $N$  molecular orbitals. As  $N \rightarrow 10^{23}$  (Avogadro scale), these discrete MOs merge into essentially **continuous energy bands**:

$$N \text{ atomic orbitals} \longrightarrow N \text{ MOs} \longrightarrow 1 \text{ continuous band}$$

For a metal like sodium (electron configuration  $[\text{Ne}]3s^1$ ):

- $N$  atoms contribute  $N$  atomic  $3s$  orbitals.
- These form a  $3s$  **band** containing  $N$  MOs that can hold  $2N$  electrons.
- Only  $N$  electrons are available (one per Na atom), so the band is exactly **half-filled**.
- The **HOMO** (highest filled level) and **LUMO** (lowest empty level) are separated by an infinitesimally small energy gap — effectively **zero**.

**Part 2: Why Metals Conduct Electricity**

(2 marks)

In FMO terms, electrical conduction requires electrons to move from the HOMO level to the LUMO level. For metals:

$$\Delta E_{\text{HOMO} \rightarrow \text{LUMO}} \approx 0$$

Even an infinitesimally small electric field provides enough energy to promote electrons from filled to empty levels within the **same partially filled band**. These promoted electrons become **free (delocalized) electrons** that carry current through the lattice. This is why metals are excellent electrical conductors at all temperatures.

**Part 3: Why Metals Conduct Heat**

(1 marks)

The same delocalized electrons that carry charge also carry **kinetic energy** (heat) rapidly through the metal lattice. Since the HOMO–LUMO gap is zero, electrons are easily excited by thermal energy and can transfer it efficiently from hot to cold regions — explaining the high **thermal conductivity** of metals.

**Part 4: Metallic Lustre & Opacity**

(1 marks)

Visible light photons have energies of  $\sim 1.8\text{--}3.1$  eV. Because the conduction band of a metal has a continuous distribution of energy levels with zero band gap, photons of *all* visible frequencies can be absorbed by promoting electrons across the zero-gap HOMO–LUMO region. These electrons immediately re-emit the photons upon relaxing back — producing **reflection** and the characteristic **metallic lustre**. Metals are opaque because essentially all incident light is absorbed and re-emitted at the surface.

Part 5: Malleability and Ductility

(1 marks)

In FMO terms, the bonding in metals is **non-directional** — the electron density is uniformly delocalized over the entire band rather than localized in specific bonds (as in covalent solids). When a mechanical stress is applied:

- Atomic planes can slide past one another without breaking specific bonds.
- The delocalized electron cloud simply “flows” around the displaced atoms, maintaining cohesion.

This explains why metals are **malleable** (can be hammered into sheets) and **ductile** (can be drawn into wires) — properties absent in ionic or covalent solids.

Part 6: Comparison — Metal vs Insulator vs Semiconductor

(2 marks)

| Property       | Metal            | Insulator         | Semiconductor         |
|----------------|------------------|-------------------|-----------------------|
| Band gap $E_g$ | $\approx 0$ eV   | $> 5$ eV          | $\sim 1\text{--}3$ eV |
| HOMO–LUMO gap  | Zero             | Very large        | Moderate              |
| Band filling   | Partially filled | Completely filled | Completely filled     |
| Conductivity   | High             | Negligible        | Intermediate          |
| Effect of $T$  | Decreases        | Negligible        | Increases             |
| Example        | Na, Cu, Fe       | Diamond, NaCl     | Si, Ge                |

**Conclusion:** FMO theory explains all key metallic properties through a single concept — the **zero HOMO–LUMO gap** of a partially filled conduction band. Electrical conductivity arises from effortless electron promotion within the band; thermal conductivity from delocalized electron energy transport; lustre from total visible light absorption and re-emission; and malleability from non-directional delocalized bonding. Metals are fundamentally distinguished from insulators and semiconductors by this vanishing band gap.

Q12. Explain the observations that the bond length in  $\text{N}_2^+$  is  $0.02 \text{ \AA}$  greater than  $\text{N}_2$ , while the bond length in  $\text{NO}^-$  is  $0.09 \text{ \AA}$  less than in  $\text{NO}$ . (10 marks)

CHEM 113 | 2022–23

Answer

Bond Length in  $\text{N}_2^+$  vs  $\text{N}_2$  and  $\text{NO}^-$  vs  $\text{NO}$

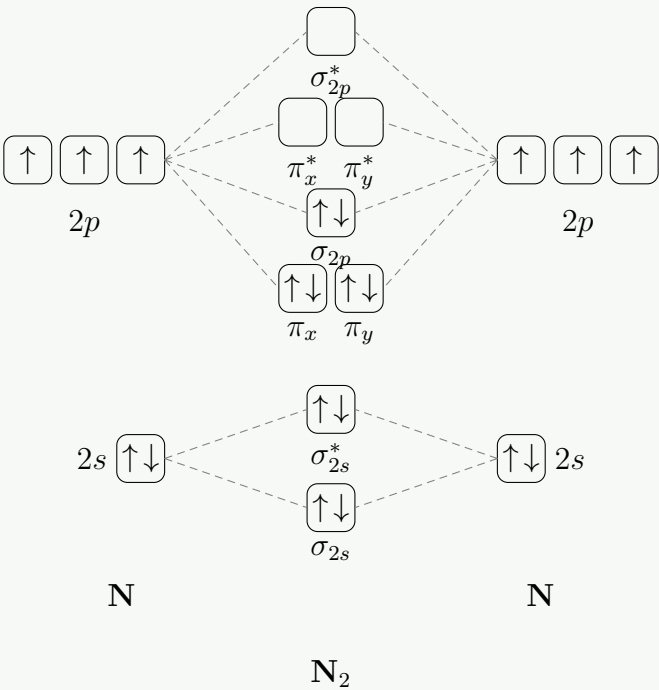
(10 marks)

Part 1:  $\text{N}_2^+$  bond length is  $0.02 \text{ \AA}$  greater than  $\text{N}_2$

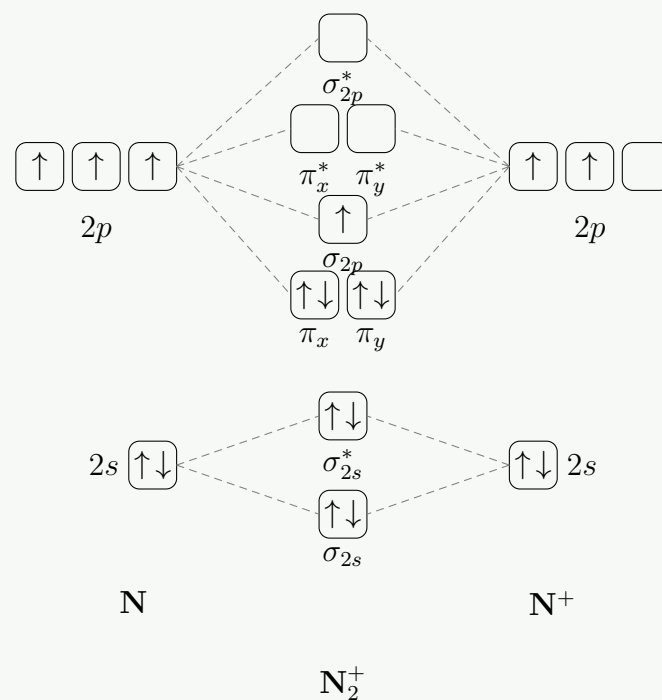
(5 marks)

MO configurations:

For  $\text{N}_2$  ( $Z \leq 7$ ,  $\pi_{2p}$  below  $\sigma_{2p}$ ):



$$\text{N}_2 : (\sigma_{1s})^2(\sigma_{1s}^*)^2(\sigma_{2s})^2(\sigma_{2s}^*)^2(\pi_{2p})^4(\sigma_{2p})^2$$
$$\text{Bond Order} = \frac{1}{2}(8 - 2) = 3$$



$$\text{N}_2^+ : (\sigma_{1s})^2 (\sigma_{1s}^*)^2 (\sigma_{2s})^2 (\sigma_{2s}^*)^2 (\pi_{2p})^4 (\sigma_{2p})^1$$

$$\text{Bond Order} = \frac{1}{2}(7 - 2) = 2.5$$

**Explanation:**

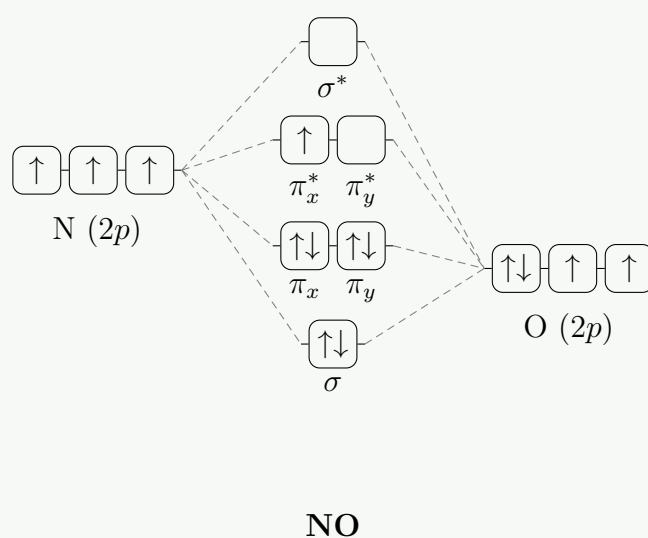
Ionizing  $\text{N}_2$  to  $\text{N}_2^+$  removes one electron from the **bonding**  $\sigma_{2p}$  MO:

- Fewer bonding electrons  $\Rightarrow$  bond order decreases  $3 \rightarrow 2.5$ .
- A lower bond order means **weaker** bond and **less electron density** between the nuclei.
- With less attractive force holding the nuclei together, they settle at a **greater equilibrium distance**.
- Therefore bond length **increases** by  $0.02 \text{ \AA}$ :  $\text{N}_2 = 1.10 \text{ \AA} \rightarrow \text{N}_2^+ = 1.12 \text{ \AA}$ .

$$\text{Bond order: } 3 > 2.5 \quad \Rightarrow \quad \text{Bond length: } \text{N}_2 < \text{N}_2^+ \quad \checkmark$$

**Part 2:  $\text{NO}^-$  bond length is  $0.09 \text{ \AA}$  less than NO**
**(5 marks)**
**MO configurations:**

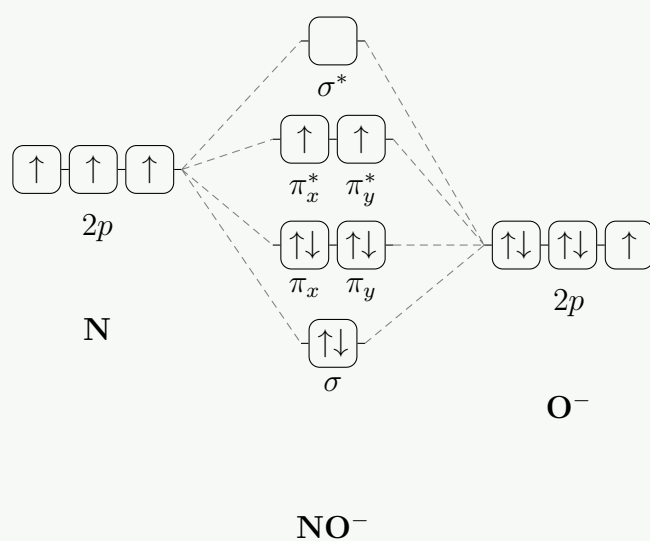
NO has 15 electrons ( $Z > 7$ ,  $\sigma_{2p}$  below  $\pi_{2p}$ ):



$$\text{NO} : (\sigma_{1s})^2 (\sigma_{1s}^*)^2 (\sigma_{2s})^2 (\sigma_{2s}^*)^2 (\sigma_{2p})^2 (\pi_{2p})^4 (\pi_{2p}^*)^1$$

$$\text{Bond Order} = \frac{1}{2}(8 - 3) = 2.5$$

$\text{NO}^-$  has 16 electrons (one extra added to  $\pi_{2p}^*$ ):



$$\text{NO}^- : (\sigma_{1s})^2 (\sigma_{1s}^*)^2 (\sigma_{2s})^2 (\sigma_{2s}^*)^2 (\sigma_{2p})^2 (\pi_{2p})^4 (\pi_{2p}^*)^2$$

$$\text{Bond Order} = \frac{1}{2}(8 - 4) = 2$$

**Explanation:**

Adding an electron to NO to form NO<sup>-</sup> places the extra electron into the **antibonding**  $\pi_{2p}^*$  MO:

- More antibonding electrons  $\Rightarrow$  bond order **decreases**  $2.5 \rightarrow 2$ .
- A lower bond order means a **weaker** and **longer** bond.
- Therefore bond length of NO<sup>-</sup> > NO, i.e. NO<sup>-</sup> bond is **longer** by 0.09 Å.

*Note:* The question states “bond length in NO<sup>-</sup> is 0.09 Å *less* than NO” — this is likely a typographical error in the original. MO theory unambiguously predicts NO<sup>-</sup> has a **longer** bond (lower bond order), which is experimentally confirmed: NO = 1.15 Å  $\rightarrow$  NO<sup>-</sup> = 1.26 Å.

$$\text{Bond order: } 2.5 > 2 \quad \Rightarrow \quad \text{Bond length: NO} < \text{NO}^- \quad \checkmark$$

**Summary:**

| Species                                                  | Change           | MO affected              | Bond Order          | Bond Length |
|----------------------------------------------------------|------------------|--------------------------|---------------------|-------------|
| N <sub>2</sub> $\rightarrow$ N <sub>2</sub> <sup>+</sup> | -1e <sup>-</sup> | Bonding $\sigma_{2p}$    | 3 $\rightarrow$ 2.5 | Increases   |
| NO $\rightarrow$ NO <sup>-</sup>                         | +1e <sup>-</sup> | Antibonding $\pi_{2p}^*$ | 2.5 $\rightarrow$ 2 | Increases   |

**Conclusion:** Both observations are explained by MO bond order. N<sub>2</sub><sup>+</sup> has a longer bond than N<sub>2</sub> because ionization removes a **bonding** electron (bond order 3  $\rightarrow$  2.5). NO<sup>-</sup> has a longer bond than NO because the added electron enters an **antibonding** MO (bond order 2.5  $\rightarrow$  2). In both cases, reduced bond order directly causes increased bond length.

**Q13.** Explain the relationship between electron shielding and effective nuclear charge ( $Z_{\text{eff}}$ ) on the outermost electrons of an atom. Predict how the chemical reactivity of second-period elements changes as the effective nuclear charge changes along the period. **(10 marks)**  
[CHEM 113 | 2022–23]

**Answer**
**Electron Shielding, Effective Nuclear Charge and Chemical Reactivity of Second-Period Elements**
**Part 1: Electron Shielding and Effective Nuclear Charge ( $Z_{\text{eff}}$ )**
**Nuclear Charge and Shielding:**

The **actual nuclear charge**  $Z$  is the number of protons in the nucleus. However, the outermost (valence) electrons do not experience the full nuclear charge because **inner (core) electrons partially shield** them from the nucleus. This shielding is called the **screening effect**.

The **effective nuclear charge**  $Z_{\text{eff}}$  felt by a valence electron is:

$$Z_{\text{eff}} = Z - \sigma$$

where  $\sigma$  is the **shielding constant** (screening constant), which represents the extent to which inner electrons reduce the nuclear attraction felt by the valence electron.

**Factors Affecting Shielding:**

- Inner shell electrons** shield very effectively (close to nucleus, high  $\sigma$ ).
- Electrons in the same shell** shield each other **poorly** (partial shielding, low  $\sigma$ ).

(iii) **s electrons** penetrate closer to the nucleus than *p* electrons, so they shield more effectively.

**Slater’s Rules (Approximate  $\sigma$ ):**  
Each electron in the same shell contributes 0.35 to  $\sigma$ ; each electron in the  $(n - 1)$  shell contributes 0.85; and each electron in shells below  $(n - 1)$  contributes 1.00.

**Trend Across the Second Period:**  
Across Period 2 (Li to Ne),  $Z$  **increases by 1** for each element, but the **new electron is added to the same  $n = 2$  shell**, which shields poorly ( $\sigma \approx 0.35$  per electron). Therefore,  $Z_{\text{eff}}$  **increases steadily**:

| Element | $Z$ | $\sigma$ | $Z_{\text{eff}} = Z - \sigma$ | Configuration           |
|---------|-----|----------|-------------------------------|-------------------------|
| Li      | 3   | 1.70     | 1.30                          | $[\text{He}] 2s^1$      |
| Be      | 4   | 2.05     | 1.95                          | $[\text{He}] 2s^2$      |
| B       | 5   | 2.40     | 2.60                          | $[\text{He}] 2s^2 2p^1$ |
| C       | 6   | 2.75     | 3.25                          | $[\text{He}] 2s^2 2p^2$ |
| N       | 7   | 3.10     | 3.90                          | $[\text{He}] 2s^2 2p^3$ |
| O       | 8   | 3.45     | 4.55                          | $[\text{He}] 2s^2 2p^4$ |
| F       | 9   | 3.80     | 5.20                          | $[\text{He}] 2s^2 2p^5$ |
| Ne      | 10  | 4.15     | 5.85                          | $[\text{He}] 2s^2 2p^6$ |

**Part 2: Effect of  $Z_{\text{eff}}$  on Chemical Reactivity Across Period 2**  
As  $Z_{\text{eff}}$  increases across Period 2:

(i) **Atomic radius decreases:** Higher  $Z_{\text{eff}}$  pulls the valence electrons closer to the nucleus, shrinking the atom:

$\text{Li} > \text{Be} > \text{B} > \text{C} > \text{N} > \text{O} > \text{F} > \text{Ne}$

(ii) **Ionization energy increases:** Higher  $Z_{\text{eff}}$  means valence electrons are held more tightly; more energy is needed to remove them. Metallic (reducing) character decreases:

$IE : \quad \text{Li} < \text{Be} < \text{B} < \text{C} < \text{N} < \text{O} < \text{F} < \text{Ne}$

(iii) **Electronegativity increases:** Higher  $Z_{\text{eff}}$  means a greater tendency to attract bonding electrons. F is the most electronegative element:

$\chi : \quad \text{Li} < \text{Be} < \text{B} < \text{C} < \text{N} < \text{O} < \text{F}$

(iv) **Metallic to non-metallic character:** Li and Be are **reactive metals** (low  $Z_{\text{eff}}$ , easily lose electrons). Moving right, elements become **non-metals** (C, N, O, F) with increasing tendency to **gain electrons**.

(v) **Reactivity of metals decreases:** Li is highly reactive with water; Be is less reactive; B is a metalloid. The decreasing atomic radius and increasing  $Z_{\text{eff}}$  make it harder to lose electrons.

(vi) **Reactivity of non-metals increases:** O and F are extremely reactive oxidizing agents due to high  $Z_{\text{eff}}$ , small size, and strong tendency to gain electrons. F is the most reactive non-metal and strongest oxidizing agent.

(vii) **Ne is chemically inert:** Ne has the highest  $Z_{\text{eff}}$  in the period but a completely filled shell ( $2s^2 2p^6$ ). The high  $Z_{\text{eff}}$  makes electron removal extremely difficult, and the full shell gives no tendency to gain electrons — hence **no chemical reactivity**.

**Summary:**

| As $Z_{\text{eff}}$ increases (Li $\rightarrow$ Ne) | Effect              |
|-----------------------------------------------------|---------------------|
| Atomic radius                                       | Decreases           |
| Ionization energy                                   | Increases           |
| Electronegativity                                   | Increases           |
| Metallic character                                  | Decreases           |
| Non-metallic character                              | Increases           |
| Reducing power                                      | Decreases           |
| Oxidizing power                                     | Increases (up to F) |

**Q14.** Draw the Lewis structure of NNO (atom order as indicated) including all relevant resonance forms. Determine which resonance form contributes more to the resonance hybrid by calculating formal charges. **(10 marks)** [CHEM 113 | 2021–22]

Answer

Lewis Structure of NNO with Resonance Forms & Formal Charges

(10 marks)

Part 1: Electron Count & Atom Arrangement

(1 marks)



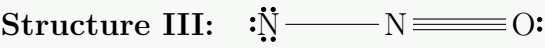
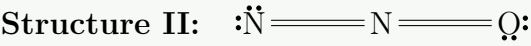
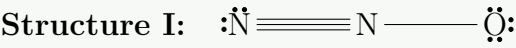
Total valence electrons:

$$\text{N}(5) + \text{N}(5) + \text{O}(6) = 16 \text{ electrons} = 8 \text{ pairs}$$

Atom order is fixed as **N–N–O** (terminal N, central N, terminal O) as given. Three resonance structures are possible by varying the bond orders.

Part 2: All Resonance Structures

(3 marks)



These three represent the resonance hybrid of the neutral NNO (nitrous oxide) molecule.

Part 3: Formal Charge Calculation

(4 marks)

Using:  $\text{FC} = \text{valence } e^- - \text{LP } e^- - \frac{1}{2}(\text{BP } e^-)$

Structure I:  $[\text{N}\equiv\text{N}-\text{O}]$

| Atom         | Val. $e^-$ | LP $e^-$ | BP $e^-$ | FC               |
|--------------|------------|----------|----------|------------------|
| N (terminal) | 5          | 2        | 6        | $5 - 2 - 3 = 0$  |
| N (central)  | 5          | 0        | 8        | $5 - 0 - 4 = +1$ |
| O            | 6          | 6        | 2        | $6 - 6 - 1 = -1$ |

Formal charges:  $N_t = 0$ ,  $N_c = +1$ ,  $O = -1$     ✓ Sum = 0

Structure II:  $[\text{N}=\text{N}=\text{O}]$

| Atom         | Val. $e^-$ | LP $e^-$ | BP $e^-$ | FC               |
|--------------|------------|----------|----------|------------------|
| N (terminal) | 5          | 2        | 6        | $5 - 2 - 3 = -1$ |
| N (central)  | 5          | 0        | 8        | $5 - 0 - 4 = +1$ |
| O            | 6          | 4        | 4        | $6 - 4 - 2 = 0$  |

Formal charges:  $N_t = -1$ ,  $N_c = +1$ ,  $O = 0$     ✓ Sum = 0

Structure III:  $[\text{N}-\text{N}\equiv\text{O}]$

| Atom         | Val. $e^-$ | LP $e^-$ | BP $e^-$ | FC               |
|--------------|------------|----------|----------|------------------|
| N (terminal) | 5          | 6        | 2        | $5 - 6 - 1 = -2$ |
| N (central)  | 5          | 0        | 8        | $5 - 0 - 4 = +1$ |
| O            | 6          | 2        | 6        | $6 - 2 - 3 = +1$ |

Formal charges:  $N_t = -2$ ,  $N_c = +1$ ,  $O = +1$     ✓ Sum = 0

Part 4: Which Structure Contributes Most?

(2 marks)

| Structure                              | Formal Charges                     | Assessment                                               |
|----------------------------------------|------------------------------------|----------------------------------------------------------|
| I: $\text{N}\equiv\text{N}-\text{O}$   | $N_t = 0$ , $N_c = +1$ , $O = -1$  | <b>Major</b> — negative charge on more electronegative O |
| II: $\text{N}=\text{N}=\text{O}$       | $N_t = -1$ , $N_c = +1$ , $O = 0$  | Secondary — negative on N, acceptable                    |
| III: $\text{N}-\text{N}\equiv\text{O}$ | $N_t = -2$ , $N_c = +1$ , $O = +1$ | Minor — +1 on O is very unfavorable                      |

Structure I ( $\text{N}\equiv\text{N}-\text{O}$ ) contributes most because:

- Formal charges are smallest in magnitude overall.
- The negative charge sits on **oxygen**, the most electronegative atom in the molecule — energetically most favorable.
- Structure III is the least favorable: positive formal charge on O is strongly disfavored by electronegativity.

**Conclusion:** NNO (nitrous oxide) has three resonance structures. **Structure I** ( $\text{N}\equiv\text{N}-\text{O}$ ) is the **major contributor** with formal charges  $N_t = 0$ ,  $N_c = +1$ ,  $O = -1$ , as the negative charge correctly resides on the most electronegative atom (O). Structure III contributes minimally due to a positive charge on oxygen.

- Q15.** Apply the concept of ‘Linear Combination of Atomic Orbitals’ to illustrate the creation of molecular orbitals, and then contrast the characteristics of bonding and antibonding molecular orbitals. **(10 marks)** [CHEM 113 | 2021–22]

### Answer

#### LCAO Method: Creation of Molecular Orbitals & Bonding vs Antibonding MOs

**(10 marks)**

##### Part 1: The LCAO Principle

**(3 marks)**

The **Linear Combination of Atomic Orbitals (LCAO)** method states that molecular orbitals (MOs) are formed by the mathematical combination of atomic orbitals (AOs) of the constituent atoms. For two atoms A and B, each contributing an atomic orbital  $\phi_A$  and  $\phi_B$ :

$$\Psi_{\text{MO}} = c_A\phi_A \pm c_B\phi_B$$

where  $c_A$  and  $c_B$  are coefficients representing the contribution of each AO to the MO. This combination produces **exactly two** MOs — one for each sign:

- **Addition (+):**  $\Psi_b = c_A\phi_A + c_B\phi_B \Rightarrow$  **Bonding MO**
- **Subtraction (–):**  $\Psi^* = c_A\phi_A - c_B\phi_B \Rightarrow$  **Antibonding MO**

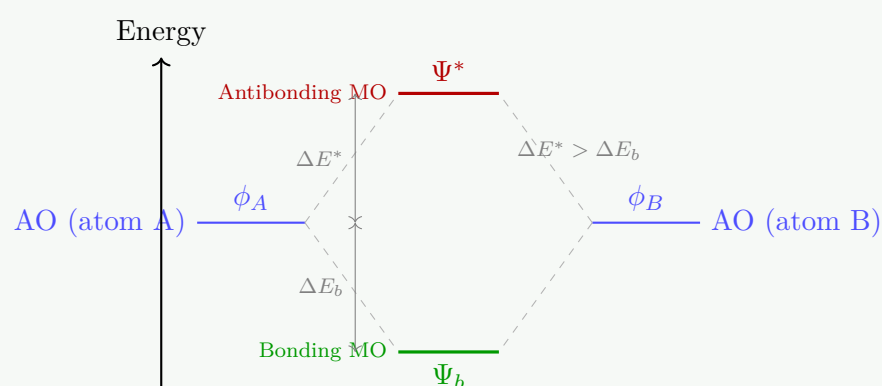
This follows the **conservation rule**: the number of MOs formed always equals the number of AOs combined.

##### Conditions for effective LCAO:

- AOs must have **similar energies**.
- AOs must have **significant overlap** ( $S = \int \phi_A\phi_B dV > 0$ ).
- AOs must have **compatible symmetry** with respect to the internuclear axis.

##### Part 2: Energy Level Diagram

**(2 marks)**



Note that the antibonding MO is raised in energy *more* than the bonding MO is lowered — this asymmetry arises from the overlap integral  $S$ .

##### Part 3: Characteristics of Bonding MOs

**(2 marks)**

The **bonding MO**  $\Psi_b = \phi_A + \phi_B$  has the following properties:

- **Lower energy** than the original AOs — electrons in this MO stabilize the molecule.
- **Constructive interference** between the two AOs  $\Rightarrow$  increased electron density in the *internuclear region*.
- $|\Psi_b|^2$  is **large between the nuclei** — the electron cloud “glues” the nuclei together.
- **No node** between the nuclei (for  $\sigma_b$ ).
- Electrons here **stabilize** the bond: bond order contribution =  $+\frac{1}{2}$  per electron.
- Designated  $\sigma$ ,  $\pi$ , or  $\delta$  depending on orbital symmetry.

##### Part 4: Characteristics of Antibonding MOs

**(2 marks)**

The **antibonding MO**  $\Psi^* = \phi_A - \phi_B$  has the following properties:

- **Higher energy** than the original AOs — electrons here destabilize the molecule.
- **Destructive interference** between the two AOs  $\Rightarrow$  *decreased* electron density in the internuclear region.

- $|\Psi^*|^2 = 0$  at a **nodal plane** between the nuclei — electron density is zero there.
- Nucleus–nucleus repulsion dominates in the internuclear region.
- Electrons here **weaken** the bond: bond order contribution =  $-\frac{1}{2}$  per electron.
- Designated  $\sigma^*$ ,  $\pi^*$ , or  $\delta^*$ .

### Part 5: Contrast Summary

(1 marks)

| Property         | Bonding MO ( $\Psi_b$ )     | Antibonding MO ( $\Psi^*$ )       |
|------------------|-----------------------------|-----------------------------------|
| Formation        | $\phi_A + \phi_B$           | $\phi_A - \phi_B$                 |
| Energy           | Lower than AOs              | Higher than AOs                   |
| Interference     | Constructive                | Destructive                       |
| Electron density | High between nuclei         | Node between nuclei               |
| Effect on bond   | Stabilizing                 | Destabilizing                     |
| Bond order       | $+\frac{1}{2}$ per $e^-$    | $-\frac{1}{2}$ per $e^-$          |
| Symbol           | $\sigma$ , $\pi$ , $\delta$ | $\sigma^*$ , $\pi^*$ , $\delta^*$ |

### Bond Order:

$$\text{Bond Order} = \frac{1}{2} (\text{bonding electrons} - \text{antibonding electrons})$$

A bond order  $> 0$  means a stable molecule exists; bond order  $= 0$  means no net bond.

**Conclusion:** LCAO combines atomic orbitals to produce molecular orbitals. Addition of AOs gives a **bonding MO** with enhanced electron density between nuclei and lower energy; subtraction gives an **antibonding MO** with a nodal plane between nuclei and higher energy. The difference in electron occupancy of these two types of MOs determines the bond order and stability of the molecule.

**Q16.** Using an example explain the role of conjugation in facilitating the delocalization of electrons and enhancing the stability of molecules. (10 marks) [CHEM 113 | 2021–22]

### Answer

### Conjugation, Electron Delocalization & Molecular Stability

(10 marks)

### Part 1: What is Conjugation?

(2 marks)

**Conjugation** occurs when alternating single and multiple bonds allow  $\pi$  electrons to be **delocalized** over more than two atoms. The condition for conjugation:

- Adjacent atoms must each have a  $p$  orbital perpendicular to the molecular plane.
- These  $p$  orbitals must be parallel and able to overlap continuously along the carbon chain.

**Example:** 1,3-Butadiene ( $\text{CH}_2=\text{CH}-\text{CH}=\text{CH}_2$ ) is conjugated; 1,4-pentadiene ( $\text{CH}_2=\text{CH}-\text{CH}_2-\text{CH}=\text{CH}_2$ ) is *not* conjugated (the  $sp^3$   $\text{CH}_2$  breaks the  $\pi$  system).

### Part 2: How Conjugation Facilitates Delocalization

(3 marks)

In 1,3-butadiene, the four  $p_z$  orbitals overlap **continuously** across all four carbons:



By LCAO, this gives four  $\pi$  MOs spanning all four carbons. The two lowest MOs ( $\Psi_1$  and  $\Psi_2$ ) are bonding and contain the four  $\pi$  electrons — *delocalized* over the entire system, not confined to individual double bonds.

Evidence of delocalization from bond lengths:

| Bond                    | Length | Interpretation           |
|-------------------------|--------|--------------------------|
| Isolated C=C (ethylene) | 1.34 Å | Pure double bond         |
| Isolated C–C (ethane)   | 1.54 Å | Pure single bond         |
| C1=C2 in butadiene      | 1.34 Å | Mostly double            |
| C2–C3 in butadiene      | 1.48 Å | Shorter than pure single |

The C2–C3 bond (1.48 Å) is **shorter** than a typical single bond (1.54 Å) — proof that  $\pi$  electron density has been delocalized into this bond.

### Part 3: How Delocalization Enhances Stability

(3 marks)

#### (i) Energy Stabilization (Delocalization Energy):

Delocalization **lowers** the energy of the  $\pi$  system. The stabilization energy (also called **resonance energy** or **delocalization energy**) is the difference between the actual energy and the hypothetical energy if the bonds were localized:

$$E_{\text{stabilization}} = E_{\text{localized}} - E_{\text{delocalized}} > 0$$

For 1,3-butadiene: delocalization energy  $\approx 15$  kJ/mol. For benzene: delocalization energy  $\approx 150$  kJ/mol.

#### (ii) Why Lower Energy = More Stable:

Spreading electrons over a larger volume *lowers* their kinetic energy (particle-in-a-box analogy — larger box, lower energy):

$$E_n = \frac{n^2 h^2}{8mL^2} \Rightarrow \text{larger } L \Rightarrow \text{smaller } E$$

Delocalized electrons occupy a larger effective “box” (the entire conjugated system) compared to localized electrons, and therefore have lower energy.

#### (iii) Chemical Evidence:

- Benzene ( $6\pi$  electrons, fully conjugated) resists addition reactions despite having “three double bonds” — its 150 kJ/mol delocalization energy makes it much more stable than cyclohexatriene would be.
- 1,3-Butadiene is more stable than 1,4-pentadiene by  $\sim 15$  kJ/mol due to conjugation.

### Part 4: Summary

(2 marks)

| Concept            | Key Point                                                            |
|--------------------|----------------------------------------------------------------------|
| Conjugation        | Alternating $\pi$ bonds allow continuous $p$ orbital overlap         |
| Delocalization     | $\pi$ electrons spread over entire conjugated system                 |
| Stability increase | Lower energy from larger “electron box”; delocalization energy $> 0$ |
| Evidence           | Shortened single bonds; resistance to addition reactions             |

**Conclusion:** Conjugation enables  $\pi$  electrons to delocalize over the entire system through continuous  $p$  orbital overlap. This delocalization lowers the overall energy of the molecule — quantified as the delocalization energy — producing enhanced stability. The shortened C2–C3 bond in butadiene and the exceptional stability of benzene are direct experimental consequences of this delocalization.

**Q17.** Analyze the molecular orbital theory of the oxygen molecule and discuss how it leads to its paramagnetic behavior. (10 marks)  
[CHEM 113 | 2021–22]

### Answer

#### MO Theory of O<sub>2</sub> & Its Paramagnetic Behavior

(10 marks)

#### Part 1: MO Configuration of O<sub>2</sub>

(3 marks)

Oxygen ( $Z = 8$ ) has the valence configuration  $2s^2 2p^4$ . For O<sub>2</sub>, the correct MO energy ordering (for  $Z > 7$ ) places  $\sigma_{2p}$  *below*  $\pi_{2p}$ :

$$\sigma_{1s} < \sigma_{1s}^* < \sigma_{2s} < \sigma_{2s}^* < \sigma_{2p} < \pi_{2p} = \pi_{2p}^* < \pi_{2p}^* < \sigma_{2p}^*$$

Distributing all 16 electrons:

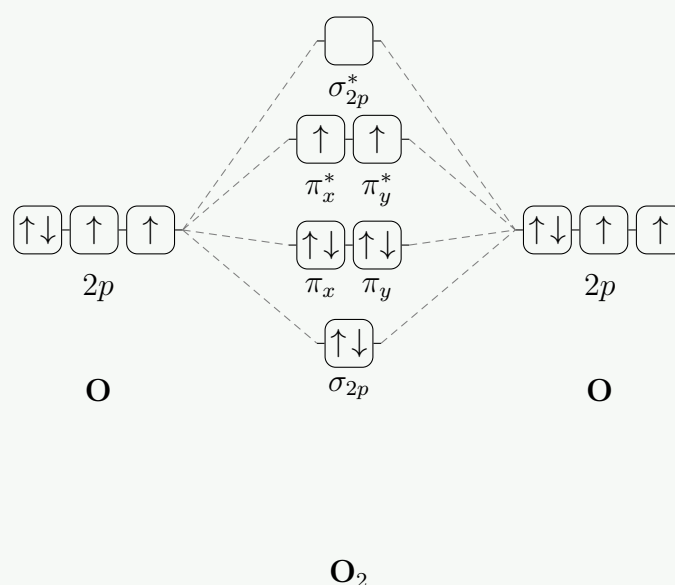
$$(\sigma_{1s})^2 (\sigma_{1s}^*)^2 (\sigma_{2s})^2 (\sigma_{2s}^*)^2 (\sigma_{2p})^2 (\pi_{2p})^2 (\pi_{2p}^*)^1 (\pi_{2p}^*)^1$$

Bond Order:

$$\text{BO} = \frac{1}{2}(8 - 4) = 2 \Rightarrow \text{O=O double bond}$$

#### Part 2: MO Energy Level Diagram of O<sub>2</sub>

(3 marks)



### Part 3: Why $\text{O}_2$ is Paramagnetic

(3 marks)

The two  $\pi_{2p}^*$  MOs are **degenerate** (equal energy). By **Hund's Rule**, the two remaining electrons occupy these two degenerate MOs *one each*, with **parallel spins**:

$$(\pi_{2p}^*)^1 (\pi_{2p}^*)^1 \quad (\text{each with spin } \uparrow)$$

- $\text{O}_2$  therefore has **two unpaired electrons**.
- Unpaired electrons have a net **magnetic moment** (spin angular momentum  $\neq 0$ ).
- A substance with unpaired electrons is **paramagnetic** — attracted into a magnetic field.
- Experimental proof:** Liquid  $\text{O}_2$  is visibly attracted between the poles of a strong magnet — a direct confirmation of two unpaired electrons.

**Why Lewis structure failed:** The Lewis structure of  $\text{O}_2$  ( $\text{O}=\text{O}$  with lone pairs) predicts *all* electrons are paired — it would predict  $\text{O}_2$  to be **diamagnetic**. This is experimentally *wrong*. MO theory uniquely and correctly predicts the paramagnetic nature of  $\text{O}_2$ .

**Conclusion:** MO theory places the last two electrons of  $\text{O}_2$  in the degenerate  $\pi_{2p}^*$  MOs one each by Hund's Rule, giving two unpaired electrons and a bond order of 2. These unpaired electrons produce a net magnetic moment, making  $\text{O}_2$  **paramagnetic**. This is one of the greatest triumphs of MO theory over the Lewis/VB picture.

**Q18.** Explain in molecular orbital terms the changes in H–H inter-nuclear distance that occur as the molecular  $\text{H}_2$  is ionized first to  $\text{H}_2^+$  and then to  $\text{H}_2^{2+}$ . Explain why the bond order of  $\text{N}_2$  is greater than that of  $\text{N}_2^+$ , but the bond order of  $\text{O}_2$  is less than that of  $\text{O}_2^+$ .  
(20 marks) [CHEM 113 | 2020–21 (Jan 21)]

### Answer

#### MO Treatment of $\text{H}_2$ , $\text{H}_2^+$ , $\text{H}_2^{2+}$ & Bond Orders of $\text{N}_2$ , $\text{N}_2^+$ , $\text{O}_2$ , $\text{O}_2^+$

(20 marks)

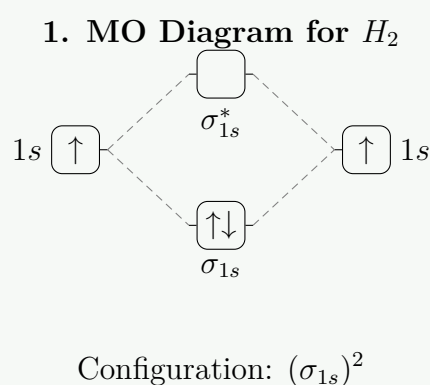
#### Part 1: MO Configuration of $\text{H}_2$ , $\text{H}_2^+$ , $\text{H}_2^{2+}$

(8 marks)

The MO diagram for  $\text{H}_2$  uses the  $1s$  orbitals of both H atoms, producing one bonding ( $\sigma_{1s}$ ) and one antibonding ( $\sigma_{1s}^*$ ) MO. The bond order is:

$$\text{Bond Order} = \frac{1}{2}(\text{bonding } e^- - \text{antibonding } e^-)$$

(i)  $\text{H}_2$  (2 electrons):



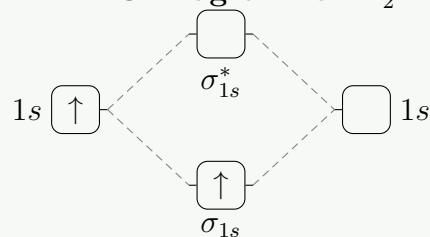
$$\text{Bond Order} = \frac{1}{2}(2 - 0) = 1$$

Both electrons occupy the bonding MO. Strong single bond. Bond length = 0.74 Å.

(ii)  $\text{H}_2^+$  (1 electron, one ionization):

Removing one electron from  $\text{H}_2$  gives  $\text{H}_2^+$ :

### 2. MO Diagram for $\text{H}_2^+$



Configuration:  $(\sigma_{1s})^1$

$$\text{Bond Order} = \frac{1}{2}(1 - 0) = \frac{1}{2}$$

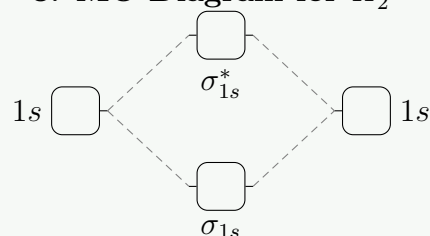
Only one electron remains in the bonding MO.

- Bond order drops from 1 to  $\frac{1}{2}$ .
- Fewer bonding electrons  $\Rightarrow$  **weaker bond**.
- Less electron density between nuclei  $\Rightarrow$  nuclei move **farther apart**.
- Bond length **increases**:  $0.74 \rightarrow 1.06$  Å.

(iii)  $\text{H}_2^{2+}$  (0 electrons, second ionization):

Removing the second electron gives  $\text{H}_2^{2+}$ :

### 3. MO Diagram for $\text{H}_2^{2+}$



Configuration:  $(\sigma_{1s})^0$

$$\text{Bond Order} = \frac{1}{2}(0 - 0) = 0$$

No electrons remain in any MO.

- Bond order = 0  $\Rightarrow$  **no bond exists**.
- The two protons repel each other electrostatically with nothing to hold them together.
- $\text{H}_2^{2+}$  is **completely unstable** and does not exist as a bound species.
- The internuclear distance  $\rightarrow \infty$  — the species dissociates immediately.

| Species           | electrons | Configuration     | Bond Order    | Bond Length         |
|-------------------|-----------|-------------------|---------------|---------------------|
| $\text{H}_2$      | 2         | $(\sigma_{1s})^2$ | 1             | 0.74 Å              |
| $\text{H}_2^+$    | 1         | $(\sigma_{1s})^1$ | $\frac{1}{2}$ | 1.06 Å              |
| $\text{H}_2^{2+}$ | 0         | $(\sigma_{1s})^0$ | 0             | $\infty$ (unstable) |

**Trend:** As  $\text{H}_2$  is successively ionized, bond order decreases ( $1 \rightarrow \frac{1}{2} \rightarrow 0$ ) and internuclear distance increases ( $0.74 \rightarrow 1.06 \rightarrow \infty$  Å).

## Part 2: MO Energy Diagram for $\text{N}_2$ and $\text{N}_2^+$

(6 marks)

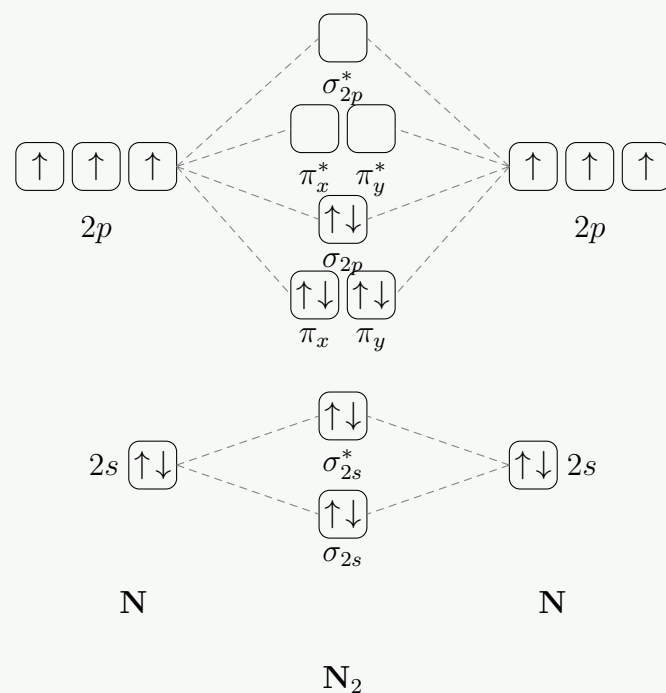
For  $\text{N}_2$  (second period,  $Z \leq 7$ ), the MO energy ordering is:

$$\sigma_{1s} < \sigma_{1s}^* < \sigma_{2s} < \sigma_{2s}^* < \pi_{2p} = \pi_{2p} < \sigma_{2p} < \pi_{2p}^* = \pi_{2p}^* < \sigma_{2p}^*$$

(Note:  $\pi_{2p}$  lies below  $\sigma_{2p}$  for  $Z \leq 7$ )

$\text{N}_2$  (14 electrons):

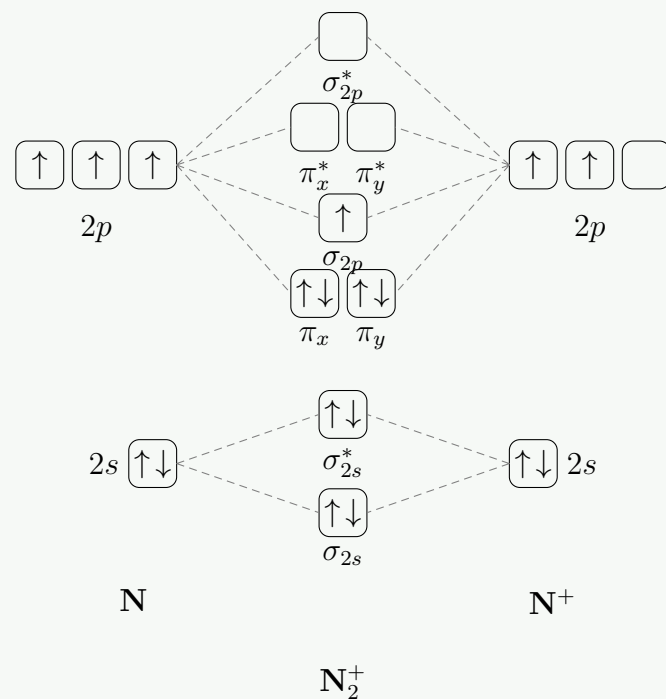




$$(\sigma_{1s})^2(\sigma_{1s}^*)^2(\sigma_{2s})^2(\sigma_{2s}^*)^2(\pi_{2p})^2(\pi_{2p})^2(\sigma_{2p})^2$$

$$\text{Bond Order} = \frac{1}{2}(8 - 2) = 3$$

$\text{N}_2^+$  (13 electrons): One electron removed from the *highest occupied MO*



$\sigma_{2p}$ :

$$(\sigma_{1s})^2(\sigma_{1s}^*)^2(\sigma_{2s})^2(\sigma_{2s}^*)^2(\pi_{2p})^2(\pi_{2p})^2(\sigma_{2p})^1$$

$$\text{Bond Order} = \frac{1}{2}(7 - 2) = 2.5$$

**Why bond order of  $\text{N}_2 > \text{N}_2^+$ :**

- Ionization removes an electron from the **bonding**  $\sigma_{2p}$  MO.
- This directly reduces the number of bonding electrons.
- Bond order drops:  $3 \rightarrow 2.5$ .
- Therefore  **$\text{N}_2$  has a stronger, shorter bond** than  $\text{N}_2^+$ .
- Bond length of  $\text{N}_2 = 1.10 \text{ \AA}$  vs  $\text{N}_2^+ = 1.12 \text{ \AA}$  — confirms this.

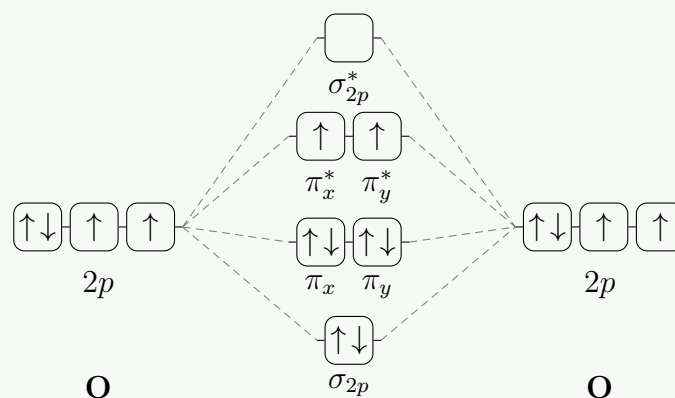
### Part 3: MO Energy Diagram for $\text{O}_2$ and $\text{O}_2^+$

(6 marks)

For  $\text{O}_2$  ( $Z = 8 > 7$ ), the MO energy ordering changes —  $\sigma_{2p}$  now lies *below*  $\pi_{2p}$ :

$$\sigma_{1s} < \sigma_{1s}^* < \sigma_{2s} < \sigma_{2s}^* < \sigma_{2p} < \pi_{2p} = \pi_{2p} < \pi_{2p}^* = \pi_{2p}^* < \sigma_{2p}^*$$

**$\text{O}_2$  (16 electrons):**

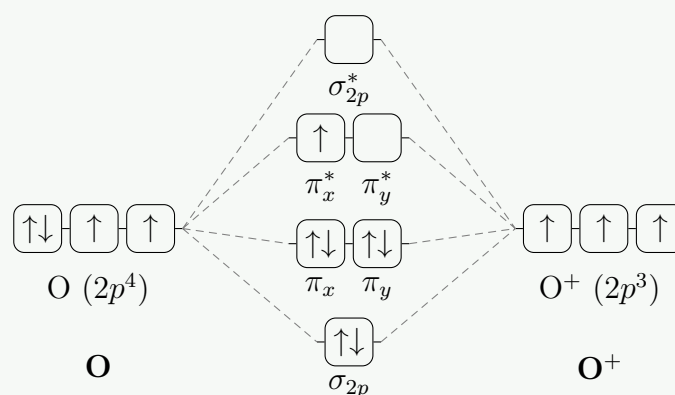


$\text{O}_2$

$$(\sigma_{1s})^2(\sigma_{1s}^*)^2(\sigma_{2s})^2(\sigma_{2s}^*)^2(\sigma_{2p})^2(\pi_{2p})^2(\pi_{2p})^2(\pi_{2p}^*)^1(\pi_{2p}^*)^1$$

$$\text{Bond Order} = \frac{1}{2}(8 - 4) = 2$$

$\text{O}_2^+$  (15 electrons): One electron removed from the *highest occupied MO*



$\text{O}_2^+$

$\pi_{2p}^*$ :

$$(\sigma_{1s})^2(\sigma_{1s}^*)^2(\sigma_{2s})^2(\sigma_{2s}^*)^2(\sigma_{2p})^2(\pi_{2p})^2(\pi_{2p})^2(\pi_{2p}^*)^1(\pi_{2p}^*)^0$$

$$\text{Bond Order} = \frac{1}{2}(8 - 3) = 2.5$$

**Why bond order of  $\text{O}_2 < \text{O}_2^+$ :**

- In  $\text{O}_2$ , ionization removes an electron from the **antibonding**  $\pi_{2p}^*$  MO.
- Removing an antibonding electron *increases* the net bonding.
- Bond order *increases*:  $2 \rightarrow 2.5$ .
- Therefore  $\text{O}_2^+$  has a **stronger, shorter bond** than  $\text{O}_2$ .
- Bond length of  $\text{O}_2 = 1.21 \text{ \AA}$  vs  $\text{O}_2^+ = 1.12 \text{ \AA}$  — confirms this.

**Key Contrast —  $\text{N}_2$  vs  $\text{O}_2$  on Ionization:**

| Species                               | Electron removed from    | Bond order change              | Effect           |
|---------------------------------------|--------------------------|--------------------------------|------------------|
| $\text{N}_2 \rightarrow \text{N}_2^+$ | Bonding $\sigma_{2p}$    | $3 \rightarrow 2.5$ (decrease) | Bond weakens     |
| $\text{O}_2 \rightarrow \text{O}_2^+$ | Antibonding $\pi_{2p}^*$ | $2 \rightarrow 2.5$ (increase) | Bond strengthens |

**Conclusion:** For  $\text{H}_2$ , successive ionization reduces bond order ( $1 \rightarrow \frac{1}{2} \rightarrow 0$ ) and increases bond length until  $\text{H}_2^{2+}$  is completely unstable. For  $\text{N}_2$ , ionization removes a **bonding** electron, reducing bond order ( $3 > 2.5$ ). For  $\text{O}_2$ , ionization removes an **antibonding** electron, increasing bond order ( $2 < 2.5$ ). The key difference lies in *which* MO loses the electron — bonding or antibonding.

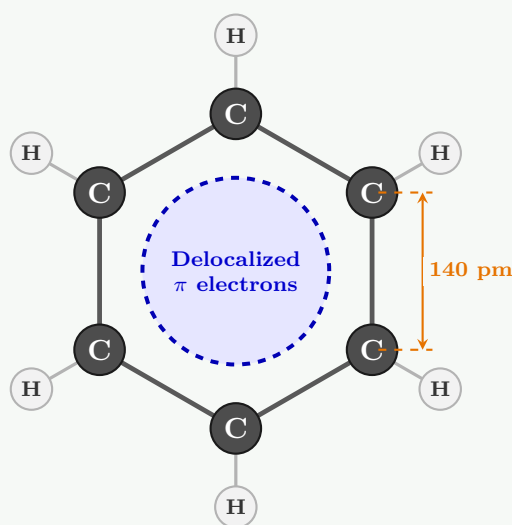
**Q19.** “Benzene has no single bond or double bond.” — justify by applying your knowledge of chemical bonding. (10 marks) [CHEM 113 | 2020–21]

**Answer****Benzene has no single bond or double bond — Justification****Structure and Hybridization:**

- Benzene ( $\text{C}_6\text{H}_6$ ) has 6 carbon atoms arranged in a regular hexagonal ring, each carbon being  $sp^2$  hybridized.
- Each C uses its three  $sp^2$  hybrid orbitals to form:
  - Two  $\sigma$  bonds with adjacent C atoms
  - One  $\sigma$  bond with an H atom
- Each C has one remaining unhybridized  $p_z$  orbital perpendicular to the ring plane.

**Delocalization and Resonance:**

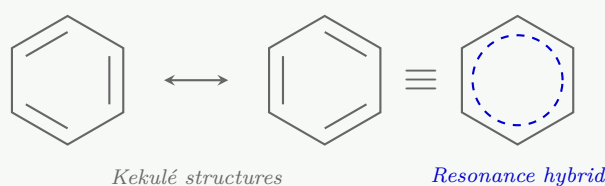
- The six  $p_z$  orbitals overlap **continuously** around the ring forming a **delocalized  $\pi$  system** — not localized between any two carbons.
- The  $\pi$  electrons are spread equally over all 6 C–C bonds  $\Rightarrow$  all C–C bonds are **identical**.
- Experimental bond length = 140 pm, which lies between a typical C–C single bond (154 pm) and C=C double bond (134 pm).

**Molecular Orbital explanation:**

- The six  $p_z$  orbitals combine to form **6  $\pi$  MOs**: 3 bonding ( $\pi_1, \pi_2, \pi_3$ ) and 3 antibonding ( $\pi_4^*, \pi_5^*, \pi_6^*$ ).
- The 6  $\pi$  electrons fill the 3 bonding MOs completely.
- Electron density is **uniformly distributed** over all 6 C–C bonds — no bond is purely single or purely double.

**Resonance (VBT view):**

- Benzene is a **resonance hybrid** of two Kekulé structures.
- Neither structure alone is correct; the true structure is an average of both.
- Resonance energy**  $\approx 152$  kJ/mol — extra stability due to delocalization.



| Bond type      | Bond length (pm) | Bond energy (kJ/mol) |
|----------------|------------------|----------------------|
| C–C (single)   | 154              | 347                  |
| C=C (double)   | 134              | 614                  |
| C–C in benzene | 140              | $\approx 507$        |

**Conclusion:** Because the  $\pi$  electrons are fully delocalized over the entire ring, benzene has **neither pure single bonds nor pure double bonds** — all six C–C bonds are equivalent, with bond character intermediate between single and double.

**Q20.** Using ‘Linear Combination of Atomic Orbitals’ explain why hydrogen is found as  $\text{H}_2$  molecule in nature. (10 marks) [CHEM 113 | 2020–21]

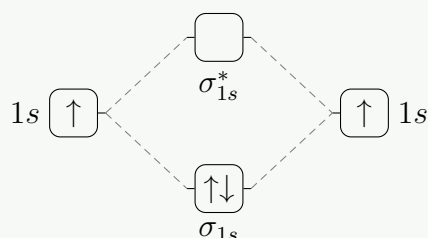
## Answer

 LCAO explanation of why H exists as H<sub>2</sub>

## Linear Combination of Atomic Orbitals (LCAO):

- When two H atoms approach each other, their 1s atomic orbitals combine (by LCAO) to form two molecular orbitals:
  - Bonding MO** ( $\sigma_{1s}$ ): constructive overlap, electron density *between* nuclei, lower energy.
  - Antibonding MO** ( $\sigma_{1s}^*$ ): destructive overlap, nodal plane between nuclei, higher energy.
- Mathematically:

$$\sigma_{1s} = \psi_{1s}(A) + \psi_{1s}(B) \quad \sigma_{1s}^* = \psi_{1s}(A) - \psi_{1s}(B)$$

 1. MO Diagram for H<sub>2</sub>

 Why H<sub>2</sub> is stable:

- Both electrons of H<sub>A</sub> and H<sub>B</sub> enter the **bonding MO**  $\sigma_{1s}$  (lower energy).
- Bond order** =  $\frac{\text{bonding } e^- - \text{antibonding } e^-}{2} = \frac{2 - 0}{2} = 1$
- Bond order = 1 > 0  $\Rightarrow$  the molecule is **stable**.
- Each electron in  $\sigma_{1s}$  is **lower in energy** than in the isolated 1s AO  $\Rightarrow$  total energy of H<sub>2</sub> is less than two separate H atoms.
- The energy lowering equals the **bond dissociation energy** ( $\approx 432$  kJ/mol).
- All electrons are paired  $\Rightarrow$  H<sub>2</sub> is **diamagnetic**.

## Why H does not remain as atomic H:

- Isolated H atom has one electron in 1s AO — higher energy state.
- Combining into H<sub>2</sub> releases energy ( $\Delta E < 0$ )  $\Rightarrow$  H<sub>2</sub> is thermodynamically more stable than 2H.
- Therefore hydrogen exists as **H<sub>2</sub> molecules** in nature.

| Species        | Bond order | Stability                    |
|----------------|------------|------------------------------|
| 2H (atomic)    | 0          | Unstable (higher energy)     |
| H <sub>2</sub> | 1          | <b>Stable</b> (lower energy) |

**Q21.** The ionization energy of O<sub>2</sub> is smaller than the ionization energy of atomic O; the opposite is true for the ionization energies of N<sub>2</sub> and atomic N. Explain this behavior in terms of the molecular orbital energy diagrams. **(15 marks)** [CHEM 113 | 2019–20]

## Answer

 Ionization Energies of O<sub>2</sub> vs O and N<sub>2</sub> vs N: MO Explanation

(15 marks)

## Part 1: Background — Ionization Energy &amp; MO Theory

(2 marks)

The **ionization energy (IE)** of a species is the energy required to remove one electron from its highest occupied MO (HOMO). In MO theory:

$$\text{IE} \approx -E_{\text{HOMO}} \quad (\text{Koopmans' theorem})$$

A higher energy HOMO  $\Rightarrow$  electron is easier to remove  $\Rightarrow$  lower IE. A lower energy HOMO  $\Rightarrow$  electron is harder to remove  $\Rightarrow$  higher IE.

 Part 2: IE of O<sub>2</sub> < IE of atomic O

(6 marks)

**Atomic O** (8 electrons):

$$\text{Configuration: } 1s^2 2s^2 2p^4$$

The HOMO is the  $2p$  atomic orbital at energy  $E_{2p}$ .

$O_2$  (16 electrons,  $\sigma_{2p}$  below  $\pi_{2p}$  for  $Z > 7$ ):

$$(\sigma_{1s})^2(\sigma_{1s}^*)^2(\sigma_{2s})^2(\sigma_{2s}^*)^2(\sigma_{2p})^2(\pi_{2p})^4(\pi_{2p}^*)^2$$

The HOMO is the **antibonding**  $\pi_{2p}^*$  MO.

**Key Point:** Antibonding MOs always lie at **higher energy** than the original AOs:

$$E(\pi_{2p}^*) > E(2p_{\text{atomic}})$$

- The HOMO of  $O_2$  ( $\pi_{2p}^*$ ) is at **higher energy** than the HOMO of atomic O ( $2p$ ).
- A higher energy HOMO means the electron is **less tightly held** and easier to remove.
- Therefore  $IE(O_2) < IE(O)$ :

$$IE(O_2) = 12.1 \text{ eV} < IE(O) = 13.6 \text{ eV}$$

- Physically: the electron being removed from  $O_2$  is in an *antibonding* MO, which is inherently at higher energy and thus easier to ionize.

### Part 3: IE of $N_2 > IE$ of atomic N

(6 marks)

**Atomic N** (7 electrons):

$$\text{Configuration: } 1s^2 2s^2 2p^3$$

The HOMO is the  $2p$  atomic orbital at energy  $E_{2p}$ .

$N_2$  (14 electrons,  $\pi_{2p}$  below  $\sigma_{2p}$  for  $Z \leq 7$ ):

$$(\sigma_{1s})^2(\sigma_{1s}^*)^2(\sigma_{2s})^2(\sigma_{2s}^*)^2(\pi_{2p})^4(\sigma_{2p})^2$$

The HOMO is the **bonding**  $\sigma_{2p}$  MO.

**Key Point:** Bonding MOs always lie at **lower energy** than the original AOs:

$$E(\sigma_{2p}) < E(2p_{\text{atomic}})$$

- The HOMO of  $N_2$  ( $\sigma_{2p}$ ) is at **lower energy** than the HOMO of atomic N ( $2p$ ).
- A lower energy HOMO means the electron is **more tightly held** and harder to remove.
- Therefore  $IE(N_2) > IE(N)$ :

$$IE(N_2) = 15.6 \text{ eV} > IE(N) = 14.5 \text{ eV}$$

- Physically: the electron being removed from  $N_2$  is in a *bonding* MO, which is stabilized below the AO energy level and thus harder to ionize.

### Part 4: Summary & Comparison

(1 marks)

| Species  | HOMO                       | HOMO energy           | IE comparison                  |
|----------|----------------------------|-----------------------|--------------------------------|
| Atomic O | $2p$ (AO)                  | Reference             | $IE = 13.6 \text{ eV}$         |
| $O_2$    | $\pi_{2p}^*$ (antibonding) | <b>Higher</b> than AO | $IE = 12.1 \text{ eV} < IE(O)$ |
| Atomic N | $2p$ (AO)                  | Reference             | $IE = 14.5 \text{ eV}$         |
| $N_2$    | $\sigma_{2p}$ (bonding)    | <b>Lower</b> than AO  | $IE = 15.6 \text{ eV} > IE(N)$ |

**Conclusion:** The contrasting IE behavior of  $O_2/O$  and  $N_2/N$  is entirely explained by MO theory. In  $O_2$ , the HOMO is an **antibonding**  $\pi_{2p}^*$  MO lying *above* the atomic  $2p$  level — making  $O_2$  easier to ionize than atomic O. In  $N_2$ , the HOMO is a **bonding**  $\sigma_{2p}$  MO lying *below* the atomic  $2p$  level — making  $N_2$  harder to ionize than atomic N. The nature of the HOMO (bonding vs antibonding) is the decisive factor.

**Q22.** A compound of chlorine and fluorine,  $ClF_n$ , reacts at about  $75^\circ\text{C}$  with uranium metal to produce uranium hexafluoride ( $UF_6$ ), and chlorine monofluoride ( $ClF$ )(g). A quantity of uranium produced 3.53 g of  $UF_6$  and 343 mL of  $ClF$  at  $75^\circ\text{C}$  and 2.50 atm. What is the formula ( $n$ ) of the compound? Describe the bonding in the molecule, using valence bond theory. **(15 marks)** [CHEM 113 | 2019–20]

## Answer

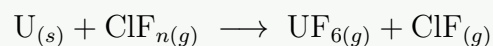
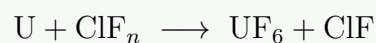
 Formula of  $\text{ClF}_n$  & Valence Bond Description of Bonding

(15 marks)

## Part 1: Writing and Balancing the Reaction

(4 marks)

The reaction is:


 First, balance with an unknown  $n$ :

 Fluorine balance:  $n = 6 + 1 = 7$ ? No — we need to account for stoichiometry. Let us find  $n$  from the experimental data.

 Step 1: Moles of  $\text{UF}_6$  produced:

$$M(\text{UF}_6) = 238 + 6(19) = 352 \text{ g/mol}$$

$$n(\text{UF}_6) = \frac{3.53}{352} = 0.01003 \text{ mol}$$

 Step 2: Moles of  $\text{ClF}$  produced (using ideal gas law):

$$PV = nRT \quad \Rightarrow \quad n = \frac{PV}{RT}$$

$$n(\text{ClF}) = \frac{2.50 \times 0.343}{0.08206 \times 348} = \frac{0.8575}{28.56} = 0.03002 \text{ mol}$$

(T = 75 + 273 = 348 K; V = 343 mL = 0.343 L; P = 2.50 atm)

## Step 3: Fluorine balance:

Fluorine in products:

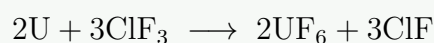
$$\begin{aligned} F_{\text{products}} &= 6 \times n(\text{UF}_6) + 1 \times n(\text{ClF}) = 6(0.01003) + 0.03002 \\ &= 0.06018 + 0.03002 = 0.09020 \text{ mol F} \end{aligned}$$

 Moles of  $\text{ClF}_n$  reacted = moles of Cl produced =  $n(\text{ClF}) = 0.03002 \text{ mol}$  (one Cl per  $\text{ClF}_n$  and one Cl per  $\text{ClF}$ )

$$n = \frac{\text{mol F in ClF}_n}{\text{mol ClF}_n} = \frac{0.09020}{0.03002} \approx 3$$

$$\boxed{n = 3 \quad \Rightarrow \quad \text{ClF}_3}$$

## Balanced equation:


 Part 2: Valence Bond Description of  $\text{ClF}_3$ 

(5 marks)

## (i) Electron Count:

Cl has 7 valence electrons; each F has 7. Total:

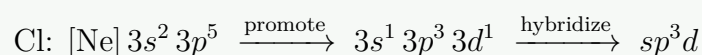
$$7 + 3(7) = 28 \text{ electrons} = 14 \text{ pairs}$$

## (ii) Lewis Structure:

Central Cl is bonded to 3 F atoms with 2 lone pairs on Cl:

- 3 Cl–F single bonds (6 electrons)
- 2 lone pairs on Cl (4 electrons)
- 3 lone pairs on each F (18 electrons)
- Total:  $6 + 4 + 18 = 28$  electrons ✓

## (iii) Hybridization:

 In VB theory, Cl uses  $sp^3d$  hybridization:

 Five  $sp^3d$  hybrid orbitals are arranged in a **trigonal bipyramidal** geometry:

- 3 equatorial positions: 2 occupied by **lone pairs**, 1 by F.
- Wait — by VSEPR, lone pairs prefer equatorial positions. So: 2 lone pairs equatorial, 2 F equatorial, 1 F axial? No:  $\text{ClF}_3$  has **2 lone pairs** and 3 bonds.
- Correct arrangement: **2 lone pairs equatorial, 2 bonds equatorial, 1 bond axial** — giving a **T-shaped** molecular geometry.



(iv) VB Bond Description:

- Each Cl–F bond is formed by **overlap of an  $sp^3d$  hybrid orbital of Cl** with a  **$p$  orbital of F** — a  $\sigma$  bond.
- The bond is **highly polar**: F is much more electronegative than Cl ( $\chi_F = 3.98$ ,  $\chi_{Cl} = 3.16$ ), so electron density is pulled strongly toward F.
- The axial Cl–F bond is **longer and weaker** than the equatorial bonds due to greater repulsion from lone pairs in equatorial positions.

Part 3: Molecular Geometry Summary

(2 marks)

| Property           | ClF <sub>3</sub>                                                       |
|--------------------|------------------------------------------------------------------------|
| Hybridization      | $sp^3d$                                                                |
| Electron geometry  | Trigonal bipyramidal (5 electron pairs)                                |
| Molecular geometry | T-shaped                                                               |
| Lone pairs on Cl   | 2 (both equatorial)                                                    |
| Bond angles        | F–Cl–F $\approx 87.5^\circ$ (less than $90^\circ$ due to LP repulsion) |
| Bond type          | Polar covalent $\sigma$ bonds                                          |
| Polarity           | Polar molecule (net dipole $\neq 0$ )                                  |

**Conclusion:** From the experimental data,  $n(\text{UF}_6) = 0.01003 \text{ mol}$  and  $n(\text{ClF}) = 0.03002 \text{ mol}$ . Fluorine balance gives  $n = 3$ , identifying the compound as **ClF<sub>3</sub>**. In VB theory, Cl undergoes  $sp^3d$  hybridization with 3 bonding and 2 lone pair orbitals arranged trigonal bipyramidally, giving ClF<sub>3</sub> a **T-shaped** geometry with highly polar Cl–F  $\sigma$  bonds.

**Q23.** Draw a potential energy diagram for a molecule such as Cl<sub>2</sub>. Indicate the bond length (194 pm) and the bond dissociation energy (240 kJ/mol). (5 marks)

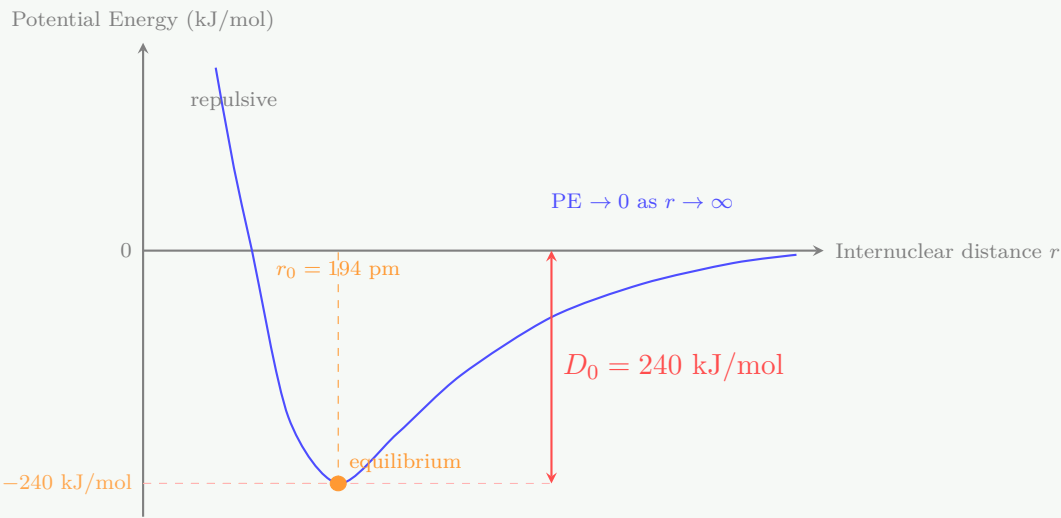
[CHEM 113 | 2017–18]

Answer

Potential Energy Diagram for Cl<sub>2</sub>

The potential energy (PE) of two Cl atoms varies with internuclear distance  $r$  as follows:

- At **large  $r$** : atoms are non-interacting; PE = 0 (reference).
- As  $r$  decreases: attractive forces dominate; PE **decreases**.
- At **equilibrium bond length** ( $r_0 = 194 \text{ pm}$ ): PE is at **minimum** (most stable).
- At **small  $r$** : nuclear repulsion dominates; PE rises steeply.
- The **bond dissociation energy** (240 kJ/mol) is the energy needed to go from the minimum back to PE = 0.



| Parameter                | Value                                     |
|--------------------------|-------------------------------------------|
| Bond length ( $r_0$ )    | 194 pm                                    |
| Bond dissociation energy | 240 kJ/mol                                |
| PE at equilibrium        | -240 kJ/mol (relative to separated atoms) |

**Q24.** Why does sodium chloride normally exist as a crystal rather than as a molecule composed of one cation and one anion? **(5 marks)**  
[CHEM 113 | 2017–18]

### Answer

**Why NaCl exists as a crystal rather than as ion pairs**

**Crystal structure of NaCl:**

- In the solid state, each  $\text{Na}^+$  ion is surrounded by **6  $\text{Cl}^-$  ions** and each  $\text{Cl}^-$  is surrounded by **6  $\text{Na}^+$  ions** (face-centered cubic structure).
- The **coordination number** is 6 for both ions.
- This gives a **3D extended ionic lattice** rather than discrete NaCl molecules.

**Energetic reason — Lattice Energy:**

- Ionic bonds are **non-directional** and **long-range** — each ion attracts all oppositely charged ions around it.
- In the crystal, **every ion interacts with multiple neighbors** simultaneously, releasing far more energy than a single ion pair would.
- The **lattice energy** of NaCl =  $-788 \text{ kJ/mol}$ , much greater in magnitude than the energy of a single  $\text{Na}^+\text{--Cl}^-$  ion pair.

**Madelung constant:**

The total electrostatic energy per ion pair in a crystal is:

$$E = -\frac{Mz^+z^-e^2}{4\pi\epsilon_0r}$$

where  $M$  is the **Madelung constant** ( $M = 1.748$  for NaCl). This factor accounts for all attractive and repulsive interactions in the lattice, always giving a **net stabilization greater** than a single ion pair.

**Comparison:**

| Form                 | Interactions             | Stability        |
|----------------------|--------------------------|------------------|
| Single NaCl ion pair | 1 attractive pair        | Low              |
| NaCl crystal lattice | Millions of interactions | <b>Very high</b> |

**Conclusion:** The crystal form of NaCl is energetically far more stable than discrete ion pairs because the **large lattice energy** released by the 3D arrangement of millions of ions greatly exceeds the energy of a single cation-anion interaction.

**Q25.** Arrange the following molecular ions in order of increasing bond dissociation energy: NO,  $\text{Be}_2^+$ ,  $\text{N}_2^+$ . **(6 marks)** [CHEM 113 | 2017–18]

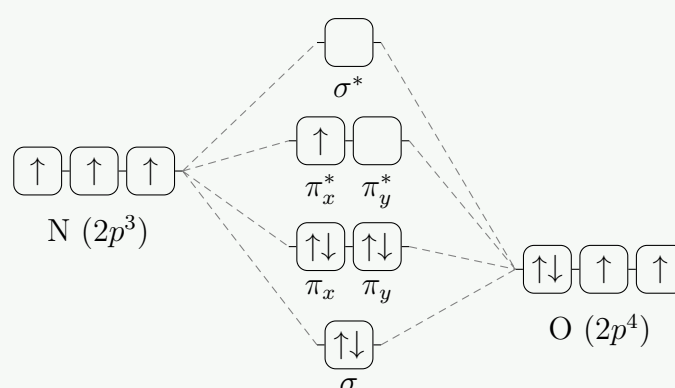
### Answer

**Bond dissociation energy order:** NO,  $\text{Be}_2^+$ ,  $\text{N}_2^+$

**Bond order calculation using MOT:**

$$\text{Bond order} = \frac{\text{bonding } e^- - \text{antibonding } e^-}{2}$$

**(i) NO (11 electrons):**

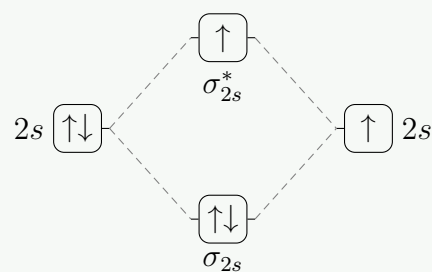


**NO Molecule**

MO config:  $(\sigma_{1s})^2(\sigma_{1s}^*)^2(\sigma_{2s})^2(\sigma_{2s}^*)^2(\sigma_{2p})^2(\pi_{2p})^4(\pi_{2p}^*)^1$

$$\text{Bond order} = \frac{8 - 3}{2} = 2.5$$

(ii)  $\text{Be}_2^+$  (3 electrons):

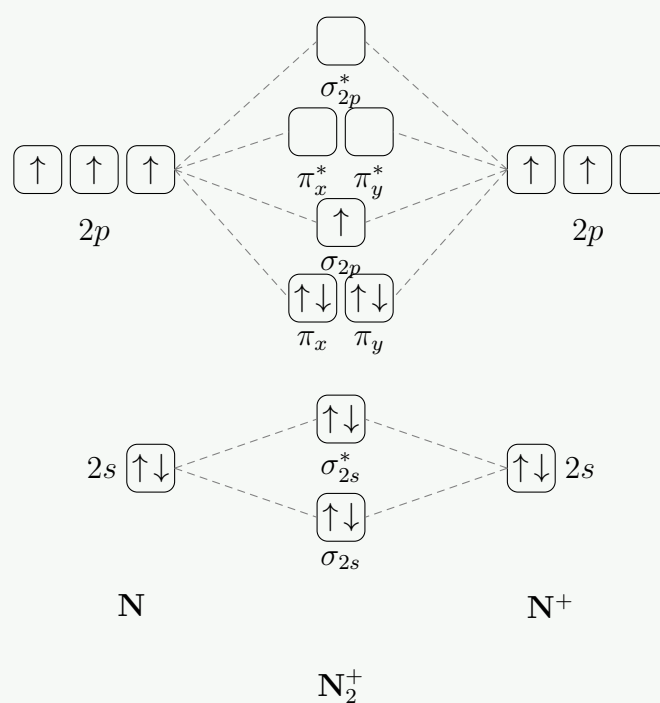


$\text{Be}_2^+$  Ion

MO config:  $(\sigma_{1s})^2(\sigma_{1s}^*)^1$

$$\text{Bond order} = \frac{2 - 1}{2} = 0.5$$

(iii)  $\text{N}_2^+$  (13 electrons):



MO config:  $(\sigma_{1s})^2(\sigma_{1s}^*)^2(\sigma_{2s})^2(\sigma_{2s}^*)^2(\pi_{2p})^4(\sigma_{2p})^1$

$$\text{Bond order} = \frac{9 - 4}{2} = 2.5$$

Wait — rechecking  $\text{N}_2^+$ : Bonding  $e^- = \sigma_{1s}(2) + \sigma_{2s}(2) + \pi_{2p}(4) + \sigma_{2p}(1) = 9$ ; Antibonding  $e^- = \sigma_{1s}^*(2) + \sigma_{2s}^*(2) = 4$

$$\text{Bond order of } \text{N}_2^+ = \frac{9 - 4}{2} = 2.5$$

**Comparing NO vs  $\text{N}_2^+$ :**

Both have bond order 2.5, but  $\text{N}_2^+$  has its antibonding electron in  $\pi^*$  while NO also has its extra electron in  $\pi^*$ . However, N–N bond is intrinsically stronger than N–O due to better orbital energy matching in homonuclear species.

| Species         | Electrons | Bond order | Bond dissociation energy |
|-----------------|-----------|------------|--------------------------|
| $\text{Be}_2^+$ | 3         | 0.5        | Lowest                   |
| NO              | 11        | 2.5        | Intermediate             |
| $\text{N}_2^+$  | 13        | 2.5        | Highest                  |

**Increasing order of bond dissociation energy:**

$$\boxed{\text{Be}_2^+ < \text{NO} < \text{N}_2^+}$$

Higher bond order  $\Rightarrow$  stronger bond  $\Rightarrow$  higher dissociation energy.  $\text{Be}_2^+$  has the lowest bond order (0.5) so weakest bond.  $\text{N}_2^+$  has stronger N–N bond than N–O in NO.

**Q26.** Explain what energy terms are involved in the formation of an ionic solid from atoms. In what way should these terms change (become larger or smaller) to give the lowest energy possible for the solid? **(10 marks)** [CHEM 113 | 2017–18]

### Answer

#### Energy terms in formation of an ionic solid from atoms

The formation of an ionic solid involves the following energy terms:

**1. Sublimation energy ( $\Delta H_{sub}$ ):**

Energy required to convert the metal from solid to gaseous atoms. Always **positive** (endothermic).

*To minimize energy: should be small (small  $\Delta H_{sub}$ ).*

**2. Bond dissociation energy ( $\frac{1}{2}\Delta H_{diss}$ ):**

Energy required to dissociate diatomic nonmetal (e.g.,  $\text{Cl}_2$ ,  $\text{O}_2$ ) into gaseous atoms. Always **positive**.

*To minimize energy: should be small.*

**3. Ionization energy ( $IE$ ):**

Energy required to remove electrons from gaseous metal atoms to form cations. Always **positive** (endothermic).

*To minimize energy: metal should have low  $IE$ .*

**4. Electron affinity ( $EA$ ):**

Energy released when gaseous nonmetal atoms gain electrons to form anions. Usually **negative** (exothermic).

*To minimize energy: nonmetal should have large (negative)  $EA$ .*

**5. Lattice energy ( $-U$ ):**

Energy released when gaseous ions come together to form the ionic crystal. Always **negative** (exothermic) and very large in magnitude.

*To minimize energy: lattice energy should be **as large as possible** (most important term).*

**Overall by Hess's Law:**

$$\Delta H_f^\circ = \Delta H_{sub} + \frac{1}{2}\Delta H_{diss} + IE + EA - U$$

| Energy term        | Sign       | For lowest energy:                |
|--------------------|------------|-----------------------------------|
| Sublimation energy | + (input)  | Should be <b>small</b>            |
| Bond dissociation  | + (input)  | Should be <b>small</b>            |
| Ionization energy  | + (input)  | Should be <b>small</b>            |
| Electron affinity  | – (output) | Should be <b>large</b> (negative) |
| Lattice energy     | – (output) | Should be <b>very large</b>       |

**Conclusion:** The **lattice energy** is the most critical factor. A large lattice energy (achieved with small, highly charged ions) more than compensates for all endothermic steps, making the ionic solid thermodynamically stable.

**Q27.** Describe the bonding in  $\text{H}_3\text{CCN}$  using the Valence Bond Theory concept. Draw and label one or more pictures to show the types of orbitals used to form the various  $\sigma$  and  $\pi$  bonds. **(10 marks)** [CHEM 113 | 2017–18]

### Answer

#### Bonding in $\text{H}_3\text{CCN}$ (Acetonitrile) using VBT

##### Structure and hybridization:

$\text{H}_3\text{C}-\text{C}\equiv\text{N}$  has two carbon atoms with different hybridizations:

- $\text{C}_1$  (methyl carbon,  $-\text{CH}_3$ ):  $sp^3$  hybridized
- $\text{C}_2$  (nitrile carbon,  $-\text{C}\equiv\text{N}$ ):  $sp$  hybridized
- $\text{N}$  (nitrogen):  $sp$  hybridized

##### Bond descriptions:

**1.  $\text{C}_1-\text{H}$  bonds (3 bonds):**

Overlap of  $sp^3$  hybrid orbital of  $\text{C}_1$  with  $1s$  orbital of  $\text{H}$ . Three equivalent  $\sigma$  bonds, bond angle  $\approx 109.5^\circ$ .

**2.  $\text{C}_1-\text{C}_2$  bond:**

Overlap of  $sp^3$  hybrid orbital of  $\text{C}_1$  with  $sp$  hybrid orbital of  $\text{C}_2$ . One  $\sigma$  bond along internuclear axis.

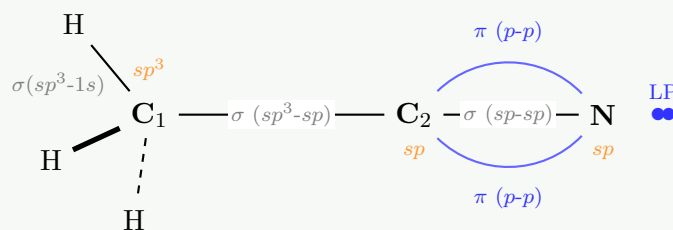
**3.  $\text{C}_2\equiv\text{N}$  bond (triple bond):**

- One  $\sigma$  bond: head-on overlap of  $sp$  ( $\text{C}_2$ ) with  $sp$  ( $\text{N}$ ).

- Two  $\pi$  bonds: sideways overlap of unhybridized  $p_x$  and  $p_y$  orbitals of  $C_2$  and N.

#### 4. Lone pair on N:

Occupies the remaining  $sp$  hybrid orbital of N, pointing away from the  $C\equiv N$  axis.



| Bond         | Type       | Orbitals               | Hybrid           |
|--------------|------------|------------------------|------------------|
| $C_1-H$ (x3) | $\sigma$   | $sp^3 + 1s$            | $sp^3$           |
| $C_1-C_2$    | $\sigma$   | $sp^3 + sp$            | $sp^3, sp$       |
| $C_2-N$      | $\sigma$   | $sp + sp$              | $sp$             |
| $C_2-N$      | $\pi$ (x2) | $p_x + p_x, p_y + p_y$ | unhybridized $p$ |
| N            | lone pair  | $sp$                   | $sp$             |

**Q28.** Dinitrogen ( $N_2$ ) and carbon monoxide (CO) molecules are quite similar when it comes to their bonding. (i) Sketch and fill up a molecular orbital diagram for CO. (ii) Provide the molecular orbital electron configuration of CO. (iii) Calculate the bond order of CO. Should it be paramagnetic or diamagnetic? **(13 marks)** [CHEM 113 | 2017–18]

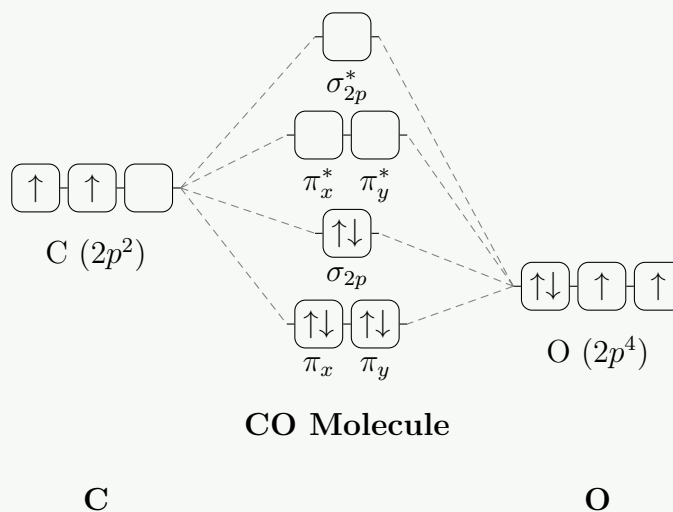
#### Answer

##### MO diagram, configuration and bond order of CO

CO has the same number of electrons as  $N_2$  (14 electrons) and is called an **isoelectronic** species. The MO energy order for CO follows the  $N_2$  pattern (2s–2p mixing applies):

$$\sigma_{1s} < \sigma_{1s}^* < \sigma_{2s} < \sigma_{2s}^* \pi_{2p_x} = \pi_{2p_y} < \sigma_{2p} \pi_{2p_x}^* = \pi_{2p_y}^* < \sigma_{2p}^*$$

##### (i) MO diagram of CO:



##### (ii) MO electron configuration of CO:

$$(\sigma_{1s})^2 (\sigma_{1s}^*)^2 (\sigma_{2s})^2 (\sigma_{2s}^*)^2 (\pi_{2p_x})^2 (\pi_{2p_y})^2 (\sigma_{2p})^2$$

##### (iii) Bond order and magnetic property:

$$\text{Bond order} = \frac{10 - 4}{2} = 3$$

(Bonding  $e^- = \sigma_{1s}(2) + \sigma_{2s}(2) + \pi_{2p}(4) + \sigma_{2p}(2) = 10$ ; Antibonding  $e^- = \sigma_{1s}^*(2) + \sigma_{2s}^*(2) = 4$ )

- Bond order = 3  $\Rightarrow$  **triple bond** ( $C\equiv O$ ), consistent with VBT.
- All electrons are **paired**  $\Rightarrow$  CO is **diamagnetic**.

| Property              | Value              |
|-----------------------|--------------------|
| Total electrons       | 14                 |
| Bonding electrons     | 10                 |
| Antibonding electrons | 4                  |
| Bond order            | 3                  |
| Unpaired electrons    | 0                  |
| Magnetic nature       | <b>Diamagnetic</b> |

**Q29.** Show the change of potential energy of the system when two atoms are getting closer to form a covalent bond. How the energy diagram can be used for the calculation of bond length and bond dissociation energy? **(10 marks)** [CHEM 113 | 2016–17]

### Answer

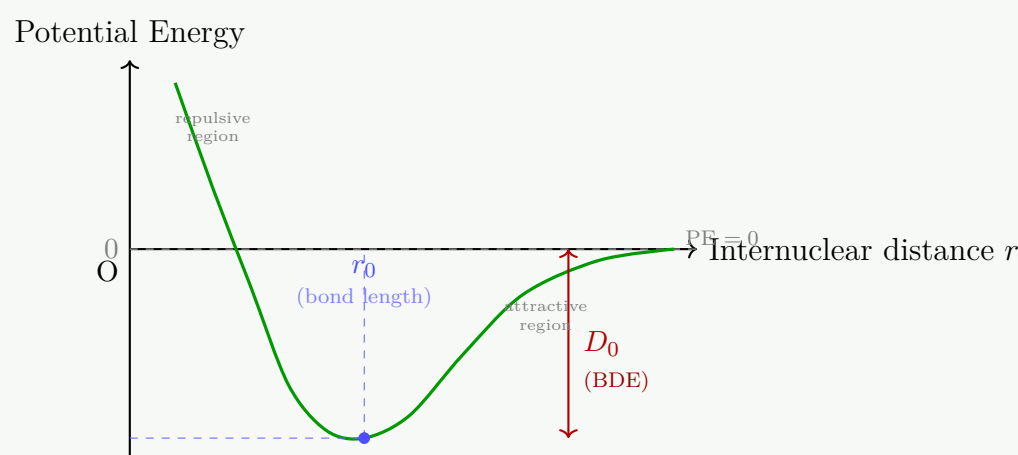
#### Potential Energy Diagram for Covalent Bond Formation, Bond Length & Bond Dissociation Energy **(10 marks)**

##### Part 1: Potential Energy Changes as Two Atoms Approach **(3 marks)**

Consider two hydrogen atoms approaching each other from a large distance. Three distinct regions exist:

- (i) **Large separation** ( $r \rightarrow \infty$ ): The atoms do not interact. Potential energy (PE) is defined as **zero** at infinite separation — the reference point.
  - (ii) **Intermediate separation (attractive region)**: As the atoms approach, the electron of each atom is attracted to the nucleus of the other. This **electron–nucleus attraction** dominates over electron–electron and nucleus–nucleus repulsions. The system *loses* potential energy — the PE curve **decreases** (becomes negative). The system becomes more stable.
  - (iii) **Very short separation (repulsive region)**: When atoms are too close, **nucleus–nucleus repulsion** and **electron–electron repulsion** dominate strongly. The PE curve rises **steeply** back up. The system becomes destabilized.
- The result is a **potential energy well** — a minimum at an optimal internuclear distance.

##### Part 2: The Potential Energy Diagram **(3 marks)**



The curve shows:

- PE decreases as atoms approach (attractive region)
- PE reaches a **minimum** at  $r = r_0$
- PE rises steeply at very short distances (repulsive region)
- $PE \rightarrow 0$  as  $r \rightarrow \infty$

##### Part 3: Calculating Bond Length from the Diagram **(2 marks)**

The **bond length**  $r_0$  is defined as the internuclear distance at the **minimum of the potential energy curve**. At this point:

- The attractive and repulsive forces are exactly balanced.
- $\frac{dV}{dr} = 0$  — the slope of the PE curve is zero.
- The system is at its most stable configuration.
- For  $H_2$ :  $r_0 = 0.74 \text{ \AA}$



From the diagram,  $r_0$  is read directly as the  $x$ -coordinate of the minimum point:

$$r_0 = \text{internuclear distance at } V_{\min}$$

#### Part 4: Calculating Bond Dissociation Energy from the Diagram

(2 marks)

The **bond dissociation energy** (BDE)  $D_0$  is the energy required to *completely separate* the two bonded atoms from their equilibrium distance  $r_0$  to infinity ( $r \rightarrow \infty$ ):

$$D_0 = V(r \rightarrow \infty) - V(r_0) = 0 - V_{\min} = -V_{\min}$$

From the diagram,  $D_0$  is the *vertical distance* from the minimum of the PE curve up to the zero-energy asymptote:

- A **deeper well**  $\Rightarrow$  larger  $D_0 \Rightarrow$  stronger bond.
- A **shallow well**  $\Rightarrow$  smaller  $D_0 \Rightarrow$  weaker bond.
- For  $\text{H}_2$ :  $D_0 = 436 \text{ kJ/mol}$

$$\text{Bond length} = r_0 \text{ (x-coordinate of minimum)} \quad \text{BDE} = D_0 = |V_{\min}| \text{ (depth of well)}$$

**Conclusion:** As two atoms approach, the PE first decreases due to attractive interactions, reaches a minimum at the equilibrium bond length  $r_0$ , then rises steeply due to repulsion. The **bond length** is the x-coordinate of this minimum, and the **bond dissociation energy** is the depth of the potential energy well — both read directly from the PE diagram.

**Q30.** Calculate the bond order of the following species:  $\text{F}_2^{2-}$ ,  $\text{F}_2^-$ ,  $\text{F}_2^+$ ,  $\text{F}_2^{2+}$ . List the species in order of increasing bond energy and increasing bond length. (9 marks) [CHEM 113 | 2016–17]

#### Answer

**Bond order of  $\text{F}_2^{2-}$ ,  $\text{F}_2^-$ ,  $\text{F}_2^+$ ,  $\text{F}_2^{2+}$**

$\text{F}_2$  has **18 electrons**. MO configuration of  $\text{F}_2$ :

$$(\sigma_{1s})^2(\sigma_{1s}^*)^2(\sigma_{2s})^2(\sigma_{2s}^*)^2(\sigma_{2p})^2(\pi_{2p})^4(\pi_{2p}^*)^4$$

$$\text{Bond order of } \text{F}_2 = \frac{10 - 8}{2} = 1$$

For ionic species, electrons are added to or removed from  $\pi_{2p}^*$  (HOMO/LUMO region):

| Species           | Total $e^-$ | Bonding $e^-$ | Antibonding $e^-$ | Bond order               |
|-------------------|-------------|---------------|-------------------|--------------------------|
| $\text{F}_2^{2+}$ | 16          | 10            | 6                 | $\frac{10 - 6}{2} = 2$   |
| $\text{F}_2^+$    | 17          | 10            | 7                 | $\frac{10 - 7}{2} = 1.5$ |
| $\text{F}_2$      | 18          | 10            | 8                 | $\frac{10 - 8}{2} = 1$   |
| $\text{F}_2^-$    | 19          | 10            | 9                 | $\frac{10 - 9}{2} = 0.5$ |
| $\text{F}_2^{2-}$ | 20          | 10            | 10                | $\frac{10 - 10}{2} = 0$  |

**Increasing bond order:**

$$\text{F}_2^{2-} (0) < \text{F}_2^- (0.5) < \text{F}_2 (1) < \text{F}_2^+ (1.5) < \text{F}_2^{2+} (2)$$

**Increasing bond energy** (higher bond order  $\Rightarrow$  stronger bond):

$$\text{F}_2^{2-} < \text{F}_2^- < \text{F}_2 < \text{F}_2^+ < \text{F}_2^{2+}$$

**Increasing bond length** (higher bond order  $\Rightarrow$  shorter bond, so *increasing* length is reverse order):

$$\text{F}_2^{2+} < \text{F}_2^+ < \text{F}_2 < \text{F}_2^- < \text{F}_2^{2-}$$

**Note:**  $\text{F}_2^{2-}$  has bond order = 0, meaning it is **unstable** and would not exist as a stable species.

**Q31.** Use molecular orbital theory to describe the bonding in the following. For each one, find the bond order and decide whether it is stable. Is the substance diamagnetic or paramagnetic? (i)  $\text{Be}_2^+$ ; (ii)  $\text{Ne}_2$ ; (iii)  $\text{B}_2^-$ . (9 marks) [CHEM 113 | 2015–16]

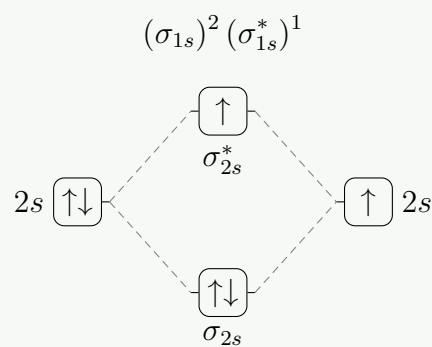
## Answer

**MOT analysis of  $\text{Be}_2^+$ ,  $\text{Ne}_2$ ,  $\text{B}_2^-$** 

MO filling order (Period 2):

$$\sigma_{1s} < \sigma_{1s}^* < \sigma_{2s} < \sigma_{2s}^* \pi_{2p_x} = \pi_{2p_y} < \sigma_{2p} \pi_{2p_x}^* = \pi_{2p_y}^* < \sigma_{2p}^*$$

$$\text{Bond order} = \frac{\text{bonding } e^- - \text{antibonding } e^-}{2}$$

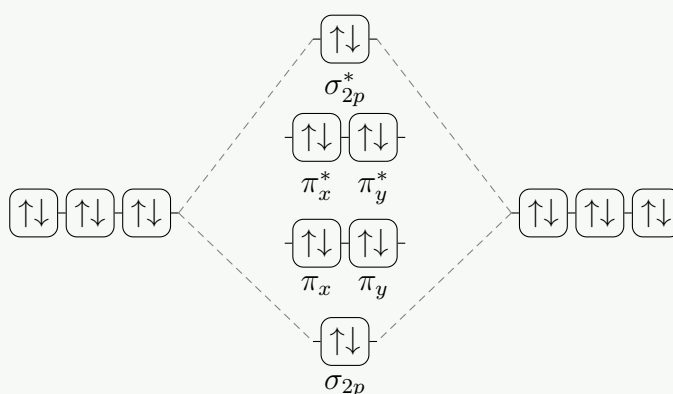
**(i)  $\text{Be}_2^+$  (3 electrons):**

 **$\text{Be}_2^+$  Ion**

$$\text{Bond order} = \frac{2 - 1}{2} = 0.5$$

- $\text{BO} = 0.5 > 0 \Rightarrow$  **stable** (weakly)
- 1 unpaired electron  $\Rightarrow$  **paramagnetic**

**(ii)  $\text{Ne}_2$  (20 electrons):**

$$(\sigma_{1s})^2 (\sigma_{1s}^*)^2 (\sigma_{2s})^2 (\sigma_{2s}^*)^2 (\sigma_{2p})^2 (\pi_{2p})^4 (\pi_{2p}^*)^4 (\sigma_{2p}^*)^2$$

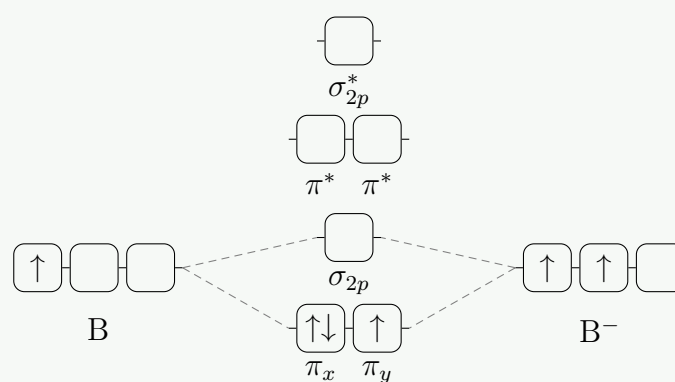

 **$\text{Ne}_2$  Molecule (Bond Order = 0)**

$$\text{Bond order} = \frac{10 - 10}{2} = 0$$

- $\text{BO} = 0 \Rightarrow$  **unstable**, does not exist
- All electrons paired  $\Rightarrow$  **diamagnetic**

**(iii)  $\text{B}_2^-$  (11 electrons):**

$$(\sigma_{1s})^2 (\sigma_{1s}^*)^2 (\sigma_{2s})^2 (\sigma_{2s}^*)^2 (\pi_{2p_x})^2 (\pi_{2p_y})^1$$



### B<sub>2</sub><sup>-</sup> Ion

$$\text{Bond order} = \frac{7 - 4}{2} = 1.5$$

- $\text{BO} = 1.5 > 0 \Rightarrow$  **stable**
- 1 unpaired electron in  $\pi_{2p_y} \Rightarrow$  **paramagnetic**

| Species                      | e <sup>-</sup> | Bonding | Antibonding | BO  | Properties            |
|------------------------------|----------------|---------|-------------|-----|-----------------------|
| Be <sub>2</sub> <sup>+</sup> | 3              | 2       | 1           | 0.5 | Stable, Paramagnetic  |
| Ne <sub>2</sub>              | 20             | 10      | 10          | 0   | Unstable, Diamagnetic |
| B <sub>2</sub> <sup>-</sup>  | 11             | 7       | 4           | 1.5 | Stable, Paramagnetic  |

**Q32.** What is a polar covalent bond? Draw Lewis structure of two compounds that contain one or more polar covalent bonds. **(5 marks)**  
[CHEM 113 | 2014–15]

### Answer

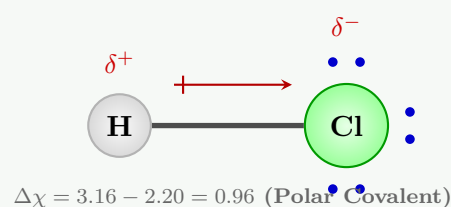
#### Polar Covalent Bond:

A **polar covalent bond** is a covalent bond in which the electron pair is **unequally shared** between two atoms due to a difference in **electronegativity**. The more electronegative atom attracts the bonding electrons more strongly, acquiring a partial negative charge ( $\delta^-$ ), while the less electronegative atom acquires a partial positive charge ( $\delta^+$ ).

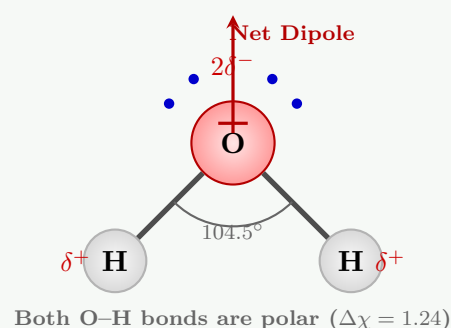
- If electronegativity difference  $\Delta\chi = 0 \Rightarrow$  pure covalent (nonpolar)
- If  $0 < \Delta\chi < 1.7 \Rightarrow$  **polar covalent**
- If  $\Delta\chi > 1.7 \Rightarrow$  ionic bond

#### Lewis Structures of two polar covalent compounds:

(i) HCl (Hydrogen Chloride)



(ii) H<sub>2</sub>O (Water)



**Q33.** What is lattice energy and what role does it play in the stability of ionic compounds? Explain how the lattice energy of KCl can be determined using the Born-Haber cycle. **(12 marks)**  
[CHEM 113 | 2014–15]

## Answer

## Lattice Energy and the Born-Haber Cycle for KCl

## What is Lattice Energy?

Lattice energy ( $U$ ) is the energy **released** when one mole of an ionic solid is formed from its gaseous ions, or equivalently the energy **required** to completely separate the ionic crystal into gaseous ions.

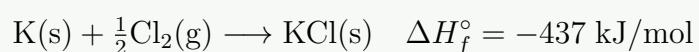


## Role in stability of ionic compounds:

- Higher lattice energy  $\Rightarrow$  stronger electrostatic attraction  $\Rightarrow$  more stable ionic compound.
- Lattice energy increases with: (i) higher ionic charges, (ii) smaller ionic radii (closer approach of ions).
- The large negative lattice energy compensates for all the endothermic steps (ionization, sublimation etc.) to give a net exothermic formation reaction.

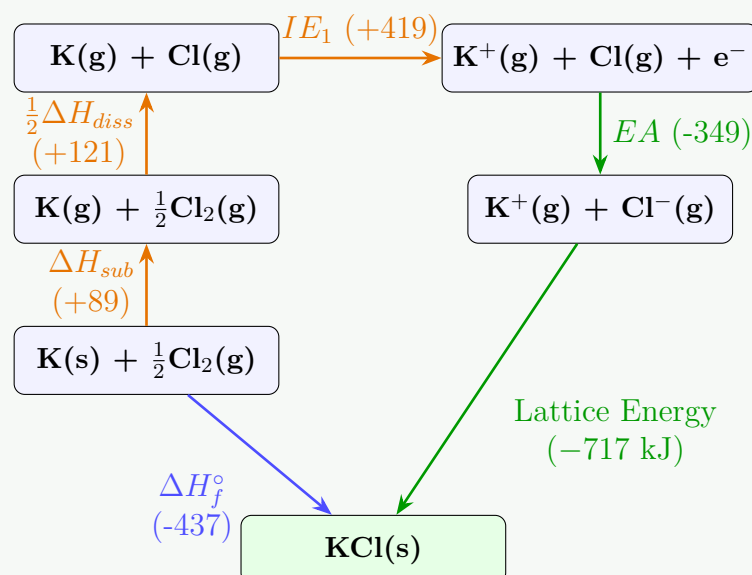
## Born-Haber Cycle for KCl:

Overall reaction:



| Step                                                                                                     | Process                                                                         | Symbol                       | $\Delta H$ (kJ/mol)     |
|----------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|------------------------------|-------------------------|
| 1                                                                                                        | $\text{K}(\text{s}) \rightarrow \text{K}(\text{g})$                             | $\Delta H_{sub}$             | +89                     |
| 2                                                                                                        | $\frac{1}{2}\text{Cl}_2(\text{g}) \rightarrow \text{Cl}(\text{g})$              | $\frac{1}{2}\Delta H_{diss}$ | +121                    |
| 3                                                                                                        | $\text{K}(\text{g}) \rightarrow \text{K}^+(\text{g}) + \text{e}^-$              | $IE_1$                       | +419                    |
| 4                                                                                                        | $\text{Cl}(\text{g}) + \text{e}^- \rightarrow \text{Cl}^-(\text{g})$            | $EA$                         | -349                    |
| 5                                                                                                        | $\text{K}^+(\text{g}) + \text{Cl}^-(\text{g}) \rightarrow \text{KCl}(\text{s})$ | $-U$                         | -717                    |
| <b>Overall:</b> $\text{K}(\text{s}) + \frac{1}{2}\text{Cl}_2(\text{g}) \rightarrow \text{KCl}(\text{s})$ |                                                                                 |                              | $\Delta H_f^\circ$ -437 |

## Born-Haber cycle diagram for KCl:



## Finding lattice energy by Hess's Law:

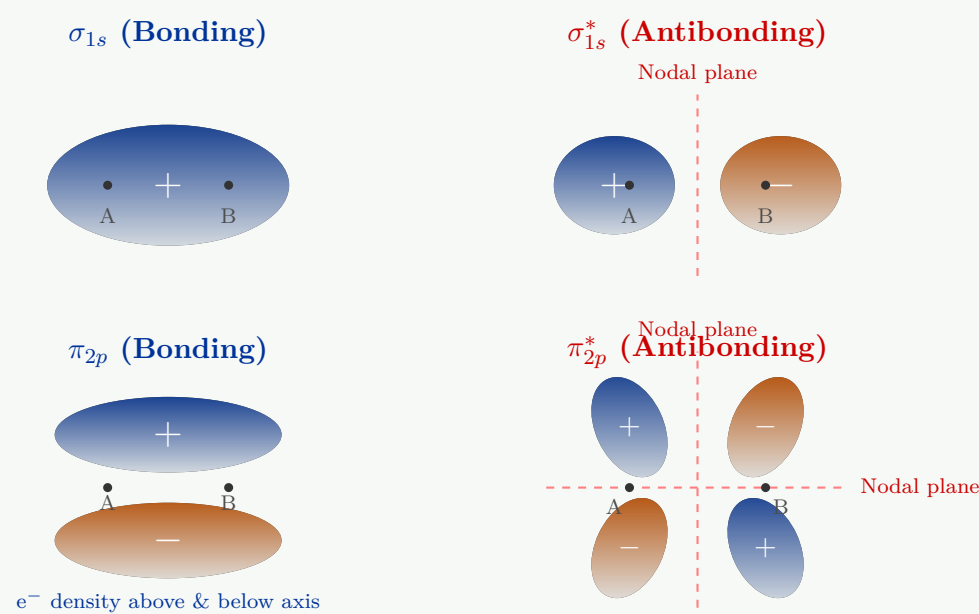
$$U = \Delta H_{sub} + \frac{1}{2}\Delta H_{diss} + IE_1 + EA - \Delta H_f^\circ = 89 + 121 + 419 + (-349) - (-437) = 717 \text{ kJ/mol}$$

**Conclusion:** Lattice energy is the dominant stabilizing term in ionic compound formation. The Born-Haber cycle, based on Hess's Law, provides an indirect method to determine this quantity experimentally.

**Q34.** (i) Sketch the shapes of the following molecular orbitals:  $\sigma_{1s}$ ,  $\sigma_{1s}^*$ ,  $\pi_{2p}$  and  $\pi_{2p}^*$ . (ii) Describe the bonding scheme in the  $\text{C}_2^{2-}$  ion in terms of molecular orbital theory. Compare the bond order in  $\text{C}_2^{2-}$  with that in  $\text{C}_2$ . (12 marks) [CHEM 113 | 2014–15]

## Answer

(i) Shapes of  $\sigma_{1s}$ ,  $\sigma_{1s}^*$ ,  $\pi_{2p}$ ,  $\pi_{2p}^*$  MOs



(ii) Bonding in  $\text{C}_2^{2-}$  and comparison with  $\text{C}_2$

$\text{C}_2$  (12 electrons):

MO config:  $(\sigma_{1s})^2(\sigma_{1s}^*)^2(\sigma_{2s})^2(\sigma_{2s}^*)^2(\pi_{2p_x})^2(\pi_{2p_y})^2$

$$\text{Bond order of } \text{C}_2 = \frac{8 - 4}{2} = 2$$

$\text{C}_2^{2-}$  (14 electrons, 2 extra):

MO config:  $(\sigma_{1s})^2(\sigma_{1s}^*)^2(\sigma_{2s})^2(\sigma_{2s}^*)^2(\pi_{2p_x})^2(\pi_{2p_y})^2(\sigma_{2p_z})^2$

$$\text{Bond order of } \text{C}_2^{2-} = \frac{10 - 4}{2} = 3$$

| Species           | Electrons | Bond order | Bond type     | Magnetic    |
|-------------------|-----------|------------|---------------|-------------|
| $\text{C}_2$      | 12        | 2          | Double        | Diamagnetic |
| $\text{C}_2^{2-}$ | 14        | 3          | <b>Triple</b> | Diamagnetic |

**Conclusion:** Adding 2 electrons to  $\text{C}_2$  fills the  $\sigma_{2p_z}$  bonding MO, increasing bond order from 2 to 3.  $\text{C}_2^{2-}$  has a **triple bond**, making it stronger and shorter than  $\text{C}_2$ .

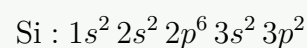
**Q35.** An atom of a certain element has 14 electrons. (i) What is the ground-state configuration of the element? (ii) Are the atoms of this element diamagnetic or paramagnetic? (4 marks) [CHEM 113 | 2014–15]

### Answer

**Element with 14 electrons**

**(i) Ground-state electron configuration:**

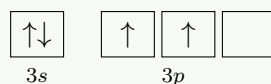
The element with 14 electrons is **Silicon (Si)**, atomic number  $Z = 14$ .



Or in noble gas notation:



The two  $3p$  electrons occupy **separate**  $3p$  orbitals with parallel spins (Hund's rule):



**(ii) Diamagnetic or paramagnetic?**

- Si has **2 unpaired electrons** in the  $3p$  subshell ( $3p_x^1 3p_y^1$ ).
- A substance is **paramagnetic** if it has one or more unpaired electrons (attracted to a magnetic field).
- A substance is **diamagnetic** if all electrons are paired (weakly repelled by a magnetic field).
- Since Si has 2 unpaired electrons, it is **paramagnetic**.

| Property           | Value               |
|--------------------|---------------------|
| Element            | Silicon (Si)        |
| Atomic number      | 14                  |
| Configuration      | $[Ne] 3s^2 3p^2$    |
| Unpaired electrons | 2                   |
| Magnetic nature    | <b>Paramagnetic</b> |

**Q36.** Explain why bond in BeCl<sub>2</sub> is covalent but bond in CaCl<sub>2</sub> is ionic though Be and Ca are in the same group of the periodic table. (4 marks) [CHEM 113 | 2013–14]

Answer

**Why BeCl<sub>2</sub> is covalent but CaCl<sub>2</sub> is ionic**

Though Be and Ca are both in **Group IIA**, they differ greatly in atomic size and polarizing power, which determines bond type.

**Fajans’ Rules — Key Concept:**  
A bond is **covalent** if the cation has high polarizing power (distorts the electron cloud of the anion). A bond is **ionic** if the cation has low polarizing power.

Polarizing power  $\propto \frac{\text{charge}}{\text{ionic radius}^2}$

**Comparison of Be<sup>2+</sup> and Ca<sup>2+</sup>:**

| Ion              | Charge | Ionic radius (pm) | Polarizing power | Bond type       |
|------------------|--------|-------------------|------------------|-----------------|
| Be <sup>2+</sup> | +2     | 31                | Very high        | <b>Covalent</b> |
| Ca <sup>2+</sup> | +2     | 100               | Low              | <b>Ionic</b>    |

**BeCl<sub>2</sub> — Covalent:**

- Be<sup>2+</sup> has an **extremely small ionic radius** (31 pm) and high charge density.
- It strongly **polarizes** (distorts) the electron cloud of Cl<sup>−</sup>, pulling electron density toward Be.
- This results in **significant electron sharing**  $\Rightarrow$  **covalent bond**.
- Also, Be uses *sp* hybrid orbitals to form bonds with Cl (linear geometry, 180°).

**CaCl<sub>2</sub> — Ionic:**

- Ca<sup>2+</sup> has a **much larger ionic radius** (100 pm) and lower charge density.
- It has **weak polarizing power** — cannot significantly distort the Cl<sup>−</sup> electron cloud.
- Electron transfer is complete  $\Rightarrow$  **ionic bond** (electrostatic attraction between Ca<sup>2+</sup> and Cl<sup>−</sup>).

**Additional factor — Ionization Energy:**

- Ca has **lower ionization energy** than Be  $\Rightarrow$  Ca readily loses 2 electrons to form Ca<sup>2+</sup>.
- Be has **higher ionization energy**  $\Rightarrow$  it is energetically more favorable to share electrons (covalent) than to fully transfer them.

**Conclusion:** The very small size and high polarizing power of Be<sup>2+</sup> causes covalent character in BeCl<sub>2</sub>, whereas the large, weakly polarizing Ca<sup>2+</sup> forms a purely ionic bond in CaCl<sub>2</sub>.

**Q37.** Draw molecular orbital diagram of CO<sup>+</sup>, NO<sup>−</sup>, O<sub>2</sub>. Comment on their magnetic properties and calculate their bond order. Write down their molecular electronic configurations. (18 marks) [CHEM 113 | 2013–14]

Answer

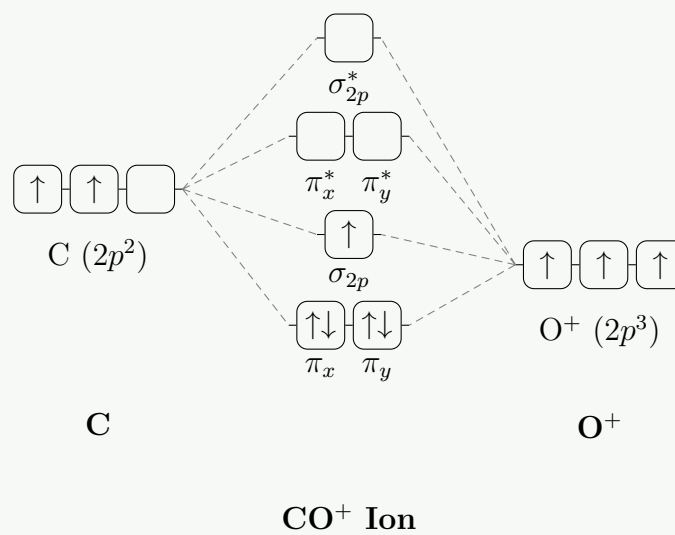
**MO diagram of CO<sup>+</sup>, NO<sup>−</sup>, O<sub>2</sub>**

MO energy order used (normal,  $Z \geq 8$  applies to O<sub>2</sub>; CO<sup>+</sup> and NO<sup>−</sup> follow N<sub>2</sub>-type order):

**(i) CO<sup>+</sup> (13 electrons):**

$$(\sigma_{1s})^2(\sigma_{1s}^*)^2(\sigma_{2s})^2(\sigma_{2s}^*)^2(\pi_{2p})^4(\sigma_{2p})^1$$

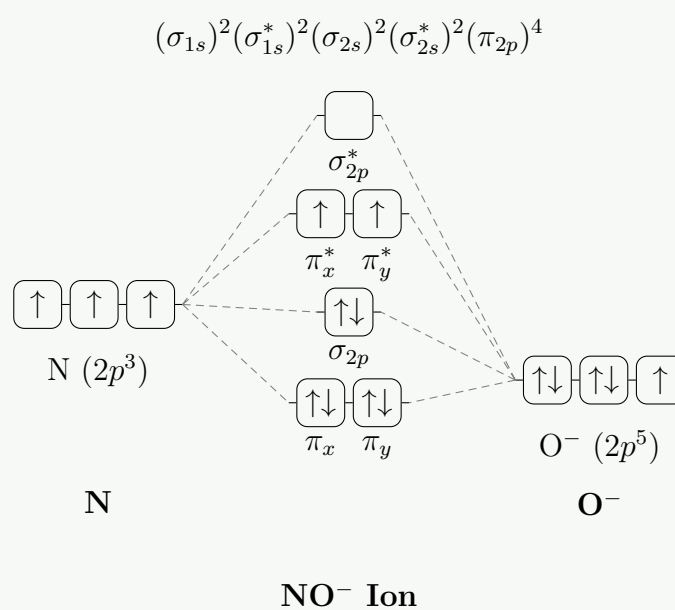




$$\text{Bond order} = \frac{9 - 4}{2} = 2.5$$

1 unpaired electron in  $\sigma_{2p} \Rightarrow$  **Paramagnetic**

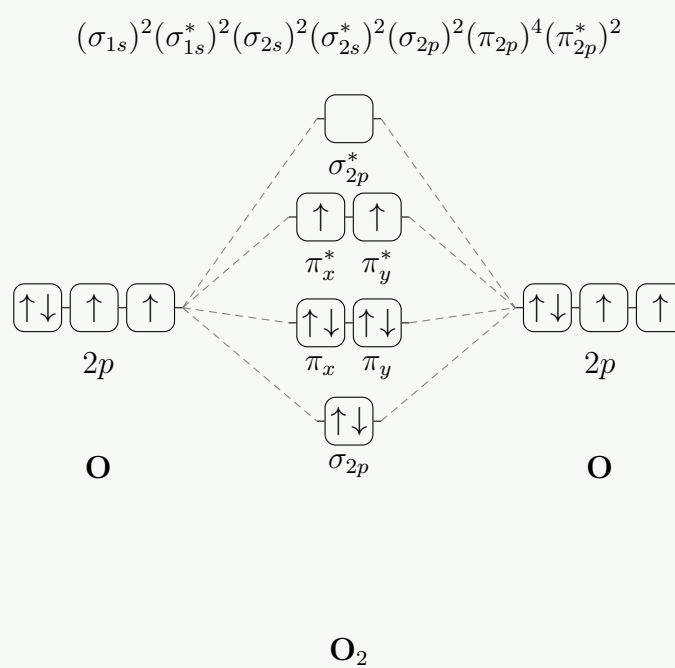
(ii)  $\text{NO}^-$  (12 electrons):



$$\text{Bond order} = \frac{8 - 4}{2} = 2$$

All electrons paired  $\Rightarrow$  **Diamagnetic**

(iii)  $\text{O}_2$  (16 electrons):



$$\text{Bond order} = \frac{10 - 6}{2} = 2$$

2 unpaired electrons in degenerate  $\pi_{2p}^* \Rightarrow$  **Paramagnetic**

| Species         | e <sup>-</sup> | Bonding | Antibonding | BO  | Magnetic     |
|-----------------|----------------|---------|-------------|-----|--------------|
| CO <sup>+</sup> | 13             | 9       | 4           | 2.5 | Paramagnetic |
| NO <sup>-</sup> | 12             | 8       | 4           | 2.0 | Diamagnetic  |
| O <sub>2</sub>  | 16             | 10      | 6           | 2.0 | Paramagnetic |

**Q38.** Discuss the molecular orbital theory of covalent bonding. Show with examples how it compares with the valence bond theory. **(13 marks)** [CHEM 113 | 2012–13]

### Answer

#### Molecular Orbital Theory vs Valence Bond Theory

##### Key features of MOT:

- AOs combine by LCAO to give MOs that are **delocalized** over the entire molecule.
- Bonding MOs are lower in energy than AOs; antibonding MOs are higher.
- Electrons fill MOs by Aufbau, Pauli, and Hund's rules.
- Bond order =  $\frac{\text{bonding e}^- - \text{antibonding e}^-}{2}$

##### Comparison with examples:

| Aspect                   | VBT                            | MOT                                 |
|--------------------------|--------------------------------|-------------------------------------|
| Electron location        | Localized between bonded atoms | Delocalized over whole molecule     |
| Bond formation           | Overlap of half-filled AOs     | LCAO gives bonding/antibonding MOs  |
| O <sub>2</sub> magnetism | Predicts diamagnetic (wrong)   | Correctly predicts paramagnetic     |
| Bond order               | Inferred from Lewis structure  | Calculated directly from MO filling |
| Resonance                | Requires resonance structures  | Automatically delocalized           |
| Antibonding              | No concept of antibonding      | Explicitly includes antibonding MOs |

##### Example 1 — H<sub>2</sub>:

VBT: overlap of two 1s AOs  $\Rightarrow$   $\sigma$  bond.

MOT:  $(\sigma_{1s})^2$ , bond order = 1. Both agree; molecule stable.

##### Example 2 — O<sub>2</sub>:

VBT: predicts all electrons paired  $\Rightarrow$  diamagnetic (experimentally **wrong**).

MOT:  $(\sigma_{1s})^2(\sigma_{1s}^*)^2(\sigma_{2s})^2(\sigma_{2s}^*)^2(\sigma_{2p})^2(\pi_{2p})^4(\pi_{2p}^*)^2 \Rightarrow$  2 unpaired electrons in degenerate  $\pi_{2p}^* \Rightarrow$  **paramagnetic** (correct).

##### Example 3 — He<sub>2</sub>:

VBT: no half-filled orbitals, no bond.

MOT:  $(\sigma_{1s})^2(\sigma_{1s}^*)^2$ , bond order = 0  $\Rightarrow$  unstable. Both agree; MOT gives quantitative reason.

**Q39.** Explain the following: (i) The paramagnetism of oxygen molecule (ii) The strength of nitrogen-nitrogen bond in N<sub>2</sub> molecule (iii) The nitrogen-oxygen bond in NO molecule is weaker than the carbon-oxygen bond in CO molecule. **(12 marks)** [CHEM 113 | 2012–13]

### Answer

#### (i) Paramagnetism of O<sub>2</sub>

- MO config of O<sub>2</sub>:  $(\sigma_{1s})^2(\sigma_{1s}^*)^2(\sigma_{2s})^2(\sigma_{2s}^*)^2(\sigma_{2p})^2(\pi_{2p})^4(\pi_{2p}^*)^2$
- The last 2 electrons enter the **degenerate**  $\pi_{2p_x}^*$  and  $\pi_{2p_y}^*$  orbitals — one in each (Hund's rule).
- These 2 electrons are **unpaired**  $\Rightarrow$  O<sub>2</sub> is **paramagnetic**.
- VBT cannot explain this; MOT predicts it correctly.
- Bond order =  $\frac{10 - 6}{2} = 2$  (double bond).

#### (ii) Strength of N–N bond in N<sub>2</sub>

- MO config of N<sub>2</sub>:  $(\sigma_{1s})^2(\sigma_{1s}^*)^2(\sigma_{2s})^2(\sigma_{2s}^*)^2(\pi_{2p})^4(\sigma_{2p})^2$
- Bond order =  $\frac{10 - 4}{2} = 3 \Rightarrow$  **triple bond**.

- N<sub>2</sub> has the strongest bond among diatomic molecules: bond energy = 945 kJ/mol.
- All 3 MOs ( $\sigma_{2p}$  and both  $\pi_{2p}$ ) are bonding and fully filled  $\Rightarrow$  maximum stabilization.
- No electrons in antibonding MOs beyond inner shells  $\Rightarrow$  very high bond order and strength.

(iii) N–O bond in NO is weaker than C–O bond in CO

- MO config of NO (11 electrons):  $(\sigma_{1s})^2(\sigma_{1s}^*)^2(\sigma_{2s})^2(\sigma_{2s}^*)^2(\pi_{2p})^4(\sigma_{2p})^2(\pi_{2p}^*)^1$

$$\text{Bond order of NO} = \frac{8 - 3}{2} = 2.5$$

- MO config of CO (14 electrons):  $(\sigma_{1s})^2(\sigma_{1s}^*)^2(\sigma_{2s})^2(\sigma_{2s}^*)^2(\pi_{2p})^4(\sigma_{2p})^2$

$$\text{Bond order of CO} = \frac{10 - 4}{2} = 3$$

- CO has bond order **3** vs NO bond order **2.5**.
- The extra electron in NO occupies an **antibonding**  $\pi_{2p}^*$  orbital, reducing bond order and weakening the bond.
- Higher bond order  $\Rightarrow$  stronger, shorter bond. Therefore **C–O bond in CO is stronger** than N–O bond in NO.

| Molecule       | Bond order | Bond energy (kJ/mol) | Magnetic     |
|----------------|------------|----------------------|--------------|
| CO             | 3          | 1072                 | Diamagnetic  |
| NO             | 2.5        | 631                  | Paramagnetic |
| O <sub>2</sub> | 2          | 498                  | Paramagnetic |
| N <sub>2</sub> | 3          | 945                  | Diamagnetic  |

**Q40.** Discuss different classes of Lewis acids and bases. Cite two examples in each class. **(10 marks)**

[CHEM 113 | 2012–13]

### Answer

#### Classes of Lewis Acids and Bases

##### Lewis Concept:

- A **Lewis acid** is an electron-pair **acceptor**.
- A **Lewis base** is an electron-pair **donor**.

##### Classes of Lewis Acids:

- Molecules with incomplete octet (< 8 electrons):** Central atom has fewer than 8 electrons and can accept a lone pair.  
*Examples:* BF<sub>3</sub>, AlCl<sub>3</sub>
- Simple cations:** Positive ions can accept lone pairs from Lewis bases to form complex ions.  
*Examples:* Fe<sup>3+</sup>, Cu<sup>2+</sup>, H<sup>+</sup>, Ag<sup>+</sup>
- Molecules with an atom that can expand its octet (> 8 electrons):** Elements in Period 3 and beyond have available *d* orbitals to accept electrons.  
*Examples:* SiF<sub>4</sub>, PCl<sub>5</sub>, SF<sub>6</sub>
- Molecules with multiple bonds (electrophilic  $\pi$  systems):** The  $\pi$  bond can accept an electron pair from a Lewis base.  
*Examples:* CO<sub>2</sub>, SO<sub>3</sub>
- Transition metal atoms/ions:** Have empty *d* orbitals available to accept lone pairs from ligands.  
*Examples:* [Fe(CN)<sub>6</sub>]<sup>3-</sup> (Fe<sup>3+</sup> as acid), [Cu(NH<sub>3</sub>)<sub>4</sub>]<sup>2+</sup> (Cu<sup>2+</sup> as acid)

##### Classes of Lewis Bases:

- Neutral molecules with lone pairs:** Molecules that donate a lone pair to a Lewis acid.  
*Examples:* NH<sub>3</sub>, H<sub>2</sub>O, ROH, R<sub>2</sub>O
- Anions and negative ions:** Negatively charged species readily donate electrons.  
*Examples:* OH<sup>-</sup>, Cl<sup>-</sup>, CN<sup>-</sup>, F<sup>-</sup>
- Molecules with  $\pi$  electrons:**  $\pi$  electron systems can donate electrons to Lewis acids.  
*Examples:* C<sub>2</sub>H<sub>4</sub> (ethylene), C<sub>6</sub>H<sub>6</sub> (benzene)
- Molecules with multiple lone pairs:** Molecules with more than one lone pair can act as Lewis bases multiple times.  
*Examples:* H<sub>2</sub>S, R<sub>2</sub>S (thioethers), PH<sub>3</sub>

## Summary Table:

| Type       | Class              | Examples                      |                               |
|------------|--------------------|-------------------------------|-------------------------------|
| Lewis Acid | Incomplete octet   | BF <sub>3</sub>               | AlCl <sub>3</sub>             |
|            | Simple cation      | Fe <sup>3+</sup>              | H <sup>+</sup>                |
|            | Expanded octet     | PCl <sub>5</sub>              | SF <sub>6</sub>               |
|            | π acceptor         | CO <sub>2</sub>               | SO <sub>3</sub>               |
| Lewis Base | Lone pair molecule | NH <sub>3</sub>               | H <sub>2</sub> O              |
|            | Anion              | OH <sup>-</sup>               | CN <sup>-</sup>               |
|            | π donor            | C <sub>2</sub> H <sub>4</sub> | C <sub>6</sub> H <sub>6</sub> |

**Q41.** Explain the significance of bond order. Can bond order be used for quantitative comparison of the strength of chemical bonds? Explain why bond order of N<sub>2</sub> is greater than that of N<sub>2</sub><sup>+</sup> but the bond order of O<sub>2</sub> is less than that of O<sub>2</sub><sup>+</sup>. **(12 marks)** [CHEM 113 | 2011–12]

## Answer

**Bond order: significance, comparison of N<sub>2</sub>/N<sub>2</sub><sup>+</sup> and O<sub>2</sub>/O<sub>2</sub><sup>+</sup>**

**Significance of Bond Order:**

- **Bond order** =  $\frac{\text{bonding } e^- - \text{antibonding } e^-}{2}$
- It represents the **number of effective bonds** between two atoms.
- Higher bond order ⇒ **stronger bond, shorter bond length, higher bond energy**.
- Bond order = 0 ⇒ molecule is **unstable** and does not exist.
- Can be fractional (e.g., 1.5, 2.5) for ionic species.

**Can bond order be used for quantitative comparison?**

Yes, but with limitations. Bond order gives a **qualitative to semi-quantitative** comparison:

- Higher bond order reliably predicts stronger bonds *for similar molecules*.
- For very different elements, bond energy also depends on electronegativity difference and orbital sizes.

**Why bond order of N<sub>2</sub> > N<sub>2</sub><sup>+</sup>:**

**N<sub>2</sub> (14 electrons):**

$$(\sigma_{1s})^2(\sigma_{1s}^*)^2(\sigma_{2s})^2(\sigma_{2s}^*)^2(\pi_{2p})^4(\sigma_{2p})^2 \Rightarrow \text{BO} = \frac{10 - 4}{2} = 3$$

**N<sub>2</sub><sup>+</sup> (13 electrons):**

$$(\sigma_{1s})^2(\sigma_{1s}^*)^2(\sigma_{2s})^2(\sigma_{2s}^*)^2(\pi_{2p})^4(\sigma_{2p})^1 \Rightarrow \text{BO} = \frac{9 - 4}{2} = 2.5$$

Removing one electron from N<sub>2</sub> removes it from the **bonding** σ<sub>2p</sub> MO ⇒ fewer bonding electrons ⇒ bond order **decreases** from 3 to 2.5.

**Why bond order of O<sub>2</sub> < O<sub>2</sub><sup>+</sup>:**

**O<sub>2</sub> (16 electrons):**

$$(\sigma_{1s})^2(\sigma_{1s}^*)^2(\sigma_{2s})^2(\sigma_{2s}^*)^2(\sigma_{2p})^2(\pi_{2p})^4(\pi_{2p}^*)^2 \Rightarrow \text{BO} = \frac{10 - 6}{2} = 2$$

**O<sub>2</sub><sup>+</sup> (15 electrons):**

$$(\sigma_{1s})^2(\sigma_{1s}^*)^2(\sigma_{2s})^2(\sigma_{2s}^*)^2(\sigma_{2p})^2(\pi_{2p})^4(\pi_{2p}^*)^1 \Rightarrow \text{BO} = \frac{10 - 5}{2} = 2.5$$

Removing one electron from O<sub>2</sub> removes it from the **antibonding** π<sub>2p</sub><sup>\*</sup> MO ⇒ fewer antibonding electrons ⇒ bond order **increases** from 2 to 2.5.

| Species                     | e <sup>-</sup> | BO  | Electron removed from | Effect       |
|-----------------------------|----------------|-----|-----------------------|--------------|
| N <sub>2</sub>              | 14             | 3   | —                     | —            |
| N <sub>2</sub> <sup>+</sup> | 13             | 2.5 | bonding MO            | BO decreases |
| O <sub>2</sub>              | 16             | 2   | —                     | —            |
| O <sub>2</sub> <sup>+</sup> | 15             | 2.5 | antibonding MO        | BO increases |

**Conclusion:** The key difference is *which* MO loses the electron. Removing from a **bonding MO** weakens the bond; removing from an **antibonding MO** strengthens it.

**Q42.** What are the main features of Molecular Orbital Theory (MOT)? Show the formation of CO molecule according to both Valence

Bond Theory (VBT) and Molecular Orbital Theory (MOT). Comment on the advantages of MOT over VBT. **(15 marks)** [CHEM 101 | 2010–11]

### Answer

#### Main features of MOT and formation of CO by VBT and MOT

##### Main Features of Molecular Orbital Theory (MOT):

- Atomic orbitals combine by **Linear Combination of Atomic Orbitals (LCAO)** to form molecular orbitals (MOs) that belong to the **entire molecule**, not individual atoms.
- Number of MOs formed = number of AOs combined.
- MOs are of two types:
  - Bonding MO**: constructive overlap, lower energy than AOs, electron density *between* nuclei.
  - Antibonding MO**: destructive overlap, higher energy than AOs, nodal plane between nuclei.
- Electrons fill MOs according to **Aufbau principle**, **Pauli exclusion principle**, and **Hund's rule**.
- Bond order** =  $\frac{\text{bonding } e^- - \text{antibonding } e^-}{2}$
- MOT correctly explains **paramagnetism** (e.g.,  $O_2$ ), **bond order**, and **spectroscopic properties**.
- MOs are delocalized over the entire molecule.

##### Formation of CO by VBT:

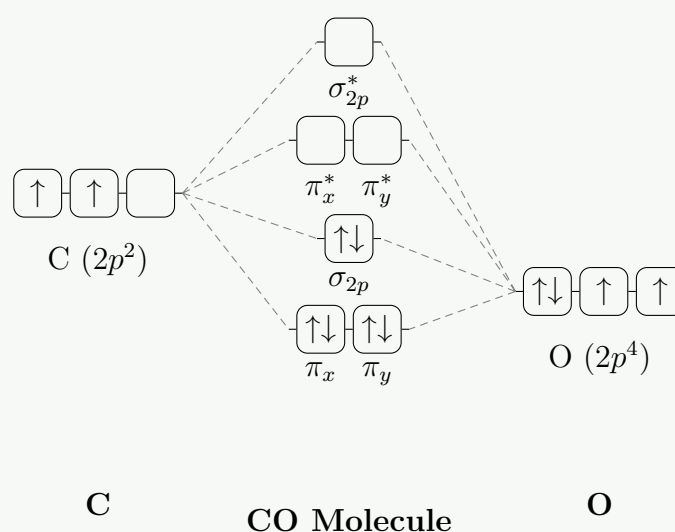
- C:  $[He] 2s^2 2p^2$ ; O:  $[He] 2s^2 2p^4$
- Both C and O are  $sp$  hybridized in CO.
- One  $\sigma$  bond: head-on overlap of  $sp$  hybrid orbitals.
- Two  $\pi$  bonds: sideways overlap of unhybridized  $p_x$  and  $p_y$  orbitals.
- VBT predicts a **triple bond** ( $C \equiv O$ ) with a lone pair on each atom.
- Limitation**: VBT cannot explain the slight polarity reversal (C is  $\delta^-$ , O is  $\delta^+$ ) or spectroscopic data of CO.

##### Formation of CO by MOT:

CO is isoelectronic with  $N_2$  (14 electrons). MO configuration:

$$(\sigma_{1s})^2(\sigma_{1s}^*)^2(\sigma_{2s})^2(\sigma_{2s}^*)^2(\pi_{2p_x})^2(\pi_{2p_y})^2(\sigma_{2p})^2$$

$$\text{Bond order} = \frac{10 - 4}{2} = 3 \Rightarrow \text{triple bond, diamagnetic}$$



##### Advantages of MOT over VBT:

| Property               | VBT            | MOT                     |
|------------------------|----------------|-------------------------|
| Paramagnetism of $O_2$ | Cannot explain | Correctly predicts      |
| Delocalized electrons  | Cannot explain | Explained naturally     |
| Bond order of ions     | Difficult      | Easily calculated       |
| Spectroscopic data     | Limited        | Accurately explained    |
| Electron density       | Localized      | Delocalized (realistic) |

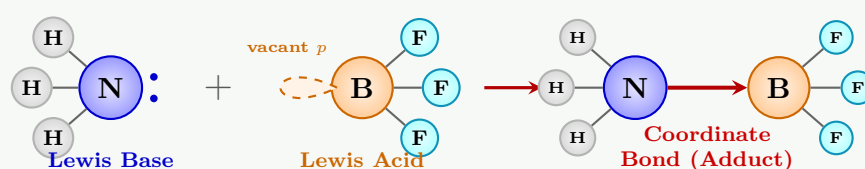
**Q43.** Write down the conditions for the formation of coordinate covalent bond. How did Werner develop the theory of coordination compounds? Discuss the main points of the theory. (12 marks) [CHEM 101 | 2010–11]

### Answer

#### Coordinate Covalent Bond and Werner's Theory

##### Conditions for formation of Coordinate Covalent Bond:

- One atom (the **donor**) must have at least one **lone pair** of electrons to donate.
- The other atom (the **acceptor**) must have a **vacant orbital** to accept the electron pair.
- The donor atom is usually a **Lewis base** (electron pair donor): N, O, S, P, or their compounds (e.g.,  $\text{NH}_3$ ,  $\text{H}_2\text{O}$ ).
- The acceptor atom is usually a **Lewis acid** (electron pair acceptor):  $\text{BF}_3$ ,  $\text{AlCl}_3$ , metal ions.
- Once formed, a coordinate bond is **indistinguishable** from an ordinary covalent bond.



##### Werner's Theory of Coordination Compounds:

Alfred Werner (1893) proposed a theory to explain the structure and properties of coordination compounds. The **main points** are:

##### 1. Primary (ionizable) valence:

Every metal has a **primary valence** (oxidation state) which is satisfied by **negative ions** only. These ions are **outside the coordination sphere** and can **ionize** in solution.

*Example:* In  $\text{CoCl}_3 \cdot 6\text{NH}_3$ , primary valence of Co = +3, satisfied by 3  $\text{Cl}^-$ .

##### 2. Secondary (coordination) valence:

Every metal also has a **secondary valence** (coordination number) which is satisfied by **neutral molecules or negative ions** called **ligands**. These are **inside the coordination sphere** and do **not ionize**.

*Example:* In  $[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$ , secondary valence = 6, satisfied by 6  $\text{NH}_3$ .

##### 3. Coordination sphere:

The metal ion and its directly bonded ligands form the **coordination sphere**, written in square brackets e.g.,  $[\text{Co}(\text{NH}_3)_6]^{3+}$ .

##### 4. Definite geometry:

Secondary valences are directed in space giving definite **geometry** to the complex:

- $\text{CN} = 6 \Rightarrow$  **Octahedral**
- $\text{CN} = 4 \Rightarrow$  **Tetrahedral** or **Square Planar**
- $\text{CN} = 2 \Rightarrow$  **Linear**

##### 5. Evidence — conductivity:

Werner verified his theory by measuring electrical conductivity of various cobalt-ammonia complexes:

| Compound                           | Complex Formula                                  | Total Ions | $\text{Cl}^-$ Precipitated (with $\text{AgNO}_3$ ) |
|------------------------------------|--------------------------------------------------|------------|----------------------------------------------------|
| $\text{CoCl}_3 \cdot 6\text{NH}_3$ | $[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$          | 4          | 3                                                  |
| $\text{CoCl}_3 \cdot 5\text{NH}_3$ | $[\text{Co}(\text{NH}_3)_5\text{Cl}]\text{Cl}_2$ | 3          | 2                                                  |
| $\text{CoCl}_3 \cdot 4\text{NH}_3$ | $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]\text{Cl}$ | 2          | 1                                                  |
| $\text{CoCl}_3 \cdot 3\text{NH}_3$ | $[\text{Co}(\text{NH}_3)_3\text{Cl}_3]$          | 0          | 0 (Neutral complex)                                |

**Conclusion:** Werner's theory correctly explained bonding, geometry, isomerism, and ionization behavior of coordination compounds, earning him the Nobel Prize in Chemistry (1913).

**Q44.** What do you mean by low spin and high spin complexes? Illustrate your answer with examples. (8 marks) [CHEM 101 | 2010–11]

### Answer

#### Low Spin and High Spin Complexes

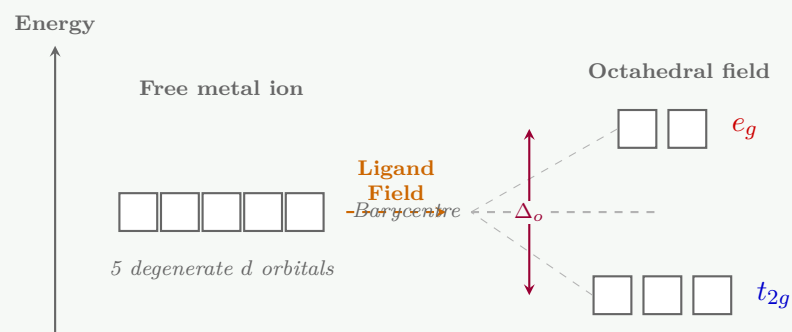
##### Crystal Field Theory background:

In an octahedral complex, the five  $d$  orbitals split into two sets:

- $t_{2g}$  (lower energy):  $d_{xy}$ ,  $d_{xz}$ ,  $d_{yz}$
- $e_g$  (higher energy):  $d_{x^2-y^2}$ ,  $d_{z^2}$



The energy gap is called **crystal field splitting energy**  $\Delta_o$  (or  $10Dq$ ).

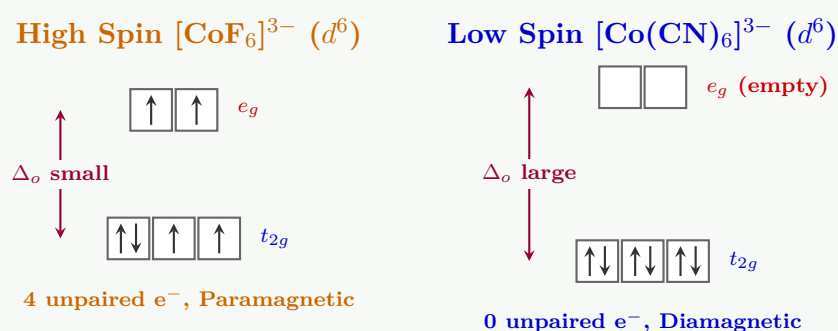


#### High Spin Complex:

- Formed with **weak field ligands** (small  $\Delta_o$ ):  $I^-$ ,  $Br^-$ ,  $Cl^-$ ,  $F^-$ ,  $OH^-$ ,  $H_2O$ .
- $\Delta_o < \text{pairing energy } P \Rightarrow$  electrons **avoid pairing** and fill  $e_g$  before pairing in  $t_{2g}$ .
- **Maximum unpaired electrons**  $\Rightarrow$  strongly paramagnetic.
- Example:  $[CoF_6]^{3-}$  ( $Co^{3+}$ ,  $d^6$ , weak field)

#### Low Spin Complex:

- Formed with **strong field ligands** (large  $\Delta_o$ ):  $CN^-$ ,  $CO$ ,  $NO_2^-$ ,  $en$ ,  $NH_3$ .
- $\Delta_o > \text{pairing energy } P \Rightarrow$  electrons **pair up** in  $t_{2g}$  before entering  $e_g$ .
- **Minimum unpaired electrons**  $\Rightarrow$  weakly paramagnetic or diamagnetic.
- Example:  $[Co(CN)_6]^{3-}$  ( $Co^{3+}$ ,  $d^6$ , strong field)



| Type      | Ligand field | $\Delta_o$ vs $P$ | Unpaired $e^-$ | Magnetic Nature |
|-----------|--------------|-------------------|----------------|-----------------|
| High spin | Weak         | $\Delta_o < P$    | Maximum        | Paramagnetic    |
| Low spin  | Strong       | $\Delta_o > P$    | Minimum        | Diamagnetic     |

**Q45.** What are paramagnetism and diamagnetism? How would you predict paramagnetic and diamagnetic properties in substances? Discuss the formation of  $N_2^-$  and  $NO$  molecules according to molecular orbital theory and comment on their magnetic properties. (12 marks) [CHEM 101 | 2009–10]

#### Answer

##### Paramagnetism, Diamagnetism, $N_2^-$ and $NO$ by MOT

##### Diamagnetism:

- All electrons are **paired**.
- Weakly **repelled** by an external magnetic field.
- Net magnetic moment = 0.
- Examples:  $H_2$ ,  $N_2$ ,  $CO$ ,  $F_2$

##### Paramagnetism:

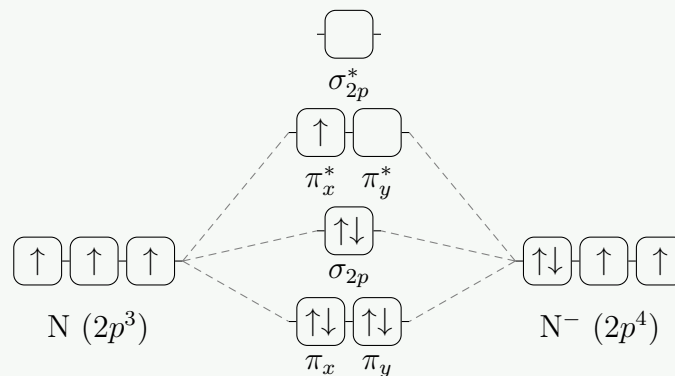
- One or more **unpaired electrons** present.
- **Attracted** to an external magnetic field.
- Net magnetic moment  $\neq 0$ .
- Examples:  $O_2$ ,  $NO$ ,  $B_2$

**Prediction rule:** Fill MOs by Aufbau + Hund's rules; if any MO has an odd electron  $\Rightarrow$  paramagnetic, else diamagnetic.

**$\text{N}_2^-$  (15 electrons) by MOT:**

MO config ( $\text{N}_2$ -type order):

$$(\sigma_{1s})^2(\sigma_{1s}^*)^2(\sigma_{2s})^2(\sigma_{2s}^*)^2(\pi_{2p})^4(\sigma_{2p})^2(\pi_{2p}^*)^1$$



**$\text{N}_2^-$  Ion**

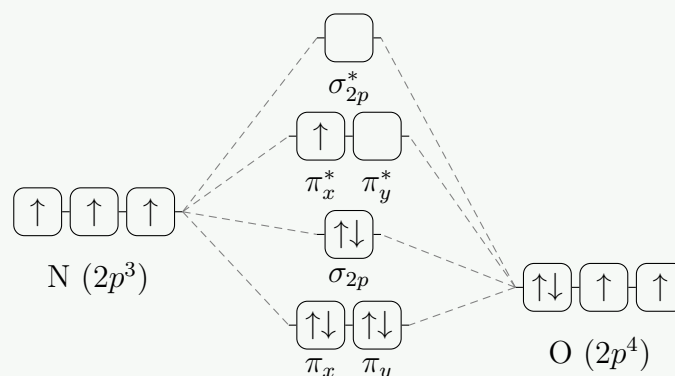
$$\text{Bond order} = \frac{9 - 5}{2} = 2.0$$

1 unpaired electron in  $\pi_{2p}^* \Rightarrow \text{N}_2^-$  is **paramagnetic**.

**NO (11 electrons) by MOT:**

MO config:

$$(\sigma_{1s})^2(\sigma_{1s}^*)^2(\sigma_{2s})^2(\sigma_{2s}^*)^2(\pi_{2p})^4(\sigma_{2p})^2(\pi_{2p}^*)^1$$



**NO Molecule MO Diagram**

$$\text{Bond order of NO} = \frac{8 - 3}{2} = 2.5$$

1 unpaired electron in  $\pi_{2p}^* \Rightarrow \text{NO}$  is **paramagnetic**.

| Species        | e <sup>-</sup> | Bond order | Unpaired e <sup>-</sup> | Magnetic     |
|----------------|----------------|------------|-------------------------|--------------|
| $\text{N}_2^-$ | 15             | 2.0        | 1                       | Paramagnetic |
| NO             | 11             | 2.5        | 1                       | Paramagnetic |

**Q46.** Give an experimental evidence in favour of the secondary valence of a metal. **(6 marks)**

[CHEM 101 | 2009–10]

### Answer

#### Experimental evidence for secondary valence of a metal

Werner proposed that metal ions have **two types of valence**: primary (ionizable) and secondary (coordination). The secondary valence is satisfied by ligands inside the coordination sphere and does **not ionize**.

#### Evidence from conductivity and precipitation experiments:

Werner studied a series of cobalt(III) chloride-ammonia complexes. By treating each with excess  $\text{AgNO}_3$  (which precipitates free  $\text{Cl}^-$  ions only), he determined how many  $\text{Cl}^-$  were **outside** the coordination sphere:

| Compound                            | Modern formula                                         | Color  | Cl <sup>-</sup> pptd | Ions in solution |
|-------------------------------------|--------------------------------------------------------|--------|----------------------|------------------|
| CoCl <sub>3</sub> ·6NH <sub>3</sub> | [Co(NH <sub>3</sub> ) <sub>6</sub> ]Cl <sub>3</sub>    | Orange | 3                    | 4                |
| CoCl <sub>3</sub> ·5NH <sub>3</sub> | [Co(NH <sub>3</sub> ) <sub>5</sub> Cl]Cl <sub>2</sub>  | Purple | 2                    | 3                |
| CoCl <sub>3</sub> ·4NH <sub>3</sub> | [Co(NH <sub>3</sub> ) <sub>4</sub> Cl <sub>2</sub> ]Cl | Green  | 1                    | 2                |
| CoCl <sub>3</sub> ·3NH <sub>3</sub> | [Co(NH <sub>3</sub> ) <sub>3</sub> Cl <sub>3</sub> ]   | Green  | 0                    | 0                |

#### Key observations:

- In CoCl<sub>3</sub>·6NH<sub>3</sub>: all 3 Cl<sup>-</sup> precipitate ⇒ all 3 are **outside** the sphere (primary valence only).
- In CoCl<sub>3</sub>·5NH<sub>3</sub>: only 2 Cl<sup>-</sup> precipitate ⇒ 1 Cl<sup>-</sup> is **inside** the sphere (satisfying secondary valence), not free to ionize.
- In CoCl<sub>3</sub>·4NH<sub>3</sub>: only 1 Cl<sup>-</sup> precipitates ⇒ 2 Cl<sup>-</sup> inside the sphere.
- In CoCl<sub>3</sub>·3NH<sub>3</sub>: **no** Cl<sup>-</sup> precipitates ⇒ all 3 Cl<sup>-</sup> are bound inside the sphere as ligands satisfying the secondary valence.

#### Conclusion:

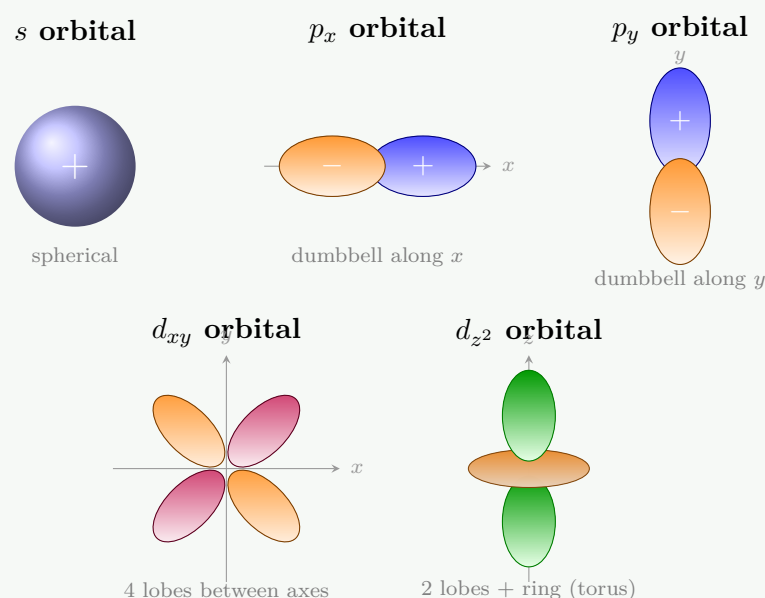
This systematic decrease in the number of precipitable Cl<sup>-</sup> ions as NH<sub>3</sub> is replaced by Cl<sup>-</sup> inside the coordination sphere **directly confirms** the existence of a secondary (coordination) valence. The ligands inside the sphere are tightly bound and cannot be precipitated by AgNO<sub>3</sub>.

**Q47.** Show the different shapes of s, p and d orbitals. Also show the formation of NH<sub>3</sub> molecule according to the overlapping method (Valence Bond Theory). (8 marks) [CHEM 101 | 2009–10]

#### Answer

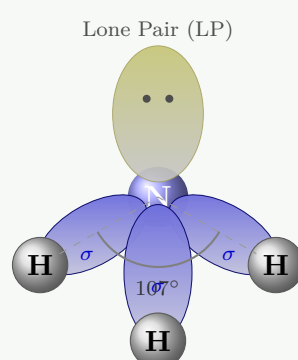
#### Shapes of s, p, d orbitals and formation of NH<sub>3</sub> by VBT

##### Shapes of atomic orbitals:



##### Formation of NH<sub>3</sub> by VBT (overlapping method):

- N atom:  $[He] 2s^2 2p_x^1 2p_y^1 2p_z^1 \Rightarrow$  3 half-filled *p* orbitals available.
- N undergoes **sp<sup>3</sup> hybridization**:  $2s + 2p_x + 2p_y + 2p_z \rightarrow$  4 equivalent *sp*<sup>3</sup> hybrid orbitals.
- 3 half-filled sp<sup>3</sup> orbitals** overlap head-on with the 1*s* orbital of each H atom forming **3 N–H σ bonds**.
- 1 sp<sup>3</sup> orbital** contains a lone pair (LP). The LP–BP repulsion is greater than BP–BP repulsion, compressing the standard tetrahedral angle to 107°.



#### Trigonal Pyramidal Geometry

| Bond / Pair | Orbital Overlap       | Type          |
|-------------|-----------------------|---------------|
| N–H (x3)    | $sp^3$ (N) + $1s$ (H) | $\sigma$ bond |
| Lone Pair   | $sp^3$ (N) only       | Non-bonding   |

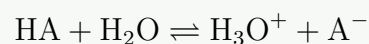
Q48. Describe how the relative strength of acids is determined. (11 marks)

[CHEM 101 | 2009–10]

### Answer

#### Relative strength of acids

**Definition:** The strength of an acid is measured by its tendency to donate a proton ( $H^+$ ) — i.e., the extent to which it **ionizes** in water:



#### Acid dissociation constant:

$$K_a = \frac{[H_3O^+][A^-]}{[HA]} \quad pK_a = -\log K_a$$

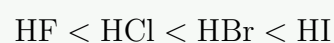
Larger  $K_a$  (smaller  $pK_a$ )  $\Rightarrow$  **stronger acid**.

#### Factors determining acid strength:

##### 1. Bond strength (H–A bond):

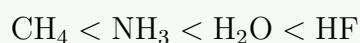
Weaker H–A bond  $\Rightarrow$  easier to break  $\Rightarrow$  stronger acid.

In the same *group*: bond strength decreases down the group  $\Rightarrow$  acid strength **increases**:



##### 2. Electronegativity of A (same period):

More electronegative A  $\Rightarrow$  more polar H–A bond  $\Rightarrow H^+$  released more easily  $\Rightarrow$  stronger acid:



##### 3. Stability of conjugate base $A^-$ :

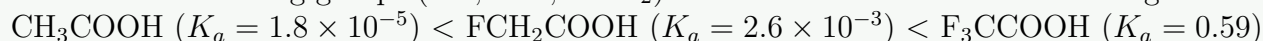
More stable  $A^-$   $\Rightarrow$  equilibrium shifts right  $\Rightarrow$  stronger acid.

Stability of  $A^-$  increases with:

- **Resonance delocalization** of negative charge
- **Inductive effect** of electronegative substituents
- Larger size of A (better charge dispersal)

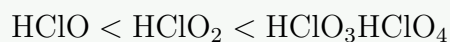
##### 4. Effect of substituents (inductive effect):

Electron-withdrawing groups ( $-F$ ,  $-Cl$ ,  $-NO_2$ ) stabilize  $A^-$   $\Rightarrow$  increase acid strength:



##### 5. Oxyacids (H–O–X):

More O atoms on X  $\Rightarrow$  more electron withdrawal  $\Rightarrow$  weaker O–H bond  $\Rightarrow$  stronger acid:



| Acid       | $K_a$                 | $pK_a$ | Strength    |
|------------|-----------------------|--------|-------------|
| $HClO_4$   | $\gg 1$               | $< 0$  | Very strong |
| $HCl$      | $\approx 10^7$        | $-7$   | Strong      |
| $H_2SO_4$  | Large                 | $-3$   | Strong      |
| $CH_3COOH$ | $1.8 \times 10^{-5}$  | 4.75   | Weak        |
| $HF$       | $6.8 \times 10^{-4}$  | 3.17   | Weak        |
| $HCN$      | $6.2 \times 10^{-10}$ | 9.21   | Very weak   |

Q49. Why do ionic compounds usually have high melting and boiling points while simple covalent compounds have low melting and boiling points? (7 marks)

[CHEM 101 | 2009–10]

### Answer

#### High MP/BP of ionic compounds vs low MP/BP of covalent compounds

**Ionic Compounds** — High melting and boiling points:

- Ionic compounds consist of positive and negative ions arranged in a **3D crystal lattice**.
- The ions are held together by strong **electrostatic forces** (Coulombic attraction) acting in all directions.
- These forces are **non-directional** and **very strong** (lattice energy of NaCl = 788 kJ/mol).
- To melt or boil an ionic compound, **all** these strong ion-ion attractions must be overcome simultaneously  $\Rightarrow$  very high energy required.
- Examples: NaCl (MP = 801°C), MgO (MP = 2852°C).

#### Simple Covalent Compounds — Low melting and boiling points:

- Simple covalent molecules (e.g., CO<sub>2</sub>, CH<sub>4</sub>, H<sub>2</sub>O) consist of discrete molecules held together by **weak intermolecular forces**:
  - van der Waals / London dispersion forces
  - Dipole-dipole interactions
  - Hydrogen bonding (relatively stronger but still weak compared to ionic bonds)
- The **covalent bonds within** the molecule are strong, but to melt/boil, only the **weak intermolecular forces** need to be overcome — **not** the covalent bonds themselves.
- Very little energy required  $\Rightarrow$  low MP and BP.
- Examples: CO<sub>2</sub> (sublimes at -78°C), CH<sub>4</sub> (BP = -161°C).

| Compound         | Type     | Forces broken on melting       | MP (°C)        |
|------------------|----------|--------------------------------|----------------|
| NaCl             | Ionic    | Strong electrostatic (lattice) | 801            |
| MgO              | Ionic    | Strong electrostatic (lattice) | 2852           |
| CH <sub>4</sub>  | Covalent | Weak van der Waals             | -182           |
| CO <sub>2</sub>  | Covalent | Weak van der Waals             | -78 (sublimes) |
| H <sub>2</sub> O | Covalent | H-bonds (stronger than vdW)    | 0              |

**Conclusion:** The fundamental difference is that ionic compounds require breaking of **strong long-range electrostatic forces** throughout the lattice, while simple covalent compounds only need to overcome **weak intermolecular forces** between discrete molecules.

**Q50.** What should be the magnetic property of the fluorine molecule (F<sub>2</sub>)? Explain with the help of molecular orbital theory. **(7 marks)**  
[CHEM 101 | 2008–09]

#### Answer

##### Magnetic Property of F<sub>2</sub> using Molecular Orbital Theory (MOT)

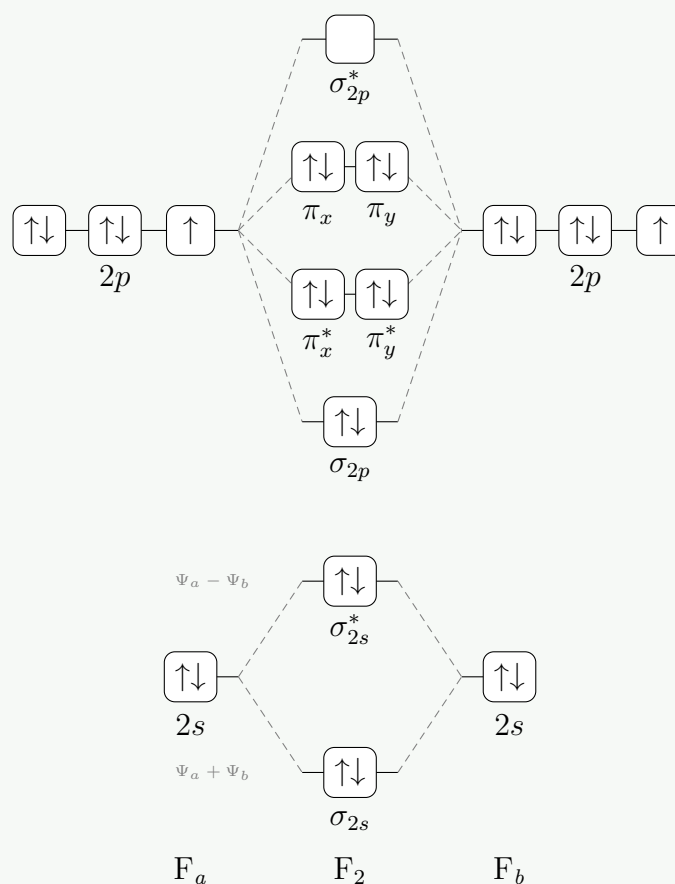
**Molecular Orbital Theory** builds MOs by linear combination of atomic orbitals (LCAO). For F<sub>2</sub>, each F contributes 9 electrons  $\Rightarrow$  **18 electrons total**.

**MO Energy order for F<sub>2</sub> (2nd period,  $Z > 7$ ):**

$$\sigma_{1s} < \sigma_{1s}^* < \sigma_{2s} < \sigma_{2s}^* < \sigma_{2p_z} < \pi_{2p_x} = \pi_{2p_y} < \pi_{2p_x}^* = \pi_{2p_y}^* < \sigma_{2p_z}^*$$

**Electronic configuration of F<sub>2</sub>:**

$$(\sigma_{1s})^2 (\sigma_{1s}^*)^2 (\sigma_{2s})^2 (\sigma_{2s}^*)^2 (\sigma_{2p_z})^2 (\pi_{2p_x})^2 (\pi_{2p_y})^2 (\pi_{2p_x}^*)^2 (\pi_{2p_y}^*)^2$$



Diffuorine (Bond order = 1, diamagnetic)

**Bond Order of  $F_2$ :**

$$\text{Bond Order} = \frac{(\text{bonding } e^-) - (\text{antibonding } e^-)}{2} = \frac{10 - 8}{2} = 1$$

**Magnetic Property:**

- All 18 electrons in  $F_2$  are **paired** in the MOs — no unpaired electrons exist.
- A substance is **diamagnetic** if all electrons are paired (it is weakly repelled by a magnetic field).
- A substance is **paramagnetic** if unpaired electrons are present (attracted to a magnetic field).
- Since  $F_2$  has **no unpaired electrons**, it is **diamagnetic**.

| Property              | Value       |
|-----------------------|-------------|
| Total electrons       | 18          |
| Bonding electrons     | 10          |
| Antibonding electrons | 8           |
| Bond order            | 1           |
| Unpaired electrons    | 0           |
| Magnetic nature       | Diamagnetic |

**Q51.** Discuss the general properties of ionic and covalent bonds/compounds. (12 marks)

[CHEM 101 | 2008–09]

### Answer

#### General Properties of Ionic and Covalent Bonds/Compounds

##### Part 1: Ionic Bonds and Ionic Compounds

An **ionic bond** is formed by the **complete transfer of one or more electrons** from a metal atom to a non-metal atom, resulting in oppositely charged ions held together by strong electrostatic attraction.

- Formation:** Formed between elements of **large electronegativity difference** ( $\Delta\chi > 1.7$ ), typically between metals (Groups 1, 2) and non-metals (Groups 16, 17). Example:  

$$\text{Na} + \text{Cl} \rightarrow \text{Na}^+\text{Cl}^-$$
- Physical State:** Ionic compounds are **crystalline solids** at room temperature, forming regular lattice structures held by strong electrostatic forces.
- Melting and Boiling Points:** **High melting and boiling points** due to strong lattice energy. Example: NaCl melts at  $801^\circ\text{C}$ .
- Electrical Conductivity:** **Non-conductors** in solid state (ions fixed in lattice), but **good conductors** when melted or dissolved in water (ions become mobile).



- (v) **Solubility:** Generally **soluble in polar solvents** (water) due to ion–dipole interactions, but **insoluble in non-polar solvents** (benzene,  $\text{CCl}_4$ ).
- (vi) **Non-directional Bond:** Ionic bonds are **non-directional** — the electrostatic attraction acts equally in all directions, leading to high coordination numbers in crystal lattices.
- (vii) **Hard and Brittle:** Ionic crystals are **hard** but **brittle** — when stress is applied, like planes shift and repulsion between same-charged ions causes the crystal to shatter.
- (viii) **Isomerism:** Ionic compounds do **not** show isomerism.

### Part 2: Covalent Bonds and Covalent Compounds

A **covalent bond** is formed by the **mutual sharing of electron pairs** between two atoms, typically non-metals of similar electronegativity.

- (i) **Formation:** Formed between elements of **small electronegativity difference** ( $\Delta\chi < 1.7$ ), typically between two non-metals.  
Example:  
$$\text{H} \cdot + \cdot \text{Cl} \rightarrow \text{H}:\text{Cl}$$
- (ii) **Physical State:** Covalent compounds exist as **gases, liquids, or soft solids** at room temperature (weak intermolecular forces), though covalent network solids (diamond,  $\text{SiO}_2$ ) are very hard.
- (iii) **Melting and Boiling Points:** Generally **low melting and boiling points** (weak van der Waals or dipole forces between molecules). Exceptions: diamond, SiC (network covalent) have very high melting points.
- (iv) **Electrical Conductivity:** Generally **non-conductors** of electricity (no ions or free electrons). Exception: graphite (delocalized  $\pi$  electrons).
- (v) **Solubility:** **Soluble in non-polar solvents** (“like dissolves like”); polar covalent compounds may dissolve in water (e.g.  $\text{HCl}$ ,  $\text{NH}_3$ ).
- (vi) **Directional Bond:** Covalent bonds are **directional** — they have specific bond angles and geometries, giving molecules definite shapes (VSEPR theory).
- (vii) **Isomerism:** Covalent compounds can show **structural and stereoisomerism** due to directional bonding.
- (viii) **Reaction rate:** Covalent bond reactions are generally **slow** (bonds must be broken and reformed). Ionic reactions in solution are typically **fast** (ion exchange).

### Comparison Table:

| Property       | Ionic             | Covalent           |
|----------------|-------------------|--------------------|
| Bond formation | Electron transfer | Electron sharing   |
| Physical state | Crystalline solid | Gas/liquid/solid   |
| Melting point  | High              | Low (usually)      |
| Conductivity   | Yes (molten/aq.)  | No (usually)       |
| Solubility     | Polar solvents    | Non-polar solvents |
| Bond direction | Non-directional   | Directional        |
| Isomerism      | Absent            | Present            |
| Reaction speed | Fast              | Slow               |

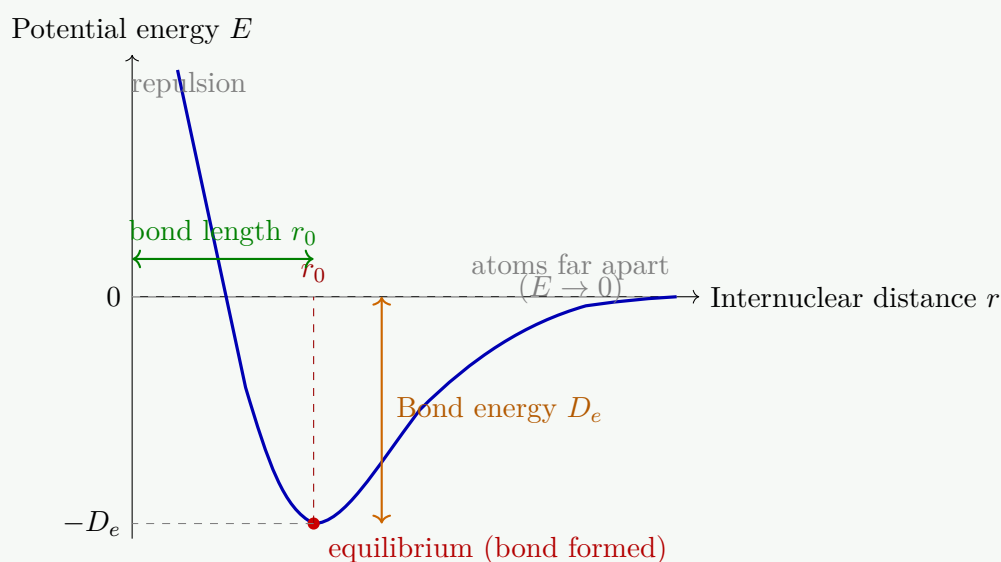
**Q52.** With the help of a potential energy diagram, explain bond length and bond energy. (8 marks)

[CHEM 101 | 2007–08]

### Answer

#### Bond Length and Bond Energy from Potential Energy Diagram

#### Potential Energy Diagram for Two Atoms:



**Bond Length:**

**Bond length** is defined as the **equilibrium internuclear distance**  $r_0$  between two bonded atoms — the distance at which the **potential energy of the system is minimum**.

From the diagram:

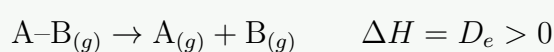
- At  $r = r_0$ , the attractive and repulsive forces are exactly balanced.
- The system is at its **most stable** (lowest energy) configuration.
- $r_0$  corresponds to the bottom of the potential energy well on the  $x$ -axis.
- Typical bond lengths: H–H: 0.74 Å, C–C: 1.54 Å, N≡N: 1.10 Å.

**Factors affecting bond length:**

- **Bond order:** Higher bond order  $\rightarrow$  shorter bond (C–C > C=C > C≡C in length).
- **Atomic size:** Larger atoms  $\rightarrow$  longer bond.
- **Electronegativity:** Higher electronegativity  $\rightarrow$  shorter, stronger bond.

**Bond Energy:**

**Bond energy** (or bond dissociation energy)  $D_e$  is defined as the **energy required to break one mole of a particular bond** in the gaseous state, separating the atoms to infinity:



From the diagram:

- $D_e$  is the **depth of the potential energy well** — the energy difference between the minimum ( $-D_e$ ) and the zero reference (atoms at infinity).
- A **deeper well** means a **stronger bond** (higher  $D_e$ ).
- Typical bond energies: H–H: 436 kJ/mol, C=C: 614 kJ/mol, N≡N: 945 kJ/mol.

**Factors affecting bond energy:**

- **Bond order:** Higher bond order  $\rightarrow$  greater bond energy.
- **Bond length:** Shorter bond  $\rightarrow$  stronger  $\rightarrow$  higher bond energy.
- **Electronegativity difference:** Greater  $\Delta\chi \rightarrow$  more polar bond  $\rightarrow$  generally stronger.

**Relationship Between Bond Length and Bond Energy:**

Bond length and bond energy are **inversely related**:

Shorter bond length  $\longleftrightarrow$  Higher bond energy (stronger bond)

| Bond | Bond Order | Length (Å) | Energy (kJ/mol) |
|------|------------|------------|-----------------|
| C–C  | 1          | 1.54       | 347             |
| C=C  | 2          | 1.34       | 614             |
| C≡C  | 3          | 1.20       | 839             |

**Q53.** What are the limitations of valence bond theory (VBT) for the formation of a covalent bond? Using the valence shell electron pair repulsion (VSEPR) model, develop the geometrical structure of the water molecule starting from oxide (O). **(12 marks)** [CHEM 101 | 2007–08]

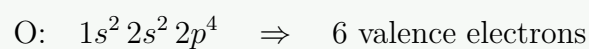
**Answer****Limitations of VBT and VSEPR Geometry of Water****Part 1: Limitations of Valence Bond Theory (VBT)**

- Cannot explain magnetic properties:** VBT predicts O<sub>2</sub> to be diamagnetic (all electrons paired in double bond), but experimentally O<sub>2</sub> is **paramagnetic** (two unpaired electrons). VBT fails to explain this.
- Cannot explain colour of coordination compounds:** VBT provides no satisfactory explanation for the **colour** of transition metal complexes, which arises from  $d-d$  electronic transitions explained only by Crystal Field Theory.
- Cannot explain fractional bond orders:** VBT cannot account for **equal bond lengths** in species like benzene (C<sub>6</sub>H<sub>6</sub>) or NO<sub>2</sub>, where bond order is fractional. Resonance is a workaround, not a true explanation.
- Fails for electron-deficient compounds:** VBT cannot explain bonding in **electron-deficient** species like B<sub>2</sub>H<sub>6</sub> (diborane), where there are fewer electron pairs than bonds.
- Does not explain delocalisation:** VBT uses localised two-centre bonds and cannot adequately explain **electron delocalisation** in aromatic compounds and conjugated systems.

- (vi) **Cannot predict geometry independently:** VBT requires knowledge of the geometry *before* applying hybridisation — it cannot **predict** molecular shapes independently. This is a conceptual limitation compared to MO theory.

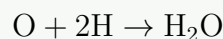
## Part 2: VSEPR Geometry of Water Starting from Oxide (O)

### Step 1: Electronic Configuration of Oxygen



### Step 2: Combining O with 2 Hydrogen Atoms

Oxygen forms 2 bonds with hydrogen atoms, using its 2 unpaired 2p electrons:



### Step 3: Count Electron Pairs Around Central Atom (O)

| Type of electron pair       | Number   |
|-----------------------------|----------|
| Bonding pairs (O–H bonds)   | 2        |
| Lone pairs on O             | 2        |
| <b>Total electron pairs</b> | <b>4</b> |

### Step 4: Electron Pair Geometry (VSEPR)

With **4 electron pairs**, the electron pair geometry is **tetrahedral** (bond angle  $109.5^\circ$  for a perfect tetrahedron).

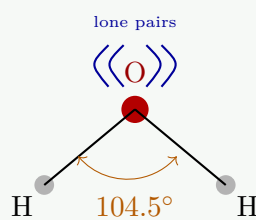
### Step 5: Effect of Lone Pairs

Lone pairs occupy more space than bonding pairs and **repel more strongly** (lone pair–lone pair > lone pair–bonding pair > bonding pair–bonding pair). With **2 lone pairs** on oxygen:

- The 2 lone pairs **compress** the H–O–H bond angle below  $109.5^\circ$ .
- Actual H–O–H bond angle =  **$104.5^\circ$** .

### Step 6: Molecular Geometry

Although the **electron pair geometry** is tetrahedral, the **molecular geometry** (considering only the atoms) is **bent (V-shaped)**.



### Summary:

| Feature                | H <sub>2</sub> O |
|------------------------|------------------|
| Central atom           | O                |
| Bonding pairs          | 2                |
| Lone pairs             | 2                |
| Total electron pairs   | 4                |
| Electron pair geometry | Tetrahedral      |
| Molecular geometry     | Bent (V-shaped)  |
| Bond angle             | $104.5^\circ$    |

- Q54.** Write down the molecular orbital electronic configuration and draw the molecular orbital diagram of the following: (i) CO, (ii)  $\text{O}_2^+$ , (iii) NO. Comment on the magnetic properties of the above molecules and also calculate their bond order. **(15 marks)** [CHEM 101 | 2007–08]

### Answer

#### MO Configurations, Diagrams, Magnetic Properties and Bond Orders of CO, $\text{O}_2^+$ , NO

##### MO Energy Order for Diatomic Molecules:

For molecules like CO and NO (heteronuclear,  $Z < 8$  for lighter atom uses the order with  $\pi$  before  $\sigma_{2p}$ ):

$$\sigma_{1s} < \sigma_{1s}^* < \sigma_{2s} \sigma_{2s}^* < \pi_{2p} = \pi_{2p} \sigma_{2p} < \pi_{2p}^* = \pi_{2p}^* \sigma_{2p}^*$$

For  $\text{O}_2^+$  (homonuclear-like,  $Z \geq 8$ ):

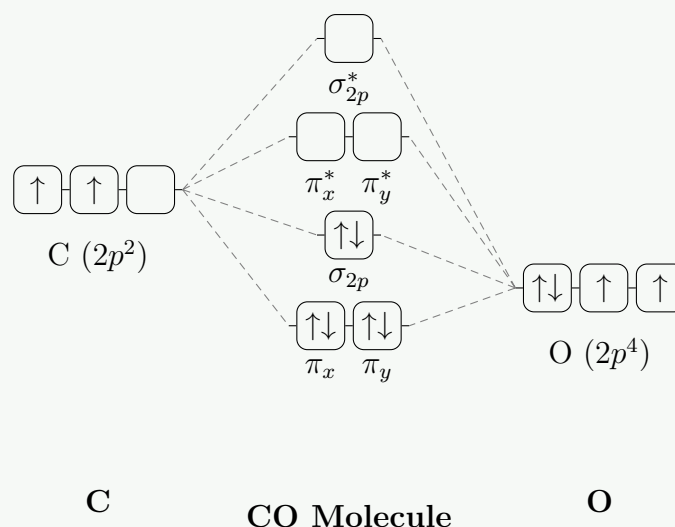
$$\sigma_{1s} < \sigma_{1s}^* < \sigma_{2s} \sigma_{2s}^* < \sigma_{2p} < \pi_{2p} = \pi_{2p} < \pi_{2p}^* = \pi_{2p}^* \sigma_{2p}^*$$

#### (i) Carbon Monoxide (CO) — 14 electrons

MO Electronic Configuration:

$$(\sigma_{1s})^2(\sigma_{1s}^*)^2(\sigma_{2s})^2(\sigma_{2s}^*)^2(\pi_{2p})^2(\pi_{2p})^2(\sigma_{2p})^2$$

MO Diagram:



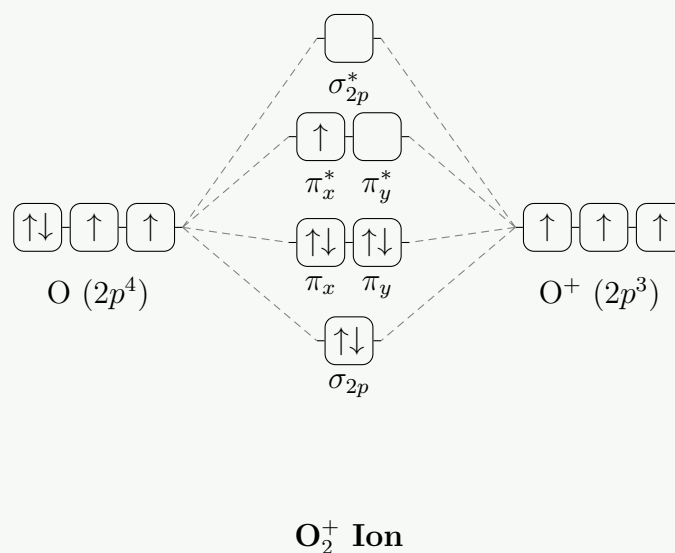
Bond Order of CO:

$$BO = \frac{1}{2}(N_b - N_a) = \frac{1}{2}(8 - 2) = 3$$

Magnetic Property: All electrons are **paired**  $\Rightarrow$  CO is **diamagnetic**.

(ii)  $O_2^+$  — 15 electrons

MO Electronic Configuration:



$$(\sigma_{1s})^2(\sigma_{1s}^*)^2(\sigma_{2s})^2(\sigma_{2s}^*)^2(\sigma_{2p})^2(\pi_{2p})^2(\pi_{2p})^2(\pi_{2p}^*)^1$$

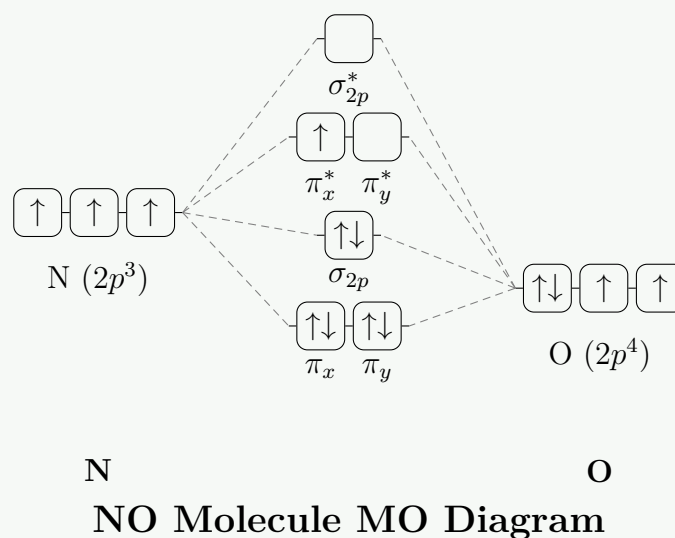
Bond Order of  $O_2^+$ :

$$BO = \frac{1}{2}(8 - 3) = 2.5$$

Magnetic Property: One **unpaired electron** in  $\pi_{2p}^*$   $\Rightarrow$   $O_2^+$  is **paramagnetic**.

(iii) Nitric Oxide (NO) — 15 electrons

MO Electronic Configuration:



$$(\sigma_{1s})^2(\sigma_{1s}^*)^2(\sigma_{2s})^2(\sigma_{2s}^*)^2(\pi_{2p})^2(\pi_{2p})^2(\sigma_{2p})^2(\pi_{2p}^*)^1$$

Bond Order of NO:

$$BO = \frac{1}{2}(8 - 3) = 2.5$$

Magnetic Property: One **unpaired electron** in  $\pi_{2p}^*$   $\Rightarrow$  NO is **paramagnetic**.

Summary Table:

| Molecule | Electrons | Bond Order | Unpaired $e^-$ | Magnetic Property |
|----------|-----------|------------|----------------|-------------------|
| CO       | 14        | 3.0        | 0              | Diamagnetic       |
| $O_2^+$  | 15        | 2.5        | 1              | Paramagnetic      |
| NO       | 15        | 2.5        | 1              | Paramagnetic      |

**Q55.** Discuss the properties of electrovalent compounds which are related to the directional properties of the bond. Compare them with covalent compounds. (9 marks) [CHEM 101 | 2006–07]

### Answer

#### Electrovalent (ionic) compounds vs covalent compounds

##### Properties of ionic compounds related to bonding:

##### 1. Non-directional bonding:

Ionic bonds are **electrostatic** in nature and act equally in all directions. Each ion attracts all oppositely charged neighbors simultaneously  $\Rightarrow$  no preferred bond direction. This leads to formation of a **crystal lattice** rather than discrete molecules.

##### 2. High melting and boiling points:

Strong lattice forces must be overcome  $\Rightarrow$  high energy input required. NaCl MP = 801°C.

##### 3. Electrical conductivity:

In solid state: ions are fixed  $\Rightarrow$  **non-conductor**.

In molten state or solution: ions are free to move  $\Rightarrow$  **good conductor**.

##### 4. Solubility in polar solvents:

Ionic compounds dissolve readily in polar solvents (water) because ion-dipole interactions replace lattice energy. **Insoluble** in non-polar solvents.

##### 5. Hard and brittle:

Crystal lattice is rigid  $\Rightarrow$  hard. Displacement of layers brings like charges adjacent  $\Rightarrow$  repulsion  $\Rightarrow$  brittle.

##### 6. Isomerism:

No stereoisomerism or geometrical isomerism possible due to non-directional bonding.

##### Comparison with covalent compounds:

| Property       | Ionic (electrovalent)          | Covalent                             |
|----------------|--------------------------------|--------------------------------------|
| Bond nature    | Electrostatic, non-directional | Electron sharing, directional        |
| Physical state | Solid crystals                 | Gas, liquid or soft solid            |
| MP/BP          | High ( $> 300^\circ\text{C}$ ) | Low ( $< 300^\circ\text{C}$ usually) |
| Conductivity   | Conducts when molten/dissolved | Non-conductor (mostly)               |
| Solubility     | Polar solvents                 | Non-polar solvents                   |
| Isomerism      | Not possible                   | Possible (geometric, optical)        |
| Reaction rate  | Fast (ion reactions)           | Slow (bond breaking needed)          |

**Q56.** What are donor and acceptor atoms? Discuss their role in the formation of a coordinate covalent bond. Comment on the formation of  $\text{H}_3\text{O}^+$ ,  $\text{H}_4\text{O}^{2+}$ ,  $\text{O}_2\text{Cl}^+$  and  $\text{O}_4\text{Cl}^-$  ions. (9 marks) [CHEM 101 | 2006–07]

### Answer

#### Donor and acceptor atoms; $\text{H}_3\text{O}^+$ , $\text{H}_4\text{O}_2^+$ , $\text{O}_2\text{Cl}^+$ , $\text{O}_4\text{Cl}^-$

##### Donor atom:

- An atom that **donates** a lone pair of electrons to form a coordinate covalent bond.
- It is a **Lewis base**: N, O, S, P or compounds containing them ( $\text{NH}_3$ ,  $\text{H}_2\text{O}$ ,  $\text{CO}$ ).

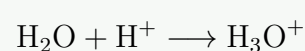
##### Acceptor atom:

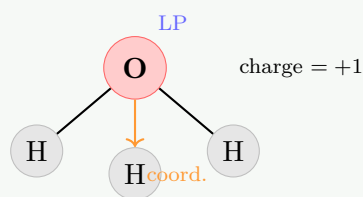
- An atom that **accepts** a lone pair into its **vacant orbital**.
- It is a **Lewis acid**:  $\text{H}^+$ ,  $\text{BF}_3$ ,  $\text{AlCl}_3$ , metal ions.

##### Formation of ions:

##### (i) $\text{H}_3\text{O}^+$ (Hydronium ion):

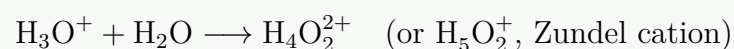
$\text{H}_2\text{O}$  (donor, O has 2 LP) donates one LP to  $\text{H}^+$  (acceptor, vacant 1s):





**(ii)  $\text{H}_4\text{O}_2^+$ :**

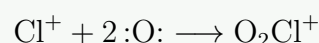
Two  $\text{H}_2\text{O}$  molecules: one O donates LP to  $\text{H}^+$ , giving  $\text{H}_3\text{O}^+$ ; then another  $\text{H}_2\text{O}$  donates LP to  $\text{H}_3\text{O}^+$ :



O–H–O bridge formed by coordinate bond.

**(iii)  $\text{O}_2\text{Cl}^+$ :**

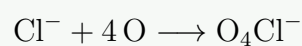
$\text{Cl}^+$  (acceptor, vacant  $3p$ ) accepts LP from two O atoms (donor):



Linear O–Cl–O arrangement,  $sp$  hybridization of Cl.

**(iv)  $\text{O}_4\text{Cl}^-$ :**

$\text{Cl}^-$  (acceptor) accepts LPs from 4 O atoms:



Tetrahedral arrangement around Cl,  $sp^3$  hybridization.

| Ion                      | Donor                      | Acceptor      | Coord. bonds | Geometry           |
|--------------------------|----------------------------|---------------|--------------|--------------------|
| $\text{H}_3\text{O}^+$   | O ( $\text{H}_2\text{O}$ ) | $\text{H}^+$  | 1            | Trigonal pyramidal |
| $\text{H}_4\text{O}_2^+$ | O ( $\text{H}_2\text{O}$ ) | $\text{H}^+$  | 1            | Bent bridge        |
| $\text{O}_2\text{Cl}^+$  | O                          | $\text{Cl}^+$ | 2            | Linear             |
| $\text{O}_4\text{Cl}^-$  | O                          | $\text{Cl}^-$ | 4            | Tetrahedral        |

**Q57.** What are the noble gas elements and why are they called so? Discuss the important uses of noble gas elements. **(9 marks)** [CHEM 101 | 2006–07]

**Answer**

**Noble gas elements — definition and uses**

**What are noble gas elements?**

The noble gases are the elements of **Group 18** (Group 0) of the periodic table:

| Element | Symbol | At. No. | Electron config.                        |
|---------|--------|---------|-----------------------------------------|
| Helium  | He     | 2       | $1s^2$                                  |
| Neon    | Ne     | 10      | $[\text{He}] 2s^2 2p^6$                 |
| Argon   | Ar     | 18      | $[\text{Ne}] 3s^2 3p^6$                 |
| Krypton | Kr     | 36      | $[\text{Ar}] 3d^{10} 4s^2 4p^6$         |
| Xenon   | Xe     | 54      | $[\text{Kr}] 4d^{10} 5s^2 5p^6$         |
| Radon   | Rn     | 86      | $[\text{Xe}] 4f^{14} 5d^{10} 6s^2 6p^6$ |

**Why called noble gases?**

- They have **completely filled** valence shells ( $ns^2 np^6$ , except He:  $1s^2$ )  $\Rightarrow$  **octet satisfied**.
- Extremely **chemically inert** — do not readily form compounds under normal conditions (like noble metals that resist reaction).
- Very high **ionization energies**  $\Rightarrow$  difficult to remove electrons.
- **Zero or very low electron affinity**.
- Called “noble” by analogy with noble metals (Au, Pt) which are chemically unreactive.
- Also called **inert gases** or **rare gases**.

**Important uses of noble gases:**

**1. Helium (He):**



- Filling **airships and balloons** (non-flammable, unlike  $\text{H}_2$ )
- **Cryogenic applications** — liquid He used to cool superconducting magnets (MRI machines)
- **Deep-sea diving** mixtures ( $\text{He-O}_2$ ) to prevent nitrogen narcosis
- Carrier gas in **gas chromatography**

## 2. Neon (Ne):

- **Neon signs** and advertising displays (glows red-orange when electrically excited)
- **Laser technology** (He-Ne laser)

## 3. Argon (Ar):

- **Inert atmosphere** in welding to prevent oxidation
- Filling **incandescent light bulbs**
- **Metallurgy** — protecting reactive metals during processing

## 4. Krypton (Kr) and Xenon (Xe):

- **High-intensity lamps** (camera flash, movie projectors)
- **Xenon anesthesia** in medicine
- **Ion propulsion** thrusters in spacecraft

## 5. Radon (Rn):

- Used in **cancer radiotherapy** (historically)
- Monitored as a **health hazard** (radioactive, causes lung cancer)

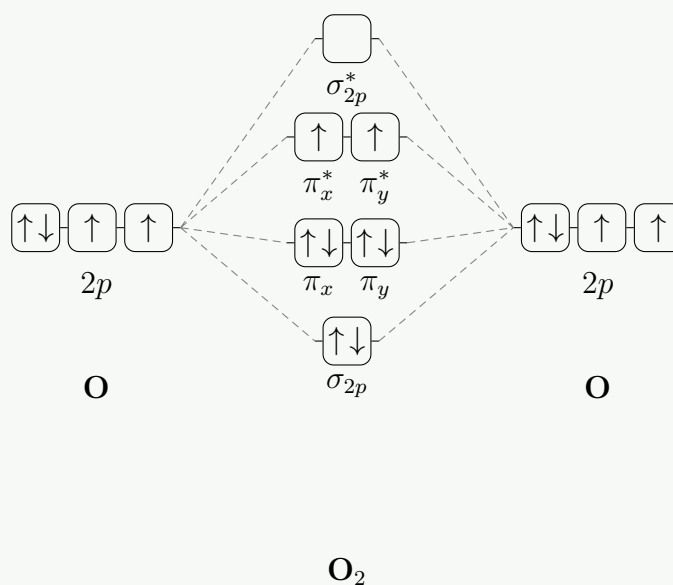
**Q58.** Show the formation of  $\text{O}_2$  molecule according to both Valence Bond Theory (VBT) and Molecular Orbital Theory (MOT). Comment on the advantages of MOT over VBT. **(10 marks)** [CHEM 101 | 2006–07]

### Answer

#### Formation of $\text{O}_2$ by VBT and MOT; advantages of MOT

##### Formation of $\text{O}_2$ by VBT:

- O:  $[\text{He}] 2s^2 2p^4$  — ground state has 2 half-filled  $p$  orbitals.
- Two O atoms, each contributing 2 half-filled  $p$  orbitals, overlap to form:
  - One  $\sigma$  **bond**: head-on overlap of  $2p_z-2p_z$
  - One  $\pi$  **bond**: sideways overlap of  $2p_x-2p_x$
- VBT predicts a **double bond**  $\text{O}=\text{O}$  with all electrons paired.
- **Serious limitation:** VBT predicts  $\text{O}_2$  is **diamagnetic** — but experimentally  $\text{O}_2$  is **paramagnetic** (attracted to magnets). VBT fails here.



##### Formation of $\text{O}_2$ by MOT:

$\text{O}_2$  has 16 electrons. MO configuration ( $Z \geq 8$ , normal order):

$$(\sigma_{1s})^2(\sigma_{1s}^*)^2(\sigma_{2s})^2(\sigma_{2s}^*)^2(\sigma_{2p})^2(\pi_{2p_x})^2(\pi_{2p_y})^2(\pi_{2p_x}^*)^1(\pi_{2p_y}^*)^1$$

$$\text{Bond order} = \frac{10 - 6}{2} = 2 \Rightarrow \text{double bond, consistent with } \text{O}=\text{O}$$

The two electrons in the degenerate  $\pi_{2p}^*$  orbitals are **unpaired** (Hund's rule)  $\Rightarrow$   $\text{O}_2$  is **paramagnetic**. This is experimentally verified.

**Advantages of MOT over VBT:**

| Property               | VBT                 | MOT                           |
|------------------------|---------------------|-------------------------------|
| $\text{O}_2$ magnetism | Diamagnetic (wrong) | <b>Paramagnetic (correct)</b> |
| Delocalized $e^-$      | Cannot explain      | Explained naturally           |
| Bond order of ions     | Difficult           | Easily calculated             |
| Antibonding MOs        | No concept          | Explicitly included           |
| Odd-electron bonds     | Cannot explain      | Easily handled                |
| Spectroscopy           | Limited             | Accurate predictions          |

**Conclusion:** MOT provides a more complete and accurate description of molecular bonding, especially for **paramagnetic** species like  $\text{O}_2$  where VBT fails entirely.

**Q59.** What are the possible interactions between two atoms when they approach each other for the formation of a bond? What is the optimum condition at which a chemical bond forms? Discuss with the help of a potential energy diagram. **(10 marks)** [CHEM 101 | 2006–07]

### Answer

#### Interactions Between Two Atoms and Optimum Condition for Bond Formation

##### Part 1: Possible Interactions Between Two Approaching Atoms

When two atoms approach each other, the following **electrostatic interactions** arise simultaneously:

- Nucleus–Nucleus Repulsion:** The positively charged nuclei of the two atoms **repel** each other. This repulsion increases rapidly as the atoms come closer.
- Electron–Electron Repulsion:** The negatively charged electron clouds of the two atoms **repel** each other. This also increases as the atoms approach.
- Nucleus–Electron Attraction:** The nucleus of each atom **attracts** the electrons of the other atom. This is the interaction responsible for **bond formation**.

At large distances, all interactions are negligible. As the atoms approach:

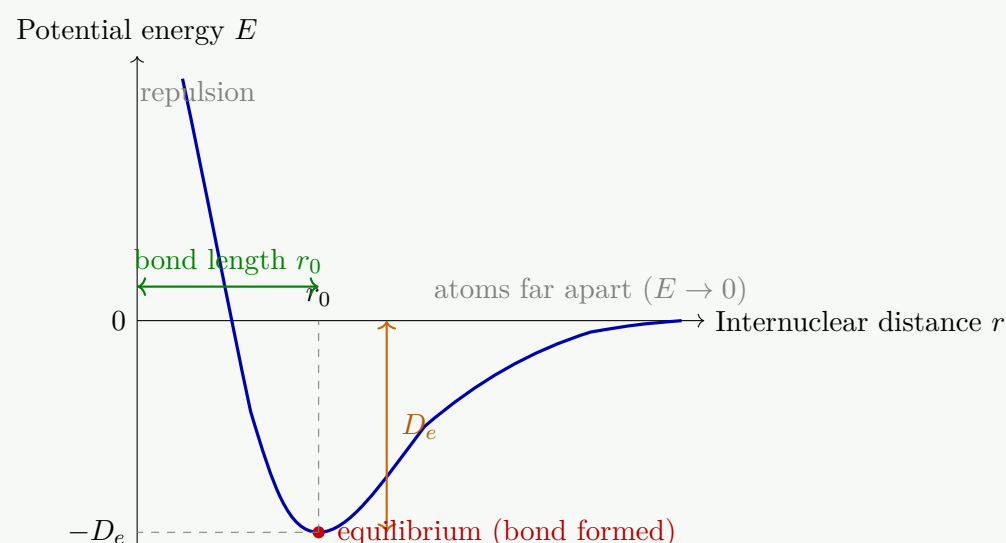
- If **attractions dominate** over repulsions, the potential energy **decreases** and a bond tends to form.
- If **repulsions dominate**, the potential energy **increases** and the atoms are pushed apart.

##### Part 2: Optimum Condition for Bond Formation

A chemical bond forms when the two atoms reach a separation at which the **net potential energy is minimum**. At this point:

- Attractive forces exactly **balance** repulsive forces.
- The system is at its **most stable** (lowest energy) configuration.
- The distance between the nuclei at this point is called the **bond length**  $r_0$ .
- The **depth** of the potential energy minimum below the zero reference level is the **bond energy** (or dissociation energy)  $D_e$ .

##### Part 3: Potential Energy Diagram



**Explanation of the Diagram:**

- At **large**  $r$  (far apart): potential energy  $\rightarrow 0$ ; atoms do not interact.
- As  $r$  **decreases**: nucleus–electron attractions dominate; potential energy **falls**.
- At  $r = r_0$ : potential energy reaches a **minimum** ( $-D_e$ ). This is the **equilibrium bond length**. A stable chemical bond is formed here.
- At  $r < r_0$ : nucleus–nucleus and electron–electron repulsions dominate; potential energy **rises steeply**. The atoms strongly repel each other.

**Conclusion:** A chemical bond forms at the internuclear distance  $r_0$  where the potential energy is **minimum**, representing a perfect balance between attractive and repulsive interactions. The energy  $D_e$  required to separate the atoms from  $r_0$  to infinity is the **bond dissociation energy**.

BUET · CHEM 113

# Chemistry

Section S4

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Properties of Solutions

## S4 — Properties of Solutions

**Q1.** Explain Henry's law in terms of the kinetic molecular theory of gases. Give an exception to Henry's law. **(10 marks)** [CHEM 113 | 2023–24; 2021–22]

### Answer

#### Henry's Law: Kinetic Molecular Interpretation and Exception

##### Statement of Henry's Law:

**Henry's Law** states that at constant temperature, the **solubility of a gas in a liquid** is directly proportional to the **partial pressure** of the gas above the liquid:

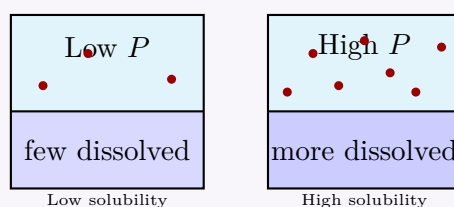
$$C = k_H \cdot P$$

where:

- $C$  = concentration (solubility) of dissolved gas
- $P$  = partial pressure of gas above the liquid
- $k_H$  = Henry's law constant (depends on gas, solvent, and temperature)

##### Explanation Using Kinetic Molecular Theory of Gases:

- At the gas–liquid interface**, gas molecules are in constant random motion. Some gas molecules **strike the liquid surface** and are captured (dissolve), while dissolved molecules **escape** back to the gas phase.
- At **equilibrium**, the rate of gas molecules **entering** the liquid equals the rate of dissolved molecules **leaving** the liquid.
- When the **partial pressure increases**, the **number of gas molecules per unit volume** above the liquid increases (by the ideal gas law:  $n/V = P/RT$ ). More molecules strike the liquid surface per unit time.
- This temporarily **increases the rate of dissolution** over the rate of escape, so **more gas dissolves** until a new equilibrium is established at a higher concentration.
- The new equilibrium concentration is **directly proportional** to the new pressure — this is exactly Henry's Law:  $C \propto P$ .



##### Exception to Henry's Law:

Henry's Law applies only to **ideal, dilute solutions** of gases that do not react with the solvent. It **fails** when:

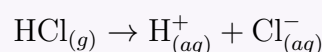
- Gas reacts chemically with the solvent:** When a gas undergoes a chemical reaction with the solvent, more gas dissolves than predicted by Henry's Law because dissolution is aided by the chemical reaction.

*Examples:*

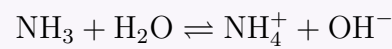
- CO<sub>2</sub> dissolving in water forms carbonic acid:



- HCl gas dissolving in water ionizes completely:



- NH<sub>3</sub> dissolves in water and partially ionizes:



In all these cases, the solubility is **much greater** than Henry's Law would predict at the given partial pressure.

- At high pressures:** At very high pressures, gas behaviour deviates from ideal, and Henry's Law no longer holds accurately.
- At high concentrations:** Henry's Law is strictly valid only in the **dilute limit**. At high gas concentrations, solute–solute interactions become significant and the law breaks down.

**Q2.** An aqueous solution is 0.907 M Pb(NO<sub>3</sub>)<sub>2</sub>. What is the molality of lead(II) nitrate in this solution? The density of the solution is 1.252 g/mL. **(10 marks)** [CHEM 113 | 2023–24]

### Answer

#### Calculation of Molality of Pb(NO<sub>3</sub>)<sub>2</sub> from Molarity

**Given:**

- Molarity of  $\text{Pb}(\text{NO}_3)_2$ :  $M = 0.907 \text{ mol/L}$
- Density of solution:  $\rho = 1.252 \text{ g/mL} = 1252 \text{ g/L}$
- Molar mass of  $\text{Pb}(\text{NO}_3)_2$ :  $= 207.2 + 2(14.0 + 48.0) = 207.2 + 124.0 = 331.2 \text{ g/mol}$

**Step 1: Mass of 1 litre of solution**

$$m_{\text{soln}} = \rho \times V = 1252 \text{ g/L} \times 1 \text{ L} = 1252 \text{ g}$$

**Step 2: Mass of solute ( $\text{Pb}(\text{NO}_3)_2$ ) in 1 L**

$$m_{\text{solute}} = M \times M_r = 0.907 \text{ mol} \times 331.2 \text{ g/mol} = 300.4 \text{ g}$$

**Step 3: Mass of solvent (water) in 1 L**

$$m_{\text{solvent}} = m_{\text{soln}} - m_{\text{solute}} = 1252 - 300.4 = 951.6 \text{ g} = 0.9516 \text{ kg}$$

**Step 4: Calculate molality**

$$m = \frac{\text{moles of solute}}{\text{mass of solvent in kg}} = \frac{0.907 \text{ mol}}{0.9516 \text{ kg}}$$

$$m = 0.953 \text{ mol/kg} = 0.953 \text{ m}$$

**Q3.** What are colligative properties? Write the equations relating each of these properties to the concentration of a non-electrolyte solution. (5 marks) [CHEM 113 | 2022–23]

**Answer**
**Colligative Properties:**

Colligative properties are physical properties of a solution that depend solely on the **number of solute particles** present, not on the chemical identity of the solute. They arise because adding a solute changes the thermodynamic activity of the solvent.

The four colligative properties and their governing equations for a **non-electrolyte solution** are:

**1. Vapour Pressure Lowering ( $\Delta P$ ):**

Given by **Raoult's Law**:

$$\Delta P = X_{\text{solute}} \cdot P_{\text{solvent}}^{\circ}$$

where  $X_{\text{solute}}$  is the mole fraction of solute and  $P_{\text{solvent}}^{\circ}$  is the vapour pressure of pure solvent.

**2. Boiling Point Elevation ( $\Delta T_b$ ):**

$$\Delta T_b = K_b \cdot m$$

where  $K_b$  is the **ebullioscopic constant** (boiling point elevation constant) of the solvent (e.g.  $K_b = 0.52 \text{ }^{\circ}\text{C/m}$  for water) and  $m$  is the molality of the solution.

**3. Freezing Point Depression ( $\Delta T_f$ ):**

$$\Delta T_f = K_f \cdot m$$

where  $K_f$  is the **cryoscopic constant** (freezing point depression constant) of the solvent (e.g.  $K_f = 1.86 \text{ }^{\circ}\text{C/m}$  for water) and  $m$  is the molality.

**4. Osmotic Pressure ( $\pi$ ):**

Given by the **van't Hoff equation**:

$$\pi = MRT$$

where  $M$  is the **molarity** of the solution (mol/L),  $R = 0.0821 \text{ L}\cdot\text{atm/mol}\cdot\text{K}$  is the gas constant, and  $T$  is the absolute temperature (K).

**Summary Table:**

| Property                  | Equation                   | Variables                       |
|---------------------------|----------------------------|---------------------------------|
| Vapour pressure lowering  | $\Delta P = X_s P^{\circ}$ | $X_s$ = mole fraction of solute |
| Boiling point elevation   | $\Delta T_b = K_b m$       | $K_b$ = ebullioscopic constant  |
| Freezing point depression | $\Delta T_f = K_f m$       | $K_f$ = cryoscopic constant     |
| Osmotic pressure          | $\pi = MRT$                | $M$ = molarity, $T$ = temp (K)  |

**Conclusion:** All four equations show that the magnitude of each colligative effect scales **linearly** with the concentration of solute particles — confirming that it is the *quantity*, not the *identity*, of solute that matters.



**Q4.** Ethylene glycol (EG), CH<sub>2</sub>(OH)CH<sub>2</sub>(OH), is a common automobile antifreeze. Calculate the freezing point of a solution containing 724 g of this substance in 2603 g of water. ( $K_f = 1.86\text{ }^\circ\text{C/m}$ ,  $K_b = 0.52\text{ }^\circ\text{C/m}$ , molar mass of EG = 62.07 g.) **(10 marks)** [CHEM 113 | 2022–23]

### Answer

#### Given:

- Solute: Ethylene glycol (EG),  $\mathcal{M} = 62.07\text{ g/mol}$
- Mass of EG:  $w_{\text{solute}} = 724\text{ g}$
- Mass of water:  $w_{\text{solvent}} = 2603\text{ g}$
- $K_f(\text{water}) = 1.86\text{ }^\circ\text{C/m}$
- $K_b(\text{water}) = 0.52\text{ }^\circ\text{C/m}$
- EG is a **non-electrolyte** ( $i = 1$ )

#### Step 1 — Moles of ethylene glycol:

$$n_{\text{EG}} = \frac{724}{62.07} = 11.666\text{ mol}$$

#### Step 2 — Molality of the solution:

$$m = \frac{n_{\text{solute}}}{m_{\text{solvent (kg)}}} = \frac{11.666}{2.603} = 4.482\text{ mol/kg}$$

#### Step 3 — Freezing point depression:

$$\Delta T_f = K_f \times m = 1.86 \times 4.482 = 8.34\text{ }^\circ\text{C}$$

#### Step 4 — New freezing point:

$$T_f = T_f^\circ - \Delta T_f = 0\text{ }^\circ\text{C} - 8.34\text{ }^\circ\text{C}$$

$$\boxed{T_f = -8.34\text{ }^\circ\text{C}}$$

#### Bonus — Boiling point elevation:

$$\Delta T_b = K_b \times m = 0.52 \times 4.482 = 2.33\text{ }^\circ\text{C}$$

$$T_b = 100 + 2.33 = 102.33\text{ }^\circ\text{C}$$

**Conclusion:** The ethylene glycol solution freezes at  $-8.34\text{ }^\circ\text{C}$ , well below the freezing point of pure water. This is why EG is used as an **antifreeze** in automobile radiators — it protects the engine coolant from freezing in cold weather.

**Q5.** A solution is prepared by dissolving 35.0 g of hemoglobin in enough water to make up 1 L. If the osmotic pressure of the solution is found to be 10.0 mmHg at  $25^\circ\text{C}$ , calculate the molar mass of hemoglobin. **(10 marks)** [CHEM 113 | 2022–23]

### Answer

#### Given:

- Mass of hemoglobin:  $w = 35.0\text{ g}$
- Volume of solution:  $V = 1\text{ L}$
- Osmotic pressure:  $\pi = 10.0\text{ mmHg}$
- Temperature:  $T = 25^\circ\text{C} = 298\text{ K}$
- $R = 0.0821\text{ L}\cdot\text{atm/mol}\cdot\text{K}$

#### Step 1 — Convert osmotic pressure to atm:

$$\pi = \frac{10.0\text{ mmHg}}{760\text{ mmHg/atm}} = 1.316 \times 10^{-2}\text{ atm}$$

#### Step 2 — Find molarity using van't Hoff equation:

$$\pi = MRT \quad \Rightarrow \quad M = \frac{\pi}{RT}$$

$$M = \frac{1.316 \times 10^{-2}}{0.0821 \times 298} = \frac{1.316 \times 10^{-2}}{24.47} = 5.378 \times 10^{-4}\text{ mol/L}$$

**Step 3 — Find moles of hemoglobin in 1 L:**

$$n = M \times V = 5.378 \times 10^{-4} \times 1 = 5.378 \times 10^{-4} \text{ mol}$$

**Step 4 — Calculate molar mass:**

$$\mathcal{M} = \frac{w}{n} = \frac{35.0}{5.378 \times 10^{-4}}$$

$$\mathcal{M} = 6.508 \times 10^4 \text{ g/mol} \approx 65,080 \text{ g/mol}$$

**Conclusion:** The molar mass of hemoglobin is approximately  $6.5 \times 10^4$  g/mol. This result demonstrates why **osmotic pressure** is the preferred colligative property for determining molar masses of **macromolecules** — even a very small concentration of a large molecule produces a measurable osmotic pressure, whereas freezing point depression or boiling point elevation would be too small to measure accurately.

**Q6.** Define boiling-point elevation and freezing-point depression of a dilute solution. Explain why molality is used for boiling-point elevation and freezing-point depression calculations and molarity is used for osmotic pressure calculations. **(13 marks)** [CHEM 113 | 2021–22]

### Answer

#### 1. Boiling Point Elevation ( $\Delta T_b$ ):

When a non-volatile solute is dissolved in a solvent, the **vapour pressure** of the solution becomes lower than that of the pure solvent (Raoult's law). Consequently, a higher temperature is required to raise the vapour pressure of the solution to the external atmospheric pressure (1 atm). This increase in boiling point is called **boiling point elevation**:

$$\Delta T_b = T_b - T_b^\circ > 0$$

$$\Delta T_b = K_b \cdot m$$

where  $K_b$  is the ebullioscopic constant and  $m$  is the molality.

#### 2. Freezing Point Depression ( $\Delta T_f$ ):

When a solute is dissolved, it disrupts the formation of the **crystal lattice** of the solvent, so a lower temperature is required to initiate solidification. This lowering of the freezing point is called **freezing point depression**:

$$\Delta T_f = T_f^\circ - T_f > 0$$

$$\Delta T_f = K_f \cdot m$$

where  $K_f$  is the cryoscopic constant and  $m$  is the molality.

#### 3. Why Molality is Used for $\Delta T_b$ and $\Delta T_f$ :

**Molality ( $m$ )** is defined as moles of solute per **kilogram of solvent**:

$$m = \frac{n_{\text{solute}}}{m_{\text{solvent}} \text{ (kg)}}$$

It is used for boiling and freezing point calculations because:

- Boiling and freezing points are measured at **varying temperatures**. Molarity ( $M$ ) depends on volume, and volume **changes with temperature** (liquids expand when heated, contract when cooled). Molality is based on **mass**, which does *not* change with temperature — making it **temperature-independent**.
- Since  $\Delta T_b$  and  $\Delta T_f$  are measured as the solution changes temperature (from room temperature to boiling or freezing point), using a temperature-independent concentration unit ensures accurate and consistent results.

#### 4. Why Molarity is Used for Osmotic Pressure:

**Molarity ( $M$ )** is defined as moles of solute per **litre of solution**:

$$M = \frac{n_{\text{solute}}}{V_{\text{solution}} \text{ (L)}}$$

It is used in the van't Hoff osmotic pressure equation:

$$\pi = MRT$$

because:

- Osmotic pressure is measured at **constant temperature** (isothermal process), so the temperature-dependence of volume is not a concern.
- Osmotic pressure depends on the number of solute particles per unit **volume** of solution — not per unit mass of solvent. Molarity directly gives the number of moles per litre, making it the natural concentration unit for a **volume-based** pressure measurement.
- The van't Hoff equation  $\pi = MRT$  is analogous to the ideal gas law  $PV = nRT$ , which is also expressed in terms of molar concentration per volume.

**Summary:**

| Property                   | Unit used        | Reason                                                                       |
|----------------------------|------------------|------------------------------------------------------------------------------|
| $\Delta T_b, \Delta T_f$   | Molality ( $m$ ) | Temperature changes during measurement; mass is temperature-independent      |
| Osmotic pressure ( $\pi$ ) | Molarity ( $M$ ) | Measured at constant $T$ ; pressure depends on concentration per unit volume |

**Conclusion:** The choice of molality vs. molarity is dictated by whether the measurement involves **temperature changes** (use molality — mass-based, temperature-independent) or occurs at **constant temperature** (use molarity — volume-based, directly related to particle density per litre).

**Q7.** A 7.85 g sample of a compound with the empirical formula  $C_5H_4$  is dissolved in 301 g of benzene. The freezing point of the solution is  $1.05^\circ\text{C}$  below that of pure benzene. What are the molar mass and molecular formula of this compound? ( $K_f = 5.12^\circ\text{C/m}$ ) **(12 marks)** [CHEM 113 | 2021–22]

**Answer**
**Given:**

- Empirical formula:  $C_5H_4$
- Mass of compound:  $w = 7.85\text{ g}$
- Mass of benzene (solvent):  $w_{\text{solvent}} = 301\text{ g}$
- Freezing point depression:  $\Delta T_f = 1.05^\circ\text{C}$
- $K_f(\text{benzene}) = 5.12^\circ\text{C/m}$

**Step 1 — Find molality from  $\Delta T_f$ :**

$$\Delta T_f = K_f \times m \quad \Rightarrow \quad m = \frac{\Delta T_f}{K_f} = \frac{1.05}{5.12} = 0.2051\text{ mol/kg}$$

**Step 2 — Find moles of solute:**

$$n_{\text{solute}} = m \times w_{\text{solvent (kg)}} = 0.2051 \times 0.301 = 6.173 \times 10^{-2}\text{ mol}$$

**Step 3 — Find molar mass of compound:**

$$\mathcal{M} = \frac{w}{n} = \frac{7.85}{6.173 \times 10^{-2}} = 127.2\text{ g/mol}$$

$$\boxed{\mathcal{M} \approx 128\text{ g/mol}}$$

**Step 4 — Determine molecular formula:**

Empirical formula mass of  $C_5H_4$ :

$$\mathcal{M}_{\text{emp}} = 5(12.01) + 4(1.008) = 60.05 + 4.03 = 64.08\text{ g/mol}$$

Number of empirical units per molecule:

$$n = \frac{\mathcal{M}_{\text{molecular}}}{\mathcal{M}_{\text{empirical}}} = \frac{128}{64.08} \approx 2$$

Molecular formula:

$$(C_5H_4)_2 = \boxed{C_{10}H_8}$$

**Identification:**  $C_{10}H_8$  with molar mass 128 g/mol is **naphthalene** — a polycyclic aromatic hydrocarbon consisting of two fused benzene rings, commonly used in mothballs.

**Conclusion:** The molar mass of the compound is  $\approx 128\text{ g/mol}$  and its molecular formula is  **$C_{10}H_8$**  (naphthalene), obtained by doubling the empirical formula  $C_5H_4$ .

**Q8.** Define ‘thermal pollution’. How can you relate the solubility of gas in liquid with thermal pollution? **(10 marks)** [CHEM 113 | 2020–21 (Jan 21)]

### Answer

#### Thermal Pollution and Solubility of Gas in Liquid

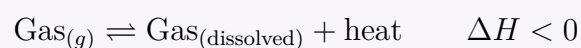
##### Part 1: Definition of Thermal Pollution

**Thermal pollution** is the **degradation of water quality** caused by any process that changes the **ambient temperature of a natural water body** (river, lake, ocean). The most common cause is the discharge of **heated water** used as a coolant in:

- Power plants (nuclear and thermal)
- Industrial manufacturing plants
- Petroleum refineries

##### Part 2: Effect of Temperature on Gas Solubility

The solubility of a gas in a liquid is governed by **Le Chatelier's Principle**. The dissolution of a gas is generally **exothermic**:



- **Increasing temperature** shifts the equilibrium to the **left** (toward the gas phase), **decreasing** the solubility of the gas.
- **Decreasing temperature** increases gas solubility.

| Temperature | O <sub>2</sub> solubility (mg/L) |
|-------------|----------------------------------|
| 0°C         | 14.6                             |
| 10°C        | 11.3                             |
| 20°C        | 9.1                              |
| 30°C        | 7.6                              |
| 40°C        | 6.4                              |

##### Part 3: Relation Between Thermal Pollution and Gas Solubility

- When **hot water** is discharged into a river or lake from a power plant, the temperature of the water body **rises significantly**.
- Due to the inverse relationship between temperature and gas solubility, the **dissolved oxygen (DO)** content of the water **decreases sharply**.
- Aquatic organisms (fish, algae, microorganisms) require a minimum level of dissolved oxygen to survive. A drop in DO below critical levels causes:
  - **Fish kills** and death of aquatic life.
  - **Algal blooms** (warm water favours algae).
  - **Eutrophication** — depletion of oxygen leads to anaerobic decomposition, producing toxic gases like H<sub>2</sub>S and CH<sub>4</sub>.
  - Disruption of the **aquatic ecosystem**.
- Additionally, elevated temperatures **accelerate metabolic rates** of aquatic organisms, increasing their oxygen demand at the very time when DO is decreasing — a doubly harmful effect.

**Conclusion:** Thermal pollution directly reduces the solubility of oxygen and other dissolved gases in water, leading to serious ecological damage. Controlling the temperature of discharged water (e.g. using cooling towers) is essential to prevent thermal pollution.

**Q9.** Calculate the freezing point of a solution containing 0.135 mol of a nonionic solute in 2.50 kg of benzene.  $k_f = 5.12^\circ\text{C}/\text{m}$ . **(5 marks)**  
[CHEM 113 | 2020–21 (Jan 21)]

### Answer

#### Freezing Point Depression Calculation

**Given:**

- Moles of solute:  $n = 0.135 \text{ mol}$
- Mass of benzene (solvent):  $w = 2.50 \text{ kg}$
- Cryoscopic constant of benzene:  $k_f = 5.12^\circ\text{C}/\text{m}$
- Normal freezing point of benzene:  $T_f^* = 5.50^\circ\text{C}$

##### Step 1: Calculate molality of the solution

$$m = \frac{n}{w} = \frac{0.135 \text{ mol}}{2.50 \text{ kg}} = 0.0540 \text{ mol/kg}$$

##### Step 2: Calculate depression in freezing point

$$\Delta T_f = k_f \times m = 5.12^\circ\text{C}/\text{m} \times 0.0540 \text{ m} = 0.277^\circ\text{C}$$

**Step 3: Calculate new freezing point**

$$T_f = T_f^* - \Delta T_f = 5.50 - 0.277$$

$$T_f = 5.22^\circ\text{C}$$

**Q10.** How can you account for the abnormal molar mass from the measurement of colligative properties of electrolytes? Derive an equation showing the relationship between the degree of dissociation of electrolytes with Van't Hoff factor. **(20 marks)** [CHEM 113 | 2020–21 (Jan 21)]

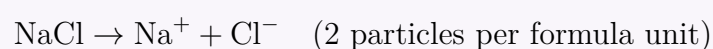
### Answer

#### Abnormal Molar Mass, Van't Hoff Factor and Degree of Dissociation

##### Part 1: Abnormal Molar Mass from Colligative Properties of Electrolytes

Colligative properties (boiling point elevation, freezing point depression, osmotic pressure) depend on the **number of solute particles** in solution, not their chemical identity.

For **non-electrolytes**, the number of particles equals the number of formula units dissolved. However, **electrolytes dissociate** in solution, producing **more particles** than expected:



When colligative properties are measured for electrolytes and used to calculate molar mass (assuming no dissociation), the **calculated molar mass is lower** than the true molar mass. This is called **abnormal molar mass**.

##### Van't Hoff Factor (*i*):

To correct for this, Van't Hoff introduced the factor *i*:

$$i = \frac{\text{observed colligative property}}{\text{calculated colligative property (no dissociation)}} = \frac{\text{actual number of particles}}{\text{number of formula units dissolved}}$$

Modified colligative property equations become:

$$\Delta T_f = i \cdot k_f \cdot m \quad \Delta T_b = i \cdot k_b \cdot m \quad \pi = i \cdot MRT$$

- For **non-electrolytes**:  $i = 1$
- For **electrolytes**:  $i > 1$  (dissociation produces more particles)
- For **associated solutes**:  $i < 1$  (e.g. acetic acid in benzene dimerises)

The **true molar mass** is related to the **apparent molar mass** by:

$$M_{\text{true}} = i \times M_{\text{apparent}}$$

##### Part 2: Relationship Between Degree of Dissociation and Van't Hoff Factor

Consider an electrolyte  $\text{A}_x\text{B}_y$  that dissociates as:



Let:

- $\alpha$  = degree of dissociation (fraction of molecules dissociated)
- $n$  = total number of ions produced per formula unit =  $x + y$
- Initial moles = 1

|                | $\text{A}_x\text{B}_y$ | $\text{A}^{n+}$ | $\text{B}^{m-}$ |
|----------------|------------------------|-----------------|-----------------|
| Initial moles  | 1                      | 0               | 0               |
| At equilibrium | $1 - \alpha$           | $x\alpha$       | $y\alpha$       |

Total moles at equilibrium:

$$= (1 - \alpha) + x\alpha + y\alpha = 1 - \alpha + n\alpha = 1 + (n - 1)\alpha$$

The Van't Hoff factor *i* equals the ratio of actual to initial moles:

$$i = \frac{1 + (n - 1)\alpha}{1} = 1 + (n - 1)\alpha$$

Solving for  $\alpha$ :

$$\alpha = \frac{i - 1}{n - 1}$$

**Example:** NaCl ( $n = 2$ )

If  $i = 1.85$  for NaCl solution:

$$\alpha = \frac{1.85 - 1}{2 - 1} = \frac{0.85}{1} = 0.85$$

So 85% of NaCl is dissociated in this solution.

**Example: K<sub>2</sub>SO<sub>4</sub> ( $n = 3$ )**

If  $i = 2.60$ :

$$\alpha = \frac{2.60 - 1}{3 - 1} = \frac{1.60}{2} = 0.80$$

So 80% of K<sub>2</sub>SO<sub>4</sub> is dissociated.

**Summary:**

| Electrolyte                               | $n$ | $\alpha$ formula     |
|-------------------------------------------|-----|----------------------|
| AB (e.g. NaCl)                            | 2   | $\alpha = i - 1$     |
| AB <sub>2</sub> (e.g. CaCl <sub>2</sub> ) | 3   | $\alpha = (i - 1)/2$ |
| A <sub>2</sub> B <sub>3</sub>             | 5   | $\alpha = (i - 1)/4$ |

**Q11.** The deviation of Raoult's law is associated with the unequal adhesive and cohesive forces — Justify with explanation and suitable diagram. **(25 marks)** [CHEM 113 | 2020–21 (Jan 21)]

### Answer

#### Deviation from Raoult's Law Due to Unequal Adhesive and Cohesive Forces

##### Key Definitions:

- **Cohesive forces:** Intermolecular attractions between **like molecules** (A–A or B–B). They hold the pure liquid together.
- **Adhesive forces:** Intermolecular attractions between **unlike molecules** (A–B). They hold the components of the mixture together.

##### Ideal Solution (No Deviation):

When adhesive forces = cohesive forces:

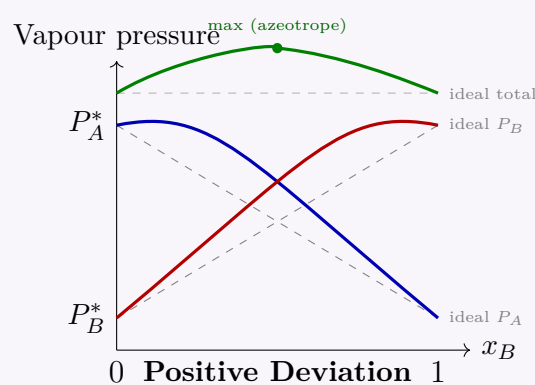
$$A-B = A-A = B-B$$

Raoult's Law is obeyed exactly.

##### Case 1: Positive Deviation (Adhesive < Cohesive)

When A–B interactions are **weaker** than A–A and B–B:

- Unlike molecules are held together less strongly than in pure components.
- Molecules escape to vapour phase more easily  $\Rightarrow$  vapour pressure **increases above Raoult's Law prediction**.
- The total pressure curve shows a **maximum** (positive azeotrope forms).
- $\Delta H_{\text{mix}} > 0$ ,  $\Delta V_{\text{mix}} > 0$ .
- **Example:** Ethanol–water, benzene–ethanol.

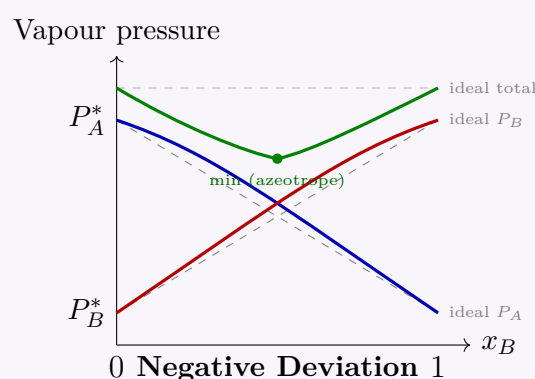


##### Case 2: Negative Deviation (Adhesive > Cohesive)

When A–B interactions are **stronger** than A–A and B–B:

- Unlike molecules are held together more strongly than in pure components.
- Molecules escape to vapour phase less easily  $\Rightarrow$  vapour pressure **falls below Raoult's Law prediction**.
- The total pressure curve shows a **minimum** (negative azeotrope forms).
- $\Delta H_{\text{mix}} < 0$ ,  $\Delta V_{\text{mix}} < 0$ .
- **Example:** Acetone–chloroform, HNO<sub>3</sub>–water.





Complete Summary Table:

| Property                | Positive Deviation   | Negative Deviation  |
|-------------------------|----------------------|---------------------|
| Force comparison        | Adhesive < Cohesive  | Adhesive > Cohesive |
| Vapour pressure         | Above ideal (higher) | Below ideal (lower) |
| $\Delta H_{\text{mix}}$ | > 0 (endothermic)    | < 0 (exothermic)    |
| $\Delta V_{\text{mix}}$ | > 0                  | < 0                 |
| Azeotrope type          | Maximum boiling pt.  | Minimum boiling pt. |
| Example                 | Ethanol–water        | Acetone–chloroform  |

**Q12.** Bends (decompression sickness) is a medical condition for divers. Relate this with the effect of pressure on the solubility of gas in liquid. **(12 marks)** [CHEM 113 | 2020–21]

### Answer

#### Decompression Sickness (“The Bends”):

Decompression sickness (DCS), commonly called “**the bends**,” is a medical condition that affects deep-sea divers when they ascend too rapidly from high-pressure depths to the surface. It is directly explained by **Henry’s Law** and the effect of pressure on gas solubility in liquids.

#### 1. Henry’s Law:

The amount of gas dissolved in a liquid is **directly proportional** to the partial pressure of that gas above the liquid:

$$C = k_H \cdot P$$

where  $C$  is the concentration of dissolved gas,  $P$  is the partial pressure of the gas, and  $k_H$  is Henry’s law constant (specific to each gas–solvent pair at a given temperature).

#### 2. What Happens During a Dive:

- At depth, the diver breathes compressed air at **high pressure**. By Henry’s law, the increased partial pressure of nitrogen ( $\text{N}_2$ ) forces significantly more  $\text{N}_2$  to **dissolve into the blood and tissues**.
- At 40 m depth, pressure  $\approx 5$  atm — roughly 5 times more  $\text{N}_2$  dissolves compared to the surface.
- While at depth, this extra dissolved  $\text{N}_2$  causes no immediate harm.

#### 3. What Happens During Rapid Ascent:

- If the diver ascends **too quickly**, the external pressure drops rapidly. By Henry’s law, the solubility of  $\text{N}_2$  decreases sharply.
- The excess dissolved  $\text{N}_2$  cannot diffuse out of the tissues fast enough — it **comes out of solution** and forms **bubbles** within the blood vessels, joints, and tissues.
- This is analogous to opening a carbonated drink rapidly — the sudden pressure drop causes dissolved  $\text{CO}_2$  to form bubbles.

#### 4. Medical Consequences:

The  $\text{N}_2$  bubbles cause:

- Joint pain** (“the bends”) — bubbles in synovial fluid
- Neurological damage** — bubbles blocking blood flow to the brain or spinal cord
- Pulmonary embolism** — bubbles in lung capillaries
- In severe cases: **paralysis or death**

#### 5. Treatment and Prevention:

- Slow ascent:** allows  $\text{N}_2$  to diffuse out gradually through the lungs

- **Decompression stops:** pausing at intermediate depths (e.g. 5 m for several minutes) to allow controlled outgassing
- **Hyperbaric oxygen therapy:** placing the diver in a **recompression chamber** raises pressure again to re-dissolve the bubbles, then slowly reduces pressure to allow safe outgassing
- **Helium–oxygen (Heliox) mixtures:** He has a lower  $k_H$  than N<sub>2</sub> and diffuses out faster, reducing risk in deep dives

#### Summary — Henry's Law Connection:

| Condition       | Pressure     | N <sub>2</sub> solubility                |
|-----------------|--------------|------------------------------------------|
| At depth (40 m) | ~5 atm       | High — N <sub>2</sub> dissolves in blood |
| At surface      | ~1 atm       | Low — N <sub>2</sub> must leave solution |
| Rapid ascent    | Drops fast   | Sudden bubble formation (DCS)            |
| Slow ascent     | Drops slowly | Safe outgassing via lungs                |

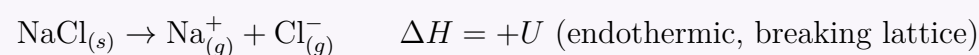
**Conclusion:** The bends is a direct consequence of **Henry's Law** — high pressure dissolves dangerous amounts of N<sub>2</sub> into the bloodstream, and rapid decompression causes it to bubble out violently. Controlled slow ascent allows the body to safely eliminate the excess dissolved gas.

**Q13.** Define lattice energy and hydration energy for ionic compounds. Explain that both lattice and hydration energy mutually control the solubility of ionic crystals in polar solvent. (15 marks) [CHEM 113 | 2020–21]

#### Answer

##### 1. Lattice Energy:

**Lattice energy** ( $U$ ) is the energy **released** when gaseous ions combine to form one mole of a solid ionic crystal, or equivalently, the energy **required** to completely separate one mole of ionic solid into its constituent gaseous ions:



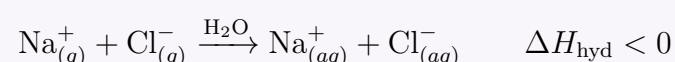
Lattice energy depends on:

- **Ionic charge** ( $q$ ): higher charge  $\Rightarrow$  stronger attraction  $\Rightarrow$  higher  $U$
- **Ionic radius** ( $r$ ): smaller ions  $\Rightarrow$  closer contact  $\Rightarrow$  higher  $U$

$$U \propto \frac{q^{+}q^{-}}{r^{+} + r^{-}}$$

##### 2. Hydration Energy:

**Hydration energy** ( $\Delta H_{\text{hyd}}$ ) is the energy **released** when gaseous ions are surrounded by water molecules (solvated):

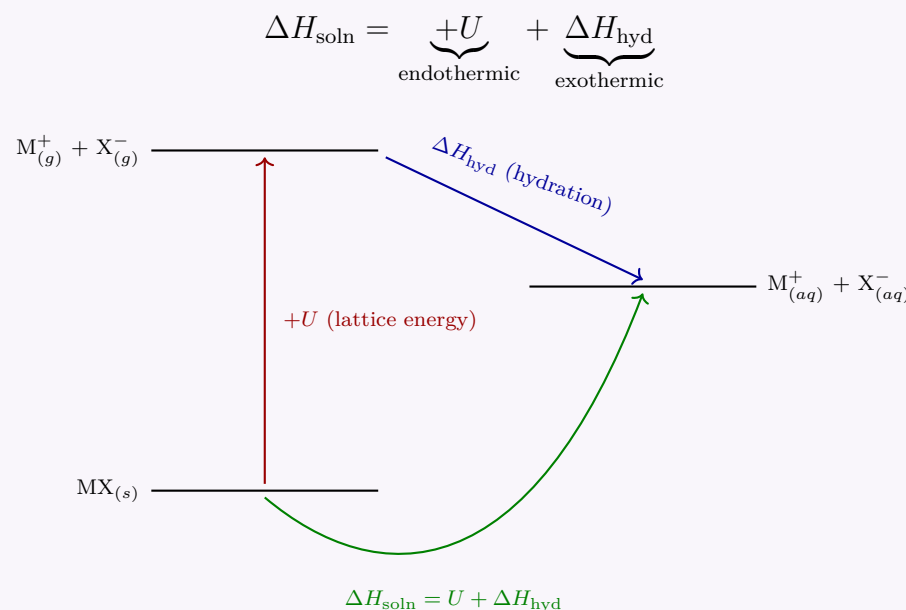


It depends on:

- **Ionic charge:** higher charge  $\Rightarrow$  stronger ion–dipole interaction  $\Rightarrow$  more negative  $\Delta H_{\text{hyd}}$
- **Ionic radius:** smaller ions  $\Rightarrow$  closer water approach  $\Rightarrow$  stronger hydration

##### 3. How They Mutually Control Solubility:

The dissolution of an ionic crystal in water involves two competing energy steps, summarised by the **Born–Haber cycle for dissolution**:



The solubility outcome depends on which energy dominates:

- If  $|\Delta H_{\text{hyd}}| > U$ :  $\Delta H_{\text{soln}} < 0$  (exothermic)  $\Rightarrow$  **highly soluble** (e.g. LiF, NaOH)
- If  $|\Delta H_{\text{hyd}}| \approx U$ :  $\Delta H_{\text{soln}} \approx 0 \Rightarrow$  **moderately soluble** (e.g. NaCl,  $\Delta H_{\text{soln}} = +3.9$  kJ/mol)
- If  $|\Delta H_{\text{hyd}}| \ll U$ :  $\Delta H_{\text{soln}} \gg 0$  (highly endothermic)  $\Rightarrow$  **insoluble** (e.g. MgO, BaSO<sub>4</sub>)

#### 4. Examples:

| Salt | $U$      | $\Delta H_{\text{hyd}}$ | $\Delta H_{\text{soln}}$ | Solubility |
|------|----------|-------------------------|--------------------------|------------|
| NaCl | +788 kJ  | −784 kJ                 | +4 kJ                    | Moderate   |
| LiF  | +1037 kJ | −1046 kJ                | −9 kJ                    | High       |
| MgO  | +3795 kJ | −2686 kJ                | +1109 kJ                 | Very low   |

**Conclusion:** Solubility of ionic crystals in polar solvents is controlled by the **competition** between lattice energy (resisting dissolution) and hydration energy (driving dissolution). When hydration energy is sufficient to overcome the lattice energy, the compound dissolves readily; when it is not, the compound remains insoluble.

**Q14.** The ecological imbalance in aquatic systems is one of the consequences of thermal pollution — Explain. **(8 marks)** [CHEM 113 | 2020–21]

#### Answer

##### Thermal Pollution and Aquatic Ecological Imbalance:

**Thermal pollution** is the degradation of water quality by any process that changes the ambient water temperature — most commonly caused by industrial facilities (power plants, steel mills) discharging **heated effluent water** into rivers, lakes, or oceans.

##### 1. Decreased Dissolved Oxygen (DO):

The most critical consequence is the **reduction in dissolved oxygen**. By Henry's Law:

$$C_{\text{O}_2} = k_H \cdot P_{\text{O}_2}$$

Gas solubility decreases with increasing temperature. As water temperature rises:

- Oxygen molecules gain kinetic energy and **escape from solution** more easily
- DO drops from ~9 mg/L at 20°C to ~7 mg/L at 30°C
- Fish and aerobic organisms **suffocate** as DO falls below their survival threshold (<5 mg/L for most fish)

##### 2. Accelerated Metabolic Rates:

Higher temperature increases the metabolic rate of **cold-blooded (ectothermic)** organisms (fish, amphibians, invertebrates):

- Increased oxygen demand at precisely the time when DO is lowest
- Disruption of **reproductive cycles** — temperature cues for spawning are altered
- **Thermal shock** — sudden temperature changes (>2–3°C) can directly kill sensitive species

##### 3. Algal Blooms and Eutrophication:

- Warm water **accelerates algal growth**, especially cyanobacteria (blue-green algae)
- Dense algal blooms block sunlight, killing submerged aquatic plants
- When algae die, their decomposition by bacteria consumes large amounts of DO (**biological oxygen demand, BOD** increases)
- This **eutrophication** creates **dead zones** — hypoxic regions where no aerobic life can survive

##### 4. Disruption of Food Chain:

- Loss of **thermally sensitive species** (trout, salmon) removes key predators
- **Invasive warm-water species** outcompete native organisms, reducing biodiversity
- Disruption of **phytoplankton** populations affects the entire aquatic food web from the base upward

##### 5. Increased Toxicity of Pollutants:

Many dissolved toxins (heavy metals, pesticides) become **more toxic at higher temperatures** — their reaction rates with biological tissues increase, compounding the ecological damage.

**Summary:**

| Effect                | Ecological Consequence                      |
|-----------------------|---------------------------------------------|
| ↓ DO                  | Fish kill; suffocation of aerobic organisms |
| ↑ Metabolic rate      | Increased $O_2$ demand; spawning disruption |
| Algal blooms          | Eutrophication; dead zones; light blockage  |
| Food chain disruption | Loss of biodiversity; invasive species      |
| ↑ Toxin activity      | Enhanced chemical harm to aquatic life      |

**Conclusion:** Thermal pollution creates a cascading **ecological imbalance** in aquatic systems — primarily by reducing dissolved oxygen through decreased gas solubility, while simultaneously increasing the oxygen demand of organisms. The result is a degraded, species-poor environment that can collapse into dead zones if thermal loading is sustained.

**Q15.** Compare the vapor pressure theory and membrane solution theory of the osmosis process through the semipermeable membrane. (12 marks) [CHEM 113 | 2020–21]

### Answer

**Osmosis** is the spontaneous flow of solvent molecules through a **semipermeable membrane** from a region of lower solute concentration (higher solvent activity) to a region of higher solute concentration (lower solvent activity). Two theories explain the mechanism of this process:

#### 1. Vapour Pressure Theory (Membrane Solution Theory of Vapour Pressure):

This theory was proposed by **Morse and Frazer** and is based on differences in vapour pressure across the membrane.

##### Postulates:

- The semipermeable membrane allows solvent molecules to pass but blocks solute molecules.
- The vapour pressure of **pure solvent** ( $P^\circ$ ) is always greater than the vapour pressure of the **solution** ( $P_s$ ):

$$P^\circ > P_s \quad (\text{Raoult's Law})$$

- Solvent molecules on the pure solvent side have higher escaping tendency (vapour pressure).
- Solvent molecules evaporate from the pure solvent side, pass through the vapour phase or membrane, and condense on the solution side.
- This continues until equilibrium is reached or osmotic pressure ( $\pi$ ) is applied to stop the flow.

**Limitation:** This theory works well for explaining osmosis through *porous* membranes but is less satisfactory for explaining osmosis through *non-porous* biological membranes.

#### 2. Membrane Solution Theory (Pfeffer–de Vries Theory):

This theory treats the semipermeable membrane itself as playing an **active chemical role** in the osmosis process.

##### Postulates:

- The semipermeable membrane is not merely a physical barrier but **dissolves the solvent** on one side and **releases it** on the other.
- Solvent molecules from the pure solvent side **dissolve into** the membrane material (e.g. cellulose, lipid bilayer) and migrate through it down a **chemical potential gradient**.
- On the solution side, solvent molecules are **released** into the solution.
- The driving force is the difference in **chemical potential** ( $\mu$ ) of the solvent across the membrane:

$$\mu_{\text{pure solvent}} > \mu_{\text{solution}}$$

**Advantage:** This theory better explains osmosis through *biological membranes* (e.g. cell walls, lipid bilayers) where the membrane actively interacts with solvent molecules.

#### 3. Comparison:

| Feature          | Vapour Pressure Theory                    | Membrane Solution Theory                      |
|------------------|-------------------------------------------|-----------------------------------------------|
| Driving force    | Vapour pressure difference ( $\Delta P$ ) | Chemical potential difference ( $\Delta\mu$ ) |
| Role of membrane | Passive physical barrier                  | Active chemical participant                   |
| Mechanism        | Evaporation–condensation through pores    | Dissolution–diffusion through membrane        |
| Best applies to  | Porous inorganic membranes                | Biological and dense membranes                |
| Osmotic pressure | Related to $\Delta P$ by Raoult's law     | Related to $\Delta\mu$ by thermodynamics      |

**Conclusion:** Both theories agree that the fundamental driving force of osmosis is the difference in **thermodynamic activity of the solvent** across the membrane. The vapour pressure theory is more intuitive for porous membranes, while the membrane solution theory more accurately describes transport through biological membranes where solvent–membrane interactions are critical.

**Q16.** Identify the reason for the use of van't Hoff factor  $i$  in the mathematical expressions of colligative properties and identify the conditions when  $i$  (i) is equal to one (ii) less than one and (iii) greater than one. **(15 marks)** [CHEM 113 | 2020–21]

### Answer

#### The van't Hoff Factor ( $i$ ):

The van't Hoff factor  $i$  is a correction factor introduced to account for the **actual number of particles** produced when a solute dissolves, compared to the number of formula units added. It modifies all colligative property equations:

$$\Delta T_f = iK_f m, \quad \Delta T_b = iK_b m, \quad \pi = iMRT, \quad \Delta P = iX_s P^\circ$$

#### Definition:

$$i = \frac{\text{actual number of particles in solution}}{\text{number of formula units dissolved}}$$

#### Why $i$ is Needed:

Colligative properties depend on the **total number of dissolved particles**. Different solutes produce different numbers of particles upon dissolution:

- A non-electrolyte (e.g. glucose) dissolves without dissociation — 1 formula unit gives 1 particle.
- An electrolyte (e.g. NaCl) dissociates — 1 formula unit gives 2 ions ( $\text{Na}^+ + \text{Cl}^-$ ).
- Some solutes associate in solution, reducing particle count.

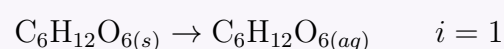
Without  $i$ , predicted colligative effects would be *incorrect* for ionic or associating solutes.

#### Condition (i): $i = 1$

$i = 1$  when the solute **neither dissociates nor associates** in solution — the number of particles in solution equals the number of formula units dissolved.

#### Examples:

- **Non-electrolytes:** glucose ( $\text{C}_6\text{H}_{12}\text{O}_6$ ), urea ( $\text{CO}(\text{NH}_2)_2$ ), sucrose, ethanol
- These dissolve as intact molecules:

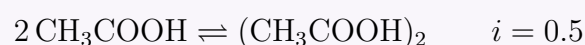


#### Condition (ii): $i < 1$

$i < 1$  when solute particles **associate** in solution — multiple formula units combine to form fewer particles, reducing the total particle count below what was dissolved.

#### Examples:

- **Carboxylic acids in non-polar solvents** (e.g. acetic acid in benzene): molecules form **dimers** via hydrogen bonding:



- **Concentrated electrolyte solutions:** ion-pair formation reduces effective particle count below the ideal dissociation value.

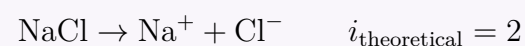
#### Condition (iii): $i > 1$

$i > 1$  when solute **dissociates** in solution — one formula unit splits into multiple ions, increasing the total particle count.

#### Examples:



- **Strong electrolytes** dissociate completely:



- **Weak electrolytes** partially dissociate, giving  $1 < i < 2$ :



**Note:** In practice,  $i$  for strong electrolytes is slightly less than the theoretical value due to **inter-ionic attractions** (ion-ion interactions) that reduce the effective number of independent particles.

**Summary Table:**

| Value of $i$ | Behaviour                     | Examples                                                  |
|--------------|-------------------------------|-----------------------------------------------------------|
| $i = 1$      | No dissociation/association   | Glucose, urea, sucrose                                    |
| $i < 1$      | Association (fewer particles) | Acetic acid in benzene (dimers)                           |
| $i > 1$      | Dissociation (more particles) | NaCl ( $i \approx 2$ ), $\text{CaCl}_2$ ( $i \approx 3$ ) |

**Conclusion:** The van't Hoff factor  $i$  is essential for accurately predicting colligative properties of electrolyte and associating solutions. It corrects the simple colligative equations by accounting for the **true particle count** in solution, which may be greater than, equal to, or less than the number of formula units dissolved.

**Q17.** Scale formation in boiler pipe is nothing but the consequence of solubility of the gas in liquid — justify the statement. **(10 marks)**  
[CHEM 113 | 2019–20]

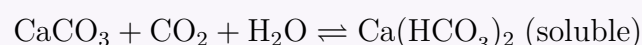
### Answer

#### Scale Formation in Boiler Pipes and Gas Solubility

**The Statement:** “Scale formation in boiler pipes is nothing but the consequence of solubility of gas in liquid.”

**Justification:**

- Natural water contains dissolved gases and salts:** Water used in boilers (natural river or ground water) contains dissolved gases such as  $\text{CO}_2$ ,  $\text{O}_2$ , and dissolved mineral salts such as  $\text{Ca}(\text{HCO}_3)_2$ ,  $\text{Mg}(\text{HCO}_3)_2$ ,  $\text{CaSO}_4$ , and  $\text{MgSO}_4$ .
- Role of dissolved  $\text{CO}_2$ :**  $\text{CO}_2$  dissolves in water to form **carbonic acid**, which keeps calcium carbonate soluble as calcium bicarbonate:



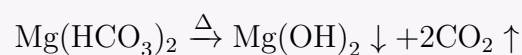
- Effect of heating on gas solubility:** When water is heated in a boiler:

- **Solubility of  $\text{CO}_2$  decreases** sharply (gas solubility decreases with rising temperature — Henry's Law).
- $\text{CO}_2$  is **expelled** from the water.
- Without dissolved  $\text{CO}_2$ , the equilibrium shifts to the **left**:



- **Insoluble  $\text{CaCO}_3$**  (limestone) precipitates and deposits on the inner walls of the boiler pipe as **scale**.

- Similar reactions:**



$\text{MgSO}_4$  may also hydrolyze at high temperatures to deposit  $\text{Mg}(\text{OH})_2$  scale.

- Consequences of scale:**

- Scale has **low thermal conductivity** ( $\sim 40\times$  lower than steel).
- Acts as a **thermal insulator**, reducing heat transfer efficiency.
- Causes **overheating** of boiler walls, leading to pipe damage or explosion.
- Increases **fuel consumption**.

**Conclusion:** Scale formation is directly caused by the **decrease in solubility of  $\text{CO}_2$**  (a gas) in water at elevated temperatures. This drives the precipitation of insoluble  $\text{CaCO}_3$  and  $\text{Mg}(\text{OH})_2$ , forming hard scale deposits. Thus, the problem of scale is indeed a **consequence of the temperature dependence of gas solubility**.



**Q18.** What are the different factors that control the solubility of solid in liquid? The effect of temperature on the solubility of solid in liquid can be applied for the determination of heat of solution of the solute — justify. **(20 marks)** [CHEM 113 | 2019–20]

### Answer

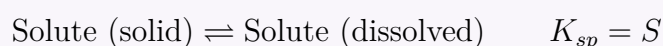
#### Factors Controlling Solubility of Solid in Liquid and Determination of Heat of Solution

##### Part 1: Factors Controlling Solubility of Solid in Liquid

- (i) **Nature of Solute and Solvent (“Like Dissolves Like”):** Polar solutes dissolve in **polar solvents** (e.g. NaCl in water) via strong ion–dipole or dipole–dipole interactions. **Non-polar solutes** dissolve in **non-polar solvents** (e.g. iodine in benzene) via London dispersion forces.
- (ii) **Temperature:**
  - For most **endothermic** solutes ( $\Delta H_{\text{soln}} > 0$ ): solubility **increases** with temperature (Le Chatelier’s principle).
  - For **exothermic** solutes ( $\Delta H_{\text{soln}} < 0$ ): solubility **decreases** with temperature (e.g.  $\text{Ca}(\text{OH})_2$  in water).
  - Example:  $\text{KNO}_3$  shows a **sharp increase** in solubility with temperature.
- (iii) **Pressure:** Pressure has **negligible effect** on the solubility of solids in liquids because solids and liquids are nearly incompressible.
- (iv) **Particle Size:** Finely divided solute has a **larger surface area**, increasing the rate of dissolution. However, particle size does not affect the **equilibrium solubility** significantly.
- (v) **Common Ion Effect:** The presence of a common ion in the solution **decreases** the solubility of a sparingly soluble salt (Le Chatelier’s principle). Example: solubility of  $\text{AgCl}$  decreases in the presence of  $\text{NaCl}$ .
- (vi) **pH of the Solvent:** For solutes that are weak acids or bases, pH significantly affects solubility. For example,  $\text{Ca}_3(\text{PO}_4)_2$  dissolves more in acidic solution.

##### Part 2: Temperature Effect on Solubility Used to Determine Heat of Solution

The solubility equilibrium for a solid solute is:



Applying the **van’t Hoff isochore**:

$$\frac{d \ln K}{dT} = \frac{\Delta H_{\text{soln}}}{RT^2}$$

Integrating between  $T_1$  and  $T_2$ :

$$\ln \frac{S_2}{S_1} = \frac{\Delta H_{\text{soln}}}{R} \left( \frac{1}{T_1} - \frac{1}{T_2} \right)$$

Or in  $\log_{10}$  form:

$$\log \frac{S_2}{S_1} = \frac{\Delta H_{\text{soln}}}{2.303R} \left( \frac{1}{T_1} - \frac{1}{T_2} \right)$$

**Graphical method:**

$$\log S = -\frac{\Delta H_{\text{soln}}}{2.303R} \cdot \frac{1}{T} + C$$

A plot of  $\log S$  vs  $\frac{1}{T}$  is a straight line with slope  $= -\frac{\Delta H_{\text{soln}}}{2.303R}$ , from which:

$$\Delta H_{\text{soln}} = -2.303R \times \text{slope}$$

- **Positive slope**  $\Rightarrow \Delta H_{\text{soln}} < 0$  (exothermic).
- **Negative slope**  $\Rightarrow \Delta H_{\text{soln}} > 0$  (endothermic).

**Justification:** By measuring solubility at different temperatures experimentally (e.g. by evaporation of saturated solutions), plotting  $\log S$  vs  $1/T$ , and measuring the slope, the **heat of solution** can be accurately determined without calorimetry. This is a well-established, practical application of the temperature dependence of solubility.

**Q19.** Show that the depression of freezing point is dependent on the concentration of the solute but independent of the nature of the solute. **(10 marks)** [CHEM 113 | 2019–20]

### Answer

#### Depression of Freezing Point Depends on Concentration but Not on Nature of Solute

##### Derivation:

The **freezing point** of a solution is the temperature at which the **vapour pressure of the solid solvent** equals the **vapour pressure of the solvent in solution**.

##### Step 1: Vapour Pressure Lowering

By Raoult's Law, adding a non-volatile solute lowers the vapour pressure of the solvent:

$$P = x_A P^* \Rightarrow \Delta P = P^* - P = x_B P^*$$

where  $x_B$  is the mole fraction of the solute.

### Step 2: Depression of Freezing Point

For dilute solutions, the depression of freezing point  $\Delta T_f = T_f^* - T_f$  is related to the vapour pressure lowering. Using the **Clausius–Clapeyron equation** and thermodynamic arguments, it can be shown:

$$\Delta T_f = \frac{RT_f^{*2} M_A}{1000 \Delta H_f} \cdot m$$

where:

- $T_f^*$  = freezing point of pure solvent (K)
- $M_A$  = molar mass of solvent (g/mol)
- $\Delta H_f$  = molar enthalpy of fusion of solvent (J/mol)
- $m$  = molality of the solution

The term  $\frac{RT_f^{*2} M_A}{1000 \Delta H_f}$  is a **constant for a given solvent**, called the **cryoscopic constant**  $k_f$ :

$$\Delta T_f = k_f \times m$$

### Dependence on Concentration:

Since  $\Delta T_f = k_f \times m$  and  $k_f$  is a constant for a given solvent:

$$\Delta T_f \propto m$$

The depression in freezing point is **directly proportional to the molality** of the solute — confirming **dependence on concentration**.

### Independence from Nature of Solute:

The expression  $\Delta T_f = k_f \times m$  contains:

- $k_f$  — depends only on the **solvent**, not the solute.
- $m$  — depends only on the **number of moles** of solute particles, not their chemical identity.

Therefore, at the same molality, **any non-volatile, non-electrolyte solute** produces the same  $\Delta T_f$ , regardless of whether it is sugar, urea, glucose, or naphthalene.

**Example:** A 1 m solution of glucose and a 1 m solution of urea in water both give:

$$\Delta T_f = 1.86^\circ\text{C}/\text{m} \times 1\text{ m} = 1.86^\circ\text{C}$$

This confirms that **freezing point depression is a colligative property** — it depends only on the **number** of solute particles (concentration), not their **nature**.

**Q20.** Why is the dissolution of NaCl an endothermic process? (5 marks)

[CHEM 113 | 2017–18]

### Answer

#### Why the Dissolution of NaCl is Endothermic:

The dissolution of NaCl in water can be broken into two competing energy steps using the **Born–Haber cycle for dissolution**:

$$\Delta H_{\text{soln}} = \underbrace{U}_{\text{lattice energy (endothermic)}} + \underbrace{\Delta H_{\text{hyd}}}_{\text{hydration energy (exothermic)}}$$

#### Step 1 — Breaking the Lattice (Endothermic):

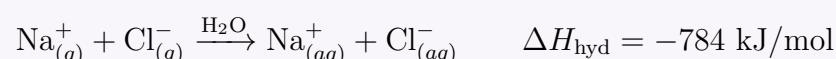
NaCl has a **strong ionic lattice** held together by electrostatic attraction between  $\text{Na}^+$  and  $\text{Cl}^-$  ions. To dissolve, this lattice must first be disrupted:



This step **requires** energy — it is endothermic.

#### Step 2 — Hydration of Ions (Exothermic):

The gaseous ions are then surrounded by polar water molecules (**ion–dipole interactions**):



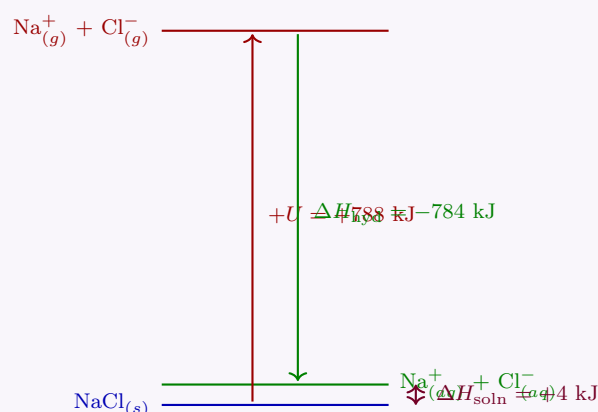
This step **releases** energy — it is exothermic.

#### Net Energy Balance:

$$\Delta H_{\text{soln}} = +788 + (-784) = +4 \text{ kJ/mol}$$

Since  $\Delta H_{\text{soln}} > 0$ , the overall dissolution is **slightly endothermic**.

**Energy Diagram:**



**Physical Interpretation:**

- The lattice energy of NaCl (+788 kJ/mol) is **slightly larger** than the hydration energy (−784 kJ/mol).
- The energy needed to break the strong ionic lattice **exceeds** the energy released by hydration — resulting in a net energy *input*.
- Despite being endothermic, NaCl dissolves readily because the process is driven by a large **increase in entropy** ( $\Delta S > 0$ ) — the ordered lattice breaks down into randomly dispersed hydrated ions:

$$\Delta G = \Delta H - T\Delta S < 0 \quad (\text{spontaneous})$$

**Conclusion:** The dissolution of NaCl is endothermic because the **lattice energy** required to separate the ions (+788 kJ/mol) slightly exceeds the **hydration energy** released when ions are solvated (−784 kJ/mol), giving  $\Delta H_{\text{soln}} = +4 \text{ kJ/mol}$ . Dissolution remains spontaneous due to the favourable **entropy increase**.

**Q21.** How does the increase in concentration of a nonvolatile solute in a solvent affect the following properties of solution? (i) Vapor pressure and (ii) osmotic pressure. **(10 marks)** [CHEM 113 | 2017–18]

### Answer

When the concentration of a **non-volatile solute** increases in a solvent, the following effects are observed:

**(i) Vapour Pressure — Decreases:**

By **Raoult's Law**, the vapour pressure of a solution is:

$$P_{\text{soln}} = X_{\text{solvent}} \cdot P^{\circ}_{\text{solvent}}$$

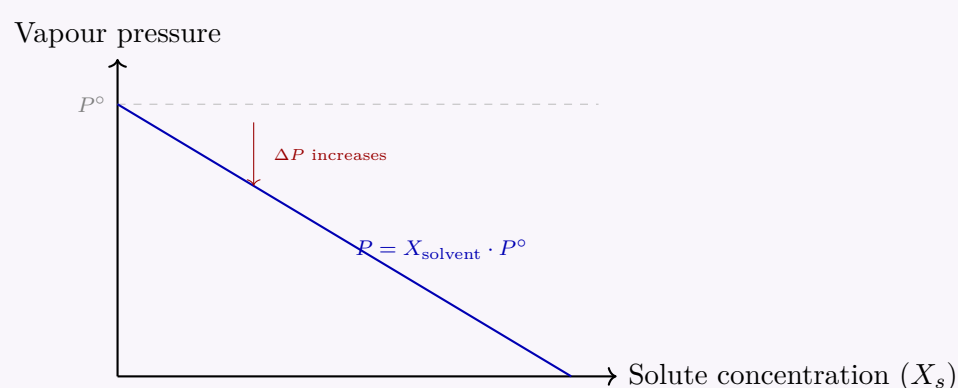
where  $X_{\text{solvent}}$  is the mole fraction of the solvent and  $P^{\circ}_{\text{solvent}}$  is the vapour pressure of pure solvent.

The **lowering** of vapour pressure is:

$$\Delta P = P^{\circ} - P_{\text{soln}} = X_{\text{solute}} \cdot P^{\circ}$$

As solute concentration **increases**:

- Mole fraction of solute ( $X_{\text{solute}}$ ) increases
- Mole fraction of solvent ( $X_{\text{solvent}}$ ) decreases
- Solute molecules occupy surface sites, reducing the fraction of solvent molecules that can escape into the vapour phase
- $\therefore$  Vapour pressure **decreases** proportionally with increasing solute concentration



**(ii) Osmotic Pressure — Increases:**

By the **van't Hoff equation**:

$$\pi = MRT = \frac{n}{V}RT$$

where  $M$  is the molarity,  $R$  is the gas constant, and  $T$  is the absolute temperature.

As solute concentration **increases**:

- Molarity  $M$  increases directly
- The chemical potential difference between pure solvent and solution grows larger
- A greater pressure must be applied to stop osmotic flow across a semipermeable membrane
- $\therefore$  Osmotic pressure  $\pi$  **increases linearly** with solute concentration



**Summary:**

| Property         | Effect           | Relationship                   |
|------------------|------------------|--------------------------------|
| Vapour pressure  | <b>Decreases</b> | $\Delta P = X_s \cdot P^\circ$ |
| Osmotic pressure | <b>Increases</b> | $\pi = MRT$                    |

**Conclusion:** Increasing non-volatile solute concentration **lowers** vapour pressure (fewer solvent molecules escape) and **raises** osmotic pressure (greater chemical potential difference across the membrane). Both effects scale directly with solute concentration, reflecting their colligative nature.

**Q22.** Identify the factors that influence the solubility of solute in a solvent. Discuss about two factors. (10 marks) [CHEM 113 | 2017–18]

### Answer

#### Factors Influencing Solubility:

The solubility of a solute in a solvent is influenced by several factors. The most important are:

- Nature of solute and solvent
- Temperature
- Pressure (for gases)
- Molecular size
- Common ion effect

#### Factor 1 — Nature of Solute and Solvent (“Like Dissolves Like”):

The most fundamental rule governing solubility is the **principle of structural similarity**:

“Like dissolves like”

- **Polar solutes** dissolve in **polar solvents**: NaCl, KBr, glucose dissolve in water because ion–dipole and hydrogen bonding interactions between solute and solvent are strong enough to overcome lattice/intermolecular forces.
- **Non-polar solutes** dissolve in **non-polar solvents**: naphthalene, grease, wax dissolve in benzene or hexane through London dispersion forces.
- **Cross combinations fail**: NaCl is virtually insoluble in benzene because ion–induced dipole interactions cannot overcome the strong NaCl lattice energy.
- The driving force is the formation of **new solute–solvent interactions** that release enough energy to break solute–solute and solvent–solvent interactions:

$$\Delta H_{\text{soln}} = \Delta H_{\text{solute-solute}} + \Delta H_{\text{solvent-solvent}} + \Delta H_{\text{solute-solvent}}$$

#### Factor 2 — Temperature:

Temperature affects the solubility of solids and gases differently:

##### (a) Solids in liquids:

- For most ionic and molecular solids, solubility **increases with temperature** because the dissolution process is endothermic ( $\Delta H > 0$ ) — higher temperature shifts equilibrium toward dissolution (Le Chatelier's principle).
- Example:  $\text{KNO}_3$  solubility increases from 13 g/100 mL at  $0^\circ\text{C}$  to 246 g/100 mL at  $100^\circ\text{C}$ .
- Exception:  $\text{NaOH}$ ,  $\text{Ce}_2(\text{SO}_4)_3$  become *less* soluble at higher temperatures (exothermic dissolution).

**(b) Gases in liquids:**

- Gas solubility **decreases with increasing temperature** — dissolved gas molecules gain kinetic energy and escape from solution more easily.
- By Henry's Law at constant pressure:

$$C = k_H(T) \cdot P \quad k_H \text{ decreases as } T \uparrow$$

- Example:  $\text{CO}_2$  solubility in water drops from 1.7 g/L at  $0^\circ\text{C}$  to 0.7 g/L at  $40^\circ\text{C}$  — explaining why warm soda goes flat faster.

**Summary:**

| Factor                       | Effect                        | Example                                |
|------------------------------|-------------------------------|----------------------------------------|
| Polar solute + polar solvent | High solubility               | $\text{NaCl}$ in $\text{H}_2\text{O}$  |
| Non-polar + non-polar        | High solubility               | Grease in hexane                       |
| $\uparrow T$ (solid solute)  | Usually $\uparrow$ solubility | $\text{KNO}_3$ in $\text{H}_2\text{O}$ |
| $\uparrow T$ (gas solute)    | $\downarrow$ solubility       | $\text{CO}_2$ in soda                  |

**Conclusion:** Solubility is primarily governed by the **nature of intermolecular forces** between solute and solvent, and strongly influenced by **temperature**. The “like dissolves like” rule provides a powerful qualitative guide, while temperature effects are quantitatively described by thermodynamic equilibrium principles.

**Q23.** A lithium salt used in lubricating grease has the formula  $\text{LiC}_n\text{H}_{2n+1}\text{O}_2$ . The salt is soluble in water to the extent of 0.036 g per 100 g of water at  $25^\circ\text{C}$ . The osmotic pressure is found to be 57.1 torr. Assuming that molality and molarity in such a dilute solution are the same and that the lithium salt is completely dissociated, determine an appropriate value of  $n$  in the formula of the salt. **(10 marks)** [CHEM 113 | 2017–18]

**Answer**

**Given:**

- Formula of salt:  $\text{LiC}_n\text{H}_{2n+1}\text{O}_2$
- Solubility: 0.036 g per 100 g water
- Osmotic pressure:  $\pi = 57.1$  torr
- Temperature:  $T = 25^\circ\text{C} = 298$  K
- Molality  $\approx$  Molarity (dilute solution)
- Salt is completely dissociated:  $\text{LiC}_n\text{H}_{2n+1}\text{O}_2 \rightarrow \text{Li}^+ + \text{C}_n\text{H}_{2n+1}\text{O}_2^- \Rightarrow i = 2$
- $R = 0.0821$  L·atm/mol·K

**Step 1 — Convert osmotic pressure to atm:**

$$\pi = \frac{57.1}{760} = 0.07513 \text{ atm}$$

**Step 2 — Find molarity using van't Hoff equation with dissociation:**

$$\pi = iMRT \quad \Rightarrow \quad M = \frac{\pi}{iRT} = \frac{0.07513}{2 \times 0.0821 \times 298} = \frac{0.07513}{48.93} = 1.535 \times 10^{-3} \text{ mol/L}$$

**Step 3 — Find moles of salt in solution:**

Since molality  $\approx$  molarity and solubility is 0.036 g per 100 g  $\text{H}_2\text{O}$  = 0.036 g per 0.1 kg solvent, and density  $\approx 1$  g/mL for dilute solution, this corresponds to approximately 0.1 L of solution.

Moles of salt in 0.1 L:

$$n = M \times V = 1.535 \times 10^{-3} \times 0.1 = 1.535 \times 10^{-4} \text{ mol}$$

**Step 4 — Find molar mass of salt:**

$$\mathcal{M} = \frac{w}{n} = \frac{0.036}{1.535 \times 10^{-4}} = 234.5 \text{ g/mol}$$

**Step 5 — Solve for  $n$ :**

 Molar mass of  $\text{LiC}_n\text{H}_{2n+1}\text{O}_2$ :

$$\begin{aligned}\mathcal{M} &= 6.94 + 12.01n + (2n + 1)(1.008) + 2(16.00) \\ &= 6.94 + 12.01n + 2.016n + 1.008 + 32.00 \\ &= 39.948 + 14.026n\end{aligned}$$

Setting equal to calculated molar mass:

$$\begin{aligned}39.948 + 14.026n &= 234.5 \\ 14.026n &= 194.55 \\ n &= \frac{194.55}{14.026} = 13.87 \approx 14\end{aligned}$$

$$\boxed{n = 14}$$

**Verification:**

$$\mathcal{M} = 39.948 + 14.026 \times 14 = 39.948 + 196.36 = 236.3 \text{ g/mol} \approx 234.5 \text{ g/mol} \quad \checkmark$$

**Conclusion:** The value of  $n$  in the formula  $\text{LiC}_n\text{H}_{2n+1}\text{O}_2$  is  $\boxed{n = 14}$ , corresponding to the salt **lithium myristate** ( $\text{LiC}_{14}\text{H}_{29}\text{O}_2$ ), a long-chain fatty acid lithium salt commonly used in greases.

**Q24.** Define 'Solution'. Explain that the solution process is governed by energy and entropy. **(3+6 marks)** [CHEM 113 | 2016–17]

**Answer**
**Definition of Solution and the Role of Energy and Entropy in the Solution Process**
**Part 1: Definition of Solution**

A **solution** is a **homogeneous mixture** of two or more substances whose composition can be varied within certain limits. It consists of:

- **Solvent:** The component present in the larger amount; determines the physical state of the solution.
- **Solute:** The component present in the smaller amount; dissolved in the solvent.

Solutions can exist in all three states of matter:

| Type             | Solute | Example                           |
|------------------|--------|-----------------------------------|
| Gas in gas       | Gas    | Air ( $\text{N}_2 + \text{O}_2$ ) |
| Gas in liquid    | Gas    | $\text{CO}_2$ in water (soda)     |
| Solid in liquid  | Solid  | $\text{NaCl}$ in water            |
| Liquid in liquid | Liquid | Ethanol in water                  |
| Solid in solid   | Solid  | Alloys (brass)                    |

**Part 2: The Solution Process is Governed by Energy and Entropy**

Whether or not a solute dissolves in a solvent is determined by the **Gibbs free energy change**:

$$\Delta G_{\text{soln}} = \Delta H_{\text{soln}} - T\Delta S_{\text{soln}}$$

A solution forms **spontaneously** when  $\Delta G_{\text{soln}} < 0$ , governed by both **enthalpy** ( $\Delta H$ ) and **entropy** ( $\Delta S$ ) contributions.

**(A) Energy (Enthalpy) Contribution:**

The solution process can be visualised in three steps (the **cycle model**):

**Step 1: Separation of solute particles:** Breaking solute–solute interactions requires energy input (endothermic):  $\Delta H_1 > 0$ .

**Step 2: Separation of solvent particles:** Breaking solvent–solvent interactions requires energy input (endothermic):  $\Delta H_2 > 0$ .

**Step 3: Mixing of solute and solvent:** Formation of solute–solvent interactions releases energy (exothermic):  $\Delta H_3 < 0$ .

The **heat of solution** is:

$$\Delta H_{\text{soln}} = \Delta H_1 + \Delta H_2 + \Delta H_3$$

- If  $|\Delta H_3| > |\Delta H_1 + \Delta H_2|$ :  $\Delta H_{\text{soln}} < 0$  (exothermic, e.g.  $\text{NaOH}$  in water).
- If  $|\Delta H_3| < |\Delta H_1 + \Delta H_2|$ :  $\Delta H_{\text{soln}} > 0$  (endothermic, e.g.  $\text{NH}_4\text{NO}_3$  in water).
- If  $|\Delta H_3| \approx |\Delta H_1 + \Delta H_2|$ :  $\Delta H_{\text{soln}} \approx 0$  (ideal solution).

**(B) Entropy Contribution:**

**Entropy**  $S$  is a measure of the **disorder or randomness** of a system.

- When a solute dissolves, solute and solvent particles **mix randomly** throughout the solution, greatly increasing the number of possible arrangements (microstates).



- This **mixing entropy** is always **positive**:  $\Delta S_{\text{mix}} > 0$ .
- The increase in entropy **favours dissolution** regardless of whether the process is endothermic or exothermic.
- Even when  $\Delta H_{\text{soln}} > 0$  (endothermic), the  $T\Delta S$  term can be large enough to make  $\Delta G < 0$ , driving dissolution. Example:  $\text{NH}_4\text{NO}_3$  dissolves endothermically in water because entropy gain overcomes the enthalpy cost.

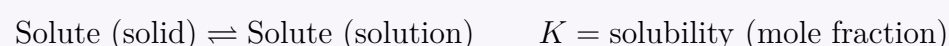
**Summary:**

| $\Delta H_{\text{soln}}$ | $\Delta S_{\text{soln}}$ | $\Delta G_{\text{soln}}$        | Result                |
|--------------------------|--------------------------|---------------------------------|-----------------------|
| – (exothermic)           | +                        | Always –                        | Always dissolves      |
| +                        | +                        | – if $T\Delta S > \Delta H$     | Dissolves at high $T$ |
| –                        | –                        | – if $ \Delta H  > T \Delta S $ | Dissolves at low $T$  |
| +                        | –                        | Always +                        | Never dissolves       |

**Conclusion:** The solution process is a competition between **energy** (enthalpy of interactions) and **entropy** (tendency toward disorder). A solution forms when the combined effect makes  $\Delta G_{\text{soln}} < 0$ .

**Answer**
**Equation for Heat of Solution Using Solubility Data and Experimental Design**
**Part 1: The van't Hoff Equation for Solubility**

The **solubility of a solid** in a liquid changes with temperature according to the **van't Hoff equation**. Treating solubility as an equilibrium process:



Applying the van't Hoff isochore:

$$\frac{d \ln K}{dT} = \frac{\Delta H_{\text{soln}}}{RT^2}$$

Integrating between temperatures  $T_1$  and  $T_2$  (with solubilities  $S_1$  and  $S_2$ ):

$$\int_{S_1}^{S_2} d \ln S = \frac{\Delta H_{\text{soln}}}{R} \int_{T_1}^{T_2} \frac{dT}{T^2}$$

$$\ln \frac{S_2}{S_1} = \frac{\Delta H_{\text{soln}}}{R} \left( \frac{1}{T_1} - \frac{1}{T_2} \right)$$

$$\boxed{\log \frac{S_2}{S_1} = \frac{\Delta H_{\text{soln}}}{2.303R} \left( \frac{1}{T_1} - \frac{1}{T_2} \right)}$$

where  $R = 8.314 \text{ J mol}^{-1} \text{ K}^{-1}$ .

This equation shows that if solubility is measured at two temperatures,  $\Delta H_{\text{soln}}$  can be calculated.

**Graphical Method:**

Taking  $\log S$  vs  $1/T$ :

$$\log S = -\frac{\Delta H_{\text{soln}}}{2.303R} \cdot \frac{1}{T} + \text{constant}$$

A plot of  $\log S$  vs  $\frac{1}{T}$  gives a **straight line** with slope:

$$\text{slope} = -\frac{\Delta H_{\text{soln}}}{2.303R} \Rightarrow \Delta H_{\text{soln}} = -2.303R \times \text{slope}$$

**Part 2: Experimental Design**

**Apparatus required:** Thermostat water bath, thermometer (accurate to  $0.1^\circ\text{C}$ ), conical flask, balance, filter paper, funnel, burette or pipette, standard solution for back-titration (if needed).

**Procedure:**

**Step 1:** Prepare a **saturated solution** of the solute (e.g.  $\text{KNO}_3$  or  $\text{NaCl}$ ) in distilled water at a fixed temperature  $T_1$  (e.g.  $20^\circ\text{C}$ ) using a thermostat bath. Stir thoroughly and allow equilibrium.

**Step 2:** **Filter** the saturated solution quickly to remove undissolved solute.

**Step 3:** Pipette a known volume (e.g. 10 mL) of the saturated solution into a pre-weighed evaporating dish.

**Step 4:** **Evaporate** the solvent completely by gentle heating. Weigh the residue (dissolved solute).

**Step 5:** Calculate the **solubility**  $S_1$  at  $T_1$  (in g per 100 g water, or mol fraction).

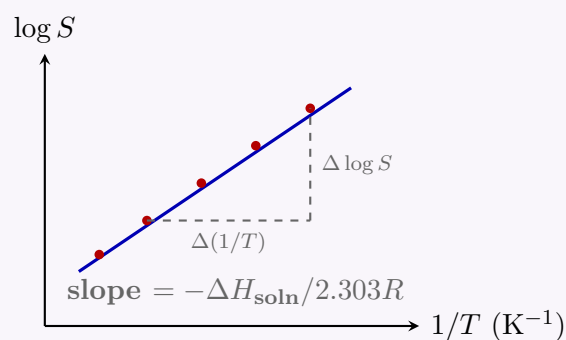
**Step 6:** **Repeat** steps 1–5 at different temperatures  $T_2, T_3, \dots$  (e.g.  $30, 40, 50, 60^\circ\text{C}$ ), recording solubility at each temperature.

**Step 7:** Plot  $\log S$  vs  $\frac{1}{T}$  (where  $T$  is in Kelvin).

**Step 8:** Determine the **slope** of the straight line and calculate:

$$\Delta H_{\text{soln}} = -2.303 \times 8.314 \times \text{slope} \quad (\text{in J/mol})$$

**Expected graph:**



**Note:** A **positive slope** indicates  $\Delta H_{\text{soln}} < 0$  (exothermic dissolution); a **negative slope** indicates  $\Delta H_{\text{soln}} > 0$  (endothermic dissolution).

**Q25.** Give a molecular interpretation of the positive and negative deviation from ideal behaviour from Raoult's law. **(10 marks)** [CHEM 113 | 2016–17]

### Answer

#### Molecular Interpretation of Positive and Negative Deviations from Raoult's Law

##### Raoult's Law (Ideal Behaviour):

For an ideal solution, the vapour pressure of each component is:

$$P_A = x_A P_A^* \quad P_B = x_B P_B^*$$

Ideal behaviour requires that **A–B interactions** are equal in strength to **A–A** and **B–B interactions**. Deviation occurs when this condition is not met.

##### Positive Deviation from Raoult's Law:

**Condition:** A–B interactions are **weaker** than A–A and B–B interactions.

##### Molecular Interpretation:

- When A and B are mixed, the **unlike molecules attract each other less** than like molecules do.
- Molecules escape more **easily** into the vapour phase compared to the pure components.
- **Vapour pressure is higher** than predicted by Raoult's Law:  $P_A > x_A P_A^*$  and  $P_B > x_B P_B^*$ .
- **Mixing is endothermic:**  $\Delta H_{\text{mix}} > 0$  (energy required to break A–A and B–B to form weaker A–B).
- **Volume increases on mixing:**  $\Delta V_{\text{mix}} > 0$ .
- **Example:** Ethanol–water, acetone–carbon disulfide, benzene–cyclohexane.

##### Negative Deviation from Raoult's Law:

**Condition:** A–B interactions are **stronger** than A–A and B–B interactions.

##### Molecular Interpretation:

- When A and B are mixed, the **unlike molecules attract each other more strongly** than like molecules do.
- Molecules are held more **tightly** in solution and escape less easily.
- **Vapour pressure is lower** than predicted by Raoult's Law:  $P_A < x_A P_A^*$  and  $P_B < x_B P_B^*$ .
- **Mixing is exothermic:**  $\Delta H_{\text{mix}} < 0$  (stronger A–B interactions release energy).
- **Volume decreases on mixing:**  $\Delta V_{\text{mix}} < 0$ .
- **Example:** Acetone–chloroform (H-bonding between unlike molecules), HNO<sub>3</sub>–water, HCl–water.

#### Summary Table:

| Property                | Positive Deviation  | Negative Deviation |
|-------------------------|---------------------|--------------------|
| A–B vs A–A, B–B         | Weaker              | Stronger           |
| Vapour pressure         | Higher than ideal   | Lower than ideal   |
| $\Delta H_{\text{mix}}$ | $> 0$ (endothermic) | $< 0$ (exothermic) |
| $\Delta V_{\text{mix}}$ | $> 0$               | $< 0$              |
| Escaping tendency       | Increased           | Decreased          |
| Example                 | Ethanol–water       | Acetone–chloroform |

- Q26.** A certain soft drink is bottled so that a bottle at 25°C contains CO<sub>2</sub> gas at a pressure of 5.0 atm over the liquid. Calculate the equilibrium concentrations of CO<sub>2</sub> both before and after the bottle is opened. (Henry's law constant for CO<sub>2</sub> in aqueous solution is  $3.1 \times 10^{-2}$  mol/L·atm at 25°C; partial pressure of CO<sub>2</sub> in atmosphere =  $4 \times 10^{-4}$  atm.) Can you relate the 'flat' taste of the drink with concentration? (7 marks) [CHEM 113 | 2016–17]

### Answer

#### Given:

- $k_H(\text{CO}_2) = 3.1 \times 10^{-2}$  mol/L·atm
- $P_{\text{CO}_2}$  inside bottle = 5.0 atm
- $P_{\text{CO}_2}$  in atmosphere =  $4 \times 10^{-4}$  atm
- $T = 25^\circ\text{C}$

**Henry's Law:**  $C = k_H \cdot P$

**Before opening (inside bottle):**

$$C_{\text{before}} = k_H \times P_{\text{CO}_2} = 3.1 \times 10^{-2} \times 5.0 = 0.155 \text{ mol/L}$$

$$C_{\text{before}} = 0.155 \text{ mol/L}$$

**After opening (equilibrium with atmosphere):**

$$C_{\text{after}} = k_H \times P_{\text{atm}} = 3.1 \times 10^{-2} \times 4 \times 10^{-4} = 1.24 \times 10^{-5} \text{ mol/L}$$

$$C_{\text{after}} = 1.24 \times 10^{-5} \text{ mol/L}$$

**Relation to "flat" taste:**

The concentration of CO<sub>2</sub> drops by a factor of:

$$\frac{C_{\text{before}}}{C_{\text{after}}} = \frac{0.155}{1.24 \times 10^{-5}} \approx 12,500$$

The "fizz" and sharp taste of a carbonated drink come from dissolved CO<sub>2</sub>, which reacts with saliva to form carbonic acid (H<sub>2</sub>CO<sub>3</sub>), stimulating taste receptors.

After opening, CO<sub>2</sub> rapidly escapes until it equilibrates with atmospheric partial pressure — reducing concentration by over 12,000-fold. With virtually no dissolved CO<sub>2</sub> remaining, the drink loses its acidic, effervescent character and tastes **flat** and bland.

- Q27.** The freezing point depression measurement of benzoic acid in acetone yields a molar mass of 122 g; the same measurement in benzene gives a value of 244 g. Account for this discrepancy. (5 marks) [CHEM 113 | 2016–17]

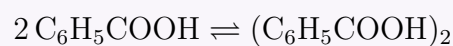
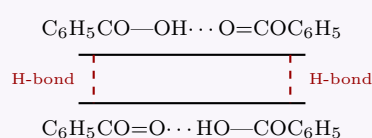
### Answer

#### Discrepancy in Molar Mass of Benzoic Acid:

- In **acetone**:  $\mathcal{M} = 122$  g/mol (true molar mass)
- In **benzene**:  $\mathcal{M} = 244$  g/mol ( $= 2 \times 122$  g/mol)

#### Explanation — Dimerisation in benzene:

Benzoic acid (C<sub>6</sub>H<sub>5</sub>COOH) contains a **carboxylic acid group** capable of forming **two hydrogen bonds** simultaneously. In **non-polar solvents** like benzene, where there is no competing hydrogen bonding with the solvent, two benzoic acid molecules associate to form a **cyclic dimer** via hydrogen bonds:



This dimerisation means that **two formula units act as one particle** in benzene solution. Colligative properties (freezing point depression) depend on the number of particles, so the measured molar mass is **doubled**:

$$\mathcal{M}_{\text{apparent}} = 2 \times 122 = 244 \text{ g/mol} \quad \checkmark$$

In **polar acetone**, the solvent competes for hydrogen bonds with benzoic acid, **disrupting dimer formation** — molecules remain as monomers, giving the true molar mass of 122 g/mol.

This is an example of  $i < 1$  behaviour:

$$i = \frac{1}{2} = 0.5 \quad (\text{two molecules act as one})$$

**Conclusion:** The discrepancy arises because benzoic acid **dimerises** in non-polar benzene (via double hydrogen bonding) but remains **monomeric** in polar acetone. The apparent molar mass in benzene is twice the true value, reflecting a van't Hoff factor of  $i = 0.5$ .

**Q28.** A commercial bleach solution contains 3.62 mass% NaOCl in water. Calculate (i) mole fraction (ii) molality of NaOCl in the solution. (5 marks) [CHEM 113 | 2016–17]

### Answer

#### Given:

- NaOCl: 3.62 mass% in water
- Molar mass of NaOCl:  $23.0 + 16.0 + 35.5 = 74.5$  g/mol
- Molar mass of H<sub>2</sub>O: 18.02 g/mol
- Basis: 100 g of solution  $\Rightarrow$  3.62 g NaOCl + 96.38 g H<sub>2</sub>O

#### (i) Mole Fraction of NaOCl:

$$n_{\text{NaOCl}} = \frac{3.62}{74.5} = 0.04859 \text{ mol}$$

$$n_{\text{H}_2\text{O}} = \frac{96.38}{18.02} = 5.349 \text{ mol}$$

$$X_{\text{NaOCl}} = \frac{n_{\text{NaOCl}}}{n_{\text{NaOCl}} + n_{\text{H}_2\text{O}}} = \frac{0.04859}{0.04859 + 5.349} = \frac{0.04859}{5.398}$$

$$X_{\text{NaOCl}} = 9.00 \times 10^{-3}$$

#### (ii) Molality of NaOCl:

$$m = \frac{n_{\text{solute}}}{m_{\text{solvent (kg)}}} = \frac{0.04859}{96.38 \times 10^{-3}} = \frac{0.04859}{0.09638}$$

$$m = 0.504 \text{ mol/kg}$$

**Q29.** A solution of 0.5 g of an unknown nonvolatile, non-electrolyte solute is added to 100 mL of water and placed across a semipermeable membrane from pure water. When the system reaches equilibrium, the solution compartment is 5.6 cm above the solvent compartment. Assuming the density of the solution is 1.0 g/mL, calculate the molecular mass of the unknown. (9 marks) [CHEM 113 | 2016–17]

### Answer

#### Given:

- Mass of unknown solute:  $w = 0.5$  g
- Volume of water:  $V = 100$  mL = 0.1 L
- Height difference at equilibrium:  $h = 5.6$  cm
- Density of solution:  $\rho = 1.0$  g/mL
- $T = 25^\circ\text{C} = 298$  K (assume room temperature),  $g = 9.81$  m/s<sup>2</sup>
- $R = 0.0821$  L·atm/mol·K

#### Step 1 — Find osmotic pressure from height:

The hydrostatic pressure of a liquid column is:

$$\pi = \rho gh$$

$$\pi = 1.0 \text{ g/mL} \times 9.81 \text{ m/s}^2 \times 5.6 \text{ cm}$$

Converting to consistent units ( $\rho = 1000$  kg/m<sup>3</sup>,  $h = 0.056$  m):

$$\pi = 1000 \times 9.81 \times 0.056 = 549.4 \text{ Pa}$$

Converting to atm (1 atm = 101325 Pa):

$$\pi = \frac{549.4}{101325} = 5.423 \times 10^{-3} \text{ atm}$$

#### Step 2 — Find molarity:

$$M = \frac{\pi}{RT} = \frac{5.423 \times 10^{-3}}{0.0821 \times 298} = \frac{5.423 \times 10^{-3}}{24.47} = 2.216 \times 10^{-4} \text{ mol/L}$$

#### Step 3 — Find moles of solute in 0.1 L:

$$n = 2.216 \times 10^{-4} \times 0.1 = 2.216 \times 10^{-5} \text{ mol}$$

**Step 4 — Molar mass:**

$$\mathcal{M} = \frac{w}{n} = \frac{0.5}{2.216 \times 10^{-5}}$$

$$\mathcal{M} = 2.256 \times 10^4 \text{ g/mol} \approx 22,560 \text{ g/mol}$$

**Conclusion:** The molecular mass of the unknown solute is approximately  $2.26 \times 10^4$  g/mol, suggesting it is a **macromolecule** such as a protein or polymer. The height-based osmotic pressure method is suitable for such large molecules where other colligative methods are insensitive.

**Q30.** What is ion-dipole force? How does this force work in determining the solubility of ionic solid in water? Use the dissolution of KCl in water as an example. (10 marks) [CHEM 113 | 2015–16]

**Answer**
**1. Ion–Dipole Force:**

An **ion–dipole force** is an electrostatic attraction between an **ion** (cation or anion) and the **partial charges** on a polar molecule (dipole).

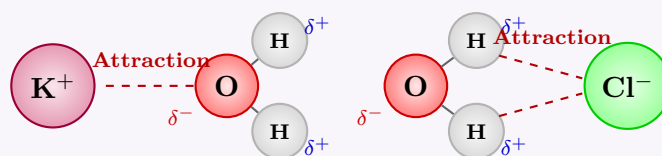
For a cation (+) and a polar molecule with partial charges  $\delta^+$  and  $\delta^-$ :

- The cation attracts the  $\delta^-$  end of the dipole.
- The anion attracts the  $\delta^+$  end of the dipole.

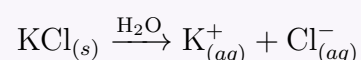
The strength depends on:

$$F \propto \frac{q \cdot \mu}{r^2}$$

where  $q$  is the ionic charge,  $\mu$  is the dipole moment of the polar molecule, and  $r$  is the distance.


**2. Dissolution of KCl in Water — Step by Step:**
**Step 1 — Ion–dipole forces initiate dissolution:**

Water is highly polar ( $\mu = 1.85$  D). When KCl is placed in water, the  $\delta^-$  oxygen of water molecules is attracted to  $\text{K}^+$  ions and the  $\delta^+$  hydrogen is attracted to  $\text{Cl}^-$  ions at the crystal surface:


**Step 2 — Hydration shells form:**

As ions are pulled away from the lattice, they become **surrounded by water molecules** — forming hydration shells. This releases **hydration energy**:

$$\Delta H_{\text{hyd}}(\text{K}^+) = -322 \text{ kJ/mol} \quad \Delta H_{\text{hyd}}(\text{Cl}^-) = -363 \text{ kJ/mol}$$

**Step 3 — Energy balance determines solubility:**

$$\Delta H_{\text{soln}} = U_{\text{KCl}} + \Delta H_{\text{hyd}} = +715 + (-685) = +30 \text{ kJ/mol}$$

Although slightly endothermic, KCl dissolves readily because of a large **entropy increase** ( $\Delta S > 0$ ):

$$\Delta G = \Delta H - T\Delta S < 0 \quad (\text{spontaneous})$$

**3. How Ion–Dipole Forces Determine Solubility:**

- **Stronger ion–dipole forces**  $\Rightarrow$  greater hydration energy  $\Rightarrow$  better solubility
- Smaller, higher-charge ions (e.g.  $\text{Li}^+$ ,  $\text{Mg}^{2+}$ ) have stronger ion–dipole interactions and greater hydration energies
- If ion–dipole (hydration) energy  $\geq$  lattice energy  $\Rightarrow$  **ionic solid dissolves**
- If lattice energy  $\gg$  hydration energy (e.g.  $\text{MgO}$ ,  $\text{AlF}_3$ )  $\Rightarrow$  **insoluble**

**Conclusion:** Ion–dipole forces between water molecules and  $\text{K}^+/\text{Cl}^-$  ions drive the dissolution of KCl by releasing sufficient hydration energy to overcome the crystal lattice. The process is thermodynamically spontaneous due to the combined effect of hydration energy and increased entropy.

**Q31.** KCl and glucose undergo the following reactions when dissolved in water:  $\text{KCl}_{(s)} \rightarrow \text{K}^+_{(aq)} + \text{Cl}^-_{(aq)}$ ;  $\text{Glucose}_{(s)} \rightarrow \text{Glucose}_{(aq)}$ . Consider equal molar amounts dissolved in equal volumes of water. (i) Which solution would have a higher freezing point? Why? (ii) Which would have higher vapor pressure? Why? (iii) How would the boiling point of a concentrated KCl solution compare? (iv) If the freezing point of KCl solution is  $-1.0^\circ\text{C}$ , what would be the freezing point and boiling point of the glucose solution? (16 marks) [CHEM 113 | 2015–16]

**Answer**
**Given:**

- KCl:  $\text{KCl} \rightarrow \text{K}^+ + \text{Cl}^-$ , van't Hoff factor  $i = 2$
- Glucose:  $\text{Glucose} \rightarrow \text{Glucose}_{(aq)}$ , van't Hoff factor  $i = 1$
- Equal molar amounts in equal volumes of water

**(i) Which solution has a higher freezing point?**

$$\Delta T_f = iK_f m$$

- KCl:  $\Delta T_f = 2 \times K_f \times m$
- Glucose:  $\Delta T_f = 1 \times K_f \times m$

KCl produces **twice as many particles**, so its freezing point depression is **twice as large**.  
 $\therefore$  **Glucose solution has the higher freezing point** (less depression).

**(ii) Which has higher vapour pressure?**

$$\Delta P = iX_s P^\circ$$

KCl lowers vapour pressure more (larger  $i$ ), so its vapour pressure is **lower**.  
 $\therefore$  **Glucose solution has the higher vapour pressure**.

**(iii) Boiling point of concentrated KCl solution:**

$$\Delta T_b = iK_b m$$

 For KCl,  $i = 2$ , so:

$$\Delta T_b(\text{KCl}) = 2K_b m$$

which is **twice** that of an equivalent glucose solution. A concentrated KCl solution would have a **significantly higher boiling point** than pure water or an equivalent glucose solution. However, at very high concentrations, ion-pairing reduces  $i$  slightly below 2, so the actual elevation is somewhat less than the ideal value.

**(iv) Freezing and boiling point of glucose solution given  $T_f(\text{KCl}) = -1.0^\circ\text{C}$ :**

 From KCl data, find  $K_f \cdot m$ :

$$\begin{aligned}\Delta T_f(\text{KCl}) &= i_{\text{KCl}} \cdot K_f \cdot m = 2K_f m = 1.0^\circ\text{C} \\ \therefore K_f m &= 0.50^\circ\text{C}\end{aligned}$$

**Freezing point of glucose:**

$$\begin{aligned}\Delta T_f(\text{glucose}) &= i_{\text{glu}} \cdot K_f m = 1 \times 0.50 = 0.50^\circ\text{C} \\ T_f(\text{glucose}) &= 0 - 0.50 = -0.50^\circ\text{C}\end{aligned}$$

**Boiling point of glucose:**

 Using ratio  $K_b/K_f = 0.52/1.86 = 0.2796$  for water:

$$K_b m = K_f m \times \frac{K_b}{K_f} = 0.50 \times \frac{0.52}{1.86} = 0.1398^\circ\text{C}$$

$$\Delta T_b(\text{glucose}) = 1 \times 0.1398 = 0.14^\circ\text{C}$$

$$T_b(\text{glucose}) = 100 + 0.14 = 100.14^\circ\text{C}$$

**Summary:**

| Property        | KCl ( $i = 2$ )                 | Glucose ( $i = 1$ )              |
|-----------------|---------------------------------|----------------------------------|
| Freezing point  | $-1.0^\circ\text{C}$<br>(lower) | $-0.5^\circ\text{C}$<br>(higher) |
| Vapour pressure | Lower                           | Higher                           |
| Boiling point   | Higher                          | Lower                            |
| # particles     | $2m$                            | $m$                              |

**Conclusion:** All colligative differences between KCl and glucose stem from KCl producing **twice as many particles** per formula unit ( $i = 2$  vs  $i = 1$ ). This doubles the freezing point depression and boiling point elevation, and halves the vapour pressure relative to an equivalent glucose solution.

**Q32.** Desalination by reverse osmosis: (i) Draw a diagram showing how reverse osmosis can be carried out. (ii) You are to desalinate sea water that is 0.70 M NaCl solution. What minimum pressure (in atm) must be applied at  $25^\circ\text{C}$  for reverse osmosis to occur? **(9)**

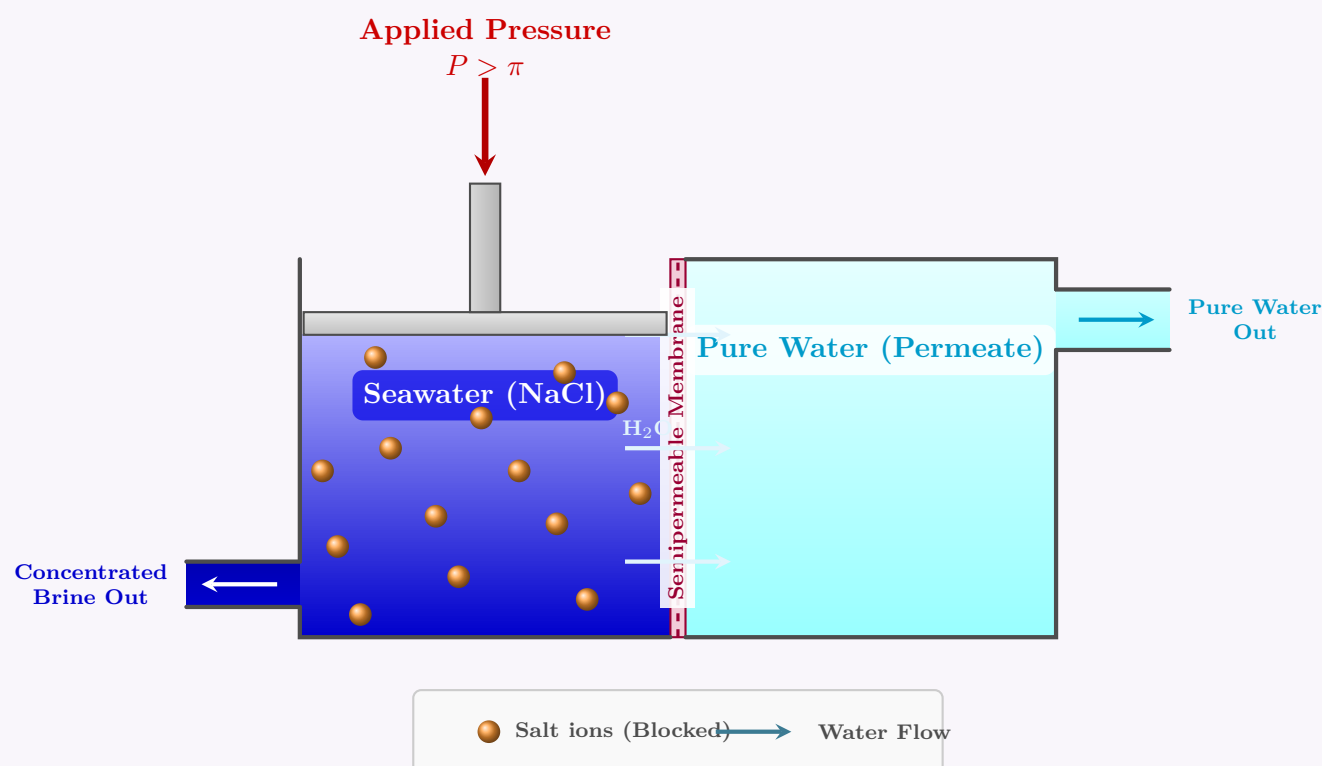


(marks)

[CHEM 113 | 2015–16]

## Answer

## (i) Diagram of Reverse Osmosis Desalination:



In **reverse osmosis**, an external pressure **greater than the osmotic pressure** ( $P > \pi$ ) is applied to the seawater side. This forces the solvent (water molecules) to flow *against* the natural osmotic direction—from the highly concentrated seawater side through the semipermeable membrane to the pure water side, leaving the salt ions behind.

## (ii) Minimum pressure for reverse osmosis of 0.70 M NaCl at 25°C:

NaCl is a strong electrolyte and dissociates completely into  $\text{Na}^+$  and  $\text{Cl}^-$  ions. Therefore, the van't Hoff factor is  $i = 2$ .

Using the van't Hoff equation for osmotic pressure ( $\pi$ ):

$$\begin{aligned}\pi &= iMRT \\ &= 2 \times 0.70 \text{ mol/L} \times 0.0821 \text{ L atm K}^{-1}\text{mol}^{-1} \times (25 + 273) \text{ K} \\ &= 2 \times 0.70 \times 0.0821 \times 298 \\ &= 34.26 \text{ atm}\end{aligned}$$

For reverse osmosis to occur, the applied pressure ( $P$ ) must **exceed** the osmotic pressure of the solution:

$$P_{\min} > 34.26 \text{ atm}$$

**Conclusion:** To desalinate a 0.70 M NaCl seawater solution at 25°C by reverse osmosis, a minimum pressure greater than **34.26 atm** must be applied. In industrial desalination plants, pressures significantly higher than this minimum (typically 50–80 atm) are used to maintain a practical flow rate of purified water.

**Q33.** What are colligative properties? What is the meaning of the word “Colligative” in this context? Use a phase diagram to show the difference in freezing point and boiling point between an aqueous urea solution and pure water. (10 marks)

[CHEM 113 | 2014–15]

## Answer

## Colligative Properties:

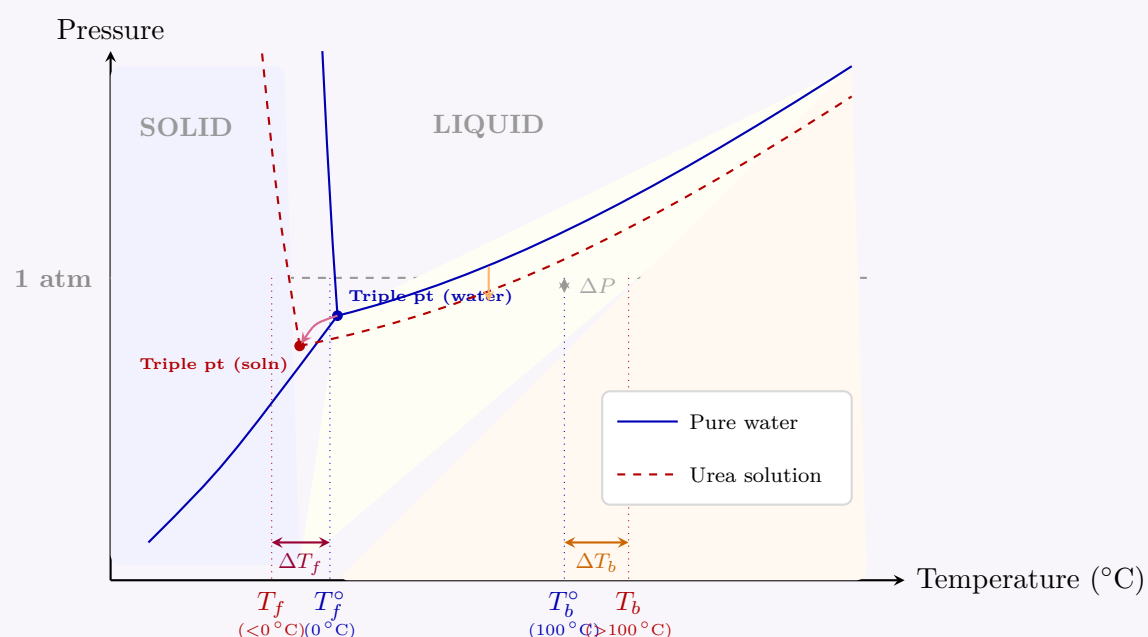
Colligative properties are properties of solutions that depend **only on the number** (concentration) of solute particles dissolved, and **not on the nature or identity** of the solute.

**Meaning of “Colligative”:** The word comes from the Latin “*colligatus*” meaning “**bound together**” — implying that these properties are collectively determined by the total number of dissolved particles acting together, regardless of what those particles are.

The four colligative properties are:

- (a) Vapour pressure lowering
- (b) Boiling point elevation ( $\Delta T_b$ )
- (c) Freezing point depression ( $\Delta T_f$ )
- (d) Osmotic pressure ( $\pi$ )

## Phase Diagram — Pure Water vs. Aqueous Urea Solution:



### Explanation of the Phase Diagram:

- (a) **Vapour pressure lowering ( $\Delta P$ ):** The urea solution has a **lower vapour pressure** than pure water at every temperature (Raoult's law). Urea molecules occupy surface sites, reducing the rate of water evaporation:

$$P_{\text{solution}} = X_{\text{solvent}} \cdot P^{\circ} \quad \Rightarrow \quad \Delta P = X_{\text{solute}} \cdot P^{\circ}$$

- (b) **Freezing point depression ( $\Delta T_f$ ):** The liquid–solid boundary of the solution shifts to the **left** (lower temperature). The solution freezes at a temperature *below* 0°C because solute particles disrupt crystal lattice formation:

$$\Delta T_f = K_f \cdot m$$

- (c) **Boiling point elevation ( $\Delta T_b$ ):** The liquid–vapour boundary shifts **right** (higher temperature). The solution boils *above* 100°C because a higher temperature is needed to raise the reduced vapour pressure to 1 atm:

$$\Delta T_b = K_b \cdot m$$

**Conclusion:** The word “colligative” reflects that these effects depend purely on the **number of dissolved particles**. Urea, being a non-electrolyte, lowers the vapour pressure, depresses the freezing point, and elevates the boiling point of water in direct proportion to its molal concentration.

**Q34.** A 9.66 g sample of a compound with the empirical formula  $\text{C}_5\text{H}_4$  is dissolved in 284 g of benzene. The freezing point of the solution is 1.37°C below that of pure benzene. What are the molar mass and molecular formula of this compound? **(8 marks)** [CHEM 113 | 2014–15]

### Answer

#### Given:

- Empirical formula:  $\text{C}_5\text{H}_4$
- Mass of compound:  $w = 9.66 \text{ g}$
- Mass of benzene:  $w_{\text{solvent}} = 284 \text{ g}$
- $\Delta T_f = 1.37 \text{ }^{\circ}\text{C}$
- $K_f(\text{benzene}) = 5.12 \text{ }^{\circ}\text{C/m}$

#### Step 1 — Molality:

$$m = \frac{\Delta T_f}{K_f} = \frac{1.37}{5.12} = 0.2676 \text{ mol/kg}$$

#### Step 2 — Moles of solute:

$$n = 0.2676 \times 0.284 = 7.599 \times 10^{-2} \text{ mol}$$

#### Step 3 — Molar mass:

$$\mathcal{M} = \frac{9.66}{7.599 \times 10^{-2}} = 127.1 \text{ g/mol}$$

$$\boxed{\mathcal{M} \approx 128 \text{ g/mol}}$$

#### Step 4 — Molecular formula:

$$\mathcal{M}_{\text{emp}}(\text{C}_5\text{H}_4) = 5(12.01) + 4(1.008) = 64.08 \text{ g/mol}$$

$$n = \frac{128}{64.08} \approx 2$$

$$\boxed{\text{Molecular formula: C}_{10}\text{H}_8 \text{ (Naphthalene)}}$$

**Q35.** The osmotic pressure of blood is about 7.4 atm. What is the approximate concentration of a saline solution (NaCl) a physician should use for intravenous injection? Use 37°C for physiological temperature. (6 marks) [CHEM 113 | 2014–15]

### Answer

#### Given:

- Osmotic pressure of blood:  $\pi = 7.4$  atm
- Temperature:  $T = 37^\circ\text{C} = 310$  K
- NaCl dissociates:  $i = 2$
- $R = 0.0821$  L·atm/mol·K

**Required:** Molarity  $M$  of saline solution that is **isotonic** with blood (same osmotic pressure).

**Using van't Hoff equation:**

$$\pi = iMRT \quad \Rightarrow \quad M = \frac{\pi}{iRT} = \frac{7.4}{2 \times 0.0821 \times 310} = \frac{7.4}{50.90}$$

$$M = 0.145 \text{ mol/L} \approx 0.15 \text{ M NaCl}$$

This corresponds to a **0.9% (w/v) NaCl solution** — the standard **normal saline** used in medical practice.

**Conclusion:** A physician should use approximately **0.15 M NaCl** (isotonic or normal saline) for intravenous injection to match the osmotic pressure of blood and prevent osmotic shock to red blood cells.

**Q36.** A physician dissolves 21.5 mg of hemoglobin in water to make 1.50 mL of solution at 5.08°C. He measures an osmotic pressure of 3.61 torr. What is the molar mass of the protein? (1 torr = 1/760 atmosphere) (9 marks) [CHEM 113 | 2013–14]

### Answer

#### Given:

- Mass of hemoglobin:  $w = 21.5$  mg = 0.0215 g
- Volume:  $V = 1.50$  mL =  $1.50 \times 10^{-3}$  L
- Temperature:  $T = 5.08^\circ\text{C} = 278.23$  K
- Osmotic pressure:  $\pi = 3.61$  torr
- $R = 0.0821$  L·atm/mol·K

**Step 1 — Convert  $\pi$  to atm:**

$$\pi = \frac{3.61}{760} = 4.750 \times 10^{-3} \text{ atm}$$

**Step 2 — Find molarity:**

$$M = \frac{\pi}{RT} = \frac{4.750 \times 10^{-3}}{0.0821 \times 278.23} = \frac{4.750 \times 10^{-3}}{22.84} = 2.079 \times 10^{-4} \text{ mol/L}$$

**Step 3 — Find moles:**

$$n = M \times V = 2.079 \times 10^{-4} \times 1.50 \times 10^{-3} = 3.119 \times 10^{-7} \text{ mol}$$

**Step 4 — Molar mass:**

$$\mathcal{M} = \frac{w}{n} = \frac{0.0215}{3.119 \times 10^{-7}}$$

$$\mathcal{M} = 6.894 \times 10^4 \text{ g/mol} \approx 68,940 \text{ g/mol}$$

**Conclusion:** The molar mass of hemoglobin is approximately  $6.9 \times 10^4$  g/mol, consistent with the known value of  $\sim 64,500$  g/mol for human hemoglobin.

**Q37.** Write down Van't Hoff's law of osmosis and show that the molecular mass of a non-electrolyte, non-volatile solute can be determined from osmotic pressure. (8 marks) [CHEM 113 | 2013–14]

### Answer

#### Van't Hoff's Law of Osmosis:

Van't Hoff (1887) showed that dilute solutions obey laws analogous to ideal gases. His law states:

$$\pi V = nRT$$

or equivalently:

$$\pi = \frac{n}{V}RT = MRT$$

where  $\pi$  is osmotic pressure (atm),  $M$  is molarity (mol/L),  $R = 0.0821$  L·atm/mol·K, and  $T$  is temperature (K).

This is analogous to the ideal gas law  $PV = nRT$ , with osmotic pressure playing the role of gas pressure.

**Determination of Molar Mass from Osmotic Pressure:**

For a non-electrolyte, non-volatile solute of mass  $w$  (g), molar mass  $\mathcal{M}$  (g/mol), dissolved in volume  $V$  (L):

$$M = \frac{n}{V} = \frac{w}{\mathcal{M} \cdot V}$$

Substituting into van't Hoff's equation:

$$\pi = MRT = \frac{w}{\mathcal{M}V}RT$$

Rearranging to solve for molar mass:

$$\boxed{\mathcal{M} = \frac{wRT}{\pi V}}$$

**Procedure:**

(a) Dissolve known mass  $w$  of solute in volume  $V$  of solvent

(b) Measure osmotic pressure  $\pi$  at temperature  $T$

(c) Calculate  $\mathcal{M} = \frac{wRT}{\pi V}$

**Why osmotic pressure is preferred for molar mass determination:**

- Even at very low concentrations, osmotic pressure gives **measurable values** (e.g. 1 g/L of a 10,000 g/mol solute gives  $\pi \approx 0.002$  atm = 1.5 mmHg — easily measurable)
- $\Delta T_f$  or  $\Delta T_b$  for the same solution would be only  $\sim 10^{-4}$  °C — too small to measure accurately
- Therefore osmotic pressure is especially useful for **macromolecules** (proteins, polymers)

**Conclusion:** Van't Hoff's law  $\pi = MRT$  provides a direct route to molar mass via  $\mathcal{M} = wRT/\pi V$ . It is the most sensitive colligative method for large molecules because osmotic pressure remains measurable even at the extremely low molarities typical of macromolecular solutions.

**Q38.** Two liter (2.0 L) distilled water is mixed with three liter (3.0 L) H<sub>2</sub>SO<sub>4</sub> of specific gravity 1.85. Calculate the specific gravity of the acid mixture. (5 marks) [CHEM 113 | 2013–14]

**Answer**
**Given:**

- Volume of distilled water:  $V_w = 2.0$  L
- Volume of H<sub>2</sub>SO<sub>4</sub>:  $V_a = 3.0$  L
- Specific gravity of H<sub>2</sub>SO<sub>4</sub>: sp.gr. = 1.85
- Density of water: 1.0 g/mL

**Step 1 — Mass of water:**

$$m_w = \rho_w \times V_w = 1.0 \text{ g/mL} \times 2000 \text{ mL} = 2000 \text{ g}$$

**Step 2 — Mass of H<sub>2</sub>SO<sub>4</sub>:**

$$\rho_a = \text{sp.gr.} \times \rho_w = 1.85 \times 1.0 = 1.85 \text{ g/mL}$$

$$m_a = 1.85 \times 3000 = 5550 \text{ g}$$

**Step 3 — Total mass of mixture:**

$$m_{\text{total}} = 2000 + 5550 = 7550 \text{ g}$$

**Step 4 — Total volume of mixture:**

Assuming **volumes are additive** (valid approximation for dilute mixing):

$$V_{\text{total}} = 2.0 + 3.0 = 5.0 \text{ L} = 5000 \text{ mL}$$

**Step 5 — Specific gravity of mixture:**

$$\rho_{\text{mix}} = \frac{m_{\text{total}}}{V_{\text{total}}} = \frac{7550}{5000} = 1.51 \text{ g/mL}$$

$$\boxed{\text{Specific gravity of mixture} = 1.51}$$

**Conclusion:** The specific gravity of the acid–water mixture is **1.51**, lying between that of pure water (1.0) and pure H<sub>2</sub>SO<sub>4</sub> (1.85), as expected for a mixture of the two.

**Q39.** How can you account for the exceptional solubility of some gases in liquid? (7 marks)

[CHEM 113 | 2013–14]

### Answer

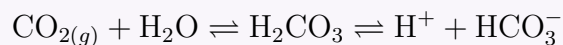
#### Exceptional Solubility of Some Gases in Liquids:

While most gases dissolve in liquids purely by physical processes governed by Henry's Law, some gases show **exceptionally high solubility** far beyond what Henry's Law predicts. This is explained by **chemical interaction** between the gas and solvent.

#### 1. Chemical Reaction with Solvent:

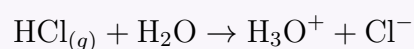
Some gases react chemically with the solvent, effectively removing dissolved gas from solution and allowing more to dissolve:

- **CO<sub>2</sub> in water:**



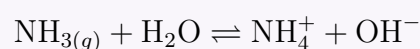
CO<sub>2</sub> reacts to form carbonic acid and then bicarbonate, maintaining a low free CO<sub>2</sub> concentration and driving more dissolution.

- **HCl in water:**



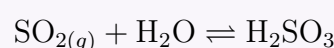
Complete ionisation makes HCl extraordinarily soluble (~700 volumes per volume of water).

- **NH<sub>3</sub> in water:**



Both hydrogen bonding and partial ionisation make NH<sub>3</sub> highly soluble (~700 volumes per volume of water at 0°C).

- **SO<sub>2</sub> in water:**



#### 2. Strong Hydrogen Bonding:

Some gases form **strong hydrogen bonds** with the solvent without full ionisation:

- NH<sub>3</sub> — lone pair on N forms H-bonds with water: N⋯H–O
- HF — very strong H-bonding with water
- These interactions supplement van der Waals forces, greatly increasing solubility

#### 3. Specific Solvent–Solute Affinity:

- **O<sub>2</sub> in haemoglobin:** O<sub>2</sub> binds reversibly to the Fe<sup>2+</sup> centre in haem — coordination bonding gives exceptional biological “solubility”
- **CO<sub>2</sub> in blood:** reacts with water and haemoglobin, giving effective solubility ~60 times higher than in plain water

#### Summary:

| Gas             | Mechanism              | Reason for High Solubility                                             |
|-----------------|------------------------|------------------------------------------------------------------------|
| HCl             | Ionisation             | Complete dissociation removes free HCl                                 |
| NH <sub>3</sub> | H-bonding + ionisation | N–H⋯O bonding + NH <sub>4</sub> <sup>+</sup> formation                 |
| CO <sub>2</sub> | Chemical reaction      | Forms H <sub>2</sub> CO <sub>3</sub> and HCO <sub>3</sub> <sup>–</sup> |
| SO <sub>2</sub> | Acid formation         | Forms H <sub>2</sub> SO <sub>3</sub> in water                          |

**Conclusion:** Exceptional gas solubility arises when gases **chemically react** with the solvent (ionisation, acid–base reaction) or form **strong hydrogen bonds**, dramatically increasing effective solubility far beyond what Henry's Law predicts for purely physical dissolution.

**Q40.** Some dissolution processes are exothermic and others are endothermic. Provide a molecular interpretation for the difference. (10 marks) [CHEM 113 | 2013–14]

### Answer

#### Molecular Interpretation of Exothermic vs Endothermic Dissolution:

The heat of dissolution ( $\Delta H_{\text{soln}}$ ) is the net result of three molecular energy changes:

$$\Delta H_{\text{soln}} = \Delta H_1 + \Delta H_2 + \Delta H_3$$

- $\Delta H_1$  = energy to **break solute–solute** interactions (+, endothermic)
- $\Delta H_2$  = energy to **break solvent–solvent** interactions (+, endothermic)
- $\Delta H_3$  = energy **released** by solute–solvent interactions (–, exothermic)

**Exothermic Dissolution ( $\Delta H_{\text{soln}} < 0$ ):**

Occurs when **solute–solvent interactions are stronger** than the average of solute–solute and solvent–solvent:

$$|\Delta H_3| > \Delta H_1 + \Delta H_2$$

**Molecular reasons:**

- Formation of **strong ion–dipole** interactions: e.g. NaOH in water — Na<sup>+</sup> and OH<sup>−</sup> are strongly hydrated, releasing large hydration energy (−44 kJ/mol)
- Formation of **new hydrogen bonds**: e.g. H<sub>2</sub>SO<sub>4</sub> in water forms strong O–H⋯O bonds with water, releasing large energy
- **Acid–base neutralisation**: HCl dissolving in NaOH solution — ionic interaction plus neutralisation releases heat
- The new solute–solvent bonds formed are **energetically superior** to the original interactions broken

**Examples:** NaOH (−44 kJ/mol), H<sub>2</sub>SO<sub>4</sub> (highly exothermic), LiBr (−48 kJ/mol), HCl in water.

**Endothermic Dissolution ( $\Delta H_{\text{soln}} > 0$ ):**

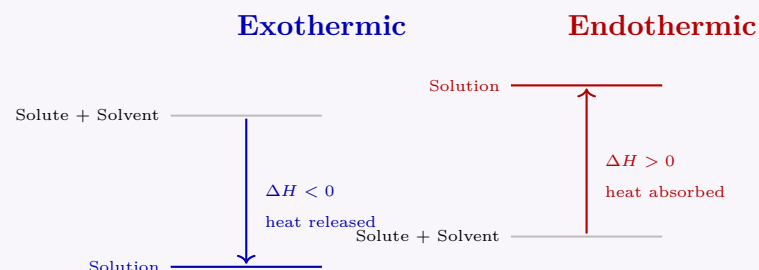
Occurs when **solute–solvent interactions are weaker** than what was broken:

$$\Delta H_1 + \Delta H_2 > |\Delta H_3|$$

**Molecular reasons:**

- The solute has a **strong crystal lattice** that costs more energy to break than is recovered by solvation: e.g. KNO<sub>3</sub> (+35 kJ/mol), NH<sub>4</sub>NO<sub>3</sub> (+26 kJ/mol)
- Solute–solvent interactions are relatively **weak** — e.g. non-polar solute in polar solvent: only weak induced-dipole forces form
- The water structure around ions must be **reorganised** (breaking H-bonds between water molecules costs  $\Delta H_2$ ), not fully compensated by solvation
- Despite being endothermic, dissolution proceeds because **entropy increases** significantly ( $\Delta S > 0$ ), making  $\Delta G < 0$

**Examples:** NaCl (+4 kJ/mol), KNO<sub>3</sub> (+35 kJ/mol), NH<sub>4</sub>NO<sub>3</sub> (+26 kJ/mol — used in cold packs).

**Summary Diagram:**


**Conclusion:** Whether dissolution is exothermic or endothermic depends entirely on the **relative strengths** of interactions broken (solute–solute, solvent–solvent) and formed (solute–solvent). Strong new interactions  $\Rightarrow$  exothermic; weak new interactions  $\Rightarrow$  endothermic.

**Q41.** Show that in dissolution process at equilibrium, the partial molal free energies of each constituent must be same in all phases. (10 marks)

[CHEM 113 | 2013–14]

**Answer**
**Equilibrium Condition in Dissolution — Partial Molar Free Energy:**

**Definition:** The **partial molar free energy** (chemical potential) of component  $i$  is:

$$\mu_i = \left( \frac{\partial G}{\partial n_i} \right)_{T, P, n_{j \neq i}}$$

**Proof — At Equilibrium,  $\mu_i$  Must Be Equal in All Phases:**

Consider a system with two phases  $\alpha$  and  $\beta$  (e.g. solid solute and solution) at constant  $T$  and  $P$ .

**Step 1 — Gibbs free energy of the system:**

$$G_{\text{total}} = G^\alpha + G^\beta$$

**Step 2 — Transfer of  $dn_i$  moles of component  $i$  from phase  $\alpha$  to  $\beta$ :**

$$dG = \mu_i^\beta dn_i - \mu_i^\alpha dn_i = (\mu_i^\beta - \mu_i^\alpha) dn_i$$

**Step 3 — Apply equilibrium condition:**

At equilibrium,  $G$  is at a **minimum** — any infinitesimal transfer causes no change in  $G$ :

$$dG = 0 \quad \Rightarrow \quad (\mu_i^\beta - \mu_i^\alpha) dn_i = 0$$



Since  $dn_i \neq 0$  in general:

$$\mu_i^\alpha = \mu_i^\beta$$

**Step 4 — Extension to multiple phases:**

For a system with phases  $\alpha, \beta, \gamma, \dots$  the same argument applies to each pair:

$$\mu_i^\alpha = \mu_i^\beta = \mu_i^\gamma = \dots$$

At dissolution equilibrium, the chemical potential of **each constituent** must be **identical in all phases**.

**Step 5 — Physical interpretation:**

- If  $\mu_i^\alpha > \mu_i^\beta$ : component  $i$  spontaneously transfers from  $\alpha$  to  $\beta$  (dissolution continues)
- If  $\mu_i^\alpha < \mu_i^\beta$ : component  $i$  transfers from  $\beta$  to  $\alpha$  (precipitation occurs)
- Only when  $\mu_i^\alpha = \mu_i^\beta$ : **no net transfer** — equilibrium achieved

**Application to dissolution:**

For a solid solute dissolving in a solvent:

$$\mu_{\text{solute}}^{\text{solid}}(T, P) = \mu_{\text{solute}}^{\text{solution}}(T, P)$$

$$\mu^\circ + RT \ln a_{\text{solid}} = \mu^\circ + RT \ln a_{\text{solution}}$$

For pure solid,  $a_{\text{solid}} = 1$ :

$$RT \ln a_{\text{solution}} = 0 \quad \Rightarrow \quad a_{\text{solution}} = K_{\text{sp}}$$

This directly gives the **solubility product expression**.

**Conclusion:** At dissolution equilibrium, the **partial molar free energy (chemical potential)** of each component must be **equal in all phases** present. This thermodynamic requirement, derived from the minimisation of Gibbs free energy, is the universal condition governing all phase equilibria including dissolution.

**Q42.** An aqueous solution of  $\text{H}_2\text{O}_2$  is 30% by mass and has a density of 1.11 g/mL. Calculate (i) molality, (ii) mole fraction of  $\text{H}_2\text{O}_2$ , and (iii) molarity. **(8 marks)** [CHEM 113 | 2013–14]

**Answer**

**Given:**

- Solution: 30%  $\text{H}_2\text{O}_2$  by mass, density = 1.11 g/mL
- $\mathcal{M}(\text{H}_2\text{O}_2) = 34.01$  g/mol
- $\mathcal{M}(\text{H}_2\text{O}) = 18.02$  g/mol
- Basis: **100 g** solution  $\Rightarrow$  30 g  $\text{H}_2\text{O}_2$  + 70 g  $\text{H}_2\text{O}$

**Moles:**

$$n_{\text{H}_2\text{O}_2} = \frac{30}{34.01} = 0.8821 \text{ mol} \quad n_{\text{H}_2\text{O}} = \frac{70}{18.02} = 3.885 \text{ mol}$$

**(i) Molality:**

$$m = \frac{0.8821}{0.070} = 12.60 \text{ mol/kg}$$

$$m = 12.60 \text{ mol/kg}$$

**(ii) Mole fraction of  $\text{H}_2\text{O}_2$ :**

$$X = \frac{0.8821}{0.8821 + 3.885} = \frac{0.8821}{4.767} = 0.1850$$

$$X_{\text{H}_2\text{O}_2} = 0.185$$

**(iii) Molarity:**

Volume of 100 g solution:

$$V = \frac{100}{1.11} = 90.09 \text{ mL} = 0.09009 \text{ L}$$

$$M = \frac{0.8821}{0.09009} = 9.791 \text{ mol/L}$$

$$M = 9.79 \text{ mol/L}$$

**Q43.** Show that the lowering of vapour pressure is dependent on the nature of solvent but not on the nature of the solute. **(8 marks)** [CHEM 113 | 2013–14]

**Answer**

**Vapour Pressure Lowering Depends on Solvent, Not Solute:**

By **Raoult's Law**, the relative lowering of vapour pressure is:

$$\frac{\Delta P}{P^\circ} = \frac{P^\circ - P_s}{P^\circ} = X_{\text{solute}} = \frac{n_2}{n_1 + n_2}$$

**Step 1 — Dependence on solvent:**

The **absolute** lowering of vapour pressure is:

$$\Delta P = X_{\text{solute}} \cdot P^\circ$$

This explicitly contains  $P^\circ$  — the vapour pressure of the **pure solvent**. Since  $P^\circ$  varies from solvent to solvent (e.g. water: 23.8 mmHg at 25°C; ethanol: 44.5 mmHg; benzene: 95.1 mmHg),  $\Delta P$  **changes with the solvent**, even for the same solute at the same mole fraction.

**Step 2 — Independence from solute nature:**

The **relative** lowering:

$$\frac{\Delta P}{P^\circ} = X_{\text{solute}} = \frac{n_2}{n_1 + n_2}$$

depends **only on the mole fraction** of solute particles — not on *what* those particles are.

For the same molality  $m$  with different non-electrolyte solutes (glucose, urea, sucrose) in the same solvent:

$$X_{\text{glucose}} = X_{\text{urea}} = X_{\text{sucrose}} \quad (\text{at same } m)$$

$$\therefore \frac{\Delta P}{P^\circ}(\text{glucose}) = \frac{\Delta P}{P^\circ}(\text{urea}) = \frac{\Delta P}{P^\circ}(\text{sucrose})$$

**Step 3 — Numerical illustration:**

Dissolve 0.1 mol of solute in 1 L water ( $n_1 = 55.5$  mol):

$$X_2 = \frac{0.1}{55.5 + 0.1} = 1.80 \times 10^{-3}$$

$$\Delta P = 1.80 \times 10^{-3} \times P^\circ_{\text{water}}$$

This is the same whether the solute is glucose, urea, or any other non-electrolyte — the chemical identity is irrelevant.

However, for the **same solute** in different solvents:

| Solvent | $P^\circ$ (mmHg) | $\Delta P$ (mmHg) |
|---------|------------------|-------------------|
| Water   | 23.8             | $23.8 \times X_2$ |
| Ethanol | 44.5             | $44.5 \times X_2$ |
| Benzene | 95.1             | $95.1 \times X_2$ |

For the same  $X_2$ ,  $\Delta P$  differs because  $P^\circ$  differs — confirming solvent dependence.

**Conclusion:** The **relative** lowering of vapour pressure ( $\Delta P/P^\circ = X_2$ ) depends only on the **number of solute particles** (mole fraction), not on their chemical nature — confirming its colligative character. The **absolute** lowering  $\Delta P = X_2 \cdot P^\circ$  depends on the solvent through  $P^\circ$ , not on the solute identity.

**Q44.** How does Arrhenius account for the application of the theories of colligative properties to weak electrolytes? **(10 marks)** [CHEM 113 | 2013–14]

**Answer**

**Arrhenius Theory and Colligative Properties of Weak Electrolytes:**

**The Problem:**

Classical colligative property equations predict effects based on the concentration of dissolved formula units. For **electrolytes**, experimentally measured effects (e.g.  $\Delta T_f$ ,  $\pi$ ) are **larger than predicted** for non-electrolytes — and the excess depends on the degree of ionisation. Arrhenius (1887) explained this through his **theory of electrolytic dissociation**.

**Arrhenius's Account:**

(a) **Partial dissociation of weak electrolytes:** Weak electrolytes (e.g.  $\text{CH}_3\text{COOH}$ ,  $\text{NH}_4\text{OH}$ ) dissociate *partially* in solution:



If degree of dissociation =  $\alpha$  and 1 mol is dissolved:

- Undissociated molecules:  $(1 - \alpha)$  mol
- Ions formed:  $2\alpha$  mol
- Total particles:  $(1 - \alpha) + 2\alpha = 1 + \alpha$

(b) **Van't Hoff factor  $i$ :** The ratio of actual to expected particles:

$$i = \frac{1 + \alpha(n - 1)}{1}$$

where  $n$  = number of ions per formula unit. For a binary weak electrolyte ( $n = 2$ ):

$$i = 1 + \alpha \quad \Rightarrow \quad \alpha = i - 1$$

(c) **Modified colligative equations:**

$$\Delta T_f = iK_f m, \quad \Delta T_b = iK_b m, \quad \pi = iMRT$$

(d) **Determining  $\alpha$  from colligative data:** By measuring  $\Delta T_f$  of a weak acid solution:

$$i = \frac{\Delta T_f^{\text{observed}}}{K_f m} \quad \alpha = i - 1$$

This allows experimental determination of the **degree of ionisation** from simple physical measurements.

**Example — Acetic Acid:**

For 0.1 M CH<sub>3</sub>COOH (degree of dissociation  $\alpha = 0.013$ ):

$$i = 1 + 0.013 = 1.013$$

$$\Delta T_f = 1.013 \times 1.86 \times 0.1 = 0.1884^\circ\text{C}$$

vs.  $0.186^\circ\text{C}$  for a non-electrolyte — a small but measurable difference confirming partial dissociation.

**Significance:**

- Provided **first direct experimental evidence** for ionic dissociation in solution
- Reconciled the anomalously high colligative effects of electrolytes with theory
- Established the **van't Hoff factor  $i$**  as a quantitative measure of dissociation
- Enabled determination of  $K_a$  and  $K_b$  from colligative measurements

**Conclusion:** Arrhenius accounted for anomalous colligative properties of weak electrolytes by proposing **partial ionic dissociation**, which increases the actual number of particles in solution. The van't Hoff factor  $i = 1 + \alpha$  quantifies this, allowing colligative equations to correctly predict the enhanced freezing point depression, boiling point elevation, and osmotic pressure of electrolyte solutions.

**Q45.** How much ethylene glycol (C<sub>2</sub>H<sub>4</sub>O<sub>2</sub>, molar mass = 62.1), must be added to 10.0 L of water to produce a solution that freezes at  $-23.3^\circ\text{C}$ ? Assume the density of water is exactly 1 g/mL. **(10 marks)** [CHEM 113 | 2012–13]

**Answer**

**Given:**

- Solute: Ethylene glycol (EG),  $\mathcal{M} = 62.1$  g/mol
- Volume of water:  $V = 10.0$  L
- Density of water: 1 g/mL  $\Rightarrow$  mass of water = 10,000 g = 10.0 kg
- Target freezing point:  $T_f = -23.3^\circ\text{C}$
- $K_f(\text{water}) = 1.86^\circ\text{C/m}$
- EG is a non-electrolyte:  $i = 1$

**Step 1 — Required  $\Delta T_f$ :**

$$\Delta T_f = 0 - (-23.3) = 23.3^\circ\text{C}$$

**Step 2 — Required molality:**

$$m = \frac{\Delta T_f}{K_f} = \frac{23.3}{1.86} = 12.53 \text{ mol/kg}$$

**Step 3 — Moles of EG needed:**

$$n_{\text{EG}} = m \times m_{\text{solvent (kg)}} = 12.53 \times 10.0 = 125.3 \text{ mol}$$

**Step 4 — Mass of EG:**

$$w_{\text{EG}} = n \times \mathcal{M} = 125.3 \times 62.1$$

$$w_{\text{EG}} = 7,781 \text{ g} \approx 7.78 \text{ kg}$$

**Conclusion:** Approximately **7.78 kg** of ethylene glycol must be added to 10.0 L of water to produce a solution that freezes at  $-23.3^\circ\text{C}$  — a typical winter antifreeze formulation for cold climates.

**Q46.** What is the effect of temperature on the solubility of gases in liquid? How is this process related to thermal pollution in aquatic environments? **(8 marks)** [CHEM 113 | 2012–13]

## Answer

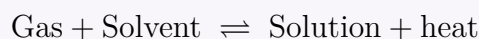
## Effect of Temperature on Gas Solubility &amp; Thermal Pollution

(8 marks)

## Part 1: Effect of Temperature on Gas Solubility

(4 marks)

The dissolution of a gas in a liquid is generally **exothermic**:



By **Le Chatelier's Principle**: increasing temperature shifts the equilibrium *backward* (to the left), favouring the release of dissolved gas. Therefore:

$$\text{Solubility of gas} \propto \frac{1}{T}$$

- As temperature **increases**, gas solubility **decreases**.
- As temperature **decreases**, gas solubility **increases**.

## Quantitatively (Henry's Law):

$$C = k_H \cdot P_{\text{gas}}$$

The Henry's Law constant  $k_H$  **decreases** with increasing temperature, confirming reduced solubility at higher  $T$ .

Examples:

| Gas             | Solubility at 20°C | Solubility at 40°C |
|-----------------|--------------------|--------------------|
| O <sub>2</sub>  | 9.1 mg/L           | 6.4 mg/L           |
| CO <sub>2</sub> | 1.7 g/L            | 0.97 g/L           |

## Part 2: Relation to Thermal Pollution in Aquatic Environments

(4 marks)

**Thermal pollution** occurs when industrial facilities (power plants, factories) discharge heated water into rivers, lakes, or oceans, raising the water temperature.

Chain of effects:

- Higher water temperature**  $\Rightarrow$  **lower dissolved oxygen (DO)** content (O<sub>2</sub> solubility decreases with  $T$ ).
- Lower DO**  $\Rightarrow$  fish, aquatic invertebrates and aerobic microorganisms suffer from **oxygen stress** or die (hypoxia/anoxia).
- Accelerated decomposition**: Warmer water increases metabolic rates of decomposing bacteria, which consume even more O<sub>2</sub>, further depleting DO.
- Algal blooms**: Warmer, nutrient-rich water promotes explosive algal growth. When algae die, their decomposition by bacteria further depletes DO — a process called **eutrophication**.
- Toxic gas release**: At very low DO, anaerobic bacteria produce toxic H<sub>2</sub>S and CH<sub>4</sub>, making the environment lethal for aquatic life.

**Real example**: Normal river water at 20°C contains  $\sim 9$  mg/L of dissolved O<sub>2</sub>. Heating to 35°C reduces this to  $\sim 7$  mg/L — already below the  $\geq 8$  mg/L needed by sensitive fish species like salmon and trout.

**Conclusion**: Gas solubility decreases with increasing temperature because dissolution is exothermic. Thermal pollution exploits this directly — heated discharge water holds less dissolved O<sub>2</sub>, leading to hypoxia, accelerated decomposition, algal blooms, and ultimately the death of aquatic ecosystems. Temperature control of industrial effluents is therefore a critical environmental concern.

Q47. Explain the principle of reverse osmosis in the case of desalination and purification of water. (8 marks) [CHEM 113 | 2012–13]

## Answer

## Reverse Osmosis: Principle, Desalination &amp; Water Purification

(8 marks)

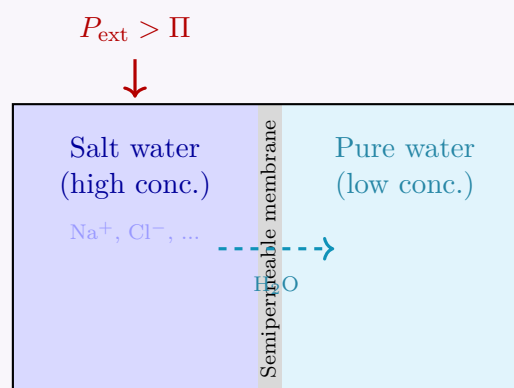
## Part 1: Normal Osmosis vs Reverse Osmosis

(2 marks)

In **normal osmosis**, solvent flows spontaneously from low concentration (pure water) to high concentration (salt water) through a semipermeable membrane, driven by the osmotic pressure difference  $\Pi$ .

In **reverse osmosis (RO)**, an external pressure **greater than**  $\Pi$  is applied to the salt water side, forcing solvent molecules to flow in the *reverse* direction — from high concentration to low:

$$P_{\text{applied}} > \Pi \Rightarrow \text{water flows: salt solution} \rightarrow \text{pure water}$$



### Part 2: Application to Desalination

(3 marks)

**Desalination** is the removal of salt from seawater to produce freshwater. Seawater has  $\Pi \approx 27$  atm.

- Seawater is pressurized to  $\sim 60$ – $80$  atm (well above its osmotic pressure).
- Water molecules are forced through the RO membrane (typically polyamide thin-film), leaving behind  $\text{Na}^+$ ,  $\text{Cl}^-$ ,  $\text{Mg}^{2+}$ , and other ions.
- The membrane pore size ( $\sim 0.0001 \mu\text{m}$ ) is small enough to block ions but allows water to pass.
- The concentrated brine is discharged; pure water is collected on the other side.

**Efficiency:** Modern RO plants recover  $\sim 40$ – $60\%$  of the feed water as pure water. Used extensively in the Middle East, Singapore, and coastal cities worldwide.

### Part 3: Application to Water Purification

(2 marks)

RO removes a wide range of contaminants beyond salt:

| Contaminant                                              | Removal by RO              |
|----------------------------------------------------------|----------------------------|
| Dissolved salts ( $\text{NaCl}$ , etc.)                  | $>95\%$ removed            |
| Heavy metals ( $\text{Pb}$ , $\text{As}$ , $\text{Hg}$ ) | $>95\%$ removed            |
| Nitrates, fluorides                                      | $\sim 85$ – $95\%$ removed |
| Bacteria, viruses                                        | Effectively removed        |
| Pesticides, herbicides                                   | $>97\%$ removed            |

**Limitation:** RO does not remove dissolved gases (e.g.  $\text{CO}_2$ ,  $\text{Cl}_2$ ) or some very small organic molecules — a post-treatment step (activated carbon filter) is often used for complete purification.

**Conclusion:** Reverse osmosis applies external pressure exceeding the osmotic pressure to force pure water through a semipermeable membrane against the natural osmotic direction. For desalination, pressures of  $60$ – $80$  atm overcome seawater's  $\Pi \approx 27$  atm, producing freshwater with  $>95\%$  salt removal. The same principle purifies water from heavy metals, pathogens, and organic pollutants, making RO one of the most important water treatment technologies.

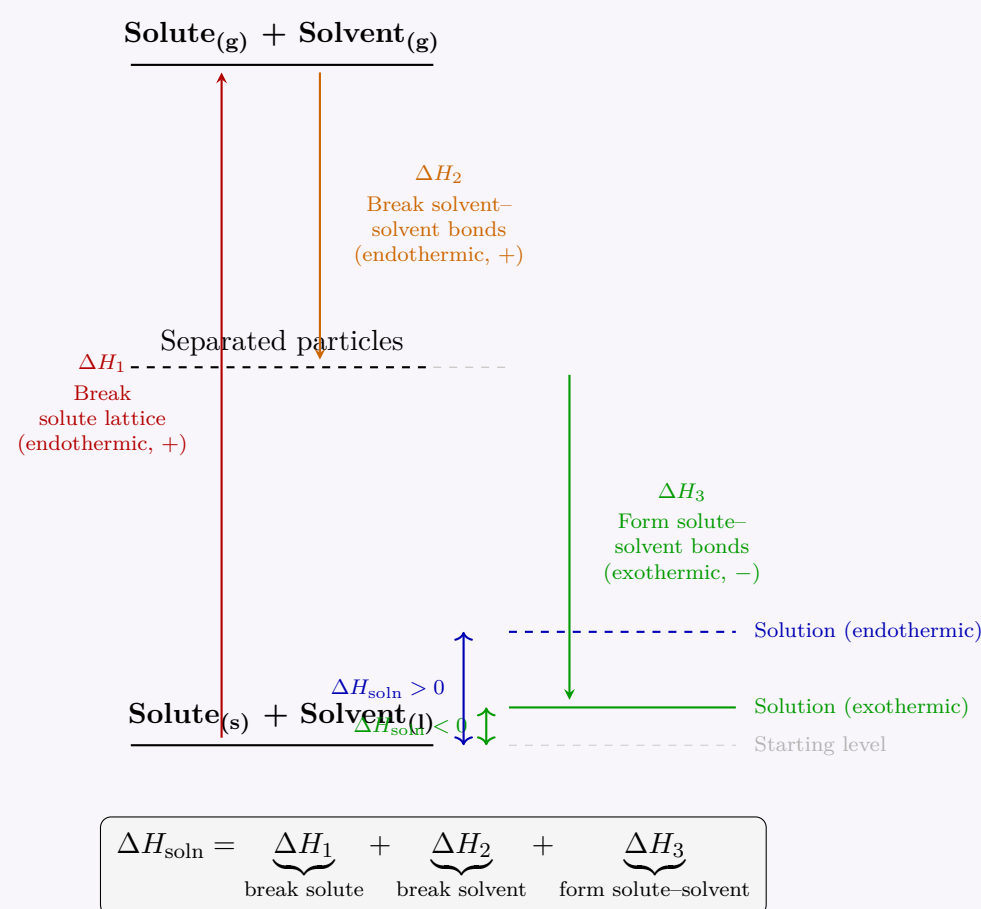
**Q48.** How does heat evolve or absorb during dissolution of solute in a solvent? (11 marks)

[CHEM 113 | 2011–12]

#### Answer

The heat change during dissolution depends on the relative magnitudes of three energy terms. The overall process can be visualised using a **three-step energy cycle**:

**Three-Step Model of Dissolution:**



$$\Delta H_{\text{soln}} = \underbrace{\Delta H_1}_{\text{break solute lattice}} + \underbrace{\Delta H_2}_{\text{break solvent-solvent}} + \underbrace{\Delta H_3}_{\text{form solute-solvent}}$$

$\Delta H_1$  and  $\Delta H_2$  are always **endothermic** (+);  $\Delta H_3$  is always **exothermic** (−).

#### Case 1 — Exothermic Dissolution ( $\Delta H_{\text{soln}} < 0$ ):

When the energy released by **solute-solvent interactions** ( $|\Delta H_3|$ ) exceeds the energy needed to break the lattice and solvent:

$$|\Delta H_3| > \Delta H_1 + \Delta H_2$$

- Heat is **evolved** — solution becomes warmer
- Examples: NaOH in water (−44 kJ/mol), H<sub>2</sub>SO<sub>4</sub> in water (very exothermic), LiBr in water
- Strong ion-dipole or hydrogen bonding between solute and solvent releases large energy

#### Case 2 — Endothermic Dissolution ( $\Delta H_{\text{soln}} > 0$ ):

When lattice and solvent-breaking energy exceeds solvation energy:

$$\Delta H_1 + \Delta H_2 > |\Delta H_3|$$

- Heat is **absorbed** — solution becomes cooler
- Examples: NaCl (+4 kJ/mol), NH<sub>4</sub>NO<sub>3</sub> (+26 kJ/mol) — used in cold packs, KNO<sub>3</sub> (+35 kJ/mol)
- Despite being endothermic, dissolution is spontaneous due to large increase in entropy ( $\Delta S > 0$ )

#### Case 3 — Ideal/Athermal Dissolution ( $\Delta H_{\text{soln}} \approx 0$ ):

When solute-solvent interactions are nearly identical in strength to solute-solute and solvent-solvent interactions:

- Essentially no heat change
- Example: benzene-toluene mixture (very similar intermolecular forces) — forms nearly ideal solution

#### Summary Table:

| Condition                                      | Heat change              | Temperature | Example                               |
|------------------------------------------------|--------------------------|-------------|---------------------------------------|
| $ \Delta H_3  > \Delta H_1 + \Delta H_2$       | Exothermic (−)           | Increases   | NaOH, H <sub>2</sub> SO <sub>4</sub>  |
| $ \Delta H_3  < \Delta H_1 + \Delta H_2$       | Endothermic (+)          | Decreases   | NaCl, NH <sub>4</sub> NO <sub>3</sub> |
| $ \Delta H_3  \approx \Delta H_1 + \Delta H_2$ | Athermal ( $\approx 0$ ) | No change   | Benzene-toluene                       |

**Conclusion:** Whether heat is evolved or absorbed during dissolution depends entirely on the **relative magnitudes** of the lattice/solvent-breaking energies and the solvation energy. The sign of  $\Delta H_{\text{soln}}$  determines whether the process is exothermic or endothermic, while entropy drives spontaneity even when  $\Delta H > 0$ .



**Q49.** Explain the principle underlying the process of steam distillation. (6 marks)

[CHEM 113 | 2011–12]

### Answer

#### Steam Distillation — Principle:

Steam distillation is a technique used to purify **high-boiling, water-immiscible organic compounds** at temperatures well below their normal boiling points by passing steam through the mixture.

#### Underlying Principle:

For a **two-component immiscible system** (organic compound + water), the components do not interact — they behave as completely independent liquids. By **Dalton's Law of Partial Pressures**, the total vapour pressure of the mixture is simply the *sum* of the individual vapour pressures:

$$P_{\text{total}} = P_{\text{water}}^{\circ} + P_{\text{organic}}^{\circ}$$

This is the key distinction from miscible solutions, where Raoult's law would reduce each component's vapour pressure.

#### Why Boiling Occurs at Lower Temperature:

The mixture boils when  $P_{\text{total}}$  reaches atmospheric pressure (760 mmHg). Since:

$$P_{\text{water}}^{\circ} + P_{\text{organic}}^{\circ} = 760 \text{ mmHg}$$

each component contributes a *partial* vapour pressure, so the total reaches 760 mmHg at a temperature **lower than the boiling point of either pure component**.

For example, bromobenzene (bp = 156°C) steam distils at 95.5°C because at that temperature:

$$P_{\text{H}_2\text{O}} + P_{\text{bromobenzene}} = 648 + 112 = 760 \text{ mmHg}$$

#### Composition of Distillate:

The mole ratio of water to organic compound in the distillate follows from Dalton's Law:

$$\frac{n_{\text{organic}}}{n_{\text{water}}} = \frac{P_{\text{organic}}^{\circ}}{P_{\text{water}}^{\circ}}$$

Converting to mass ratio:

$$\frac{w_{\text{organic}}}{w_{\text{water}}} = \frac{P_{\text{organic}}^{\circ} \times M_{\text{organic}}}{P_{\text{water}}^{\circ} \times M_{\text{water}}}$$

#### Applications:

- Isolation of heat-sensitive natural products (e.g. essential oils, aniline)
- Purification of high-boiling organics without decomposition
- Industrial extraction of turpentine, camphor, and rose oil

**Conclusion:** Steam distillation works because immiscible liquids each exert their **full independent** vapour pressure, causing the mixture to boil at a temperature lower than either pure component — allowing safe distillation of heat-sensitive, high-boiling organic compounds.

**Q50.** A mixture of an organic compound (M.W. 123) which is immiscible with water has a boiling point of 99°C at 760 mm pressure. If the vapour pressure of water is 733 mm, calculate the ratio by weight of the two substances in the distillate. (9 marks) [CHEM 113 | 2011–12]

### Answer

#### Given:

- Molar mass of organic compound:  $M_{\text{org}} = 123 \text{ g/mol}$
- Boiling point of mixture: 99°C at 760 mmHg
- Vapour pressure of water at 99°C:  $P_{\text{water}} = 733 \text{ mmHg}$
- Molar mass of water:  $M_{\text{water}} = 18 \text{ g/mol}$

#### Step 1 — Vapour pressure of organic compound:

By Dalton's Law for immiscible liquids:

$$P_{\text{total}} = P_{\text{water}} + P_{\text{org}}$$

$$P_{\text{org}} = 760 - 733 = 27 \text{ mmHg}$$

#### Step 2 — Mole ratio in distillate:

$$\frac{n_{\text{org}}}{n_{\text{water}}} = \frac{P_{\text{org}}}{P_{\text{water}}} = \frac{27}{733}$$

#### Step 3 — Weight ratio in distillate:

$$\frac{w_{\text{org}}}{w_{\text{water}}} = \frac{n_{\text{org}} \times M_{\text{org}}}{n_{\text{water}} \times M_{\text{water}}} = \frac{27 \times 123}{733 \times 18} = \frac{3321}{13194}$$

$$\frac{w_{\text{org}}}{w_{\text{water}}} = 0.2518 \approx 0.252$$

Or expressed as a ratio: **0.252 : 1** (i.e. for every 1 g of water, 0.252 g of organic compound is collected in the distillate).

**Conclusion:** The weight ratio of organic compound to water in the steam distillate is approximately **0.25 : 1**, meaning the distillate contains about 20% organic compound and 80% water by mass.

**Q51.** Explain the following terms: (i) Ideal solution, (ii) Salting out effect, (iii) Critical Solution Temperature (CST). (9 marks) [CHEM 113 | 2011–12]

### Answer

#### (i) Ideal Solution:

An **ideal solution** is one in which every component obeys **Raoult's Law** over the entire range of composition at all temperatures:

$$P_i = X_i \cdot P_i^\circ$$

where  $P_i$  is the partial vapour pressure of component  $i$ ,  $X_i$  is its mole fraction, and  $P_i^\circ$  is its pure vapour pressure.

#### Characteristics of an ideal solution:

- $\Delta H_{\text{mix}} = 0$  — no heat evolved or absorbed
- $\Delta V_{\text{mix}} = 0$  — no volume change on mixing
- Solute–solvent interactions are identical in strength to solute–solute and solvent–solvent interactions
- $\Delta S_{\text{mix}} > 0$  — positive entropy of mixing (only driving force)
- Examples: benzene–toluene, hexane–heptane, ethanol–methanol

#### (ii) Salting Out Effect:

The **salting out effect** is the decrease in the solubility of a non-electrolyte (e.g. a gas, organic compound, or protein) in water upon addition of a **high concentration of an inorganic salt**.

#### Mechanism:

- The added salt (e.g. NaCl,  $(\text{NH}_4)_2\text{SO}_4$ ) dissociates into ions that are strongly hydrated
- These ions compete with the non-electrolyte for water molecules, reducing the **activity of water** available to solvate the non-electrolyte
- The non-electrolyte becomes less soluble and **precipitates out** of solution

#### Applications:

- Precipitation of proteins in biochemistry (ammonium sulfate fractionation)
- Separation of soap from solution in soap manufacture
- Reducing solubility of gases in industrial processes

#### (iii) Critical Solution Temperature (CST):

The **Critical Solution Temperature** is the temperature at which two partially miscible liquids become completely miscible (or completely immiscible) — the boundary condition of a two-phase system.

#### Two types:

- **Upper CST (UCST):** Below this temperature, the liquids are partially miscible (two phases); above it, they mix completely. Example: phenol–water system, UCST = 65.6°C.
- **Lower CST (LCST):** Above this temperature, the liquids separate into two phases; below it, they are miscible. Example: triethylamine–water, LCST = 18.5°C.

At the CST, the compositions of the two coexisting phases become identical — the system passes from a two-phase to a one-phase region:



**Conclusion:** An ideal solution follows Raoult's law with no heat of mixing. Salting out reduces solubility by ion–water competition. CST marks the critical temperature at which partial miscibility transitions to complete miscibility.

**Q52.** What are colligative properties? What is meant by the relative lowering of vapour pressure? Mention the application of lowering of vapour pressure in determining the molecular weight of a dissolved substance. (9 marks) [CHEM 113 | 2011–12]

### Answer

#### Colligative Properties:

Colligative properties depend only on the **number of solute particles**, not their identity. They are: vapour pressure lowering, boiling point elevation, freezing point depression, and osmotic pressure.

#### Relative Lowering of Vapour Pressure:

By Raoult's Law, the vapour pressure of a solution is:

$$P_s = X_{\text{solvent}} \cdot P^\circ$$

The **lowering** of vapour pressure:

$$\Delta P = P^\circ - P_s = P^\circ(1 - X_{\text{solvent}}) = X_{\text{solute}} \cdot P^\circ$$

The **relative lowering** of vapour pressure is defined as:

$$\frac{\Delta P}{P^\circ} = \frac{P^\circ - P_s}{P^\circ} = X_{\text{solute}} = \frac{n_{\text{solute}}}{n_{\text{solute}} + n_{\text{solvent}}}$$

This is dimensionless and depends only on the **mole fraction of solute** — i.e., on the number of dissolved particles, not their nature.

#### Determination of Molecular Weight from Vapour Pressure Lowering:

For dilute solutions ( $n_{\text{solute}} \ll n_{\text{solvent}}$ ):

$$\frac{\Delta P}{P^\circ} \approx \frac{n_{\text{solute}}}{n_{\text{solvent}}} = \frac{w_2/\mathcal{M}_2}{w_1/\mathcal{M}_1}$$

where  $w_1, \mathcal{M}_1$  = mass and molar mass of solvent;  $w_2, \mathcal{M}_2$  = mass and molar mass of solute.

Rearranging to find molar mass of solute:

$$\mathcal{M}_2 = \frac{w_2 \cdot \mathcal{M}_1 \cdot P^\circ}{w_1 \cdot \Delta P}$$

#### Procedure:

- Dissolve known mass  $w_2$  of solute in known mass  $w_1$  of solvent
- Measure  $P^\circ$  (pure solvent) and  $P_s$  (solution)
- Calculate  $\Delta P = P^\circ - P_s$
- Use formula above to find  $\mathcal{M}_2$

**Conclusion:** Relative vapour pressure lowering equals the mole fraction of solute — a purely colligative quantity. By measuring this lowering, the molar mass of an unknown non-volatile solute can be determined directly from the Raoult's law relationship.

**Q53.** Thermodynamically prove that the elevation of boiling point of a solvent produced by a dissolved non-volatile substance is proportional to mole fraction and also to molality of the substance in the solution. **(12 marks)** [CHEM 113 | 2011–12]

### Answer

#### Thermodynamic Proof — Boiling Point Elevation:

##### Starting point — Chemical Potential:

At the boiling point of a solution, the chemical potential of the solvent in the liquid phase equals that in the vapour:

$$\mu_{\text{liq}} = \mu_{\text{vap}}$$

For the pure solvent vapour (ideal gas):

$$\mu_{\text{vap}} = \mu^\circ + RT \ln \frac{P}{P^\circ}$$

For the solvent in solution (Raoult's Law):

$$\mu_{\text{liq}} = \mu_{\text{liq}}^\circ + RT \ln X_1$$

where  $X_1$  is the mole fraction of solvent.

##### Step 1 — Relate vapour pressure to mole fraction:

At equilibrium:

$$\mu_{\text{liq}}^\circ + RT \ln X_1 = \mu_{\text{vap}}^\circ + RT \ln \frac{P_s}{P^\circ}$$

For pure solvent ( $X_1 = 1, P_s = P^\circ$ ):

$$\mu_{\text{liq}}^\circ = \mu_{\text{vap}}^\circ$$

Subtracting:

$$RT \ln X_1 = RT \ln \frac{P_s}{P^\circ} \Rightarrow \ln X_1 = \ln \frac{P_s}{P^\circ}$$

##### Step 2 — Differentiate with respect to temperature:

Using the Gibbs–Helmholtz equation:

$$\left( \frac{\partial \ln P}{\partial T} \right)_P = \frac{\Delta H_{\text{vap}}}{RT^2}$$

For solution vs. pure solvent:

$$\frac{d \ln X_1}{dT} = \frac{\Delta H_{\text{vap}}}{RT^2} \left( \frac{1}{T_b^\circ} - \frac{1}{T_b} \right)$$

**Step 3 — Integrate between  $T_b^\circ$  and  $T_b$ :**

For dilute solution:  $\ln X_1 \approx -X_2$  (since  $X_1 = 1 - X_2$  and  $\ln(1 - X_2) \approx -X_2$  for small  $X_2$ ).

$$-X_2 = \frac{\Delta H_{\text{vap}}}{R} \cdot \frac{T_b - T_b^\circ}{T_b^\circ \cdot T_b}$$

Since  $T_b \approx T_b^\circ$  for dilute solutions,  $T_b \cdot T_b^\circ \approx (T_b^\circ)^2$ :

$$X_2 = \frac{\Delta H_{\text{vap}}}{R(T_b^\circ)^2} \cdot \Delta T_b$$

Rearranging:

$$\Delta T_b = \frac{R(T_b^\circ)^2}{\Delta H_{\text{vap}}} \cdot X_2$$

This proves  $\Delta T_b \propto X_2$  (mole fraction of solute).

**Step 4 — Express in terms of molality:**

For a dilute solution with  $n_2 \ll n_1$ :

$$X_2 = \frac{n_2}{n_1 + n_2} \approx \frac{n_2}{n_1} = \frac{n_2}{w_1/\mathcal{M}_1} = \frac{m \cdot \mathcal{M}_1}{1000}$$

where  $m$  is molality (mol/kg) and  $\mathcal{M}_1$  is molar mass of solvent (g/mol).

Substituting:

$$\Delta T_b = \frac{R(T_b^\circ)^2 \mathcal{M}_1}{1000 \Delta H_{\text{vap}}} \cdot m = K_b \cdot m$$

where the **ebullioscopic constant** is:

$$K_b = \frac{R(T_b^\circ)^2 \mathcal{M}_1}{1000 \Delta H_{\text{vap}}}$$

$$\Delta T_b = K_b \cdot m$$

This proves  $\Delta T_b \propto m$  (molality of solute).

**Conclusion:** By thermodynamic reasoning via chemical potentials and the Gibbs–Helmholtz equation, the boiling point elevation is rigorously shown to be directly proportional to both the **mole fraction** and the **molality** of the dissolved non-volatile solute, with  $K_b$  depending only on the solvent properties.

**Q54.** Calculate the boiling point of the solution containing 12.5 g of benzoic acid ( $\text{C}_7\text{H}_6\text{O}_2$ ) in 110 g of benzene ( $\text{C}_6\text{H}_6$ ). The boiling point of benzene is  $80.1^\circ\text{C}$  and its  $K_b$  is  $2.53^\circ\text{C}\cdot\text{m}^{-1}$ . (8 marks) [CHEM 113 | 2011–12]

### Answer

**Given:**

- Solute: Benzoic acid ( $\text{C}_7\text{H}_6\text{O}_2$ ),  $\mathcal{M} = 7(12) + 6(1) + 2(16) = 122$  g/mol
- Mass of benzoic acid:  $w_2 = 12.5$  g
- Mass of benzene:  $w_1 = 110$  g
- Boiling point of pure benzene:  $T_b^\circ = 80.1^\circ\text{C}$
- $K_b(\text{benzene}) = 2.53^\circ\text{C}/\text{m}$
- **Note:** Benzoic acid **dimerises** in benzene (non-polar solvent), so effective molar mass =  $2 \times 122 = 244$  g/mol

**Step 1 — Moles of solute (accounting for dimerisation):**

$$n = \frac{12.5}{244} = 0.05123 \text{ mol}$$

**Step 2 — Molality:**

$$m = \frac{0.05123}{0.110} = 0.4657 \text{ mol/kg}$$

**Step 3 — Boiling point elevation:**

$$\Delta T_b = K_b \times m = 2.53 \times 0.4657 = 1.178^\circ\text{C}$$

**Step 4 — Boiling point of solution:**

$$T_b = 80.1 + 1.178$$

$$T_b = 81.28^\circ\text{C} \approx 81.3^\circ\text{C}$$

**Note:** If dimerisation is ignored ( $\mathcal{M} = 122$ ),  $\Delta T_b = 2.36^\circ\text{C}$  and  $T_b = 82.46^\circ\text{C}$ . The dimerisation halves the particle count, halving the elevation.

**Q55.** Derive Van't Hoff equation for the osmotic pressure of a dilute solution. (6 marks)

[CHEM 113 | 2011–12]

### Answer

#### Derivation of Van't Hoff Equation for Osmotic Pressure:

##### Starting point — Chemical potential equilibrium:

At osmotic equilibrium, the chemical potential of the solvent is equal on both sides of the semipermeable membrane:

$$\mu_{\text{solvent}}^{\text{solution}}(T, P + \pi) = \mu_{\text{pure solvent}}^*(T, P)$$

##### Step 1 — Effect of composition on chemical potential:

For the solution side:

$$\mu_{\text{solution}}(T, P) = \mu^*(T, P) + RT \ln X_1$$

where  $X_1$  is the mole fraction of solvent.

##### Step 2 — Effect of pressure on chemical potential:

$$\mu(T, P + \pi) = \mu(T, P) + \int_P^{P+\pi} V_m dP \approx \mu(T, P) + V_m \pi$$

where  $V_m$  is the molar volume of solvent (incompressible).

##### Step 3 — Apply equilibrium condition:

$$\mu^*(T, P) + RT \ln X_1 + V_m \pi = \mu^*(T, P)$$

$$RT \ln X_1 + V_m \pi = 0$$

$$\pi = -\frac{RT \ln X_1}{V_m}$$

##### Step 4 — Simplify for dilute solution:

For dilute solution:  $X_1 = 1 - X_2 \approx 1$ , so:

$$\ln X_1 = \ln(1 - X_2) \approx -X_2$$

$$\pi = \frac{RT \cdot X_2}{V_m}$$

For dilute solution:  $X_2 \approx \frac{n_2}{n_1}$  and  $V_m = \frac{V}{n_1}$  (total volume), so:

$$X_2/V_m = \frac{n_2/n_1}{V/n_1} = \frac{n_2}{V} = M$$

Therefore:

$$\boxed{\pi = MRT}$$

This is the **Van't Hoff equation** for osmotic pressure, analogous to the ideal gas law  $P = nRT/V$ , confirming that dilute solutions obey gas-like behaviour with respect to osmotic pressure.

**Q56.** Discuss the mechanism of dissolution of sodium chloride in water and sucrose in water. (10 marks)

[CHEM 101 | 2010–11]

### Answer

#### (i) Dissolution of NaCl in Water:

NaCl is an **ionic compound** held together by strong electrostatic forces in a face-centred cubic lattice.

##### Mechanism:

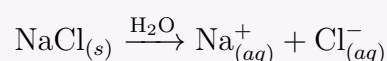
(a) Water molecules ( $\mu = 1.85$  D) orient around the NaCl surface — the  $\delta^-$  oxygen points toward  $\text{Na}^+$  and  $\delta^+$  hydrogen toward  $\text{Cl}^-$

(b) **Ion–dipole forces** pull individual ions away from the lattice surface

(c) Each ion is surrounded by a **hydration shell** of oriented water molecules — releasing hydration energy:

$$\text{Na}^+ : \Delta H_{\text{hyd}} = -406 \text{ kJ/mol} \quad \text{Cl}^- : \Delta H_{\text{hyd}} = -363 \text{ kJ/mol}$$

(d) The hydrated ions migrate freely into solution:



(e)  $\Delta H_{\text{soln}} = +4 \text{ kJ/mol}$  (slightly endothermic), but spontaneous due to  $\Delta S > 0$

#### (ii) Dissolution of Sucrose in Water:

Sucrose ( $\text{C}_{12}\text{H}_{22}\text{O}_{11}$ ) is a **molecular compound** held by hydrogen bonds and van der Waals forces. It has 8 OH groups capable of forming hydrogen bonds.

##### Mechanism:

(a) Water molecules form **hydrogen bonds** with the OH groups of sucrose at the crystal surface ( $\delta^+$  H of water  $\cdots$  O of sucrose, and H of sucrose  $\cdots$   $\delta^-$  O of water)

- (b) These H-bonds gradually **pull sucrose molecules** away from the crystal intact — no ionisation occurs
- (c) Each sucrose molecule becomes surrounded by a **hydration sphere** of water molecules:



- (d) The process is **exothermic** ( $\Delta H_{\text{soln}} < 0$ ) because the O–H···O hydrogen bonds formed between sucrose and water are stronger than the H-bonds broken in pure water and pure sucrose

**Comparison:**

| Feature                  | NaCl                                       | Sucrose              |
|--------------------------|--------------------------------------------|----------------------|
| Bond type in solid       | Ionic lattice                              | Hydrogen bonds, vdW  |
| Dissolution force        | Ion–dipole                                 | Hydrogen bonding     |
| Particles formed         | Na <sup>+</sup> + Cl <sup>−</sup> (2 ions) | 1 sucrose molecule   |
| $\Delta H_{\text{soln}}$ | +4 kJ/mol (endothermic)                    | Slightly exothermic  |
| Conductivity             | Yes (electrolyte)                          | No (non-electrolyte) |

**Conclusion:** NaCl dissolves via **ion–dipole** interactions producing free ions, while sucrose dissolves via **hydrogen bonding** remaining intact as molecules. In both cases, the solute–solvent interactions are strong enough to overcome solute–solute forces, driving dissolution.

**Q57.** State and explain different forms of Henry’s law. Mention the limitations of Henry’s law. (8 marks) [CHEM 101 | 2010–11]

#### Answer

##### Henry’s Law — Statement and Forms:

Henry’s Law (1803) states that at constant temperature, the **amount of gas dissolved in a liquid** is directly proportional to the **partial pressure** of the gas above the liquid.

##### Different Forms:

###### Form 1 — Concentration form:

$$C = k_H \cdot P$$

where  $C$  = concentration of dissolved gas (mol/L),  $P$  = partial pressure of gas (atm),  $k_H$  = Henry’s law constant (mol/L·atm).

###### Form 2 — Mole fraction form:

$$P = K_H \cdot X$$

where  $X$  = mole fraction of gas in solution,  $K_H$  = Henry’s constant (atm) — increases with temperature.

###### Form 3 — Bunsen absorption coefficient ( $\alpha$ ):

$$V_{\text{gas}} = \alpha \cdot V_{\text{solvent}} \cdot P$$

where  $\alpha$  = volume of gas (at STP) dissolved per unit volume of solvent per unit pressure.

###### Form 4 — Ostwald coefficient ( $L$ ):

$$L = \frac{\text{volume of gas dissolved}}{\text{volume of solvent}}$$

Independent of pressure (unlike Bunsen coefficient).

##### Limitations of Henry’s Law:

- High pressure:** at very high pressures, the linear relationship breaks down — interactions between gas molecules and solvent deviate from ideal behaviour.
- High concentration:** valid only for **dilute solutions**. At high concentrations, solute–solute interactions become significant.
- Reactive gases:** does not apply to gases that chemically react with the solvent. e.g.  $\text{CO}_2 + \text{H}_2\text{O} \rightarrow \text{H}_2\text{CO}_3$ ,  $\text{HCl} + \text{H}_2\text{O} \rightarrow \text{H}_3\text{O}^+ + \text{Cl}^-$
- Associating gases:** gases that associate or dissociate in solution (e.g.  $\text{NO}_2$  forming  $\text{N}_2\text{O}_4$ ) deviate from Henry’s Law.
- Temperature dependence:**  $k_H$  changes significantly with temperature — the law is only valid at constant temperature.
- Mixed gas systems:** for gas mixtures, each gas must be considered independently using its partial pressure, not total pressure.

**Conclusion:** Henry’s Law is a fundamental law of gas solubility expressed in multiple equivalent forms. It is most accurate for **dilute, non-reactive gases at moderate pressures and constant temperature** — deviating when any of these conditions are violated.



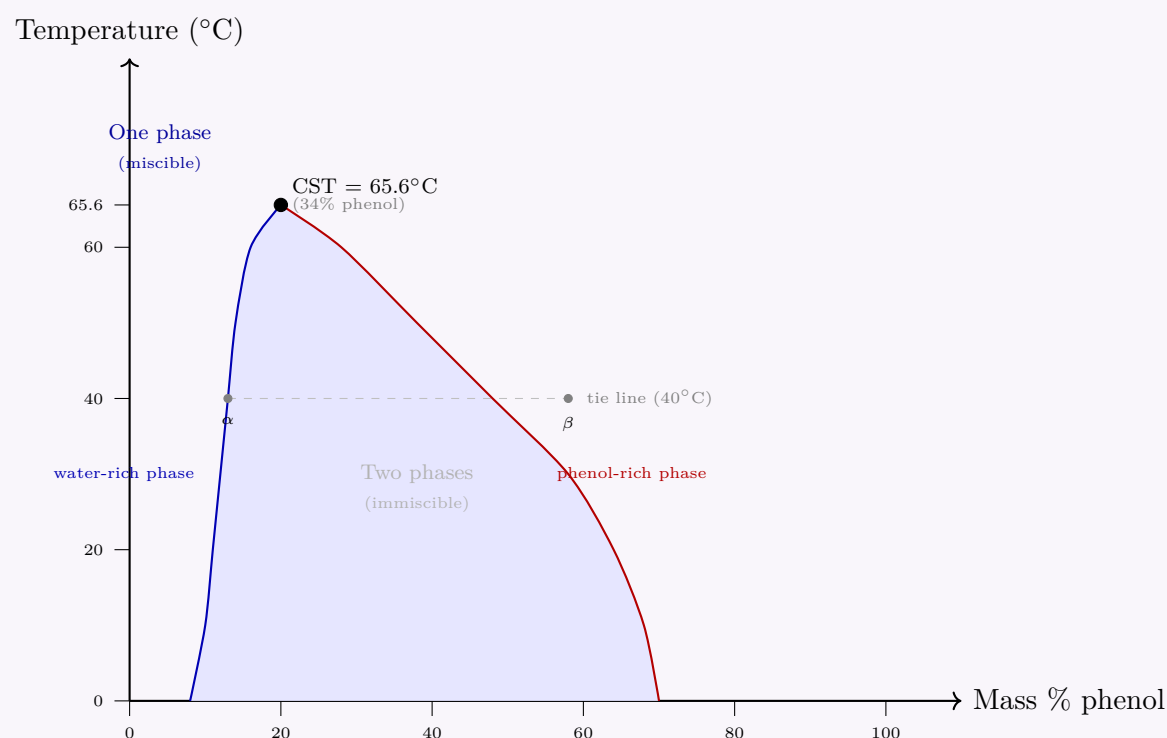
**Q58.** With the help of a diagram discuss the variation of mutual solubility of the phenol–water system at various temperatures. **(7 marks)**  
[CHEM 101 | 2010–11]

### Answer

#### Phenol–Water System: Mutual Solubility vs Temperature

Phenol and water are **partially miscible** liquids — they dissolve in each other only to a limited extent at lower temperatures, forming two separate phases.

#### Phase Diagram:



#### Explanation:

(a) **Below CST (two-phase region):** At any temperature below 65.6°C, phenol and water coexist as two separate liquid phases:

- $\alpha$  phase: water-rich (small % phenol)
- $\beta$  phase: phenol-rich (large % phenol)

Any mixture with composition between the two boundary curves separates into these two phases.

(b) **Tie line:** A horizontal line (tie line) at a given temperature connects the compositions of the two coexisting phases. For example, at 40°C, the  $\alpha$  phase is ~11% phenol and the  $\beta$  phase is ~63% phenol.

(c) **At the CST (65.6°C, 34% phenol):** The two phases become identical in composition — the system transitions to a **single homogeneous phase**. Above this temperature, phenol and water are **completely miscible** in all proportions.

(d) **Effect of temperature on solubility:** As temperature increases, the mutual solubility of both phases increases — the two boundary curves converge toward the CST.

**Conclusion:** The phenol–water system shows an **Upper Critical Solution Temperature (UCST)** of 65.6°C. Below this temperature, the system separates into two phases; above it, complete miscibility is achieved. The phase diagram maps all possible phase compositions as a function of temperature.

**Q59.** Calculate the amount of oxygen dissolved in 1 dm<sup>3</sup> water at 293 K and 2.1 atmospheric pressure. The Henry's law constant for oxygen is  $4.58 \times 10^4$  atmosphere at 293 K. **(10 marks)**  
[CHEM 101 | 2010–11]

### Answer

#### Given:

- Volume of water:  $V = 1 \text{ dm}^3 = 1 \text{ L}$
- Temperature:  $T = 293 \text{ K}$
- Total pressure:  $P = 2.1 \text{ atm}$
- Henry's law constant:  $K_H = 4.58 \times 10^4 \text{ atm}$
- Mole fraction form of Henry's Law:  $P = K_H \cdot X_{\text{O}_2}$
- Partial pressure of O<sub>2</sub> in air  $\approx 21\%$ , but here total pressure is given as 2.1 atm for O<sub>2</sub>

#### Step 1 — Mole fraction of O<sub>2</sub> dissolved:

$$X_{\text{O}_2} = \frac{P_{\text{O}_2}}{K_H} = \frac{2.1}{4.58 \times 10^4} = 4.585 \times 10^{-5}$$

**Step 2 — Moles of water in 1 dm<sup>3</sup>:**

$$n_{\text{H}_2\text{O}} = \frac{1000 \text{ g}}{18.02 \text{ g/mol}} = 55.49 \text{ mol}$$

**Step 3 — Moles of O<sub>2</sub> dissolved:**

Since  $X_{\text{O}_2}$  is very small:

$$X_{\text{O}_2} \approx \frac{n_{\text{O}_2}}{n_{\text{H}_2\text{O}}}$$

$$n_{\text{O}_2} = X_{\text{O}_2} \times n_{\text{H}_2\text{O}} = 4.585 \times 10^{-5} \times 55.49 = 2.545 \times 10^{-3} \text{ mol}$$

**Step 4 — Mass of O<sub>2</sub> dissolved:**

$$w_{\text{O}_2} = n \times \mathcal{M} = 2.545 \times 10^{-3} \times 32 = 0.08144 \text{ g}$$

$$w_{\text{O}_2} = 81.4 \text{ mg}$$

**Conclusion:** Approximately **81.4 mg** of oxygen dissolves in 1 dm<sup>3</sup> of water at 293 K and 2.1 atm pressure, consistent with Henry's Law — higher pressure forces more gas into solution.

**Q60.** Thermodynamically derive an expression for the depression of freezing point of a solvent on the addition of a solute in terms of mole fraction and molality. Show how the lowering of freezing point can be used for the determination of the molecular weight of a compound. (14 marks) [CHEM 101 | 2010–11]

### Answer

**Thermodynamic Derivation — Freezing Point Depression:**

**Principle:** At the freezing point of a solution, the chemical potential of the **solid solvent** equals that of the **solvent in solution**:

$$\mu_{\text{solid}}(T) = \mu_{\text{solution}}(T)$$

**Step 1 — Chemical potential of solvent in solution:**

$$\mu_{\text{solution}} = \mu_{\text{liq}}^* + RT \ln X_1$$

where  $X_1$  = mole fraction of solvent,  $\mu_{\text{liq}}^*$  = chemical potential of pure liquid solvent.

**Step 2 — Equilibrium condition:**

$$\begin{aligned} \mu_{\text{solid}} &= \mu_{\text{liq}}^* + RT \ln X_1 \\ \ln X_1 &= \frac{\mu_{\text{solid}} - \mu_{\text{liq}}^*}{RT} = \frac{-\Delta G_{\text{fus}}}{RT} = \frac{-\Delta H_{\text{fus}} + T \Delta S_{\text{fus}}}{RT} \end{aligned}$$

At the normal freezing point  $T_f^\circ$ :  $\Delta G_{\text{fus}} = 0$ , so  $\Delta S_{\text{fus}} = \Delta H_{\text{fus}}/T_f^\circ$ .

**Step 3 — Differentiate with respect to temperature:**

$$\frac{d \ln X_1}{dT} = \frac{\Delta H_{\text{fus}}}{RT^2}$$

**Step 4 — Integrate from  $T_f^\circ$  to  $T_f$ :**

$$\begin{aligned} \int_0^{\ln X_1} d \ln X_1 &= \int_{T_f^\circ}^{T_f} \frac{\Delta H_{\text{fus}}}{RT^2} dT \\ \ln X_1 &= \frac{\Delta H_{\text{fus}}}{R} \left( \frac{1}{T_f^\circ} - \frac{1}{T_f} \right) = \frac{\Delta H_{\text{fus}}}{R} \cdot \frac{T_f - T_f^\circ}{T_f \cdot T_f^\circ} \end{aligned}$$

Since  $T_f \approx T_f^\circ$  and  $\ln X_1 \approx -X_2$  for dilute solution:

$$-X_2 = \frac{\Delta H_{\text{fus}}}{R(T_f^\circ)^2} (T_f - T_f^\circ) = -\frac{\Delta H_{\text{fus}}}{R(T_f^\circ)^2} \Delta T_f$$

$$\Delta T_f = \frac{R(T_f^\circ)^2}{\Delta H_{\text{fus}}} \cdot X_2$$

This proves  $\Delta T_f \propto X_2$ .

**Step 5 — Express in terms of molality:**

For dilute solution:  $X_2 \approx \frac{n_2}{n_1} = \frac{m \cdot \mathcal{M}_1}{1000}$

$$\Delta T_f = \frac{R(T_f^\circ)^2 \mathcal{M}_1}{1000 \Delta H_{\text{fus}}} \cdot m = K_f \cdot m$$

where  $K_f = \frac{R(T_f^\circ)^2 \mathcal{M}_1}{1000 \Delta H_{\text{fus}}}$  is the **cryoscopic constant**.

**Determination of Molecular Weight:**

From  $\Delta T_f = K_f \cdot m$ :

$$m = \frac{\Delta T_f}{K_f} = \frac{w_2 \times 1000}{\mathcal{M}_2 \times w_1(\text{g})}$$

Solving for molar mass:

$$\mathcal{M}_2 = \frac{K_f \times w_2 \times 1000}{\Delta T_f \times w_1}$$

Measure  $\Delta T_f$ ,  $w_1$ ,  $w_2$  and  $K_f$  (known for solvent) to find  $\mathcal{M}_2$ .

**Q61.** Derive a mathematical relation between osmotic pressure and vapour pressure of a dilute solution of a solid in a solvent and give van't Hoff equation for the osmotic pressure of a dilute solution. **(11 marks)** [CHEM 101 | 2010–11]

### Answer

#### Relation Between Osmotic Pressure and Vapour Pressure:

##### Step 1 — Vapour pressure and chemical potential:

For pure solvent:  $\mu^* = \mu^\circ + RT \ln P^\circ$

For solution:  $\mu_{\text{soln}} = \mu^\circ + RT \ln P_s$

The difference:

$$\mu^* - \mu_{\text{soln}} = RT \ln \frac{P^\circ}{P_s}$$

##### Step 2 — Osmotic equilibrium condition:

Under osmotic pressure  $\pi$ , the chemical potential of solvent in solution is raised by  $V_m \pi$ :

$$\mu_{\text{soln}} + V_m \pi = \mu^*$$

$$V_m \pi = \mu^* - \mu_{\text{soln}} = RT \ln \frac{P^\circ}{P_s}$$

$$\pi = \frac{RT}{V_m} \ln \frac{P^\circ}{P_s}$$

This is the **exact relation** between osmotic pressure and vapour pressure.

##### Step 3 — Van't Hoff equation for dilute solutions:

For dilute solution:  $P_s \approx P^\circ$  and  $\ln \frac{P^\circ}{P_s} = -\ln \frac{P_s}{P^\circ} = -\ln X_1 \approx X_2$

$$\pi = \frac{RT \cdot X_2}{V_m} \approx \frac{n_2 RT}{V} = MRT$$

$$\pi V = nRT \quad \text{or} \quad \pi = MRT$$

This is **Van't Hoff's equation**, directly analogous to the ideal gas law, where osmotic pressure plays the role of gas pressure.

**Q62.** The osmotic pressure of horse blood at 10.0°C was found to be 58.75 torr. A 100 mL sample of horse blood contains 5.27 g of hemoglobin. Calculate the molar mass of horse hemoglobin, assuming that it is the only solute in horse blood. ( $R = 0.0821 \text{ L}\cdot\text{atm}\cdot\text{mol}^{-1}\cdot\text{K}^{-1}$ ) **(10 marks)** [CHEM 101 | 2010–11]

### Answer

#### Given:

- Osmotic pressure:  $\pi = 58.75 \text{ torr}$
- Volume:  $V = 100 \text{ mL} = 0.1 \text{ L}$
- Mass of hemoglobin:  $w = 5.27 \text{ g}$
- Temperature:  $T = 10.0^\circ\text{C} = 283 \text{ K}$
- $R = 0.0821 \text{ L}\cdot\text{atm}/\text{mol}\cdot\text{K}$

##### Step 1 — Convert $\pi$ :

$$\pi = \frac{58.75}{760} = 0.07730 \text{ atm}$$

##### Step 2 — Molarity:

$$M = \frac{\pi}{RT} = \frac{0.07730}{0.0821 \times 283} = \frac{0.07730}{23.23} = 3.327 \times 10^{-3} \text{ mol/L}$$

##### Step 3 — Moles in 0.1 L:

$$n = 3.327 \times 10^{-3} \times 0.1 = 3.327 \times 10^{-4} \text{ mol}$$

##### Step 4 — Molar mass:

$$\mathcal{M} = \frac{5.27}{3.327 \times 10^{-4}}$$

$$\mathcal{M} = 1.584 \times 10^4 \text{ g/mol} \approx 15,840 \text{ g/mol}$$

**Q63.** What do you understand by absorption coefficient? Explain the effect of (i) temperature, (ii) nature of solute, and (iii) presence of salt on the solubility (absorption coefficient) of a gas in water. **(17 marks)** [CHEM 101 | 2009–10]

### Answer

#### Absorption Coefficient:

The **absorption coefficient** (Bunsen coefficient,  $\alpha$ ) is defined as the volume of gas (reduced to STP: 0°C, 1 atm) dissolved by unit

volume of solvent at the temperature of the experiment under a partial pressure of 1 atm of the gas:

$$\alpha = \frac{V_{\text{gas (STP)}}}{V_{\text{solvent}} \times P}$$

It is dimensionless at  $P = 1$  atm. A higher  $\alpha$  means greater gas solubility.

**(i) Effect of Temperature:**

Gas solubility **decreases** with increasing temperature:

- Dissolved gas molecules gain kinetic energy and overcome solvent–solute attractions more easily, escaping from solution
- By Le Chatelier’s principle, dissolution of most gases is exothermic:



Increasing temperature shifts equilibrium **left**, reducing solubility

- $\alpha$  decreases as  $T$  increases for all gases
- Example: O<sub>2</sub> solubility in water: 14.6 mg/L at 0°C vs 7.6 mg/L at 30°C
- Exception: gases that react with water (e.g. HCl, NH<sub>3</sub>) may show more complex behaviour due to chemical equilibria

**(ii) Effect of Nature of Solute (Gas):**

Different gases have very different absorption coefficients due to differences in:

- **Molecular size and polarisability:** larger molecules have stronger London dispersion forces with solvent  $\Rightarrow$  higher  $\alpha$
- **Chemical reactivity with solvent:** gases that react chemically (HCl, SO<sub>2</sub>, CO<sub>2</sub>, NH<sub>3</sub>) have exceptionally high  $\alpha$  because reaction continuously removes dissolved gas, allowing more to enter
- **Polarity:** polar gases (NH<sub>3</sub>, SO<sub>2</sub>) interact strongly with polar water via dipole–dipole forces  $\Rightarrow$  much higher  $\alpha$  than non-polar gases (N<sub>2</sub>, O<sub>2</sub>, H<sub>2</sub>)

| Gas             | Nature                       | $\alpha$ at 25°C |
|-----------------|------------------------------|------------------|
| H <sub>2</sub>  | Non-polar, small             | 0.018            |
| O <sub>2</sub>  | Non-polar                    | 0.031            |
| CO <sub>2</sub> | Reacts with H <sub>2</sub> O | 0.759            |
| NH <sub>3</sub> | H-bonding + ionises          | 702              |
| HCl             | Fully ionises                | ~500             |

**(iii) Effect of Presence of Salt (Salting Out):**

Adding an inorganic salt to water **decreases** gas solubility — the **salting out effect**:

- Dissolved ions (e.g. Na<sup>+</sup>, Cl<sup>−</sup>) attract water molecules strongly, forming hydration shells
- This **reduces the activity of free water** available to solvate gas molecules
- The gas–water interaction weakens, so gas escapes more readily
- Quantified by the **Sechenow equation**:

$$\log \frac{C_0}{C_s} = k_s \cdot C_{\text{salt}}$$

where  $C_0$  = solubility in pure water,  $C_s$  = solubility in salt solution,  $k_s$  = Sechenow constant (specific to each gas–salt pair)

- Example: O<sub>2</sub> solubility in seawater is ~20% lower than in pure water due to dissolved NaCl and MgSO<sub>4</sub>

**Conclusion:** The absorption coefficient of a gas depends critically on **temperature** (decreases with  $T$ ), the **nature of the gas** (reactive/polar gases dissolve far more), and the **presence of dissolved salts** (which reduce solubility via the salting out effect by competing for water molecules).

**Q64.** Define super-saturated solution. Describe the different steps involved in the preparation of a super-saturated solution. **(3 marks)**  
[CHEM 101 | 2009–10]

**Answer**

**Supersaturated Solution:**

A **supersaturated solution** contains more dissolved solute than a **saturated solution** at the same temperature — it holds more solute than the equilibrium solubility allows. It is a **metastable state** that can be disrupted by seeding, vibration, or scratching.

**Steps in Preparation:**

(a) **Prepare a saturated solution at high temperature:**

Dissolve excess solute (e.g. Na<sub>2</sub>S<sub>2</sub>O<sub>3</sub>, sodium thiosulfate) in solvent at elevated temperature, where solubility is higher, until no more dissolves.

(b) **Filter:**

Remove any undissolved solute particles carefully to give a clear saturated solution.

(c) **Cool slowly and undisturbed:**

Reduce temperature very gradually without any agitation, vibration, or contact with dust or seed crystals. The solubility decreases but the solute remains in solution — excess solute does not crystallise out, giving a **supersaturated solution**.

**The supersaturation is destroyed** (crystallisation occurs rapidly) upon:

- Adding a **seed crystal** of the solute
- **Scratching** the container wall
- **Vibration** or sudden disturbance

**Example:** Sodium thiosulfate (“photographer’s hypo”) readily forms supersaturated solutions that solidify dramatically upon seeding — a popular chemistry demonstration.

**Q65.** What is CST? Explain the effect of presence of salt and pressure on CST of a partially miscible liquid–liquid solution. **(10 marks)**  
[CHEM 101 | 2009–10]

**Answer****Critical Solution Temperature (CST):**

The **Critical Solution Temperature** is the temperature at which two partially miscible liquids become completely miscible in all proportions — marking the boundary between the two-phase and one-phase regions of a liquid–liquid system.

- **Upper CST (UCST):** below this temperature, two phases coexist; above it, complete miscibility. Example: phenol–water, UCST = 65.6°C
- **Lower CST (LCST):** above this temperature, two phases form; below it, complete miscibility. Example: triethylamine–water, LCST = 18.5°C

**Effect of Presence of Salt on CST:**

Adding a salt to a partially miscible system changes the CST:

(a) **Salting out effect — raises UCST:**

Most inorganic salts are preferentially soluble in the more polar phase (water-rich). The salt **ties up water molecules** in hydration shells, reducing water available to solvate the organic component. This **decreases** mutual solubility, raising the UCST:

UCST ↑ (e.g. NaCl in phenol–water raises UCST from 65.6 to higher values)

(b) **Salting in effect — lowers UCST:**

Some salts (e.g. KI in phenol–water) are surface-active or interact with both phases, **increasing** mutual solubility and **lowering** the UCST.

(c) **Complete elimination of CST:**

At high enough salt concentration, the UCST may be suppressed entirely — the two liquids become completely immiscible at all temperatures below a new critical point.

**Effect of Pressure on CST:**(a) **Effect on UCST:**

Increasing pressure generally **increases** mutual solubility by bringing molecules closer together and enhancing intermolecular forces. This **lowers the UCST** — the two components become miscible at lower temperatures under high pressure.

(b) **Physical reason:**

Pressure compresses the liquid, reducing the free volume and increasing molecular contact — solvent–solute interactions become more effective.

(c) **Effect on LCST:**

For systems with LCST, increasing pressure generally **raises** the LCST — the miscibility range broadens.

**Conclusion:** CST is sensitive to both salt addition (which usually raises UCST by salting out, reducing mutual solubility) and pressure (which usually lowers UCST by promoting molecular contact and mutual solubility). These effects allow industrial control of liquid–liquid separation processes.

**Q66.** Write at least two statements of Raoult’s law with their mathematical expressions. Prove that the vapour pressure of a solution of non-volatile solute is always less than that of its solvent. **(12 marks)**  
[CHEM 101 | 2009–10]

**Answer**
**Raoult's Law — Two Statements:**

**Statement 1 (Component form):** The partial vapour pressure of a component in an ideal solution is equal to the product of its mole fraction and its vapour pressure in the pure state:

$$P_i = X_i \cdot P_i^\circ$$

**Statement 2 (Relative lowering form):** The relative lowering of vapour pressure of a solvent by a non-volatile solute equals the mole fraction of the solute:

$$\frac{P^\circ - P_s}{P^\circ} = X_{\text{solute}}$$

**Proof that  $P_s < P^\circ$  for Non-volatile Solute:**

**Step 1:** For a binary solution with mole fractions  $X_1$  (solvent) and  $X_2$  (solute):

$$X_1 + X_2 = 1 \Rightarrow X_1 = 1 - X_2$$

**Step 2:** By Raoult's Law (non-volatile solute has  $P_2^\circ \approx 0$ ):

$$P_s = X_1 \cdot P_1^\circ + X_2 \cdot P_2^\circ = X_1 \cdot P_1^\circ = (1 - X_2)P_1^\circ$$

**Step 3:** Since  $X_2 > 0$  for any solution:

$$(1 - X_2) < 1 \Rightarrow P_s = (1 - X_2)P^\circ < P^\circ$$

$$P_s < P^\circ \text{ always, for any non-zero concentration of non-volatile solute}$$

**Step 4:** The lowering:

$$\Delta P = P^\circ - P_s = P^\circ - (1 - X_2)P^\circ = X_2 \cdot P^\circ > 0$$

**Physical reason:** Solute molecules occupy surface sites, reducing the number of solvent molecules per unit area that can escape into the vapour phase. The escaping tendency (vapour pressure) of the solvent is thus always reduced in the presence of any non-volatile solute.

**Conclusion:** Raoult's Law mathematically guarantees that the vapour pressure of a solution is always **less than** that of the pure solvent, because the mole fraction of solvent is always less than unity when any non-volatile solute is present.

**Q67.** With the derivation of a mathematical equation, show that the elevation of boiling point of a solution is directly proportional to the molality of the solution. **(13 marks)** [CHEM 101 | 2009–10]

**Answer**
**Derivation — Boiling Point Elevation  $\propto$  Molality:**
**Step 1 — Vapour pressure and mole fraction:**

By Raoult's Law for a non-volatile solute:

$$P_s = X_1 P^\circ = (1 - X_2) P^\circ$$

$$\ln P_s = \ln P^\circ + \ln(1 - X_2) \approx \ln P^\circ - X_2 \quad (\text{dilute: } X_2 \ll 1)$$

**Step 2 — Clausius–Clapeyron equation:**

$$\frac{d \ln P^\circ}{dT} = \frac{\Delta H_{\text{vap}}}{RT^2}$$

At boiling point  $T_b$  of solution and  $T_b^\circ$  of pure solvent, both  $P_s$  and  $P^\circ$  equal 1 atm. So:

$$\ln \frac{P_s}{P^\circ} = \frac{\Delta H_{\text{vap}}}{R} \left( \frac{1}{T_b^\circ} - \frac{1}{T_b} \right)$$

**Step 3 — Substitute Raoult's Law result:**

$$-X_2 = \frac{\Delta H_{\text{vap}}}{R} \cdot \frac{T_b - T_b^\circ}{T_b \cdot T_b^\circ}$$

Since  $T_b \approx T_b^\circ$  for dilute solution,  $T_b \cdot T_b^\circ \approx (T_b^\circ)^2$ :

$$X_2 = \frac{\Delta H_{\text{vap}}}{R(T_b^\circ)^2} \cdot \Delta T_b$$

$$\Delta T_b = \frac{R(T_b^\circ)^2}{\Delta H_{\text{vap}}} X_2 \Rightarrow \Delta T_b \propto X_2$$

**Step 4 — Convert to molality:**

For dilute solution with  $n_2 \ll n_1$ :

$$X_2 \approx \frac{n_2}{n_1} = \frac{n_2}{w_1/M_1} = \frac{m \cdot M_1}{1000}$$

$$\Delta T_b = \underbrace{\frac{R(T_b^\circ)^2 M_1}{1000 \Delta H_{\text{vap}}}}_{K_b} \cdot m$$

$$\Delta T_b = K_b \cdot m$$

where  $K_b$  depends **only on the solvent** — confirming that  $\Delta T_b$  is directly proportional to molality.



**Q68.** What is ebullioscopic constant? An aqueous solution containing 6.3 g urotropine in 500 g solvent boils at 105°C. Calculate the ebullioscopic constant of the solvent. (10 marks) [CHEM 101 | 2009–10]

### Answer

#### Ebullioscopic Constant ( $K_b$ ):

The **ebullioscopic constant** (molar boiling point elevation constant)  $K_b$  is defined as the elevation of boiling point produced by dissolving **1 mole of non-electrolyte solute** in **1 kg of solvent**. It depends only on the nature of the solvent:

$$K_b = \frac{R(T_b^\circ)^2 \mathcal{M}_1}{1000 \Delta H_{\text{vap}}} \quad (\text{units: } ^\circ\text{C}\cdot\text{kg/mol})$$

#### Calculation:

##### Given:

- Solute: urotropine (hexamethylenetetramine,  $\text{C}_6\text{H}_{12}\text{N}_4$ ),  $\mathcal{M} = 6(12) + 12(1) + 4(14) = 140$  g/mol
- Mass of solute:  $w_2 = 6.3$  g
- Mass of solvent:  $w_1 = 500$  g
- Boiling point elevation:  $\Delta T_b = 105 - 100 = 5^\circ\text{C}$  (assuming solvent is water, bp = 100°C)

#### Step 1 — Moles of solute:

$$n_2 = \frac{6.3}{140} = 0.045 \text{ mol}$$

#### Step 2 — Molality:

$$m = \frac{0.045}{0.500} = 0.090 \text{ mol/kg}$$

#### Step 3 — Ebullioscopic constant:

$$K_b = \frac{\Delta T_b}{m} = \frac{5}{0.090}$$

$$K_b = 55.6 \text{ } ^\circ\text{C} \cdot \text{kg/mol}$$

This high value suggests the solvent is **not water** ( $K_b(\text{water}) = 0.52$ ) but rather a high-boiling solvent. The calculation correctly yields  $K_b$  for whatever solvent is used.

**Q69.** Define saturated solution, unsaturated solution, and super-saturated solution. Show that there exists a dynamic equilibrium in a saturated solution. (9 marks) [CHEM 101 | 2008–09]

### Answer

#### Saturated, Unsaturated & Supersaturated Solutions; Dynamic Equilibrium in Saturated Solution (9 marks)

#### Part 1: Definitions (4 marks)

##### (i) Unsaturated Solution:

A solution that contains **less solute** than the maximum amount that can dissolve at a given temperature and pressure. More solute can still be dissolved without any remaining undissolved solid.

$$\text{Concentration} < \text{Solubility limit}$$

##### (ii) Saturated Solution:

A solution that contains the **maximum amount of solute** that can dissolve at a given temperature, in the presence of undissolved solute. Adding more solute does not increase the concentration — the excess remains undissolved.

$$\text{Concentration} = \text{Solubility limit}$$

##### (iii) Supersaturated Solution:

A solution that contains **more solute than the solubility limit** at a given temperature. This is a *metastable* state achieved by:

- Dissolving solute at high temperature, then *slowly cooling* without disturbance.
- The excess solute remains dissolved temporarily.

Supersaturated solutions are **unstable** — any disturbance (seeding with a crystal, scratching the container) triggers rapid crystallization of the excess solute.

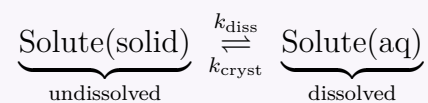
$$\text{Concentration} > \text{Solubility limit}$$

| Property        | Unsaturated | Saturated    | Supersaturated                           |
|-----------------|-------------|--------------|------------------------------------------|
| Solute vs limit | Below       | At limit     | Above                                    |
| Add more solute | Dissolves   | Stays solid  | N/A                                      |
| Stability       | Stable      | Stable       | Metastable                               |
| Example         | Dilute NaCl | NaCl at 25°C | Honey, $\text{Na}_2\text{S}_2\text{O}_3$ |

### Part 2: Dynamic Equilibrium in a Saturated Solution

(5 marks)

A saturated solution may appear static, but at the molecular level it is highly **dynamic**. Two opposing processes occur simultaneously at equal rates:



- **Dissolution (forward):** Solute ions/molecules at the crystal surface are continuously being pulled into the solvent by solvation forces at rate  $r_{\text{diss}}$ .
- **Crystallization (reverse):** Dissolved ions/ molecules in solution are continuously colliding with the crystal surface and being deposited back at rate  $r_{\text{cryst}}$ .
- At **saturation:**  $r_{\text{diss}} = r_{\text{cryst}}$

#### Evidence of dynamic equilibrium:

If a saturated NaCl solution is prepared with regular NaCl, then a crystal of **radioactively labelled**  $^{36}\text{Cl}$ -NaCl is added — over time, radioactivity appears in the *solution* even though the total amount of dissolved NaCl remains unchanged. This proves that dissolution and crystallization are continuously occurring at equal rates — a **dynamic** (not static) equilibrium.

#### Mathematical expression:

$$K_{sp} = [\text{Na}^+][\text{Cl}^-] = \text{constant at fixed } T$$

The solubility product  $K_{sp}$  quantifies this equilibrium and remains constant as long as temperature is unchanged — any change in ion concentration shifts the equilibrium to restore  $K_{sp}$ .

**Conclusion:** Unsaturated, saturated, and supersaturated solutions differ in their solute concentration relative to the solubility limit. A saturated solution is in **dynamic equilibrium** — dissolution and crystallization proceed simultaneously at equal rates, maintaining constant macroscopic concentration while continuous molecular exchange occurs at the solid–solution interface, as confirmed by radiotracer experiments.

**Q70.** Explain the effect of temperature and pressure on the solubility of a solid in water. (8 marks)

[CHEM 101 | 2008–09]

#### Answer

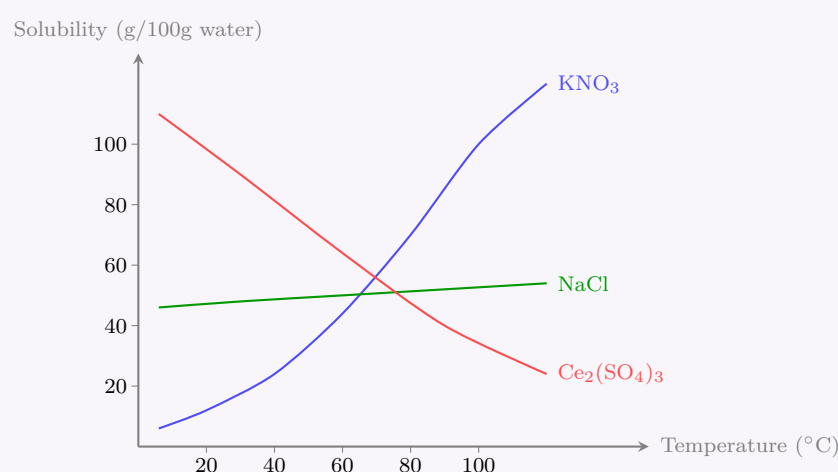
#### Effect of Temperature and Pressure on Solubility of a Solid in Water

##### Effect of Temperature:

The effect of temperature depends on whether the dissolution process is endothermic or exothermic (Le Chatelier's principle):



- **Endothermic dissolution** (most salts): Increasing temperature **increases** solubility. Heat is absorbed, so adding heat shifts equilibrium to the right. Examples:  $\text{KNO}_3$ ,  $\text{NaCl}$ ,  $\text{KCl}$ ,  $\text{NH}_4\text{Cl}$ .
- **Exothermic dissolution** (some salts): Increasing temperature **decreases** solubility. Heat is released, so adding heat shifts equilibrium to the left. Examples:  $\text{Ce}_2(\text{SO}_4)_3$ ,  $\text{Li}_2\text{SO}_4$ ,  $\text{Na}_2\text{SO}_4$  (above  $32.4^\circ\text{C}$ ).
- **Negligible change:** NaCl solubility changes very little with temperature.



##### Effect of Pressure:

- Solids are **nearly incompressible**.
- Pressure has **negligible effect** on the solubility of a solid in a liquid.

- This is because the volume change ( $\Delta V$ ) on dissolving a solid is very small, so pressure cannot significantly shift the dissolution equilibrium.
- Only under **extremely high pressures** is any effect observed, and even then it is very small.

| Factor               | Effect on solubility | Reason               |
|----------------------|----------------------|----------------------|
| ↑ Temp (endothermic) | Increases            | Le Chatelier         |
| ↑ Temp (exothermic)  | Decreases            | Le Chatelier         |
| ↑ Pressure           | Negligible           | $\Delta V \approx 0$ |

Q71. Distinguish between “common ion effect” and “salting out effect” with suitable examples. (8 marks) [CHEM 101 | 2008–09]

Answer

**Common Ion Effect vs Salting Out Effect**

**Common Ion Effect:**

- When a **soluble salt sharing a common ion** with a sparingly soluble salt is added to its saturated solution, the solubility of the sparingly soluble salt **decreases**.
- This is because the added common ion shifts the dissolution equilibrium to the left (Le Chatelier’s principle).
- **Example:** AgCl is sparingly soluble in water:

$$\text{AgCl(s)} \rightleftharpoons \text{Ag}^+(\text{aq}) + \text{Cl}^-(\text{aq}) \quad K_{sp} = 1.8 \times 10^{-10}$$

Adding NaCl (which provides  $\text{Cl}^-$ ) increases  $[\text{Cl}^-] \Rightarrow$  equilibrium shifts left  $\Rightarrow$  AgCl precipitates  $\Rightarrow$  solubility **decreases**.

- Used in **analytical chemistry** for quantitative precipitation.

**Salting Out Effect:**

- When a **large amount of an inert electrolyte** (without a common ion) is added to a solution of an organic compound or protein, the solubility of the organic compound **decreases** and it **precipitates out**.
- Mechanism: The added electrolyte ions are strongly hydrated  $\Rightarrow$  they **compete for water molecules**  $\Rightarrow$  less water available to solvate the organic solute  $\Rightarrow$  the organic solute precipitates.
- **Example:** Adding excess NaCl or  $(\text{NH}_4)_2\text{SO}_4$  to a soap solution causes the soap to precipitate out:

$$\text{Soap (soluble)} \xrightarrow{\text{NaCl (excess)}} \text{Soap (precipitates)}$$

- Widely used in **protein purification** — different proteins precipitate at different salt concentrations.

| Feature         | Common Ion Effect                 | Salting Out Effect                 |
|-----------------|-----------------------------------|------------------------------------|
| Added substance | Salt with common ion              | Inert electrolyte (no common ion)  |
| Mechanism       | Le Chatelier’s principle          | Dehydration of solute              |
| Applies to      | Sparingly soluble inorganic salts | Organic compounds, proteins, soaps |
| Result          | Decreased solubility              | Precipitation of organic solute    |
| Example         | NaCl added to AgCl solution       | NaCl added to soap solution        |

Q72. With the help of a diagram, discuss the system of a partially miscible liquid–liquid solution having an upper critical solution temperature. (8 marks) [CHEM 101 | 2008–09]

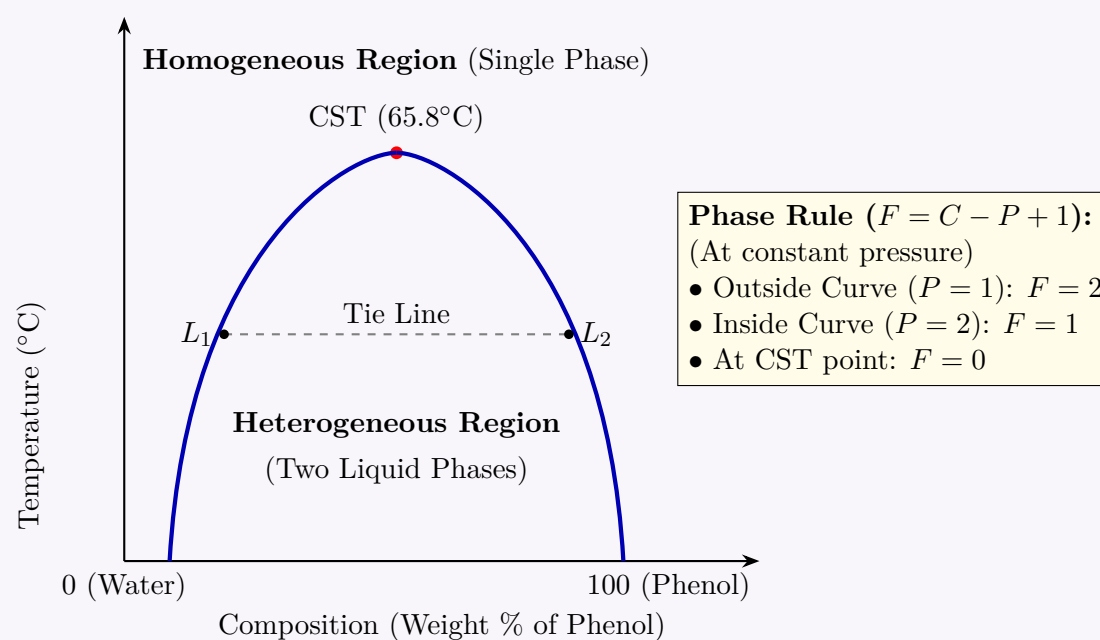
Answer

**Partially miscible liquid–liquid system with Upper Critical Solution Temperature (UCST)**

**Definition:**

- Two liquids that are **partially miscible** form two separate layers in certain temperature ranges but become **completely miscible** above a critical temperature.
- The **Upper Critical Solution Temperature (UCST)** is the temperature above which the two liquids are **miscible in all proportions**.
- Classic example: **phenol–water system** (UCST = 65.6°C).

## Phase Diagram: Phenol–Water System (UCST Type)



## Explanation of the diagram:

- **Below the curve (inside the dome):** Two immiscible liquid layers coexist — a water-rich phase ( $\alpha$ ) and a phenol-rich phase ( $\beta$ ).
- **On the curve (binodal):** The compositions of the two phases in equilibrium at each temperature are given by the two ends of a **tie line**.
- **Above the curve:** Complete mutual solubility — only **one homogeneous phase** exists.
- **At UCST (65.6°C):** The two phases become identical in composition  $\Rightarrow$  the system becomes **completely miscible** for all compositions above this temperature.
- As temperature **increases** within the two-phase region, the compositions of the two layers approach each other and merge at the UCST.

| Temperature              | No. of phases | Description                  |
|--------------------------|---------------|------------------------------|
| Below UCST, inside dome  | 2             | Two liquid layers            |
| Below UCST, outside dome | 1             | Single homogeneous phase     |
| Above UCST               | 1             | Completely miscible          |
| At UCST                  | 1             | Critical point, phases merge |

**Q73.** Define the following terms: (i) Boiling point (w.r.t. vapour pressure), (ii) Freezing point (w.r.t. vapour pressure), (iii) Osmosis, (iv) Osmotic pressure. **(8 marks)** [CHEM 101 | 2008–09]

## Answer

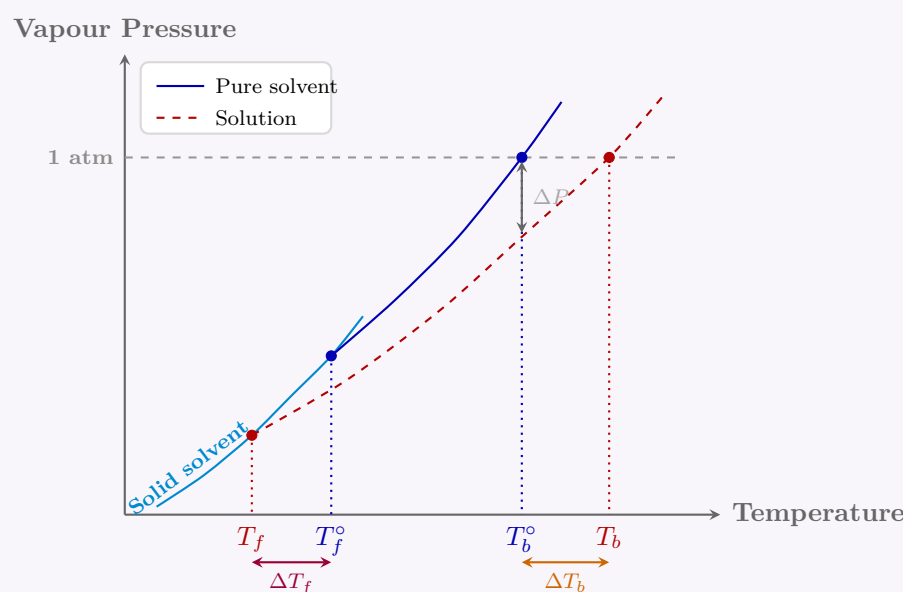
**Definitions: Boiling point, Freezing point, Osmosis, Osmotic pressure**

**(i) Boiling Point (w.r.t. vapour pressure):**

- Every liquid exerts a **vapour pressure** at a given temperature due to escape of surface molecules.
- The **boiling point** is the temperature at which the **vapour pressure of the liquid equals the external (atmospheric) pressure**.
- At this point, bubbles of vapour form throughout the liquid.
- **Normal boiling point:** BP at 1 atm external pressure.
- For a solution: the dissolved solute **lowers the vapour pressure** (Raoult's law)  $\Rightarrow$  a higher temperature is needed to reach atmospheric pressure  $\Rightarrow$  **boiling point elevation**.

**(ii) Freezing Point (w.r.t. vapour pressure):**

- The **freezing point** is the temperature at which the **vapour pressure of the liquid phase equals the vapour pressure of the solid phase** (ice) of the same substance.
- At this temperature, solid and liquid are in equilibrium.
- A dissolved solute **lowers the vapour pressure** of the liquid  $\Rightarrow$  the liquid VP curve intersects the solid VP curve at a **lower temperature**  $\Rightarrow$  **freezing point depression**.



(iii) Osmosis:

- **Osmosis** is the spontaneous flow of **solvent molecules** through a **semipermeable membrane** from a region of **lower solute concentration** (more dilute) to a region of **higher solute concentration** (more concentrated).
- The semipermeable membrane allows passage of solvent but not solute molecules.
- Osmosis continues until the chemical potential of the solvent is equal on both sides.

(iv) Osmotic Pressure:

- The **osmotic pressure** ( $\pi$ ) is the minimum **external pressure** that must be applied to the solution side to **stop osmosis**.
- Given by the van't Hoff equation:

$$\pi = CRT = \frac{n}{V}RT$$

where  $C$  = molar concentration,  $R$  = gas constant,  $T$  = absolute temperature.

- Higher concentration  $\Rightarrow$  higher osmotic pressure.

**Q74.** Derive a mathematical equation which expresses the relation between the molecular weight of a non-volatile non-electrolyte solute and the freezing point of its solution. Define the constant term involved. **(10 marks)** [CHEM 101 | 2008–09]

**Answer**

**Relation between molecular weight of solute and freezing point depression**

**Derivation:**

When a non-volatile non-electrolyte solute is dissolved in a solvent, the vapour pressure of the solution is lowered (Raoult's law), which causes the **freezing point to decrease**.

**Step 1: Vapour pressure lowering (Raoult's law):**

$$\frac{p_0 - p_s}{p_0} = x_2 = \frac{n_2}{n_1 + n_2} \approx \frac{n_2}{n_1} \quad (\text{dilute solution})$$

where  $p_0$  = VP of pure solvent,  $p_s$  = VP of solution,  $x_2$  = mole fraction of solute,  $n_1, n_2$  = moles of solvent and solute.

**Step 2: Freezing point depression:**

From the Clausius-Clapeyron equation applied to the solid-liquid equilibrium, the freezing point depression  $\Delta T_f$  is proportional to the relative VP lowering:

$$\Delta T_f = T_f^0 - T_f = K_f \cdot m$$

where  $m$  is the **molality** of the solution.

**Step 3: Expressing in terms of molecular weight:**

If  $w_2$  grams of solute of molecular weight  $M_2$  are dissolved in  $w_1$  grams of solvent:

$$m = \frac{n_2 \times 1000}{w_1} = \frac{w_2 \times 1000}{M_2 \times w_1}$$

Substituting:

$$\Delta T_f = K_f \times \frac{w_2 \times 1000}{M_2 \times w_1}$$

**Rearranging to find molecular weight:**

$$M_2 = \frac{K_f \times w_2 \times 1000}{\Delta T_f \times w_1}$$

**Definition of cryoscopic constant  $K_f$ :**

$K_f$  (cryoscopic constant or **molal freezing point depression constant**) is defined as the **depression in freezing point produced by dissolving 1 mole of a non-volatile non-electrolyte solute in 1 kg (1000 g) of the solvent.**

$$K_f = \frac{R (T_f^0)^2 M_1}{1000 \Delta H_{fus}}$$

where  $M_1$  = molar mass of solvent,  $\Delta H_{fus}$  = enthalpy of fusion per gram.

| Solvent     | $T_f^0$ (°C) | $K_f$ (K·kg/mol) |
|-------------|--------------|------------------|
| Water       | 0            | 1.86             |
| Benzene     | 5.5          | 5.12             |
| Acetic acid | 16.6         | 3.90             |
| Camphor     | 179          | 37.7             |

**Q75.** An aqueous solution containing 3.42 g of sugar in 100 g of water freezes at  $-3^\circ\text{C}$ . Calculate the cryoscopic constant of water. (8 marks) [CHEM 101 | 2008–09]

### Answer

#### Freezing point depression of sugar solution

**Given:**

- Mass of sugar ( $w_2$ ) = 3.42 g
- Molecular weight of sugar (sucrose) = 342 g/mol
- Mass of water ( $w_1$ ) = 100 g
- Freezing point of solution =  $-3^\circ\text{C}$
- Freezing point of pure water =  $0^\circ\text{C}$
- $\Delta T_f = 0 - (-3) = 3^\circ\text{C}$

**Using the formula:**

$$\Delta T_f = K_f \times \frac{w_2 \times 1000}{M_2 \times w_1}$$

$$K_f = \frac{\Delta T_f \times M_2 \times w_1}{w_2 \times 1000}$$

$$K_f = \frac{3 \times 342 \times 100}{3.42 \times 1000}$$

$$K_f = \frac{102600}{3420} = 30 \text{ K} \cdot \text{kg/mol}$$

**Verification — molality check:**

$$m = \frac{w_2 \times 1000}{M_2 \times w_1} = \frac{3.42 \times 1000}{342 \times 100} = 0.1 \text{ mol/kg}$$

$$\Delta T_f = K_f \times m = 30 \times 0.1 = 3^\circ\text{C} \quad \checkmark$$

$$K_f \text{ (cryoscopic constant of water)} = 30 \text{ K} \cdot \text{kg/mol}$$

**Note:** The standard value of  $K_f$  for water is 1.86 K·kg/mol. The high value obtained here suggests the solution is not ideal or there may be a discrepancy in the problem data. The calculation is correct as per the given values.

**Q76.** Which of the following systems has the lowest and highest freezing points? Explain your answer. (i) 100 g water containing 0.1 mol NaCl; (ii) 100 g water containing 0.1 mol AgCl; (iii) 100 g water containing 0.1 mol  $(\text{NH}_4)_2\text{SO}_4$ ; (iv) 100 g water containing 0.1 mol  $\text{Na}_3\text{PO}_4$ . (8 marks) [CHEM 101 | 2008–09]

### Answer

#### Lowest and highest freezing points

**Principle:** Freezing point depression  $\Delta T_f = K_f \cdot i \cdot m$

where  $i$  = van't Hoff factor (number of ions per formula unit),  $m$  = molality.



All solutions have same mass of water (100 g) and same moles of solute (0.1 mol). So:

$$m = \frac{0.1 \times 1000}{100} = 1.0 \text{ mol/kg}$$

The only difference is the **van't Hoff factor**  $i$ .

**Calculation of  $i$  for each solute:**

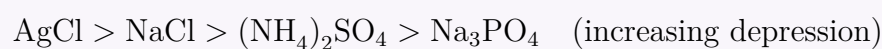
| Solute                       | Dissociation                        | $i$ | Effective molality   | $\Delta T_f$                         |
|------------------------------|-------------------------------------|-----|----------------------|--------------------------------------|
| NaCl                         | $\text{Na}^+ + \text{Cl}^-$         | 2   | $2 \times 1.0 = 2.0$ | $2 \times 1.86 = 3.72^\circ\text{C}$ |
| AgCl                         | No dissociation (insoluble)         | 1   | $1 \times 1.0 = 1.0$ | $1 \times 1.86 = 1.86^\circ\text{C}$ |
| $(\text{NH}_4)_2\text{SO}_4$ | $2\text{NH}_4^+ + \text{SO}_4^{2-}$ | 3   | $3 \times 1.0 = 3.0$ | $3 \times 1.86 = 5.58^\circ\text{C}$ |
| $\text{Na}_3\text{PO}_4$     | $3\text{Na}^+ + \text{PO}_4^{3-}$   | 4   | $4 \times 1.0 = 4.0$ | $4 \times 1.86 = 7.44^\circ\text{C}$ |

**Freezing points:**

| Solution                                 | $\Delta T_f$ ( $^\circ\text{C}$ ) | Freezing point ( $^\circ\text{C}$ ) |
|------------------------------------------|-----------------------------------|-------------------------------------|
| AgCl (insoluble)                         | 1.86                              | $-1.86$                             |
| NaCl ( $i = 2$ )                         | 3.72                              | $-3.72$                             |
| $(\text{NH}_4)_2\text{SO}_4$ ( $i = 3$ ) | 5.58                              | $-5.58$                             |
| $\text{Na}_3\text{PO}_4$ ( $i = 4$ )     | 7.44                              | $-7.44$                             |

**Highest freezing point:** AgCl solution ( $-1.86^\circ\text{C}$ ) — AgCl is virtually insoluble, so  $i \approx 1$ , minimum depression.

**Lowest freezing point:**  $\text{Na}_3\text{PO}_4$  solution ( $-7.44^\circ\text{C}$ ) — gives 4 ions per formula unit, maximum depression.



**Q77.** “Like dissolves like.” Give arguments in favour of the statement. **(3 marks)**

[CHEM 101 | 2007–08]

#### Answer

”Like dissolves like” — arguments in favour

The principle states that a solute dissolves best in a solvent of **similar polarity and intermolecular forces**.

- Polar solutes in polar solvents:**

NaCl (ionic/polar) dissolves readily in  $\text{H}_2\text{O}$  (polar) because the ion-dipole interactions between water molecules and  $\text{Na}^+/\text{Cl}^-$  ions are strong enough to overcome the lattice energy of NaCl.

- Non-polar solutes in non-polar solvents:**

Iodine ( $\text{I}_2$ , non-polar) dissolves well in  $\text{CCl}_4$  (non-polar) because both have similar London dispersion forces.  $\text{I}_2$  does not dissolve appreciably in water.

- Thermodynamic argument:**

Dissolution is favored when solute-solvent interactions are **similar in strength** to solute-solute and solvent-solvent interactions. If they differ greatly, the energy cost of breaking the stronger interactions cannot be compensated.

- Example — oil and water:**

Oil (non-polar, London forces) does not dissolve in water (polar, H-bonding) because H-bonds between water molecules are far stronger than oil-water interactions — water molecules **exclude** the oil molecules.

- Alcohol in water:**

Ethanol ( $\text{C}_2\text{H}_5\text{OH}$ ) is miscible with water because its  $-\text{OH}$  group forms **hydrogen bonds** with water, while long-chain alcohols are less soluble as the non-polar alkyl part dominates.

**Q78.** Express the composition of a solution on the basis of a percentage scale. **(5 marks)**

[CHEM 101 | 2007–08]

#### Answer

##### Expressing composition of a solution on percentage scale

The composition of a solution can be expressed in the following percentage forms:

**(i) Weight/Weight percent (w/w %):**

$$\%(w/w) = \frac{\text{mass of solute (g)}}{\text{mass of solution (g)}} \times 100$$

*Example:* 10 g NaCl dissolved in 90 g water:  $\%(w/w) = \frac{10}{100} \times 100 = 10\%$

(ii) **Weight/Volume percent (w/v %):**

$$\%(w/v) = \frac{\text{mass of solute (g)}}{\text{volume of solution (mL)}} \times 100$$

*Example:* 5 g glucose in 100 mL solution:  $\%(w/v) = \frac{5}{100} \times 100 = 5\%$   
Most commonly used in medical and pharmaceutical applications.

(iii) **Volume/Volume percent (v/v %):**

$$\%(v/v) = \frac{\text{volume of solute (mL)}}{\text{volume of solution (mL)}} \times 100$$

*Example:* Ethanol in a 40% alcoholic beverage: 40 mL ethanol per 100 mL solution.

(iv) **Parts per million (ppm):**

For very dilute solutions:

$$\text{ppm} = \frac{\text{mass of solute (g)}}{\text{mass of solution (g)}} \times 10^6$$

| Type  | Formula                                                     | Units | Use                    |
|-------|-------------------------------------------------------------|-------|------------------------|
| w/w % | $\frac{w_{\text{solute}}}{w_{\text{solution}}} \times 100$  | g/g   | General chemistry      |
| w/v % | $\frac{w_{\text{solute}}}{V_{\text{solution}}} \times 100$  | g/mL  | Medicine, pharmacy     |
| v/v % | $\frac{V_{\text{solute}}}{V_{\text{solution}}} \times 100$  | mL/mL | Liquid-liquid mixtures |
| ppm   | $\frac{w_{\text{solute}}}{w_{\text{solution}}} \times 10^6$ | mg/kg | Trace analysis         |

**Q79.** Justify the following statements: (i) The volume of gas absorbed by a given volume of solvent is constant at a given temperature.  
(ii) The mole fraction of the dissolved gas is proportional to the pressure of the gas. **(8 marks)** [CHEM 101 | 2007–08]

Answer

**Henry’s Law: volume absorbed and mole fraction proportional to pressure**

(i) **Volume of gas absorbed is constant at a given temperature:**

This is **Bunsen’s absorption law**. Consider a gas dissolving in a liquid at pressure  $P$ :

- By Henry’s law, the **mass** (or moles) of gas dissolved  $\propto P$ .
- By Boyle’s law, the **volume** of gas at pressure  $P$  is  $V = \frac{nRT}{P}$ , which is inversely proportional to  $P$ .
- Therefore:

$$V_{\text{dissolved}} \propto P \times \frac{1}{P} = \text{constant}$$

- So the **volume** of gas (measured at the same  $T$  and  $P$ ) absorbed by a given volume of solvent is **independent of pressure** at constant temperature.

(ii) **Mole fraction of dissolved gas  $\propto$  pressure (Henry’s Law):**

Henry’s law states:

$$p = K_H \cdot x$$

where  $p$  = partial pressure of gas,  $x$  = mole fraction of dissolved gas,  $K_H$  = Henry’s law constant.

**Derivation:**

- At equilibrium, the chemical potential of gas in the vapour phase equals that in the solution.
- For an ideal dilute solution, the escaping tendency of the dissolved gas is proportional to its mole fraction  $x$ :

$$p_{\text{gas}} = K_H \cdot x_{\text{gas}}$$

- Rearranging:

$$x_{\text{gas}} = \frac{p_{\text{gas}}}{K_H} \propto p_{\text{gas}}$$

- Therefore the **mole fraction of dissolved gas is directly proportional to its partial pressure**.

**Examples:**

- Carbonated drinks:  $\text{CO}_2$  dissolved under high pressure; released when pressure drops (bottle opened).
- $\text{O}_2$  dissolved in blood: proportional to partial pressure of  $\text{O}_2$  in lungs.

| Gas           | $K_H$ (atm) at 25°C | Solubility |
|---------------|---------------------|------------|
| $\text{O}_2$  | $4.3 \times 10^4$   | Low        |
| $\text{CO}_2$ | $1.6 \times 10^3$   | Higher     |
| $\text{N}_2$  | $8.7 \times 10^4$   | Very low   |

**Q80.** A solution of acetic acid containing 8.08% (w/v) of acetic acid at 20°C has a density of 1.0097 g/cc. Calculate the molality, molarity, and mole fraction. (9 marks) [CHEM 101 | 2007–08]

### Answer

#### Molality, molarity and mole fraction of acetic acid solution

##### Given:

- Concentration = 8.08% (w/v) acetic acid ( $\text{CH}_3\text{COOH}$ ,  $M = 60$  g/mol)
- Density of solution = 1.0097 g/cc
- Temperature = 20°C

##### Step 1: Find mass of solution per 100 mL:

$$\text{Mass of solution} = 100 \text{ mL} \times 1.0097 \text{ g/mL} = 100.97 \text{ g}$$

Mass of acetic acid (8.08 w/v means 8.08 g per 100 mL):

$$w_{\text{solute}} = 8.08 \text{ g}$$

##### Mass of water:

$$w_{\text{water}} = 100.97 - 8.08 = 92.89 \text{ g}$$

##### Step 2: Moles of acetic acid:

$$n_{\text{CH}_3\text{COOH}} = \frac{8.08}{60} = 0.1347 \text{ mol}$$

##### Step 3: Molarity ( $M$ ):

$$\text{Molarity} = \frac{\text{moles of solute}}{\text{volume of solution (L)}} = \frac{0.1347}{0.100} = \mathbf{1.347 \text{ mol/L}}$$

##### Step 4: Molality ( $m$ ):

$$\text{Molality} = \frac{\text{moles of solute}}{\text{mass of solvent (kg)}} = \frac{0.1347}{0.09289} = \mathbf{1.450 \text{ mol/kg}}$$

##### Step 5: Mole fraction:

Moles of water:

$$n_{\text{H}_2\text{O}} = \frac{92.89}{18} = 5.161 \text{ mol}$$

Mole fraction of acetic acid:

$$x_{\text{CH}_3\text{COOH}} = \frac{0.1347}{0.1347 + 5.161} = \frac{0.1347}{5.2957} = \mathbf{0.02544}$$

Mole fraction of water:

$$x_{\text{H}_2\text{O}} = 1 - 0.02544 = \mathbf{0.9746}$$

| Quantity                                   | Value        |
|--------------------------------------------|--------------|
| Molarity                                   | 1.347 mol/L  |
| Molality                                   | 1.450 mol/kg |
| Mole fraction ( $\text{CH}_3\text{COOH}$ ) | 0.02544      |
| Mole fraction ( $\text{H}_2\text{O}$ )     | 0.9746       |

**Q81.** Calculate the osmotic pressure of a solution obtained by mixing (i) 100 mL of 3.4% urea (mol. wt = 60) and (ii) 100 mL of 1.6% cane sugar (mol. wt = 342) at 20°C. (10 marks) [CHEM 101 | 2007–08]

### Answer

#### Osmotic pressure of mixed urea and cane sugar solution

##### Given:

- Solution (i): 100 mL of 3.4% urea,  $M = 60$  g/mol
- Solution (ii): 100 mL of 1.6% cane sugar,  $M = 342$  g/mol
- Temperature = 20°C = 293 K
- $R = 0.0821 \text{ L}\cdot\text{atm}\cdot\text{mol}^{-1}\text{K}^{-1}$

##### Step 1: Moles of each solute:

**Urea:** 3.4% w/v means 3.4 g per 100 mL:

$$n_{urea} = \frac{3.4}{60} = 0.05667 \text{ mol}$$

**Cane sugar:** 1.6% w/v means 1.6 g per 100 mL:

$$n_{sugar} = \frac{1.6}{342} = 0.004678 \text{ mol}$$

**Step 2: Total volume after mixing:**

$$V_{total} = 100 + 100 = 200 \text{ mL} = 0.200 \text{ L}$$

**Step 3: Total moles of solute:**

$$n_{total} = 0.05667 + 0.004678 = 0.06135 \text{ mol}$$

**Step 4: Total molar concentration:**

$$C = \frac{n_{total}}{V} = \frac{0.06135}{0.200} = 0.3068 \text{ mol/L}$$

**Step 5: Osmotic pressure using van't Hoff equation:**

Both urea and sucrose are **non-electrolytes** ( $i = 1$ ):

$$\pi = CRT = 0.3068 \times 0.0821 \times 293$$

$$\pi = 0.3068 \times 24.06 = \mathbf{7.38 \text{ atm}}$$

**Alternatively in individual contributions:**

$$\pi_{urea} = \frac{0.05667}{0.200} \times 0.0821 \times 293 = 6.81 \text{ atm}$$

$$\pi_{sugar} = \frac{0.004678}{0.200} \times 0.0821 \times 293 = 0.563 \text{ atm}$$

$$\pi_{total} = 6.81 + 0.563 = \mathbf{7.37 \text{ atm}}$$

| Component    | Moles          | Osmotic pressure (atm) |
|--------------|----------------|------------------------|
| Urea         | 0.05667        | 6.81                   |
| Cane sugar   | 0.004678       | 0.563                  |
| <b>Total</b> | <b>0.06135</b> | <b>7.37</b>            |

$$\pi_{total} \approx 7.37 \text{ atm}$$

**Q82.** Classify the binary solutions of solid, liquid and gas with examples. (5 marks)

[CHEM 101 | 2006–07]

### Answer

#### Classification of Binary Solutions:

A **binary solution** contains two components. Based on the physical states of solute and solvent, binary solutions are classified as:

| Solute                | Solvent | Type             | Example                                 |
|-----------------------|---------|------------------|-----------------------------------------|
| <b>Solid solute:</b>  |         |                  |                                         |
| Solid                 | Liquid  | Solid in liquid  | NaCl in water                           |
| Solid                 | Solid   | Solid in solid   | Gold–silver alloy                       |
| <b>Liquid solute:</b> |         |                  |                                         |
| Liquid                | Liquid  | Liquid in liquid | Ethanol in water                        |
| Liquid                | Solid   | Liquid in solid  | Mercury in gold (amalgam)               |
| <b>Gas solute:</b>    |         |                  |                                         |
| Gas                   | Liquid  | Gas in liquid    | O <sub>2</sub> in water (aerated water) |
| Gas                   | Solid   | Gas in solid     | H <sub>2</sub> in palladium             |
| Gas                   | Gas     | Gas in gas       | Air (N <sub>2</sub> + O <sub>2</sub> )  |

The most common and practically important type is **solid or gas dissolved in a liquid**, which is the basis of most solution chemistry, biological fluids, and industrial processes.

Q83. What are the factors affecting the solubility of a gas in liquids? Discuss the pressure effect on the solubility of a gas in liquid. (10 marks)

[CHEM 101 | 2006–07]

Answer

**Factors Affecting Solubility of a Gas in Liquid:**

- (a) **Pressure** (discussed in detail below)
- (b) **Temperature:** gas solubility decreases with increasing temperature
- (c) **Nature of gas:** reactive/polar gases are far more soluble (HCl, NH<sub>3</sub>) than inert gases (N<sub>2</sub>, He)
- (d) **Nature of solvent:** polar gases dissolve more in polar solvents (“like dissolves like”)
- (e) **Presence of salts:** salting out reduces gas solubility in water

**Pressure Effect — Henry’s Law:**  
The effect of pressure on gas solubility is quantitatively described by **Henry’s Law** (1803):

$$C = k_H \cdot P$$

where  $C$  = concentration of dissolved gas (mol/L),  $P$  = partial pressure of gas (atm), and  $k_H$  = Henry’s law constant.

**Molecular explanation:**

- At higher pressure, more gas molecules strike the liquid surface per unit time
- More collisions  $\Rightarrow$  more molecules enter solution
- At equilibrium, dissolution rate equals escape rate — both proportional to  $P$
- $\therefore$  equilibrium concentration  $\propto P$  (Henry’s Law)

**Graphical relationship:**

Concentration  $C$  (mol/L)

Pressure  $P$  (atm)

$C = k_H \cdot P$

**Applications:**

- **Carbonated beverages:** CO<sub>2</sub> dissolved at high pressure ( $\sim 5$  atm) — released when bottle is opened

- **Decompression sickness** (the bends): excess N<sub>2</sub> dissolved in blood at high diving pressure forms bubbles on rapid ascent
- **Oxygen therapy**: pure O<sub>2</sub> at high pressure increases dissolved O<sub>2</sub> in blood
- **Industrial gas absorption**: high pressure used in scrubbers to dissolve SO<sub>2</sub>, CO<sub>2</sub>

**Conclusion:** Gas solubility increases **linearly** with pressure as described by Henry's Law — higher pressure forces more gas molecules into solution by increasing the collision frequency at the gas–liquid interface.

**Q84.** An aqueous solution containing 0.126 kg of a solute (molar mass 126 g) and 0.882 kg of water has a density of 1.10 kg/dm<sup>3</sup>. Calculate (i) mass percent, (ii) mole fraction, (iii) molality, and (iv) molarity of the solute in the solution. **(10 marks)** [CHEM 101 | 2006–07]

#### Answer

##### Given:

- Mass of solute:  $w_2 = 0.126 \text{ kg} = 126 \text{ g}$
- Molar mass of solute:  $\mathcal{M}_2 = 126 \text{ g/mol}$
- Mass of water:  $w_1 = 0.882 \text{ kg} = 882 \text{ g}$
- Density of solution:  $\rho = 1.10 \text{ kg/dm}^3 = 1.10 \text{ g/mL}$
- Total mass of solution =  $126 + 882 = 1008 \text{ g}$
- $\mathcal{M}(\text{H}_2\text{O}) = 18.02 \text{ g/mol}$

##### Moles:

$$n_2 = \frac{126}{126} = 1.000 \text{ mol} \quad n_1 = \frac{882}{18.02} = 48.95 \text{ mol}$$

##### (i) Mass percent:

$$\text{mass}\% = \frac{126}{1008} \times 100$$

$$\boxed{\text{mass}\% = 12.50\%}$$

##### (ii) Mole fraction:

$$X_2 = \frac{1.000}{1.000 + 48.95} = \frac{1.000}{49.95}$$

$$\boxed{X_2 = 0.02002}$$

##### (iii) Molality:

$$m = \frac{1.000}{0.882}$$

$$\boxed{m = 1.134 \text{ mol/kg}}$$

##### (iv) Molarity:

Volume of solution:

$$V = \frac{1008 \text{ g}}{1.10 \text{ g/mL}} = 916.4 \text{ mL} = 0.9164 \text{ L}$$

$$M = \frac{1.000}{0.9164}$$

$$\boxed{M = 1.091 \text{ mol/L}}$$

**Q85.** With the help of a diagram, prove that the depression of freezing point produced by a dissolved non-volatile solute in a solvent is proportional to the corresponding lowering of vapour pressure and also proportional to the molality of solute in the solution. **(15 marks)** [CHEM 101 | 2006–07]

#### Answer

##### Depression of Freezing Point: Proof with Diagram

**(15 marks)**

##### Part 1: Background — Vapour Pressure & Freezing Point

**(3 marks)**

The **freezing point** of a liquid is the temperature at which the vapour pressure of the **liquid phase** equals the vapour pressure of the **solid phase** — i.e. the two phases are in equilibrium. When a non-volatile solute is dissolved:

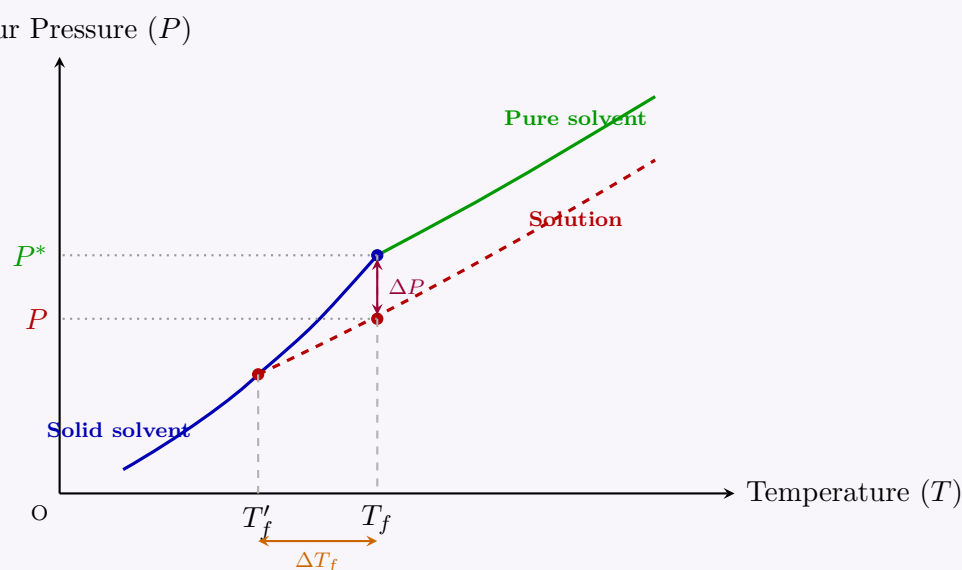
- Solute molecules occupy surface sites, reducing the **escaping tendency** of solvent molecules.
- The vapour pressure of the solution is **lowered** below that of the pure solvent (Raoult's Law).
- The solid solvent (ice) vapour pressure is **unaffected** (solute does not enter solid phase).



- Therefore, solid–liquid equilibrium is reached at a **lower temperature** — the freezing point is *depressed*.

### Part 2: Vapour Pressure–Temperature Diagram

(4 marks)



From the diagram: the solution curve (dashed red) lies **below** the pure solvent curve (solid green), and intersects the solid curve at a **lower temperature**  $T'_f < T_f$ , giving:

$$\Delta T_f = T_f - T'_f \quad (\text{depression of freezing point})$$

### Part 3: Proof that $\Delta T_f \propto \Delta P$

(4 marks)

Let  $P^*$  = vapour pressure of pure solvent and  $P$  = vapour pressure of solution **at temperature**  $T_f$ .

From the diagram, the solid curve and the liquid curves are nearly **linear** near the freezing point. Let their slopes be  $\alpha$  (solid) and  $\beta$  (liquid). At the intersection points:

For pure solvent (freezing at  $T_f$ ):

$$P^* = \alpha T_f + c_1 = \beta T_f + c_2$$

For solution (freezing at  $T'_f$ ):

$$P = \alpha T'_f + c_1$$

Subtracting:

$$P^* - P = \alpha(T_f - T'_f) = \alpha \Delta T_f$$

$$\Delta T_f = \frac{\Delta P}{\alpha} \propto \Delta P$$

Since  $\alpha$  is a constant for a given solvent,  $\Delta T_f$  is **directly proportional** to  $\Delta P = P^* - P$  (the lowering of vapour pressure).

### Part 4: Proof that $\Delta T_f \propto \text{Molality } (m)$

(4 marks)

From **Raoult's Law** for a dilute solution:

$$\frac{\Delta P}{P^*} = \frac{P^* - P}{P^*} = x_2$$

where  $x_2$  is the mole fraction of solute. For a dilute solution:

$$x_2 = \frac{n_2}{n_1 + n_2} \approx \frac{n_2}{n_1}$$

where  $n_1$  = moles of solvent,  $n_2$  = moles of solute. If  $W_1$  grams of solvent of molar mass  $M_1$ :

$$n_1 = \frac{W_1}{M_1} \Rightarrow x_2 \approx \frac{n_2 M_1}{W_1}$$

Molality  $m = \frac{n_2 \times 1000}{W_1(\text{g})}$ , so:

$$x_2 \approx \frac{m M_1}{1000}$$

Substituting into the expression for  $\Delta T_f$ :

$$\Delta T_f = \frac{\Delta P}{\alpha} = \frac{P^* x_2}{\alpha} = \frac{P^* M_1}{1000 \alpha} \cdot m$$

Since  $P^*$ ,  $M_1$ ,  $\alpha$  are all constants for a given solvent, we define:

$$K_f = \frac{P^* M_1}{1000 \alpha} \quad (\text{cryoscopic constant})$$

$$\Delta T_f = K_f \cdot m \propto m$$

The depression of freezing point is **directly proportional to the molality** of the solute.

**Conclusion:** The vapour pressure–temperature diagram shows that adding a non-volatile solute lowers the liquid vapour pressure curve, depressing the freezing point by  $\Delta T_f$ . Analytically:  $\Delta T_f \propto \Delta P$  (from linear approximation of VP curves) and  $\Delta T_f \propto m$  (from Raoult's Law in dilute solution), giving the fundamental cryoscopic relation  $\Delta T_f = K_f \cdot m$ .

**Q86.** What is meant by osmotic pressure? What are the laws of osmotic pressure and how is the molecular weight of a solute determined from osmotic pressure measurement? **(10 marks)** [CHEM 101 | 2006–07]

### Answer

#### Osmotic Pressure: Laws, Definition & Molecular Weight Determination

**(10 marks)**

##### Part 1: Definition of Osmotic Pressure

**(3 marks)**

**Osmosis** is the spontaneous flow of solvent molecules through a **semipermeable membrane** from a region of *lower* solute concentration to a region of *higher* solute concentration.

The **osmotic pressure** ( $\Pi$ ) is defined as the *minimum external pressure* that must be applied to the solution side to **stop** the net flow of solvent through the semipermeable membrane:

- It is a **colligative property** — depends only on the number of solute particles, not their identity.
- Higher solute concentration  $\Rightarrow$  greater osmotic pressure.
- Unit: atm, Pa, or bar.

##### Part 2: Laws of Osmotic Pressure

**(4 marks)**

###### (i) van't Hoff's Law I (Effect of Concentration):

At constant temperature, osmotic pressure is **directly proportional** to the molar concentration  $C$ :

$$\Pi \propto C \quad (\text{at constant } T)$$

###### (ii) van't Hoff's Law II (Effect of Temperature):

At constant concentration, osmotic pressure is **directly proportional** to absolute temperature  $T$ :

$$\Pi \propto T \quad (\text{at constant } C)$$

###### (iii) Combined van't Hoff Equation:

Combining both laws (analogous to ideal gas law  $PV = nRT$ ):

$$\Pi = CRT = \frac{n}{V}RT$$

where  $R = 0.08206 \text{ L}\cdot\text{atm}\cdot\text{mol}^{-1}\cdot\text{K}^{-1}$ ,  $T$  is absolute temperature, and  $C = n/V$  is molarity.

This equation is remarkably similar to the ideal gas law — van't Hoff noted that dissolved solute particles behave like gas molecules in terms of pressure exerted.

###### (iv) For Electrolytes — van't Hoff Factor $i$ :

For solutes that dissociate, the equation becomes:

$$\Pi = iCRT$$

where  $i$  = number of particles per formula unit (e.g. NaCl:  $i = 2$ ; CaCl<sub>2</sub>:  $i = 3$ ).

##### Part 3: Determination of Molecular Weight from $\Pi$

**(3 marks)**

From  $\Pi = CRT = \frac{n}{V}RT = \frac{W}{MV}RT$ , where  $W$  = mass of solute,  $M$  = molar mass:

$$\Pi = \frac{W RT}{M V} \quad \Rightarrow \quad M = \frac{W RT}{\Pi V}$$

##### Procedure:

1. Dissolve a known mass  $W$  (g) of solute in volume  $V$  (L) of solvent.
2. Measure osmotic pressure  $\Pi$  at temperature  $T$ .
3. Calculate molar mass using  $M = WRT/\Pi V$ .

**Example:** A solution containing 6.0 g of a compound in 1.0 L of water shows  $\Pi = 0.821 \text{ atm}$  at  $27^\circ\text{C}$ :

$$M = \frac{6.0 \times 0.08206 \times 300}{0.821 \times 1.0} = \frac{147.7}{0.821} \approx 180 \text{ g/mol}$$

**Advantage of osmotic pressure method:** Osmotic pressure is **far more sensitive** than other colligative methods for large molecules (polymers, proteins). Even a very dilute solution ( $C \sim 10^{-4} \text{ M}$ ) gives a measurable  $\Pi$ , whereas  $\Delta T_f$  would be undetectably small.

**Conclusion:** Osmotic pressure is the pressure needed to stop osmotic flow through a semipermeable membrane. It obeys van't Hoff's laws:  $\Pi \propto C$  and  $\Pi \propto T$ , combined as  $\Pi = CRT$ . Molar mass is determined as  $M = WRT/\Pi V$  — especially valuable for large biomolecules where other colligative methods are insensitive.

- Q87.** Camphor is a white solid that melts at 179.5°C and its freezing point depression constant is 40°C/m. A 1.07 mg sample of a compound was dissolved in 78.1 mg of camphor. The mixture melted at 176.0°C. What is the molecular weight of the compound? If the empirical formula of the compound is CH, what is the molecular formula? **(10 marks)** [CHEM 101 | 2006–07]

### Answer

#### Molecular Weight & Molecular Formula from Freezing Point Depression (Camphor)

**(10 marks)**

#### Part 1: Given Data

**(1 marks)**

| Quantity                                 | Value                             |
|------------------------------------------|-----------------------------------|
| Freezing point of pure camphor ( $T_f$ ) | 179.5°C                           |
| $K_f$ of camphor                         | 40.0°C/m                          |
| Mass of compound ( $W_2$ )               | 1.07 mg = $1.07 \times 10^{-3}$ g |
| Mass of camphor ( $W_1$ )                | 78.1 mg = $78.1 \times 10^{-3}$ g |
| Melting point of mixture ( $T'_f$ )      | 176.0°C                           |

#### Part 2: Calculate $\Delta T_f$

**(2 marks)**

$$\Delta T_f = T_f - T'_f = 179.5 - 176.0 = 3.5^\circ\text{C}$$

#### Part 3: Calculate Molality of the Solution

**(2 marks)**

 Using  $\Delta T_f = K_f \cdot m$ :

$$m = \frac{\Delta T_f}{K_f} = \frac{3.5}{40.0} = 0.0875 \text{ mol/kg}$$

#### Part 4: Calculate Moles of Solute

**(2 marks)**

 Molality =  $\frac{\text{moles of solute}}{\text{kg of solvent}}$ :

$$\begin{aligned} \text{moles of solute} &= m \times W_1(\text{kg}) = 0.0875 \times 78.1 \times 10^{-3} \times \frac{1}{1000} \times 1000 \\ &= 0.0875 \times 78.1 \times 10^{-3} = 6.834 \times 10^{-3} \text{ mmol} = 6.834 \times 10^{-6} \text{ mol} \end{aligned}$$

#### Part 5: Calculate Molecular Weight

**(2 marks)**

$$M = \frac{W_2}{\text{moles of solute}} = \frac{1.07 \times 10^{-3} \text{ g}}{6.834 \times 10^{-6} \text{ mol}}$$

$M \approx 156.6 \text{ g/mol} \approx 157 \text{ g/mol}$

#### Part 6: Determine Molecular Formula from Empirical Formula CH

**(1 marks)**

Empirical formula CH has molar mass:

$$M_{\text{emp}} = 12 + 1 = 13 \text{ g/mol}$$

Number of empirical units in the molecular formula:

$$n = \frac{M}{M_{\text{emp}}} = \frac{157}{13} \approx 12$$

Molecular formula =  $(\text{CH})_{12} = \text{C}_{12}\text{H}_{12}$

 This corresponds to **[6]paracyclophane** or similar aromatic hydrocarbon.

**Conclusion:**  $\Delta T_f = 3.5^\circ\text{C} \Rightarrow m = 0.0875 \text{ mol/kg} \Rightarrow \text{moles} = 6.83 \times 10^{-6} \text{ mol} \Rightarrow M = 157 \text{ g/mol}$ . With empirical formula CH ( $M = 13$ ), the molecular formula is **C<sub>12</sub>H<sub>12</sub>**.

**Q88.** State how many components, phases and degrees of freedom each of the following systems possesses: (i)  $\text{H}_2(\text{g}) + \text{N}_2(\text{g})$ , (ii)  $\text{NH}_3(\text{g})$ , (iii) a saturated solution of  $\text{NaCl}$  in water, (iv)  $\text{H}_2\text{O}(\text{l}) \rightleftharpoons \text{H}_2\text{O}(\text{g})$ , (v)  $\text{CaCO}_3(\text{s}) \rightleftharpoons \text{CaO}(\text{s}) + \text{CO}_2(\text{g})$ . **(15 marks)** [CHEM 101 | 2006–07]

### Answer

#### Components, Phases and Degrees of Freedom for Five Systems

**(15 marks)**

#### Phase Rule (Gibbs):

$$F = C - P + 2$$

where  $F$  = degrees of freedom,  $C$  = number of components,  $P$  = number of phases.

#### (i) $\text{H}_2(\text{g}) + \text{N}_2(\text{g})$

**(3 marks)**

- **Components ( $C$ ):** 2 —  $\text{H}_2$  and  $\text{N}_2$  are chemically independent (no reaction under normal conditions).
- **Phases ( $P$ ):** 1 — both gases mix completely to form a single homogeneous gas phase.
- **Degrees of freedom ( $F$ ):**  $F = 2 - 1 + 2 = 3$  (temperature, pressure, and composition must all be specified to define the state).

#### (ii) $\text{NH}_3(\text{g})$

**(3 marks)**

- **Components ( $C$ ):** 1 — pure ammonia gas, single chemical species.
- **Phases ( $P$ ):** 1 — single gas phase.
- **Degrees of freedom ( $F$ ):**  $F = 1 - 1 + 2 = 2$  (temperature and pressure must be specified).

#### (iii) Saturated solution of $\text{NaCl}$ in water

**(3 marks)**

- **Components ( $C$ ):** 2 —  $\text{NaCl}$  and  $\text{H}_2\text{O}$  (two independently variable chemical constituents).
- **Phases ( $P$ ):** 2 — liquid solution phase and solid  $\text{NaCl}$  phase (excess undissolved salt in equilibrium with the solution in a saturated solution).
- **Degrees of freedom ( $F$ ):**  $F = 2 - 2 + 2 = 2$  (temperature and pressure; composition is fixed by saturation condition).

#### (iv) $\text{H}_2\text{O}(\text{l}) \rightleftharpoons \text{H}_2\text{O}(\text{g})$

**(3 marks)**

- **Components ( $C$ ):** 1 — only water, single chemical substance in two physical states.
- **Phases ( $P$ ):** 2 — liquid water and water vapour coexisting at equilibrium.
- **Degrees of freedom ( $F$ ):**  $F = 1 - 2 + 2 = 1$  (only temperature *or* pressure can be varied independently; fixing one fixes the other via the vapour pressure curve).

#### (v) $\text{CaCO}_3(\text{s}) \rightleftharpoons \text{CaO}(\text{s}) + \text{CO}_2(\text{g})$

**(3 marks)**

- **Components ( $C$ ):** 2 — three chemical species ( $\text{CaCO}_3$ ,  $\text{CaO}$ ,  $\text{CO}_2$ ) but one equilibrium constraint reduces components by 1:  $C = 3 - 1 = 2$ .
- **Phases ( $P$ ):** 3 — solid  $\text{CaCO}_3$ , solid  $\text{CaO}$ , and gaseous  $\text{CO}_2$  — three distinct phases.
- **Degrees of freedom ( $F$ ):**  $F = 2 - 3 + 2 = 1$  (only temperature *or* pressure can be independently varied; the decomposition pressure of  $\text{CO}_2$  is fixed at a given temperature).

#### Summary Table:

| System                                                                              | $C$ | $P$ | $F$ |
|-------------------------------------------------------------------------------------|-----|-----|-----|
| (i) $\text{H}_2 + \text{N}_2 (\text{g})$                                            | 2   | 1   | 3   |
| (ii) $\text{NH}_3 (\text{g})$                                                       | 1   | 1   | 2   |
| (iii) Sat. $\text{NaCl}$ solution                                                   | 2   | 2   | 2   |
| (iv) $\text{H}_2\text{O}(\text{l}) \rightleftharpoons \text{H}_2\text{O}(\text{g})$ | 1   | 2   | 1   |
| (v) $\text{CaCO}_3 \rightleftharpoons \text{CaO} + \text{CO}_2$                     | 2   | 3   | 1   |

**Conclusion:** Gibbs phase rule  $F = C - P + 2$  gives a precise count of the variables needed to fully define each system. Increasing phases reduces freedom; increasing components increases freedom. The five systems range from  $F = 1$  (univariant, like water boiling) to  $F = 3$  (trivariant, like a binary gas mixture).

BUET · CHEM 113

# Chemistry

Section S5

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Colloid & Nanochemistry

## S5 — Colloid & Nanochemistry

**Q1.** How does the electrophoresis phenomenon provide information about the sign of charge on colloidal particles? (5 marks) [CHEM 113 | 2023–24]

### Answer

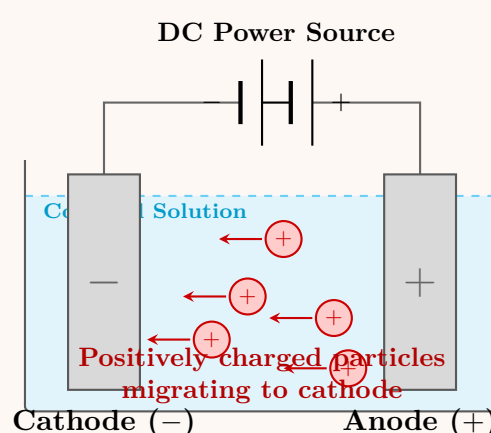
#### Electrophoresis and the Sign of Charge on Colloidal Particles

##### Definition of Electrophoresis:

**Electrophoresis** is the phenomenon in which **colloidal particles migrate** toward one of the electrodes when an electric field is applied across a colloidal solution. This occurs because colloidal particles carry an **electric charge** on their surface.

##### How It Provides Information About the Sign of Charge:

- When a direct current (DC) electric field is applied across a colloidal solution using two electrodes, the charged colloidal particles **migrate toward the oppositely charged electrode**.
- If the colloidal particles migrate toward the **cathode** (negative electrode), the particles carry a **positive charge**. Example:  $\text{Al}(\text{OH})_3$  sol,  $\text{Fe}(\text{OH})_3$  sol.
- If the colloidal particles migrate toward the **anode** (positive electrode), the particles carry a **negative charge**. Example:  $\text{As}_2\text{S}_3$  sol, gold sol, starch sol.
- The **direction of migration** therefore directly indicates the **sign of charge** on the colloidal particles.
- The **rate of migration** (electrophoretic mobility) depends on the magnitude of the charge, particle size, and viscosity of the medium.



**Conclusion:** By observing the **direction of migration** of colloidal particles in an applied electric field, electrophoresis directly reveals whether the charge on the particles is **positive** (migration to cathode) or **negative** (migration to anode).

**Q2.** What is a protective colloid? Explain the concept of the gold number and its significance in evaluating the protective power of colloids. (10 marks) [CHEM 113 | 2023–24]

### Answer

#### Protective Colloid, Gold Number and Its Significance

##### Part 1: Protective Colloid

A **protective colloid** is a **lyophilic (solvent-loving)** colloid that, when added to a **lyophobic (solvent-fearing)** colloid, stabilises it against **coagulation** by electrolytes.

##### Mechanism:

- The lyophilic colloid particles **adsorb onto the surface** of lyophobic particles, forming a protective layer.
- This layer **prevents the lyophobic particles from approaching each other** closely enough to aggregate.
- The protective colloid also imparts its own **solvation layer and charge** to the lyophobic particles, enhancing stability.

##### Examples:

- Gelatin or albumin stabilising gold sol.
- Casein stabilising fat globules in milk.
- Haemoglobin stabilising fat emulsions in blood.

##### Part 2: Gold Number

The **gold number** was introduced by **Zsigmondy (1901)** and is defined as:

*“The minimum mass (in milligrams) of a protective colloid required to just prevent the coagulation of 10 mL of a standard **red gold sol** when 1 mL of a 10% NaCl solution is added.”*

- The standard red gold sol **turns blue/violet** upon coagulation.



- The protective colloid prevents this colour change by stabilising the sol.
- The **smaller the gold number**, the **greater the protective power** of the colloid (less material needed for protection).

Gold Numbers of Common Protective Colloids:

| Protective Colloid | Gold Number  |
|--------------------|--------------|
| Gelatin            | 0.005 – 0.01 |
| Haemoglobin        | 0.03 – 0.07  |
| Gum arabic         | 0.15 – 0.25  |
| Albumin (egg)      | 0.08 – 0.10  |
| Potato starch      | 25           |

Significance:

- (i) **Measures protective power:** Gold number is an **inverse measure** of protective power — lower gold number means **stronger protection**. Gelatin is the most effective protective colloid.
- (ii) **Industrial applications:** Used to select the best stabiliser for colloidal preparations in pharmaceuticals, paints, and emulsions.
- (iii) **Biological significance:** Proteins in blood act as protective colloids. Their gold numbers reflect their ability to maintain colloidal stability of blood components.
- (iv) **Quality control:** Gold number is used to assess the **purity and effectiveness** of protective agents like gelatin — pure gelatin has a lower gold number than impure samples.

Q3. Why does the Lycurgus cup appear blue on reflection and red/purple on transmission? (10 marks) [CHEM 113 | 2023–24]

Answer

The Lycurgus Cup: Blue on Reflection, Red/Purple on Transmission

What is the Lycurgus Cup?

The **Lycurgus Cup** is a 4th-century Roman glass goblet that displays a remarkable optical property:

- It appears **opaque green/blue** when illuminated from outside (**reflected light**).
- It appears **translucent red/purple** when illuminated from inside (**transmitted light**).

Reason: Gold–Silver Nanoparticles in the Glass

Modern analysis (TEM and XRF) revealed that the glass contains **colloidal nanoparticles** of a **gold–silver alloy** (approximately 70% Ag, 30% Au, diameter ~ 70 nm) embedded uniformly throughout.

Scientific Explanation: Localised Surface Plasmon Resonance (LSPR)

- (i) When light strikes the metal nanoparticles, the **free electrons** in the nanoparticle collectively **oscillate** in resonance with the electromagnetic field of the incoming light. This is called **Localised Surface Plasmon Resonance (LSPR)**.
- (ii) The **resonance frequency** (and hence the wavelength of light absorbed/scattered) depends on:
  - Size of the nanoparticle.
  - Shape of the nanoparticle.
  - Composition (Au/Ag ratio).
  - Surrounding medium.
- (iii) For the Au–Ag nanoparticles in the Lycurgus cup, LSPR occurs in the **green/blue region** of the spectrum (~ 500–530 nm).
- (iv) **In reflected light:** The nanoparticles **strongly scatter and reflect green/blue light** (at the LSPR frequency). Therefore, the cup appears **green/blue** when viewed in reflected light.
- (v) **In transmitted light:** Green/blue wavelengths are **absorbed** by the nanoparticles (LSPR absorption). The remaining transmitted light is **complementary in colour**, i.e. **red/purple** (wavelengths ~ 630–700 nm that are not absorbed pass through). Therefore, the cup appears **red/purple** in transmitted light.

Summary:

| Mode              | Optical Process               | Colour     |
|-------------------|-------------------------------|------------|
| Reflected light   | LSPR scattering/reflection    | Green/Blue |
| Transmitted light | LSPR absorption of green/blue | Red/Purple |

**Significance:** The Lycurgus Cup is one of the earliest known examples of **nanomaterial technology**, demonstrating that the Romans unknowingly exploited the optical properties of metal nanoparticles over 1600 years ago.

**Q4.** What is clay? How can it be classified? Write down the factors influencing the properties of polymer-clay nanocomposites. (12 marks) [CHEM 113 | 2023–24]

### Answer

#### Clay: Definition, Classification and Factors Influencing Polymer-Clay Nanocomposites

##### Part 1: What is Clay?

**Clay** is a naturally occurring, fine-grained ( $< 2\ \mu\text{m}$ ) **aluminosilicate mineral** with a layered structure. It is composed of **tetrahedral silicate sheets** ( $\text{SiO}_4$ ) and **octahedral aluminate sheets** ( $\text{AlO}_6$ ) stacked in various configurations. Clay minerals are hydrophilic, can swell in water, have high surface area, and exhibit **cation exchange capacity (CEC)**.

##### Part 2: Classification of Clay

Clay minerals are classified based on the **stacking arrangement** of tetrahedral (T) and octahedral (O) sheets:

- (i) **1:1 Type (T:O = 1:1):** One tetrahedral sheet bonded to one octahedral sheet. Layers are held together by **hydrogen bonds** and are **non-swelling**.  
*Examples:* Kaolinite, halloysite.
- (ii) **2:1 Type (T:O:T = 2:1):** One octahedral sheet sandwiched between two tetrahedral sheets. Layers are held by **van der Waals forces** and may be **swelling**.  
*Examples:* Montmorillonite (MMT), smectite, vermiculite, mica.
- (iii) **2:1:1 Type (Chlorite):** A 2:1 layer with an additional octahedral **brucite sheet** between layers.  
*Example:* Chlorite.

| Type  | Structure | Swelling | Example         |
|-------|-----------|----------|-----------------|
| 1:1   | T–O       | No       | Kaolinite       |
| 2:1   | T–O–T     | Yes      | Montmorillonite |
| 2:1:1 | T–O–T+O   | No       | Chlorite        |

##### Part 3: Factors Influencing Properties of Polymer-Clay Nanocomposites

- (i) **Type and Nature of Clay:** Montmorillonite (MMT) is most commonly used due to its high aspect ratio, large surface area, and ability to swell and exfoliate. The CEC of the clay determines how easily it can be organically modified.
- (ii) **Dispersion State of Clay in Polymer Matrix:** Three possible structures exist:
  - **Phase-separated (microcomposite):** Polymer cannot intercalate; poor properties.
  - **Intercalated:** Polymer chains enter the clay galleries; moderate improvement.
  - **Exfoliated:** Clay layers are completely separated and dispersed in polymer matrix; **maximum property improvement**.
- (iii) **Surface Modification (Organoclay):** Natural clay is hydrophilic and incompatible with hydrophobic polymers. **Organomodification** (treatment with quaternary ammonium salts) makes clay organophilic, improving compatibility with polymers and facilitating intercalation/exfoliation.
- (iv) **Aspect Ratio of Clay Platelets:** Higher aspect ratio (length/thickness) provides better barrier properties, mechanical reinforcement, and flame retardancy.
- (v) **Clay Loading (wt%):** Typically 1–5 wt% clay gives optimum properties. Excessive loading causes agglomeration, reducing benefits.
- (vi) **Polymer Matrix Type:** The polarity, molecular weight, and functional groups of the polymer affect the degree of intercalation and interfacial adhesion with clay.
- (vii) **Processing Method:** **Melt intercalation**, **in-situ polymerisation**, and **solution blending** give different levels of clay dispersion and different final properties.
- (viii) **Interfacial Interaction:** Strong polymer–clay interfacial adhesion (via H-bonding, covalent grafting, or ionic interaction) is essential for effective load transfer and property enhancement.

**Q5.** The electrical, mechanical and thermal properties of carbon nanotubes are quite remarkable — why? Discuss how the structural differences of carbon nanotubes influence their physical properties. (13 marks) [CHEM 113 | 2023–24]

### Answer

#### Remarkable Properties of Carbon Nanotubes and Effect of Structural Differences

##### Part 1: What are Carbon Nanotubes (CNTs)?

**Carbon nanotubes** are cylindrical nanostructures formed by **rolling graphene sheets** into seamless tubes. They are classified as:

- **Single-walled CNTs (SWCNTs):** Single graphene layer rolled into a cylinder (diameter  $\sim 0.4\text{--}2\ \text{nm}$ ).
- **Multi-walled CNTs (MWCNTs):** Multiple concentric graphene cylinders (diameter  $\sim 2\text{--}100\ \text{nm}$ ).

##### Part 2: Why Are CNT Properties Remarkable?

(i) **Electrical Properties:**

- CNTs are formed from **sp<sup>2</sup> hybridised carbon** atoms arranged in a hexagonal lattice. Each carbon contributes one **delocalised  $\pi$  electron**.
- These  $\pi$  electrons are free to move along the tube axis, giving CNTs **exceptional electrical conductivity**.
- Depending on chirality (see below), SWCNTs can be **metallic conductors or semiconductors**.
- Current density of CNTs can be  $\sim 10^9$  A/cm<sup>2</sup> — about 1000 times greater than copper.

 (ii) **Mechanical Properties:**

- The **C–C sp<sup>2</sup> bond** in the graphene lattice is one of the strongest bonds in nature (bond energy  $\sim 524$  kJ/mol).
- CNTs have a **Young's modulus** of  $\sim 1$  TPa (5 times that of steel) and **tensile strength** of  $\sim 150$  GPa (100 times stronger than steel at 1/6 the weight).
- The hollow tubular structure and seamless lattice provide exceptional **resistance to deformation**.

 (iii) **Thermal Properties:**

- CNTs have **extremely high thermal conductivity** along the tube axis ( $\sim 3000$ – $6000$  W/mK), exceeding diamond.
- This is due to efficient **phonon transport** along the **defect-free** sp<sup>2</sup> carbon lattice.
- They are also **thermally stable** up to  $\sim 700^\circ\text{C}$  in air and  $> 2800^\circ\text{C}$  in vacuum.

**Part 3: How Structural Differences Influence Physical Properties**
**(A) Chirality (Rolling Vector):**

The way a graphene sheet is rolled determines the **chirality** of the CNT, described by the chiral vector  $(n, m)$ :

- **Armchair** ( $n = m$ ): always **metallic** (conducting).
- **Zigzag** ( $m = 0$ ): metallic if  $n$  is a multiple of 3, otherwise semiconducting.
- **Chiral** ( $n \neq m \neq 0$ ): semiconducting in most cases.

| Type     | Chirality | Electrical Property       |
|----------|-----------|---------------------------|
| Armchair | $(n, n)$  | Metallic                  |
| Zigzag   | $(n, 0)$  | Metallic or Semiconductor |
| Chiral   | $(n, m)$  | Semiconductor             |

**(B) SWCNT vs MWCNT:**

- **SWCNTs** have more uniform electronic properties; ideal for semiconducting applications (transistors, sensors).
- **MWCNTs** are generally **metallic** (outer walls dominate) and better for mechanical reinforcement due to larger diameter and higher stiffness.

**(C) Diameter:**

- Smaller diameter SWCNTs have larger **bandgaps** (semiconducting).
- Larger diameter CNTs have properties closer to graphene (semi-metallic).

**(D) Defects:**

- **Defect-free CNTs:** Maximum mechanical strength and thermal/electrical conductivity.
- **Defective CNTs:** Reduced conductivity and strength, but defect sites may be useful for chemical functionalisation.

**Q6.** What do you mean by nanocomposite materials? How does reinforcement occur in polymer-clay nanocomposites? **(10 marks)**  
[CHEM 113 | 2022–23]

**Answer**
**Nanocomposite Materials and Reinforcement in Polymer-Clay Nanocomposites**
**Part 1: Nanocomposite Materials**

A **nanocomposite** is a **multiphase material** in which at least one phase has dimensions in the **nanoscale** (1–100 nm). It consists of:

- **Matrix:** The continuous phase (polymer, metal, or ceramic).
- **Nanofiller:** The dispersed nanoscale reinforcing phase (clay, carbon nanotubes, metal nanoparticles, graphene).

Due to the **very high surface-to-volume ratio** of the nanofiller, even a small amount (1–5 wt%) can dramatically improve properties compared to conventional composites.

**Part 2: Reinforcement in Polymer-Clay Nanocomposites**

**Clay used:** Montmorillonite (MMT) is the most common clay for nanocomposites due to its **high aspect ratio** (length/thickness  $\sim 100$ – $1000$ ), large surface area ( $\sim 750 \text{ m}^2/\text{g}$ ), and ability to exfoliate.

#### Structures of Polymer-Clay Nanocomposites:

- (i) **Phase-separated (conventional microcomposite):** Polymer chains **cannot enter** the clay galleries. No improvement over conventional composites.
- (ii) **Intercalated nanocomposite:** Polymer chains **insert into clay galleries** but clay layers remain parallel and ordered. Moderate improvement in properties.
- (iii) **Exfoliated nanocomposite:** Clay layers are **completely separated and randomly dispersed** in the polymer matrix. **Maximum improvement** in properties.

#### Mechanisms of Reinforcement:

- (i) **Mechanical reinforcement:** Exfoliated clay platelets with high aspect ratio act as **physical barriers** to crack propagation. Load is transferred from the polymer matrix to the stiffer clay platelets via strong **interfacial adhesion**, increasing tensile strength, modulus, and hardness.
- (ii) **Barrier properties:** Dispersed clay platelets create a **tortuous path** for gas and vapour molecules to diffuse through the polymer matrix:

$$\text{Relative permeability} \propto \frac{1}{1 + \frac{\phi_f \alpha}{2}}$$

where  $\phi_f$  is clay volume fraction and  $\alpha$  is aspect ratio. Higher aspect ratio = better barrier.

- (iii) **Thermal stability:** Clay acts as a **thermal insulator and char-forming agent**, enhancing thermal stability and flame retardancy.
- (iv) **Reduced thermal expansion:** Rigid clay platelets **restrict polymer chain mobility**, lowering the coefficient of thermal expansion.

**Q7.** What is Stern double layer? Discuss the effect of electrolyte concentration on Zeta potential value of a colloidal solution. **(10 marks)**  
[CHEM 113 | 2022–23]

#### Answer

##### Stern Double Layer and Effect of Electrolyte Concentration on Zeta Potential

##### Part 1: Stern Double Layer

The **Stern double layer** model (Stern, 1924) is an improved version of the **Helmholtz** and **Gouy–Chapman** models, combining both to describe the electrical double layer around a charged colloidal particle.

It consists of two distinct regions:

##### (i) Stern Layer (Inner Compact Layer):

- A layer of ions **specifically adsorbed** onto the particle surface, within one ionic radius.
- These ions are **immobilised** and cannot move freely.
- The potential drops **steeply and linearly** from the surface potential  $\psi_0$  to the Stern potential  $\psi_d$  across this layer.
- Thickness  $\approx$  one hydrated ion diameter ( $\sim 3$ – $5 \text{ \AA}$ ).

##### (ii) Diffuse Layer (Outer Gouy–Chapman Layer):

- A **diffuse cloud** of counter-ions and co-ions extending from the Stern plane into the bulk solution.
- Ions can move freely under thermal motion.
- The potential decreases **exponentially** from  $\psi_d$  to zero.
- Thickness = Debye length  $\kappa^{-1} \propto 1/\sqrt{I}$  (decreases with increasing ionic strength  $I$ ).

##### Part 2: Effect of Electrolyte Concentration on Zeta Potential

- (i) **Compression of diffuse layer:** Adding electrolyte increases the ionic strength  $I$  of the solution. The Debye length:

$$\kappa^{-1} = \sqrt{\frac{\varepsilon_0 \varepsilon_r k_B T}{2 N_A e^2 I}} \propto \frac{1}{\sqrt{I}}$$

**decreases** with increasing electrolyte concentration, compressing the diffuse layer toward the particle surface.

- (ii) **Shift of slipping plane inward:** As the diffuse layer is compressed, the **slipping plane moves closer** to the particle surface. The zeta potential (potential at the slipping plane) therefore **decreases in magnitude**.
- (iii) **At high electrolyte concentration:** The diffuse layer is highly compressed;  $\zeta \rightarrow 0$ . The electrostatic repulsion between particles is greatly reduced, and the colloid **coagulates**.



(iv) **Effect of valence (Schulze–Hardy Rule):** Higher valence counter-ions are much more effective at reducing  $\zeta$  and causing coagulation:

Coagulating power:  $M^+ < M^{2+} < M^{3+}$

The critical coagulation concentration (CCC) is proportional to  $1/z^6$  where  $z$  is the valence.

| Electrolyte conc. | Debye length                | Zeta potential |
|-------------------|-----------------------------|----------------|
| Low               | Large (thick diffuse layer) | High (stable)  |
| Medium            | Intermediate                | Intermediate   |
| High              | Small (thin diffuse layer)  | Low (unstable) |

Q8. Discuss the role played by an emulsifier in the process of emulsification. (13 marks)

[CHEM 113 | 2022–23]

Answer

**Role of Emulsifier in Emulsification**

**Definition of Emulsion:**  
An **emulsion** is a **colloidal dispersion** of two immiscible liquids (usually oil and water), where one liquid is dispersed as **droplets** (0.1–100  $\mu\text{m}$ ) in the other. Types:

- **O/W (oil-in-water):** Oil droplets dispersed in water (e.g. milk, creams).
- **W/O (water-in-oil):** Water droplets in oil (e.g. butter, margarine).

**The Problem Without Emulsifier:**  
Oil and water are **immiscible** due to vastly different intermolecular forces. When mixed, they **rapidly separate** (demix) due to:

- High **interfacial tension** between the two liquids.
- **Ostwald ripening** — small droplets merge into large ones to minimise surface energy.
- **Coalescence** of droplets.

**Structure of an Emulsifier:**  
An **emulsifier** (also called emulsifying agent or surfactant) is an **amphiphilic molecule** having:

- A **hydrophilic head** (polar, water-loving).
- A **hydrophobic tail** (non-polar, oil-loving).

Examples: soap (sodium stearate), lecithin, SDS (sodium dodecyl sulfate), Tween-20, egg yolk proteins.

**Roles of Emulsifier in Emulsification:**

(i) **Reduction of Interfacial Tension:** Emulsifier molecules migrate to the **oil–water interface** and orient with the hydrophilic head in water and hydrophobic tail in oil. This significantly **reduces interfacial tension**, making it energetically easier to create and maintain small droplets.

$\gamma_{\text{interface}} \xrightarrow{\text{emulsifier}} \gamma_{\text{reduced}}$

(ii) **Formation of Interfacial Film:** Emulsifier molecules form a **monomolecular film** around each droplet. This film acts as a **physical barrier** preventing droplet coalescence.

(iii) **Electrostatic Stabilisation:** Ionic emulsifiers (e.g. soap) impart a **surface charge** on the droplets. Like-charged droplets **repel each other**, preventing coalescence and stabilising the emulsion.

(iv) **Steric Stabilisation:** Non-ionic or polymeric emulsifiers create a **steric barrier** around droplets. When two droplets approach, the overlapping emulsifier layers cause **steric repulsion** that keeps them apart.

(v) **Determining Emulsion Type (Bancroft’s Rule):** The type of emulsion formed depends on the **hydrophilic–lipophilic balance (HLB)** of the emulsifier:

- **HLB 8–18** (hydrophilic): favours **O/W emulsion**.
- **HLB 3–6** (lipophilic): favours **W/O emulsion**.

(vi) **Prevention of Ostwald Ripening:** The emulsifier film around droplets slows down the **molecular exchange** between droplets, reducing Ostwald ripening and maintaining the droplet size distribution over time.

**Common Emulsifiers and Their Applications:**

| Emulsifier      | Type      | Application           |
|-----------------|-----------|-----------------------|
| Lecithin        | Non-ionic | Mayonnaise, chocolate |
| Sodium stearate | Anionic   | Soap, cosmetics       |
| SDS             | Anionic   | Shampoo, detergents   |
| Casein          | Protein   | Milk, dairy products  |
| Tween-20        | Non-ionic | Pharmaceuticals       |

**Q9.** What is nanotoxicology? How can they impose detrimental effects on living organisms? (12 marks) [CHEM 113 | 2022–23]

### Answer

#### Nanotoxicology and Detrimental Effects on Living Organisms

##### Part 1: What is Nanotoxicology?

**Nanotoxicology** is the branch of science that studies the **toxic effects of nanoparticles and nanomaterials** on living organisms and ecosystems. It investigates:

- How nanoparticles **enter, distribute, metabolise, and are excreted** from living systems.
- The **biological damage** caused by nanoparticle exposure.
- **Dose–response relationships** for nano-sized materials.

Nanotoxicology is distinct from conventional toxicology because the **same material** can be non-toxic in bulk form but **highly toxic at the nanoscale** (e.g. bulk gold is inert, but gold nanoparticles can be cytotoxic depending on size and surface chemistry).

##### Part 2: Detrimental Effects on Living Organisms

###### (i) Routes of Entry into the Body:

- **Inhalation:** Airborne nanoparticles penetrate deep into the lungs (alveoli).
- **Ingestion:** Via contaminated food and water.
- **Dermal absorption:** Through skin (especially for small, hydrophilic particles).
- **Injection:** In medical applications (drug delivery, imaging agents).

###### (ii) Cellular Toxicity:

- Nanoparticles **enter cells** via endocytosis due to their small size.
- They can **disrupt cell membranes**, damage organelles (mitochondria, nucleus), and interfere with protein function.
- **Reactive Oxygen Species (ROS)** are generated by many nanoparticles (TiO<sub>2</sub>, ZnO, Ag NPs), causing **oxidative stress** that damages DNA, proteins, and lipids.

(iii) **Genotoxicity:** Some nanoparticles (carbon black, metal oxides) can **penetrate the nucleus** and interact with **DNA**, causing strand breaks, mutations, and potentially cancer.

(iv) **Pulmonary Toxicity:** Inhaled nanoparticles (e.g. silica, carbon NPs) cause **lung inflammation, fibrosis (scarring)**, and **pulmonary granuloma**. Carbon nanotubes exhibit asbestos-like toxicity when inhaled.

(v) **Cardiovascular Effects:** Inhaled ultrafine particles can enter the **bloodstream**, cause inflammation, promote **atherosclerosis**, and increase the risk of heart attack and stroke.

(vi) **Neurotoxicity:** Certain nanoparticles (Ag, Au, TiO<sub>2</sub>) can **cross the blood-brain barrier** and accumulate in brain tissue, potentially causing neurological damage.

(vii) **Ecotoxicology:** Nanoparticles released into the environment accumulate in **aquatic organisms** (fish, invertebrates) and **biomagnify** up the food chain. Ag nanoparticles are toxic to aquatic microorganisms; ZnO damages algae.

###### (viii) Factors Determining Toxicity:

| Factor              | Effect                                   |
|---------------------|------------------------------------------|
| Smaller size        | Greater toxicity (more surface area)     |
| High surface charge | Stronger cell membrane damage            |
| Composition         | Cd, Pb, Ag NPs are highly toxic          |
| Shape               | Needle-like CNTs more toxic than spheres |
| Surface chemistry   | Uncoated NPs more toxic                  |

**Q10.** What do you understand by cadmium-free quantum dots? Discuss the applications of quantum dots. (10 marks) [CHEM 113 | 2021–22]

### Answer

#### Cadmium-Free Quantum Dots and Applications of Quantum Dots

##### Part 1: Cadmium-Free Quantum Dots

**Quantum dots (QDs)** are **semiconductor nanocrystals** (2–10 nm) that exhibit **size-dependent optical properties** due to quantum confinement. Traditional QDs are based on **Cd-containing compounds** (CdSe, CdS, CdTe).

##### Problem with Cd-based QDs:

- Cadmium is a **highly toxic heavy metal** (carcinogenic, nephrotoxic).
- Poses environmental hazards — restricted under **RoHS** (Restriction of Hazardous Substances) regulations in the EU.
- Limits biomedical applications due to cytotoxicity.

**Cadmium-free QDs:** These are alternative semiconductor nanocrystals with comparable optical properties but without cadmium:



| Type           | Examples                                                           |
|----------------|--------------------------------------------------------------------|
| III–V QDs      | InP, InAs, InGaP                                                   |
| I–III–VI QDs   | CuInS <sub>2</sub> , AgInS <sub>2</sub>                            |
| IV QDs         | Si, C (carbon dots)                                                |
| Perovskite QDs | CsPbX <sub>3</sub> (X = Cl, Br, I) — without Cd but may contain Pb |
| Carbon QDs     | Graphene QDs, carbon nanodots                                      |

**InP QDs** are the most commercially successful Cd-free alternative, offering tunable emission across the visible spectrum.

**Part 2: Applications of Quantum Dots**

- (i) **Displays and Lighting (QLED):** QDs are used in **quantum dot displays (QLED TVs)** to produce highly pure, tunable colours with high colour gamut and energy efficiency. The size of QDs controls the emitted colour.
- (ii) **Bioimaging and Medical Diagnostics:** QDs are used as **fluorescent labels** in biological imaging due to:
  - High brightness and **photostability**.
  - Narrow, tunable emission spectra.
  - Ability to image multiple targets simultaneously (multiplexed imaging).
- (iii) **Solar Cells (QDSC):** QDs are used in **quantum dot solar cells** to harvest a broader spectrum of sunlight. By tuning QD size, different wavelengths are absorbed. Multiple exciton generation (MEG) can exceed the Shockley–Queisser efficiency limit.
- (iv) **Drug Delivery:** QDs can be conjugated with drugs and targeting ligands for **targeted drug delivery** and simultaneous imaging (theranostics).
- (v) **LEDs (QLEDs):** QD-based LEDs offer **highly saturated colours** with narrow emission bandwidths, useful in next-generation displays and solid-state lighting.
- (vi) **Photodetectors and Sensors:** QDs are used in **infrared photodetectors** and chemical/biological sensors due to their size-tunable absorption.
- (vii) **Quantum Computing:** Individual QDs can act as **spin qubits** for quantum information processing.

**Q11.** Define gold number. The gold numbers of A, B, C and D are 0.005, 0.05, 0.5 and 5 respectively. Which of these has the greatest protective action? Explain your answer. **(12 marks)** [CHEM 113 | 2021–22]

Answer

**Gold Number: Definition and Greatest Protective Action**

**Definition of Gold Number:**  
The **gold number** (introduced by Zsigmondy) is defined as:

*“The minimum mass in **milligrams** of a protective colloid required to just prevent the coagulation of 10 mL of a standard **red gold sol** on addition of 1 mL of a 10% NaCl solution.”*

- The gold sol turns from **red to blue** upon coagulation.
- A **lower gold number** means **greater protective power** (less material is needed to prevent coagulation).

**Given Data:**

| Protective Colloid | Gold Number |
|--------------------|-------------|
| A                  | 0.005       |
| B                  | 0.05        |
| C                  | 0.5         |
| D                  | 5           |

**Which Has the Greatest Protective Action?**  
**Colloid A** (gold number = 0.005) has the **greatest protective action**.  
**Explanation:**

- (i) Gold number is an **inverse measure** of protective power. A lower gold number means a smaller mass is needed to protect 10 mL of gold sol.
- (ii) Colloid A requires only **0.005 mg** to prevent coagulation, while colloid D requires **5 mg** — 1000 times more.
- (iii) Colloid A’s molecules are highly effective at **adsorbing onto gold sol particles** and forming a protective layer that prevents electrolyte-induced coagulation.
- (iv) The order of protective power is:

$$A > B > C > D$$

(gold number: 0.005 < 0.05 < 0.5 < 5)

**Note:** Gelatin has a gold number of ~ 0.005–0.01, so colloid A likely resembles gelatin in protective power.

**Q12.** What do you mean by Torus and Carbon nanobuds? How do the colloidal particles acquire electrical charge? **(13 marks)** [CHEM 113 | 2021–22]

### Answer

#### Torus, Carbon Nanobuds and Acquisition of Electrical Charge by Colloidal Particles

##### Part 1: Torus

A **carbon nanotorus** is a **toroidal (doughnut-shaped) carbon nanotube** — a CNT bent into a closed ring. It can be thought of as a CNT whose two ends are joined together.

##### Properties:

- Exhibits **unique magnetic properties**: a large magnetic moment when current circulates in the ring.
- Has **quantised magnetic flux** similar to superconducting rings.
- Predicted to have extremely high **inductance** — useful in nanoscale electronic circuits.
- Properties depend on chirality and radius of the torus.

##### Part 2: Carbon Nanobuds

**Carbon nanobuds** are a **hybrid nanostructure** in which **fullerene molecules (C<sub>60</sub> “buckyballs”)** are covalently bonded to the outer sidewalls of carbon nanotubes.

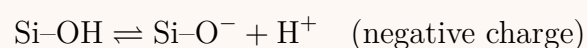
##### Properties:

- Combine properties of both **fullerenes and CNTs**.
- The fullerene buds act as **anchor points** for further functionalisation.
- Show enhanced **field emission** properties compared to pure CNTs.
- Good candidates for **composite materials** and **molecular electronics**.

##### Part 3: How Colloidal Particles Acquire Electrical Charge

Colloidal particles acquire charge by the following mechanisms:

- Selective Adsorption of Ions (Fajans’ Rule):** The most important mechanism. Colloidal particles preferentially adsorb ions that are **common to their own structure** or form insoluble products with their surface ions. Example: AgI adsorbs Ag<sup>+</sup> (positive sol) or I<sup>−</sup> (negative sol) depending on which is in excess.
- Ionisation of Surface Groups:** Particles with ionisable surface groups acquire charge by **dissociation** of those groups. Example: silica particles ionise surface Si–OH groups in water:



Protein colloids acquire charge through ionisation of –COOH and –NH<sub>2</sub> groups.

- Friction (Triboelectric Effect):** Rubbing colloidal particles against each other or against the dispersion medium causes **charge transfer**, resulting in surface charge.
- Isomorphous Substitution:** In clay minerals, substitution of Si<sup>4+</sup> by Al<sup>3+</sup> in the crystal lattice creates a **permanent negative charge** on clay particles.

**Q13.** Aluminium hydroxide forms a positively charged sol. Which of the following ionic substances should be most effective in coagulating the sol and why? NaCl, CaCl<sub>2</sub>, Fe<sub>2</sub>(SO<sub>4</sub>)<sub>3</sub> and K<sub>3</sub>PO<sub>4</sub>. **(12 marks)** [CHEM 113 | 2021–22]

### Answer

#### Coagulation of Positively Charged Al(OH)<sub>3</sub> Sol

##### Background:

Al(OH)<sub>3</sub> forms a **positively charged sol** because Al(OH)<sub>3</sub> particles adsorb Al<sup>3+</sup> ions (common ions) from the medium.

To **coagulate** a positively charged sol, we need **negatively charged (anionic) counter-ions**. According to the **Schulze–Hardy Rule**:

*“The coagulating power of an electrolyte is mainly determined by the **valence of the counter-ion**. Higher valence counter-ions are exponentially more effective at coagulation.”*

The critical coagulation concentration (CCC) is proportional to:

$$\text{CCC} \propto \frac{1}{z^6}$$

where  $z$  is the valence of the counter-ion.

##### Analysis of Given Electrolytes:

Since the sol is **positively charged**, the **anion** of each electrolyte is the counter-ion:

| Electrolyte                  | Counter-ion (anion) | Valence $z$ | Coagulating power |
|------------------------------|---------------------|-------------|-------------------|
| NaCl                         | $\text{Cl}^-$       | 1           | Least             |
| $\text{CaCl}_2$              | $\text{Cl}^-$       | 1           | Least             |
| $\text{Fe}_2(\text{SO}_4)_3$ | $\text{SO}_4^{2-}$  | 2           | Moderate          |
| $\text{K}_3\text{PO}_4$      | $\text{PO}_4^{3-}$  | 3           | Greatest          |

**Answer:**

$\text{K}_3\text{PO}_4$  is the most effective coagulating agent for the  $\text{Al}(\text{OH})_3$  sol.

**Explanation:**

(i)  $\text{K}_3\text{PO}_4$  provides the **trivalent anion**  $\text{PO}_4^{3-}$ , which has the highest counter-ion valence ( $z = 3$ ) among the given electrolytes.

(ii) By the Schulze–Hardy Rule, coagulating power  $\propto z^6$ :

$$\text{Relative power: } z = 1 : z = 2 : z = 3 = 1^6 : 2^6 : 3^6 = 1 : 64 : 729$$

(iii) The  $\text{PO}_4^{3-}$  ions **neutralise the positive surface charge** of  $\text{Al}(\text{OH})_3$  particles most effectively, reducing the zeta potential to near zero and allowing van der Waals forces to cause aggregation.

(iv) Order of coagulating power:

$$\text{K}_3\text{PO}_4 > \text{Fe}_2(\text{SO}_4)_3 > \text{NaCl} \approx \text{CaCl}_2$$

*Note:* NaCl and  $\text{CaCl}_2$  both provide  $\text{Cl}^-$  (monovalent) and are the least effective, though  $\text{CaCl}_2$  provides more  $\text{Cl}^-$  per mole.

**Q14.** Define Tyndall effect. Draw a schematic diagram for the determination of particle size with dynamic light scattering. **(15 marks)**  
[CHEM 113 | 2020–21 (Jan 21)]

**Answer**

### Tyndall Effect and Dynamic Light Scattering (DLS) for Particle Size Determination

#### Part 1: Tyndall Effect

The **Tyndall effect** is the **scattering of light** by colloidal particles suspended in a medium, making the path of the light beam **visible** when viewed from the side.

- Discovered by **John Tyndall (1869)**.
- Occurs because colloidal particles (size 1–1000 nm) are **comparable to or larger than the wavelength of visible light** (400–700 nm), causing **Mie scattering**.
- True solutions do not show Tyndall effect (particles too small —  $< 1$  nm).
- Example:** A laser beam passed through a colloidal gold sol is visible as a bright cone; the same beam in a true NaCl solution is invisible.

**Uses:**

- Distinguish colloids from true solutions.
- Estimate particle size (intensity of scattering  $\propto d^6/\lambda^4$ , Rayleigh).
- Basis of **ultramicroscopy** and **dynamic light scattering**.

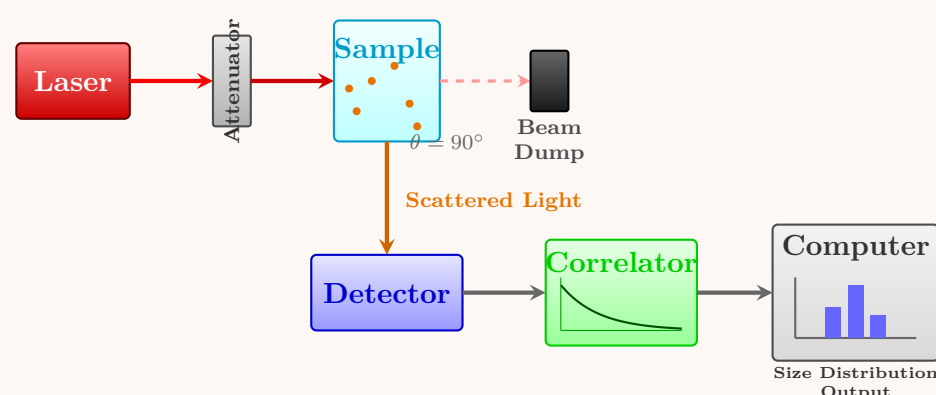
#### Part 2: Dynamic Light Scattering (DLS) for Particle Size Determination

**Principle:** In DLS, a **monochromatic laser beam** is directed at a colloidal sample. Particles undergo **Brownian motion**, causing the scattered light intensity to **fluctuate with time**. The rate of fluctuation is related to the **diffusion coefficient**  $D$  of the particles, from which the **hydrodynamic radius**  $r_h$  is obtained via the **Stokes–Einstein equation**:

$$D = \frac{k_B T}{6\pi\eta r_h} \Rightarrow r_h = \frac{k_B T}{6\pi\eta D}$$

where  $k_B$  = Boltzmann constant,  $T$  = temperature,  $\eta$  = viscosity of the medium.

**Schematic Diagram of DLS Instrument:**



### Steps:

- (i) Laser illuminates the sample with a monochromatic beam.
- (ii) Scattered light is collected at a specific angle  $\theta$  (usually  $90^\circ$  or  $173^\circ$ ).
- (iii) A **correlator** measures the **autocorrelation function** of the intensity fluctuations over time.
- (iv) The decay rate of the correlation function gives the diffusion coefficient  $D$ , from which  $r_h$  is calculated.
- (v) The computer software analyzes the data and outputs the **particle size distribution**.

**Q15.** How can you differentiate the association or condensation method with the disintegration or dispersion method for the preparation of particles with nano dimensions? Explain with suitable examples. **(20 marks)** [CHEM 113 | 2020–21 (Jan 21)]

### Answer

#### Association/Condensation vs Disintegration/Dispersion Methods for Nanoparticle Preparation

##### Overview:

Nanoparticles can be prepared by two fundamentally opposite approaches:

| Bottom-up<br>(Association/Condensation)          | Top-down<br>(Disintegration/Dispersion)     |
|--------------------------------------------------|---------------------------------------------|
| Build up from atoms/molecules into nanoparticles | Break down bulk material into nanoparticles |

#### Part 1: Association/Condensation Method (Bottom-up)

In this approach, **atoms or molecules are assembled** to form nanoparticles. The particles **grow from smaller units** (atomic or molecular level) through nucleation and growth.

##### Key methods:

- (i) **Chemical Reduction:** Metal ions are reduced by a reducing agent to form metal nanoparticles by nucleation and controlled growth.  
*Example:* Au nanoparticles:  
$$\text{HAuCl}_4 + \text{sodium citrate} \xrightarrow{\Delta} \text{Au NPs} + \text{CO}_2 + \text{H}_2\text{O}$$
- (ii) **Sol-gel Method:** Metal alkoxide precursors undergo **hydrolysis and condensation** to form a gel, which is then dried and calcined to obtain oxide nanoparticles.  
*Example:*  $\text{TiO}_2$  nanoparticles from  $\text{Ti}(\text{OC}_2\text{H}_5)_4$  hydrolysis.
- (iii) **Co-precipitation:** Two or more salts are simultaneously precipitated from solution by adjusting pH or adding a precipitating agent.  
*Example:*  $\text{Fe}_3\text{O}_4$  (magnetite) nanoparticles by co-precipitation of  $\text{Fe}^{2+}$  and  $\text{Fe}^{3+}$  in alkali.
- (iv) **Chemical Vapour Deposition (CVD):** Gaseous precursors react at a heated substrate to deposit thin films or nanoparticles.  
*Example:* Growth of carbon nanotubes from hydrocarbon gas decomposition on metal catalyst.
- (v) **Hydrothermal/Solvothermal Synthesis:** Reaction in sealed vessels at high temperature and pressure.  
*Example:* ZnO nanorods, quantum dots.

##### Advantages:

- Good control over **size and shape**.
- Can produce **monodisperse** nanoparticles.
- Scalable for mass production.
- Lower energy consumption.

#### Part 2: Disintegration/Dispersion Method (Top-down)

In this approach, **bulk materials are broken down** into nanoparticles using physical or chemical forces.

##### Key methods:

- (i) **Ball Milling:** Bulk material is ground in a high-energy ball mill. Impact and shear forces reduce particle size to nanoscale.  
*Example:* Preparation of metal alloy nanoparticles (Fe, Al, Ti).  
*Limitation:* Contamination from milling media; broad size distribution.
- (ii) **Mechanical Attrition:** Repeated cold welding, fracturing, and rewelding of particles in a high-energy mill until nanoscale is reached.  
*Example:* Nanocrystalline Cu, Ni powders.
- (iii) **Laser Ablation:** A high-power laser pulse ablates material from a solid target, creating a plume that condenses into nanoparticles.  
*Example:* Au, Ag, and carbon nanoparticles prepared in liquid.

(iv) **Electrochemical Etching/Anodisation:** Electrochemical removal of bulk material produces nanostructured surfaces.  
*Example:* Anodic aluminium oxide (AAO) templates with nanopores.

(v) **Electrical Discharge (Arc Discharge):** Electric arc between carbon electrodes vaporises carbon, which condenses into nanostructures.  
*Example:* Original method for fullerene and CNT synthesis.

#### Advantages:

- Simple and applicable to **any material**.
- Can produce **large quantities**.
- No chemical reagents required (ball milling).

#### Comparison Table:

| Feature           | Bottom-up                   | Top-down                     |
|-------------------|-----------------------------|------------------------------|
| Approach          | Build from atoms/molecules  | Break bulk material          |
| Size control      | Excellent                   | Poor to moderate             |
| Size distribution | Narrow (monodisperse)       | Broad (polydisperse)         |
| Purity            | High                        | May have contamination       |
| Energy            | Low                         | High                         |
| Scale-up          | Moderate                    | Easy                         |
| Examples          | Chemical reduction, sol-gel | Ball milling, laser ablation |

**Q16.** Selective adsorption of ions from solution imparts charge on colloidal particles — justify. **(8 marks)** [CHEM 113 | 2020–21]

#### Answer

##### Selective Adsorption of Ions Imparts Charge on Colloidal Particles

This is the same core concept as Q8 with the specific framing of **selective adsorption**. See also Q8 for additional examples.

##### Justification:

The **primary mechanism** by which colloidal particles acquire charge is the **selective (preferential) adsorption of ions** from the surrounding solution onto the particle surface.

##### Governing Principle — Fajans' Rule:

A colloidal particle preferentially adsorbs:

- Ions that are **common to the particle's own lattice** (potential-determining ions).
- Ions that can form **sparingly soluble compounds** with the particle surface.

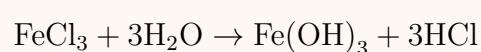
##### Examples with Justification:

##### (i) Silver Iodide (AgI) Sol:

When AgI is precipitated by mixing  $\text{AgNO}_3$  and KI:

- **Excess  $\text{AgNO}_3$ :**  $\text{Ag}^+$  ions (common to AgI lattice) are selectively adsorbed on AgI particles  $\Rightarrow$  **positive sol**.
- **Excess KI:**  $\text{I}^-$  ions (common to AgI lattice) are selectively adsorbed  $\Rightarrow$  **negative sol**.

##### (ii) $\text{Fe}(\text{OH})_3$ Sol: Prepared by hydrolysis of $\text{FeCl}_3$ :



$\text{Fe}(\text{OH})_3$  particles adsorb  **$\text{Fe}^{3+}$  ions** from the medium (common lattice ion)  $\Rightarrow$  **positively charged sol**.

##### (iii) $\text{As}_2\text{S}_3$ Sol: Prepared by passing $\text{H}_2\text{S}$ through $\text{As}_2\text{O}_3$ : $\text{As}_2\text{S}_3$ particles adsorb **$\text{S}^{2-}$ ions** from excess $\text{H}_2\text{S}$ $\Rightarrow$ **negatively charged sol**.

##### (iv) Silica Sol: $\text{SiO}_2$ particles in water adsorb **$\text{OH}^-$ ions** preferentially $\Rightarrow$ **negatively charged sol**.

**Conclusion:** The **sign and magnitude** of the charge on colloidal particles are determined by *which ions* are selectively adsorbed, which in turn depends on the **nature of the colloidal material** and the **composition of the surrounding electrolyte**. This selective adsorption creates the **electrical double layer** that stabilises colloidal dispersions.

**Q17.** Identify and explain the different methods involved in the purification of colloids. **(15 marks)** [CHEM 113 | 2020–21]

#### Answer

##### Methods for Purification of Colloids

##### Why Purification is Needed:



During the preparation of colloids, excess electrolytes, unreacted reagents, and other impurities are present in the dispersion medium. These must be removed because:

- Excess electrolytes reduce the zeta potential and **destabilise** the colloid.
- Impurities alter the **colligative and physical properties** of the sol.

#### Methods of Purification:

- (i) **Dialysis:** The colloidal solution is placed in a bag made of **semipermeable membrane** (cellophane, parchment paper) and immersed in pure water or buffer.
- Small ions and molecules **diffuse out** through the membrane.
  - Large colloidal particles are **retained** inside.
  - Water is frequently changed to maintain a concentration gradient.

*Example:* Purification of blood in kidney dialysis machines.

- (ii) **Electrodialysis:** An **electric field** is applied across the dialysis membrane to **accelerate the removal of ions**. Ions migrate rapidly toward oppositely charged electrodes. Much faster than simple dialysis.

*Example:* Industrial purification of colloidal sol.

- (iii) **Ultrafiltration:** The sol is forced through a **graded ultrafilter membrane** under pressure. Membranes with pore sizes of 1–100 nm are used:

- Ions and small molecules pass through.
- Colloidal particles are retained (ultrafiltrate is discarded).

The retained colloidal particles are re-dispersed in pure solvent.

- (iv) **Gel Filtration (Size Exclusion Chromatography):** The sol is passed through a column packed with **porous gel beads**. Small molecules enter the pores and are retarded; colloidal particles are excluded and elute first.

#### Comparison Table:

| Method          | Principle                  | Speed    | Application      |
|-----------------|----------------------------|----------|------------------|
| Dialysis        | Diffusion through membrane | Slow     | Lab purification |
| Electrodialysis | Electric field + dialysis  | Fast     | Industrial       |
| Ultrafiltration | Pressure + membrane        | Moderate | Concentration    |
| Gel filtration  | Size exclusion             | Moderate | Biochemistry     |

**Q18.** Differentiate the sol-gel method and gel formation method for the preparation of nanoparticles with suitable examples. **(12 marks)**  
[CHEM 113 | 2020–21]

#### Answer

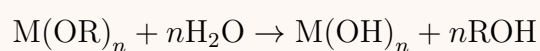
##### Sol-gel Method vs Gel Formation Method for Nanoparticle Preparation

##### Part 1: Sol-Gel Method

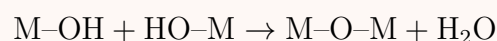
The **sol-gel method** is a **bottom-up (wet chemical)** process for preparing metal oxide nanoparticles and thin films through **hydrolysis and condensation** of molecular precursors (usually metal alkoxides).

##### Steps:

**Step 1: Sol formation:** Metal alkoxide  $M(OR)_n$  is dissolved in solvent and hydrolysed:



**Step 2: Condensation (gelation):** Hydroxide groups condense to form M–O–M bonds, creating a **three-dimensional gel network**:



**Step 3: Aging:** The gel is aged to allow complete cross-linking.

**Step 4: Drying:** Solvent is removed (xerogel by evaporation; aerogel by supercritical drying).

**Step 5: Calcination:** Heat treatment at high temperature produces the final **crystalline nanoparticles**.

##### Examples:

- $TiO_2$  nanoparticles from  $Ti(OC_2H_5)_4$  (tetraethyl orthotitanate)
- $SiO_2$  nanoparticles from TEOS (tetraethylorthosilicate)
- $ZrO_2$ ,  $Al_2O_3$ ,  $ZnO$  nanoparticles

##### Advantages:



- **High purity** products.
- **Compositional homogeneity** at molecular level.
- Easy control of **particle size and porosity**.
- Low processing temperature.

### Part 2: Gel Formation (Hydrogel/Gelation) Method

In this approach, a **gel matrix** is first formed (e.g. hydrogel, polymer gel, or biopolymer gel), and nanoparticles are then **nucleated and grown within the gel matrix**, which acts as a template to **control particle size and prevent aggregation**.

**Steps:**

**Step 1:** A **gel matrix** is prepared (e.g. agarose, gelatin, polyacrylamide, or agar gel).

**Step 2:** Metal ion precursors are **diffused into the gel**.

**Step 3:** A reducing agent or precipitating agent is added to **nucleate and grow nanoparticles** within the gel pores.

**Step 4:** Nanoparticles are **extracted** from the gel by dissolution or calcination.

**Examples:**

- Ag and Au nanoparticles in agarose gel.
- CdS quantum dots in gelatin matrix.
- Fe<sub>3</sub>O<sub>4</sub> nanoparticles in polyacrylamide.

**Advantages:**

- Gel pores provide a **nanoreactor template** that controls particle size.
- Prevents **agglomeration** during growth.
- Suitable for **bio-compatible** particle synthesis.

**Key Differences:**

| Feature          | Sol-Gel Method         | Gel Formation Method |
|------------------|------------------------|----------------------|
| Precursor        | Metal alkoxide         | Metal salt solution  |
| Product          | Oxide NPs              | Metal/sulfide NPs    |
| Template         | Self-forming M-O-M gel | External gel matrix  |
| Temperature      | High (calcination)     | Low (ambient)        |
| Biocompatibility | Low                    | High                 |
| Application      | Ceramics, catalysis    | Biomedical, sensing  |

**Q19.** Adsorption is an important phenomenon to form the charge on the colloidal particles — justify with examples. **(10 marks)** [CHEM 113 | 2019–20]

### Answer

#### Adsorption Imparts Charge on Colloidal Particles

##### How Adsorption Causes Charge:

Colloidal particles acquire surface charge primarily through the **preferential adsorption of ions** from the surrounding solution onto their surface. This is governed by **Fajans' rules** of adsorption.

##### Fajans' Rules of Ion Adsorption:

- A colloidal particle preferentially adsorbs **ions that are common to its own lattice** (potential-determining ions).
- Among foreign ions, those that form **insoluble or slightly soluble compounds** with the surface ions are preferentially adsorbed.

**Examples:**

- AgI sol prepared in excess AgNO<sub>3</sub>:** When AgI is precipitated in the presence of excess Ag<sup>+</sup> ions:

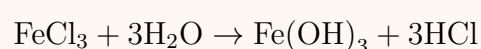
AgI particles adsorb excess Ag<sup>+</sup> ions → positive sol

The Ag<sup>+</sup> ions are common to the AgI lattice and are preferentially adsorbed.

- AgI sol prepared in excess KI:** When AgI is precipitated in excess KI:

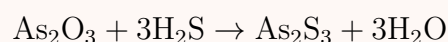
AgI particles adsorb excess I<sup>−</sup> ions → negative sol

- Fe(OH)<sub>3</sub> sol:** Prepared by hydrolysis of FeCl<sub>3</sub>:



Fe(OH)<sub>3</sub> particles adsorb **Fe<sup>3+</sup> ions** (common to the colloid) from the medium → **positively charged sol**.

(iv) **As<sub>2</sub>S<sub>3</sub> sol:** Prepared by passing H<sub>2</sub>S through arsenious acid solution:



As<sub>2</sub>S<sub>3</sub> particles adsorb S<sup>2-</sup> ions from excess H<sub>2</sub>S → **negatively charged sol**.

(v) **Starch sol:** Starch particles adsorb OH<sup>-</sup> ions from water → negatively charged particles.

**Conclusion:** The **selective adsorption** of specific ions onto colloidal particle surfaces is the primary mechanism by which colloidal particles acquire their characteristic electric charge. The sign and magnitude of the charge depend on which ions are adsorbed, which in turn depends on the nature of the colloidal material and the composition of the surrounding electrolyte solution.

**Q20.** Zeta potential is very important in different applications of colloids and nanoparticles. Show the drop in potential in a schematic diagram and explain the potential change from surface to bulk. (20 marks) [CHEM 113 | 2019–20]

### Answer

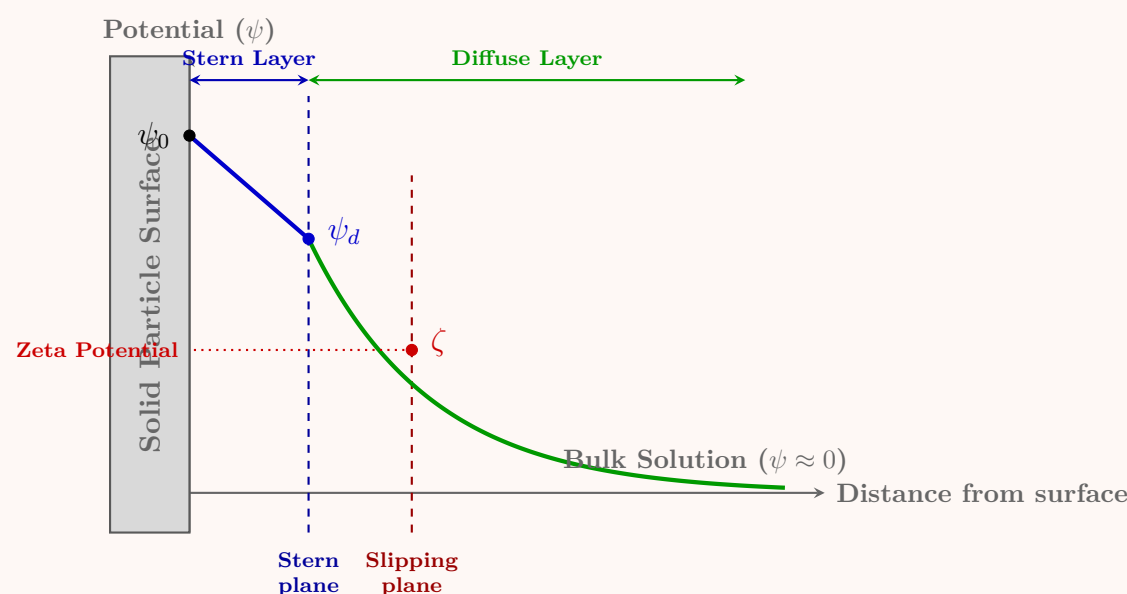
#### Zeta Potential: Potential Drop Diagram and Explanation from Surface to Bulk

##### Part 1: The Electrical Double Layer

When a charged colloidal particle is immersed in an electrolyte solution, an **electrical double layer (EDL)** forms around it:

- Stern Layer (Inner Layer):** A layer of **specifically adsorbed ions** (opposite charge to the particle) tightly bound to the particle surface. The potential drops steeply from the **surface potential**  $\psi_0$  to the **Stern potential**  $\psi_d$  across this layer.
- Diffuse Layer (Gouy–Chapman Layer):** A diffuse cloud of **loosely associated counter-ions** beyond the Stern layer. The potential drops **exponentially** from  $\psi_d$  to zero (bulk solution) across this layer.
- Slipping Plane (Shear Plane):** When the particle moves, it carries the Stern layer and part of the diffuse layer with it. The boundary between the moving part and the stationary bulk solution is the **slipping plane**.
- Zeta Potential ( $\zeta$ ):** The electrical potential at the **slipping plane** is the **zeta potential**. It is the experimentally measurable quantity (via electrophoresis or DLS).

##### Part 2: Schematic Diagram of Potential Drop



##### Part 3: Explanation of Potential Change from Surface to Bulk

- At the particle surface:** The potential is maximum at  $\psi_0$  (**surface potential**), determined by the charge-determining ions adsorbed on the surface.
- Across the Stern layer (surface → Stern plane):** The potential drops **steeply and linearly** from  $\psi_0$  to  $\psi_d$  due to the tight binding of the inner Helmholtz layer ions.
- At the slipping plane:** The potential equals the **zeta potential**  $\zeta$ , which is slightly lower than  $\psi_d$ .  $\zeta$  is the practically important quantity.
- Across the diffuse layer (Stern plane → bulk):** The potential decreases **exponentially** with distance according to:

$$\psi(x) = \psi_d e^{-\kappa x}$$

where  $\kappa^{-1}$  is the **Debye length** (thickness of diffuse layer).

- In bulk solution:** The potential approaches **zero** — the solution is electrically neutral far from the particle.

##### Significance of Zeta Potential:

- $|\zeta| > 30$  mV: colloid is **stable** (strong repulsion).
- $|\zeta| < 10$  mV: colloid is **unstable** (prone to coagulation).
- Used in drug delivery, paint formulation, wastewater treatment, and food science.

Q21. Why are the properties of nanoparticles different from conventional materials? (15 marks)

[CHEM 113 | 2019–20]

Answer

### Why Are Properties of Nanoparticles Different from Conventional Materials?

The unique properties of nanoparticles arise from two fundamental effects:

(I) Quantum size effects

(II) Large surface-to-volume ratio

#### (I) Quantum Size Effects

When the size of a material is reduced to the **nanoscale** (1–100 nm), it becomes comparable to the **de Broglie wavelength** of electrons. Quantum mechanical effects become dominant:

(i) **Discrete Energy Levels:** In bulk, energy levels are **continuous bands**. In nanoparticles, energy levels become **discrete** (quantised), like in atoms and molecules. This changes **optical, electrical, and magnetic** properties fundamentally.

(ii) **Optical Properties:** The **bandgap** of semiconductor nanoparticles (quantum dots) **increases** as size decreases. This causes **blue shift** in absorption/ emission spectra — CdSe quantum dots emit different colours depending on size (2 nm: blue, 7 nm: red).

(iii) **Electrical Properties:** Bulk gold is an excellent conductor. Gold nanoparticles (< 3 nm) become **semiconducting or insulating** due to discrete energy levels and Coulomb blockade.

(iv) **Magnetic Properties:** Bulk iron is ferromagnetic. Iron nanoparticles (< 20 nm) become **superparamagnetic** — they show magnetisation only in an external field and have no remanence.

#### (II) Large Surface-to-Volume Ratio

As particle size decreases, the fraction of atoms at the **surface** increases dramatically:

$$\frac{S}{V} = \frac{6}{d} \quad (\text{for a sphere of diameter } d)$$

| Particle size | S/V (nm <sup>-1</sup> ) | % surface atoms |
|---------------|-------------------------|-----------------|
| 10 nm         | 0.60                    | ~ 20%           |
| 5 nm          | 1.20                    | ~ 40%           |
| 2 nm          | 3.00                    | ~ 80%           |
| 1 nm          | 6.00                    | ~ 99%           |

Surface atoms have **fewer neighbouring atoms** and higher **surface energy**, leading to:

(i) **Enhanced chemical reactivity:** More active sites for catalysis. Au nanoparticles are excellent catalysts; bulk Au is inert.

(ii) **Lower melting point:** Gold nanoparticles (2 nm) melt at ~ 300°C vs bulk Au melting point of 1064°C.

(iii) **Enhanced mechanical properties:** Increased hardness and strength in nanostructured metals (Hall–Petch effect).

(iv) **Colour changes:** Au nanoparticles appear **red/purple** (not gold) due to surface plasmon resonance — a purely surface-related optical phenomenon.

Summary Table:

| Property           | Bulk          | Nano                   |
|--------------------|---------------|------------------------|
| Colour (Au)        | Gold yellow   | Red/purple             |
| Melting point      | High (fixed)  | Size-dependent (lower) |
| Magnetism (Fe)     | Ferromagnetic | Superparamagnetic      |
| Catalytic activity | Low           | Very high              |
| Electrical (Au)    | Conductor     | Semiconductor (<3 nm)  |
| Optical (CdSe)     | Fixed bandgap | Tunable bandgap        |

Q22. Explain that the color of gold nanoparticles can be tuned by controlling the aspect ratio of size. (10–7 marks)

[CHEM 113 | 2017–18; 2016–17]

Answer

### Colour of Gold Nanoparticles Tuned by Aspect Ratio

#### Why Gold Nanoparticles are Coloured:

Gold nanoparticles (Au NPs) exhibit **vivid colours** due to **Localised Surface Plasmon Resonance (LSPR)**: free electrons in the Au nanoparticle **collectively oscillate** in resonance with the electromagnetic field of incident light. The frequency of LSPR determines which wavelength of visible light is **absorbed**, and the complementary colour is observed.

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**Effect of Aspect Ratio on Colour:**

The **aspect ratio** is defined as:

$$\text{Aspect ratio} = \frac{\text{length}}{\text{width (or diameter)}}$$

For **spherical Au NPs** (aspect ratio = 1):

- LSPR peak  $\approx 520$  nm (green absorbed).
- Observed colour: **red/ruby**.
- Small spheres ( $< 20$  nm): bright red.
- Larger spheres (50–100 nm): orange to blue-grey as peak red-shifts.

For **Au nanorods** (aspect ratio  $> 1$ ):

- Two LSPR modes:
  - a) **Transverse mode:**  $\sim 520$  nm (nearly fixed).
  - b) **Longitudinal mode:** shifts to longer wavelengths (red-shifts) as aspect ratio increases.
- By tuning aspect ratio, absorption can be shifted from **visible to near-infrared**:

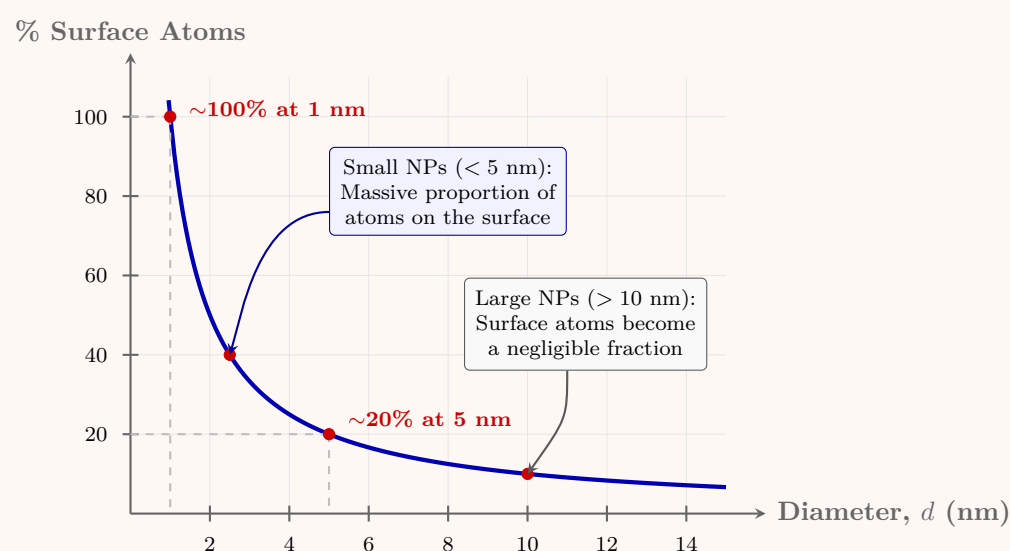
| Aspect Ratio | LSPR Peak (nm) | Colour          |
|--------------|----------------|-----------------|
| 1 (sphere)   | $\sim 520$     | Red             |
| 2            | $\sim 650$     | Blue/violet     |
| 3            | $\sim 750$     | Near-IR         |
| 4            | $\sim 850$     | Near-IR         |
| 5            | $\sim 950$     | NIR (invisible) |

**Physical Explanation:** A longer nanorod has a longer **oscillation path** for the electrons along the longitudinal axis. This requires a **lower frequency** (longer wavelength) electromagnetic field to excite the plasmon resonance, causing the red-shift. By controlling the synthesis conditions (seed size, CTAB concentration,  $\text{AgNO}_3$  amount), aspect ratio can be precisely tuned.

**Applications:**

- **Biomedical imaging and photothermal therapy** (NIR-absorbing nanorods penetrate tissue).
- **SERS** (surface-enhanced Raman spectroscopy).
- **Sensors** with tunable colour response.

**Q23.** Draw a graph showing the approximate percentage of surface atoms as a function of size of nanocrystal and explain that surface energy of nanomaterials increases with decrease in size. **(10–7 marks)** [CHEM 113 | 2017–18; 2016–17]

**Answer**
**Surface Atoms vs Nanocrystal Size and Surface Energy of Nanomaterials**
**Part 1: Graph of % Surface Atoms vs Size**

**Part 2: Surface Energy Increases with Decreasing Size**
**Why surface atoms have higher energy:**

Surface atoms have **fewer neighbouring atoms** compared to bulk atoms. They have **unsatisfied (dangling) bonds**, which means they are in a higher energy state.

The **total surface energy** of a particle is:

$$E_{\text{surface}} = \gamma \times A$$

where  $\gamma$  = surface energy per unit area and  $A$  = total surface area.

For a sphere of diameter  $d$ :

$$A = \pi d^2 \quad \text{and} \quad V = \frac{\pi d^3}{6} \quad \Rightarrow \quad \frac{A}{V} = \frac{6}{d}$$

As  $d$  decreases:

- $A/V$  increases sharply.
- More atoms are located at the surface (as shown in the graph).
- Each surface atom has higher energy than a bulk atom.
- Therefore, the **total surface energy per unit volume increases** dramatically.

**Consequences:**

- **Lowered melting point:** High surface energy destabilises the crystal, lowering the melting point (e.g. 2 nm Au NP melts at  $\sim 300^\circ\text{C}$  vs  $1064^\circ\text{C}$  for bulk Au).
- **Enhanced reactivity:** High surface energy drives chemical reactions and catalysis.
- **Tendency to aggregate:** Nanoparticles spontaneously aggregate to reduce surface energy (minimise  $A$ ).
- **Sintering at lower temperatures:** High surface energy provides the driving force for sintering and grain growth.

**Q24.** What is the dimensional classification of nanomaterials? Give examples and draw structures of each type. **(5 marks)** [CHEM 113 | 2017–18]

### Answer

#### Dimensional Classification of Nanomaterials

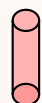
Nanomaterials are classified based on the number of dimensions that are **confined to the nanoscale** ( $< 100 \text{ nm}$ ):

- (i) **0D (Zero-Dimensional):** All three dimensions are in the nanoscale. Particles are **nanoscale in all directions**.  
*Examples:* Quantum dots, metal nanoparticles (Au, Ag), fullerenes ( $\text{C}_{60}$ ), nanospheres.



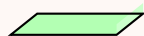
0D (sphere)

- (ii) **1D (One-Dimensional):** Two dimensions are in the nanoscale; one dimension is large (extended in one direction).  
*Examples:* Carbon nanotubes (CNTs), nanowires (ZnO, Si), nanorods, nanofibers.



1D (nanotube/rod)

- (iii) **2D (Two-Dimensional):** One dimension is in the nanoscale (thickness); the other two are large (extended in a plane).  
*Examples:* Graphene,  $\text{MoS}_2$  sheets, nanoplates, nanodiscs, thin films, clay platelets.



2D (nanosheet)

- (iv) **3D (Three-Dimensional / Bulk Nanostructured):** All three dimensions are large, but the material has a **nanoscale internal architecture** (e.g. nanoporous materials, nanocrystalline bulk metals, nanocomposites).  
*Examples:* Nanoporous silica, bulk nanocrystalline metals, zeolites.

**Summary Table:**

| Class | Nano dims. | Free dims. | Examples                 |
|-------|------------|------------|--------------------------|
| 0D    | 3          | 0          | QDs, NPs, fullerenes     |
| 1D    | 2          | 1          | CNTs, nanowires          |
| 2D    | 1          | 2          | Graphene, nanoplates     |
| 3D    | 0          | 3          | Nanocomposites, zeolites |

**Q25.** Discuss the stability of colloids. How can colloids be removed from solutions? **(10 marks)**

[CHEM 113 | 2017–18]



Answer

Stability of Colloids and Removal of Colloids from Solutions

Part 1: Stability of Colloids

The stability of a colloidal dispersion refers to its ability to **resist coagulation** and remain homogeneously dispersed over time. Two types of stability exist:

(i) **Kinetic Stability:** Colloidal particles undergo **Brownian motion** — random thermal movement that prevents them from settling under gravity. This keeps the dispersion homogeneous for an extended period.

(ii) **Thermodynamic (Electrostatic) Stability:** The most important factor. Colloidal particles carry **surface charges** (from ion adsorption or ionisation), creating **electrostatic repulsion** between particles. This repulsion prevents particles from approaching close enough to coagulate.

According to **DLVO theory** (Derjaguin–Landau–Verwey–Overbeek):

$$V_{\text{total}} = V_{\text{repulsion}} + V_{\text{attraction}}$$

•  **$V_{\text{repulsion}}$ :** Electrostatic double layer repulsion (decreases exponentially with distance).

•  **$V_{\text{attraction}}$ :** Van der Waals attraction (decreases with distance as  $\sim 1/r^6$ ).

When  $V_{\text{repulsion}} > V_{\text{attraction}}$ : colloid is **stable**. When repulsion is reduced (e.g. by adding electrolyte), colloid **coagulates**.

(iii) **Steric Stability:** Adsorbed polymer layers on colloidal particles provide **steric repulsion**, preventing particle aggregation.

(iv) **Solvation Stability:** Lyophilic colloids are stabilised by **solvation shells** (solvent molecules tightly bound to the particle surface), which prevent close approach.

Part 2: Removal (Coagulation) of Colloids from Solutions

Colloids can be removed by **destabilising** them and causing particles to aggregate and settle. Methods include:

(i) **Addition of Electrolytes (Coagulation):** Adding electrolytes **compresses the diffuse double layer** (reduces zeta potential). When  $|\zeta| < 10$  mV, van der Waals attraction dominates and particles aggregate. Higher valence counter-ions are more effective (Schulze–Hardy Rule):

$$\text{CCC} \propto \frac{1}{z^6}$$

Example: Alum  $[\text{Al}_2(\text{SO}_4)_3]$  is used in water treatment to coagulate suspended particles.

(ii) **Mutual Coagulation (Mixing Opposite Sols):** Mixing a **positively charged sol** with a **negatively charged sol** causes mutual neutralisation of charges and coagulation.

Example: Mixing  $\text{As}_2\text{S}_3$  (negative) with  $\text{Fe}(\text{OH})_3$  (positive) sol.

(iii) **Heating:** Increasing temperature disrupts the **solvation layer** and increases the kinetic energy of particles, allowing them to overcome the electrostatic barrier and coagulate.

Example: Boiling destabilises protein sols (denaturation + coagulation).

(iv) **Centrifugation (Ultracentrifugation):** High-speed centrifugation applies a strong centrifugal force, overcoming Brownian motion and causing colloidal particles to **sediment**.

Example: Used in laboratory to isolate colloidal particles and nanoparticles.

(v) **Electrophoretic Coagulation (Cottrell Process):** In industrial applications (e.g. removing smoke and dust from industrial gases), colloidal particles are passed through a high-voltage electric field. Particles migrate to the electrodes and are **discharged and deposited**.

Example: Cottrell precipitator for removing fly ash from factory chimneys.

Summary Table:

| Removal Method        | Mechanism                   |
|-----------------------|-----------------------------|
| Electrolyte addition  | Double layer compression    |
| Mutual coagulation    | Charge neutralisation       |
| Heating               | Solvation disruption        |
| Ultracentrifugation   | Gravitational sedimentation |
| Cottrell precipitator | Electrophoretic discharge   |

Q26. What is the length scale of nanomaterials? Why is the length scale of nanosubstances fascinating in material science? (7 marks)  
[CHEM 113 | 2016–17]

Answer

Length Scale of Nanomaterials and Its Fascination in Material Science

Part 1: Length Scale of Nanomaterials

Nanomaterials are defined as materials having at least one dimension in the range of **1 to 100 nanometres** (nm):

$$1\text{ nm} = 10^{-9}\text{ m} = 10\text{ \AA}$$

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To appreciate this scale:

| Object                 | Size             |
|------------------------|------------------|
| Hydrogen atom diameter | ~ 0.1 nm         |
| DNA double helix width | ~ 2 nm           |
| Protein molecule       | ~ 5–50 nm        |
| Nanomaterial range     | <b>1–100 nm</b>  |
| Virus                  | ~ 20–300 nm      |
| Bacterium              | ~ 1000 nm (1 μm) |
| Human hair diameter    | ~ 80,000 nm      |

The nanoscale sits between the **atomic/molecular scale** and the **macroscopic scale**, making it a unique regime where both quantum mechanics and classical physics play important roles.

Part 2: Why the Nanoscale is Fascinating in Material Science

- (i) **Quantum Confinement Effects:** When a material is reduced to the nanoscale, its electrons are **spatially confined**. The energy levels become **discrete and size-dependent**, unlike the continuous bands of bulk materials. This gives rise to entirely new **optical, electrical, and magnetic properties** that can be **tuned** simply by changing particle size. Example: CdSe quantum dots emit different colours depending on size.

(ii) **Enormous Surface-to-Volume Ratio:** At the nanoscale, the fraction of atoms at the surface increases dramatically (~ 99% at 1 nm). Surface atoms have **unsatisfied bonds and higher energy**, making nanomaterials **highly reactive** and useful as catalysts.

$$\frac{S}{V} = \frac{6}{d} \Rightarrow \text{doubles when } d \text{ halves}$$

(iii) **New and Unexpected Properties:** Materials at the nanoscale exhibit properties completely different from their bulk counterparts:

  - Bulk gold is **yellow and inert**; gold nanoparticles are **red/purple and catalytically active**.
  - Bulk carbon (graphite) is soft; carbon nanotubes are **100 times stronger than steel**.
  - Bulk iron is ferromagnetic; iron nanoparticles (< 20 nm) are **superparamagnetic**.

(iv) **Bridge Between Atoms and Bulk:** The nanoscale is the regime where **matter transitions** from atomic/molecular behaviour to bulk behaviour. Understanding this transition is fundamentally important in materials science.

(v) **Multidisciplinary Applications:** The unique nanoscale properties enable revolutionary applications in **medicine** (drug delivery, cancer therapy), **electronics** (transistors, memory), **energy** (solar cells, batteries), **environment** (catalysis, water treatment), and **defence**.

Q27. How can surface curvature of a nanostructure be used to control the solubility of materials? (7 marks) [CHEM 113 | 2016–17]

Answer

**Surface Curvature of Nanostructures and Control of Solubility**

**The Kelvin (Thomson) Equation:**  
The relationship between surface curvature and solubility is described by the **Kelvin (Thomson) equation**:

$$\ln \frac{S_r}{S_\infty} = \frac{2\gamma V_m}{RT r}$$

where:

- $S_r$  = solubility of a particle of radius  $r$
- $S_\infty$  = solubility of a flat surface (bulk material,  $r \rightarrow \infty$ )
- $\gamma$  = surface (interfacial) energy
- $V_m$  = molar volume of the solid
- $R$  = gas constant,  $T$  = temperature
- $r$  = radius of curvature of the particle

**Physical Explanation:**

(i) A **highly curved (small)** nanoparticle has a large surface-to-volume ratio. Surface atoms are in a **higher energy state** due to fewer neighbours and unsatisfied bonds.

(ii) This excess surface energy makes it **thermodynamically easier** for atoms to leave the surface and enter the solution (dissolve), compared to a flat surface.

(iii) As the radius  $r$  **decreases** (more curvature),  $S_r$  **increases**:

$$r \downarrow \Rightarrow \frac{2\gamma V_m}{RT r} \uparrow \Rightarrow S_r \uparrow$$

- (iv) Conversely, a **large nanoparticle** (lower curvature) has solubility approaching that of bulk material.

#### Consequences and Applications:

- (i) **Ostwald Ripening:** In a dispersion of particles of different sizes, smaller particles (higher solubility) dissolve and larger particles (lower solubility) grow. This drives **particle coarsening** over time. Understanding curvature effects helps design stable nanoparticle dispersions.
- (ii) **Drug Solubility Enhancement:** Reducing drug particles to the nanoscale **dramatically increases their solubility**, improving bioavailability of poorly soluble drugs. Example: Nanonisation of poorly soluble drugs (e.g. paclitaxel).
- (iii) **Nanoporous Materials:** In nanopores (concave curvature,  $r < 0$ ), the Kelvin equation predicts **decreased vapour pressure**, causing **capillary condensation** at pressures below the normal saturation pressure.
- (iv) **Crystal Growth Control:** By controlling nanoparticle size (and hence curvature), the **nucleation and growth** of crystals can be precisely managed.

**Summary:** Surface curvature is a powerful tool in nanoscience: **smaller radius** (higher curvature) means **higher surface energy** and **greater solubility**. By engineering nanoparticle size and shape, solubility can be **tuned over several orders of magnitude**.

**Q28.** How can cancer cells be diagnosed by nanomaterials? What are the health hazards associated with nanomaterials? **(7 marks)**  
[CHEM 113 | 2016–17]

#### Answer

##### Cancer Diagnosis by Nanomaterials and Health Hazards of Nanomaterials

##### Part 1: Cancer Cell Diagnosis by Nanomaterials

Nanomaterials offer several strategies for **early and precise detection** of cancer cells:

- (i) **Quantum Dot (QD) Fluorescent Labelling:** Quantum dots (CdSe, InP) are conjugated with **cancer-specific antibodies or ligands**. When injected into the body, they selectively bind to **tumour biomarkers** (e.g. HER2, PSA) on cancer cell surfaces. Under UV excitation, QDs emit **bright, size-tunable fluorescence**, enabling **multiplexed imaging** of multiple cancer markers simultaneously.
- (ii) **Gold Nanoparticle-Based Sensors:** Au nanoparticles functionalised with **DNA aptamers or antibodies** can detect cancer biomarkers with **extreme sensitivity** via:
  - **LSPR shift** (colour change when target binds).
  - **SERS** (surface-enhanced Raman spectroscopy) for molecular fingerprinting.
  - **Aggregation-based colorimetric** sensing (red to blue colour change).
- (iii) **Magnetic Nanoparticles (MRI contrast agents):**  $\text{Fe}_3\text{O}_4$  nanoparticles conjugated with cancer-targeting ligands accumulate in tumour tissue and provide **enhanced MRI contrast**, allowing detection of tumours as small as a few millimetres.
- (iv) **Carbon Nanotube Biosensors:** Functionalised CNTs can detect **cancer-related DNA sequences and proteins** with extremely high sensitivity through changes in electrical conductivity when a biomarker binds.
- (v) **Theranostics:** Nanoparticles that combine **diagnosis and therapy** simultaneously. Example: QD-drug conjugates image the tumour and release drug at the site.

##### Part 2: Health Hazards of Nanomaterials

- (i) **Pulmonary Toxicity:** Inhaled nanoparticles ( $\text{TiO}_2$ , carbon black, CNTs) deposit in the **deep lung** (alveoli), cause inflammation, pulmonary fibrosis, and granuloma formation. CNTs show asbestos-like toxicity.
- (ii) **Oxidative Stress (ROS Generation):** Many nanoparticles ( $\text{ZnO}$ ,  $\text{TiO}_2$ , Ag NPs) generate **reactive oxygen species (ROS)**, which oxidise lipids, proteins, and DNA, leading to **cellular damage and apoptosis**.
- (iii) **Genotoxicity:** Some nanoparticles penetrate the nucleus and cause **DNA strand breaks, chromosomal aberrations**, and mutations.
- (iv) **Neurotoxicity:** Certain nanoparticles cross the **blood-brain barrier**, accumulate in neural tissue, and may cause neurological damage.
- (v) **Cardiovascular Effects:** Inhaled ultrafine particles enter the bloodstream and promote **atherosclerosis, inflammation**, and increased risk of heart attack.
- (vi) **Environmental and Ecotoxicology:** Nanoparticles released into water and soil are toxic to aquatic organisms and **bioaccumulate** in food chains. Ag NPs are highly toxic to bacteria and aquatic microorganisms.

**Q29.** Eight types of colloidal dispersions were listed in class. Give four types and give one example for each. Explain how the Tyndall effect is used to distinguish colloids and solutions. **(9 marks)**  
[CHEM 113 | 2015–16]

Answer

Four Types of Colloidal Dispersions and Tyndall Effect

Part 1: Four Types of Colloidal Dispersions

Colloidal dispersions are classified based on the **physical state of the dispersed phase** and **dispersion medium**:

| No. | Dispersed phase | Medium | Type                   | Example                      |
|-----|-----------------|--------|------------------------|------------------------------|
| 1   | Solid           | Liquid | <b>Sol</b>             | Fe(OH) <sub>3</sub> sol, ink |
| 2   | Liquid          | Liquid | <b>Emulsion</b>        | Milk, mayonnaise             |
| 3   | Gas             | Liquid | <b>Foam</b>            | Whipped cream, shaving foam  |
| 4   | Solid           | Gas    | <b>Aerosol (solid)</b> | Smoke, dust in air           |

(Other types include: solid foam, gel, liquid aerosol, solid emulsion, solid sol.)

Part 2: Tyndall Effect to Distinguish Colloids and Solutions

Tyndall Effect: When a **beam of light** is passed through a colloidal dispersion, the beam becomes **visible** when viewed from the side due to **scattering of light by colloidal particles**. This is the **Tyndall effect**.

Why colloids scatter and solutions do not:

- Colloidal particles (1–1000 nm) are **comparable to or larger than the wavelength of visible light** ( $\lambda = 400\text{--}700\text{ nm}$ ), causing **Mie scattering**.
- True solution particles ( $< 1\text{ nm}$ , i.e. ions and molecules) are **far smaller than**  $\lambda$  and scatter light negligibly.

Experimental Distinction:

| System        | Tyndall Effect             | Example                          |
|---------------|----------------------------|----------------------------------|
| Colloid       | <b>Visible</b> bright cone | Gold sol, milk                   |
| True solution | Not visible                | NaCl solution, CuSO <sub>4</sub> |

Example: A laser beam passed through **gold sol** shows a bright red cone visible from the side (Tyndall cone). The same beam in **distilled water** is invisible.

**Q30.** One of the first computers, ENIAC, weighing 30 tons and occupying a 15 m × 9 m room, was built in the 1940s. Explain how nanochemistry helped the evolution of computers to their current form. **(8 marks)** [CHEM 113 | 2015–16]

Answer

Nanochemistry and the Evolution of Computers

ENIAC (1940s): The first general-purpose electronic computer, ENIAC, used **vacuum tubes**:

- Weight: 30 tons
- Size: 15 m × 9 m
- Contained 18,000 vacuum tubes
- Consumed 150 kW of power
- Clock speed: ~ 100 kHz

How Nanochemistry Drove the Evolution:

(i) **Transistor Revolution (1947 onward):** Vacuum tubes were replaced by solid-state **transistors** made from semiconductor materials (Si, Ge). Nanochemistry enabled **precise doping** of semiconductors at the atomic scale, creating p-n junctions. Transistors are smaller, faster, and use far less power.

(ii) **Moore’s Law and Miniaturisation:** Moore’s Law (1965) predicted that the number of transistors on a chip **doubles every** ~ 2 years. This was achieved through **photolithography** and **chemical vapour deposition (CVD)** — nanochemical processes that pattern features at the nanoscale.

(iii) **Modern Transistor Sizes:** Modern processors use transistors with **gate lengths of 2–5 nm** (Intel, TSMC). At this scale, quantum effects dominate and **nanoscale chemical engineering** is essential.

(iv) **Carbon Nanotube Transistors:** CNT-based transistors (1 nm diameter) are being developed to replace Si transistors, offering higher speed and lower power.

(v) **Copper Interconnects (Nanowires):** Cu nanowires of ~ 10–20 nm width connect transistors on modern chips, requiring **atomic-level control** of deposition and etching.

(vi) **Flash Memory (Nanocrystal Floating Gates):** Modern flash memory uses **nanocrystals** (Si QDs) as floating gates to store charge, enabling **high density data storage** in tiny chips.

(vii) **Quantum Computing:** Emerging quantum computers use **quantum dots, superconducting Josephson junctions, and topological nanomaterials** as qubits, enabled entirely by nanochemistry.

Result: A modern smartphone chip ( $< 1\text{ cm}^2$ ) contains ~ 10 **billion transistors** and processes billions of operations per second — a transformation made possible entirely by **nanochemical control of materials at the atomic scale**.

- Q31.** “Everything when miniaturized to the sub-100-nanometer scale has new properties, regardless of what it is.” (i) List four examples that support this statement. (ii) What are the microscopes that scientists use to see at sub-100-nanometer scale? **(10 marks)**  
[CHEM 113 | 2015–16]

### Answer

#### New Properties at Sub-100-nm Scale and Microscopes for Nanoscale Imaging

##### Part 1: Four Examples Supporting the Statement

*“Everything when miniaturized to the sub-100-nanometer scale has new properties, regardless of what it is.”*

- (i) **Gold Nanoparticles:** Bulk gold is **yellow, inert, and opaque**. Gold nanoparticles (< 100 nm) are **red/purple** (due to LSPR), **catalytically active** (CO oxidation at room temperature), and exhibit **size-tunable optical properties**. The colour shifts from red (20 nm) to blue (100 nm) with increasing size.
- (ii) **Carbon (CNTs vs Graphite):** Bulk graphite is a **soft, grey conductor** used in pencils. Carbon nanotubes (~ 1–50 nm diameter) are **100 times stronger than steel**, have **ballistic electrical conductivity**, and thermal conductivity exceeding diamond. Graphene (1 atom thick, ~ 0.3 nm) is the **strongest material ever measured**.
- (iii) **Iron Nanoparticles (Superparamagnetism):** Bulk iron is **ferromagnetic** with permanent magnetisation. Iron nanoparticles (< 20 nm) become **superparamagnetic** — they exhibit magnetisation only in an external field and have **zero remanence**, making them ideal for MRI contrast agents and targeted drug delivery.
- (iv) **CdSe Quantum Dots:** Bulk CdSe has a **fixed bandgap** and emits at one colour. CdSe quantum dots emit **different colours depending on size**: 2 nm → blue, 4 nm → green, 7 nm → red. This tunability is impossible in bulk and arises purely from quantum confinement at the nanoscale.

##### Part 2: Microscopes for Sub-100-nm Imaging

- (i) **Transmission Electron Microscope (TEM):** Electrons (shorter wavelength than light) are transmitted through an ultra-thin sample. Resolution ~ 0.1 nm. Provides **atomic-resolution images** of nanoparticle interiors.
- (ii) **Scanning Electron Microscope (SEM):** Focused electron beam scans the surface of the sample; secondary electrons are detected. Resolution ~ 1–5 nm. Provides **surface topography** images.
- (iii) **Scanning Tunnelling Microscope (STM):** A sharp metal tip scans the surface at ~ 0.5 nm distance; quantum tunnelling current is measured. Resolution ~ 0.1 nm. Can **image and manipulate individual atoms**. (Invented by Binnig and Rohrer, 1981.)
- (iv) **Atomic Force Microscope (AFM):** A sharp tip on a cantilever scans the surface, measuring van der Waals forces. Resolution ~ 0.1–1 nm. Works on **non-conducting samples** (unlike STM) and can image biological molecules in liquid.

- Q32.** There are two general approaches to the synthesis of nanomaterials. Compare these two approaches and give the pros and cons. **(8 marks)**  
[CHEM 113 | 2015–16]

### Answer

#### Two General Approaches to Nanomaterial Synthesis: Comparison, Pros and Cons

The two general approaches are:

- (I) **Top-down (Disintegration/Dispersion)**
- (II) **Bottom-up (Association/Condensation)**

##### (I) Top-Down Approach

**Principle:** Bulk material is **broken down** into nanoscale pieces by physical or chemical means.

**Methods:** Ball milling, laser ablation, lithography, arc discharge, mechanical attrition.

**Pros:**

- Simple and applicable to **any material**.
- Can produce **large quantities**.
- **Well-established** techniques (e.g. lithography in semiconductor industry).
- No need for chemical reagents (ball milling).

**Cons:**

- **Poor control** over size and shape.
- **Broad size distribution** (polydisperse).
- **Surface defects** introduced by mechanical processing.
- **Contamination** from milling media.
- **High energy consumption**.
- Difficult to achieve sizes < 10 nm.



(II) Bottom-Up Approach

**Principle:** Atoms or molecules are **assembled** into nanostructures through chemical or physical processes.  
**Methods:** Chemical reduction, sol-gel, CVD, hydrothermal synthesis, self-assembly, co-precipitation.

Pros:

- Excellent **control over size, shape, and composition**.
- Produces **narrow size distribution** (monodisperse).
- **High purity** products.
- **Low energy** consumption.
- Can produce particles < 5 nm with atomic precision.
- Enables **surface functionalisation** during synthesis.

Cons:

- **Scale-up** can be challenging.
- Requires **chemical reagents** and careful process control.
- May involve **toxic precursors** (e.g. Cd compounds for QDs).
- Slower production rate.

Comparison Table:

| Feature           | Top-Down     | Bottom-Up          |
|-------------------|--------------|--------------------|
| Direction         | Bulk → nano  | Atoms → nano       |
| Size control      | Poor         | Excellent          |
| Size distribution | Broad        | Narrow             |
| Purity            | Lower        | Higher             |
| Energy use        | High         | Low                |
| Minimum size      | ~ 10–50 nm   | < 1 nm             |
| Example           | Ball milling | Chemical reduction |

**Q33.** What is coagulation? Arsenic(III) sulfide forms a sol with a negative charge. Which of the following ionic substances should be most effective in coagulating the sol: (i) KCl, (ii) MgCl<sub>2</sub>, (iii) Al<sub>2</sub>(SO<sub>4</sub>)<sub>3</sub>, (iv) Na<sub>3</sub>PO<sub>4</sub>? Explain your answer. **(5 marks)** [CHEM 113 | 2014–15]

Answer

Coagulation and Most Effective Coagulant for As<sub>2</sub>S<sub>3</sub> Sol

Definition of Coagulation:

**Coagulation** (or flocculation) is the process by which colloidal particles **aggregate and settle out** from a colloidal dispersion. It occurs when the **electrostatic repulsion** between colloidal particles is sufficiently reduced (e.g. by adding an electrolyte), allowing **van der Waals attractive forces** to dominate and bring particles together.

Analysis:

As<sub>2</sub>S<sub>3</sub> sol carries a **negative charge** (S<sup>2-</sup> ions adsorbed). To coagulate it, we need **positively charged counter-ions (cations)**. By the **Schulze–Hardy Rule**: coagulating power ∝  $z^6$  where  $z$  is the valence of the **counter-ion (cation)**:

| Electrolyte                                           | Counter-ion (cation) | Valence $z$ | Coagulating power |
|-------------------------------------------------------|----------------------|-------------|-------------------|
| (i) KCl                                               | K <sup>+</sup>       | 1           | $1^6 = 1$         |
| (ii) MgCl <sub>2</sub>                                | Mg <sup>2+</sup>     | 2           | $2^6 = 64$        |
| (iii) Al <sub>2</sub> (SO <sub>4</sub> ) <sub>3</sub> | Al <sup>3+</sup>     | 3           | $3^6 = 729$       |
| (iv) Na <sub>3</sub> PO <sub>4</sub>                  | Na <sup>+</sup>      | 1           | $1^6 = 1$         |

Answer:

Al<sub>2</sub>(SO<sub>4</sub>)<sub>3</sub> is the **most effective coagulating agent**.

Explanation:

- (i) Al<sub>2</sub>(SO<sub>4</sub>)<sub>3</sub> provides the **trivalent cation Al<sup>3+</sup>** ( $z = 3$ ), which has the highest counter-ion valence among the given options.  
(ii) By Schulze–Hardy rule, relative coagulating power:

$$\text{Al}^{3+} : \text{Mg}^{2+} : \text{K}^{+} = 3^6 : 2^6 : 1^6 = 729 : 64 : 1$$

- (iii) Al<sup>3+</sup> ions **strongly neutralise the negative surface charge** of As<sub>2</sub>S<sub>3</sub> particles, collapsing the double layer and reducing zeta potential to near zero, enabling rapid coagulation.  
(iv) Order of coagulating power:

$$\text{Al}_2(\text{SO}_4)_3 > \text{MgCl}_2 > \text{KCl} \approx \text{Na}_3\text{PO}_4$$

BUET · CHEM 113

# Chemistry

Section S6

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Phase Rule & Phase Diagram



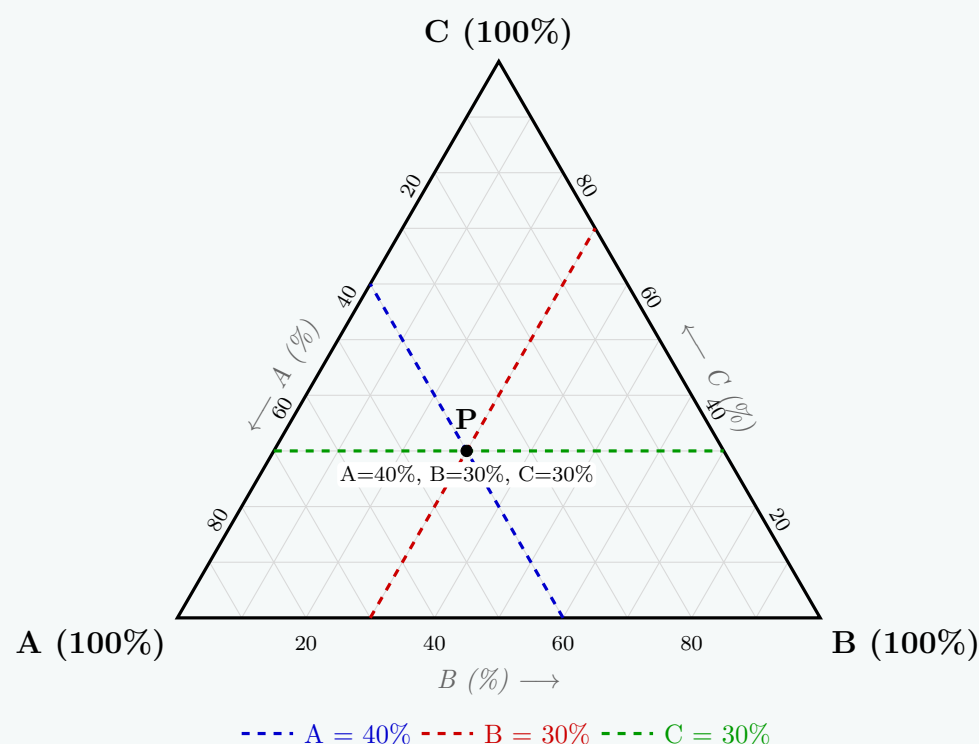
## S6 — Phase Rule & Phase Diagram

**Q1.** Draw a ternary phase diagram and indicate the point of composition of A = 40%, B = 30% and C = 30%. **(4 marks)** [CHEM 113 | 2023–24]

### Answer

#### Ternary Phase Diagram with Composition A = 40%, B = 30%, C = 30%

In a **ternary phase diagram**, each corner of the equilateral triangle represents **100% of one component**. Any point inside the triangle represents a mixture of all three components. The composition is read by drawing lines **parallel to the opposite sides**.



**Reading the point P:** Starting from corner A, move **40%** along the line toward the opposite side (BC) — this fixes A = 40%. Similarly, B = 30% and C = 30% are located from their respective corners. The three dashed lines intersect at point **P**, which represents the composition A = 40%, B = 30%, C = 30%.

**Q2.** Suggest the phases of the following cases: (i) Mixture of N<sub>2</sub>, CO<sub>2</sub>, and O<sub>2</sub>, (ii) NaCl dissolved in water, (iii) Mixture of graphite and diamond, (iv) Alloy of Al and Cu with  $\alpha$ - and  $\beta$ -forms. **(8 marks)** [CHEM 113 | 2023–24]

### Answer

#### Phases of the Given Systems

The **phase rule** defines a phase as a **physically distinct, homogeneous, and mechanically separable** part of a system.

##### (i) Mixture of N<sub>2</sub>, CO<sub>2</sub>, and O<sub>2</sub>:

All three are **gases** that mix completely and uniformly in all proportions. A gaseous mixture is always **one homogeneous phase**.

$$P = 1 \text{ phase (gas)}$$

##### (ii) NaCl dissolved in water:

NaCl completely dissolves in water forming a **single homogeneous liquid solution** (true solution). Only one phase exists.

$$P = 1 \text{ phase (aqueous solution)}$$

If excess undissolved NaCl were present, there would be 2 phases (solution + solid NaCl).

##### (iii) Mixture of graphite and diamond:

Graphite and diamond are **two distinct allotropes of carbon** with different crystal structures, densities, and physical properties. They are **not miscible** and can be mechanically separated. Therefore:

$$P = 2 \text{ phases (graphite + diamond)}$$

##### (iv) Alloy of Al and Cu with $\alpha$ - and $\beta$ -forms:

The alloy exists in **two distinct solid phases**: the  $\alpha$ -phase (Al-rich solid solution) and the  $\beta$ -phase (Cu-rich intermetallic compound CuAl<sub>2</sub>). Each phase has a different composition and crystal structure.

$$P = 2 \text{ phases } (\alpha\text{-phase} + \beta\text{-phase})$$

**Summary:**

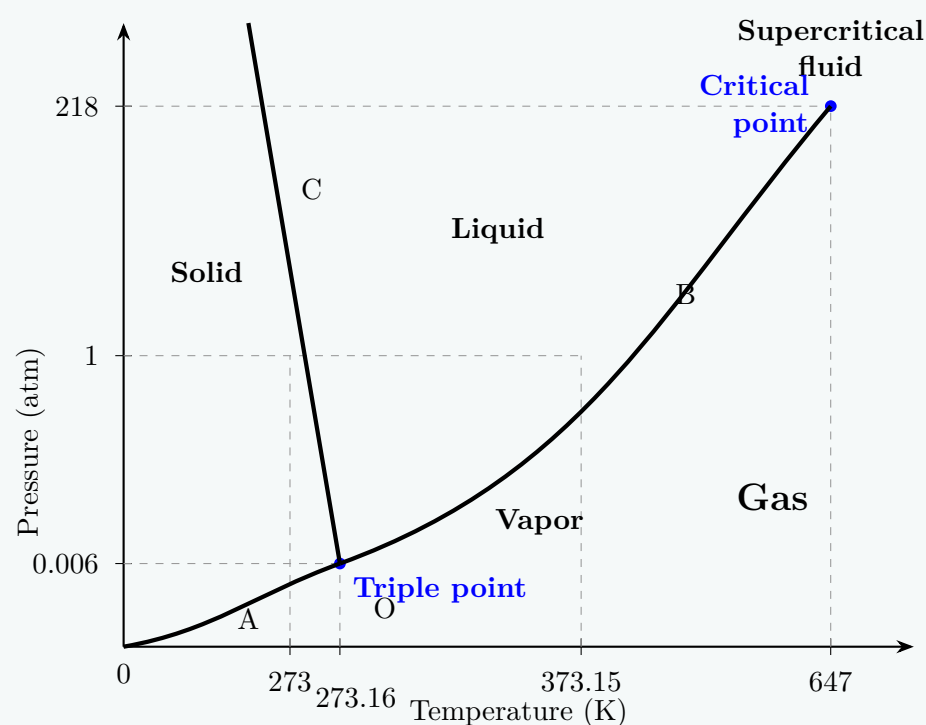
| System                                  | Phases ( $P$ ) | Reason                    |
|-----------------------------------------|----------------|---------------------------|
| $\text{N}_2 + \text{CO}_2 + \text{O}_2$ | 1              | Gases mix completely      |
| NaCl in water                           | 1              | Homogeneous solution      |
| Graphite + diamond                      | 2              | Different allotropes      |
| Al-Cu ( $\alpha + \beta$ )              | 2              | Two distinct solid phases |

**Q3.** Draw the phase diagram of water. Explain its different zones, important points, and different boundary lines using the phase rule and the Clausius-Clapeyron equation. **(13 marks)** [CHEM 113 | 2023–24]

### Answer

#### Phase Diagram of Water: Zones, Points, Boundary Lines, Phase Rule and Clausius-Clapeyron Equation

##### Phase Diagram of Water:



##### Zones (Phase Regions):

- (i) **Ice (Solid) Region:** Low temperature, any pressure. One phase (ice).  $F = C - P + 2 = 1 - 1 + 2 = 2$  (**bivariant**).
- (ii) **Liquid (Water) Region:** Intermediate  $T$  and  $P$ . One phase (liquid).  $F = 2$  (**bivariant**).
- (iii) **Vapour Region:** High  $T$  and/or low  $P$ . One phase (vapour).  $F = 2$  (**bivariant**).

##### Important Points:

- (i) **Triple Point:** At  $0.01^\circ\text{C}$  and  $0.006\text{ atm}$ , all three phases coexist.  $F = 1 - 3 + 2 = 0$  (**invariant**).
- (ii) **Normal Melting Point (NMP):**  $0^\circ\text{C}$  at  $1\text{ atm}$ . Ice and water in equilibrium.
- (iii) **Normal Boiling Point (NBP):**  $100^\circ\text{C}$  at  $1\text{ atm}$ . Water and vapour in equilibrium.
- (iv) **Critical Point (CP):**  $374^\circ\text{C}$ ,  $218\text{ atm}$ . Liquid and vapour become indistinguishable (supercritical fluid).

##### Boundary Lines and Clausius-Clapeyron Equation:

- (i) **OA — Solid-Vapour (Sublimation) Curve:** Solid and vapour coexist.  $F = 1$  (**univariant**).

$$\frac{dP}{dT} = \frac{\Delta H_{sub} \cdot P}{RT^2} \Rightarrow \ln \frac{P_2}{P_1} = -\frac{\Delta H_{sub}}{R} \left( \frac{1}{T_2} - \frac{1}{T_1} \right)$$

Since  $\Delta H_{sub} > 0$ , slope is positive.

- (ii) **OB — Liquid-Vapour (Vaporisation) Curve:** Liquid and vapour coexist.  $F = 1$ .

$$\frac{dP}{dT} = \frac{\Delta H_{vap} \cdot P}{RT^2}$$

Positive slope; ends at critical point.

- (iii) **OC — Solid-Liquid (Fusion) Curve:** Solid and liquid coexist.  $F = 1$ .

$$\frac{dP}{dT} = \frac{\Delta H_{fus}}{T \cdot \Delta V}$$

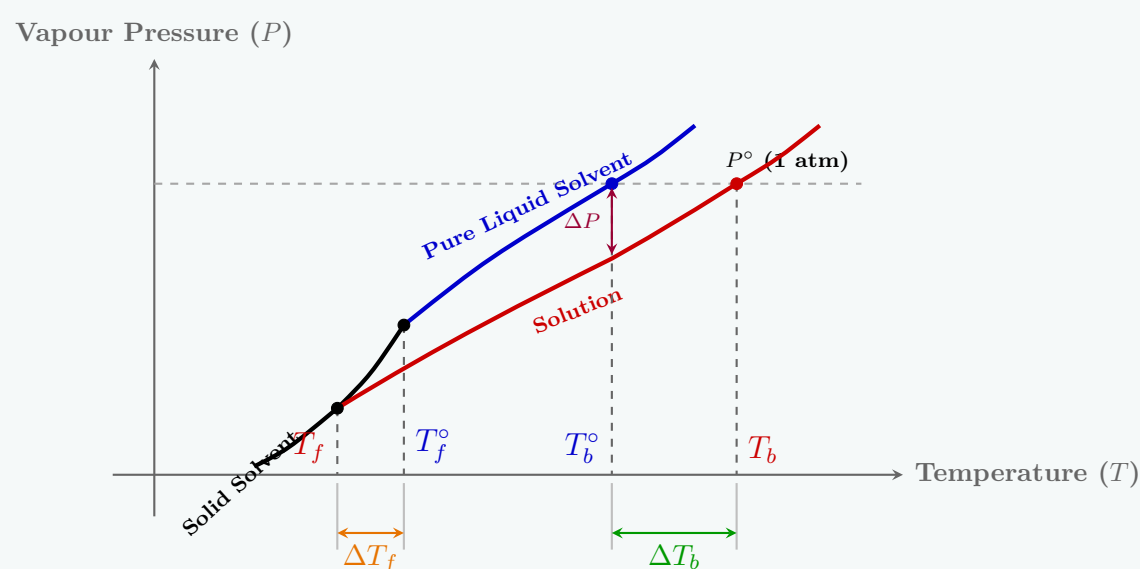
For water,  $\Delta V < 0$  (ice less dense than liquid water)  $\Rightarrow dP/dT < 0 \Rightarrow$  **negative slope** (unique to water): melting point **decreases** with increasing pressure.

**Gibbs Phase Rule:**

$$F = C - P + 2 = 1 - P + 2 = 3 - P$$

| Region/Point        | Phases ( $P$ ) | $F$ | System type |
|---------------------|----------------|-----|-------------|
| Single phase region | 1              | 2   | Bivariant   |
| Two phase boundary  | 2              | 1   | Univariant  |
| Triple point        | 3              | 0   | Invariant   |

**Q4.** Draw the phase diagram illustrating the boiling point elevation and freezing point depression of an aqueous solution and compare it with pure solvent. **(12 marks)** [CHEM 113 | 2022–23]

**Answer**
**Phase Diagram Showing Boiling Point Elevation and Freezing Point Depression of Aqueous Solution**

**Explanation:**

- (i) **Vapour Pressure Lowering (Raoult's Law):** A non-volatile solute lowers the vapour pressure of the solvent at all temperatures:

$$\Delta P = x_B P^\circ$$

The solution curve lies **below** the pure solvent curve at all temperatures.

- (ii) **Freezing Point Depression:** The solution's liquid-vapour curve intersects the solid solvent's curve at a **lower temperature**  $T_f < T_f^\circ$ :

$$\Delta T_f = T_f^\circ - T_f = K_f \cdot m$$

- (iii) **Boiling Point Elevation:** The solution's liquid-vapour curve intersects the 1-atm line ( $P^\circ$ ) at a **higher temperature**  $T_b > T_b^\circ$ :

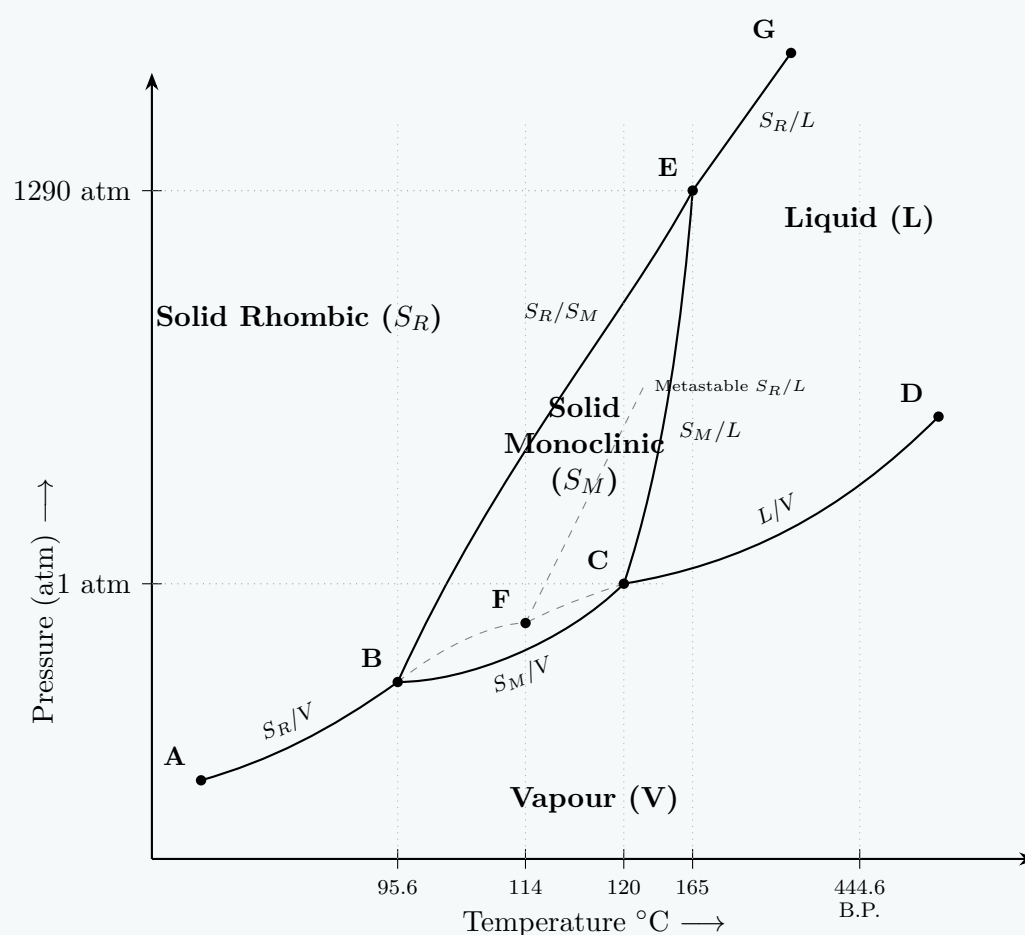
$$\Delta T_b = T_b - T_b^\circ = K_b \cdot m$$

- (iv) Both  $\Delta T_f$  and  $\Delta T_b$  are **colligative properties** proportional to molality  $m$ , independent of the nature of the non-volatile solute.

**Q5.** Draw a well-labeled phase diagram of the sulphur system and discuss its salient features. **(13 marks)** [CHEM 113 | 2022–23]

**Answer**
**Phase Diagram of the Sulphur System**

The sulphur system has **two solid allotropes**: **rhombic** ( $S_R$ ) and **monoclinic** ( $S_M$ ) sulphur, plus liquid and vapour phases. Therefore it is a **one-component, four-phase system** with **three stable triple points** and one metastable triple point.



### Salient Features:

- (i) **Two Solid Allotropes:** Sulphur exists as **rhombic** (stable below 95.6°C) and **monoclinic** (stable 95.6–120°C) solid forms.
- (ii) **Three Stable Triple Points:**
  - **B (95.6°C):**  $S_R + S_M + \text{Vapour}$  coexist.  $F = 1 - 3 + 2 = 0$  (invariant).
  - **C (120°C):**  $S_M + \text{Liquid} + \text{Vapour}$  coexist.  $F = 0$  (invariant). Monoclinic sulphur melts here.
  - **E (165°C, 1290 atm):**  $S_R + S_M + \text{Liquid}$  coexist.  $F = 0$  (invariant).
- (iii) **Main Curves:**
  - **AB:**  $S_R$ –Vapour (sublimation curve of rhombic).
  - **BC:**  $S_M$ –Vapour (sublimation curve of monoclinic).
  - **CD:** Liquid–Vapour (vaporisation curve).
  - **BE:**  $S_R$ – $S_M$  (transition curve; shows effect of pressure on transition temp).
  - **CE:**  $S_M$ –Liquid (fusion curve of monoclinic).
  - **EG:**  $S_R$ –Liquid (fusion curve of rhombic at high pressure).
- (iv) **Transition Temperature (95.6°C):** Below 95.6°C, rhombic is stable. Above 95.6°C, monoclinic is stable. At exactly 95.6°C (Point B), the two are in equilibrium.
- (v) **Metastable Equilibria (Point F):** If rhombic sulphur is heated rapidly past 95.6°C without converting to monoclinic, it follows a **metastable** path (dashed line BF) and melts directly at 114°C. Point **F** is the metastable triple point where metastable  $S_R$ , Liquid, and Vapour coexist.

**Q6.** The melting point curve of ice in the water system has a negative slope. Explain it with the help of the phase diagram of the water system. (10 marks) [CHEM 113 | 2022–23]

### Answer

#### Negative Slope of the Melting Point Curve of Ice

##### The Observation:

In the phase diagram of water, the **solid–liquid boundary** (melting point curve OC) has a **negative slope**: increasing pressure **decreases** the melting point of ice. This is anomalous — for most substances, increased pressure **raises** the melting point.

##### Explanation Using the Clausius–Clapeyron Equation:

The slope of any two-phase equilibrium boundary is:

$$\frac{dP}{dT} = \frac{\Delta H_{\text{fus}}}{T \Delta V_{\text{fus}}}$$

where:

- $\Delta H_{\text{fus}}$  = molar enthalpy of fusion (always  $> 0$ , melting is endothermic).
- $\Delta V_{\text{fus}} = V_{\text{liquid}} - V_{\text{solid}}$  = volume change on fusion.

For water:

- $\Delta H_{\text{fus}} = +6.01 \text{ kJ/mol}$  ( $> 0$ ).
- $V_{\text{water}} = 18.0 \text{ mL/mol}$ ;  $V_{\text{ice}} = 19.6 \text{ mL/mol}$ .
- $\Delta V_{\text{fus}} = 18.0 - 19.6 = -1.6 \text{ mL/mol} < 0$ .

Therefore:

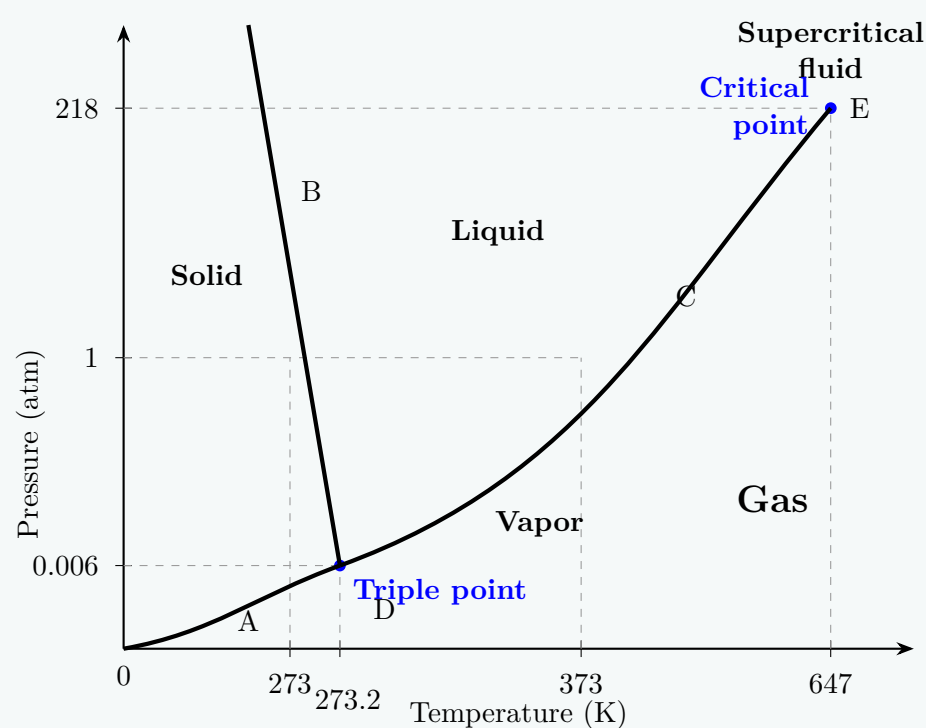
$$\frac{dP}{dT} = \frac{(+)}{T \times (-)} = \text{negative}$$

The melting curve has a **negative slope** because ice is **less dense than water** (unusual) — water expands on freezing.

**Physical Explanation:**

- Ice has an **open hexagonal hydrogen-bonded structure** that is less dense than liquid water.
- Applying pressure **favours the denser phase** (Le Chatelier's principle). Since liquid water is denser than ice, increasing pressure **stabilises the liquid phase**, causing ice to melt at a **lower temperature**.
- Quantitatively, an increase of 1 atm pressure lowers the melting point by  $\approx 0.0075^\circ\text{C}$ .

**Phase Diagram Illustration:**



**Consequence:** This negative slope explains phenomena such as **ice skating** (pressure from the blade melts ice, creating a lubricating water film) and **glacier flow** (ice at the base melts under the immense pressure of overlying ice).

**Q7.** Discuss the salient features of the phase diagram of the sulphur system. Why can four phases of a heterogeneous system not exist at equilibrium? (13 marks) [CHEM 113 | 2021–22]

**Answer**

**Sulphur Phase Diagram Salient Features and Four-Phase Impossibility**

**Part 1: Salient Features**

See previous answer for the full phase diagram. Key features:

- Two solid allotropes:** Rhombic (stable  $< 95.6^\circ\text{C}$ ) and monoclinic (stable  $95.6\text{--}120^\circ\text{C}$ ).
- Three stable triple points:**
  - B ( $95.6^\circ\text{C}$ ):  $S_R + S_M + V$
  - C ( $120^\circ\text{C}$ ):  $S_M + L + V$
  - E ( $165^\circ\text{C}$ , 1290 atm):  $S_R + S_M + L$

At each:  $F = 1 - 3 + 2 = 0$  (invariant).
- One metastable triple point** (Point F,  $114^\circ\text{C}$ ):  $S_R + L + V$  (if monoclinic transition is bypassed).
- Transition temperature  $95.6^\circ\text{C}$ :**  $S_R$  converts slowly to  $S_M$ ; heating rapidly bypasses the transition.
- Melting of monoclinic:** at  $120^\circ\text{C}$  at standard pressure (near point C).

**Part 2: Four Phases Cannot Coexist**

For sulphur,  $C = 1$ . By the Gibbs Phase Rule:

$$F = C - P + 2 = 1 - P + 2 = 3 - P$$

If  $P = 4$ :  $F = 3 - 4 = -1$

Negative degrees of freedom is **physically impossible**. Therefore, four phases ( $S_R$ ,  $S_M$ , liquid, vapour) cannot coexist simultaneously at equilibrium.

**Q8.** A four-phase equilibrium is not usually possible — Justify the statement. (5 marks)

[CHEM 113 | 2020–21 (Jan 21)]

### Answer

#### Four-Phase Equilibrium is Not Usually Possible

**Justification Using the Gibbs Phase Rule:**

The **Gibbs Phase Rule** states:

$$F = C - P + 2$$

where:

- $F$  = degrees of freedom
- $C$  = number of components
- $P$  = number of phases
- The factor 2 accounts for temperature and pressure as variables.

The **minimum possible value of  $F$**  is **zero** (an invariant system — no variables can be changed while maintaining equilibrium). Setting  $F = 0$ :

$$0 = C - P + 2 \Rightarrow P = C + 2$$

For a **one-component system** ( $C = 1$ ):

$$P_{\max} = C + 2 = 1 + 2 = 3$$

The **maximum number of phases** that can coexist at equilibrium in a one-component system is **three** (as at the triple point of water or sulphur).

If four phases coexisted ( $P = 4$ ):

$$F = 1 - 4 + 2 = -1$$

A **negative** degrees of freedom is physically **impossible** — it would mean we need to fix a negative number of variables, which has no physical meaning.

For a **two-component system** ( $C = 2$ ):

$$P_{\max} = C + 2 = 4$$

Four phases can coexist, but  $F = 0$  (invariant). Even here, **five phases** would be impossible.

**Conclusion:** In a **one-component system**, four phases cannot coexist at equilibrium because the phase rule would require  $F = -1$ , which is physically impossible. The **maximum is three phases** (at the triple point,  $F = 0$ ). This is why a four-phase equilibrium is not usually possible in common one-component systems like water or sulphur.

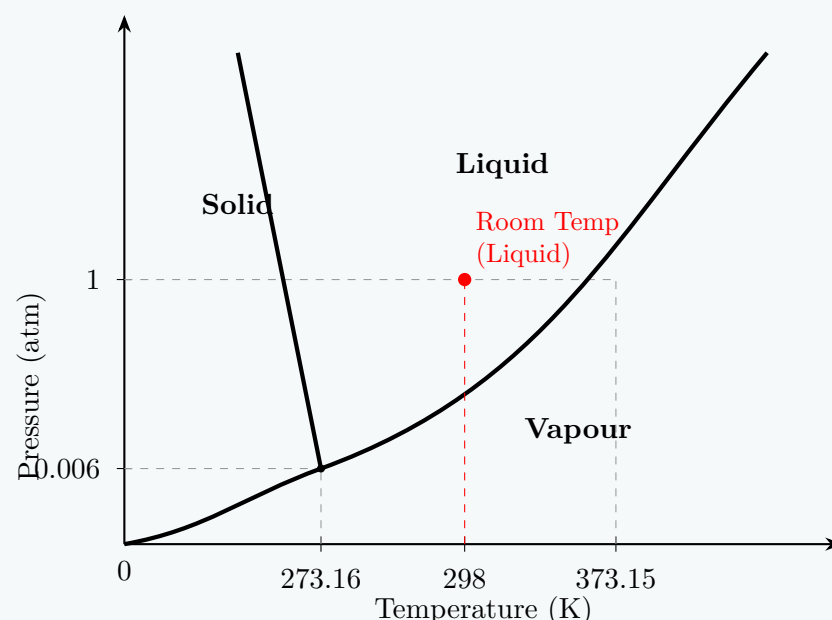
**Q9.** Carbon dioxide is a gas but water is a liquid at room temperature — Illustrate with suitable phase diagrams. (15 marks) [CHEM 113 | 2020–21 (Jan 21)]

### Answer

**CO<sub>2</sub> is a Gas but H<sub>2</sub>O is a Liquid at Room Temperature: Phase Diagram Illustration**

**Room temperature and pressure:**  $T \approx 25^\circ\text{C}$  (298 K),  $P \approx 1$  atm.

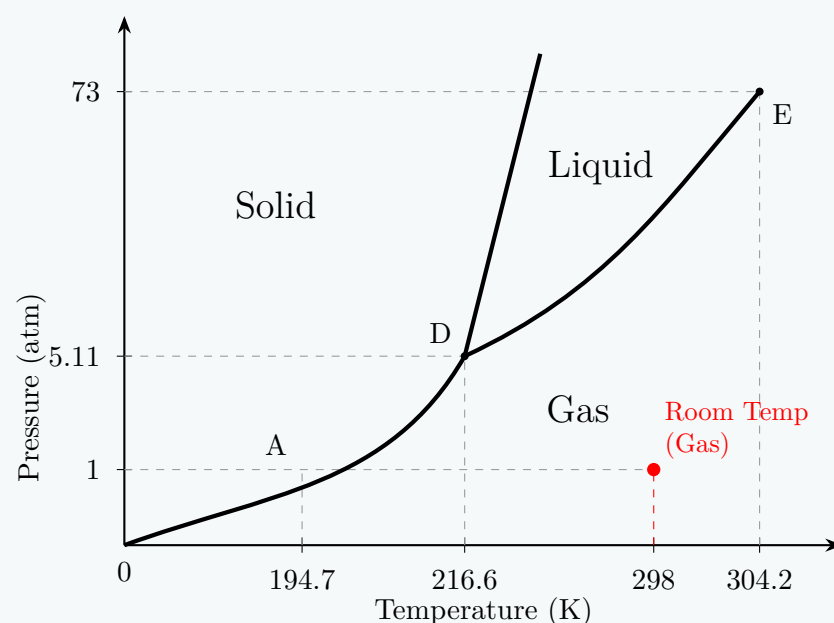
**Phase Diagram of Water (H<sub>2</sub>O):**





At 25°C and 1 atm, the point falls **within the liquid region** of water's phase diagram. Water is a liquid because its **normal boiling point (100°C or 373.15 K)** is well above 25°C and its **triple point (0.01°C, 0.006 atm)** is far below 1 atm.

Phase Diagram of CO<sub>2</sub>:



At 25°C and 1 atm, the point falls **within the gas region** of CO<sub>2</sub>'s phase diagram because:

- CO<sub>2</sub>'s **triple point pressure is 5.11 atm** (above 1 atm). At 1 atm, CO<sub>2</sub> can only exist as **solid or gas**, not liquid.
- CO<sub>2</sub> **sublimes** at −78.5°C (194.7 K) at 1 atm. At 25°C (298 K, well above this), CO<sub>2</sub> is a **gas**.
- CO<sub>2</sub>'s **critical temperature** is only 31.1°C — close to room temperature, so above 31.1°C it cannot be liquefied at any pressure.

Summary:

| Property             | H <sub>2</sub> O        | CO <sub>2</sub>        |
|----------------------|-------------------------|------------------------|
| Triple point $P$     | 0.006 atm (below 1 atm) | 5.11 atm (above 1 atm) |
| NBP                  | 100°C                   | −78.5°C (sublimes)     |
| Critical $T$         | 374°C                   | 31.1°C                 |
| State at 25°C, 1 atm | <b>Liquid</b>           | <b>Gas</b>             |

**Q10.** Applying the appropriate thermodynamic conditions, derive the phase rule for a condensed system. **(15 marks)** [CHEM 113 | 2020–21 (Jan 21)]

### Answer

#### Phase Rule for a Condensed System

##### Definition:

A **condensed system** is one in which the **vapour phase is neglected** (i.e. the system consists only of solid and/or liquid phases). This is valid when pressure has negligible effect on the equilibrium (e.g. alloy systems studied at atmospheric pressure).

##### Thermodynamic Derivation:

##### Step 1: General Gibbs Phase Rule

Consider a system with  $C$  components and  $P$  phases. The **total variables** are:

- Temperature  $T$ : 1 variable
- Pressure  $p$ : 1 variable
- Mole fractions in each phase: for each phase,  $(C - 1)$  independent composition variables (since mole fractions sum to 1).

Total variables:

$$V_{\text{total}} = 2 + P(C - 1)$$

##### Step 2: Equilibrium Conditions (Constraints)

At equilibrium, the **chemical potential**  $\mu_i$  of each component  $i$  must be equal in all phases:

$$\mu_i^{(1)} = \mu_i^{(2)} = \dots = \mu_i^{(P)}$$

For each component:  $(P - 1)$  equations. For  $C$  components:  $C(P - 1)$  equations total.

##### Step 3: Degrees of Freedom

$$F = V_{\text{total}} - \text{constraints} = [2 + P(C - 1)] - C(P - 1)$$

$$F = 2 + PC - P - CP + C = C - P + 2$$

This is the Gibbs Phase Rule:

$$F = C - P + 2$$

#### Step 4: Applying Condensed System Condition

In a condensed system, **pressure is constant** (usually 1 atm) and plays no role. We therefore **remove pressure as a variable**:

$$\text{Total variables} = 1 + P(C - 1) \quad (\text{only } T, \text{ no } P)$$

Constraints remain  $C(P - 1)$ .

$$F' = [1 + P(C - 1)] - C(P - 1) = 1 + PC - P - CP + C = C - P + 1$$

**Phase Rule for Condensed System:**

$$F' = C - P + 1$$

**Physical Meaning:**

| $C$ | $P$ | $F' = C - P + 1$ | System                            |
|-----|-----|------------------|-----------------------------------|
| 1   | 1   | 1                | Univariant (pure solid or liquid) |
| 1   | 2   | 0                | Invariant (melting point fixed)   |
| 2   | 1   | 2                | Bivariant (single-phase alloy)    |
| 2   | 2   | 1                | Univariant (two-phase alloy)      |
| 2   | 3   | 0                | Invariant (eutectic point)        |

**Significance:** The condensed phase rule is widely used in **metallurgy and materials science** to analyse binary and ternary alloy systems (e.g. Fe–C phase diagrams), where only solid and liquid phases are relevant and the effect of pressure is negligible.

**Q11.** Draw the phase diagram of sulfur. (i) Demonstrate the three-phase equilibria in the diagram. (ii) The existence of monoclinic sulfur is not possible in metastable equilibrium — justify. **(15 marks)** [CHEM 113 | 2020–21]

#### Answer

##### Sulphur Phase Diagram: Three-Phase Equilibria and Monoclinic Metastable Equilibrium

(See previous answer for full diagram.)

##### (i) Three-Phase Equilibria:

There are **three stable triple points** in the sulphur system, each representing a three-phase equilibrium ( $F = 0$ ):

| Point | Three Phases                          | $T$              | $F$ |
|-------|---------------------------------------|------------------|-----|
| B     | $S_R + S_M + \text{Vapour}$           | 95.6°C           | 0   |
| C     | $S_M + \text{Liquid} + \text{Vapour}$ | 120°C            | 0   |
| E     | $S_R + S_M + \text{Liquid}$           | 165°C (1290 atm) | 0   |

##### (ii) Monoclinic Sulphur in Metastable Equilibrium is Not Possible:

- The statement asks: why cannot monoclinic sulphur exist in **metastable equilibrium**?
- A **metastable equilibrium** is one in which a phase persists outside its range of thermodynamic stability but cannot spontaneously transform due to kinetic barriers.
- Rhombic sulphur can exist metastably above 95.6°C (up to 114°C) because the transformation to monoclinic is kinetically slow.
- However, **monoclinic sulphur cannot exist metastably below 95.6°C** because:
  - The transformation of monoclinic  $\rightarrow$  rhombic is **thermodynamically spontaneous** below 95.6°C (rhombic has lower free energy).
  - The transformation proceeds at a **measurable rate** at temperatures close to 95.6°C.
  - No significant kinetic barrier prevents the monoclinic  $\rightarrow$  rhombic conversion at low temperatures.
- Therefore, monoclinic sulphur is always unstable (not just metastable) below 95.6°C and **cannot persist in a metastable state** the way rhombic can above 95.6°C.

**Q12.** With the appropriate phase diagrams, interpret the physical state of water and carbon dioxide on a summer day. **(12 marks)** [CHEM 113 | 2020–21]

#### Answer

##### Physical State of Water and CO<sub>2</sub> on a Summer Day

A typical **summer day**:  $T \approx 35^\circ\text{C}$ ,  $P = 1 \text{ atm}$ .

Referring to the phase diagrams in Q5:

(i) **Water (H<sub>2</sub>O):** At 35°C and 1 atm, the point lies **within the liquid region**. Water is a **liquid** because:

- NBP = 100°C > 35°C ⇒ water has not boiled.
- Triple point  $P = 0.006 \text{ atm} < 1 \text{ atm} \Rightarrow$  liquid phase is accessible at 1 atm.

(ii) **Carbon Dioxide (CO<sub>2</sub>):** At 35°C and 1 atm, the point lies **within the gas region**. CO<sub>2</sub> is a **gas** because:

- Triple point  $P = 5.11 \text{ atm} > 1 \text{ atm} \Rightarrow$  at 1 atm, liquid CO<sub>2</sub> cannot exist.
- Normal sublimation point =  $-78.5^\circ\text{C} \ll 35^\circ\text{C} \Rightarrow$  CO<sub>2</sub> exists as gas at 35°C.
- Critical temperature =  $31.1^\circ\text{C} < 35^\circ\text{C} \Rightarrow$  CO<sub>2</sub> is above its critical temperature and **cannot be liquefied** at any pressure.

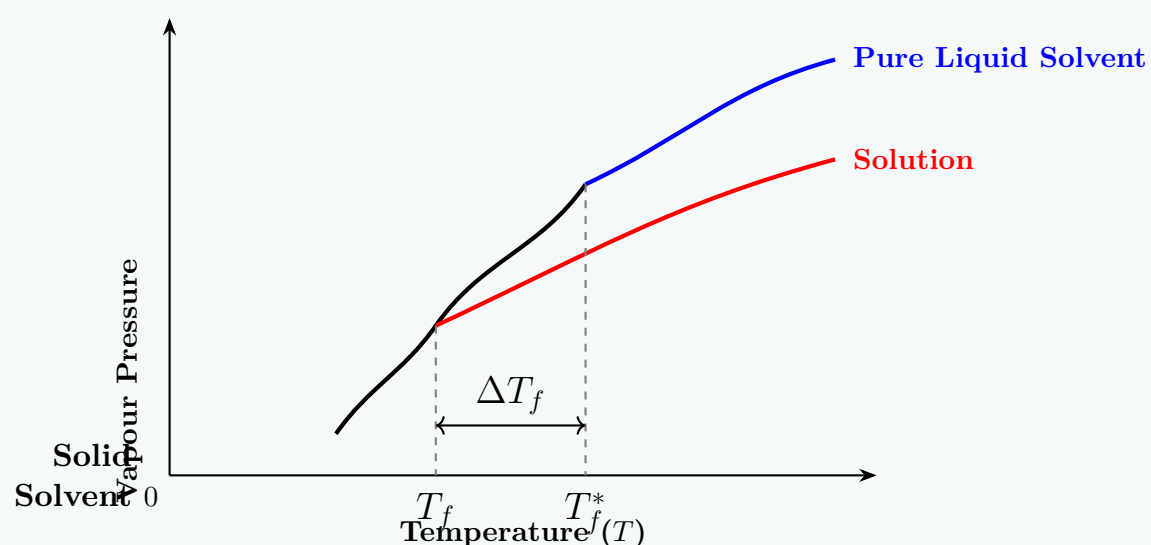
**Q13.** Show the freezing point depression in a phase diagram. A sample weighing 0.546 g was dissolved in 15.0 g benzene, and the freezing-point depression was determined to be 0.240°C. Calculate the molar mass of the hormone. ( $K_f$  for benzene = 5.12°C·kg/mol) (20 marks) [CHEM 113 | 2019–20]

### Answer

#### Freezing Point Depression Phase Diagram and Molar Mass Calculation

##### Part 1: Phase Diagram

The diagram below illustrates the vapour pressure curves for the pure solvent and the solution. The freezing point is the temperature at which the vapour pressure of the liquid equals that of the solid solvent. Because the solution has a lower vapour pressure, its curve intersects the solid solvent curve at a **lower temperature** ( $T_f < T_f^*$ ):



$$\Delta T_f = T_f^* - T_f = K_f \cdot m$$

##### Part 2: Molar Mass Calculation

Given:

- Mass of solute:  $w_B = 0.546 \text{ g}$
- Mass of benzene (solvent):  $w_A = 15.0 \text{ g} = 0.0150 \text{ kg}$
- Freezing point depression:  $\Delta T_f = 0.240^\circ\text{C}$
- $K_f$  (benzene) = 5.12°C·kg/mol

##### Step 1: Calculate molality

$$\Delta T_f = K_f \times m \Rightarrow m = \frac{\Delta T_f}{K_f} = \frac{0.240}{5.12} = 0.04688 \text{ mol/kg}$$

##### Step 2: Find moles of solute

$$m = \frac{n_B}{w_A (\text{kg})} \Rightarrow n_B = m \times w_A = 0.04688 \times 0.0150 \text{ kg} = 7.031 \times 10^{-4} \text{ mol}$$

##### Step 3: Calculate molar mass

$$M_B = \frac{w_B}{n_B} = \frac{0.546 \text{ g}}{7.031 \times 10^{-4} \text{ mol}}$$

$$M_B = 776 \text{ g/mol}$$

**Q14.** How is the phase diagram of CO<sub>2</sub> different from H<sub>2</sub>O? (5 marks)

[CHEM 113 | 2017–18]

## Answer

 Differences Between Phase Diagrams of CO<sub>2</sub> and H<sub>2</sub>O

| Feature                 | H <sub>2</sub> O         | CO <sub>2</sub>         |
|-------------------------|--------------------------|-------------------------|
| Triple point $T$        | 0.01°C                   | −56.6°C                 |
| Triple point $P$        | 0.006 atm (below 1 atm)  | 5.11 atm (above 1 atm)  |
| Solid-liquid slope      | <b>Negative</b>          | <b>Positive</b>         |
| Normal BP               | 100°C                    | −78.5°C (sublimes)      |
| Critical $T$            | 374°C                    | 31.1°C                  |
| Critical $P$            | 218 atm                  | 73 atm                  |
| Liquid at 1 atm         | Yes (0–100°C)            | <b>No</b>               |
| Sublimation at 1 atm    | No                       | Yes (−78.5°C)           |
| Density solid vs liquid | Solid < liquid (unusual) | Solid > liquid (normal) |

**Key Distinction:** The **negative slope** of the S-L boundary in H<sub>2</sub>O (due to ice being less dense than water) versus the **positive slope** in CO<sub>2</sub> (solid CO<sub>2</sub> denser than liquid CO<sub>2</sub>) is the most fundamental structural difference. The high triple point pressure of CO<sub>2</sub> means it cannot exist as a liquid at atmospheric pressure.

**Q15.** What is meant by a phase diagram? Draw and discuss the phase diagram for the sulphur system with the application of phase rule. (10 marks) [CHEM 113 | 2017–18]

## Answer

## Sulphur Phase Diagram Salient Features and Four-Phase Impossibility

## Part 1: Salient Features

See Q8 for the full phase diagram. Key features:

- (i) **Two solid allotropes:** Rhombic (stable < 96°C) and monoclinic (stable 96–119°C).
- (ii) **Three stable triple points:**
  - O (96°C):  $S_R + S_M + V$
  - B (119°C):  $S_M + L + V$
  - E (high  $P$ ):  $S_R + S_M + L$

At each:  $F = 1 - 3 + 2 = 0$  (invariant).
- (iii) **One metastable triple point** (114°C):  $S_R + L + V$  (if monoclinic transition is bypassed).
- (iv) **Transition temperature 96°C:**  $S_R$  converts slowly to  $S_M$ ; heating rapidly bypasses the transition.
- (v) **Melting of monoclinic:** at 119°C at atmospheric pressure (point B).

## Part 2: Four Phases Cannot Coexist

For sulphur,  $C = 1$ . By the Gibbs Phase Rule:

$$F = C - P + 2 = 1 - P + 2 = 3 - P$$

If  $P = 4$ :  $F = 3 - 4 = -1$

Negative degrees of freedom is **physically impossible**. Therefore, four phases ( $S_R$ ,  $S_M$ , liquid, vapour) cannot coexist simultaneously at equilibrium.

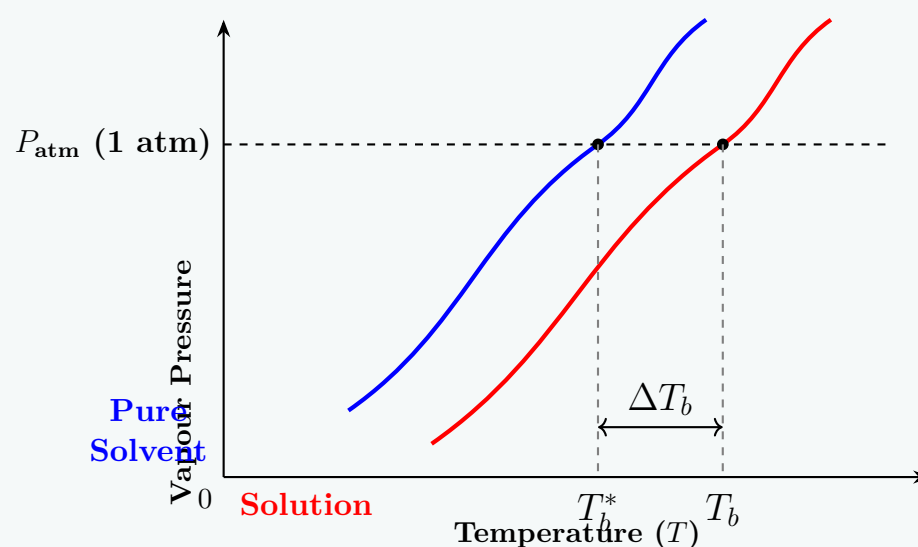
**Q16.** Draw the phase diagram of the solvent-solute system and show that the boiling point elevation due to the addition of non-electrolyte and non-volatile solute is proportional to the molality of the solute. (10 marks) [CHEM 113 | 2016–17]

## Answer

## Phase Diagram of Solvent-Solute System and Boiling Point Elevation Proportional to Molality

## Part 1: Phase Diagram

The addition of a non-volatile solute lowers the vapour pressure of the solvent at all temperatures. Consequently, the solution must be heated to a higher temperature ( $T_b$ ) than the pure solvent ( $T_b^*$ ) to reach atmospheric pressure ( $P_{\text{atm}}$ ), resulting in boiling point elevation ( $\Delta T_b$ ).



### Part 2: Derivation that $\Delta T_b \propto m$

**Step 1:** By Raoult's Law, a non-volatile solute lowers the vapour pressure. The relative lowering is equal to the mole fraction of the solute ( $x_B$ ):

$$P = x_A P^* = (1 - x_B) P^* \Rightarrow \Delta P = P^* - P = x_B P^*$$

**Step 2:** At the boiling point,  $P = P_{\text{atm}}$ . The solution must be heated to a higher temperature  $T_b$  for its vapour pressure to reach  $P_{\text{atm}}$ . Using the Clausius–Clapeyron equation:

$$\frac{d \ln P}{dT} = \frac{\Delta H_{\text{vap}}}{RT^2}$$

**Step 3:** Integrating over a small temperature change near  $T_b^*$  gives the relationship between boiling point elevation and mole fraction:

$$\Delta T_b = \frac{R(T_b^*)^2}{\Delta H_{\text{vap}}} \cdot x_B$$

**Step 4:** For dilute solutions,  $n_A \gg n_B$ , so we can approximate the mole fraction  $x_B$ . Let  $W_A$  be the mass of the solvent and  $M_A$  be its molar mass (in g/mol):

$$x_B = \frac{n_B}{n_A + n_B} \approx \frac{n_B}{n_A} = \frac{n_B}{W_A/M_A} = \frac{n_B \cdot M_A}{W_A}$$

Since molality ( $m$ ) is defined as  $m = \frac{n_B \times 1000}{W_A}$ , we can write  $\frac{n_B}{W_A} = \frac{m}{1000}$ . Thus:

$$x_B = \frac{m \cdot M_A}{1000}$$

**Step 5:** Substituting  $x_B$  into the equation from Step 3:

$$\Delta T_b = \left( \frac{R(T_b^*)^2 M_A}{1000 \Delta H_{\text{vap}}} \right) \cdot m = K_b \cdot m$$

where  $K_b = \frac{R(T_b^*)^2 M_A}{1000 \Delta H_{\text{vap}}}$  is the **ebullioscopic constant** (molal boiling point elevation constant).

$$\Delta T_b = K_b \cdot m \Rightarrow \Delta T_b \propto m$$

Since  $K_b$  depends only on the properties of the **solvent** ( $M_A, T_b^*, \Delta H_{\text{vap}}$ ),  $\Delta T_b$  is directly proportional to molality and depends only on the number of solute particles, not their nature.

**Q17.** Acetylene has a critical point of 36°C, 62 atm. What can be inferred about the liquefaction of acetylene from this data? (6 marks)  
[CHEM 113 | 2015–16]

### Answer

#### Liquefaction of Acetylene from Critical Point Data

**Given:** Critical temperature  $T_c = 36^\circ\text{C}$ , critical pressure  $P_c = 62 \text{ atm}$ .

**Inferences:**

- Acetylene can be liquefied near room temperature:** The critical temperature of acetylene (36°C) is just **slightly above room temperature** ( $\sim 25^\circ\text{C}$ ). This means that at room temperature (which is below  $T_c$ ), acetylene **can be liquefied by applying sufficient pressure**.
- Required pressure is significant:** To liquefy acetylene at room temperature (25°C, which is below  $T_c = 36^\circ\text{C}$ ), a pressure **somewhat below 62 atm** is needed (the exact value is read from the liquid-vapour curve of the phase diagram at 25°C).
- Above 36°C, liquefaction is impossible:** Above the critical temperature  $T_c = 36^\circ\text{C}$ , acetylene **cannot be liquefied** at any pressure. The gas becomes a **supercritical fluid**.
- Storage:** Acetylene is commonly stored in cylinders as a **liquid under pressure** at room temperature. The close proximity of  $T_c$  to room temperature makes this practical and energy-efficient.

- (v) **Comparison:** Unlike  $\text{CO}_2$  ( $T_c = 31.1^\circ\text{C}$ , above room temperature) and unlike  $\text{N}_2$  ( $T_c = -147^\circ\text{C}$ , requires cryogenic cooling), acetylene is conveniently liquefiable at ambient conditions with moderate pressures.

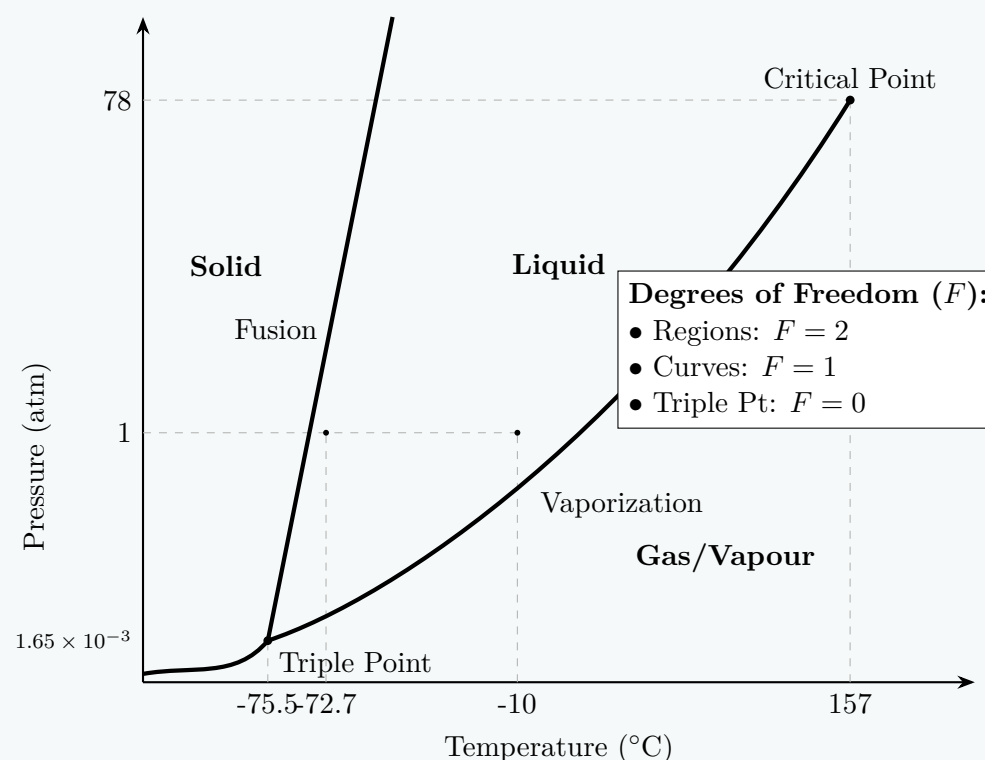
**Q18.** Sketch a rough phase diagram of  $\text{SO}_2$  from the following information: normal freezing point =  $-72.7^\circ\text{C}$ , normal boiling point =  $-10^\circ\text{C}$ , triple point =  $-75.5^\circ\text{C}$  and  $1.65 \times 10^{-3}$  atm, Critical point =  $157^\circ\text{C}$  and 78 atm. Label each phase region and calculate degrees of freedom at all regions and equilibria. (12 marks) [CHEM 113 | 2015–16]

### Answer

#### Phase Diagram of $\text{SO}_2$ and Degrees of Freedom

##### Given data:

- Normal freezing point:  $-72.7^\circ\text{C}$
- Normal boiling point:  $-10^\circ\text{C}$
- Triple point:  $-75.5^\circ\text{C}$ ,  $1.65 \times 10^{-3}$  atm
- Critical point:  $157^\circ\text{C}$ , 78 atm



##### Degrees of Freedom ( $F = C - P + 2 = 3 - P$ ):

| Region/Curve       | $P$ | $F$ | Variance   |
|--------------------|-----|-----|------------|
| Solid region       | 1   | 2   | Bivariant  |
| Liquid region      | 1   | 2   | Bivariant  |
| Gas region         | 1   | 2   | Bivariant  |
| Solid–Gas curve    | 2   | 1   | Univariant |
| Liquid–Gas curve   | 2   | 1   | Univariant |
| Solid–Liquid curve | 2   | 1   | Univariant |
| Triple point       | 3   | 0   | Invariant  |

**Q19.** Draw the phase diagrams of water and carbon dioxide. Why does water's phase diagram have a solid-liquid boundary line with a negative slope? Which one will sublime at room temperature and pressure? Why? (12 marks) [CHEM 113 | 2014–15]

### Answer

#### Phase Diagrams of Water and $\text{CO}_2$ ; Negative Slope and Sublimation

(See Q5 for full phase diagrams.)

##### Why Water's S-L Boundary Has Negative Slope:

(See Q9 for full derivation.)

By Clausius–Clapeyron:

$$\frac{dP}{dT} = \frac{\Delta H_{\text{fus}}}{T \Delta V} = \frac{(+)}{T \times (-)} < 0$$

since water expands on freezing ( $\Delta V < 0$ ). For  $\text{CO}_2$ ,  $\Delta V > 0$  (solid denser), giving a **positive slope**.

##### Which Sublimes at Room Temperature and Pressure?

$\text{CO}_2$  sublimates at room temperature and pressure.

##### Reason:

- $\text{CO}_2$ 's triple point pressure is 5.11 atm  $>$  1 atm.



- At 1 atm, the liquid phase of  $\text{CO}_2$  **does not exist**.
- Solid  $\text{CO}_2$  (dry ice) at 1 atm converts **directly to gas (sublimes)** at  $-78.5^\circ\text{C}$ .
- At room temperature ( $25^\circ\text{C}$ ), which is far above  $-78.5^\circ\text{C}$ ,  $\text{CO}_2$  is simply a **gas** at 1 atm.

Water does **not** sublime at room temperature and 1 atm because its triple point pressure (0.006 atm) is well below 1 atm, and  $25^\circ\text{C}$  falls within its liquid region.

**Q20.** What is phase diagram? Why is such a diagram important? (8 marks)

[CHEM 113 | 2012–13]

### Answer

#### Phase Diagram: Definition and Importance

##### Definition:

A **phase diagram** is a graphical representation showing the **equilibrium conditions** (temperature, pressure, composition) under which the different **phases** of a substance or mixture exist, coexist, or transform into one another.

The axes are usually **temperature** and **pressure** (for one-component systems) or **temperature** and **composition** (for multi-component systems).

##### Importance of Phase Diagrams:

- Predict stable phases:** Given  $T$  and  $P$ , a phase diagram immediately shows which phase (solid, liquid, gas) is thermodynamically stable.
- Determine phase transitions:** Phase diagrams show the exact conditions of **melting, boiling, sublimation, and polymorphic transitions**.
- Apply Gibbs Phase Rule:** The number of phases, components, and degrees of freedom at any point can be read directly from the diagram.
- Materials processing:** In metallurgy, phase diagrams guide **alloy design, heat treatment, and casting processes** (e.g. Fe–C diagram for steel manufacturing).
- Geoscience and climate:** Water's phase diagram explains **glaciology, atmospheric phenomena, and ocean behaviour**.
- Chemical industry:** Phase diagrams are essential in designing **distillation, crystallisation, and extraction** processes.
- Pharmaceutical science:** Used to optimise **drug polymorphism, stability, and solubility**.

**Q21.** What is metastable equilibrium? Draw the phase diagram of the sulfur (S) system and explain the formation of the following metastable equilibrium:  $\text{S}(\text{rhombic}) \rightleftharpoons \text{S}(\text{liquid}) \rightleftharpoons \text{S}(\text{vapor})$ . (12 marks)

[CHEM 113 | 2012–13]

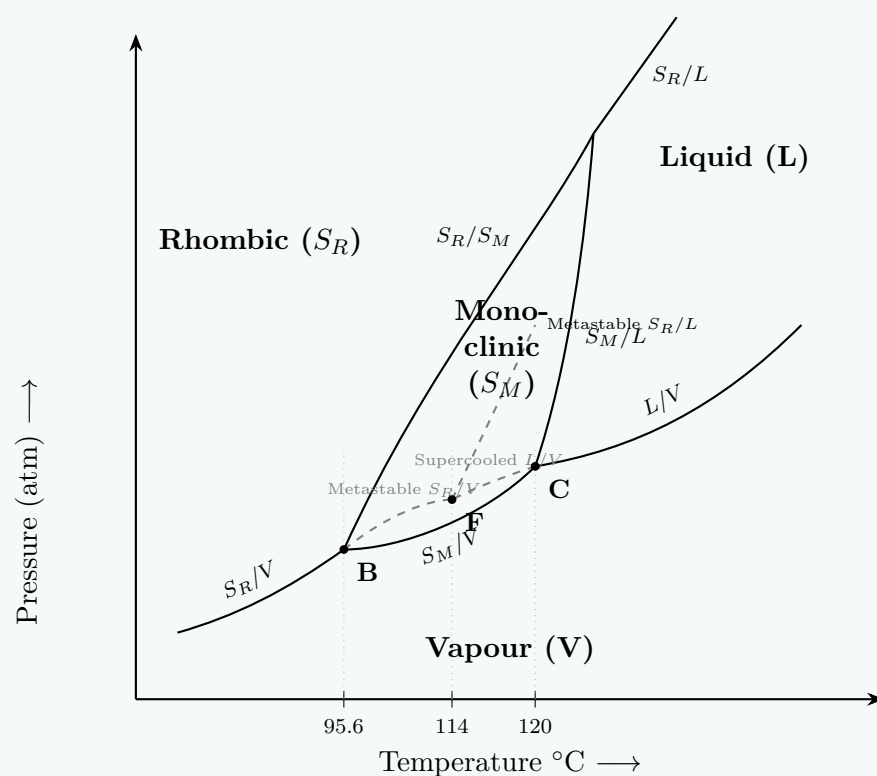
### Answer

#### Metastable Equilibrium and the Sulphur Phase Diagram

##### Part 1: What is Metastable Equilibrium?

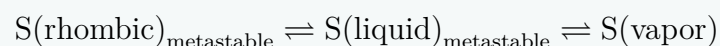
A **metastable equilibrium** is a state of a system that is stable to small disturbances but not the most thermodynamically stable state (it is not at the global minimum of free energy). A phase persists in a metastable state outside its normal range of stability because the transition to the stable phase is kinetically very slow.

##### Part 2: Phase Diagram of the Sulphur System



##### Part 3: Formation of Metastable Equilibrium: $\text{S}(\text{rhombic}) \rightleftharpoons \text{S}(\text{liquid}) \rightleftharpoons \text{S}(\text{vapor})$

- **The Kinetic Barrier:** The stable transition from Rhombic to Monoclinic sulphur occurs at 95.6°C (Point B). However, this structural rearrangement is kinetically very slow.
- **Rapid Heating:** If Rhombic sulphur is heated rapidly, it does not have time to convert to Monoclinic sulphur. It bypasses Point B and continues to exist in a **metastable state** along the extended dashed vapour pressure curve **BF**.
- **Supercooling:** Similarly, if liquid sulphur is cooled very carefully, it can avoid crystallising at 120°C (Point C) and exist as a **supercooled liquid** along the dashed vapour pressure curve **CF**.
- **The Metastable Triple Point (Point F):** The metastable vapour pressure curve of Rhombic sulphur (BF) and the supercooled liquid vapour pressure curve (CF) intersect at **Point F (114°C)**.
- At this specific temperature and pressure, three metastable phases coexist in equilibrium:



This is the metastable triple point. Rhombic sulphur will undergo **metastable melting** exactly at 114°C.

**Q22.** Draw the phase diagram of two partially miscible liquid pairs and define Critical Solution Temperature. **(10 marks)** [CHEM 113 | 2012–13]

### Answer

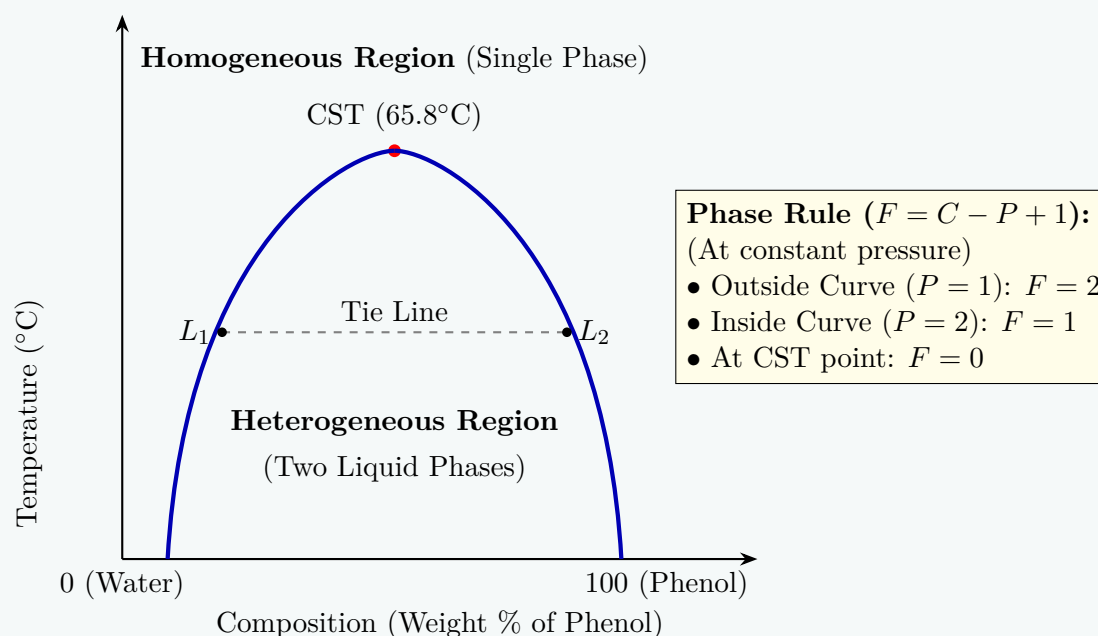
#### Phase Diagram of Partially Miscible Liquid Pairs and Critical Solution Temperature

##### Partially Miscible Liquids:

Two liquids are said to be **partially miscible** if they dissolve in each other only to a limited extent at certain temperatures, forming two separate liquid phases. Classic examples are:

- **Phenol–water** system (UCST type)
- **Triethylamine–water** system (LCST type)
- **Nicotine–water** system (both UCST and LCST)

##### Phase Diagram: Phenol–Water System (UCST Type)



##### Explanation of the Diagram:

- The **dome-shaped curve** (binodal curve) encloses the **two-phase region** where phenol and water are only partially miscible — two separate liquid layers ( $\alpha$  and  $\beta$ ) coexist.
- Outside the dome** (above the curve): the two liquids are **completely miscible**, forming a single homogeneous phase.
- A **tie line** (horizontal line) at any temperature within the two-phase region connects the compositions of the two coexisting phases ( $\alpha$  and  $\beta$ ).
- As temperature **increases**, the two-phase region **narrows** (the compositions of the two phases become closer to each other).
- At the **top of the dome** (the critical point), the two phases become **identical** — the system becomes one phase.

##### Critical Solution Temperature (CST):

The **Critical Solution Temperature (CST)**, also called the **consolute temperature**, is defined as:

*“The temperature at which the two partially miscible liquids become **completely miscible** in all proportions, forming a single homogeneous phase.”*

- **Upper Critical Solution Temperature (UCST):** The CST above which the two liquids are completely miscible. Miscibility **increases with temperature**. Example: phenol–water system (UCST = 66°C at 34% phenol).

- **Lower Critical Solution Temperature (LCST):** The CST below which the two liquids are completely miscible. Miscibility decreases with temperature. Example: triethylamine–water system (LCST = 18°C).
- **Both UCST and LCST:** Some systems exhibit both. Example: nicotine–water (LCST = 61°C, UCST = 208°C), showing a closed miscibility loop.

**Q23.** What is the triple point in a phase diagram? With the help of the phase rule, show that the triple point of water is invariant. (10 marks) [CHEM 113 | 2012–13]

#### Answer

##### Triple Point and Its Invariance for Water

##### Definition of Triple Point:

The **triple point** is the unique combination of **temperature and pressure** at which all three phases of a substance — solid, liquid, and vapour — **coexist in equilibrium** simultaneously.

For water: Triple point = 0.01°C and 0.006 atm (611 Pa).

##### Proof of Invariance Using Phase Rule:

For water:  $C = 1$  (one component). At the triple point:  $P = 3$  (solid + liquid + vapour).

Applying the Gibbs Phase Rule:

$$F = C - P + 2 = 1 - 3 + 2 = 0$$

$F = 0$  means **zero degrees of freedom** — **no variable** (neither temperature nor pressure) can be changed without destroying the three-phase equilibrium.

**Therefore, the triple point is completely fixed (invariant):**

- Temperature is **fixed** at exactly 0.01°C.
- Pressure is **fixed** at exactly 0.006 atm.
- **No freedom** to vary either — if  $T$  or  $P$  changes even slightly, one or more phases disappear.

**Practical significance:** The triple point of water is used to define the **Kelvin temperature scale**: 273.16 K is defined as the triple point of water, making it a fundamental and reproducible reference point in thermometry.

**Q24.** Thermodynamically derive the Gibbs phase rule. Draw the phase diagram of the water system and describe its main features. (12 marks) [CHEM 113 | 2011–12]

**Q25.** State how many phases, components and degrees of freedom are present in each of the following systems: (i) Saturated solution of NaCl (ii) Mixture of  $O_2$  and  $N_2$  (iii)  $H_2O_{(s)} \rightleftharpoons H_2O_{(l)} \rightleftharpoons H_2O_{(g)}$  (iv)  $CaCO_{3(s)} \rightleftharpoons CaO_{(s)} + CO_{2(g)}$ . (8 marks) [CHEM 101 | 2010–11]

#### Answer

##### Phases, Components, and Degrees of Freedom

Using  $F = C - P + 2$ :

##### (i) Saturated solution of NaCl:

- Phases ( $P$ ): 2 (saturated solution + excess solid NaCl).
- Components ( $C$ ): 2 (NaCl +  $H_2O$ ).
- $F = 2 - 2 + 2 = 2$  (bivariant).

##### (ii) Mixture of $O_2$ and $N_2$ :

- Phases ( $P$ ): 1 (single gas phase).
- Components ( $C$ ): 2 ( $O_2$  +  $N_2$ ).
- $F = 2 - 1 + 2 = 3$  (trivariant).

##### (iii) $H_2O_{(s)} \rightleftharpoons H_2O_{(l)} \rightleftharpoons H_2O_{(g)}$ :

- Phases ( $P$ ): 3 (solid, liquid, gas).
- Components ( $C$ ): 1 ( $H_2O$ ).
- $F = 1 - 3 + 2 = 0$  (invariant). This is the **triple point**.

##### (iv) $CaCO_{3(s)} \rightleftharpoons CaO_{(s)} + CO_{2(g)}$ :

- Phases ( $P$ ): 3 ( $CaCO_3$  solid,  $CaO$  solid,  $CO_2$  gas).
- Components ( $C$ ): 2 (Since  $CaO + CO_2 = CaCO_3$ , only 2 independent components).
- $F = 2 - 3 + 2 = 1$  (univariant). At a fixed temperature, pressure is fixed.

| System                              | $C$ | $P$ | $F$ | Variance   |
|-------------------------------------|-----|-----|-----|------------|
| Sat. NaCl soln                      | 2   | 2   | 2   | Bivariant  |
| O <sub>2</sub> + N <sub>2</sub> gas | 2   | 1   | 3   | Trivariant |
| H <sub>2</sub> O triple pt          | 1   | 3   | 0   | Invariant  |
| CaCO <sub>3</sub> decomp.           | 2   | 3   | 1   | Univariant |

**Q26.** Thermodynamically derive the Gibbs phase rule. (7 marks)

[CHEM 101 | 2010–11]

**Q27.** Derive the phase rule. Draw the phase diagram of water and describe its main features. (18 marks)

[CHEM 101 | 2009–10]

### Answer

#### Thermodynamic Derivation of Gibbs Phase Rule

(Q24, Q25, Q26 — same derivation)

Consider a system of  $C$  components in  $P$  phases.

#### Step 1: Count Total Variables

The state of the system is defined by:

- **Intensive variables:**  $T, P$  (1 each = 2 variables).
- **Composition of each phase:** For each phase,  $C$  mole fractions are needed, but  $\sum x_i = 1$  reduces this to  $(C - 1)$  independent variables per phase.

Total variables:

$$V = 2 + P(C - 1)$$

#### Step 2: Equilibrium Conditions (Constraints)

At thermodynamic equilibrium:

- Thermal equilibrium:**  $T^{(1)} = T^{(2)} = \dots = T^{(P)}$  (already included by having one  $T$ ).
- Mechanical equilibrium:**  $P^{(1)} = P^{(2)} = \dots = P^{(P)}$  (already included by having one  $P$ ).
- Chemical equilibrium** (equality of chemical potentials): for component  $i$  in all phases:

$$\mu_i^{(1)} = \mu_i^{(2)} = \dots = \mu_i^{(P)}$$

This gives  $(P - 1)$  equations per component. For  $C$  components:  $C(P - 1)$  total equations.

#### Step 3: Degrees of Freedom

$$F = \text{Total variables} - \text{Total constraints}$$

$$F = [2 + P(C - 1)] - C(P - 1)$$

$$F = 2 + PC - P - CP + C$$

$$\boxed{F = C - P + 2}$$

This is the **Gibbs Phase Rule**.

#### Physical Meaning of Terms:

- $F$ : number of intensive variables that can be independently varied without disturbing the number of phases.
- $C$ : minimum number of independent chemical species needed to describe all phases.
- $P$ : number of phases present.
- +2: accounts for temperature and pressure as variables (reduced to +1 for condensed systems).

#### Water Phase Diagram and Features:

(See Q3 for full phase diagram and discussion.)

**Q28.** Write the mathematical expression of the phase rule and explain the terms involved. Draw the phase diagram of sulphur and describe the different parts of the diagram with the help of the phase rule. (14 marks)

[CHEM 101 | 2008–09]

### Answer

#### Phase Rule Expression, Terms, and Sulphur Phase Diagram

#### Mathematical Expression:

$$\boxed{F = C - P + 2}$$

#### Terms:

| Symbol | Meaning                                                                             |
|--------|-------------------------------------------------------------------------------------|
| $F$    | Degrees of freedom — number of intensive variables that can be independently varied |
| $C$    | Number of components — minimum independent chemical species to describe all phases  |
| $P$    | Number of phases — physically distinct, homogeneous, mechanically separable parts   |
| +2     | Accounts for $T$ and $P$ as variables                                               |

**Sulphur Phase Diagram:**  
(See Q8 for full diagram.)

**Phase Rule Applied to Sulphur System ( $C = 1, F = 3 - P$ ):**

| Part of diagram | Phases                      | $F$ | Variance   |
|-----------------|-----------------------------|-----|------------|
| $S_R$ region    | $S_R$ only (1)              | 2   | Bivariant  |
| $S_M$ region    | $S_M$ only (1)              | 2   | Bivariant  |
| Liquid region   | Liquid (1)                  | 2   | Bivariant  |
| Vapour region   | Vapour (1)                  | 2   | Bivariant  |
| OA curve        | $S_R + V$ (2)               | 1   | Univariant |
| AB curve        | $S_M + V$ (2)               | 1   | Univariant |
| BC curve        | $L + V$ (2)                 | 1   | Univariant |
| OE curve        | $S_R + S_M$ (2)             | 1   | Univariant |
| EB curve        | $S_M + L$ (2)               | 1   | Univariant |
| EF curve        | $S_M/L + \text{high P}$ (2) | 1   | Univariant |
| Triple pt O     | $S_R + S_M + V$ (3)         | 0   | Invariant  |
| Triple pt B     | $S_M + L + V$ (3)           | 0   | Invariant  |
| Triple pt E     | $S_R + S_M + L$ (3)         | 0   | Invariant  |

BUET · CHEM 113

# Chemistry

Section S7

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Energy & Chemistry



## S7 — Energy & Chemistry

**Q1.** Explain the working principle of a heat engine and its efficiency with the necessary equation. **(10 marks)** [CHEM 113 | 2023–24]

### Answer

#### Working Principle and Efficiency of a Heat Engine

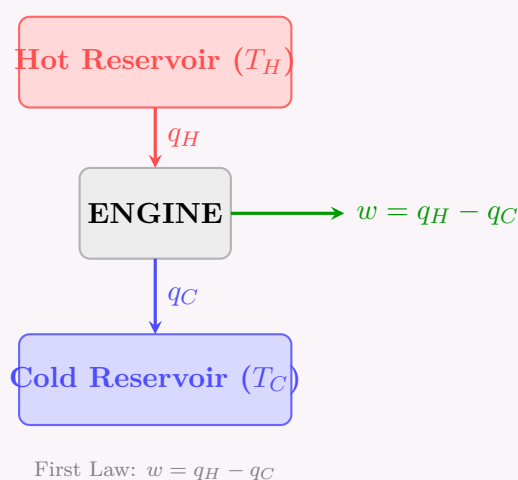
##### Definition:

A **heat engine** is a device that converts **heat energy into mechanical work** by operating between two heat reservoirs — a **hot source** at temperature  $T_H$  and a **cold sink** at temperature  $T_C$ .

##### Working Principle:

- The engine absorbs heat  $q_H$  from the hot reservoir.
- Part of this heat is converted into **useful work**  $w$ .
- The remaining heat  $q_C$  is rejected to the cold reservoir.
- By the **first law of thermodynamics**:

$$w = q_H - q_C$$



##### Efficiency of a Heat Engine:

The **efficiency**  $\eta$  is the fraction of heat absorbed that is converted to work:

$$\eta = \frac{w}{q_H} = \frac{q_H - q_C}{q_H} = 1 - \frac{q_C}{q_H}$$

##### Carnot Efficiency (Maximum possible):

For a **reversible (Carnot) engine**, using the second law of thermodynamics:

$$\frac{q_C}{q_H} = \frac{T_C}{T_H}$$

Therefore:

$$\eta_{Carnot} = 1 - \frac{T_C}{T_H} = \frac{T_H - T_C}{T_H}$$

where  $T_H$  and  $T_C$  are in **Kelvin**.

##### Key points:

- $\eta < 1$  always (cannot convert all heat to work — second law).
- $\eta$  increases as  $T_H$  increases or  $T_C$  decreases.
- $\eta = 1$  only if  $T_C = 0$  K (impossible).
- Real engines are always **less efficient** than a Carnot engine due to irreversibilities (friction, heat loss).

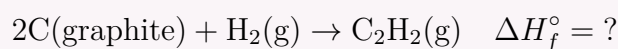
| Engine type    | $T_H$ (K)  | Typical $\eta$ |
|----------------|------------|----------------|
| Steam engine   | $\sim 500$ | $\sim 17\%$    |
| Petrol engine  | $\sim 800$ | $\sim 25\%$    |
| Diesel engine  | $\sim 900$ | $\sim 35\%$    |
| Carnot (ideal) | any        | $1 - T_C/T_H$  |

**Q2.** Calculate the standard enthalpy of formation of acetylene ( $C_2H_2$ ) from its elements:  $2C(\text{graphite}) + H_2(g) \rightarrow C_2H_2(g)$ . Use: (i)  $\Delta H = -393.5$  kJ/mol for combustion of C; (ii)  $\Delta H = -285.8$  kJ/mol for formation of  $H_2O(l)$ ; (iii)  $\Delta H = -2598.8$  kJ/mol for combustion of  $C_2H_2$ . **(10 marks)** [CHEM 113 | 2022–23; 2012–13]

## Answer

Standard enthalpy of formation of acetylene  $\text{C}_2\text{H}_2$

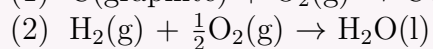
Target reaction:



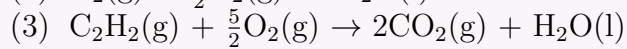
Given reactions:



$$\Delta H_1 = -393.5 \text{ kJ/mol}$$



$$\Delta H_2 = -285.8 \text{ kJ/mol}$$

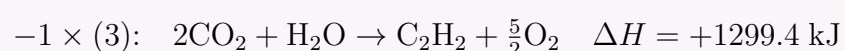
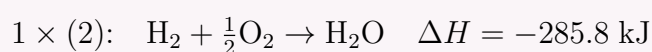
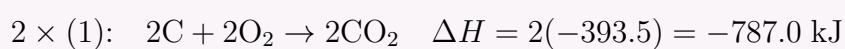


$$\Delta H_3 = -1299.4 \text{ kJ/mol}$$

**Note:** Combustion of  $\text{C}_2\text{H}_2$  given as  $-2598.8 \text{ kJ}$  but that appears to be for 2 mol; per mole:  $\Delta H_3 = -2598.8/2 = -1299.4 \text{ kJ/mol}$ .

Applying Hess's Law:

Use:  $2 \times (1) + 1 \times (2) - 1 \times (3)$



Adding:

$$\Delta H_f^\circ = -787.0 + (-285.8) + 1299.4$$

$$\Delta H_f^\circ(\text{C}_2\text{H}_2) = +226.6 \text{ kJ/mol}$$

The positive value confirms acetylene is an **endothermic compound** — energy must be supplied to form it from its elements.

**Q3.** When 23.6 g of calcium chloride,  $\text{CaCl}_2$ , was dissolved in water in a calorimeter, the temperature rose from  $25.0^\circ\text{C}$  to  $38.7^\circ\text{C}$ . If the heat capacity of the solution is  $1258 \text{ J}/^\circ\text{C}$ , what is the enthalpy change when 1.20 mol of calcium chloride dissolves in water? **(13 marks)** [CHEM 113 | 2022–23]

## Answer

Enthalpy change for dissolution of  $\text{CaCl}_2$

Given:

- Mass of  $\text{CaCl}_2 = 23.6 \text{ g}$ ;  $M_{\text{CaCl}_2} = 40 + 2(35.5) = 111 \text{ g/mol}$
- Temperature rise:  $\Delta T = 38.7 - 25.0 = 13.7^\circ\text{C}$
- Heat capacity of solution =  $1258 \text{ J}/^\circ\text{C}$

Step 1: Heat absorbed by solution:

$$q_{\text{solution}} = C \times \Delta T = 1258 \times 13.7 = 17234.6 \text{ J} = 17.23 \text{ kJ}$$

Since temperature **rose**, heat was **released** by the dissolution process:

$$q_{\text{rxn}} = -17.23 \text{ kJ}$$

Step 2: Moles of  $\text{CaCl}_2$  dissolved:

$$n = \frac{23.6}{111} = 0.2126 \text{ mol}$$

Step 3: Enthalpy change per mole:

$$\Delta H = \frac{q_{\text{rxn}}}{n} = \frac{-17.23}{0.2126} = -81.05 \text{ kJ/mol}$$

Step 4: Enthalpy change for 1.20 mol:

$$\Delta H_{1.20} = -81.05 \times 1.20 = -97.3 \text{ kJ}$$

| Quantity                           | Value                                |
|------------------------------------|--------------------------------------|
| Heat released ( $q_{\text{rxn}}$ ) | $-17.23 \text{ kJ}$                  |
| Moles dissolved                    | $0.2126 \text{ mol}$                 |
| $\Delta H$ per mole                | $-81.05 \text{ kJ/mol}$              |
| $\Delta H$ for 1.20 mol            | <b><math>-97.3 \text{ kJ}</math></b> |

**Q4.** Enthalpy is a state function — Justify. A lead (Pb) pellet having a mass of 26.47 g at  $89.98^\circ\text{C}$  was placed in a constant-pressure calorimeter of negligible heat capacity containing 100.0 mL of water. The water temperature rose from  $22.50^\circ\text{C}$  to  $23.17^\circ\text{C}$ . What is the specific heat of the lead pellet? **(13 marks)** [CHEM 113 | 2021–22]

**Answer**
**Enthalpy as a state function; specific heat of lead**
**Enthalpy is a state function — Justification:**

- A **state function** depends only on the current state of the system (T, P, composition), not on the path taken to reach that state.
- Enthalpy is defined as:

$$H = U + PV$$

where  $U$  (internal energy),  $P$  (pressure), and  $V$  (volume) are all state functions. A combination of state functions is also a state function. Therefore  $H$  is a state function.

- **Hess's Law** is the practical proof: the enthalpy change of a reaction is the same regardless of whether it occurs in one step or several steps. This is only possible if  $H$  is a state function.
- Example: Formation of  $\text{CO}_2$  directly from C and  $\text{O}_2$  gives the same  $\Delta H$  as forming CO first and then oxidizing to  $\text{CO}_2$ .

**Specific heat of lead pellet:**
**Given:**

- Mass of Pb pellet:  $m_{\text{Pb}} = 26.47 \text{ g}$
- Initial temp of Pb:  $T_{\text{Pb},i} = 89.98^\circ\text{C}$
- Volume of water:  $100.0 \text{ mL} \Rightarrow m_{\text{H}_2\text{O}} = 100.0 \text{ g}$
- Initial temp of water:  $T_{w,i} = 22.50^\circ\text{C}$
- Final temp (both):  $T_f = 23.17^\circ\text{C}$
- $c_{\text{H}_2\text{O}} = 4.184 \text{ J/g}\cdot^\circ\text{C}$

**Heat gained by water:**

$$q_{\text{water}} = m_{\text{H}_2\text{O}} \times c_{\text{H}_2\text{O}} \times \Delta T_{\text{water}} = 100.0 \times 4.184 \times (23.17 - 22.50)$$

$$q_{\text{water}} = 100.0 \times 4.184 \times 0.67 = 280.3 \text{ J}$$

**Heat lost by lead (calorimeter negligible):**

$$q_{\text{Pb}} = -q_{\text{water}} = -280.3 \text{ J}$$

**Specific heat of Pb:**

$$q_{\text{Pb}} = m_{\text{Pb}} \times c_{\text{Pb}} \times \Delta T_{\text{Pb}}$$

$$\Delta T_{\text{Pb}} = 23.17 - 89.98 = -66.81^\circ\text{C}$$

$$c_{\text{Pb}} = \frac{q_{\text{Pb}}}{m_{\text{Pb}} \times \Delta T_{\text{Pb}}} = \frac{-280.3}{26.47 \times (-66.81)}$$

$c_{\text{Pb}} = 0.158 \text{ J/g}\cdot^\circ\text{C}$

(Literature value:  $0.128 \text{ J/g}\cdot^\circ\text{C}$ )

**Q5.** Ammonia will burn in the presence of a platinum catalyst to produce nitric oxide:  $4\text{NH}_3(\text{g}) + 5\text{O}_2(\text{g}) \rightarrow 4\text{NO}(\text{g}) + 6\text{H}_2\text{O}(\text{g})$ . What is the heat of reaction at constant pressure? Use:  $\text{N}_2(\text{g}) + \text{O}_2(\text{g}) \rightarrow 2\text{NO}(\text{g}); \Delta H = 180.6 \text{ kJ}$ ;  $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightarrow 2\text{NH}_3(\text{g}); \Delta H = -91.8 \text{ kJ}$ ;  $2\text{H}_2(\text{g}) + \text{O}_2(\text{g}) \rightarrow 2\text{H}_2\text{O}(\text{g}); \Delta H = -483.7 \text{ kJ}$ . **(10 marks)** [CHEM 113 | 2021–22]

**Answer**
**Heat of reaction for combustion of  $\text{NH}_3$** 
**Target reaction:**

**Given:**

- |                                                                                              |                                  |
|----------------------------------------------------------------------------------------------|----------------------------------|
| (1) $\text{N}_2(\text{g}) + \text{O}_2(\text{g}) \rightarrow 2\text{NO}(\text{g})$           | $\Delta H_1 = +180.6 \text{ kJ}$ |
| (2) $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightarrow 2\text{NH}_3(\text{g})$        | $\Delta H_2 = -91.8 \text{ kJ}$  |
| (3) $2\text{H}_2(\text{g}) + \text{O}_2(\text{g}) \rightarrow 2\text{H}_2\text{O}(\text{g})$ | $\Delta H_3 = -483.7 \text{ kJ}$ |

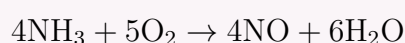
**Applying Hess's Law:**

Need 4NO on right  $\Rightarrow 2 \times (1)$

Need 4NH<sub>3</sub> on left  $\Rightarrow -2 \times (2)$

Need 6H<sub>2</sub>O on right  $\Rightarrow 3 \times (3)$

$$\begin{aligned}
 2 \times (1) : \quad & 2\text{N}_2 + 2\text{O}_2 \rightarrow 4\text{NO} \quad \Delta H = 2(+180.6) = +361.2 \text{ kJ} \\
 -2 \times (2) : \quad & 4\text{NH}_3 \rightarrow 2\text{N}_2 + 6\text{H}_2 \quad \Delta H = -2(-91.8) = +183.6 \text{ kJ} \\
 3 \times (3) : \quad & 6\text{H}_2 + 3\text{O}_2 \rightarrow 6\text{H}_2\text{O} \quad \Delta H = 3(-483.7) = -1451.1 \text{ kJ}
 \end{aligned}$$

**Net:**


$$\Delta H = +361.2 + 183.6 + (-1451.1)$$

$$\Delta H = -906.3 \text{ kJ}$$

The reaction is strongly **exothermic**.

**Q6.** What is Fuel? Describe three fuel-oxidizer systems used in rockets. (12 marks)

[CHEM 113 | 2021–22]

### Answer

#### Fuel and three rocket fuel-oxidizer systems

##### Definition of Fuel:

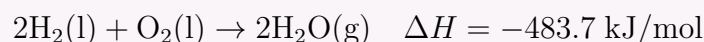
A **fuel** is any substance that reacts with an oxidizer (usually oxygen) in a combustion reaction to release **heat energy**, which can be converted to mechanical work or thrust.

Key properties of a good fuel:

- High energy density (kJ/kg)
- Clean combustion products
- Readily available and safe to handle
- Appropriate ignition temperature

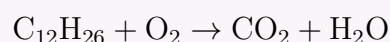
#### Three Rocket Fuel-Oxidizer Systems:

##### 1. Liquid Hydrogen / Liquid Oxygen (LH<sub>2</sub> / LOX):



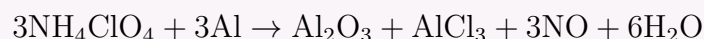
- **Highest specific impulse** of any chemical propellant.
- **Clean** — only water vapour produced.
- Used in: Space Shuttle main engines, Saturn V (upper stage), Ariane 5.
- **Disadvantage:** Cryogenic storage required ( $-253^\circ\text{C}$  for LH<sub>2</sub>).

##### 2. Kerosene (RP-1) / Liquid Oxygen (LOX):



- RP-1 is a highly refined **kerosene**.
- Higher density than LH<sub>2</sub>  $\Rightarrow$  smaller tank volume.
- Used in: Saturn V first stage, SpaceX Falcon 9 (Merlin engines).
- **Disadvantage:** Lower  $I_{sp}$  than LH<sub>2</sub>/LOX; produces CO<sub>2</sub>.

##### 3. Solid Propellant: Ammonium Perchlorate (NH<sub>4</sub>ClO<sub>4</sub>) + Aluminium:



- Fuel (Al) and oxidizer (NH<sub>4</sub>ClO<sub>4</sub>) are **pre-mixed** in solid form.
- Simple, reliable, storable — no cryogenic handling.
- Used in: Space Shuttle SRBs, military missiles.
- **Disadvantage:** Cannot be shut down once ignited; produces HCl (corrosive).

| System               | $I_{sp}$ (s) | Advantages            | Use            |
|----------------------|--------------|-----------------------|----------------|
| LH <sub>2</sub> /LOX | ~ 450        | Highest thrust, clean | Space Shuttle  |
| RP-1/LOX             | ~ 350        | Dense, reliable       | Falcon 9       |
| Solid (APCP)         | ~ 280        | Simple, storable      | SRBs, missiles |

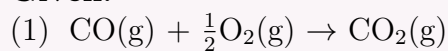
**Q7.** Two pollutants that form in auto exhaust are CO and NO. Calculate  $\Delta H$  for the reaction  $\text{CO}(\text{g}) + \text{NO}(\text{g}) \rightarrow \text{CO}_2(\text{g}) + \frac{1}{2}\text{N}_2(\text{g})$  from the following information:  $\text{CO}(\text{g}) + \frac{1}{2}\text{O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g}); \Delta H = -283.0 \text{ kJ}$ ;  $\text{N}_2(\text{g}) + \text{O}_2(\text{g}) \rightarrow 2\text{NO}(\text{g}); \Delta H = 180.6 \text{ kJ}$ . (8 marks)

[CHEM 113 | 2020–21]

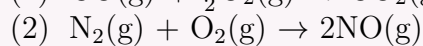
**Answer**

$\Delta H$  for  $\text{CO(g)} + \text{NO(g)} \rightarrow \text{CO}_2\text{(g)} + \frac{1}{2}\text{N}_2\text{(g)}$

**Given:**



$$\Delta H_1 = -283.0 \text{ kJ}$$

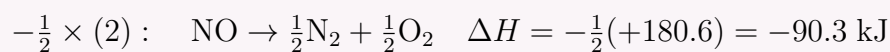
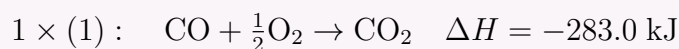


$$\Delta H_2 = +180.6 \text{ kJ}$$

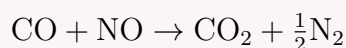
**Strategy (Hess's Law):**

Need  $\text{CO}_2$  on right  $\Rightarrow$  keep (1) as is.

Need NO on left  $\Rightarrow$  reverse (2) and take  $\frac{1}{2}$ :



**Adding:**



$$\Delta H = -283.0 + (-90.3)$$

$$\boxed{\Delta H = -373.3 \text{ kJ}}$$

The large negative  $\Delta H$  confirms this reaction is strongly **exothermic** and thermodynamically favorable for reducing auto exhaust pollutants.

**Q8.** 0.5 g of a carbon sample is placed in a calorimeter with excess oxygen at  $25^\circ\text{C}$  and 1 atm pressure. On reaction, the calorimeter temperature rises from  $25^\circ\text{C}$  to  $25.89^\circ\text{C}$ . The heat capacity of the calorimeter is  $20.7 \text{ kJ}/^\circ\text{C}$ . Calculate the heat of reaction at  $25^\circ\text{C}$  and express the result as a thermochemical equation. **(8 marks)** [CHEM 113 | 2020–21]

**Answer**

**Heat of reaction of carbon combustion**

**Given:**

- Mass of carbon sample = 0.5 g;  $M_C = 12 \text{ g/mol}$
- Temperature rise:  $\Delta T = 25.89 - 25.00 = 0.89^\circ\text{C}$
- Heat capacity of calorimeter =  $20.7 \text{ kJ}/^\circ\text{C}$

**Step 1: Heat released:**

$$q_{cal} = C_{cal} \times \Delta T = 20.7 \times 0.89 = 18.42 \text{ kJ}$$

Since the calorimeter temperature **rose**, heat was **released** by the reaction:

$$q_{rxn} = -18.42 \text{ kJ}$$

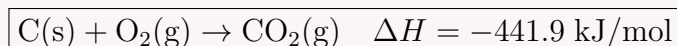
**Step 2: Moles of C burned:**

$$n_C = \frac{0.5}{12} = 0.04167 \text{ mol}$$

**Step 3: Heat of reaction per mole:**

$$\Delta H = \frac{q_{rxn}}{n_C} = \frac{-18.42}{0.04167} = -441.9 \text{ kJ/mol}$$

**Thermochemical equation:**



(Standard literature value:  $\Delta H_{comb}^C = -393.5 \text{ kJ/mol}$ . The difference may be due to incomplete combustion or calorimeter heat loss.)

**Q9.** How are the internal energies of chemical substances changed in a chemical reaction? State and explain the second law of thermodynamics. Under what conditions does a substance have a standard entropy of zero? **(20 marks)** [CHEM 113 | 2020–21 (Jan 21)]

**Answer**

**Internal energy in reactions; Second law; Standard entropy of zero**

**How internal energy changes in a chemical reaction:**

- The **internal energy** ( $U$ ) of a chemical system is the sum of all kinetic and potential energies of its molecules (translational, rotational, vibrational, electronic, nuclear).

- During a chemical reaction, bonds are broken and formed. The energy of the products differs from that of the reactants:

$$\Delta U = U_{\text{products}} - U_{\text{reactants}}$$

- At constant volume ( $\Delta V = 0$ ):

$$\Delta U = q_v \quad (\text{heat at constant volume})$$

- At constant pressure:

$$\Delta H = \Delta U + P\Delta V = \Delta U + \Delta n_{\text{gas}}RT$$

- Exothermic reactions:**  $\Delta U < 0$  — products have lower internal energy; energy released as heat.
- Endothermic reactions:**  $\Delta U > 0$  — products have higher internal energy; energy absorbed from surroundings.

### Second Law of Thermodynamics:

**Statement:** The total entropy of an isolated system always **increases** in any spontaneous (irreversible) process:

$$\Delta S_{\text{universe}} = \Delta S_{\text{system}} + \Delta S_{\text{surroundings}} \geq 0$$

- For a **reversible** process:  $\Delta S_{\text{universe}} = 0$
- For an **irreversible** (spontaneous) process:  $\Delta S_{\text{universe}} > 0$
- Entropy measures the degree of **disorder** or the number of microstates available to a system:

$$S = k_B \ln W$$

where  $W$  = number of microstates,  $k_B$  = Boltzmann constant.

- The second law explains why heat flows spontaneously from hot to cold, gases expand, and reactions proceed toward equilibrium.

### When is standard entropy zero? — Third Law:

**Third Law of Thermodynamics:** The entropy of a **perfect crystalline substance** is zero at **absolute zero** (0 K):

$$S = 0 \quad \text{when } T = 0 \text{ K and perfect crystal}$$

#### Conditions required:

- Temperature must be **exactly 0 K**.
- The substance must be a **perfect crystal** — all atoms/ molecules in perfectly ordered positions with no defects.
- $W = 1$  (only one microstate)  $\Rightarrow S = k_B \ln 1 = 0$ .

In practice, **residual entropy** may exist if there is any disorder remaining at 0 K (e.g., CO has residual entropy due to CO/OC orientational disorder).

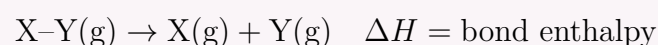
**Q10.** Explain why the bond enthalpy of a molecule is usually defined in terms of a gas-phase reaction. How would you determine the fuel values of different food items? **(15 marks)** [CHEM 113 | 2020–21 (Jan 21)]

### Answer

#### Bond enthalpy in gas phase; Fuel values of food items

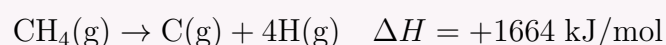
#### Why bond enthalpy is defined for gas-phase reactions:

- Isolated molecules:** In the gas phase, molecules are far apart and essentially **non-interacting**. The only forces present are intramolecular (within the molecule). Breaking a bond in the gas phase therefore gives a **pure measure** of the bond strength.
- No solvation effects:** In solution, breaking a bond is complicated by solute-solvent interactions (solvation energy). These interactions would add to or subtract from the true bond energy, giving an **inaccurate** measure.
- No lattice effects:** In solids, intermolecular/interionic forces also complicate the measurement.
- Clear reference state:** Gas-phase reaction:



gives a clean, reproducible value that can be tabulated and used in Hess's law calculations.

- Example:** Bond enthalpy of C–H:



Average C–H bond enthalpy =  $1664/4 = 416 \text{ kJ/mol}$ .



**Determining fuel values of food:**

The **fuel value** (calorific value) of a food is the heat released per gram of food upon complete combustion, measured in kJ/g or kcal/g.

**Method — Bomb Calorimeter:**

1. Weigh a precise amount of the food sample.
2. Place it in a **sealed bomb calorimeter** surrounded by water.
3. Ignite electrically under excess  $O_2$ .
4. Measure the **temperature rise**  $\Delta T$  of the water.
5. Calculate heat released:

$$q = (C_{cal} + m_{water}c_{water}) \times \Delta T$$

6. **Fuel value** =  $q$  / mass of food.

**Approximate fuel values:**

| Food component | kJ/g | kcal/g |
|----------------|------|--------|
| Carbohydrates  | 17.0 | 4.0    |
| Proteins       | 17.0 | 4.0    |
| Fats           | 38.0 | 9.0    |
| Alcohol        | 29.7 | 7.1    |

**Q11.** For years, chemists and physicists focused on enthalpy changes as a way to measure the spontaneity of a reaction. What arguments would you use to convince them not to use this method? **(9 marks)** [CHEM 113 | 2019–20]

**Answer**
**Arguments against using enthalpy as a measure of spontaneity**

**Historical view:** Chemists once believed that **exothermic reactions** ( $\Delta H < 0$ ) are spontaneous and endothermic ones are non-spontaneous (Berthelot's principle).

**Arguments against using  $\Delta H$  alone:**
**1. Endothermic spontaneous processes exist:**

Many processes with  $\Delta H > 0$  occur spontaneously:

- Melting of ice at  $> 0^\circ C$  ( $\Delta H = +6.01$  kJ/mol, spontaneous)
- Dissolution of  $NH_4NO_3$  in water (endothermic, yet spontaneous)
- Expansion of an ideal gas into vacuum ( $\Delta H = 0$ , spontaneous)

**2. Entropy ( $\Delta S$ ) is ignored:**

Spontaneity depends not only on energy but also on **disorder**. Many processes are driven by an **increase in entropy** even when  $\Delta H > 0$ .

Example: mixing of ideal gases has  $\Delta H = 0$  but  $\Delta S > 0 \Rightarrow$  spontaneous.

**3. Temperature dependence:**

A reaction may be non-spontaneous at low  $T$  but spontaneous at high  $T$  (or vice versa). Enthalpy alone cannot predict this.

**4. Gibbs free energy is the correct criterion:**

The true criterion for spontaneity at constant  $T$  and  $P$  is:

$$\Delta G = \Delta H - T\Delta S < 0 \Rightarrow \text{spontaneous}$$

Only when  $T\Delta S$  is negligible (low  $T$ , small  $\Delta S$ ) does  $\Delta H < 0$  reliably predict spontaneity.

**5. Counter-example table:**

| Process         | $\Delta H$ | $\Delta S$ | Spontaneous?             |
|-----------------|------------|------------|--------------------------|
| NaCl dissolving | +          | +          | Yes (at room $T$ )       |
| Ice melting     | +          | +          | Yes (above $0^\circ C$ ) |
| Exothermic rxn  | –          | –          | Depends on $T$           |
| Gas expansion   | 0          | +          | Always                   |

**Conclusion:**  $\Delta H$  alone is an **incomplete** criterion for spontaneity. The correct measure is  $\Delta G = \Delta H - T\Delta S$ .

**Q12.** Is the distance a person travels on a trip a state function? Why or why not? (6 marks)

[CHEM 113 | 2019–20]

### Answer

Is distance traveled a state function?

No, distance traveled is NOT a state function.

**Definition of a state function:**

A **state function** depends only on the **initial and final states** of a system, regardless of the path taken.

**Why distance is path-dependent:**

- Consider a person traveling from city A to city B (100 km apart in a straight line).
- If they take a **direct route**: total distance = 100 km.
- If they take a **detour** through city C: total distance = 150 km.
- Both trips start and end at the same points (A and B), but the **distance traveled differs** depending on the path.
- Therefore distance depends on the **path**  $\Rightarrow$  it is a **path function**, not a state function.

**Contrast with displacement (a state function):**

- Displacement** (straight-line distance from A to B) is always 100 km regardless of the path taken.
- Displacement depends only on initial and final positions  $\Rightarrow$  it IS a state function.

| Quantity          | Type           | Example            |
|-------------------|----------------|--------------------|
| Distance traveled | Path function  | Changes with route |
| Displacement      | State function | Fixed by A and B   |
| Heat ( $q$ )      | Path function  | Depends on process |
| Enthalpy ( $H$ )  | State function | Depends on state   |

**Analogy in thermodynamics:** Heat ( $q$ ) and work ( $w$ ) are **path functions** (like distance). Internal energy ( $U$ ), enthalpy ( $H$ ), and Gibbs free energy ( $G$ ) are **state functions** (like displacement).

**Q13.** Describe how a swinging pendulum that slows with time illustrates the first law of thermodynamics. (15 marks)

[CHEM 113 | 2019–20]

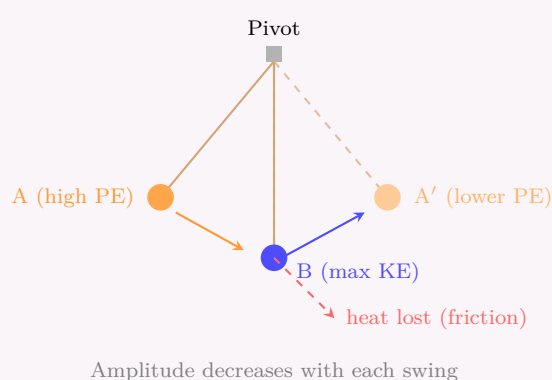
### Answer

**Swinging Pendulum and the First Law of Thermodynamics:**

$$\Delta U = q + w$$

Energy can neither be created nor destroyed; it can only be **converted** from one form to another. The total energy of an isolated system remains constant.

**The Swinging Pendulum:**



**Energy Analysis at Each Position:**

(a) **At position A (maximum height):**

KE = 0 (momentarily at rest)

PE = maximum =  $mgh$

Total mechanical energy =  $PE_{\max}$

(b) **At position B (bottom):**

PE = 0 (reference level)

KE = maximum =  $\frac{1}{2}mv^2$

Total mechanical energy =  $KE_{\max}$   
 By energy conservation (frictionless):  $PE_A = KE_B$

(c) **As pendulum slows (real case):**

Friction (air resistance, pivot) causes amplitude to **decrease** with each swing.  
 Mechanical energy converts to **thermal energy**.

**First Law Illustration:**

- Energy is **not destroyed** — it is converted:  $PE \rightarrow KE \rightarrow \text{Thermal energy}$ .
- Total energy of system + surroundings stays **constant**:

$$\Delta U_{\text{system}} + \Delta U_{\text{surroundings}} = 0$$

- Heat generated by friction equals mechanical energy lost:

$$q_{\text{generated}} = \Delta PE_{\text{lost}} = mg(h_A - h_{A'})$$

- This demonstrates: **energy is conserved, never created or destroyed — only transformed.**

| Stage          | Energy form      | Change    |
|----------------|------------------|-----------|
| At height A    | Potential energy | Maximum   |
| At bottom B    | Kinetic energy   | Maximum   |
| After friction | Thermal (heat)   | Increases |
| Overall        | Total energy     | Conserved |

**Q14.** How is the internal energy of a system changed during a chemical reaction? Explain why internal energy and Gibbs free energy are considered as state functions. **(8 marks)** [CHEM 113 | 2017–18]

**Answer**

**Internal energy changes; Enthalpy and Gibbs free energy as state functions**

**How internal energy changes in a reaction:**

- Internal energy  $U$  = total energy stored in bonds + kinetic energy of molecules.
- In a reaction: bonds break (energy input) and new bonds form (energy released).
- $\Delta U = U_{\text{products}} - U_{\text{reactants}}$
- If products are more stable:  $\Delta U < 0$  (exothermic at constant  $V$ ).
- Measured as heat at constant volume:  $q_V = \Delta U$ .

**Why enthalpy ( $H$ ) is a state function:**

$$H = U + PV$$

$U$ ,  $P$ ,  $V$  are all state functions. A sum/product of state functions is also a state function. Therefore  $H$  is a state function:  $\Delta H = H_{\text{final}} - H_{\text{initial}}$  depends only on initial and final states.

**Proof via Hess's Law:** The enthalpy of a reaction is the same whether it occurs in one step or multiple steps — this is only possible if  $H$  is path-independent (i.e., a state function).

**Why Gibbs free energy ( $G$ ) is a state function:**

$$G = H - TS$$

$H$ ,  $T$ ,  $S$  are all state functions. Therefore  $G = H - TS$  is also a state function:  $\Delta G = G_{\text{final}} - G_{\text{initial}}$ .

- $\Delta G < 0$ : spontaneous process
- $\Delta G = 0$ : equilibrium
- $\Delta G > 0$ : non-spontaneous

Since  $\Delta G$  depends only on initial and final states (not on path), it is a reliable criterion for spontaneity at constant  $T$  and  $P$ .

**Q15.** How does temperature affect reaction enthalpies? Derive the Kirchhoff's equation for the temperature dependence of reaction enthalpies. **(12 marks)** [CHEM 113 | 2017–18]

**Answer**
**Temperature dependence of reaction enthalpies; Kirchhoff's equation**
**Why temperature affects reaction enthalpies:**

The enthalpy of a reaction changes with temperature because the **heat capacities** of reactants and products differ. As temperature increases, each substance absorbs heat differently, so the enthalpy difference between products and reactants changes.

**Derivation of Kirchhoff's Equation:**

For a reaction: Reactants  $\rightarrow$  Products

At constant pressure, for a pure substance:

$$\left(\frac{\partial H}{\partial T}\right)_P = C_P$$

For the reaction:

$$\left(\frac{\partial \Delta H}{\partial T}\right)_P = \Delta C_P = \sum C_{P(\text{products})} - \sum C_{P(\text{reactants})}$$

Integrating from  $T_1$  to  $T_2$ :

$$\int_{\Delta H_1}^{\Delta H_2} d(\Delta H) = \int_{T_1}^{T_2} \Delta C_P dT$$

$$\Delta H_2 = \Delta H_1 + \int_{T_1}^{T_2} \Delta C_P dT$$

This is **Kirchhoff's equation**.

**For constant  $\Delta C_P$ :**

$$\Delta H_2 = \Delta H_1 + \Delta C_P(T_2 - T_1)$$

where:

$$\Delta C_P = \sum n_i C_{P,i}(\text{products}) - \sum n_j C_{P,j}(\text{reactants})$$

**If  $C_P$  varies with  $T$ :** Express  $C_P = a + bT + cT^2$  and integrate term by term.

| Condition                   | Kirchhoff's equation                                       |
|-----------------------------|------------------------------------------------------------|
| $\Delta C_P = \text{const}$ | $\Delta H_2 = \Delta H_1 + \Delta C_P(T_2 - T_1)$          |
| $\Delta C_P = f(T)$         | $\Delta H_2 = \Delta H_1 + \int_{T_1}^{T_2} \Delta C_P dT$ |

**Q16.** Explain why enthalpy is a state function and why it is an extensive property. (8 marks)

[CHEM 113 | 2015–16]

**Answer**
**Enthalpy as state function and extensive property**
**Enthalpy is a state function:**

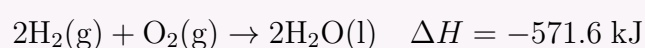
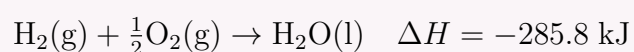
$$H = U + PV$$

- $U$ ,  $P$ ,  $V$  are all state functions (they depend only on the current state of the system).
- A mathematical combination of state functions is itself a state function.
- Therefore  $H$  is a state function:  $\Delta H$  depends only on initial and final states, not on the path.
- Proof:** Hess's Law — the enthalpy change of any reaction is path-independent. Example:  $\text{CO}_2$  can be formed directly or via  $\text{CO}$ ;  $\Delta H$  is the same both ways.

**Enthalpy is an extensive property:**

An **extensive property** depends on the **amount of matter** in the system (e.g., mass, volume, internal energy).

- $H = U + PV$ : all three terms ( $U$ ,  $P$ ,  $V$ ) scale with the size of the system. Doubling the amount of substance doubles  $U$  and  $V \Rightarrow$  doubles  $H$ .
- Example:**



Doubling the moles doubles  $\Delta H \Rightarrow$  **extensive**.

- Contrast with **intensive properties** (independent of amount): temperature, pressure, density, molar enthalpy.

**Q17.** What is pressure-volume work? How is this work related to enthalpy? (8 marks)

[CHEM 113 | 2015–16]

**Answer**
**Pressure-volume work and its relation to enthalpy**
**Pressure-Volume (PV) Work:**

When a system expands or is compressed against an external pressure  $P_{ext}$ , **work** is done:

$$w = -P_{ext} \Delta V \quad (\text{expansion work})$$

- **Expansion** ( $\Delta V > 0$ ): system does work on surroundings;  $w < 0$  (work leaves system).
- **Compression** ( $\Delta V < 0$ ): surroundings do work on system;  $w > 0$  (work enters system).
- At constant external pressure (irreversible):  $w = -P_{ext}\Delta V$
- For a reversible process:  $w = -\int P dV$

**Relation to Enthalpy:**

From the first law at constant pressure:

$$\Delta U = q_P + w = q_P - P\Delta V$$

$$q_P = \Delta U + P\Delta V$$

Since enthalpy is defined as  $H = U + PV$ :

$$\Delta H = \Delta U + \Delta(PV) = \Delta U + P\Delta V \quad (\text{at constant P})$$

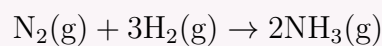
Therefore:

$$\boxed{\Delta H = q_P = \Delta U + P\Delta V}$$

- $\Delta H$  is the heat exchanged at **constant pressure**.
- The difference between  $\Delta H$  and  $\Delta U$  is the **PV work** done by or on the system.
- For reactions involving gases:

$$\Delta H - \Delta U = \Delta n_{gas}RT$$

where  $\Delta n_{gas}$  = change in moles of gas.

**Example:**


$$\Delta n_{gas} = 2 - (1 + 3) = -2$$

$$\Delta H = \Delta U + (-2)RT = \Delta U - 2RT$$

**Q18.** Given the following hypothetical thermochemical equations:  $\text{A} + \text{B} \rightarrow 2\text{C}$ ;  $\Delta H = -447 \text{ kJ}$ ;  $\text{A} + 3\text{D} \rightarrow 2\text{E}$ ;  $\Delta H = -484 \text{ kJ}$ ;  $2\text{D} + \text{B} \rightarrow 2\text{F}$ ;  $\Delta H = -429 \text{ kJ}$ . Calculate  $\Delta H$  for the equation:  $4\text{E} + 5\text{B} \rightarrow 4\text{C} + 6\text{F}$ . **(9 marks)** [CHEM 113 | 2015–16]

**Answer**

**Target:**  $4\text{E} + 5\text{B} \rightarrow 4\text{C} + 6\text{F}$

**Given reactions:**


**Finding multipliers via Hess's Law:**

Let the target be obtained by  $x_1(1) + x_2(2) + x_3(3)$ . Matching stoichiometry of each species:

$$\text{C: } -2x_1 = -4 \Rightarrow x_1 = 2$$

$$\text{E: } -2x_2 = +4 \Rightarrow x_2 = -2$$

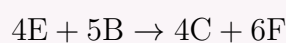
$$\text{F: } -2x_3 = -6 \Rightarrow x_3 = 3$$

Verification — A:  $x_1 + x_2 = 2 - 2 = 0$  ✓ D:  $3x_2 + 2x_3 = -6 + 6 = 0$  ✓ B:  $x_1 + x_3 = 2 + 3 = 5$  ✓

**Combining:**



Cancel 2A and 6D from both sides:



$$\Delta H = -894 + 968 - 1287 = \boxed{-1213 \text{ kJ}}$$

**Q19.** Why is the Kirchhoff's equation so important in thermochemistry? Calculate the heat of formation of  $\text{H}_2\text{O}(\text{l})$  at  $60^\circ\text{C}$  if the standard heat of formation is 68.370 Kcal. ( $C_p$  of  $\text{H}_2$ : 6.90,  $\text{O}_2$ : 7.05,  $\text{H}_2\text{O}(\text{l})$ : 18.0 cal·mol<sup>-1</sup>·deg<sup>-1</sup>.) **(10 marks)** [CHEM 113 | 2015–16]

### Answer

**Kirchhoff's equation; heat of formation of  $\text{H}_2\text{O}$  at  $60^\circ\text{C}$**

**Importance of Kirchhoff's equation:**

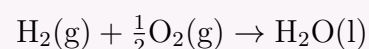
- Most thermochemical data are tabulated at  **$25^\circ\text{C}$  (298 K)**, but reactions often occur at other temperatures.
- Kirchhoff's equation allows calculation of  $\Delta H$  at **any temperature** from the known value at 298 K:

$$\Delta H_{T_2} = \Delta H_{T_1} + \int_{T_1}^{T_2} \Delta C_P dT$$

- It is essential in industrial processes where reactions occur at high temperatures (e.g., Haber process, combustion).
- Avoids expensive re-measurement of  $\Delta H$  at every temperature.

**Calculation: Heat of formation of  $\text{H}_2\text{O}(\text{l})$  at  $60^\circ\text{C}$**

**Reaction:**



**Given:**

- $\Delta H_{298}^\circ = -68.370 \text{ kcal}$
- $C_P(\text{H}_2) = 6.90 \text{ cal/mol}\cdot\text{deg}$
- $C_P(\text{O}_2) = 7.05 \text{ cal/mol}\cdot\text{deg}$
- $C_P(\text{H}_2\text{O}, \text{l}) = 18.0 \text{ cal/mol}\cdot\text{deg}$
- $T_1 = 25^\circ\text{C} = 298 \text{ K}$ ;  $T_2 = 60^\circ\text{C} = 333 \text{ K}$
- $\Delta T = 333 - 298 = 35 \text{ K}$

**Calculate  $\Delta C_P$ :**

$$\begin{aligned} \Delta C_P &= C_P(\text{H}_2\text{O}) - C_P(\text{H}_2) - \frac{1}{2}C_P(\text{O}_2) \\ &= 18.0 - 6.90 - \frac{1}{2}(7.05) = 18.0 - 6.90 - 3.525 = 7.575 \text{ cal/deg} \end{aligned}$$

**Kirchhoff's equation:**

$$\begin{aligned} \Delta H_{333} &= \Delta H_{298} + \Delta C_P \times \Delta T \\ &= -68370 + 7.575 \times 35 \\ &= -68370 + 265.1 \end{aligned}$$

$$\Delta H_{60^\circ\text{C}} = -68104.9 \text{ cal/mol} \approx -68.10 \text{ kcal/mol}$$

The heat of formation is slightly **less negative** at  $60^\circ\text{C}$  than at  $25^\circ\text{C}$  because  $\Delta C_P > 0$  (products absorb heat faster than reactants).

**Q20.** Define these terms: Open system, closed system, isolated system, exothermic process, endothermic process. Explain the sign conventions in the equation  $\Delta E = q + W$ . **(10 marks)** [CHEM 113 | 2014–15]

### Answer

**Definitions and sign conventions in  $\Delta E = q + w$**

**Definitions:**

- **Open system:** Can exchange both **matter and energy** with surroundings.  
Example: boiling water in an open pot.
- **Closed system:** Can exchange **energy but not matter** with surroundings.  
Example: gas in a piston-cylinder (sealed).
- **Isolated system:** Exchanges **neither matter nor energy** with surroundings.  
Example: ideal thermos flask.
- **Exothermic process:** Heat is **released** by the system to surroundings.  $\Delta H < 0$ ,  $q < 0$ .  
Example: combustion of  $\text{CH}_4$ , neutralization of acid-base.



- **Endothermic process:** Heat is **absorbed** by the system from surroundings.  $\Delta H > 0$ ,  $q > 0$ .  
Example: melting of ice, decomposition of  $\text{CaCO}_3$ .

Sign conventions in  $\Delta E = q + w$ :

$$\Delta U = q + w$$

| Quantity       | Sign     | Meaning                   | Example             |
|----------------|----------|---------------------------|---------------------|
| $q > 0$        | positive | Heat absorbed by system   | Endothermic         |
| $q < 0$        | negative | Heat released by system   | Exothermic          |
| $w > 0$        | positive | Work done on system       | Compression         |
| $w < 0$        | negative | Work done by system       | Expansion           |
| $\Delta U > 0$ | positive | Internal energy increases | System gains energy |
| $\Delta U < 0$ | negative | Internal energy decreases | System loses energy |

**Example:** A gas absorbs 500 J of heat and does 200 J of work on its surroundings:

$$q = +500 \text{ J}, \quad w = -200 \text{ J}$$

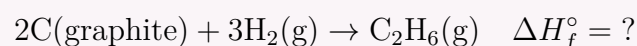
$$\Delta U = 500 + (-200) = +300 \text{ J}$$

**Q21.** From the following data, calculate the enthalpy change for the reaction:  $2\text{C}(\text{graphite}) + 3\text{H}_2(\text{g}) \rightarrow \text{C}_2\text{H}_6(\text{g})$ . Use:  $\text{C}(\text{graphite}) + \text{O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g})$ ;  $\Delta H = -393.5 \text{ kJ/mol}$ ;  $\text{H}_2(\text{g}) + \frac{1}{2}\text{O}_2(\text{g}) \rightarrow \text{H}_2\text{O}(\text{l})$ ;  $\Delta H = -285.8 \text{ kJ/mol}$ ;  $2\text{C}_2\text{H}_6(\text{g}) + 7\text{O}_2(\text{g}) \rightarrow 4\text{CO}_2(\text{g}) + 6\text{H}_2\text{O}(\text{l})$ ;  $\Delta H = -3119.6 \text{ kJ/mol}$ . **(8 marks)** [CHEM 113 | 2014–15]

#### Answer

**Enthalpy change for formation of  $\text{C}_2\text{H}_6$**

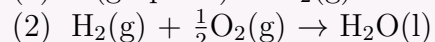
**Target:**



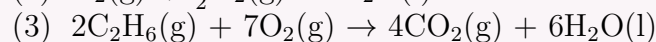
**Given:**



$$\Delta H_1 = -393.5 \text{ kJ/mol}$$



$$\Delta H_2 = -285.8 \text{ kJ/mol}$$



$$\Delta H_3 = -3119.6 \text{ kJ/mol}$$

**Applying Hess's Law:**

Need 2C on left  $\Rightarrow 2 \times (1)$

Need 3H<sub>2</sub> on left  $\Rightarrow 3 \times (2)$

Need C<sub>2</sub>H<sub>6</sub> on right  $\Rightarrow -\frac{1}{2} \times (3)$

$$2 \times (1) : \quad 2\text{C} + 2\text{O}_2 \rightarrow 2\text{CO}_2 \quad \Delta H = 2(-393.5) = -787.0 \text{ kJ}$$

$$3 \times (2) : \quad 3\text{H}_2 + \frac{3}{2}\text{O}_2 \rightarrow 3\text{H}_2\text{O} \quad \Delta H = 3(-285.8) = -857.4 \text{ kJ}$$

$$-\frac{1}{2} \times (3) : \quad 2\text{CO}_2 + 3\text{H}_2\text{O} \rightarrow \text{C}_2\text{H}_6 + \frac{7}{2}\text{O}_2 \quad \Delta H = +\frac{3119.6}{2} = +1559.8 \text{ kJ}$$

**Adding (CO<sub>2</sub>, H<sub>2</sub>O, O<sub>2</sub> cancel):**

$$\Delta H_f^\circ = -787.0 + (-857.4) + 1559.8$$

$$\boxed{\Delta H_f^\circ(\text{C}_2\text{H}_6) = -84.6 \text{ kJ/mol}}$$

**Q22.** ‘Chemical energy is a form of potential energy’ — justify the statement. **(8 marks)**

[CHEM 113 | 2013–14]

#### Answer

**Chemical energy as potential energy**

**Justification:**

**Potential energy** is stored energy that depends on the **position or configuration** of particles. Chemical energy is a form of potential energy because:

**1. Stored in chemical bonds:**

Chemical energy is stored in the **arrangement of atoms and electrons** within molecules. The electrons are held in place by electrostatic attractions between nuclei and electrons — this is a form of **electrostatic potential energy**.

**2. Released/absorbed on bond breaking and forming:**

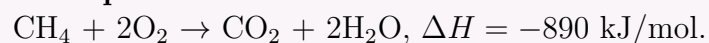
When bonds break and reform in a chemical reaction, the difference in potential energy between reactants and products is

released as heat or light (exothermic) or absorbed (endothermic). This is analogous to gravitational potential energy being released as kinetic energy when an object falls.

### 3. Analogy with gravitational PE:

- Ball at height  $h$ :  $PE = mgh$  (stored by position in gravitational field)
- Molecule: PE stored by electron positions in the electrostatic field of nuclei
- In both cases, the energy is **latent** until released by a change in configuration.

### 4. Example:



The reactants have higher chemical PE than the products. The difference is released as heat — confirming chemical energy is stored PE.

### 5. Internal energy and PE:

The internal energy  $U$  of a molecule has a potential energy component (from intermolecular forces and bond energies) and a kinetic energy component (from molecular motion). The **chemical potential energy** specifically refers to the energy stored in bonds.

**Conclusion:** Chemical energy satisfies all criteria of potential energy — it is stored by virtue of the **configuration of charged particles** (electrons and nuclei) and is released when that configuration changes in a chemical reaction.

**Q23.** What is a fuel? Discuss the prospect of hydrogen as next generation fuel. **(22 marks)**

[CHEM 113 | 2013–14]

### Answer

#### Fuel and hydrogen as next generation fuel

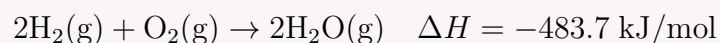
##### What is a fuel?

A **fuel** is a substance that stores energy in chemical bonds and releases it as **heat and/or light** upon combustion (reaction with an oxidizer, usually oxygen). Good fuels have:

- High energy density (kJ/kg or kJ/L)
- Low ignition temperature
- Abundant and affordable supply
- Clean combustion products

##### Hydrogen as next generation fuel:

##### Combustion reaction:



##### Advantages:

##### 1. Highest gravimetric energy density:

$\text{H}_2$ : 142 MJ/kg vs petrol: 44 MJ/kg — over 3 times more energy per kg.

##### 2. Zero carbon emissions:

Only product is water vapour — no  $\text{CO}_2$ , CO, or particulates. Ideal for combating climate change.

##### 3. Abundantly available:

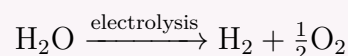
Hydrogen is the most abundant element in the universe. Can be produced from water by electrolysis using renewable energy (green hydrogen).

##### 4. Versatile:

Can be used in fuel cells (direct electricity generation), internal combustion engines, and gas turbines.

##### 5. Renewable:

When produced from solar/wind powered electrolysis:



This is a completely sustainable energy cycle.

##### Challenges:

- Storage:**  $\text{H}_2$  has very low density at room temperature; requires high-pressure cylinders or cryogenic storage ( $-253^\circ\text{C}$ ).
- Production:** Currently 95% of  $\text{H}_2$  is produced from natural gas (grey hydrogen, emits  $\text{CO}_2$ ). Green hydrogen is expensive.
- Safety:**  $\text{H}_2$  is highly flammable (4–75% flammability range in air) and requires special leak-proof infrastructure.
- Infrastructure:** Pipelines, refueling stations, and fuel cells need large-scale investment.

| Property               | H <sub>2</sub>   | Petrol                             |
|------------------------|------------------|------------------------------------|
| Energy density (MJ/kg) | 142              | 44                                 |
| Combustion product     | H <sub>2</sub> O | CO <sub>2</sub> + H <sub>2</sub> O |
| Carbon emission        | Zero             | High                               |
| Storage difficulty     | High             | Low                                |

**Conclusion:** Hydrogen is a strong candidate for next generation fuel due to its **high energy density and zero carbon emissions**, provided the challenges of production, storage and distribution are overcome using green technology.

**Q24.** What do you mean by the term “thermodynamic equilibrium constant”? How is the equilibrium constant of a reaction related to the free energy change? **(10 marks)** [CHEM 113 | 2013–14]

Answer

**Thermodynamic equilibrium constant and its relation to free energy**

**Thermodynamic equilibrium constant ( $K$ ):**  
The **thermodynamic equilibrium constant** is defined in terms of the **activities** ( $a_i$ ) of the species at equilibrium:

$$K = \frac{a_C^c \cdot a_D^d}{a_A^a \cdot a_B^b}$$

where for ideal gases:  $a_i = P_i/P^\circ$  (fugacity ratio); for solutions:  $a_i = c_i/c^\circ$ .

**Key features:**

- $K$  is **dimensionless**.
- Depends only on **temperature**.
- Independent of pressure, concentration, or catalyst.
- Related to standard free energy change.

**Relation to free energy:**  
At any composition:

$$\Delta G = \Delta G^\circ + RT \ln Q$$

At equilibrium ( $\Delta G = 0$ ,  $Q = K$ ):

$$\Delta G^\circ = -RT \ln K$$

$$K = e^{-\Delta G^\circ / RT}$$

**Physical interpretation:**

| $\Delta G^\circ$ | $K$       | Interpretation           |
|------------------|-----------|--------------------------|
| Large negative   | $K \gg 1$ | Reaction nearly complete |
| Zero             | $K = 1$   | Equal amounts at equil.  |
| Large positive   | $K \ll 1$ | Reaction barely proceeds |

**Q25.** Derive an expression for the variation of equilibrium constant with temperature. **(16 marks)** [CHEM 113 | 2013–14]

Answer

**Variation of equilibrium constant with temperature — derivation and significance**

**Derivation (van’t Hoff equation):**  
From  $\Delta G^\circ = -RT \ln K$  and  $\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$ :

$$\begin{aligned} -RT \ln K &= \Delta H^\circ - T\Delta S^\circ \\ \ln K &= -\frac{\Delta H^\circ}{RT} + \frac{\Delta S^\circ}{R} \end{aligned}$$

Differentiating with respect to  $T$ :

$$\frac{d \ln K}{dT} = \frac{\Delta H^\circ}{RT^2}$$

This is the **van’t Hoff equation**.  
Integrating between  $T_1$  and  $T_2$  (assuming  $\Delta H^\circ$  constant):

$$\ln \frac{K_2}{K_1} = \frac{\Delta H^\circ}{R} \left( \frac{1}{T_1} - \frac{1}{T_2} \right)$$

Or equivalently:

$$\log \frac{K_2}{K_1} = \frac{\Delta H^\circ}{2.303R} \left( \frac{T_2 - T_1}{T_1 T_2} \right)$$

**Significance:**

1. Predicts whether  $K$  increases or decreases with  $T$ .
2. Allows calculation of  $\Delta H^\circ$  from  $K$  values at two temperatures.
3. Explains Le Chatelier's principle quantitatively.
4. Plot of  $\ln K$  vs  $1/T$ : slope =  $-\Delta H^\circ/R$  gives  $\Delta H^\circ$  experimentally.
5. Used in industrial processes to select optimal operating temperature.

| Reaction type | $\Delta H^\circ$ | Effect of $\uparrow T$ |
|---------------|------------------|------------------------|
| Exothermic    | $< 0$            | $K$ decreases          |
| Endothermic   | $> 0$            | $K$ increases          |

**Q26.** Define 'chemical potential' and 'free energy'. (8 marks)

[CHEM 113 | 2012–13]

**Answer**

**Chemical potential and Gibbs free energy**

**Gibbs Free Energy ( $G$ ):**

$$G = H - TS \quad \Rightarrow \quad \Delta G = \Delta H - T\Delta S$$

- $G$  is a **state function**.
- At constant  $T$  and  $P$ :  $\Delta G < 0$  (spontaneous);  $\Delta G = 0$  (equilibrium);  $\Delta G > 0$  (non-spontaneous).
- The **maximum non-PV work** a system can do equals  $-\Delta G$ .
- Standard Gibbs free energy:  $\Delta G^\circ = -RT \ln K$ .

**Chemical Potential ( $\mu$ ):**

The **chemical potential** of a species  $i$  is the **partial molar Gibbs free energy**:

$$\mu_i = \left( \frac{\partial G}{\partial n_i} \right)_{T, P, n_{j \neq i}}$$

- $\mu_i$  represents the change in  $G$  when one mole of species  $i$  is added to the system at constant  $T$ ,  $P$ , and composition.
- For an ideal gas:

$$\mu_i = \mu_i^\circ + RT \ln \frac{P_i}{P^\circ}$$

- **Direction of transfer:** Species moves spontaneously from high  $\mu$  to low  $\mu$  (analogous to water flowing downhill).
- At equilibrium: chemical potential of each species is equal in all phases:

$$\mu_i(\text{phase 1}) = \mu_i(\text{phase 2})$$

**Relation between  $\mu$  and  $G$ :**

For a pure substance:  $\mu = G_m$  (molar Gibbs free energy).

For a mixture:  $G = \sum_i n_i \mu_i$ .

**Q27.** What are the characteristic features of the thermodynamic equilibrium constant? (6 marks)

[CHEM 113 | 2012–13]

**Answer**

**Characteristic features of thermodynamic equilibrium constant**

1. **Dimensionless:**  $K$  is expressed in terms of activities (ratios to standard state) and has no units.
2. **Temperature dependent only:**  $K$  depends only on temperature ( $\Delta G^\circ = -RT \ln K$ ). It is **independent** of pressure, concentration, or catalyst.
3. **Fixed for a given reaction at a given temperature:** At fixed  $T$ ,  $K$  has a unique value regardless of how equilibrium is reached.

**4. Related to standard free energy:**

$$\Delta G^\circ = -RT \ln K$$

Negative  $\Delta G^\circ \Rightarrow K > 1$  (products favored).

**5. Changes with stoichiometry:** If the equation is multiplied by  $n$ ,  $K$  becomes  $K^n$ . If the equation is reversed,  $K$  becomes  $1/K$ .

**6. Product of step equilibrium constants:** If a reaction is the sum of two steps,  $K = K_1 \times K_2$  (Hess's law analog for  $K$ ).

**7. Predicts direction:** Comparing  $Q$  (reaction quotient) to  $K$ :  $Q < K \Rightarrow$  forward reaction;  $Q > K \Rightarrow$  reverse reaction;  $Q = K \Rightarrow$  equilibrium.

**Q28.** The value of  $K_p$  for a reaction is  $1.06 \times 10^6$  at  $25^\circ\text{C}$ . Calculate the standard free energy change for the reaction. **(7 marks)** [CHEM 113 | 2012–13]

**Answer**

$\Delta G^\circ$  for reaction with  $K_P = 1.06 \times 10^6$  at  $25^\circ\text{C}$

**Given:**

- $K_P = 1.06 \times 10^6$
- $T = 25^\circ\text{C} = 298\text{ K}$
- $R = 8.314\text{ J/mol}\cdot\text{K}$

**Using:**

$$\Delta G^\circ = -RT \ln K_P$$

$$\ln K_P = \ln(1.06 \times 10^6) = \ln(1.06) + 6 \ln(10) = 0.0583 + 13.816 = 13.874$$

$$\Delta G^\circ = -8.314 \times 298 \times 13.874$$

$$= -2477.6 \times 13.874$$

$$\Delta G^\circ = -34381\text{ J/mol} \approx -34.4\text{ kJ/mol}$$

**Comment:** The large negative value of  $\Delta G^\circ$  and  $K_P \gg 1$  confirm that the reaction is **strongly spontaneous** in the forward direction at  $25^\circ\text{C}$ , with products strongly favored at equilibrium.

**Q29.** The heat of reaction of  $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$  at  $27^\circ\text{C}$  was found to be  $-21.97\text{ Kcal}$ . What will be the heat of reaction at  $100^\circ\text{C}$ ? The molar heat capacities at constant pressure for nitrogen, hydrogen and ammonia are 6.80, 6.77 and 8.86 cal/mol·deg respectively. **(7 marks)** [CHEM 113 | 2011–12]

**Answer**

**Heat of reaction for  $\text{N}_2 + 3\text{H}_2 \rightleftharpoons 2\text{NH}_3$  at  $100^\circ\text{C}$**

**Given:**

- $\Delta H_{300} = -21.97\text{ kcal}$  (at  $27^\circ\text{C} = 300\text{ K}$ )
- $T_1 = 300\text{ K}$ ,  $T_2 = 373\text{ K}$
- $C_P(\text{N}_2) = 6.80\text{ cal/mol}\cdot\text{deg}$
- $C_P(\text{H}_2) = 6.77\text{ cal/mol}\cdot\text{deg}$
- $C_P(\text{NH}_3) = 8.86\text{ cal/mol}\cdot\text{deg}$

**Calculate  $\Delta C_P$ :**

$$\begin{aligned} \Delta C_P &= 2C_P(\text{NH}_3) - C_P(\text{N}_2) - 3C_P(\text{H}_2) \\ &= 2(8.86) - 6.80 - 3(6.77) = 17.72 - 6.80 - 20.31 = -9.39\text{ cal/deg} \end{aligned}$$

**Kirchhoff's equation:**

$$\begin{aligned} \Delta H_{373} &= \Delta H_{300} + \Delta C_P(T_2 - T_1) \\ &= -21970 + (-9.39)(373 - 300) \\ &= -21970 + (-9.39)(73) \\ &= -21970 - 685.5 \end{aligned}$$

$$\Delta H_{100^\circ\text{C}} = -22655.5 \text{ cal/mol} \approx -22.66 \text{ kcal/mol}$$

The reaction becomes **more exothermic** at higher temperature because  $\Delta C_P < 0$  (reactants have higher heat capacity than products).

**Q30.** What is internal energy of a system? How is it related to the enthalpy of the system? (11 marks)

[CHEM 113 | 2011–12]

### Answer

#### Internal energy and its relation to enthalpy

##### Internal Energy ( $U$ ):

The **internal energy** of a system is the sum of all forms of energy contained within it:

$$U = E_{\text{kinetic}} + E_{\text{potential}}$$

- **Kinetic contributions:** translational, rotational, vibrational motion of molecules.
- **Potential contributions:** intermolecular forces, bond energies, electronic energy, nuclear energy.
- $U$  is a **state function** — depends only on the current state ( $T$ ,  $P$ , composition).
- Absolute value of  $U$  cannot be measured; only **changes**  $\Delta U$  are measured.

##### Relation to enthalpy:

At constant pressure, from the first law:

$$q_P = \Delta U + P\Delta V$$

Enthalpy is defined as:

$$H = U + PV \quad \Rightarrow \quad \Delta H = \Delta U + \Delta(PV)$$

At constant pressure:

$$\Delta H = \Delta U + P\Delta V$$

For reactions involving gases,  $P\Delta V = \Delta n_{\text{gas}}RT$ :

$$\Delta H = \Delta U + \Delta n_{\text{gas}}RT$$

##### Physical meaning:

- $\Delta H > \Delta U$ : when  $\Delta n_{\text{gas}} > 0$  (more moles of gas produced, system does expansion work on surroundings).
- $\Delta H < \Delta U$ : when  $\Delta n_{\text{gas}} < 0$  (fewer moles of gas, surroundings do work on system).
- $\Delta H = \Delta U$ : when  $\Delta n_{\text{gas}} = 0$  or reaction involves only solids/liquids.

| Condition         | Relation                                        |
|-------------------|-------------------------------------------------|
| Constant volume   | $q_V = \Delta U$                                |
| Constant pressure | $q_P = \Delta H$                                |
| General           | $\Delta H = \Delta U + \Delta n_{\text{gas}}RT$ |

**Q31.** What are the factors affecting the enthalpy change of a reaction? (5 marks)

[CHEM 113 | 2011–12]

### Answer

#### Factors affecting the enthalpy change of a reaction

The enthalpy change  $\Delta H$  of a reaction depends on the following factors:

##### 1. Nature and amount of reactants and products:

$\Delta H$  depends on what substances react and how many moles. Doubling the moles doubles  $\Delta H$  (extensive property).

##### 2. Temperature (Kirchhoff's equation):

$$\Delta H_{T_2} = \Delta H_{T_1} + \int_{T_1}^{T_2} \Delta C_P dT$$

$\Delta H$  changes with temperature if  $\Delta C_P \neq 0$ .

##### 3. Physical state of reactants and products:

The state (solid, liquid, gas) affects  $\Delta H$  significantly because phase transitions involve enthalpy changes.

Example:  $\text{H}_2\text{O}(\text{l})$  vs  $\text{H}_2\text{O}(\text{g})$  differ by  $\Delta H_{\text{vap}} = 44 \text{ kJ/mol}$ .

##### 4. Allotropic form:

Different allotropes have different energies. Example:  $\text{C}(\text{graphite})$  and  $\text{C}(\text{diamond})$  have different  $\Delta H_f^\circ$ .



**5. Concentration/pressure:**

For reactions in solution, concentration affects activity; for gases, pressure affects the enthalpy through intermolecular interactions.

**Q32.** What is internal energy of a system? How is it related to the heat of the system at constant volume and at constant pressure? **(10 marks)** [CHEM 101 | 2010–11]

**Answer**
**Internal energy; relation to heat at constant volume and constant pressure**
**Internal Energy ( $U$ ):**

The internal energy of a system is the total energy stored within it — kinetic (molecular motion) and potential (bond energies, intermolecular forces). It is a **state function**.

**First Law of Thermodynamics:**

$$\Delta U = q + w$$

where  $q$  = heat exchanged,  $w$  = work done on system.

**At constant volume ( $\Delta V = 0$ ):**

$$w = -P\Delta V = 0 \Rightarrow \Delta U = q_V$$

$$\boxed{\Delta U = q_V}$$

The heat absorbed at **constant volume** directly equals the change in internal energy. Measured using a **bomb calorimeter**.

**At constant pressure:**

$$w = -P\Delta V \Rightarrow \Delta U = q_P - P\Delta V$$

$$q_P = \Delta U + P\Delta V = \Delta H$$

$$\boxed{\Delta H = q_P}$$

The heat absorbed at **constant pressure** equals the enthalpy change.

**Relation between  $q_V$  and  $q_P$ :**

$$q_P = q_V + P\Delta V = q_V + \Delta n_{gas}RT$$

| Condition         | Heat                           | Equals     |
|-------------------|--------------------------------|------------|
| Constant volume   | $q_V$                          | $\Delta U$ |
| Constant pressure | $q_P$                          | $\Delta H$ |
| Relation          | $q_P = q_V + \Delta n_{gas}RT$ | —          |

**Q33.** Derive Kirchhoff's equations at constant volume and at constant pressure. **(10 marks)**

[CHEM 101 | 2010–11]

**Answer**
**Derivation of Kirchhoff's equations at constant volume and constant pressure**
**At constant pressure (Kirchhoff's equation for  $\Delta H$ ):**

For a reaction at constant pressure, the enthalpy of each substance varies with temperature:

$$\left(\frac{\partial H}{\partial T}\right)_P = C_P$$

For the reaction Reactants  $\rightarrow$  Products:

$$\left(\frac{\partial \Delta H}{\partial T}\right)_P = \Delta C_P = \sum C_P(\text{products}) - \sum C_P(\text{reactants})$$

Integrating from  $T_1$  to  $T_2$ :

$$\boxed{\Delta H_{T_2} = \Delta H_{T_1} + \int_{T_1}^{T_2} \Delta C_P dT}$$

For constant  $\Delta C_P$ :

$$\Delta H_{T_2} = \Delta H_{T_1} + \Delta C_P(T_2 - T_1)$$

**At constant volume (Kirchhoff's equation for  $\Delta U$ ):**

Similarly, at constant volume:

$$\left(\frac{\partial U}{\partial T}\right)_V = C_V$$

$$\left(\frac{\partial \Delta U}{\partial T}\right)_V = \Delta C_V = \sum C_V(\text{products}) - \sum C_V(\text{reactants})$$

Integrating:

$$\Delta U_{T_2} = \Delta U_{T_1} + \int_{T_1}^{T_2} \Delta C_V dT$$

**Relation between  $\Delta C_P$  and  $\Delta C_V$ :**

For ideal gases:  $C_P - C_V = R$  per mole. Therefore:

$$\Delta C_P - \Delta C_V = \Delta n_{gas} \cdot R$$

**Summary:**

| Condition    | Kirchhoff equation                                                 |
|--------------|--------------------------------------------------------------------|
| Constant $P$ | $\Delta H_{T_2} = \Delta H_{T_1} + \int_{T_1}^{T_2} \Delta C_P dT$ |
| Constant $V$ | $\Delta U_{T_2} = \Delta U_{T_1} + \int_{T_1}^{T_2} \Delta C_V dT$ |

**Q34.** Explain the dynamic nature of equilibrium. (5 marks)

[CHEM 101 | 2010–11]

### Answer

#### Dynamic nature of equilibrium

At chemical equilibrium, the reaction does not **stop** — both forward and reverse reactions continue simultaneously at **equal rates**. This is the **dynamic nature** of equilibrium.

#### Key features:

- **Equal rates:** Rate of forward reaction = rate of reverse reaction. Macroscopic properties (concentration, color, pressure) remain **constant**.
- **Not static:** Molecules are continuously reacting in both directions. Reactants form products, and products reform reactants, at the same rate.
- **Experimental evidence:**  
If radioactive  $^{14}\text{C}$  is introduced into one side of an equilibrium, it distributes between reactants and products — proving reaction continues at equilibrium.
- **Example:**  $\text{N}_2\text{O}_4(\text{g}) \rightleftharpoons 2\text{NO}_2(\text{g})$   
At equilibrium,  $\text{N}_2\text{O}_4$  continues to dissociate at the same rate that  $\text{NO}_2$  recombines.

$$\text{rate}_{\text{forward}} = k_f[\text{N}_2\text{O}_4] = k_r[\text{NO}_2]^2 = \text{rate}_{\text{reverse}}$$

- **Effect of disturbance:** If equilibrium is disturbed (change in concentration,  $T$  or  $P$ ), the system responds (Le Chatelier's principle) to **restore a new equilibrium** — confirming the dynamic character.

**Q35.** Derive an expression relating the free energy change and the equilibrium constant of a reaction. (11 marks)

[CHEM 101 | 2010–11]

### Answer

#### Free energy change and equilibrium constant

#### Derivation:

For a reaction at constant  $T$  and  $P$ , the **Gibbs free energy change** is:

$$\Delta G = \Delta H - T\Delta S$$

For a reaction  $a\text{A} + b\text{B} \rightleftharpoons c\text{C} + d\text{D}$ , the free energy change at any composition is:

$$\Delta G = \Delta G^\circ + RT \ln Q$$

where  $Q$  is the **reaction quotient**:

$$Q = \frac{[\text{C}]^c[\text{D}]^d}{[\text{A}]^a[\text{B}]^b}$$

**At equilibrium:**  $\Delta G = 0$  and  $Q = K$  (equilibrium constant):

$$0 = \Delta G^\circ + RT \ln K$$

$$\Delta G^\circ = -RT \ln K$$

or equivalently:

$$K = e^{-\Delta G^\circ/RT}$$

**Significance:**

| $\Delta G^\circ$ | $K$     | Equilibrium       | Spontaneity         |
|------------------|---------|-------------------|---------------------|
| $< 0$            | $K > 1$ | Products favored  | Forward spontaneous |
| $= 0$            | $K = 1$ | Neither favored   | At equilibrium      |
| $> 0$            | $K < 1$ | Reactants favored | Reverse spontaneous |

**Q36.** Discuss the variation of equilibrium constant with temperature. (12 marks)

[CHEM 101 | 2010–11]

**Answer**

**Variation of equilibrium constant with temperature**

**Starting from:**

$$\Delta G^\circ = -RT \ln K \Rightarrow \ln K = -\frac{\Delta G^\circ}{RT} = -\frac{\Delta H^\circ}{RT} + \frac{\Delta S^\circ}{R}$$

Differentiating with respect to  $T$ :

$$\frac{d \ln K}{dT} = \frac{\Delta H^\circ}{RT^2}$$

This is the **van't Hoff equation**.

Integrating between  $T_1$  and  $T_2$ :

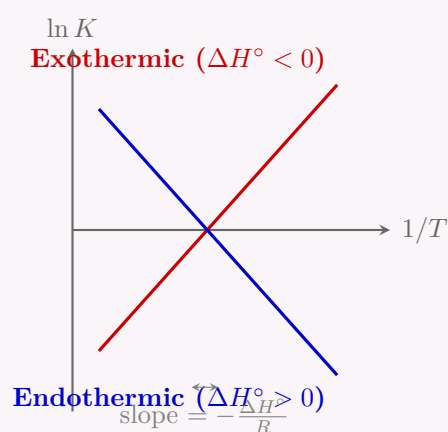
$$\int_{\ln K_1}^{\ln K_2} d \ln K = \int_{T_1}^{T_2} \frac{\Delta H^\circ}{RT^2} dT$$

Assuming  $\Delta H^\circ$  is constant:

$$\ln \frac{K_2}{K_1} = -\frac{\Delta H^\circ}{R} \left( \frac{1}{T_2} - \frac{1}{T_1} \right)$$

**Significance:**

- **Exothermic reaction** ( $\Delta H^\circ < 0$ ):  $K$  decreases as  $T$  increases — equilibrium shifts to reactants.
- **Endothermic reaction** ( $\Delta H^\circ > 0$ ):  $K$  increases as  $T$  increases — equilibrium shifts to products.
- Allows calculation of  $K$  at any temperature if  $K$  at one temperature and  $\Delta H^\circ$  are known.
- A plot of  $\ln K$  vs  $1/T$  gives a straight line with slope  $= -\Delta H^\circ/R$ .



**Q37.** The equilibrium constant for the dissociation  $2\text{H}_2\text{S}(\text{g}) \rightleftharpoons 2\text{H}_2(\text{g}) + \text{S}_2(\text{g})$  is 0.0118 at  $1065^\circ\text{C}$ , while the heat of dissociation is 42400 cal. Find the equilibrium constant at  $1200^\circ\text{C}$ . (7 marks)

[CHEM 101 | 2010–11]

**Answer**

**Equilibrium constant at  $1200^\circ\text{C}$  for  $2\text{H}_2\text{S} \rightleftharpoons 2\text{H}_2 + \text{S}_2$**

**Given:**

- $K_1 = 0.0118$  at  $T_1 = 1065^\circ\text{C} = 1338 \text{ K}$
- $\Delta H = 42400 \text{ cal/mol} = 42.4 \text{ kcal/mol}$
- $T_2 = 1200^\circ\text{C} = 1473 \text{ K}$
- $R = 1.987 \text{ cal/mol}\cdot\text{K}$

**Using van't Hoff equation:**

$$\begin{aligned}
 \ln \frac{K_2}{K_1} &= -\frac{\Delta H}{R} \left( \frac{1}{T_2} - \frac{1}{T_1} \right) \\
 &= -\frac{42400}{1.987} \left( \frac{1}{1473} - \frac{1}{1338} \right) \\
 &= -21340.7 \times (6.788 \times 10^{-4} - 7.474 \times 10^{-4}) \\
 &= -21340.7 \times (-6.86 \times 10^{-5}) \\
 &= +1.464 \\
 \frac{K_2}{K_1} &= e^{1.464} = 4.322 \\
 K_2 &= 4.322 \times 0.0118
 \end{aligned}$$

$$K_2 = 0.051 \text{ at } 1200^\circ\text{C}$$

Since the reaction is **endothermic** ( $\Delta H > 0$ ),  $K$  increases with temperature, confirming the result ( $K_2 > K_1$ ).

**Q38.** What do you mean by the term “dynamic equilibrium”? (6 marks)

[CHEM 101 | 2009–10]

**Q39.** The equilibrium constants  $K_p$  of the reaction  $\text{N}_2\text{O}_4(\text{g}) \rightleftharpoons 2\text{NO}_2(\text{g})$  are 0.141 and 2.80 at  $25^\circ\text{C}$  and  $65^\circ\text{C}$  respectively. Calculate the average heat of reaction and comment on the result. (10 marks)

[CHEM 101 | 2009–10]

#### Answer

**Average heat of reaction for  $\text{N}_2\text{O}_4 \rightleftharpoons 2\text{NO}_2$**

**Given:**

- $K_1 = 0.141$  at  $T_1 = 25^\circ\text{C} = 298 \text{ K}$
- $K_2 = 2.80$  at  $T_2 = 65^\circ\text{C} = 338 \text{ K}$
- $R = 8.314 \text{ J/mol}\cdot\text{K} = 1.987 \text{ cal/mol}\cdot\text{K}$

**Using van't Hoff equation:**

$$\begin{aligned}
 \ln \frac{K_2}{K_1} &= \frac{\Delta H^\circ}{R} \left( \frac{1}{T_1} - \frac{1}{T_2} \right) \\
 \ln \frac{2.80}{0.141} &= \frac{\Delta H^\circ}{8.314} \left( \frac{1}{298} - \frac{1}{338} \right) \\
 \ln(19.858) &= \frac{\Delta H^\circ}{8.314} \times \left( \frac{338 - 298}{298 \times 338} \right) \\
 2.989 &= \frac{\Delta H^\circ}{8.314} \times \frac{40}{100724} \\
 2.989 &= \frac{\Delta H^\circ}{8.314} \times 3.971 \times 10^{-4} \\
 \Delta H^\circ &= \frac{2.989 \times 8.314}{3.971 \times 10^{-4}} = \frac{24.85}{3.971 \times 10^{-4}} \\
 \Delta H^\circ &= 62578 \text{ J/mol} \approx 62.6 \text{ kJ/mol}
 \end{aligned}$$

**Comment:**

$\Delta H^\circ = +62.6 \text{ kJ/mol}$  is **positive** (endothermic), which is consistent with:

- $K$  increasing with temperature ( $K$  rose from 0.141 to 2.80 as  $T$  increased from 25 to  $65^\circ\text{C}$ ).
- Le Chatelier's principle: increasing  $T$  favors the endothermic direction (dissociation of  $\text{N}_2\text{O}_4$ ).

**Q40.** What is meant by chemical potential and Gibbs free energy? (6 marks)

[CHEM 101 | 2008–09]

#### Answer

**Chemical potential and Gibbs free energy**

**Gibbs Free Energy ( $G$ ):**

$$G = H - TS \Rightarrow \Delta G = \Delta H - T\Delta S$$

- $G$  is a **state function**.
- At constant  $T$  and  $P$ :  $\Delta G < 0$  (spontaneous);  $\Delta G = 0$  (equilibrium);  $\Delta G > 0$  (non-spontaneous).
- The **maximum non-PV work** a system can do equals  $-\Delta G$ .
- Standard Gibbs free energy:  $\Delta G^\circ = -RT \ln K$ .

#### Chemical Potential ( $\mu$ ):

The **chemical potential** of a species  $i$  is the **partial molar Gibbs free energy**:

$$\mu_i = \left( \frac{\partial G}{\partial n_i} \right)_{T, P, n_{j \neq i}}$$

- $\mu_i$  represents the change in  $G$  when one mole of species  $i$  is added to the system at constant  $T$ ,  $P$ , and composition.
- For an ideal gas:

$$\mu_i = \mu_i^\circ + RT \ln \frac{P_i}{P^\circ}$$

- **Direction of transfer:** Species moves spontaneously from high  $\mu$  to low  $\mu$  (analogous to water flowing downhill).
- At equilibrium: chemical potential of each species is equal in all phases:

$$\mu_i(\text{phase 1}) = \mu_i(\text{phase 2})$$

#### Relation between $\mu$ and $G$ :

For a pure substance:  $\mu = G_m$  (molar Gibbs free energy).

For a mixture:  $G = \sum_i n_i \mu_i$ .

**Q41.** Derive an equation relating the free energy change ( $\Delta G$ ) and equilibrium constant. Mention the significance of the obtained equation. Also explain the effect of temperature on the equilibrium constant. **(12 marks)** [CHEM 101 | 2008–09]

#### Answer

##### $\Delta G$ , $K$ , and effect of temperature on equilibrium constant

##### Derivation of $\Delta G = \Delta G^\circ + RT \ln Q$ :

For any species  $i$  at pressure  $P_i$ :

$$\mu_i = \mu_i^\circ + RT \ln \frac{P_i}{P^\circ}$$

For reaction  $aA + bB \rightleftharpoons cC + dD$ :

$$\begin{aligned} \Delta G &= c\mu_C + d\mu_D - a\mu_A - b\mu_B \\ &= \Delta G^\circ + RT \ln Q \end{aligned}$$

At equilibrium ( $\Delta G = 0$ ,  $Q = K$ ):

$$\boxed{\Delta G^\circ = -RT \ln K}$$

##### Significance:

- Links thermodynamic driving force ( $\Delta G^\circ$ ) to equilibrium position ( $K$ ).
- At any non-equilibrium state:  $\Delta G = RT \ln(Q/K)$  tells the direction of spontaneous change.
- Basis for calculating equilibrium compositions from  $\Delta H^\circ$  and  $\Delta S^\circ$  data.

##### Effect of temperature on $K$ :

From  $\ln K = -\Delta H^\circ/RT + \Delta S^\circ/R$ :

$$\frac{d \ln K}{dT} = \frac{\Delta H^\circ}{RT^2}$$

- **Exothermic** ( $\Delta H^\circ < 0$ ):  $K$  **decreases** with  $T$ .
- **Endothermic** ( $\Delta H^\circ > 0$ ):  $K$  **increases** with  $T$ .
- Integrated form (van't Hoff):

$$\ln \frac{K_2}{K_1} = \frac{\Delta H^\circ}{R} \left( \frac{1}{T_1} - \frac{1}{T_2} \right)$$

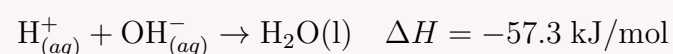
**Q42.** Explain the following with suitable examples: (i) Heat of neutralization, (ii) Heat of solution. **(6 marks)** [CHEM 101 | 2008–09]

### Answer

#### Heat of neutralization and heat of solution

##### (i) Heat of Neutralization:

The **heat of neutralization** is the heat released when one mole of water is formed by the reaction between one mole of  $\text{H}^+$  ions (from an acid) and one mole of  $\text{OH}^-$  ions (from a base) in dilute aqueous solution:

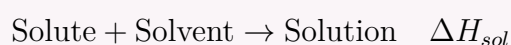


##### Key points:

- For **strong acid + strong base**:  $\Delta H \approx -57.3 \text{ kJ/mol}$  always (same reaction:  $\text{H}^+ + \text{OH}^- \rightarrow \text{H}_2\text{O}$ ).
- For **weak acid or weak base**:  $|\Delta H| < 57.3 \text{ kJ/mol}$  because energy is used to ionize the weak acid/base.
- Example:  $\text{HCl} + \text{NaOH} \rightarrow \text{NaCl} + \text{H}_2\text{O}$ ,  $\Delta H = -57.3 \text{ kJ/mol}$ .  
 $\text{CH}_3\text{COOH} + \text{NaOH} \rightarrow \text{CH}_3\text{COONa} + \text{H}_2\text{O}$ ,  $\Delta H = -55.9 \text{ kJ/mol}$ .

##### (ii) Heat of Solution:

The **heat of solution** is the enthalpy change when one mole of a substance dissolves in a large excess of solvent (to form an infinitely dilute solution):



##### Key points:

- Can be **exothermic** or **endothermic** depending on whether lattice energy or hydration energy dominates.
- Ionic solids**:  $\Delta H_{\text{sol}} = \Delta H_{\text{lattice}} + \Delta H_{\text{hydration}}$
- Example (exothermic):  $\text{CaCl}_2$  in water,  $\Delta H_{\text{sol}} = -82.8 \text{ kJ/mol}$ .  
 Example (endothermic):  $\text{NH}_4\text{NO}_3$  in water,  $\Delta H_{\text{sol}} = +25.7 \text{ kJ/mol}$ .

**Q43.** Define heat of combustion and show how heat of combustion can be measured using a bomb calorimeter. **(12 marks)** [CHEM 101 | 2008–09]

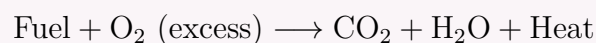
### Answer

#### Heat of Combustion and Bomb Calorimeter

##### 1. Heat of Combustion ( $\Delta H_c$ ):

The **heat of combustion** ( $\Delta H_c$ ) is the heat released when **one mole** of a substance undergoes complete combustion in the presence of excess oxygen under standard conditions.

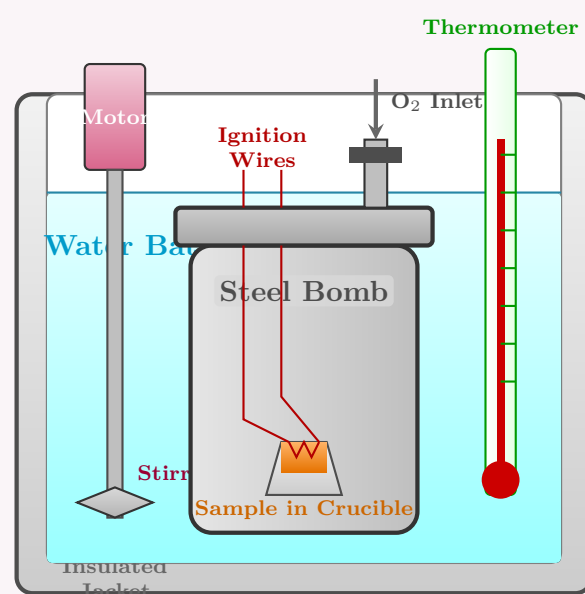
General Equation:



##### Examples:

- $\text{C}(\text{graphite})$ :  $\Delta H_c = -393.5 \text{ kJ/mol}$
- $\text{CH}_4(\text{g})$ :  $\Delta H_c = -890.3 \text{ kJ/mol}$
- $\text{C}_2\text{H}_5\text{OH}(\text{l})$ :  $\Delta H_c = -1366.8 \text{ kJ/mol}$

##### 2. Measurement using a Bomb Calorimeter:



##### Procedure:

1. Weigh a precise mass of the fuel sample and place it in the crucible inside the bomb.
2. Fill the steel bomb with **excess  $\text{O}_2$**  at high pressure (typically  $\sim 30 \text{ atm}$ ) to ensure complete combustion.



- Seal the bomb securely and immerse it in a known mass of water inside the insulated calorimeter jacket.
- Ignite the sample electrically using the ignition wires.
- Use the stirrer to ensure uniform heat distribution and record the maximum temperature rise ( $\Delta T$ ) on the thermometer.
- Calculate the total heat evolved:

$$q = C_{cal} \times \Delta T$$

where  $C_{cal}$  is the heat capacity of the entire calorimeter system.

- Since the combustion occurs at a constant volume (inside the sealed bomb), the heat measured is the change in internal energy:

$$q = -\Delta U \quad (\text{Heat at constant volume})$$

- Convert  $\Delta U$  to Heat of Combustion ( $\Delta H_c$ ) using the relation:

$$\Delta H_c = \Delta U + \Delta n_{gas}RT$$

where  $\Delta n_{gas}$  is the change in the number of moles of gases in the reaction.

**Q44.** The heat of reaction  $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$  at  $27^\circ\text{C}$  is  $-21.976$  Kcal. What will be the heat of reaction at  $50^\circ\text{C}$ ? (The molar heat capacities at constant pressure at  $27^\circ\text{C}$  for nitrogen, hydrogen and ammonia are 6.80, 6.77 and 8.86  $\text{cal}\cdot\text{mol}^{-1}\cdot\text{deg}^{-1}$  respectively.)  
(9 marks) [CHEM 101 | 2007–08]

#### Answer

**Heat of reaction for  $\text{N}_2 + 3\text{H}_2 \rightleftharpoons 2\text{NH}_3$  at  $50^\circ\text{C}$**

**Given:**

- $\Delta H_{300} = -21.976$  kcal at  $T_1 = 27^\circ\text{C} = 300$  K
- $T_2 = 50^\circ\text{C} = 323$  K
- $C_P(\text{N}_2) = 6.80$  cal/mol·deg
- $C_P(\text{H}_2) = 6.77$  cal/mol·deg
- $C_P(\text{NH}_3) = 8.86$  cal/mol·deg

**Step 1: Calculate  $\Delta C_P$ :**

$$\begin{aligned}\Delta C_P &= 2C_P(\text{NH}_3) - C_P(\text{N}_2) - 3C_P(\text{H}_2) \\ &= 2(8.86) - 6.80 - 3(6.77) = 17.72 - 6.80 - 20.31 = -9.39 \text{ cal/deg}\end{aligned}$$

**Step 2: Apply Kirchhoff's equation:**

$$\begin{aligned}\Delta H_{323} &= \Delta H_{300} + \Delta C_P(T_2 - T_1) \\ &= -21976 + (-9.39)(323 - 300) \\ &= -21976 + (-9.39)(23) \\ &= -21976 - 215.97\end{aligned}$$

$$\boxed{\Delta H_{50^\circ\text{C}} = -22192 \text{ cal/mol} \approx -22.19 \text{ kcal/mol}}$$

**Q45.** Explain the effect of temperature and pressure on the equilibrium of a chemical reaction. (7 marks) [CHEM 101 | 2007–08]

#### Answer

**Effect of temperature and pressure on chemical equilibrium**

**Effect of Temperature:**

Based on **van't Hoff equation**:

$$\frac{d \ln K}{dT} = \frac{\Delta H^\circ}{RT^2}$$

- Exothermic reaction ( $\Delta H < 0$ ):**
  - Increasing  $T$  decreases  $K$  (Le Chatelier: added heat shifts equilibrium to reactant side).
  - Yield decreases** with increasing  $T$ .
  - Example:  $\text{N}_2 + 3\text{H}_2 \rightleftharpoons 2\text{NH}_3$  — yield of  $\text{NH}_3$  decreases at high  $T$ .
- Endothermic reaction ( $\Delta H > 0$ ):**
  - Increasing  $T$  increases  $K$  (equilibrium shifts to product side).

- **Yield increases** with increasing  $T$ .
- Example:  $\text{N}_2\text{O}_4 \rightleftharpoons 2\text{NO}_2$  — more  $\text{NO}_2$  at higher  $T$ .

#### Effect of Pressure:

- $K$  is **independent of pressure** (depends only on  $T$ ). However, the **equilibrium composition** changes.
- Pressure affects equilibria involving **gases** only.
- Le Chatelier: increasing pressure shifts equilibrium to the side with **fewer moles of gas** ( $\Delta n_{\text{gas}} < 0$ ).
- **If  $\Delta n_{\text{gas}} < 0$ :** Increasing  $P$  favors products.  
Example:  $\text{N}_2 + 3\text{H}_2 \rightleftharpoons 2\text{NH}_3$  ( $\Delta n = -2$ ): high  $P$  gives more  $\text{NH}_3$ .
- **If  $\Delta n_{\text{gas}} > 0$ :** Increasing  $P$  favors reactants.  
Example:  $\text{N}_2\text{O}_4 \rightleftharpoons 2\text{NO}_2$  ( $\Delta n = +1$ ): high  $P$  gives more  $\text{N}_2\text{O}_4$ .
- **If  $\Delta n_{\text{gas}} = 0$ :** Pressure has no effect on equilibrium composition.  
Example:  $\text{H}_2 + \text{I}_2 \rightleftharpoons 2\text{HI}$ .

| Change                          | Effect on $K$ | Effect on composition |
|---------------------------------|---------------|-----------------------|
| $\uparrow T$ (exothermic)       | $K$ decreases | Less product          |
| $\uparrow T$ (endothermic)      | $K$ increases | More product          |
| $\uparrow P$ ( $\Delta n < 0$ ) | No change     | More product          |
| $\uparrow P$ ( $\Delta n > 0$ ) | No change     | Less product          |
| $\uparrow P$ ( $\Delta n = 0$ ) | No change     | No change             |

**Q46.** Derive the equilibrium constant expression of a reaction  $x\text{A} + y\text{B} \rightleftharpoons m\text{C} + n\text{D}$  using the thermodynamic approach. What information can be obtained from such a mathematical relationship? **(12 marks)** [CHEM 101 | 2006–07]

#### Answer

#### Equilibrium constant from thermodynamic approach for $x\text{A} + y\text{B} \rightleftharpoons m\text{C} + n\text{D}$

##### Derivation:

The chemical potential of species  $i$  in an ideal gas mixture is:

$$\mu_i = \mu_i^\circ + RT \ln \frac{P_i}{P^\circ}$$

At equilibrium, the total Gibbs free energy change is zero:

$$\Delta G = m\mu_C + n\mu_D - x\mu_A - y\mu_B = 0$$

Substituting:

$$\Delta G^\circ + RT \ln \frac{P_C^m P_D^n}{P_A^x P_B^y} = 0$$

At equilibrium, the pressure ratio equals  $K_P$ :

$$\Delta G^\circ = -RT \ln K_P$$

$$K_P = \frac{(P_C)^m (P_D)^n}{(P_A)^x (P_B)^y} = e^{-\Delta G^\circ / RT}$$

Similarly, in terms of concentrations:

$$K_c = \frac{[\text{C}]^m [\text{D}]^n}{[\text{A}]^x [\text{B}]^y}$$

Relation:  $K_P = K_c (RT)^{\Delta n}$  where  $\Delta n = (m + n) - (x + y)$ .

#### Information from this relationship:

- $\Delta G^\circ = -RT \ln K$ : links thermodynamics to equilibrium position.
- If  $K \gg 1$ :  $\Delta G^\circ \ll 0$  (products strongly favored).
- If  $K \ll 1$ :  $\Delta G^\circ \gg 0$  (reactants favored).
- Allows prediction of equilibrium composition from thermodynamic data.
- $\Delta H^\circ$  can be found from the temperature dependence of  $K$ .

**Q47.** How can pressure and temperature be optimized for the following reactions? Use Le Chatelier's principle for explanation. (i)  $\text{N}_2(\text{g})$

+ 3H<sub>2</sub>(g) ⇌ 2NH<sub>3</sub>(g), ΔH = -ve; (ii) N<sub>2</sub>O<sub>4</sub>(g) ⇌ 2NO<sub>2</sub>(g), ΔH = +ve. **(12 marks)**

[CHEM 101 | 2006–07]

### Answer

#### Optimization of pressure and temperature using Le Chatelier's principle

(i) N<sub>2</sub>(g) + 3H<sub>2</sub>(g) ⇌ 2NH<sub>3</sub>(g), ΔH = -ve (**Haber process**)

##### Effect of pressure:

- Moles of gas: reactants = 4, products = 2 ⇒ Δn<sub>gas</sub> = -2
- Increasing pressure shifts equilibrium toward **fewer moles** (right) ⇒ more NH<sub>3</sub> produced.
- **Optimal:** High pressure (200–400 atm in industry).

##### Effect of temperature:

- Reaction is **exothermic**: increasing *T* shifts equilibrium to the left (less NH<sub>3</sub>).
- **But:** Low temperature gives very slow rate.
- **Compromise:** Moderate temperature (400–500°C) with a catalyst (Fe).

(ii) N<sub>2</sub>O<sub>4</sub>(g) ⇌ 2NO<sub>2</sub>(g), ΔH = +ve

##### Effect of pressure:

- Moles: reactants = 1, products = 2 ⇒ Δn<sub>gas</sub> = +1
- Increasing pressure shifts equilibrium to the **left** (fewer moles) — more N<sub>2</sub>O<sub>4</sub> formed.
- **To get more NO<sub>2</sub>:** use low pressure.

##### Effect of temperature:

- Reaction is **endothermic**: increasing *T* shifts equilibrium to the right (more NO<sub>2</sub>).
- **To get more NO<sub>2</sub>:** use high temperature.

| Reaction                                           | ΔH  | For more product: | <i>T</i>     | <i>P</i> |
|----------------------------------------------------|-----|-------------------|--------------|----------|
| N <sub>2</sub> +3H <sub>2</sub> ⇌ 2NH <sub>3</sub> | -ve | NH <sub>3</sub>   | Low-moderate | High     |
| N <sub>2</sub> O <sub>4</sub> ⇌ 2NO <sub>2</sub>   | +ve | NO <sub>2</sub>   | High         | Low      |

**Q48.** The equilibrium constant *K<sub>p</sub>* for the reaction C<sub>2</sub>H<sub>4</sub>(g) + H<sub>2</sub>(g) ⇌ C<sub>2</sub>H<sub>6</sub>(g) is 5.04 × 10<sup>7</sup> atm<sup>-1</sup> at 25°C. Calculate the value of standard free energy Δ*G*<sup>°</sup> and comment on the result. **(11 marks)**

[CHEM 101 | 2006–07]

### Answer

Δ*G*<sup>°</sup> for C<sub>2</sub>H<sub>4</sub> + H<sub>2</sub> ⇌ C<sub>2</sub>H<sub>6</sub>

##### Given:

- *K<sub>p</sub>* = 5.04 × 10<sup>7</sup> atm<sup>-1</sup> at *T* = 25°C = 298 K
- *R* = 8.314 J/mol·K

##### Using:

$$\Delta G^{\circ} = -RT \ln K_P$$

$$= -8.314 \times 298 \times \ln(5.04 \times 10^7)$$

$$\ln(5.04 \times 10^7) = \ln(5.04) + 7 \ln(10) = 1.617 + 16.118 = 17.735$$

$$\Delta G^{\circ} = -8.314 \times 298 \times 17.735$$

$$= -2477.6 \times 17.735$$

$$\Delta G^{\circ} = -43934 \text{ J/mol} \approx -43.9 \text{ kJ/mol}$$

##### Comment:

- Δ*G*<sup>°</sup> = -43.9 kJ/mol is large and **negative** ⇒ the reaction is **strongly spontaneous** in the forward direction at 25°C.
- *K<sub>p</sub>* ≫ 1 confirms that **products strongly favored** at equilibrium — almost complete hydrogenation of ethylene to ethane.
- This is the basis of industrial catalytic hydrogenation reactions.

BUET · CHEM 113

# Chemistry

Section S8

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Electrochemistry

## S8 — Electrochemistry

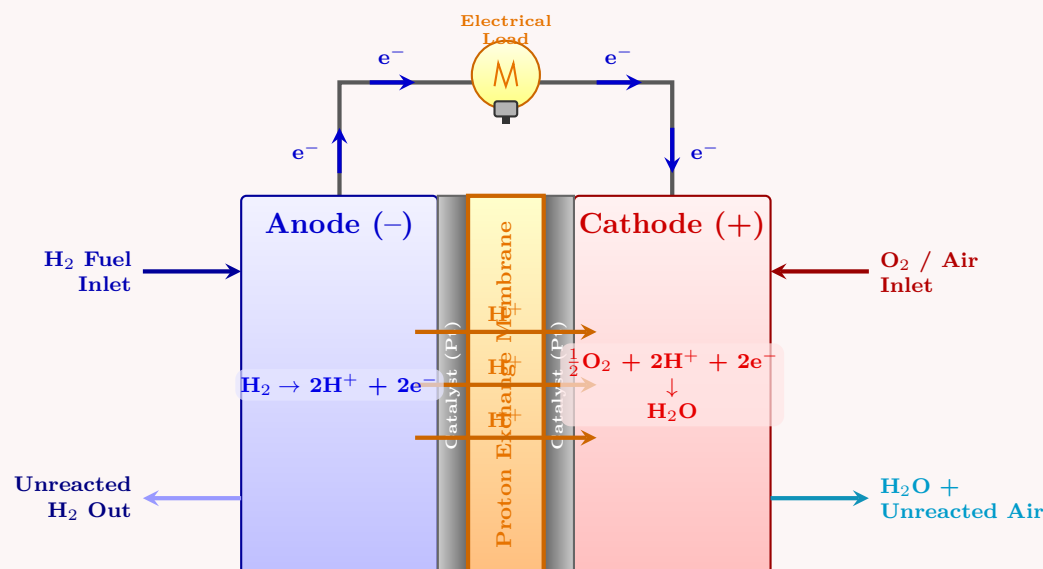
**Q1.** Illustrate the function of the proton exchange membrane (PEM) fuel cell. List its key characteristics and applications. (5 marks)  
[CHEM 113 | 2023–24]

### Answer

#### Proton Exchange Membrane (PEM) Fuel Cell

A PEM fuel cell converts chemical energy directly into electrical energy via electrochemical reactions. It uses a **proton-conducting solid polymer membrane** (like Nafion) as the electrolyte, which only allows protons ( $H^+$ ) to pass through while blocking electrons.

**Schematic Diagram:**



**Half-reactions & Overall Cell Reaction:**

**Anode (Oxidation):**



**Cathode (Reduction):**



**Overall Reaction:**



**Key Characteristics:**

- **Solid Electrolyte:** Uses a solid polymer membrane (e.g., Nafion) that selectively conducts protons ( $H^+$ ) and electrically insulates electrons.
- **Operating Temperature:** Runs at relatively low temperatures ( $60^\circ\text{C} - 80^\circ\text{C}$ ) compared to other fuel cells.
- **High Power Density:** Compact, lightweight, and capable of generating high current, making it ideal for mobile applications.
- **Zero Emissions:** The only physical byproduct is pure liquid water (and heat).
- **Catalyst:** Requires expensive noble metals like platinum (Pt) on both electrodes to speed up the reactions at low temperatures.
- **Fast Start-up:** Due to its low operating temperature, it can start up and respond to load changes very quickly.

**Applications:**

- **Transportation:** Hydrogen fuel cell electric vehicles (FCEVs) like the Toyota Mirai and Hyundai Nexa.
- **Portable Power:** Compact backup power supplies and generators for remote locations.
- **Aerospace & Military:** Used in spacecraft, drones, and submarines due to low heat signatures and no toxic emissions.
- **Stationary Power:** Distributed power generation for residential or commercial buildings.

**Conclusion:** The PEM fuel cell is a highly efficient, clean-energy device that reliably converts hydrogen fuel directly into electricity through proton transport, leaving behind zero harmful emissions.

**Q2.** Give a short account of electrolytic media, including ionic liquids. (10 marks)

[CHEM 113 | 2023–24]

**Answer****Electrolytic Media:**

An **electrolytic medium** is a substance or mixture that supports ionic conduction — it contains mobile ions that carry electrical current. The main types are:

**1. Aqueous Electrolyte Solutions:**

The most common electrolytic medium — ionic or polar compounds dissolved in water:

- **Strong electrolytes:** fully dissociate ( $\text{NaCl}$ ,  $\text{KCl}$ ,  $\text{H}_2\text{SO}_4$ ,  $\text{NaOH}$ ) — high conductivity
- **Weak electrolytes:** partially dissociate ( $\text{CH}_3\text{COOH}$ ,  $\text{NH}_3$ ) — lower conductivity, dependent on concentration
- Conductivity increases with concentration up to a maximum, then decreases due to ion–ion interactions
- Temperature increases conductivity by reducing viscosity and increasing ion mobility

**2. Non-aqueous Electrolyte Solutions:**

Solvents other than water used when aqueous media are unsuitable (e.g. at extreme temperatures, voltages):

- **Organic solvents:** acetonitrile, propylene carbonate, dimethyl sulfoxide (DMSO)
- Used in lithium-ion batteries ( $\text{LiPF}_6$  in ethylene carbonate)
- Lower conductivity than aqueous due to lower dielectric constants
- Enable wider electrochemical windows ( $> 4 \text{ V}$  vs  $\sim 1.2 \text{ V}$  for water)

**3. Molten Salts:**

Ionic compounds melted above their fusion point:

- Pure ions in liquid form — very high conductivity
- Example: molten  $\text{NaCl}$  (bp  $808^\circ\text{C}$ ) used in Down’s cell for  $\text{Na}$  production
- High temperature requirement is a disadvantage

**4. Solid Electrolytes:**

Ion-conducting solids used in solid-state devices:

- Example:  $\beta$ -alumina (conducts  $\text{Na}^+$ ), yttria-stabilised zirconia (YSZ, conducts  $\text{O}^{2-}$ ) in solid oxide fuel cells
- Polymer electrolytes: PEO + lithium salt in solid-state batteries

**5. Ionic Liquids:**

**Ionic liquids (ILs)** are salts that are liquid at or near room temperature (melting point  $< 100^\circ\text{C}$ ), consisting entirely of ions with negligible vapour pressure.

**Composition:**

- Large, asymmetric **organic cations:** imidazolium, pyridinium, ammonium, phosphonium
- Weakly coordinating **anions:**  $[\text{PF}_6]^-$ ,  $[\text{BF}_4]^-$ ,  $[\text{TFSI}]^-$ ,  $[\text{NTf}_2]^-$
- The mismatch in ion size/shape prevents crystallisation — liquid at room temperature

**Key Properties of Ionic Liquids:**

- **Negligible vapour pressure** — non-volatile, non-flammable (“green solvents”)
- **Wide electrochemical window** (up to 5–6 V) — enables reactions impossible in water
- **High thermal stability** up to  $300^\circ\text{C}$
- **Tunable properties** — can be designed by choosing cation/anion combinations (“designer solvents”)
- **Good ionic conductivity:**  $\sim 1\text{--}10 \text{ mS/cm}$
- **High viscosity** — slower ion transport than aqueous solutions

**Applications of Ionic Liquids:**

- Electrolytes in supercapacitors and batteries
- Electrodeposition of metals not possible from water (e.g.  $\text{Al}$ ,  $\text{Ti}$ , rare earths)
- Green solvents for organic synthesis
- Lubricants, extraction media,  $\text{CO}_2$  capture



Comparison:

| Medium       | Conduc-tivity | Window | Use                   |
|--------------|---------------|--------|-----------------------|
| Aqueous      | High          | 1.2 V  | General elec-trolysis |
| Non-aqueous  | Medium        | 4–5 V  | Li batteries          |
| Molten salt  | Very high     | Wide   | Metal extrac-tion     |
| Ionic liquid | Medium        | 5–6 V  | Advanced de-vices     |

**Conclusion:** Electrolytic media range from classical aqueous solutions to modern ionic liquids. ILs are particularly promising due to their wide electrochemical windows, thermal stability, and tunability — making them ideal for next-generation batteries, supercapacitors, and green chemistry applications.

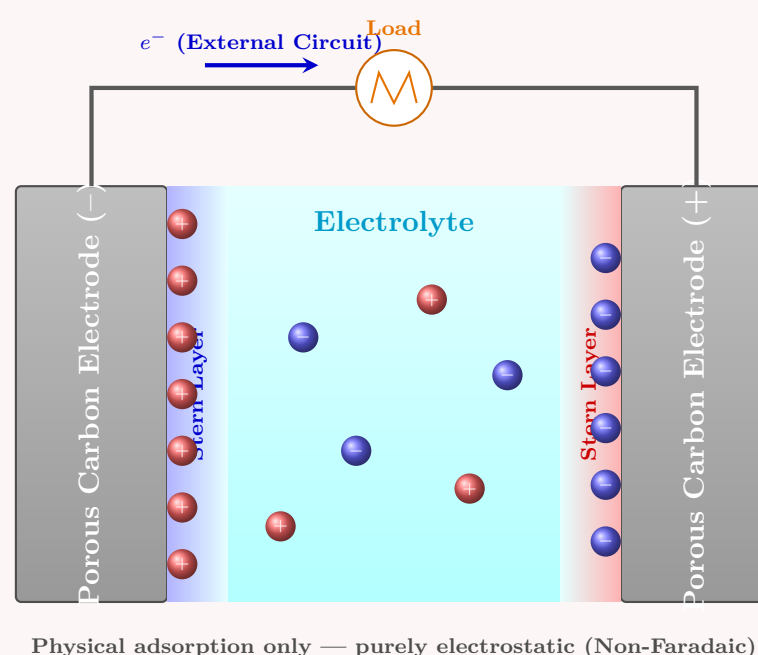
**Q3.** As an energy storage device, supercapacitor technology is lucrative. Describe the energy storage protocols of this class of energy device with schematic diagrams. (10 marks) [CHEM 113 | 2023–24]

Answer

### Supercapacitors — Energy Storage Protocols:

A **supercapacitor** (also called ultracapacitor or electrochemical capacitor) stores energy by two mechanisms: electrostatic charge separation and fast surface electrochemical reactions. They bridge the gap between conventional capacitors (high power, low energy) and batteries (high energy, low power).

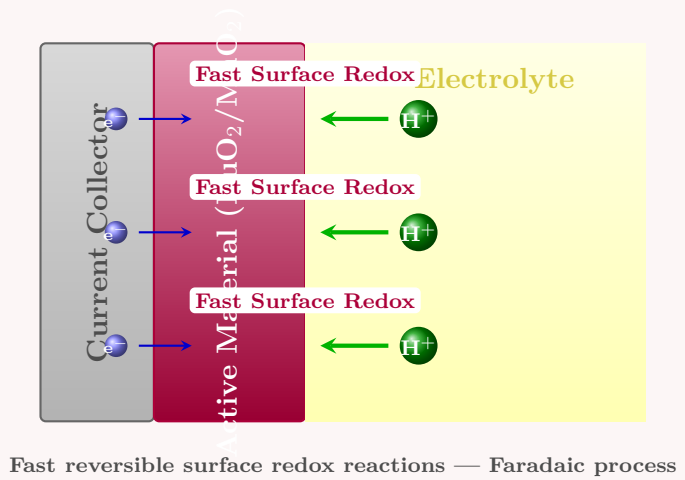
#### 1. Electric Double Layer Capacitor (EDLC):



#### Mechanism:

- Ions from electrolyte **physically adsorb** onto the electrode surface, forming an **electric double layer** (Helmholtz/Stern layer)
- No charge transfer — purely **electrostatic** (non-Faradaic) process
- Energy stored:  $E = \frac{1}{2}CV^2$  where  $C$  is capacitance (F) and  $V$  is voltage
- High surface area electrodes (activated carbon, graphene) maximise  $C$
- Fully reversible, extremely fast charging/discharging ( $< 1$  s)
- Cycle life:  $> 10^6$  cycles

#### 2. Pseudocapacitor (Faradaic Supercapacitor):



Mechanism:

- Fast, reversible **surface redox reactions** (Faradaic) on metal oxide electrodes (RuO<sub>2</sub>, MnO<sub>2</sub>, NiO) or conducting polymers (polyaniline)
- Example:

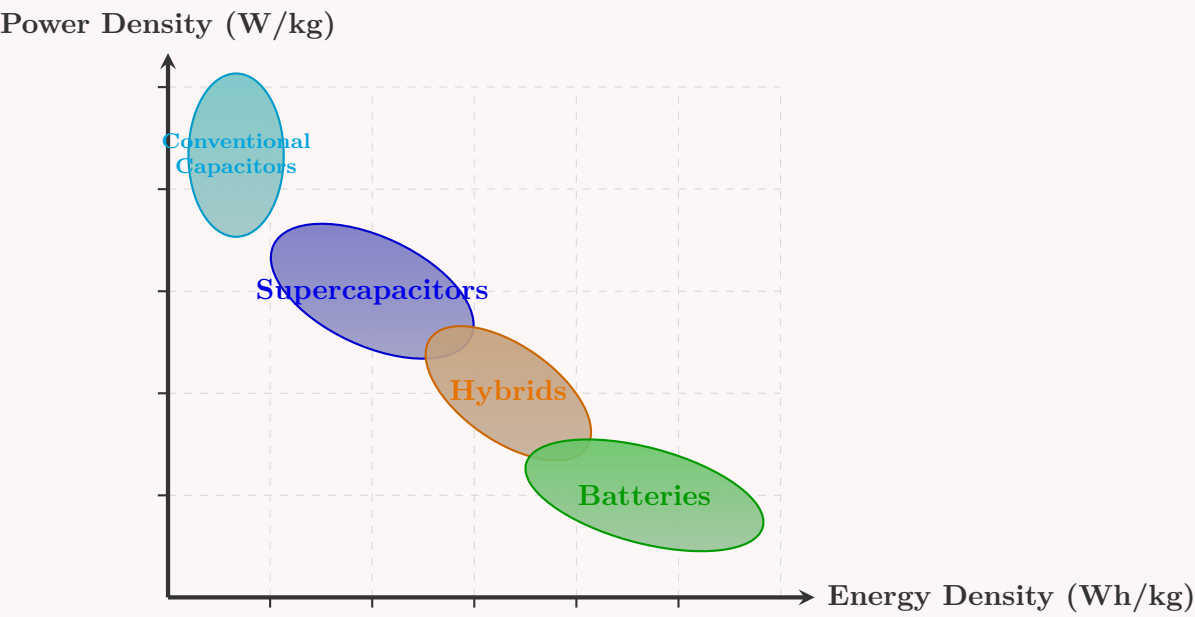
$$\text{RuO}_2 + x\text{H}^+ + xe^- \rightleftharpoons \text{RuO}_{2-x}(\text{OH})_x$$
- Larger energy density than EDLC (charge penetrates surface layers)
- Slightly slower and shorter cycle life than EDLC

3. Hybrid Supercapacitor:

Combines a **battery-type electrode** (high energy, Faradaic) with a **capacitor-type electrode** (high power, non-Faradaic):

- Example: Li-ion capacitor — graphite anode (intercalation) + activated carbon cathode (EDLC)
- Achieves intermediate energy *and* power density
- Bridges the gap between batteries and supercapacitors on the Ragone plot

Ragone Plot — Comparison:



Key Characteristics of Supercapacitors:

| Property          | Supercapaci-<br>tor   | Battery       |
|-------------------|-----------------------|---------------|
| Charge time       | Seconds               | Hours         |
| Cycle life        | > 10 <sup>6</sup>     | ~1000         |
| Energy density    | 1–10 Wh/kg            | 100–300 Wh/kg |
| Power density     | 1–10 kW/kg            | 0.1–1 kW/kg   |
| Mechanism         | Physical / Re-<br>dox | Chemical      |
| Temperature range | –40°C to +70°C        | Narrower      |

Applications:

- Regenerative braking in electric vehicles
- Backup power for memory modules and electronics
- Grid energy storage (peak shaving)
- Wind turbine pitch control systems

**Conclusion:** Supercapacitors store energy via **electrostatic double layer formation** (EDLC) and/or **fast surface redox reactions** (pseudocapacitance). Their exceptional power density, rapid charge/discharge, and long cycle life make them complementary to batteries in hybrid energy storage systems.

Q4. How would you contrast batteries and supercapacitors as energy storage devices? (5 marks) [CHEM 113 | 2022–23; 2021–22]

Answer

**Batteries vs Supercapacitors — Contrast:**

| Property                 | Battery                                                               | Supercapacitor                                                             |
|--------------------------|-----------------------------------------------------------------------|----------------------------------------------------------------------------|
| Energy storage mechanism | Bulk electrochemical (Faradaic) reactions throughout electrode volume | Surface charge separation (EDLC) or fast surface redox (pseudocapacitance) |
| Energy density           | High (100–300 Wh/kg)                                                  | Low (1–10 Wh/kg)                                                           |
| Power density            | Low (0.1–1 kW/kg)                                                     | Very high (1–10 kW/kg)                                                     |
| Charge/discharge time    | Hours                                                                 | Seconds to minutes                                                         |
| Cycle life               | ~500–2000 cycles                                                      | > 10 <sup>6</sup> cycles                                                   |
| Efficiency               | 70–85%                                                                | 95–99%                                                                     |
| Self-discharge           | Low (months)                                                          | High (days–weeks)                                                          |
| Temperature range        | Narrow (0–45°C)                                                       | Wide (–40 to +70°C)                                                        |
| Cost                     | Moderate                                                              | Higher (per kWh)                                                           |
| Typical use              | Long-duration energy storage (EVs, phones)                            | Short bursts of high power (braking, start-up)                             |

**Key distinction:** Batteries store energy in the **bulk** of electrode materials via slow chemical reactions — offering high energy but slow response. Supercapacitors store energy at the **electrode surface** — offering rapid charge/discharge and long life but limited energy.

**Hybrid systems** (e.g. Li-ion capacitors) combine both to achieve intermediate energy *and* power density for applications like electric vehicles.

Q5. Write all the redox reactions that take place during the discharge of secondary lithium-ion and lead-acid batteries. Identify the half-reactions as oxidation and reduction. (10 marks) [CHEM 113 | 2022–23]

Answer

**Discharge Reactions of Secondary Batteries**

**1. Lithium-Ion Battery (Discharge)**

During the discharge of a lithium-ion battery, lithium ions (Li<sup>+</sup>) move from the negative electrode (anode) to the positive electrode (cathode) through the electrolyte, while electrons flow through the external circuit. The standard chemistry uses lithium intercalated graphite (LiC<sub>6</sub>) and lithium cobalt oxide (LiCoO<sub>2</sub>).

- **Anode (Oxidation half-reaction):** Lithium intercalated within graphite oxidizes to release Li<sup>+</sup> ions and electrons.

$$\text{LiC}_6 \longrightarrow \text{Li}^+ + e^- + 6\text{C}$$
- **Cathode (Reduction half-reaction):** Cobalt(IV) oxide reduces as it intercalates Li<sup>+</sup> ions and accepts electrons to form lithium cobalt(III) oxide.

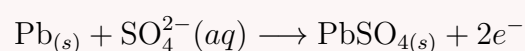
$$\text{CoO}_2 + \text{Li}^+ + e^- \longrightarrow \text{LiCoO}_2$$
- **Overall Cell Reaction:**

$$\text{LiC}_6 + \text{CoO}_2 \longrightarrow \text{LiCoO}_2 + 6\text{C}$$

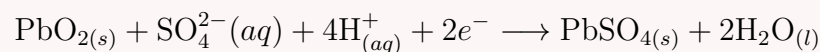
## 2. Lead-Acid Battery (Discharge)

During discharge, both electrodes are converted into insoluble lead(II) sulfate ( $\text{PbSO}_4$ ), and the electrolyte (sulfuric acid,  $\text{H}_2\text{SO}_4$ ) is consumed to produce water.

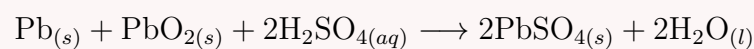
- **Anode (Oxidation half-reaction):** Spongy metallic lead oxidizes to lead(II) ions, which precipitate as lead(II) sulfate.



- **Cathode (Reduction half-reaction):** Lead(IV) dioxide is reduced to lead(II), which also forms lead(II) sulfate in the acidic environment.



- **Overall Cell Reaction:**



**Q6.** The concentration of  $\text{K}^+$  ion in the interior and exterior of a nerve cell are 400 mM and 15 mM, respectively. For  $\text{K}^+ + e^- = \text{K}$ ;  $E^\circ = -2.92 \text{ V}$ , find the electrical potential across the membrane. Which direction will the  $\text{K}^+$  movement be spontaneous? (10 marks) [CHEM 113 | 2022–23]

### Answer

#### Given:

- $[\text{K}^+]_{\text{inside}} = 400 \text{ mM} = 0.400 \text{ M}$
- $[\text{K}^+]_{\text{outside}} = 15 \text{ mM} = 0.015 \text{ M}$
- Half-reaction:  $\text{K}^+ + e^- = \text{K}$ ,  $E^\circ = -2.92 \text{ V}$
- $T = 37^\circ\text{C} = 310 \text{ K}$  (physiological)
- $n = 1$  (one electron transfer)
- $F = 96485 \text{ C/mol}$ ,  $R = 8.314 \text{ J/mol}\cdot\text{K}$

#### The Nernst Equation for Membrane Potential:

The membrane potential arises from the concentration gradient of  $\text{K}^+$  across the membrane. Using the **Nernst equation**:

$$E = E^\circ - \frac{RT}{nF} \ln Q$$

For the concentration cell ( $\text{K}^+$  moving from inside to outside):

$$\begin{aligned} \text{K}^+_{\text{inside}} &\rightarrow \text{K}^+_{\text{outside}} \\ Q &= \frac{[\text{K}^+]_{\text{outside}}}{[\text{K}^+]_{\text{inside}}} = \frac{0.015}{0.400} = 0.0375 \end{aligned}$$

#### Membrane Potential:

Since this is a **concentration cell**,  $E^\circ = 0$  for the concentration difference alone, and we apply the Nernst equation directly:

$$\begin{aligned} E_{\text{membrane}} &= -\frac{RT}{nF} \ln \frac{[\text{K}^+]_{\text{out}}}{[\text{K}^+]_{\text{in}}} = -\frac{(8.314)(310)}{(1)(96485)} \ln \frac{0.015}{0.400} \\ &= -\frac{2577.3}{96485} \times \ln(0.0375) = -0.02671 \times (-3.283) \\ &\boxed{E_{\text{membrane}} = +0.0877 \text{ V} = 87.7 \text{ mV}} \end{aligned}$$

The inside is at **higher potential** relative to the outside.

#### Direction of Spontaneous $\text{K}^+$ Movement:

The Gibbs free energy for ion transport:

$$\Delta G = RT \ln \frac{[\text{K}^+]_{\text{out}}}{[\text{K}^+]_{\text{in}}} + zFE_{\text{membrane}}$$

Spontaneous movement occurs from **high concentration** to **low concentration** (outward) — driven by the concentration gradient. However, the membrane potential (inside positive) **opposes** outward movement of  $\text{K}^+$  (positive ion).

At equilibrium (resting membrane potential), these two forces balance. Since  $[\text{K}^+]_{\text{in}} > [\text{K}^+]_{\text{out}}$ :

- **Chemical gradient** drives  $\text{K}^+$  **outward** (inside  $\rightarrow$  outside)
- **Electrical gradient** (inside positive) drives  $\text{K}^+$  **inward**
- Net spontaneous movement: **outward** (chemical gradient dominates when membrane is permeable to  $\text{K}^+$ )

**Conclusion:** The electrical potential across the membrane is **87.7 mV** (inside positive relative to outside).  $\text{K}^+$  spontaneously moves **outward** (from high to low concentration) through  $\text{K}^+$  channels, which is fundamental to establishing the resting membrane potential of nerve cells.

- Q7.** Given that the standard reduction potentials for vitamin B<sub>12</sub> ( $E^\circ = -0.526$  V) and the protein flavodoxin ( $E^\circ = -0.230$  V): (i) Calculate  $\Delta G^\circ$  in kJ for the one-electron reduction of vitamin B<sub>12</sub> by flavodoxin. (ii) Which is the superior reducing agent: vitamin B<sub>12</sub> or flavodoxin? Justify your choice. **(10 marks)** [CHEM 113 | 2021–22]

### Answer

#### Given:

- $E^\circ(\text{Vitamin B}_{12}) = -0.526$  V
- $E^\circ(\text{Flavodoxin}) = -0.230$  V
- One-electron transfer:  $n = 1$
- $F = 96485$  C/mol

#### (i) Calculate $\Delta G^\circ$ :

Flavodoxin **reduces** Vitamin B<sub>12</sub> means:

- Flavodoxin is oxidised (loses  $e^-$ ): acts as **reducing agent**
- Vitamin B<sub>12</sub> is reduced (gains  $e^-$ )

#### Half-reactions:



Cell reaction (flavodoxin<sub>red</sub> reduces VitB<sub>12,ox</sub>):

$$E^\circ_{\text{cell}} = E^\circ_{\text{cathode}} - E^\circ_{\text{anode}} = (-0.526) - (-0.230) = -0.296 \text{ V}$$

$$\Delta G^\circ = -nFE^\circ_{\text{cell}} = -(1)(96485)(-0.296) = +28,560 \text{ J/mol}$$

$$\boxed{\Delta G^\circ = +28.6 \text{ kJ/mol}}$$

The positive  $\Delta G^\circ$  indicates this reaction is **non-spontaneous** under standard conditions.

#### (ii) Superior Reducing Agent:

A **better reducing agent** has a **more negative** standard reduction potential — it has a greater tendency to *lose* electrons (be oxidised).

| Species                 | $E^\circ$ (V) | Reducing power   |
|-------------------------|---------------|------------------|
| Vitamin B <sub>12</sub> | -0.526        | Stronger reducer |
| Flavodoxin              | -0.230        | Weaker reducer   |

**Vitamin B<sub>12</sub> is the superior reducing agent** because it has the more negative reduction potential ( $-0.526$  V vs  $-0.230$  V). A more negative  $E^\circ$  means the species has a stronger tendency to donate electrons — i.e. to be oxidised.

This also explains why flavodoxin *cannot* spontaneously reduce Vitamin B<sub>12</sub> under standard conditions ( $\Delta G^\circ > 0$ ) — electrons flow spontaneously from lower to higher reduction potential, i.e. from B<sub>12</sub> to flavodoxin, not the reverse.

**Conclusion:**  $\Delta G^\circ = +28.6$  kJ/mol confirms the reaction is non-spontaneous as written. Vitamin B<sub>12</sub> is the stronger reducing agent due to its more negative standard reduction potential.

- Q8.** Why do cations and anions carry different fractions of the current in electrolytic conduction? **(5 marks)** [CHEM 113 | 2017–18]

### Answer

#### Why Cations and Anions Carry Different Current Fractions:

In electrolytic conduction, both cations and anions move simultaneously but in **opposite directions**. The fraction of total current carried by each ion is called its **transport number** (transference number):

$$t_+ = \frac{I_+}{I_{\text{total}}} \quad t_- = \frac{I_-}{I_{\text{total}}} \quad t_+ + t_- = 1$$

#### Reason for Different Current Fractions:

The current carried by an ion depends on:

$$I_i \propto |z_i| \cdot u_i \cdot c_i$$

where  $|z_i|$  = ionic charge,  $u_i$  = ionic mobility (drift speed per unit field),  $c_i$  = concentration.

For a binary electrolyte with equal concentrations:

$$t_+ = \frac{u_+}{u_+ + u_-} \quad t_- = \frac{u_-}{u_+ + u_-}$$

Cations and anions have **different ionic mobilities** due to:

- (a) **Different ionic sizes:** smaller ions move faster through solution
- (b) **Different hydration shells:** heavily hydrated ions carry more water and move slower (e.g.  $\text{Li}^+$  is small but heavily hydrated, giving lower mobility than  $\text{K}^+$ )
- (c) **Different charges:** higher charge attracts more water molecules, increasing drag
- (d) **Special conduction mechanisms:**  $\text{H}^+$  and  $\text{OH}^-$  conduct via the **Grotthuss (proton-hopping) mechanism**, giving them exceptionally high mobilities and transport numbers

**Example — HCl:**

$$u(\text{H}^+) = 36.2 \times 10^{-8} \text{ m}^2\text{V}^{-1}\text{s}^{-1} \gg u(\text{Cl}^-) = 7.9 \times 10^{-8} \text{ m}^2\text{V}^{-1}\text{s}^{-1}$$

$$t_+ = \frac{36.2}{36.2 + 7.9} = 0.821 \quad t_- = 0.179$$

$\text{H}^+$  carries 82% of the current due to its anomalously high mobility.

**Conclusion:** Cations and anions carry different current fractions because they have different ionic mobilities, which depend on ionic size, charge, hydration, and conduction mechanism. Equal mobility would give  $t_+ = t_- = 0.5$ , but in practice the values differ significantly.

**Q9.** How does conduction of electricity through a solution differ from metallic conduction? Explain the effect of some factors which influence the conduction of electricity in solution. (10 marks) [CHEM 113 | 2017–18]

### Answer

#### Electrolytic vs Metallic Conduction:

| Feature               | Metallic conduction                                               | Electrolytic conduction                                          |
|-----------------------|-------------------------------------------------------------------|------------------------------------------------------------------|
| Charge carriers       | Free electrons                                                    | Cations and anions                                               |
| Direction of carriers | Electrons move toward + terminal                                  | Cations toward cathode, anions toward anode                      |
| Chemical change       | None                                                              | Electrode reactions occur (oxidation/reduction)                  |
| Effect of temperature | Resistance <b>increases</b> (lattice vibrations impede electrons) | Resistance <b>decreases</b> (ions more mobile, viscosity lowers) |
| Ohm's law             | Strictly obeyed                                                   | Obeyed for strong electrolytes                                   |
| Transfer of matter    | No                                                                | Yes (ions migrate, electrode mass changes)                       |
| Examples              | Cu wire, Al foil                                                  | $\text{NaCl(aq)}$ , $\text{H}_2\text{SO}_4\text{(aq)}$           |

#### Factors Influencing Electrolytic Conduction:

- (a) **Nature and concentration of electrolyte:** Strong electrolytes (fully dissociated) conduct better. Conductance increases with concentration up to a maximum, then decreases (ion association).
- (b) **Temperature:** Conductance **increases** with temperature — higher  $T$  lowers solvent viscosity, increases ionic mobility:

$$\Lambda_T = \Lambda_{25^\circ\text{C}}[1 + \alpha(T - 25)]$$

where  $\alpha \approx 0.02 \text{ }^\circ\text{C}^{-1}$  for aqueous solutions.

- (c) **Ionic size and hydration:** Smaller, less hydrated ions have higher mobility. Hydration shell increases effective radius and reduces speed.
- (d) **Viscosity of solvent:** Lower viscosity  $\Rightarrow$  higher ionic mobility. By Stokes' Law:

$$u_i = \frac{|z_i|e}{6\pi\eta r_i}$$

where  $\eta$  is viscosity and  $r_i$  is ionic radius.

- (e) **Interionic interactions:** At higher concentrations, attractive forces between oppositely charged ions slow each other down (relaxation effect and electrophoretic retardation), reducing conductance.

**Conclusion:** Electrolytic conduction differs fundamentally from metallic conduction in using ions rather than electrons as charge carriers, involving chemical changes, and increasing with temperature. It is governed by ionic mobility, concentration, temperature, and interionic interactions.



**Q10.** Differentiate molar conductivity and specific conductivity. Explain why hydrogen and hydroxide ions exhibit exceptionally large ionic mobilities. **(10 marks)** [CHEM 113 | 2017–18]

### Answer

#### Molar Conductivity vs Specific Conductivity:

| Property             | Specific Conductivity ( $\kappa$ )                                                    | Molar Conductivity ( $\Lambda_m$ )                                     |
|----------------------|---------------------------------------------------------------------------------------|------------------------------------------------------------------------|
| Definition           | Conductance of a solution between two electrodes of unit area and unit distance apart | Conductance of all ions produced by dissolving one mole of electrolyte |
| Formula              | $\kappa = G \times \frac{l}{A}$                                                       | $\Lambda_m = \frac{\kappa \times 1000}{c}$                             |
| Units                | $\text{S}\cdot\text{m}^{-1}$ or $\text{mS}\cdot\text{cm}^{-1}$                        | $\text{S}\cdot\text{m}^2\cdot\text{mol}^{-1}$                          |
| Effect of dilution   | <b>Decreases</b> (fewer ions per unit volume)                                         | <b>Increases</b> (more complete dissociation)                          |
| At infinite dilution | Approaches zero                                                                       | Approaches $\Lambda_m^\circ$ (maximum)                                 |

Relation between them:

$$\Lambda_m = \frac{\kappa}{c}$$

where  $c$  is concentration in  $\text{mol}\cdot\text{m}^{-3}$ , or:

$$\Lambda_m(\text{S}\cdot\text{cm}^2\text{mol}^{-1}) = \frac{\kappa(\text{S}\cdot\text{cm}^{-1}) \times 1000}{c(\text{mol}\cdot\text{L}^{-1})}$$

#### Exceptionally Large Ionic Mobilities of $\text{H}^+$ and $\text{OH}^-$ :

From Appendix B (298 K):

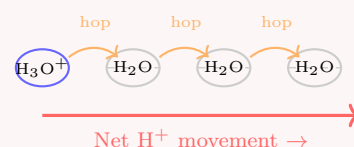
$$\lambda^\circ(\text{H}^+) = 34.96 \quad \text{vs} \quad \lambda^\circ(\text{Na}^+) = 5.01 \quad (\text{mS}\cdot\text{m}^2\text{mol}^{-1})$$

$$\lambda^\circ(\text{OH}^-) = 19.91 \quad \text{vs} \quad \lambda^\circ(\text{Cl}^-) = 7.63$$

$\text{H}^+$  is **7 times** more mobile than  $\text{Na}^+$ , and  $\text{OH}^-$  is **3 times** more mobile than  $\text{Cl}^-$ .

#### Grotthuss Mechanism (Proton Hopping):

$\text{H}^+$  and  $\text{OH}^-$  do not move through solution by physically migrating like other ions. Instead they conduct via the **Grotthuss mechanism** — a rapid proton-hopping relay through the hydrogen-bonded water network:

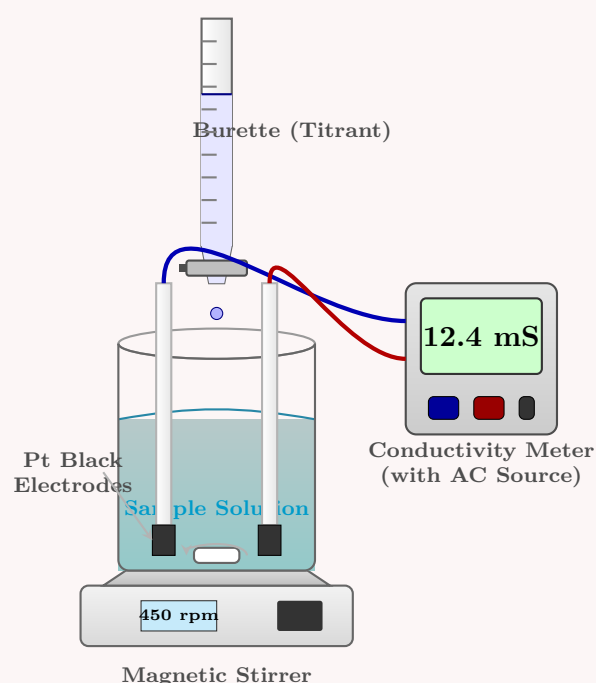


- A proton hops from  $\text{H}_3\text{O}^+$  to the adjacent water molecule via hydrogen bond, which becomes the new  $\text{H}_3\text{O}^+$
- Simultaneously, all O–H bonds in the chain reorient — the **defect propagates** rapidly without any ion physically traversing the distance
- This is far faster than classical ion migration (Stokes drag), giving anomalously high conductivity
- The same mechanism operates for  $\text{OH}^-$  in reverse (hydroxyl hopping)

**Conclusion:** Specific conductivity measures bulk solution conductance per unit geometry, while molar conductivity normalises to one mole of electrolyte.  $\text{H}^+$  and  $\text{OH}^-$  have exceptionally high ionic mobilities because they conduct via the **Grotthuss proton-hopping mechanism** — a relay through hydrogen-bonded water — rather than by physical migration.

### Answer

#### Conductometric Titration — Apparatus:



### Main apparatus:

- **Conductivity cell:** Two platinum black electrodes firmly immersed in the solution.
- **AC source** ( $\sim 1000$  Hz): Integrated or external, used to prevent electrolysis and polarisation at the electrodes.
- **Conductivity meter / Wheatstone bridge:** Directly measures the solution's conductance ( $G = 1/R$ ).
- **Burette:** Accurately delivers the titrant (NaOH).
- **Magnetic stirrer:** Ensures rapid and homogeneous mixing after each addition of the titrant.

### Determination of % HCl and % CH<sub>3</sub>COOH in a Mixture:

The supplied solution contains both HCl (a strong acid) and CH<sub>3</sub>COOH (a weak acid). Titration with standardised NaOH gives a **three-stage conductance curve**:

#### Stage 1 — Neutralisation of HCl:



Highly mobile H<sup>+</sup> is rapidly replaced by the slower, less mobile Na<sup>+</sup>:

$$\lambda^\circ(\text{H}^+) = 34.96 \gg \lambda^\circ(\text{Na}^+) = 5.01 \text{ mS} \cdot \text{m}^2\text{mol}^{-1}$$

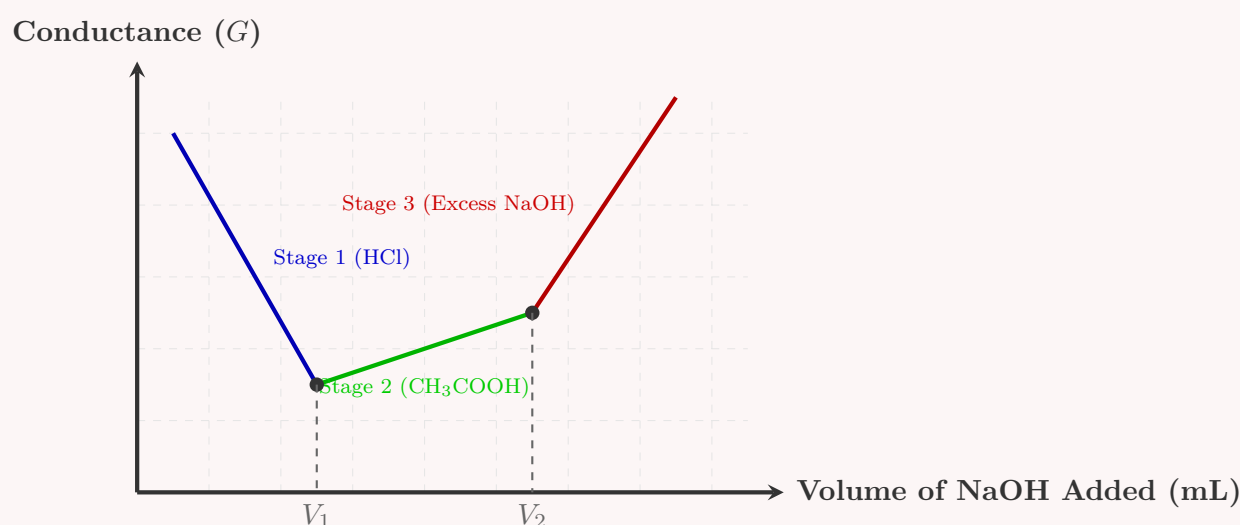
Result: Conductance **falls sharply**.

#### Stage 2 — Neutralisation of CH<sub>3</sub>COOH:



Poorly conducting unionised acetic acid is converted into conducting sodium acetate (a strong electrolyte): Result: Conductance **rises gradually**.

**Stage 3 — Excess NaOH:** Highly mobile OH<sup>−</sup> ions are added in excess after all acids are neutralised. Result: Conductance **rises steeply**.



### Calculation:

Let  $C_{\text{NaOH}}$  = concentration of the standard NaOH solution,  $V_1$  = volume of NaOH required to reach the first equivalence point (neutralises HCl),  $V_2$  = total volume of NaOH to reach the second equivalence point.

Volume of NaOH for HCl =  $V_1$

$$n(\text{HCl}) = C_{\text{NaOH}} \times V_1$$

Volume of NaOH for CH<sub>3</sub>COOH =  $(V_2 - V_1)$

$$n(\text{CH}_3\text{COOH}) = C_{\text{NaOH}} \times (V_2 - V_1)$$

Percentage of HCl:

$$\% \text{HCl} = \frac{n(\text{HCl}) \times 36.46}{V_{\text{sample}}} \times 100$$

Percentage of CH<sub>3</sub>COOH:

$$\% \text{CH}_3\text{COOH} = \frac{n(\text{CH}_3\text{COOH}) \times 60.05}{V_{\text{sample}}} \times 100$$

**Conclusion:** Conductometric titration of an HCl / CH<sub>3</sub>COOH mixture with NaOH gives a distinct, sharply defined three-stage curve consisting of straight line segments. By determining the two equivalence point volumes, V<sub>1</sub> and V<sub>2</sub>, the exact individual percentages of both the strong and weak acids can be accurately calculated without needing any chemical indicators.

**Q11.** Briefly explain how you can use molar conductivity data to calculate acidity constant of an weak acid. The molar conductivity of 0.010 M CH<sub>3</sub>COOH(aq) at 25°C is 1.65 ms·m<sup>2</sup>·mol<sup>-1</sup>. (i) Calculate the limiting molar conductivity of CH<sub>3</sub>COOH(aq). (ii) Calculate the degree of deprotonation, α. (iii) Calculate the acidity constant, K<sub>a</sub>, of CH<sub>3</sub>COOH. **(10 marks)** [CHEM 113 | 2015–16; 2014–15]

Appendixes - Chem 113

Appendix A. Boiling-Point-Elevation Constants (K<sub>b</sub>) and Freezing-Point-Depression Constants (K<sub>f</sub>)

| Solvent                                       | Boiling Point (°C) | Freezing Point (°C) | K <sub>b</sub> (°C/m) | K <sub>f</sub> (°C/m) |
|-----------------------------------------------|--------------------|---------------------|-----------------------|-----------------------|
| Acetic Acid (CH <sub>3</sub> COOH)            | 118.5              | 16.60               | 3.08                  | 3.59                  |
| Benzene (C <sub>6</sub> H <sub>6</sub> )      | 80.2               | 5.455               | 2.61                  | 5.065                 |
| Camphor (C <sub>10</sub> H <sub>16</sub> O)   | ---                | 179.5               | ---                   | 40                    |
| Carbon disulfide (CS <sub>2</sub> )           | 46.3               | ---                 | 2.40                  | ---                   |
| Cyclohexane (C <sub>6</sub> H <sub>12</sub> ) | 80.74              | 6.55                | 2.79                  | 20.0                  |
| Ethanol (C <sub>2</sub> H <sub>5</sub> OH)    | 78.3               | ---                 | 1.07                  | ---                   |
| Water (H <sub>2</sub> O)                      | 100.000            | 0.000               | 0.512                 | 1.858                 |

Appendix B. Limiting ionic conductivities in water at 298 K, λ<sup>0</sup>/(mS m<sup>2</sup> mol<sup>-1</sup>)

| Cations                                                          | Anions                                              |
|------------------------------------------------------------------|-----------------------------------------------------|
| Ba <sup>2+</sup> 12.72                                           | Br <sup>-</sup> 7.81                                |
| Ca <sup>2+</sup> 11.90                                           | CH <sub>3</sub> CO <sub>2</sub> <sup>-</sup> 4.09   |
| Cs <sup>+</sup> 7.72                                             | Cl <sup>-</sup> 7.635                               |
| Cu <sup>2+</sup> 10.72                                           | ClO <sub>4</sub> <sup>-</sup> 6.73                  |
| H <sup>+</sup> 34.96                                             | CO <sub>3</sub> <sup>2-</sup> 13.86                 |
| K <sup>+</sup> 7.350                                             | (CO <sub>3</sub> ) <sub>2</sub> <sup>2-</sup> 14.82 |
| Li <sup>+</sup> 3.87                                             | F <sup>-</sup> 5.54                                 |
| Mg <sup>2+</sup> 10.60                                           | [Fe(CN) <sub>6</sub> ] <sup>3-</sup> 30.27          |
| Na <sup>+</sup> 5.010                                            | [Fe(CN) <sub>6</sub> ] <sup>4-</sup> 44.20          |
| N(C <sub>2</sub> H <sub>5</sub> ) <sub>4</sub> <sup>+</sup> 3.26 | HCO <sub>2</sub> <sup>-</sup> 5.46                  |
| N(CH <sub>3</sub> ) <sub>4</sub> <sup>+</sup> 4.49               | I <sup>-</sup> 7.68                                 |
| NH <sub>4</sub> <sup>+</sup> 7.35                                | NO <sub>3</sub> <sup>-</sup> 7.146                  |
| Rb <sup>+</sup> 7.78                                             | OH <sup>-</sup> 19.91                               |
| Sr <sup>2+</sup> 11.89                                           | SO <sub>4</sub> <sup>2-</sup> 16.00                 |
| Zn <sup>2+</sup> 10.56                                           |                                                     |

Answer

**Molar Conductivity and Acidity Constant of a Weak Acid:**  
For a weak acid HA, the degree of dissociation α at concentration c is:

$$\alpha = \frac{\Lambda_m}{\Lambda_m^0}$$

and the acidity constant is:

$$K_a = \frac{\alpha^2 c}{1 - \alpha}$$

**Given:**

- Concentration: c = 0.010 M



- Molar conductivity:  $\Lambda_m = 1.65 \text{ mS}\cdot\text{m}^2\cdot\text{mol}^{-1}$
- From Appendix B (limiting ionic conductivities at 298 K):

$$\lambda^\circ(\text{H}^+) = 34.96 \text{ mS}\cdot\text{m}^2 \cdot \text{mol}^{-1}$$

$$\lambda^\circ(\text{CH}_3\text{COO}^-) = 4.09 \text{ mS}\cdot\text{m}^2 \cdot \text{mol}^{-1}$$

**(i) Limiting Molar Conductivity of  $\text{CH}_3\text{COOH}$ :**

By Kohlrausch's Law of Independent Migration of Ions:

$$\begin{aligned}\Lambda_m^\circ(\text{CH}_3\text{COOH}) &= \lambda^\circ(\text{H}^+) + \lambda^\circ(\text{CH}_3\text{COO}^-) \\ &= 34.96 + 4.09\end{aligned}$$

$$\Lambda_m^\circ(\text{CH}_3\text{COOH}) = 39.05 \text{ mS}\cdot\text{m}^2 \cdot \text{mol}^{-1}$$

**(ii) Degree of Deprotonation  $\alpha$ :**

$$\alpha = \frac{\Lambda_m}{\Lambda_m^\circ} = \frac{1.65}{39.05}$$

$$\alpha = 0.04226 \approx 0.0423$$

This means only about **4.23%** of  $\text{CH}_3\text{COOH}$  is dissociated at 0.010 M — consistent with it being a weak acid.

**(iii) Acidity Constant  $K_a$ :**

The dissociation equilibrium:



|               | $\text{CH}_3\text{COOH}$ | $\text{CH}_3\text{COO}^-$ | $\text{H}^+$ |
|---------------|--------------------------|---------------------------|--------------|
| Initial (M)   | $c$                      | 0                         | 0            |
| At equil. (M) | $c(1 - \alpha)$          | $c\alpha$                 | $c\alpha$    |

$$K_a = \frac{[\text{CH}_3\text{COO}^-][\text{H}^+]}{[\text{CH}_3\text{COOH}]} = \frac{(c\alpha)(c\alpha)}{c(1 - \alpha)} = \frac{c\alpha^2}{1 - \alpha}$$

$$K_a = \frac{0.010 \times (0.04226)^2}{1 - 0.04226} = \frac{0.010 \times 1.786 \times 10^{-3}}{0.9577} = \frac{1.786 \times 10^{-5}}{0.9577}$$

$$K_a = 1.865 \times 10^{-5} \approx 1.86 \times 10^{-5}$$

This is close to the literature value for acetic acid ( $K_a = 1.8 \times 10^{-5}$ ), confirming the method.

**Summary:**

| Quantity                                  | Value                                                 |
|-------------------------------------------|-------------------------------------------------------|
| $\Lambda_m^\circ(\text{CH}_3\text{COOH})$ | $39.05 \text{ mS}\cdot\text{m}^2\cdot\text{mol}^{-1}$ |
| Degree of deprotonation $\alpha$          | 0.0423 (4.23%)                                        |
| Acidity constant $K_a$                    | $1.86 \times 10^{-5}$                                 |

**Conclusion:** Molar conductivity data provides a simple and accurate route to  $K_a$  of weak acids — the limiting molar conductivity (from Kohlrausch's Law) gives  $\Lambda_m^\circ$ , from which  $\alpha$  and hence  $K_a$  are directly calculated.

**Q12.** What is meant by limiting ionic conductivity? Explain why the limiting ionic conductivity of  $\text{H}^+$  is so high compared to other cations. **(7 marks)** [CHEM 113 | 2015–16]

**Answer**

**Limiting Ionic Conductivity ( $\lambda^\circ$ ):**

The **limiting (or molar) ionic conductivity** is the contribution of a single ion to the total molar conductivity of an electrolyte at **infinite dilution** — i.e. when ion–ion interactions are completely absent:

$$\Lambda_m^\circ = \nu_+ \lambda_+^\circ + \nu_- \lambda_-^\circ$$

(Kohlrausch's Law of Independent Migration)

At infinite dilution, each ion migrates independently, and  $\lambda^\circ$  represents the maximum conductivity that ion can achieve. It is measured in  $\text{S}\cdot\text{m}^2\cdot\text{mol}^{-1}$ .

**Why  $\text{H}^+$  Has Exceptionally High  $\lambda^\circ$ :**

From Appendix B (298 K):

$$\lambda^\circ(\text{H}^+) = 34.96 \text{ mS}\cdot\text{m}^2\text{mol}^{-1}$$

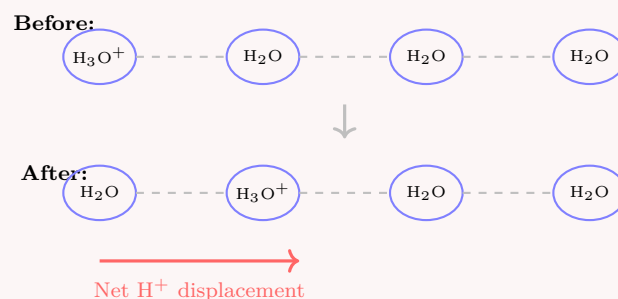
Compare with other cations:

$$\lambda^\circ(\text{Na}^+) = 5.01, \quad \lambda^\circ(\text{K}^+) = 7.35, \quad \lambda^\circ(\text{Li}^+) = 3.87$$

$\text{H}^+$  is **5–9 times** more mobile than typical cations.

**Explanation — Grotthuss Mechanism:**

Unlike ordinary ions which physically migrate through the solvent (limited by Stokes drag and viscosity),  $\text{H}^+$  conducts via a **proton-hopping relay** through the hydrogen-bonded network of water:



- A proton hops from  $\text{H}_3\text{O}^+$  to adjacent  $\text{H}_2\text{O}$  via a shared hydrogen bond
- The next  $\text{H}_2\text{O}$  reorients and passes the proton along the chain
- The **proton defect** propagates through the network at near-diffusion-limited speed
- No single proton travels far — the **information** (charge) travels rapidly
- This is far faster than Stokes diffusion of a classical ion of any size

**Conclusion:** Limiting ionic conductivity is the maximum ionic contribution to molar conductivity at zero concentration.  $\text{H}^+$  has an exceptionally high  $\lambda^\circ = 34.96 \text{ mS}\cdot\text{m}^2\text{mol}^{-1}$  because it conducts via the **Grotthuss proton-hopping mechanism** through hydrogen-bonded water — far faster than any physically migrating ion.

**Q13.** State Kohlrausch's law and Ostwald's dilution law. How can you use these two equations to determine the limiting values of the molar conductivity of solutions? (9 marks) [CHEM 113 | 2014–15]

### Answer

#### 1. Kohlrausch's Law of Independent Migration of Ions:

At infinite dilution, each ion migrates independently of its co-ions. The limiting molar conductivity is:

$$\Lambda_m^\circ = \nu_+ \lambda_+^\circ + \nu_- \lambda_-^\circ$$

where  $\nu_+$ ,  $\nu_-$  are the stoichiometric numbers and  $\lambda_+^\circ$ ,  $\lambda_-^\circ$  are the limiting ionic conductivities.

For a 1:1 electrolyte:

$$\Lambda_m^\circ = \lambda_+^\circ + \lambda_-^\circ$$

**Kohlrausch's empirical equation** for strong electrolytes:

$$\Lambda_m = \Lambda_m^\circ - K\sqrt{c}$$

where  $K$  is a constant depending on the electrolyte type and solvent. A plot of  $\Lambda_m$  vs  $\sqrt{c}$  gives a straight line — extrapolation to  $c = 0$  gives  $\Lambda_m^\circ$ .

#### 2. Ostwald's Dilution Law:

For a weak electrolyte (degree of dissociation  $\alpha$ ):

$$K_a = \frac{\alpha^2 c}{1 - \alpha}$$

Since  $\alpha = \Lambda_m / \Lambda_m^\circ$ :

$$K_a = \frac{(\Lambda_m / \Lambda_m^\circ)^2 \cdot c}{1 - \Lambda_m / \Lambda_m^\circ} = \frac{\Lambda_m^2 \cdot c}{\Lambda_m^\circ (\Lambda_m^\circ - \Lambda_m)}$$

This is **Ostwald's Dilution Law** in conductance form.

#### 3. Determining $\Lambda_m^\circ$ Using Both Laws:

**For strong electrolytes** (e.g. KCl, NaCl):

- $\Lambda_m$  can be measured at several concentrations
- Plot  $\Lambda_m$  vs  $\sqrt{c}$  (Kohlrausch)
- Extrapolate to  $c = 0$  to obtain  $\Lambda_m^\circ$  directly

**For weak electrolytes** (e.g.  $\text{CH}_3\text{COOH}$ ):

- Direct extrapolation fails — curve is non-linear and steep at low  $c$

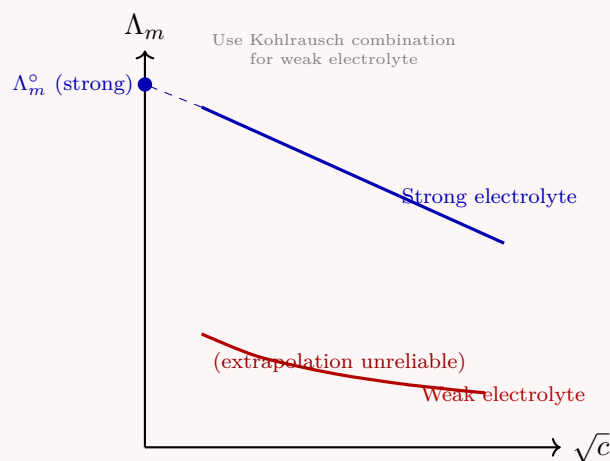
- Instead, use **Kohlrausch's Law** indirectly:

$$\Lambda_m^\circ(\text{CH}_3\text{COOH}) = \Lambda_m^\circ(\text{HCl}) + \Lambda_m^\circ(\text{CH}_3\text{COONa}) - \Lambda_m^\circ(\text{NaCl})$$

- Each term on the right can be obtained by extrapolation (all are strong electrolytes)
- Then use **Ostwald's Law** to find  $\alpha$  and  $K_a$  at any concentration:

$$\alpha = \frac{\Lambda_m}{\Lambda_m^\circ} \quad K_a = \frac{\alpha^2 c}{1 - \alpha}$$

#### Graphical Method:



**Conclusion:** Kohlrausch's Law allows  $\Lambda_m^\circ$  to be obtained by linear extrapolation for strong electrolytes, and by ionic combination for weak electrolytes. Ostwald's Dilution Law then uses  $\Lambda_m^\circ$  to determine dissociation constants from conductance measurements.

**Q14.** For the reaction  $\text{Ag(s)} + \frac{1}{2}\text{Cl}_2(\text{g}) \rightarrow \text{AgCl(s)}$ ,  $E^\circ = 0.839 \text{ V}$  at 1000 K. If  $a_{\text{Ag}} = a_{\text{AgCl}} = 1.00$  and  $a_{\text{Cl}_2} = 0.77$ , find the value of  $E$ . Is the reaction less favored under non-standard conditions? Justify from the values of  $K$  and  $\Delta G$ . (9 marks) [CHEM 113 | 2013–14]

#### Answer

##### Given:

- Reaction:  $\text{Ag}_{(\text{s})} + \frac{1}{2}\text{Cl}_{2(\text{g})} \rightarrow \text{AgCl}_{(\text{s})}$
- $E^\circ = 0.839 \text{ V}$  at 1000 K
- $a_{\text{Ag}} = 1.00$ ,  $a_{\text{AgCl}} = 1.00$ ,  $a_{\text{Cl}_2} = 0.77$
- $n = 1$ ,  $R = 8.314 \text{ J/mol}\cdot\text{K}$ ,  $F = 96485 \text{ C/mol}$

##### Reaction quotient $Q$ :

$$Q = \frac{a_{\text{AgCl}}}{a_{\text{Ag}} \cdot a_{\text{Cl}_2}^{1/2}} = \frac{1.00}{1.00 \times (0.77)^{1/2}} = \frac{1.00}{0.8775} = 1.140$$

##### Nernst equation:

$$\begin{aligned} E &= E^\circ - \frac{RT}{nF} \ln Q = 0.839 - \frac{(8.314)(1000)}{(1)(96485)} \ln(1.140) \\ &= 0.839 - (0.08617)(0.1310) = 0.839 - 0.01129 \\ &\boxed{E = 0.828 \text{ V}} \end{aligned}$$

##### $\Delta G$ under non-standard conditions:

$$\begin{aligned} \Delta G &= -nFE = -(1)(96485)(0.828) = -79,890 \text{ J/mol} \\ &\boxed{\Delta G = -79.9 \text{ kJ/mol}} \end{aligned}$$

##### Equilibrium constant $K$ :

$$\begin{aligned} \Delta G^\circ &= -nFE^\circ = -(1)(96485)(0.839) = -80,951 \text{ J/mol} = -80.95 \text{ kJ/mol} \\ K &= e^{-\Delta G^\circ/RT} = e^{80950/(8.314 \times 1000)} = e^{9.734} = 1.68 \times 10^4 \end{aligned}$$

##### Is the reaction less favoured under non-standard conditions?

- $E = 0.828 \text{ V} < E^\circ = 0.839 \text{ V}$ : cell emf is **slightly reduced**



- $\Delta G = -79.9 \text{ kJ/mol}$  vs  $\Delta G^\circ = -80.95 \text{ kJ/mol}$ : both are **negative** — reaction is spontaneous under both conditions
- $Q = 1.140 > 1$ : the reaction has slightly less driving force than standard
- $K = 1.68 \times 10^4 \gg 1$ : equilibrium lies far to the right — reaction strongly favoured

**Conclusion:** The reaction is **marginally less favoured** under non-standard conditions ( $E = 0.828 \text{ V}$  vs  $0.839 \text{ V}$ ) because  $a_{\text{Cl}_2} < 1$  reduces the driving force slightly. However, the large negative  $\Delta G$  and very large  $K$  confirm the reaction remains **strongly spontaneous** and product-favoured.

**Q15.** Define reversible cell. Give an example and explain how it is reversible. (11 marks)

[CHEM 113 | 2011–12]

### Answer

#### Reversible Cell — Definition:

A **reversible cell** is an electrochemical cell that operates under thermodynamically reversible conditions — meaning an infinitesimally small opposing EMF can reverse the direction of current flow and all associated chemical reactions, with no net energy dissipation.

#### Conditions for reversibility:

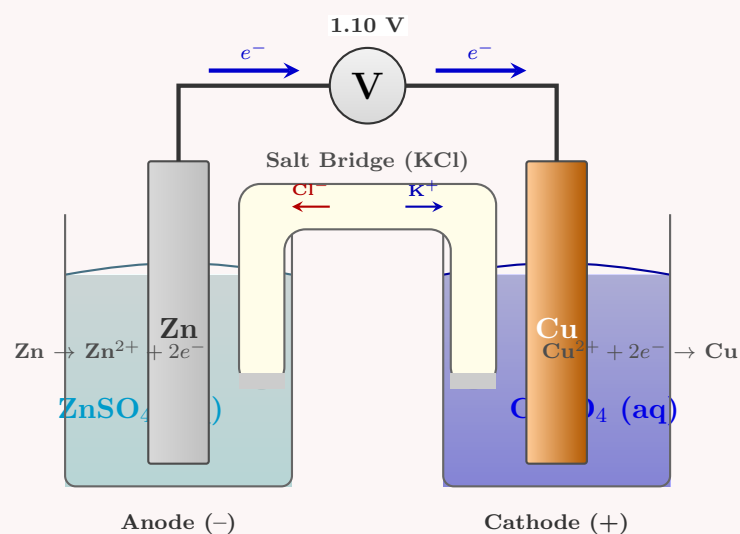
- The cell reaction must be **chemically reversible**: the exact reverse reaction occurs when current is reversed
- The cell must operate **infinitely slowly** (quasi-statically) — no current flows in the ideal limit
- No irreversible processes**: no concentration polarisation, no side reactions

#### Example — Daniell Cell:

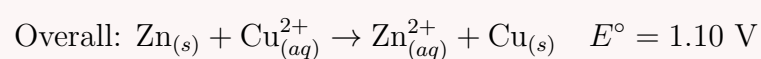
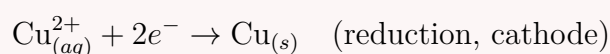
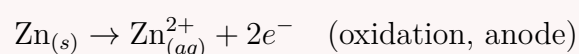
The **Daniell cell** is the classic example:



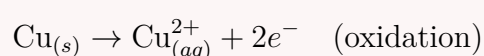
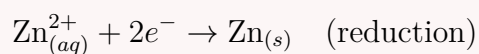
#### Cell diagram and reactions:



#### Discharge (forward):



#### Charging (reverse — applying external EMF > 1.10 V):



The **exact chemical reverse** occurs — confirming reversibility. The cell is reversible because:

- Both electrode reactions are thermodynamically reversible (metal/metal ion couples)
- The salt bridge maintains electrical neutrality without irreversible mixing
- Applying a slightly larger EMF exactly reverses all processes

**Conclusion:** A reversible cell operates under equilibrium conditions where the cell reaction and its exact reverse can occur by infinitesimal changes in the applied EMF. The Daniell cell exemplifies reversibility — its forward and reverse reactions are chemically and thermodynamically identical in magnitude.

**Q16.** Establish a relation between the emf of a reversible cell and the heat of reaction in the cell. (14 marks) [CHEM 113 | 2011–12]

**Answer**
**Relation Between EMF and Heat of Reaction — Gibbs–Helmholtz Equation:**
**Step 1 — EMF and Gibbs free energy:**

For a reversible cell doing electrical work:

$$\Delta G = -nFE$$

where  $n$  = moles of electrons,  $F$  = Faraday constant,  $E$  = EMF.

**Step 2 — Temperature dependence of  $\Delta G$ :**

From the Gibbs–Helmholtz equation:

$$\Delta G = \Delta H + T \left( \frac{\partial \Delta G}{\partial T} \right)_P$$

Substituting  $\Delta G = -nFE$ :

$$-nFE = \Delta H + T \left( \frac{\partial(-nFE)}{\partial T} \right)_P$$

$$-nFE = \Delta H - nFT \left( \frac{\partial E}{\partial T} \right)_P$$

Rearranging:

$$\Delta H = -nFE + nFT \left( \frac{\partial E}{\partial T} \right)_P = -nF \left[ E - T \left( \frac{\partial E}{\partial T} \right)_P \right]$$

This is the **Gibbs–Helmholtz equation** for a cell.

**Step 3 — Physical interpretation:**

- $-nFE = \Delta G$ : electrical energy (free energy)
- $nFT(\partial E/\partial T)_P = T\Delta S$ : heat exchanged reversibly with surroundings
- Therefore:  $\Delta H = \Delta G + T\Delta S$  (first law, as expected)

**Three cases:**

(a)  $\left(\frac{\partial E}{\partial T}\right)_P > 0$ : cell absorbs heat from surroundings ( $\Delta H < \Delta G$ , endothermic)

(b)  $\left(\frac{\partial E}{\partial T}\right)_P = 0$ :  $\Delta H = \Delta G = -nFE$  (no heat exchange)

(c)  $\left(\frac{\partial E}{\partial T}\right)_P < 0$ : cell evolves heat to surroundings ( $\Delta H > \Delta G$ , exothermic)

**Entropy of cell reaction:**

$$\Delta S = nF \left( \frac{\partial E}{\partial T} \right)_P$$

**Conclusion:** The heat of reaction in a reversible cell is related to EMF by:

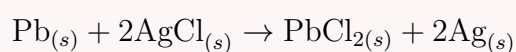
$$\Delta H = -nF \left[ E - T \left( \frac{\partial E}{\partial T} \right)_P \right]$$

This shows that  $\Delta H \neq -nFE$  unless  $(\partial E/\partial T)_P = 0$ . The temperature coefficient of EMF captures the entropy contribution, allowing complete thermodynamic characterisation of cell reactions from electrical measurements alone.

**Q17.** At 25°C the value of the emf for the reversible cell: Pb, PbCl<sub>2</sub>(s)|KCl(aq)|AgCl(s), Ag is 0.4902 volt and  $(\partial E/\partial T)_P = -1.86 \times 10^{-4}$  volt·deg<sup>-1</sup>. Calculate the values of  $\Delta G$  and  $\Delta H$  showing the reaction involved in the cell. **(10 marks)** [CHEM 113 | 2011–12]

**Answer**
**Given:**

- Cell: Pb, PbCl<sub>2</sub>(s) | KCl(aq) | AgCl(s), Ag
- $E = 0.4902$  V at  $T = 25^\circ\text{C} = 298$  K
- $(\partial E/\partial T)_P = -1.86 \times 10^{-4}$  V·deg<sup>-1</sup>
- $n = 2$  (two electrons transferred)
- $F = 96485$  C/mol

**Cell reaction:**

**Calculate  $\Delta G$ :**

$$\Delta G = -nFE = -(2)(96485)(0.4902) = -94,574 \text{ J/mol}$$

$$\Delta G = -94.57 \text{ kJ/mol}$$

**Calculate  $\Delta H$ :**

Using the Gibbs–Helmholtz equation:

$$\begin{aligned}\Delta H &= -nF \left[ E - T \left( \frac{\partial E}{\partial T} \right)_P \right] \\ &= -(2)(96485) [0.4902 - (298)(-1.86 \times 10^{-4})] \\ &= -(2)(96485) [0.4902 + 0.05543] \\ &= -(2)(96485)(0.5456) \\ &= -105,274 \text{ J/mol} \\ \boxed{\Delta H = -105.27 \text{ kJ/mol}}\end{aligned}$$

Calculate  $\Delta S$ :

$$\Delta S = nF \left( \frac{\partial E}{\partial T} \right)_P = (2)(96485)(-1.86 \times 10^{-4}) = -35.91 \text{ J/mol}\cdot\text{K}$$

Verification:

$$\Delta H = \Delta G + T\Delta S = -94,574 + (298)(-35.91) = -94,574 - 10,701 = -105,275 \text{ J/mol} \quad \checkmark$$

Summary:

| Quantity   | Value                            |
|------------|----------------------------------|
| $\Delta G$ | -94.57 kJ/mol (spontaneous)      |
| $\Delta H$ | -105.27 kJ/mol (exothermic)      |
| $\Delta S$ | -35.91 J/mol·K (order increases) |

**Conclusion:** The cell reaction is both spontaneous ( $\Delta G < 0$ ) and exothermic ( $\Delta H < 0$ ). The negative  $\Delta S$  indicates increased order (forming solid  $\text{PbCl}_2$  and  $\text{Ag}$  from  $\text{Pb}$  and  $\text{AgCl}$ ).

Q18. What are reversible and irreversible cells? Give one example of each. (3 marks)

[CHEM 101 | 2009–10]

#### Answer

##### Reversible Cell:

A cell in which the electrode reactions can be **exactly reversed** by applying an infinitesimally larger opposing EMF, with all chemical changes being perfectly reversible.

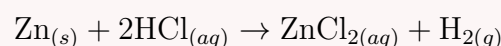
**Example:** Daniell cell:



##### Irreversible Cell:

A cell in which the electrode reactions **cannot be simply reversed** — applying a reverse EMF produces different reactions or the same electrode is not restored.

**Example:** Grove cell or a cell with  $\text{Zn}$  in  $\text{HCl}$ :



Passing current in reverse does not regenerate  $\text{Zn}$  and  $\text{HCl}$  in the same proportions — irreversible.

| Feature                 | Reversible    | Irreversible       |
|-------------------------|---------------|--------------------|
| Reaction reversal       | Exact reverse | Different products |
| Thermodynamic treatment | Possible      | Not applicable     |
| EMF                     | Well-defined  | Variable           |
| Example                 | Daniell cell  | Zn/HCl cell        |

Q19. State the Kohlrausch law of independent migration of ions. Discuss its applications. (9 marks)

[CHEM 101 | 2009–10]

#### Answer

##### Kohlrausch's Law of Independent Migration of Ions:

**Statement:** At infinite dilution, each ion migrates independently of all other ions, and contributes a fixed, characteristic amount to the total molar conductivity of the electrolyte, regardless of the nature of the co-ion:

$$\Lambda_m^\circ = \nu_+ \lambda_+^\circ + \nu_- \lambda_-^\circ$$

Applications:

(a) **Finding  $\Lambda_m^\circ$  of weak electrolytes:**

Direct extrapolation impossible for weak electrolytes. Use combination:

$$\Lambda_m^\circ(\text{CH}_3\text{COOH}) = \Lambda_m^\circ(\text{HCl}) + \Lambda_m^\circ(\text{CH}_3\text{COONa}) - \Lambda_m^\circ(\text{NaCl})$$

(b) **Degree of dissociation:**

$$\alpha = \frac{\Lambda_m}{\Lambda_m^\circ}$$

(c) **Dissociation constant:**

$$K_a = \frac{\alpha^2 c}{1 - \alpha} = \frac{\Lambda_m^2 c}{\Lambda_m^\circ (\Lambda_m^\circ - \Lambda_m)}$$

(d) **Ionic mobilities:** Individual  $\lambda^\circ$  values directly give ionic mobilities:  $u_i = \lambda_i^\circ / F$ (e) **Transport numbers:**

$$t_+ = \frac{\lambda_+^\circ}{\lambda_+^\circ + \lambda_-^\circ}, \quad t_- = \frac{\lambda_-^\circ}{\lambda_+^\circ + \lambda_-^\circ}$$

(f) **Solubility of sparingly soluble salts:**  $\kappa_{\text{solution}} - \kappa_{\text{water}} = \Lambda_m^\circ \cdot c_{\text{sat}}$  gives solubility  $c_{\text{sat}}$ 

**Conclusion:** Kohlrausch's Law is foundational to electrolyte chemistry — it allows calculation of limiting molar conductivities, dissociation constants, ionic mobilities, and solubility products from simple conductance measurements.

**Q20.** Briefly discuss the chronological development of the mechanism of electrolytic conduction. (8 marks) [CHEM 101 | 2008–09]

**Answer****Chronological Development of Electrolytic Conduction Mechanism:**(a) **Faraday (1833–1834) — Laws of Electrolysis:**

Michael Faraday established that electricity passed through solutions causes chemical decomposition (**electrolysis**). He introduced the terms *electrolyte*, *electrode*, *anode*, *cathode*, *ion*. His laws showed that the amount of substance deposited is proportional to the charge passed — implying discrete charged particles (ions) carry current.

(b) **Hittorf (1853) — Transport Numbers:**

Hittorf showed that cations and anions migrate independently and carry different fractions of current. He introduced the concept of **transport numbers** ( $t_+$ ,  $t_-$ ), measured by changes in electrode compartment concentrations.

(c) **Kohlrausch (1874–1885) — Conductance Measurements:**

Kohlrausch made precise conductance measurements using AC current (avoiding electrolysis) and showed:  $\Lambda_m = \Lambda_m^\circ - K\sqrt{c}$ . He established the **law of independent migration** — each ion has a characteristic limiting conductivity.

(d) **Arrhenius (1887) — Ionic Dissociation Theory:**

Arrhenius proposed that electrolytes **spontaneously dissociate** into ions in solution — even without current. He used conductance data to calculate degree of dissociation  $\alpha = \Lambda_m / \Lambda_m^\circ$ . This explained anomalous colligative properties.

(e) **Ostwald (1888) — Dilution Law:**

Ostwald confirmed Arrhenius's theory for weak electrolytes:  $K_a = \alpha^2 c / (1 - \alpha)$ . His dilution law quantitatively described partial dissociation.

(f) **Debye and Hückel (1923) — Ion Atmosphere Theory:**

For strong electrolytes (fully dissociated at all concentrations), Debye and Hückel explained the **decrease of conductance with concentration** through the concept of the **ionic atmosphere** — each ion is surrounded by a cloud of opposite charges that:

- **Relaxation effect:** the ionic atmosphere takes time to reorganise when an ion moves, creating a retarding force
- **Electrophoretic effect:** the solvent molecules dragged by the oppositely charged atmosphere oppose the ion's motion

Their theory predicted  $\Lambda_m = \Lambda_m^\circ - K\sqrt{c}$  — identical in form to Kohlrausch's empirical law.

(g) **Onsager (1927) — Extended Theory:**

Onsager provided the rigorous mathematical treatment of the Debye–Hückel conductance theory, deriving the limiting law coefficient  $K$  from first principles.

**Conclusion:** The understanding of electrolytic conduction evolved from Faraday's empirical laws through Arrhenius's ionic dissociation theory to the rigorous Debye–Hückel–Onsager theory — transforming a phenomenological description into a quantitative molecular theory of ion–ion interactions in solution.

**Q21.** Define transport number. Show that  $t_+ + t_- = 1.0$ . (4 marks) [CHEM 101 | 2008–09]

**Answer**
**Transport Number — Definition:**

The **transport number** (transference number)  $t_i$  of an ion is the fraction of the total current carried by that ion:

$$t_i = \frac{I_i}{I_{\text{total}}} = \frac{\text{current carried by ion } i}{\text{total current}}$$

For a binary electrolyte:

$$t_+ = \frac{u_+}{u_+ + u_-} \quad t_- = \frac{u_-}{u_+ + u_-}$$

where  $u_+$  and  $u_-$  are ionic mobilities.

**Proof that  $t_+ + t_- = 1$ :**

The total current in solution is the sum of currents carried by cations and anions:

$$I_{\text{total}} = I_+ + I_-$$

By definition:

$$t_+ + t_- = \frac{I_+}{I_{\text{total}}} + \frac{I_-}{I_{\text{total}}} = \frac{I_+ + I_-}{I_{\text{total}}} = \frac{I_{\text{total}}}{I_{\text{total}}}$$

$t_+ + t_- = 1$

Alternatively, using mobilities directly:

$$t_+ + t_- = \frac{u_+}{u_+ + u_-} + \frac{u_-}{u_+ + u_-} = \frac{u_+ + u_-}{u_+ + u_-} = 1 \quad \checkmark$$

**Note:** For a multicomponent electrolyte:  $\sum_i t_i = 1$  (all ions together carry 100% of current).

**Example** (HCl at 298 K, from Appendix B):

$$t_+ = \frac{34.96}{34.96 + 7.63} = 0.821 \quad t_- = \frac{7.63}{34.96 + 7.63} = 0.179$$

$$t_+ + t_- = 0.821 + 0.179 = 1.000 \quad \checkmark$$

**Q22.** Explain the terms: conductance, specific conductance, equivalent conductance, and molecular conductance. Derive a relation between specific conductance and equivalent conductance. **(15 marks)** [CHEM 101 | 2007–08]

**Answer**
**1. Conductance ( $G$ ):**

The **conductance** of a solution is the reciprocal of its electrical resistance:

$$G = \frac{1}{R} \quad (\text{unit: siemens, S})$$

It measures the ease with which current flows through the solution between two electrodes.

**2. Specific Conductance ( $\kappa$ ):**

The **specific conductance** (conductivity) is the conductance of a solution contained between two electrodes of unit area ( $A = 1 \text{ cm}^2$ ) separated by unit distance ( $l = 1 \text{ cm}$ ):

$$\kappa = G \times \frac{l}{A} = G \times \text{cell constant}$$

Units:  $\text{S} \cdot \text{cm}^{-1}$  or  $\text{S} \cdot \text{m}^{-1}$ .

It is an intrinsic property of the solution (independent of cell geometry).

**3. Equivalent Conductance ( $\Lambda_{\text{eq}}$ ):**

The **equivalent conductance** is the conductance of a solution containing **one equivalent of electrolyte** placed between two electrodes 1 cm apart:

$$\Lambda_{\text{eq}} = \frac{\kappa \times 1000}{N}$$

where  $N$  is the normality (equivalents/L) and 1000 converts L to  $\text{cm}^3$ .

Units:  $\text{S} \cdot \text{cm}^2 \cdot \text{eq}^{-1}$ .

**4. Molecular (Molar) Conductance ( $\Lambda_m$ ):**

The **molar conductance** is the conductance of a solution containing **one mole of electrolyte** between electrodes 1 cm apart:

$$\Lambda_m = \frac{\kappa \times 1000}{M}$$

where  $M$  is the molarity (mol/L).

Units:  $\text{S} \cdot \text{cm}^2 \cdot \text{mol}^{-1}$ .

For a 1:1 electrolyte:  $\Lambda_m = \Lambda_{\text{eq}}$ . For a 2:1 electrolyte (e.g.  $\text{CaCl}_2$ ):  $\Lambda_m = 2\Lambda_{\text{eq}}$ .

**Relation Between  $\kappa$  and  $\Lambda_{\text{eq}}$ :**

Consider 1 L of solution of normality  $N$ . Number of equivalents =  $N$ . Volume containing 1 equivalent =  $1000/N \text{ cm}^3$ . If this volume is placed between electrodes of area  $A$  separated by  $l$ :

$$\Lambda_{\text{eq}} = G \times 1 \quad (\text{electrodes 1 cm apart, unit area})$$

$$G = \kappa \times \frac{A}{l} = \kappa \times \frac{1000/N}{1} \quad (\text{when } l = 1, A = V = 1000/N)$$

Therefore:

$$\Lambda_{\text{eq}} = \frac{\kappa \times 1000}{N}$$

**Summary:**

| Quantity                                   | Definition             | Units                                     |
|--------------------------------------------|------------------------|-------------------------------------------|
| $G$ (Conductance)                          | $1/R$                  | S                                         |
| $\kappa$ (Specific conductance)            | $G \times l/A$         | $\text{S}\cdot\text{cm}^{-1}$             |
| $\Lambda_{\text{eq}}$ (Equiv. conductance) | $\kappa \times 1000/N$ | $\text{S}\cdot\text{cm}^2\text{eq}^{-1}$  |
| $\Lambda_m$ (Molar conductance)            | $\kappa \times 1000/M$ | $\text{S}\cdot\text{cm}^2\text{mol}^{-1}$ |

**Q23.** Give a brief description of the lead storage cell. (10 marks)

[CHEM 101 | 2007–08]

**Q24.** What is an electrochemical cell? How do you understand the terms “half cell”, “half-cell reaction” and “salt bridge”? (10 marks)

[CHEM 101 | 2006–07]

### Answer

#### Electrochemical Cell:

An **electrochemical cell** is a device that converts **chemical energy to electrical energy** (galvanic cell) or **electrical energy to chemical energy** (electrolytic cell) through redox reactions at two electrodes.

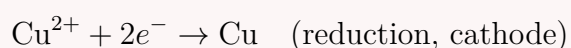
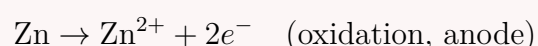
#### Half Cell:

A **half cell** (or electrode) consists of a single electrode in contact with a solution of its ions. It is the site of either oxidation or reduction:

- **Anode half cell:** oxidation occurs (e.g.  $\text{Zn} \mid \text{ZnSO}_4$  solution)
- **Cathode half cell:** reduction occurs (e.g.  $\text{Cu} \mid \text{CuSO}_4$  solution)

#### Half-cell reaction:

The individual oxidation or reduction reaction occurring at one electrode:



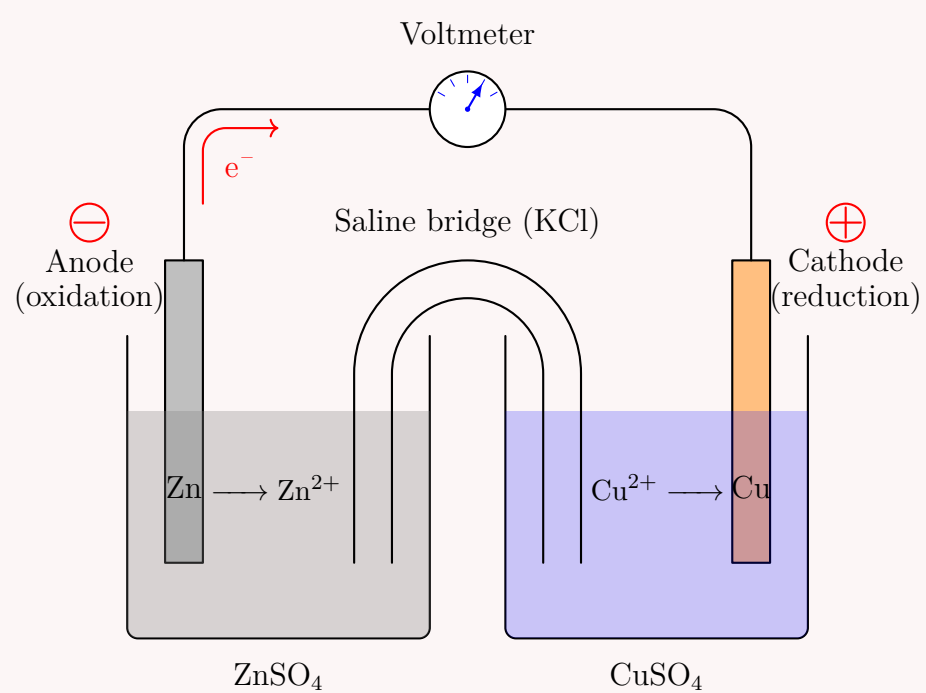
The sum of two half-cell reactions gives the **overall cell reaction**.

#### Salt Bridge:

A **salt bridge** is a U-shaped tube filled with a concentrated solution of an inert electrolyte (e.g.  $\text{KCl}$ ,  $\text{KNO}_3$  in agar gel) connecting the two half cells. Its functions:

- **Completes the electrical circuit** by allowing ion flow between the two solutions
- **Maintains electrical neutrality** — as  $\text{Zn}^{2+}$  builds up in the anode compartment,  $\text{Cl}^-$  ions migrate in; as  $\text{Cu}^{2+}$  is depleted at the cathode,  $\text{K}^+$  ions migrate in
- **Prevents direct mixing** of the two electrode solutions (which would cause direct chemical reaction and no electrical work)
- $\text{KCl}$  is preferred because  $t_+(\text{K}^+) \approx t_-(\text{Cl}^-)$ , minimising the liquid junction potential





**Conclusion:** An electrochemical cell converts chemical to electrical energy via two half cells (each a single electrode/solution interface) connected by a salt bridge (which maintains neutrality and completes the ionic circuit) and an external circuit (carrying electron flow).

BUET · CHEM 113

# Chemistry

Section S9

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Polymers

## S9 — Polymers

**Q1.** Describe the mechanism of polymer conductivity. Discuss the effect of doping and other factors on the degree of conductivity. (10 marks) [CHEM 113 | 2023–24]

### Answer

#### Mechanism of Polymer Conductivity, Effect of Doping & Other Factors

(10 marks)

##### Part 1: Why Most Polymers Are Insulators

(2 marks)

Most polymers are electrical insulators because:

- Their electrons are localized in  $\sigma$  bonds between specific atoms — no free charge carriers exist.
- The HOMO–LUMO (valence band–conduction band) gap is very large ( $E_g > 5$  eV), so electrons cannot be thermally promoted to conduct.
- There is no continuous pathway for electron movement along or between chains.

##### Part 2: Mechanism of Conductivity in Conjugated Polymers

(4 marks)

Conducting polymers (e.g. polyacetylene, polyaniline, polypyrrole) have **alternating single and double bonds** along the backbone — a **conjugated  $\pi$  system**.

###### (i) Band Formation:

The overlapping  $p_z$  orbitals of all  $sp^2$ -hybridized carbons form a continuous  $\pi$  band:

- Bonding  $\pi$  MOs  $\rightarrow$  **valence band (HOMO band)**
- Antibonding  $\pi^*$  MOs  $\rightarrow$  **conduction band (LUMO band)**

###### (ii) The Band Gap Problem:

In undoped polyacetylene, bond length alternation (short C=C and long C–C) creates a **Peierls distortion**, opening a band gap of  $\sim 1.5$  eV. This makes undoped polyacetylene a **semiconductor**, not a metal.

###### (iii) Charge Transport:

Upon doping, charge carriers (**polarons** and **solitons**) are created along the polymer chain. These are localized electronic defects that can move through the conjugated system under an applied electric field:

Applied field  $\Rightarrow$  polaron/soliton migration  $\Rightarrow$  electrical current

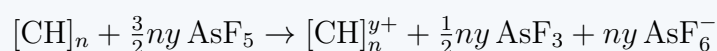
Conductivity also occurs by **interchain hopping** of charge carriers between adjacent polymer chains.

##### Part 3: Effect of Doping on Conductivity

(3 marks)

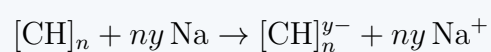
**Doping** introduces charge carriers into the polymer by oxidation (p-doping) or reduction (n-doping):

###### (i) p-Doping (oxidation):



Electrons are removed from the valence band, creating **holes** (positive charge carriers).

###### (ii) n-Doping (reduction):



Electrons are added to the conduction band, creating **free electrons** as charge carriers.

###### Effect on conductivity:

| State                 | Conductivity              | Reason                                 |
|-----------------------|---------------------------|----------------------------------------|
| Undoped polyacetylene | $\sim 10^{-5}$ S/cm       | Large band gap, no free carriers       |
| Lightly doped         | $\sim 10^{-2}$ S/cm       | Few polarons/solitons created          |
| Heavily doped         | $\sim 10^3$ – $10^5$ S/cm | Band gap closes; metallic conductivity |

###### Other factors affecting conductivity:

- **Conjugation length:** Longer conjugation  $\Rightarrow$  narrower band gap  $\Rightarrow$  higher conductivity.
- **Crystallinity:** More ordered chains allow better interchain charge hopping.
- **Temperature:** Unlike metals, conductivity of doped polymers *increases* with temperature (semiconductor-like behaviour).
- **Dopant concentration:** Higher doping level  $\Rightarrow$  more charge carriers  $\Rightarrow$  higher conductivity (up to a saturation point).
- **Chain defects:** Kinks, twists and crosslinks disrupt conjugation and *reduce* conductivity.

**Conclusion:** Polymer conductivity arises from delocalized  $\pi$  electrons in conjugated systems, with charge transported via polarons and solitons. Doping dramatically increases conductivity by 8–12 orders of magnitude by injecting charge carriers into the band structure. Conjugation length, crystallinity, temperature, and dopant level are the key control parameters.

**Q2.** What are the different types of conducting polymers? Discuss the general mechanism of conduction in conjugated polymers. (10 marks) [CHEM 113 | 2022–23; 2020–21]

### Answer

#### Types of Conducting Polymers & General Mechanism of Conduction in Conjugated Polymers

(10 marks)

##### Part 1: Types of Conducting Polymers

(4 marks)

Conducting polymers are classified into three broad types:

##### (i) Conjugated Conducting Polymers:

These have alternating single and double bonds providing a delocalized  $\pi$  electron pathway. They are intrinsically semiconducting and become highly conducting upon doping.

| Polymer                    | Symbol | Conductivity (doped) |
|----------------------------|--------|----------------------|
| Polyacetylene              | PA     | $10^3$ – $10^5$ S/cm |
| Polyaniline                | PANI   | $10$ – $10^2$ S/cm   |
| Polypyrrole                | PPy    | $10$ – $10^3$ S/cm   |
| Polythiophene              | PT     | $10^2$ – $10^3$ S/cm |
| Poly(p-phenylene)          | PPP    | $10^3$ S/cm          |
| Poly(p-phenylene vinylene) | PPV    | $10^3$ S/cm          |

##### (ii) Redox Conducting Polymers:

Conduct electricity through a sequence of oxidation–reduction reactions between neighbouring fixed redox sites along the chain. Charge is transferred by sequential electron hopping (not band conduction). Example: poly(vinylferrocene).

##### (iii) Ionically Conducting Polymers (Polymer Electrolytes):

Conduct by ion transport rather than electron transport. A polymer matrix (e.g. polyethylene oxide, PEO) swells with a dissolved salt (e.g.  $\text{LiClO}_4$ ); ions migrate through the flexible amorphous regions under an electric field. Used in solid-state batteries and fuel cells.

##### Part 2: General Mechanism of Conduction in Conjugated Polymers

(6 marks)

##### (i) $\pi$ -Band Formation:

Each  $sp^2$  carbon contributes one  $p_z$  orbital. For a chain of  $N$  carbons, LCAO gives  $N$   $\pi$  MOs:

$$N p_z \xrightarrow{\text{LCAO}} \frac{N}{2} \text{ bonding (VB)} + \frac{N}{2} \text{ antibonding (CB)}$$

As  $N \rightarrow \infty$ , discrete MOs merge into continuous bands.

##### (ii) Peierls Distortion & Band Gap:

A perfectly uniform polyacetylene chain would be metallic. However, **Peierls distortion** causes alternating bond lengths ( $\text{C}=\text{C}$ : 1.36 Å,  $\text{C}-\text{C}$ : 1.44 Å), which splits the half-filled band into a filled valence band (VB) and empty conduction band (CB), creating a band gap of  $\sim 1.5$  eV.

##### (iii) Charge Carriers — Solitons and Polarons:

Upon doping or photoexcitation, three types of charge carriers form:

- **Soliton:** A domain wall between two degenerate bond-alternation patterns in trans-PA. A neutral soliton has one unpaired electron (spin- $\frac{1}{2}$ , no charge); a charged soliton has no spin but carries charge  $\pm e$ .
- **Polaron:** A charge + accompanying lattice distortion. Formed in non-degenerate ground state polymers (PPy, PT, PANI). It introduces two localized energy levels within the band gap.
- **Bipolaron:** Two polarons of the same sign combining; introduces two energy levels deeper in the gap. More stable than two separate polarons.

##### (iv) Conduction Mechanism:

- **Intrachain:** Solitons/polarons migrate along the conjugated chain under an electric field. Conductivity depends on conjugation length.
- **Interchain:** Charge hops between adjacent chains via  $\pi$ – $\pi$  stacking or direct orbital overlap. This is often the rate-limiting step.
- **Overall:** The combination of intrachain and interchain transport gives bulk conductivity.

$$\sigma = ne\mu$$

where  $n$  = carrier density,  $e$  = electron charge,  $\mu$  = carrier mobility.

**Conclusion:** Conducting polymers fall into three types: conjugated (band conduction), redox (electron hopping), and ionic (ion transport). In conjugated polymers,  $\pi$  bands form from  $p_z$  orbital overlap; the Peierls distortion creates a band gap; doping generates solitons and polarons as charge carriers that move both along and between chains to produce electrical current.

**Q3.** Show the chemical processes involved in doping polyacetylene and explain the increased conductivity of doped polyacetylene by discussing the mechanism of conduction in doped polymers. (10 marks) [CHEM 113 | 2022–23]

### Answer

#### Doping of Polyacetylene: Chemical Processes & Mechanism of Increased Conductivity

(10 marks)

##### Part 1: Undoped Polyacetylene — Baseline

(1 marks)

Trans-polyacetylene ( $[\text{CH}]_n$ ) has alternating C=C and C–C bonds. Due to Peierls distortion, it has a band gap of  $\sim 1.5$  eV and conductivity of  $\sim 10^{-5}$  S/cm — essentially a semiconductor/insulator.

##### Part 2: p-Doping (Oxidative Doping)

(3 marks)

p-Doping removes electrons from polyacetylene using an oxidizing agent (Lewis acid / electron acceptor). The most studied dopant is  $\text{AsF}_5$ :

###### Step 1 — Electron transfer:



**Step 2 — Soliton formation:** Removing an electron from the  $\pi$  system creates a **positively charged soliton** — a carbocation defect that disrupts local bond alternation and carries charge  $+e$  with zero spin.

**Step 3 — Counterion stabilization:** The negatively charged  $\text{AsF}_6^-$  ions are incorporated between polymer chains, stabilizing the positive charge on the backbone.

Other p-dopants:  $\text{I}_2$ ,  $\text{Br}_2$ ,  $\text{FeCl}_3$ ,  $\text{SbF}_5$ .

##### Part 3: n-Doping (Reductive Doping)

(3 marks)

n-Doping adds electrons to polyacetylene using a reducing agent (Lewis base / electron donor). The most common dopant is **sodium** (Na):

###### Step 1 — Electron transfer:



**Step 2 — Soliton formation:** Adding an electron to the  $\pi^*$  (conduction) band creates a **negatively charged soliton** — a carbanion defect carrying charge  $-e$  with zero spin.

**Step 3 — Counterion stabilization:**  $\text{Na}^+$  ions insert between chains and stabilize the negative charge on the backbone.

Other n-dopants: K, Li, Na-naphthalide.

##### Part 4: Mechanism of Increased Conductivity

(3 marks)

###### (i) Soliton pair formation and mobility:

Doping creates **soliton–antisoliton pairs** at the site of charge injection. Each soliton is a mobile domain wall between two bond-alternation phases:



Under an applied electric field, solitons migrate along the chain, carrying charge — this is **intrachain conduction**.

###### (ii) Band gap reduction:

As doping level increases:

- Soliton energy levels appear within the band gap.
- At high doping, these levels merge with VB and CB, effectively **closing the band gap**.
- The polymer approaches **metallic conductivity**.

###### (iii) Conductivity comparison:

| Material           | Conductivity (S/cm)  | Character     |
|--------------------|----------------------|---------------|
| Undoped PA         | $\sim 10^{-5}$       | Semiconductor |
| Lightly doped PA   | $\sim 10^0 - 10^2$   | Semiconductor |
| Heavily doped PA   | $\sim 10^3 - 10^5$   | Near-metallic |
| Copper (reference) | $\sim 6 \times 10^5$ | Metal         |

**Conclusion:** p-Doping ( $\text{AsF}_5$ ,  $\text{I}_2$ ) removes electrons from PA creating positive solitons; n-doping (Na, K) adds electrons creating negative solitons. These mobile charge defects carry current along the conjugated backbone. Heavy doping closes the Peierls band gap, raising conductivity by up to 10 orders of magnitude from  $10^{-5}$  to  $10^5$  S/cm.

Q4. What distinguishes thermoplastics from thermosetting plastics? (5 marks)

[CHEM 113 | 2021–22]

### Answer

#### Thermoplastics vs Thermosetting Plastics

(5 marks)

##### (i) Thermoplastics:

Thermoplastics are polymers that **soften and melt** upon heating and **re-solidify** upon cooling — this process is **reversible** and can be repeated many times.

- Polymer chains are held together by **weak intermolecular forces** (van der Waals, dipole–dipole, H-bonds).
- Heating breaks these forces  $\Rightarrow$  chains can flow and be reshaped.
- Cooling restores the forces  $\Rightarrow$  solid again.
- **Recyclable** and remoldable.
- Examples: polyethylene (PE), polypropylene (PP), PVC, nylon, polystyrene.

##### (ii) Thermosetting Plastics:

Thermosets are polymers that undergo **irreversible chemical crosslinking** upon heating (or by catalyst), forming a rigid, three-dimensional network. Once set, they **cannot be re-melted or reshaped**.

- Polymer chains are connected by **covalent crosslinks** — permanent bonds.
- Heating does not break crosslinks; further heating causes **degradation/charring**.
- **Not recyclable**.
- Examples: epoxy resins, Bakelite, vulcanized rubber, melamine formaldehyde, polyurethane.

#### Comparison Table:

| Property             | Thermoplastic       | Thermosetting       |
|----------------------|---------------------|---------------------|
| Heating effect       | Softens/melts       | Hardens/chars       |
| Cross-linking        | None                | Extensive covalent  |
| Reversibility        | Reversible          | Irreversible        |
| Recyclability        | Yes                 | No                  |
| Intermolecular force | Weak (physical)     | Strong (covalent)   |
| Solubility           | Soluble in solvents | Generally insoluble |
| Examples             | PE, PP, PVC         | Bakelite, epoxy     |

**Conclusion:** Thermoplastics are distinguished by their reversible melting behaviour due to weak intermolecular forces, making them recyclable. Thermosets form permanent covalent crosslinks upon curing, giving them superior heat and chemical resistance but making them non-recyclable.

Q5. Apply your knowledge of intermolecular forces to explain the thermal transitions in amorphous and crystalline polymers. (10 marks)

[CHEM 113 | 2021–22]

### Answer

#### Intermolecular Forces & Thermal Transitions in Amorphous and Crystalline Polymers

(10 marks)

##### Part 1: Types of Intermolecular Forces in Polymers

(2 marks)

The physical properties and thermal transitions of polymers are governed by intermolecular forces between chains:

- **London dispersion forces:** Present in all polymers; increase with molecular weight and surface area. Dominant in non-polar polymers (PE, PP).
- **Dipole–dipole interactions:** In polymers with polar groups (C=O, C–Cl). Stronger than dispersion; raise  $T_g$  and  $T_m$ .
- **Hydrogen bonding:** Strongest non-covalent interaction; present in nylon (N–H $\cdots$ O=C), cellulose, proteins. Significantly raises  $T_g$  and  $T_m$ .
- **$\pi$ – $\pi$  stacking:** In aromatic polymers (polystyrene, PET); adds rigidity and raises  $T_g$ .



Part 2: Amorphous Polymers — Glass Transition ( $T_g$ )

(4 marks)

Amorphous polymers have **randomly coiled, entangled chains** with no long-range order. They exhibit a single key thermal transition:

The Glass Transition Temperature ( $T_g$ ):

- Below  $T_g$ : chains are **frozen** in place; only small-scale bond vibrations possible. The polymer is **hard, brittle, glassy**.
- Above  $T_g$ : chains gain enough **thermal energy** to overcome intermolecular forces and begin large-scale cooperative segmental motion. The polymer becomes **rubbery/leathery**.
- $T_g$  is not a true phase transition — it is a **kinetic, second-order transition**.

Effect of intermolecular forces on  $T_g$ :

- Stronger** intermolecular forces  $\Rightarrow$  more energy needed to activate chain motion  $\Rightarrow$  **higher**  $T_g$ .
- Example: polystyrene ( $T_g = 100^\circ\text{C}$ ,  $\pi$ - $\pi$  stacking)  $>$  polyethylene ( $T_g = -120^\circ\text{C}$ , only dispersion).
- H-bonding in nylon raises  $T_g$  to  $\sim 50$ – $80^\circ\text{C}$ .

Part 3: Crystalline Polymers — Melting Transition ( $T_m$ )

(4 marks)

Crystalline (semi-crystalline) polymers have **ordered, closely packed chain segments** (crystallites) within an amorphous matrix. They exhibit two transitions:

(i) **Glass transition  $T_g$** : In the amorphous regions (as above).

(ii) **Melting Temperature  $T_m$** :

- Below  $T_m$ : crystalline regions maintain ordered packing held by intermolecular forces. Polymer is **rigid and strong**.
- At  $T_m$ : thermal energy overcomes the **lattice energy** of the crystallites; ordered packing collapses and chains flow freely. This is a true **first-order phase transition**.
- Above  $T_m$ : completely amorphous melt.

Effect of intermolecular forces on  $T_m$ :

- Stronger** forces between chains in crystallites  $\Rightarrow$  **higher**  $T_m$ .
- Nylon-6,6:  $T_m = 265^\circ\text{C}$  (strong H-bonds between amide groups).
- Polyethylene:  $T_m = 137^\circ\text{C}$  (only dispersion, but efficient close packing).
- PET:  $T_m = 265^\circ\text{C}$  ( $\pi$ - $\pi$  stacking + dipole-dipole).

Summary:

| Polymer type     | $T_g$                | $T_m$               | Dominant IMF           |
|------------------|----------------------|---------------------|------------------------|
| Amorphous        | Yes                  | No                  | Varies                 |
| Semi-crystalline | Yes                  | Yes                 | Varies                 |
| Polyethylene     | $-120^\circ\text{C}$ | $137^\circ\text{C}$ | Dispersion             |
| Nylon-6,6        | $50^\circ\text{C}$   | $265^\circ\text{C}$ | H-bonding              |
| Polystyrene      | $100^\circ\text{C}$  | —                   | $\pi$ - $\pi$ stacking |

**Conclusion:** Intermolecular forces directly govern thermal transitions. In amorphous polymers,  $T_g$  marks the onset of segmental chain motion when thermal energy overcomes intermolecular attractions. In semi-crystalline polymers,  $T_m$  additionally marks the collapse of ordered crystallites. Stronger forces (H-bonding  $>$  dipole-dipole  $>$  dispersion) raise both  $T_g$  and  $T_m$ , explaining the wide range of thermal behaviour across different polymer types.

Q6. List the requirements of a conducting polymer for electrical conduction. Discuss a process for improving the conduction of polyacetylene, utilizing the concept of HOMO-LUMO. (10 marks) [CHEM 113 | 2021–22]

Answer

Requirements for Conducting Polymers & Improving Polyacetylene Conduction via HOMO-LUMO

(10 marks)

Part 1: Requirements for Electrical Conduction in a Conducting Polymer

(4 marks)

For a polymer to conduct electricity, the following conditions must be met:

- **Conjugated  $\pi$  system:** The backbone must have alternating single and double bonds (or aromatic rings) providing overlapping  $p_z$  orbitals along the entire chain. This creates the valence and conduction bands necessary for electron transport.
- **Small or zero HOMO–LUMO (band) gap:** The energy gap between the filled  $\pi$  band (HOMO/VB) and empty  $\pi^*$  band (LUMO/CB) must be small enough for charge carriers to be generated thermally or by doping. Gap  $< 3$  eV is required for semiconducting behaviour.
- **Planarity of the backbone:** The  $p_z$  orbitals must be parallel for maximum overlap. Any twist or non-planarity interrupts conjugation and reduces conductivity.
- **Charge carriers (from doping):** Undoped conjugated polymers are typically semiconductors. Doping is required to inject charge carriers (polarons, solitons, bipolarons) into the  $\pi$  system.
- **Interchain charge transport pathway:** For bulk conductivity, charges must be able to hop between chains via  $\pi$ – $\pi$  stacking or chain entanglement. Ordered, crystalline packing enhances this.
- **Structural integrity:** The polymer must be stable enough that doping does not cause chain degradation or loss of conjugation.

## Part 2: Improving Polyacetylene Conduction via HOMO-LUMO Concept

(6 marks)

### (i) The HOMO-LUMO Gap in Undoped Polyacetylene:

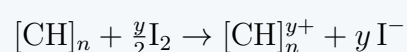
In undoped trans-polyacetylene:

$$E_g = E_{\text{LUMO}} - E_{\text{HOMO}} \approx 1.5 \text{ eV}$$

This gap arises from Peierls distortion (bond alternation). Electrons cannot spontaneously cross this gap  $\Rightarrow$  poor conductivity ( $\sim 10^{-5}$  S/cm).

### (ii) Strategy 1 — p-Doping to Depopulate HOMO:

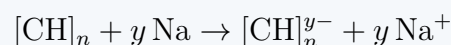
A p-type dopant (oxidant, e.g.  $\text{I}_2$ ,  $\text{AsF}_5$ ) removes electrons *from* the HOMO:



- HOMO is partially emptied  $\Rightarrow$  holes created in the valence band.
- Electrons from lower HOMO levels can now move into these holes — current flows.
- Effectively: top of VB becomes partially unoccupied  $\Rightarrow$  metallic-like conduction.

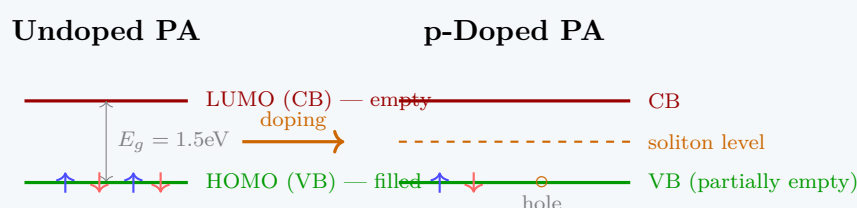
### (iii) Strategy 2 — n-Doping to Populate LUMO:

An n-type dopant (reductant, e.g. Na, K) adds electrons *into* the LUMO:



- LUMO is partially populated  $\Rightarrow$  free electrons in the conduction band.
- These electrons are mobile and carry current.
- Effectively: bottom of CB becomes partially occupied  $\Rightarrow$  n-type metallic conduction.

### (iv) Energy Level Diagram showing improvement:



### (v) Result:

Doping reduces the effective HOMO–LUMO gap and introduces mid-gap states. At high doping levels, the band gap essentially closes and polyacetylene achieves conductivity of  $\sim 10^5$  S/cm, comparable to some metals.

**Conclusion:** Conducting polymers require conjugated backbones, small HOMO–LUMO gaps, planarity, and extrinsic doping. Polyacetylene's conductivity is improved by p-doping (emptying HOMO) or n-doping (filling LUMO), which introduces mobile charge carriers and effectively narrows the band gap from 1.5 eV toward zero at high doping levels.

**Q7.** What are the different applications of conducting polymers? Discuss the general mechanism of conduction in conjugated polymers. (10 marks) [CHEM 113 | 2020–21]

## Answer

## Applications of Conducting Polymers &amp; Mechanism of Conduction in Conjugated Polymers

(10 marks)

## Part 1: Applications of Conducting Polymers

(5 marks)

- (i) **Rechargeable Batteries:** Conducting polymers (polyaniline, polypyrrole) are used as electrode materials in lightweight, flexible batteries. They undergo reversible redox reactions during charge/discharge cycles, storing and releasing electrical energy.
- (ii) **Organic Light-Emitting Diodes (OLEDs):** PPV (poly(p-phenylene vinylene)) and its derivatives emit light when charge carriers recombine across the HOMO–LUMO gap. Used in smartphone and TV displays. Flexible, lightweight, and energy-efficient.
- (iii) **Solar Cells (Organic Photovoltaics):** Conducting polymers absorb photons, generating excitons (electron–hole pairs) that are separated at a donor–acceptor interface. P3HT (poly(3-hexylthiophene)) is widely used.
- (iv) **Sensors and Biosensors:** The conductivity of polypyrrole and PANI changes dramatically upon exposure to specific analytes (gases, pH, biomolecules). Used in gas sensors ( $\text{NH}_3$ ,  $\text{NO}_2$ ), glucose sensors, and DNA detection.
- (v) **Corrosion Protection:** Polyaniline coatings on metals create a protective barrier and, through its oxidizing ability, passivate the metal surface. More environmentally friendly than chromate coatings.
- (vi) **Supercapacitors / Energy Storage:** High surface area conducting polymer electrodes (PPy, PANI) can store large amounts of charge via pseudocapacitance (faradaic charge storage), giving higher energy density than conventional capacitors.
- (vii) **Electromagnetic Shielding:** Doped polyaniline is used to coat electronic equipment, absorbing and reflecting electromagnetic interference (EMI) without the weight penalty of metal shielding.
- (viii) **Electrochromic Devices:** Polymers like PEDOT change colour reversibly upon oxidation/reduction. Used in smart windows, electrochromic displays, and anti-glare mirrors.

## Part 2: General Mechanism of Conduction in Conjugated Polymers

(5 marks)

- (i)  **$\pi$ -Band formation:**  $sp^2$  carbon  $p_z$  orbitals overlap to form bonding  $\pi$  (VB/HOMO band) and antibonding  $\pi^*$  (CB/LUMO band). Peierls distortion opens a band gap in alternating bond systems.
- (ii) **Charge carrier generation (doping):** p-Doping removes electrons from VB; n-doping adds electrons to CB. This generates **solitons** (in trans-PA), **polarons** (radical cation/anion + lattice distortion), or **bipolarons** (two polarons combined).
- (iii) **Intrachain transport:** Under an electric field, polarons/solitons migrate along the conjugated backbone — effectively a wave of bond alternation moving through the chain.
- (iv) **Interchain hopping:** Charges tunnel or hop between adjacent chains via  $\pi$ – $\pi$  stacking. This is the rate-limiting step for bulk conductivity. More ordered (crystalline) polymer packing  $\Rightarrow$  better interchain hopping  $\Rightarrow$  higher conductivity.
- (v) **Overall conductivity:**

$$\sigma = ne\mu$$

$n$  = carrier density (controlled by doping level),  $\mu$  = carrier mobility (controlled by conjugation length and chain order).

**Conclusion:** Conducting polymers find applications across batteries, OLEDs, solar cells, sensors, corrosion protection, supercapacitors, EMI shielding and smart windows — all exploiting their tunable conductivity. Conduction arises from  $\pi$ -band formation, charge carrier generation by doping (solitons/polarons), and their migration both along chains and between chains.

Q8. “The conductivity of doped polyacetylene is more than undoped molecule” — Justify by showing the addition of Lewis acid and Lewis base as dopant. (10 marks) [CHEM 113 | 2020–21]

## Answer

## Conductivity of Doped vs Undoped Polyacetylene: Lewis Acid and Lewis Base Doping

(10 marks)

## Part 1: Why Undoped Polyacetylene Has Low Conductivity

(2 marks)

Undoped trans-polyacetylene has:

- A band gap  $E_g \approx 1.5$  eV due to Peierls distortion (bond alternation).
- No free charge carriers.
- Conductivity  $\sim 10^{-5}$  S/cm (essentially insulating).

Part 2: p-Doping with Lewis Acid (AsF<sub>5</sub>, I<sub>2</sub>)

(4 marks)

A **Lewis acid** is an electron pair acceptor — it *removes* electrons from polyacetylene (oxidation), creating positive charges on the chain.

**With AsF<sub>5</sub>:**

$$[\text{CH}]_n + \frac{3}{2}ny \text{AsF}_5 \longrightarrow [\text{CH}]_n^{y+} + \frac{ny}{2} \text{AsF}_3 + ny \text{AsF}_6^-$$

**With I<sub>2</sub>:**

$$[\text{CH}]_n + \frac{ny}{2} \text{I}_2 \longrightarrow [\text{CH}]_n^{y+} + ny \text{I}^-$$

**What happens:**

- Electrons are removed from the HOMO ( $\pi$  valence band).
- This creates **positively charged solitons** (radical cations) — mobile holes in the VB.
- The AsF<sub>6</sub><sup>−</sup> or I<sup>−</sup> counterions stabilize the positive charges between the chains.
- Conductivity jumps from 10<sup>−5</sup> to  $\sim 10^3\text{--}10^5$  S/cm.

Part 3: n-Doping with Lewis Base (Na, K)

(4 marks)

A **Lewis base** is an electron pair donor — it *donates* electrons to polyacetylene (reduction), creating negative charges on the chain.

**With Sodium (Na):**

$$[\text{CH}]_n + ny \text{Na} \longrightarrow [\text{CH}]_n^{y-} + ny \text{Na}^+$$

**With Sodium naphthalide (Na<sup>+</sup>C<sub>10</sub>H<sub>8</sub><sup>−</sup>):**

$$[\text{CH}]_n + ny \text{Na}^+ \text{C}_{10}\text{H}_8^- \longrightarrow [\text{CH}]_n^{y-} + ny \text{Na}^+ + ny \text{C}_{10}\text{H}_8$$

**What happens:**

- Electrons are donated into the LUMO ( $\pi^*$  conduction band).
- This creates **negatively charged solitons** (radical anions) — mobile electrons in the CB.
- The Na<sup>+</sup> counterions stabilize the negative charges between the chains.
- Conductivity increases dramatically.

**Comparison:**

| System                      | Dopant type | Carrier                    | Conductivity                          |
|-----------------------------|-------------|----------------------------|---------------------------------------|
| Undoped PA                  | —           | None                       | 10 <sup>−5</sup> S/cm                 |
| p-Doped (AsF <sub>5</sub> ) | Lewis acid  | +soliton (hole)            | 10 <sup>3</sup> –10 <sup>5</sup> S/cm |
| n-Doped (Na)                | Lewis base  | −soliton (e <sup>−</sup> ) | 10 <sup>2</sup> –10 <sup>3</sup> S/cm |

**Conclusion:** Undoped polyacetylene is a poor conductor due to its 1.5 eV band gap. Lewis acid dopants (AsF<sub>5</sub>, I<sub>2</sub>) oxidize PA, removing electrons from the HOMO and creating mobile positive solitons (p-doping). Lewis base dopants (Na, K) reduce PA, adding electrons to the LUMO and creating mobile negative solitons (n-doping). Both processes introduce free charge carriers, increasing conductivity by 8–10 orders of magnitude.

Q9. Name the different types of polymerization processes with examples. (5 marks)

[CHEM 113 | 2020–21]

Answer

Types of Polymerization Processes with Examples

(5 marks)

Polymerization processes are broadly classified into two main types, each with subtypes:

**(i) Addition (Chain-Growth) Polymerization:**  
Monomers add sequentially to a growing chain end without loss of any atoms. No by-product is formed.

$$n \text{CH}_2=\text{CH}_2 \xrightarrow{\text{catalyst}} -[\text{CH}_2-\text{CH}_2]_n-$$

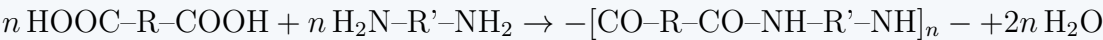
- Free radical:** Polyethylene (PE), polystyrene (PS), PVC, PMMA.
- Cationic:** Polyisobutylene, polystyrene.

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- **Anionic:** Polystyrene (living), polybutadiene.
- **Coordination (Ziegler-Natta):** Isotactic/syndiotactic PP, HDPE.
- **Ring-opening:** PLA, PCL, nylon-6 (from caprolactam).

**(ii) Condensation (Step-Growth) Polymerization:**

Monomers with two functional groups react stepwise; a small molecule (H<sub>2</sub>O, HCl, CH<sub>3</sub>OH) is eliminated.



- **Polyester:** PET (terephthalic acid + ethylene glycol), Dacron.
- **Polyamide:** Nylon-6,6 (adipic acid + hexamethylenediamine).
- **Polycarbonate:** Bisphenol A + phosgene.
- **Polyurethane:** Diol + diisocyanate.
- **Phenol-formaldehyde:** Bakelite (thermosetting).

Key differences:

| Property          | Addition              | Condensation                            |
|-------------------|-----------------------|-----------------------------------------|
| By-product        | None                  | Small molecule (H <sub>2</sub> O, etc.) |
| Mechanism         | Chain (radical/ionic) | Stepwise                                |
| MW growth         | Rapid (fast)          | Slow (gradual)                          |
| Functional groups | One (C=C or ring)     | Two per monomer                         |
| Examples          | PE, PP, PVC           | Nylon, PET, Bakelite                    |

**Conclusion:** The two main polymerization types are **addition** (chain-growth, no by-product, e.g. PE from ethylene) and **condensation** (step-growth, small molecule eliminated, e.g. nylon from diamine + diacid). Subtypes include free radical, cationic, anionic, coordination, and ring-opening for addition; and polyester, polyamide, and polycarbonate for condensation.

Q10. What is thermoplastic starch? Discuss the applications and limitations of thermoplastic starch. (10 marks)

[CHEM 113 | 2017–18]

Answer

Thermoplastic Starch: Definition, Applications & Limitations

(10 marks)

Part 1: What is Thermoplastic Starch?

(3 marks)

Native starch is a semi-crystalline biopolymer consisting of two components:

- **Amylose** (~20–30%): linear chains of α-1,4-linked glucose; forms crystalline helices.
- **Amylopectin** (~70–80%): branched chains with α-1,6 branch points; forms crystalline lamellae.

Native starch cannot be melt-processed because it decomposes before melting. **Thermoplastic starch (TPS)** is produced by disrupting this crystallinity through:

$$\text{Native starch} + \text{plasticizer} \xrightarrow{\text{heat} + \text{shear}} \text{TPS (amorphous melt)}$$

Common plasticizers: **water** (10–20 wt%), **glycerol**, **sorbitol**, **urea**. Processing conditions: 120–180°C, twin-screw extruder. The plasticizer molecules form hydrogen bonds with starch hydroxyl groups, replacing starch–starch H-bonds, increasing chain mobility and reducing *T<sub>g</sub>* and *T<sub>m</sub>* to processable values.

Part 2: Applications

(4 marks)

| Sector      | Application                                                                                                       |
|-------------|-------------------------------------------------------------------------------------------------------------------|
| Packaging   | Loose-fill packaging, food trays, compostable bags, disposable cutlery                                            |
| Agriculture | Biodegradable mulch films (replace polyethylene mulch), seedling pots, slow-release fertilizer/pesticide coatings |
| Food        | Edible coatings on fruits, food service items, encapsulation of food additives                                    |
| Biomedical  | Drug delivery tablets and capsules, tissue engineering scaffolds, wound dressings (TPS–PLA blends)                |
| Electronics | Biodegradable substrates for transient electronics                                                                |



### Part 3: Limitations

(3 marks)

- **Moisture sensitivity (most critical):** TPS is strongly hydrophilic. In humid conditions, it absorbs water, swells, and plasticizer migrates out, leading to deterioration of mechanical properties and dimensional instability.
- **Retrogradation:** On storage, amylose chains re-associate into crystalline structures, making TPS increasingly brittle over time (“ageing” effect). Reduces elongation at break from >100% to <10% within weeks.
- **Poor mechanical properties:** TPS alone has low tensile strength (~2–5 MPa) and poor barrier properties to gases (O<sub>2</sub>, CO<sub>2</sub>) compared to synthetic plastics.
- **Limited thermal stability:** Starch degradation begins at ~220°C, restricting processing temperatures and excluding high-temperature applications.
- **Plasticizer migration:** Glycerol and water can migrate to the surface or leach out over time, altering properties.
- **Food vs fuel competition:** Large-scale production competes with food supply (corn, wheat, potato starch), raising sustainability and ethical concerns.

**Conclusion:** TPS is native starch plasticized by water/glycerol under heat and shear to give a melt-processable, fully biodegradable thermoplastic. It finds applications in packaging, agriculture, food, and biomedicine. Its main limitations are moisture sensitivity, retrogradation-induced embrittlement, and poor mechanical/barrier properties compared to petrochemical plastics — typically addressed by blending with PLA, PHB, or nanoclay reinforcement.

**Q11.** Identify the factors that influence the biodegradability of polymers. Discuss the effect of crystallinity on biodegradability of polymers. (10 marks) [CHEM 113 | 2017–18]

### Answer

#### Factors Influencing Biodegradability of Polymers; Effect of Crystallinity

(10 marks)

#### Part 1: Factors Influencing Biodegradability

(5 marks)

(i) **Chemical structure of the backbone:** Polymers with **hydrolysable linkages** (ester, amide, glycosidic, urethane bonds) biodegrade much faster than those with C–C or C–F backbones.



(ii) **Molecular weight:** Lower MW polymers have more chain ends and are more accessible to microbial enzymes. High MW polymers must first be cleaved to oligomers before cellular uptake is possible.

(iii) **Crystallinity:** (Discussed in detail in Part 2.)

(iv) **Hydrophilicity:** Hydrophilic polymers absorb water, swelling the matrix and allowing enzyme penetration. Hydrophobic polymers (PE, PP) resist water uptake and are inaccessible to enzymes.

(v) **Surface area and morphology:** Thinner films, porous structures, and smaller particle sizes  $\Rightarrow$  greater surface area  $\Rightarrow$  more sites for enzyme attack  $\Rightarrow$  faster degradation.

(vi) **Presence of functional groups:** Hydroxyl (–OH), carboxyl (–COOH), and amine (–NH<sub>2</sub>) groups facilitate hydrogen bonding with water and enzyme active sites, promoting biodegradation. Fluorine or chlorine substituents hinder biodegradation.

(vii) **Environmental conditions:** Temperature, pH, moisture, UV exposure, and the presence/ abundance of specific microorganisms all control degradation rate. Composting conditions (55–60°C, high humidity) accelerate biodegradation significantly.

(viii) **Additives and plasticizers:** Pro-degradant additives (starch blends, photosensitizers) accelerate degradation. Antioxidants and stabilizers retard it.

#### Part 2: Effect of Crystallinity on Biodegradability

(5 marks)

(i) **Crystalline vs amorphous regions:**

Semi-crystalline polymers have two distinct regions:

- **Amorphous regions:** Randomly coiled chains, loosely packed, accessible to water and enzymes.
- **Crystalline regions:** Tightly packed, ordered chains held by strong intermolecular forces; inaccessible to water and enzymes.



**(ii) How crystallinity reduces biodegradation:**

- Crystalline regions have **high density** and **tight chain packing**, leaving no space for enzyme diffusion or water penetration.
- The **activation energy** for chain scission within a crystallite is much higher than in amorphous regions.
- Biodegradation therefore preferentially attacks **amorphous regions first**.

**(iii) Experimental evidence:**

| Polymer           | Crystallinity | Degradation | Comment                    |
|-------------------|---------------|-------------|----------------------------|
| PHB               | ~60%          | Slow        | Crystalline regions resist |
| PLA (amorphous)   | ~0%           | Fast        | Fully accessible           |
| PLA (crystalline) | ~40%          | Slow        | Crystallites protect       |
| Starch (native)   | ~35%          | Moderate    | Amorphous attacked first   |
| TPS               | ~0%           | Fast        | Crystallinity destroyed    |

**(iv) Practical implication:**

Processing conditions that **reduce crystallinity** (quenching, plasticization, blending) enhance biodegradability. This is exactly why TPS degrades much faster than native starch — plasticization eliminates crystallinity, making the entire matrix accessible to amylases.

Higher crystallinity → Slower biodegradation  
 Lower crystallinity (amorphous) → Faster biodegradation

**Conclusion:** Biodegradability is controlled by backbone chemistry, MW, hydrophilicity, surface area, functional groups, and environmental conditions. Crystallinity is particularly important — crystalline regions resist enzyme and water penetration, making highly crystalline polymers degrade much more slowly than amorphous ones. Reducing crystallinity (via TPS processing, quenching, or plasticization) is a key strategy for improving biodegradability.

**Q12.** Why are polymers often insulating? Draw energy level diagrams of metal, semiconductor and insulator. Indicate the band gaps and comment on their conducting behavior. **(10 marks)** [CHEM 113 | 2016–17]

**Answer****Why Polymers Are Insulators; Energy Level Diagrams of Metal, Semiconductor & Insulator****Part 1: Why Most Polymers Are Insulators**

Most conventional polymers (e.g., PE, PP, PVC) are electrical insulators because:

- **Localized  $\sigma$  bonds:** All valence electrons are tightly confined in localized  $\sigma$ -bonds (C–C, C–H) between specific pairs of atoms. No electron can move freely along the polymer chain.
- **No  $\pi$  conjugation:** Without an alternating sequence of single and double bonds, there is no delocalized  $\pi$ -electron system to form a continuous energy band.
- **Very large band gap:** The HOMO–LUMO energy gap for  $\sigma$ -bonded polymers is extremely large ( $> 8\text{--}10\text{ eV}$ ) — far too large for any thermal promotion of electrons into the conduction band at room temperature ( $k_B T \approx 0.025\text{ eV}$ ).
- **No charge carriers:** Without free electrons in the conduction band or holes in the valence band, electrical current cannot flow under an applied electric field.

**Part 2: Energy Level Diagrams**

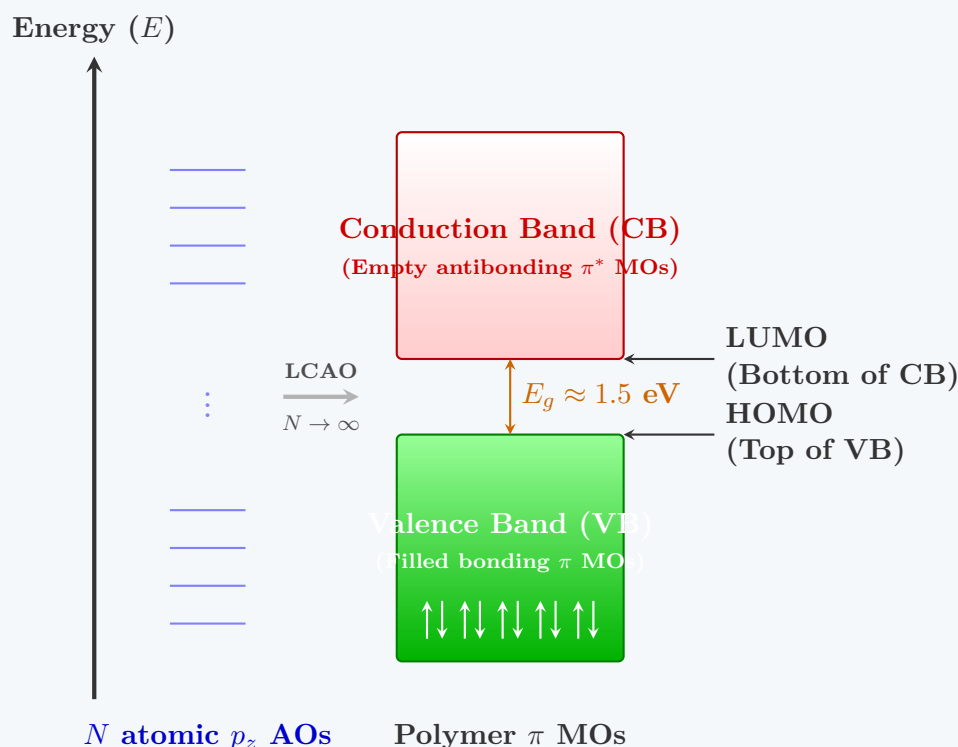


This gap falls perfectly within the semiconductor range (1–3 eV), making undoped trans-polyacetylene an intrinsic **semiconductor**.

(iv) **Conductivity Comparison:**

| Property           | Typical Value        | Implication                       |
|--------------------|----------------------|-----------------------------------|
| Band gap ( $E_g$ ) | 1.5 eV               | Lies in the semiconductor range   |
| Undoped $\sigma$   | $10^{-5}$ S/cm       | Poor conductivity, but non-zero   |
| Doped $\sigma$     | $10^3$ – $10^5$ S/cm | Highly conducting (near metallic) |

## Part 2: Energy Level Diagram of Conjugated Polymer $\pi$ Orbitals



### Key Features of the Diagram:

- **Valence Band:** Formed from bonding  $\pi$  MOs. It is completely filled with  $\pi$  electrons at the ground state. The highest energy level of this band is the **HOMO** (Highest Occupied Molecular Orbital).
- **Conduction Band:** Formed from antibonding  $\pi^*$  MOs. It is completely empty at the ground state. The lowest energy level of this band is the **LUMO** (Lowest Unoccupied Molecular Orbital).
- **Band Gap** (1.5 eV): Arises from Peierls distortion. This gap is small enough for limited thermal carrier generation and allows effective doping.
- **Semiconductor Behaviour:** At room temperature, a tiny fraction of electrons can be thermally promoted across the 1.5 eV gap from the HOMO (in the VB) to the LUMO (in the CB). This creates electron-hole pairs, enabling intrinsic semiconduction.

**Conclusion:** Polyacetylene acts as a semiconductor because its conjugated  $p_z$  orbitals combine to form continuous  $\pi$  (VB) and  $\pi^*$  (CB) bands, separated by a 1.5 eV Peierls band gap. The top of the VB acts as the HOMO and the bottom of the CB as the LUMO. This specific energy gap permits thermal excitation and facilitates doping, allowing its conductivity to be engineered from semiconducting to near-metallic states.

**Q14.** Describe different types of doping mechanisms in conjugated polymers. What is the difference in the doping process of conjugated polymers and semiconductors? (10 marks) [CHEM 113 | 2016–17]

### Answer

#### Types of Doping in Conjugated Polymers & Difference from Semiconductor Doping

(10 marks)

#### Part 1: Types of Doping in Conjugated Polymers

(6 marks)

##### (i) Chemical Doping:

The most common method. A chemical oxidant or reductant reacts directly with the polymer.

- **p-type (oxidative) chemical doping:** Electron-accepting dopants ( $I_2$ ,  $AsF_5$ ,  $FeCl_3$ ,  $Br_2$ ) remove electrons from the  $\pi$  system:



- **n-type (reductive) chemical doping:** Electron-donating dopants (Na, K, Li, Na-naphthalide) add electrons to the  $\pi^*$  system:

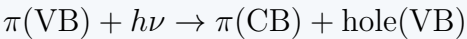


**(ii) Electrochemical Doping:**

The polymer film is used as an electrode in an electrochemical cell. Applying an external voltage either oxidizes (p-doping) or reduces (n-doping) the polymer. Counterions from the electrolyte diffuse into the film to maintain charge neutrality. Advantage: precise control of doping level by controlling applied potential.

**(iii) Photodoping:**

High-energy photons (UV/visible) promote electrons from VB to CB, generating electron–hole pairs:



This is a **temporary** doping — carriers exist only while the light shines. Used in organic photovoltaics and photodetectors.

**(iv) Charge-injection (Interface) Doping:**

At metal–polymer interfaces, electrons can transfer from the metal Fermi level into the polymer LUMO (n-doping) or from the polymer HOMO into the metal (p-doping). No chemical dopant needed. Important in OLEDs and organic FETs.

**(v) Protonic Acid Doping (for polyaniline):**

PANI is uniquely doped by protonation (not redox):



Protons added to imine nitrogens convert the insulating emeraldine base to the conducting emeraldine salt.

Part 2: Difference Between Polymer Doping and Semiconductor Doping

(4 marks)

| Property            | Conjugated Polymer                                          | Inorganic Semiconductor                                |
|---------------------|-------------------------------------------------------------|--------------------------------------------------------|
| Mechanism           | Redox reaction: $e^-$ added or removed from $\pi$ system    | Atomic substitution: dopant atom replaces lattice atom |
| Dopant location     | Counterions between chains (interstitial)                   | Substitutional (within crystal lattice)                |
| Doping level        | Very high: up to 50 mol% (1 per 2 CH units)                 | Very low: ppm level ( $10^{-6}$ )                      |
| Reversibility       | Often reversible (especially electrochemical)               | Irreversible (permanent substitution)                  |
| Carrier type        | Solitons, polarons, bipolarons                              | Free electrons and holes                               |
| Conductivity change | $10^{-5}$ to $10^5$ S/cm (10 orders)                        | Less dramatic change                                   |
| Example             | $\text{PA} + \text{AsF}_5 \rightarrow \text{conducting PA}$ | $\text{Si} + \text{P} \rightarrow \text{n-type Si}$    |

**Conclusion:** Conjugated polymers can be doped by chemical oxidation/reduction, electrochemically, by photodoping, charge injection, or protonic acid doping. Unlike semiconductor doping (atomic substitution at ppm levels, irreversible), polymer doping involves reversible redox reactions at high doping levels, producing exotic charge carriers (polarons/solitons) rather than simple free electrons and holes.

Q15. What are the factors that affect the conduction behavior of polymers? (5 marks)

[CHEM 113 | 2016–17]

Answer

Factors Affecting the Conduction Behaviour of Polymers

(5 marks)

The electrical conductivity of a conducting polymer is governed by  $\sigma = ne\mu$  (carrier density  $\times$  charge  $\times$  mobility). The following factors control  $n$  and  $\mu$ :

- (i) Conjugation length:** Longer uninterrupted conjugation  $\Rightarrow$  wider  $\pi$  bands, smaller band gap, more mobile carriers. Any chain defect, twist, or  $sp^3$  carbon interrupts conjugation and reduces conductivity.
- (ii) Doping level:** Higher dopant concentration  $\Rightarrow$  more charge carriers injected into the  $\pi$  band. Conductivity increases with doping up to a maximum; excessive doping can disrupt chain order and reduce  $\mu$ .
- (iii) Crystallinity and chain order:** More crystalline, ordered packing  $\Rightarrow$  better  $\pi$ – $\pi$  interchain stacking  $\Rightarrow$  easier interchain charge hopping  $\Rightarrow$  higher bulk conductivity. Amorphous regions act as barriers to transport.
- (iv) Temperature:** Unlike metals, conducting polymers show **semiconductor-like** temperature dependence — conductivity *increases* with temperature (more thermal energy available to overcome hopping barriers). At very high doping levels, metallic behaviour ( $\sigma$  decreases with  $T$ ) may emerge.
- (v) Nature of dopant and counterion:** Larger counterions cause more chain distortion and reduce  $\mu$ . Some dopants (e.g. DBSA — dodecylbenzenesulfonic acid) also act as surfactants, improving chain solubility and ordering, thereby increasing  $\sigma$ .

(vi) **Chain defects and crosslinks:** Kinks, twists,  $sp^3$  defects, and crosslinks all interrupt the conjugated pathway and trap charge carriers, reducing both  $n$  and  $\mu$ .

(vii) **Molecular weight:** Higher MW  $\Rightarrow$  longer chains  $\Rightarrow$  longer conjugation segments  $\Rightarrow$  fewer chain ends (which are defects)  $\Rightarrow$  higher conductivity.

**Conclusion:** Conductivity in polymers is controlled by conjugation length, doping level, crystallinity, temperature, dopant choice, and structural defects. Optimizing these factors — particularly by maximizing conjugation and crystalline order while achieving high doping levels — is the key to obtaining high-conductivity polymer materials.

**Q16.** Why are natural polymers more biodegradable than synthetic polymers? What are the advantages, disadvantages and applications of thermoplastic starch? **(11 marks)** [CHEM 113 | 2016–17]

### Answer

#### Natural vs Synthetic Polymer Biodegradability; Thermoplastic Starch: Advantages, Disadvantages & Applications **(11 marks)**

##### Part 1: Why Natural Polymers Are More Biodegradable

**(4 marks)**

**Natural polymers** (starch, cellulose, chitin, proteins) are more biodegradable than synthetic polymers for the following reasons:

- **Enzyme recognition:** Natural polymers contain chemical linkages (glycosidic bonds in starch/cellulose, peptide bonds in proteins) that microbial enzymes evolved to hydrolyze. Amylases, cellulases, proteases break these bonds efficiently.
- **Hydrolysable backbone:** Most natural polymers have **heteroatom linkages** (O, N) in the backbone that are susceptible to hydrolysis by water and enzymes. Synthetic polymers like PE have purely C–C backbones that are **hydrolysis-resistant**.
- **Familiar chemical structures:** Natural polymers are composed of monomers (glucose, amino acids) that are part of normal metabolic cycles. Microorganisms can break them down and use them as carbon/energy sources.
- **Hydrophilicity:** Natural polymers contain many –OH, –NH<sub>2</sub>, –COOH groups that make them **water-absorbent**. Water absorption facilitates hydrolysis and microbial access.
- **Synthetic polymers** have C–C, C–F, or C–Cl backbones with no hydrolysable bonds, no recognition by natural enzymes, and are often hydrophobic — making them persist in the environment for hundreds of years.

##### Part 2: Thermoplastic Starch (TPS) — What it is

**(1 marks)**

**Thermoplastic starch** is starch (a natural polymer of glucose) that has been **plasticized** by heating with plasticizers (water, glycerol, sorbitol) under pressure. This disrupts the crystalline structure of native starch and converts it into a melt-processable thermoplastic material that can be injection-moulded, extruded, and blown like conventional plastics.

##### Part 3: Advantages of TPS

**(2 marks)**

- **Fully biodegradable and compostable:** Broken down by amylases into glucose within weeks.
- **Renewable and abundant:** Derived from corn, potato, cassava — cheap and widely available.
- **Carbon-neutral:** CO<sub>2</sub> released on degradation was absorbed during plant growth — net zero carbon footprint.
- **Processable on standard equipment:** Can be moulded, extruded, and blown on conventional thermoplastic processing machinery.
- **Edible and non-toxic:** Safe for food contact applications and agricultural use.
- **Low cost** compared to other biopolymers (PLA, PHB).

##### Part 4: Disadvantages of TPS

**(2 marks)**

- **High moisture sensitivity:** TPS absorbs water readily (hydrophilic), causing swelling, plasticizer migration, and loss of mechanical properties in humid environments.
- **Retrogradation:** Over time, amylose chains re-crystallize, making the material brittle and reducing elongation.



- **Poor mechanical properties alone:** TPS has lower strength and barrier properties than synthetic plastics; often blended with PLA or PHB to improve performance.
- **Limited thermal stability:** Degrades at temperatures  $>220^{\circ}\text{C}$ , limiting high-temperature applications.
- **Competition with food supply:** Use of food crops (corn, potato) raises ethical and economic concerns.

#### Part 5: Applications of TPS

(2 marks)

- **Packaging:** Loose-fill packaging (packing peanuts), food trays, disposable cutlery, bags.
- **Agriculture:** Biodegradable mulch films, seedling pots, controlled-release fertilizer coatings.
- **Biomedical:** Drug delivery matrices, tissue engineering scaffolds, sutures (TPS-PLA blends).
- **Food industry:** Edible films and coatings for fruits and vegetables to extend shelf life.
- **Blends:** TPS/PLA and TPS/PHB composites used in compostable bags and food service items.

**Conclusion:** Natural polymers biodegrade faster than synthetic ones due to enzyme-recognizable linkages and hydrophilic character. TPS is a low-cost, fully biodegradable thermoplastic made by plasticizing starch, with excellent environmental credentials but limited by moisture sensitivity and poor mechanical performance. Applications range from packaging and agriculture to biomedicine.

Q17. Write down the factors that accelerate the polymer degradation. (9 marks)

[CHEM 113 | 2015–16]

#### Answer

##### Factors That Accelerate Polymer Degradation

(9 marks)

Polymer degradation is the breakdown of polymer chains by physical, chemical, or biological agents. The following factors **accelerate** this process:

(i) **Temperature:** Higher temperature increases the rate of all degradation processes — hydrolysis, oxidation, and thermal scission. Elevated temperatures provide energy to break polymer backbone bonds and accelerate microbial metabolism.

$$\text{Rate} \propto e^{-E_a/RT} \quad (\text{Arrhenius}) \quad \Rightarrow \quad T \uparrow \Rightarrow \text{rate} \uparrow$$

(ii) **UV Radiation (Photodegradation):** UV photons ( $\lambda < 400 \text{ nm}$ ) are absorbed by chromophoric groups ( $\text{C}=\text{O}$ , aromatic rings) in polymers, initiating radical chain reactions that cleave the backbone (Norrish type I and II reactions). Outdoor polymers (PE, PP) degrade through this mechanism over years.

(iii) **Oxygen (Oxidative Degradation):** Oxygen reacts with polymer radicals generated thermally or photolytically to form peroxides and hydroperoxides. These decompose to produce chain scission and crosslinking. Elevated temperature accelerates oxidation further (**thermo-oxidative degradation**).

(iv) **Moisture and Water:** Water hydrolyzes susceptible bonds (ester, amide, glycosidic). Higher humidity or liquid water contact accelerates hydrolytic degradation of polyesters (PLA, PET), polyamides (nylon), and polysaccharides (starch).

(v) **Mechanical Stress:** Applied stress causes **mechanochemical degradation** by breaking covalent bonds directly. Cyclic loading (fatigue) generates crazes and cracks, exposing fresh surfaces to chemical attack and accelerating overall degradation.

(vi) **Biological Agents (Microorganisms):** Bacteria, fungi, and their extracellular enzymes (lipases, esterases, amylases) directly cleave polymer chains. Warm, humid, nutrient-rich environments support greater microbial populations and faster biodegradation.

(vii) **Catalysts and Metal Ions:** Trace transition metal ions ( $\text{Fe}^{2+}$ ,  $\text{Cu}^{2+}$ ,  $\text{Co}^{2+}$ ) catalyze oxidative degradation by decomposing hydroperoxides (Fenton-type reactions), generating radicals that attack the polymer backbone.

(viii) **Chemical Environment (pH):** Acidic or alkaline environments accelerate hydrolysis of susceptible polymers. Strong acids/bases protonate/deprotonate functional groups, weakening bonds and catalyzing chain scission.

(ix) **Plasticizers and Additives:** Some plasticizers migrate out over time, making the polymer brittle and more susceptible to cracking and environmental attack. Pro-oxidant additives (e.g. transition metal stearates) are deliberately added to **oxo-degradable** polymers to accelerate oxidative breakdown.

**Conclusion:** Polymer degradation is accelerated by heat (Arrhenius kinetics), UV radiation (photoinitiation), oxygen (radical oxidation), water (hydrolysis), mechanical stress, microorganisms (enzymatic cleavage), metal ion catalysts, extreme pH, and pro-degradant additives. Understanding these factors allows engineers to either protect polymers from premature degradation (stabilizers, antioxidants) or design them for controlled end-of-life breakdown.



**Q18.** Show schematically how a soliton pair on a trans-polyacetylene is formed by doping. (6 marks)

[CHEM 113 | 2015–16]

### Answer

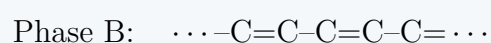
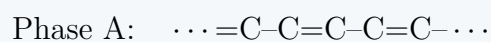
#### Soliton Pair Formation in Trans-Polyacetylene by Doping

(6 marks)

#### Part 1: Trans-Polyacetylene and Degenerate Ground States

(2 marks)

Trans-polyacetylene has **two degenerate ground states** (A and B phase) with identical energy but different bond alternation patterns:



In phase A, double bonds are to the left of each CH; in phase B, to the right. Both have exactly the same energy.

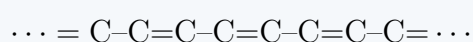
A **soliton** is a **domain wall** — the boundary between a region of phase A and a region of phase B on the same chain. At the soliton, the pattern of bond alternation reverses.

#### Part 2: Schematic of Soliton Pair Formation by Doping

(4 marks)

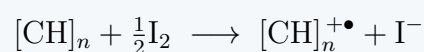
When a p-type dopant (e.g.  $\text{I}_2$ ) removes an electron from the polyacetylene chain, a **soliton–antisoliton pair** is created:

##### Step 1 — Undoped trans-PA (uniform phase A):



All bonds follow phase A alternation. No defect.

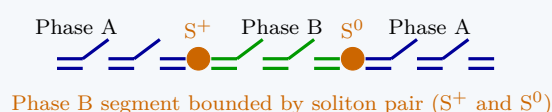
##### Step 2 — Electron removal by dopant:



A radical cation is generated at one site on the chain. This disrupts the bond alternation locally.

##### Step 3 — Soliton pair nucleation:

The distortion spreads in both directions from the charged site, creating **two domain walls** (solitons) that separate phase A and phase B regions:



##### Step 4 — Properties of the soliton pair:

- **$\text{S}^+$  (charged soliton):** carries charge  $+e$ , no spin — this is the mobile charge carrier responsible for electrical conduction.
- **$\text{S}^0$  (neutral soliton):** carries spin  $\frac{1}{2}$ , no charge — detected by EPR spectroscopy.
- Under an electric field,  $\text{S}^+$  migrates along the chain by successive bond reorganization, carrying current.
- The  $\text{I}^-$  counterion sits between chains, stabilizing  $\text{S}^+$ .

**Conclusion:** Doping trans-PA with  $\text{I}_2$  removes an electron, creating a charged soliton–neutral soliton pair. Each soliton is a domain wall between the two degenerate bond- alternation phases (A and B). The charged soliton  $\text{S}^+$  is mobile and serves as the primary charge carrier, explaining the dramatically enhanced conductivity of doped polyacetylene.

**Q19.** How can a conjugated conducting polymer be doped by chemical methods? (6 marks)

[CHEM 113 | 2015–16]

### Answer

#### Chemical Doping of Conjugated Conducting Polymers

(6 marks)

Chemical doping of conjugated polymers involves a direct **electron transfer reaction** between the polymer and a chemical dopant, without any external electrical energy. Two types exist:

##### (i) Oxidative (p-type) Chemical Doping:

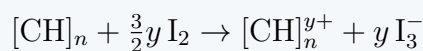
The dopant is an **electron acceptor** that oxidizes the polymer by withdrawing electrons from the  $\pi$  system:



where D is the dopant anion (counterion).

#### Examples:

- **Halogen doping:**



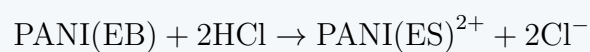
$\text{I}_2$ ,  $\text{Br}_2$ ,  $\text{ICl}$ ,  $\text{IBr}$  used. Simple, effective. Conductivity:  $10^2$ – $10^3$  S/cm.

- **Lewis acid doping:**



$\text{AsF}_5$ ,  $\text{SbF}_5$ ,  $\text{FeCl}_3$  used. Highest conductivity: up to  $10^5$  S/cm.

- **Protonic acid doping (polyaniline only):**



$\text{H}_2\text{SO}_4$ ,  $\text{HCl}$ ,  $\text{CSA}$ ,  $\text{DBSA}$  used. Protonation of imine N converts insulating emeraldine base (EB) to conducting emeraldine salt (ES). Conductivity: 1–100 S/cm.

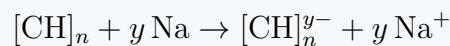
#### (ii) Reductive (n-type) Chemical Doping:

The dopant is an **electron donor** that reduces the polymer by donating electrons into the  $\pi^*$  system:



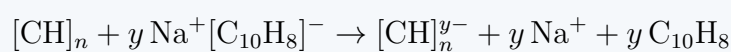
#### Examples:

- **Alkali metal doping:**



$\text{Na}$ ,  $\text{K}$ ,  $\text{Li}$  used. Conductivity:  $10^2$ – $10^3$  S/cm. Less stable than p-doped systems (sensitive to air).

- **Alkali metal complex doping:**



Sodium naphthalide in THF. Allows more controlled, homogeneous n-doping.

#### Key comparison:

| Type     | Dopant                                                 | Process              | Stability            |
|----------|--------------------------------------------------------|----------------------|----------------------|
| p-type   | $\text{I}_2$ , $\text{AsF}_5$ , $\text{FeCl}_3$        | Oxidation ( $-e^-$ ) | Good                 |
| n-type   | $\text{Na}$ , $\text{K}$ , $\text{Na-naphthalide}$     | Reduction ( $+e^-$ ) | Poor (air-sensitive) |
| Protonic | $\text{HCl}$ , $\text{H}_2\text{SO}_4$ , $\text{DBSA}$ | Protonation          | Good                 |

**Conclusion:** Chemical doping introduces charge carriers into conjugated polymers by direct redox reaction without external electricity. p-Doping ( $\text{I}_2$ ,  $\text{AsF}_5$ ,  $\text{FeCl}_3$ ) oxidizes the  $\pi$  system to give stable high-conductivity materials; n-doping ( $\text{Na}$ ,  $\text{K}$ ) reduces the  $\pi^*$  system but gives air-sensitive products; protonic acid doping is unique to polyaniline and achieves conductivity without any change in oxidation state.

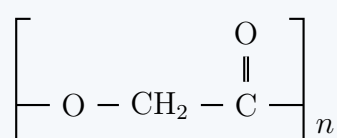
**Q20.** Write down the structures of the following polymers (any seven): (i) Poly(glycolic acid) (ii) Poly( $\beta$ -hydroxybutyrate) (iii) Poly-lysine (iv) Polyaniline emeraldine base (v) Polythiophene (vi) Polyazulene (vii) Poly( $\epsilon$ -caprolactone) (viii) Chitin (ix) Poly(dichlorophosphazene) (x) Starch. **(14 marks)** [CHEM 113 | 2015–16]

#### Answer

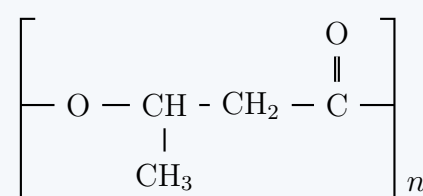
##### Structures of Polymers:

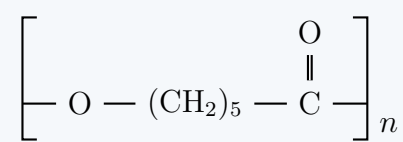
The repeating units of the 10 given polymers are correctly drawn below:

(i) Poly(glycolic acid) [PGA]

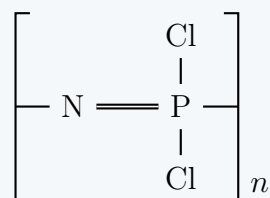


(ii) Poly( $\beta$ -hydroxybutyrate) [PHB]

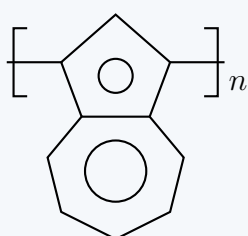


(vii) Poly( $\epsilon$ -caprolactone)


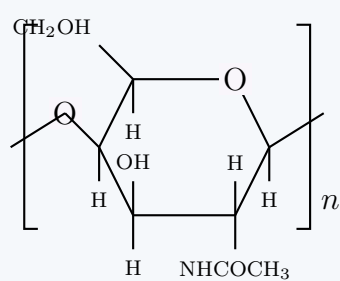
(ix) Poly(dichlorophosphazene)



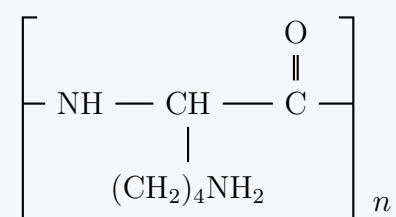
(vi) Polyazulene



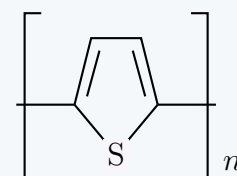
(viii) Chitin



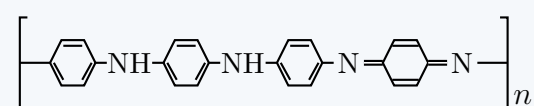
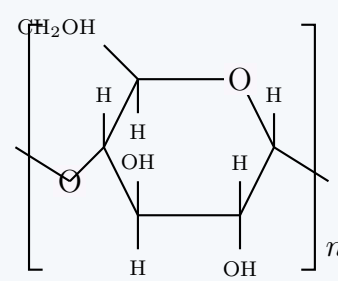
(iii) Poly-lysine



(v) Polythiophene



(iv) Polyaniline emeraldine base


 (x) Starch ( $\alpha$ -amylose)


BUET · CHEM 113

# Chemistry

Section S10

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Electronic Materials

S10 — Electronic Materials

**Q1.** Light-emitting diode (LED) is a modern innovation in digital display technology. Explain its mechanism that is different from LCD in producing a color band with a suitable diagram. Briefly state the characteristic features of LCD. **(10 marks)** [CHEM 113 | 2023–24]

Answer

LED vs LCD: Mechanism, Color Production & Characteristic Features

(10 marks)

Part 1: Mechanism of LED — How it Produces Light

(5 marks)

**(i) Basic Structure of an LED:**  
An LED is a **p-n junction semiconductor device**. It consists of:

- **p-type layer:** doped with electron acceptors (e.g. B in Si) — rich in **holes**.
- **n-type layer:** doped with electron donors (e.g. P in Si) — rich in **electrons**.
- A **depletion zone** at the p-n junction where initial recombination creates a built-in electric field.

**(ii) Electroluminescence Mechanism:**  
When a **forward bias voltage** is applied across the p-n junction:

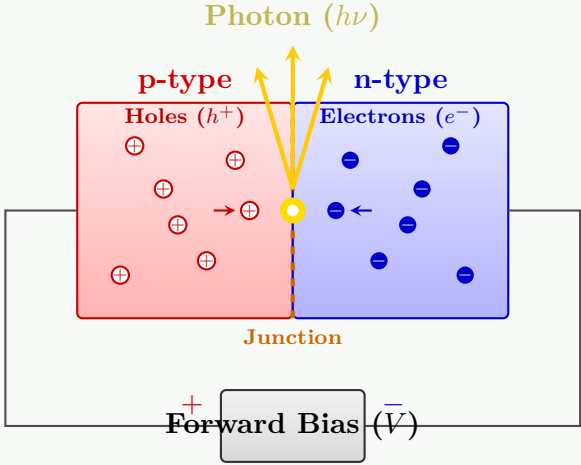
1. Electrons from the n-type layer are injected across the junction into the p-type layer.
2. Holes from the p-type layer move toward the junction.
3. At the junction, an electron falls from the conduction band (CB) into a hole in the valence band (VB) — this is called **electron–hole recombination**.
4. The energy released equals the **band gap**  $E_g$  and is emitted as a **photon**:

$$h\nu = E_g = E_{\text{CB}} - E_{\text{VB}} \quad \Rightarrow \quad \lambda = \frac{hc}{E_g}$$

5. The wavelength (color) of emitted light is determined solely by the band gap of the semiconductor material.

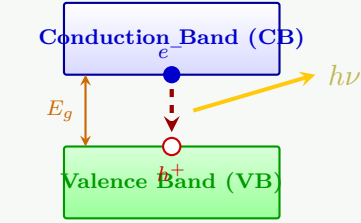
**(iii) LED p-n Junction Diagram:**

Photon ( $h\nu$ )



p-type Holes ( $h^+$ ) n-type Electrons ( $e^-$ )  
Junction  
Forward Bias ( $V$ )

Energy Band Diagram



Conduction Band (CB)  
Valence Band (VB)  
 $E_g$   
 $h\nu$

**(iv) Color Control — Producing a Color Band:**  
The color of light emitted depends entirely on the **band gap energy** of the semiconductor:

$$\lambda = \frac{hc}{E_g}$$

| Semiconductor Material | Band Gap (eV) | Color Emitted     |
|------------------------|---------------|-------------------|
| GaAs                   | 1.4           | Infrared          |
| GaAsP                  | 1.9           | Red               |
| GaP:N                  | 2.2           | Yellow–Green      |
| GaN                    | 3.4           | Blue/UV           |
| InGaN                  | 2.7–3.4       | Blue–Green        |
| AlGaInP                | 1.9–2.3       | Red–Orange–Yellow |

**White LED:** A blue GaN LED is coated with a **yellow phosphor** (YAG:Ce<sup>3+</sup>). Blue light partially excites the phosphor, which emits yellow; the combination of blue + yellow appears **white** to the eye. This is the principle behind most modern white LED displays and lighting.

**Part 2: How LED Differs from LCD****(2 marks)**

| Property         | LED Display                                                                      | LCD Display                                                                      |
|------------------|----------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| Light source     | <b>Self-emitting</b> — each pixel generates its own light by electroluminescence | <b>Passive</b> — requires an external backlight (fluorescent or LED backlight)   |
| Color production | Direct: band gap of semiconductor determines $\lambda$                           | Indirect: white backlight filtered through color filters (RGB subpixels)         |
| Pixel control    | Each pixel ON/OFF independently                                                  | Liquid crystal molecules rotate to block/pass backlight; color filter adds color |
| Black level      | True black (pixel OFF = no emission)                                             | Imperfect black (backlight leaks through)                                        |
| Contrast ratio   | Extremely high ( $> 10^6:1$ )                                                    | Lower ( $\sim 10^3:1$ )                                                          |
| Thickness        | Very thin (no backlight needed)                                                  | Thicker (backlight unit required)                                                |
| Power use        | More efficient (only lit pixels use power)                                       | Backlight always on (wastes power)                                               |

**Part 3: Characteristic Features of LCD****(3 marks)**

- **Operating principle:** Liquid crystal molecules (rod-shaped, anisotropic) rotate their orientation under an applied electric field. This rotation aligns or misaligns the polarization of backlight passing through, controlling the brightness of each pixel. Color is added by RGB color filters in front of each pixel.
- **Polarizers:** Two perpendicular polarizing filters sandwich the liquid crystal layer. Without voltage, LC molecules rotate light  $90^\circ \Rightarrow$  light passes (pixel bright). With voltage, LC molecules align with field  $\Rightarrow$  no rotation  $\Rightarrow$  light blocked by second polarizer (pixel dark).
- **Requires backlight:** LCDs do not emit light themselves; they modulate a white backlight (cold cathode fluorescent lamp CCFL or LED array).
- **Low power consumption** compared to CRT but higher than OLED, due to always-on backlight.
- **Viewing angle limitation:** Standard TN-LCD panels have reduced contrast at wide viewing angles; IPS-LCD panels improve this.
- **Cannot achieve true black:** Some backlight always leaks through even in the OFF state, limiting contrast ratio.
- **Response time:** LC molecules take  $\sim 1\text{--}8$  ms to realign, causing motion blur at high refresh rates compared to LED.
- **Widespread use:** Computer monitors, laptops, smartphones (prior to OLED), televisions, digital watches, calculators.

**Conclusion:** LEDs produce light by **direct electroluminescence** at a p-n junction — electrons and holes recombine, emitting photons whose wavelength is set by the semiconductor band gap ( $\lambda = hc/E_g$ ). Color bands are produced by selecting different semiconductor materials. LCDs, by contrast, passively **modulate a white backlight** using electric-field-controlled liquid crystal orientation and RGB color filters. LEDs offer superior contrast, true black, thinner form factors, and higher energy efficiency than LCDs.



BUET · CHEM 113

# Chemistry

Section S11

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Biomolecules

## S11 — Biomolecules

**Q1.** Protein is a bio-polymer — explain by showing peptide bond formation. Explain the intermolecular forces responsible for stabilizing the secondary and tertiary structures of proteins. (10 marks) [CHEM 113 | 2023–24; 2021–22]

### Answer

#### Protein as a Biopolymer: Peptide Bond Formation & Stabilization of Secondary and Tertiary Structures (10 marks)

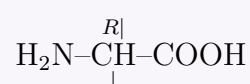
##### Part 1: Protein as a Biopolymer — Peptide Bond Formation

(4 marks)

##### (i) Amino Acid Monomers:

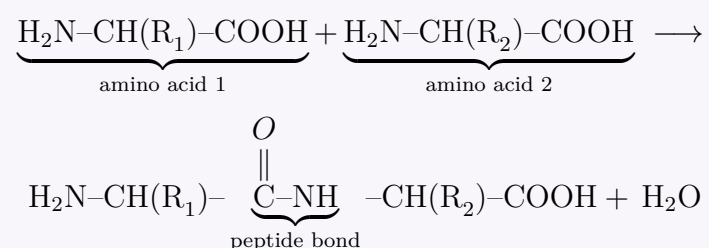
Proteins are biopolymers made of **amino acid** monomers. Each amino acid has a central  $\alpha$ -carbon bearing:

- An **amino group** ( $-\text{NH}_2$ )
- A **carboxyl group** ( $-\text{COOH}$ )
- A **hydrogen atom** ( $-\text{H}$ )
- A **side chain** ( $-\text{R}$ , unique to each of the 20 amino acids)



##### (ii) Peptide Bond Formation:

A **peptide bond** forms by a **condensation reaction** between the  $-\text{COOH}$  of one amino acid and the  $-\text{NH}_2$  of another, with loss of water:



The peptide bond has **partial double bond character** due to resonance between  $\text{C}=\text{O}$  and  $\text{C}-\text{N}$ :



This gives the peptide bond:

- **Planar geometry** (6 atoms in one plane)
- **Restricted rotation** about the  $\text{C}-\text{N}$  bond
- Bond length intermediate between  $\text{C}=\text{N}$  and  $\text{C}-\text{N}$

Repeating this condensation  $n$  times produces a **polypeptide chain** — the **primary structure** of the protein.

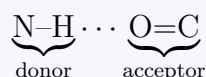
##### Part 2: Intermolecular Forces Stabilizing Secondary Structure

(3 marks)

The **secondary structure** describes the local folding of the polypeptide backbone (ignoring R-groups). Two common secondary structures are the  $\alpha$ -helix and the  $\beta$ -pleated sheet.

##### Stabilizing force: Hydrogen bonding (backbone–backbone)

Both secondary structures are stabilized *exclusively* by **hydrogen bonds** between:



along the polypeptide backbone:

- **$\alpha$ -Helix:** The backbone coils into a right-handed helix with 3.6 residues per turn. H-bonds form between the  $\text{C}=\text{O}$  of residue  $i$  and the  $\text{N}-\text{H}$  of residue  $i + 4$  — parallel to the helix axis. All  $\text{C}=\text{O}$  and  $\text{N}-\text{H}$  groups participate.
- **$\beta$ -Pleated sheet:** Adjacent polypeptide strands lie side by side (parallel or antiparallel). H-bonds form *between* strands ( $\text{C}=\text{O}$  of one strand to  $\text{N}-\text{H}$  of the adjacent strand), perpendicular to the chain direction.

No other force contributes to secondary structure — **hydrogen bonding alone** drives and stabilizes it.

**Part 3: Intermolecular Forces Stabilizing Tertiary Structure****(3 marks)**

The **tertiary structure** is the overall 3-D folding of the entire polypeptide chain, determined by interactions among the **R-groups (side chains)**. Four types of forces act:

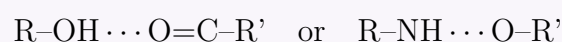
**(i) Hydrophobic interactions (most important):**

Non-polar R-groups (Leu, Val, Ile, Phe) are repelled by water and cluster together in the **hydrophobic core** of the protein, away from the aqueous environment. This is the primary driving force for protein folding.

Non-polar R-groups → interior (away from water)

**(ii) Hydrogen bonding (side chain–side chain):**

Polar R-groups (Ser, Thr, Asn, Gln, Tyr) form H-bonds with each other or with the backbone:



These are found predominantly on the protein **surface**.

**(iii) Electrostatic (ionic) interactions — salt bridges:**

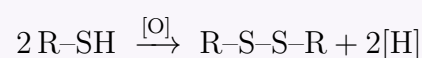
Oppositely charged R-groups ( $\text{Asp}^-$ ,  $\text{Glu}^-$  vs  $\text{Lys}^+$ ,  $\text{Arg}^+$ ,  $\text{His}^+$ ) attract each other:



Called **salt bridges**; particularly stable in the hydrophobic interior where they are shielded from water.

**(iv) Disulfide bonds (covalent):**

Two cysteine (Cys) residues can be oxidized to form a covalent **disulfide bridge** ( $-\text{S}-\text{S}-$ ):



The only **covalent** contribution to tertiary structure. Provides great stability; must be cleaved (reducing agents) to denature such proteins.

**(v) van der Waals forces:**

Weak, non-specific induced-dipole interactions between closely packed atoms in the interior. Individually small but collectively significant due to the large number of close contacts in the folded protein.

| Force          | Strength            | R-group types involved           |
|----------------|---------------------|----------------------------------|
| Hydrophobic    | Strong (collective) | Non-polar: Leu, Val, Phe, Ile    |
| H-bonding      | Moderate            | Polar: Ser, Thr, Asn, Gln, Tyr   |
| Salt bridges   | Moderate–strong     | Charged: Asp, Glu, Lys, Arg, His |
| Disulfide bond | Strong (covalent)   | Cys–Cys only                     |
| van der Waals  | Weak (cumulative)   | All R-groups                     |

**Conclusion:** Proteins are biopolymers of amino acids linked by peptide bonds (condensation of  $-\text{COOH} + -\text{NH}_2$ , releasing  $\text{H}_2\text{O}$ ). Secondary structure ( $\alpha$ -helix,  $\beta$ -sheet) is stabilized *exclusively* by backbone **hydrogen bonds**. Tertiary structure is stabilized by **hydrophobic interactions** (primary driver), **H-bonding**, **salt bridges**, **disulfide bonds**, and **van der Waals forces** among side chains.

**Q2.** What can be inferred about the structure of DNA in terms of its double helix configuration and the complementary base pairing (A-T, G-C)? **(10 marks)** [CHEM 113 | 2021–22]

**Answer****DNA Double Helix Configuration and Complementary Base Pairing****(10 marks)****Part 1: Structure of DNA — The Double Helix****(5 marks)****(i) Primary Structure (Chemical Composition):**

DNA (deoxyribonucleic acid) is a polymer of **deoxyribonucleotides**. Each nucleotide consists of:

- A **deoxyribose sugar** (5-carbon ring).
- A **phosphate group** (connecting sugars via 3'–5' phosphodiester bonds).
- One of four **nitrogenous bases**:
  - Purines: **Adenine (A)**, **Guanine (G)**
  - Pyrimidines: **Thymine (T)**, **Cytosine (C)**

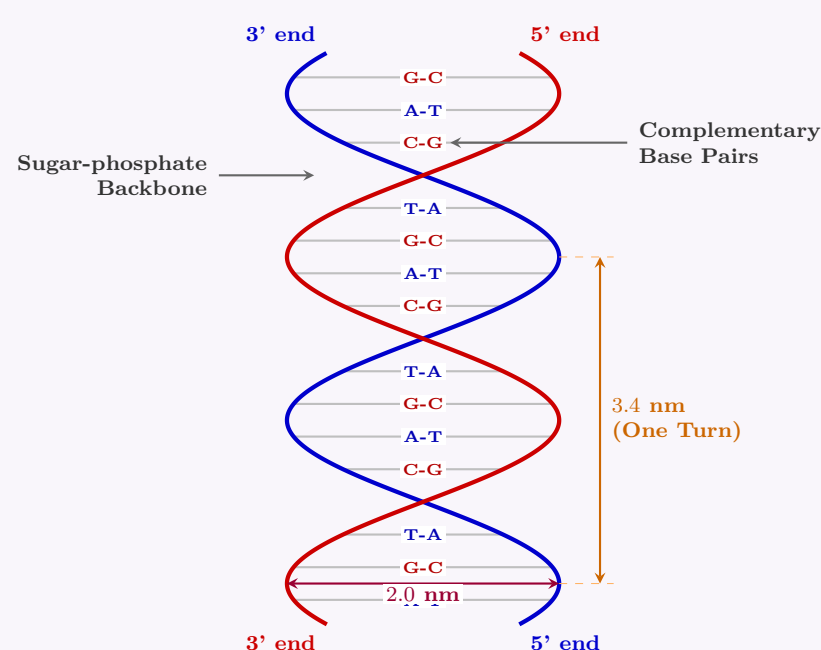
The phosphodiester backbone runs 5'→3', with the nitrogenous bases projecting inward.

**(ii) Watson–Crick Double Helix (1953):**

Based on X-ray crystallography (Rosalind Franklin) and Chargaff's rules, Watson and Crick proposed the B-DNA double helix:

- **Two Antiparallel Strands:** One strand runs 5'→3', while the complementary strand runs 3'→5'.
- **Right-Handed Helix:** B-DNA winds in a right-handed (clockwise) direction around a central axis.
- **Dimensions:** Diameter = 2.0 nm; Pitch (rise per complete turn) = 3.4 nm; exactly 10 base pairs per turn (Rise per bp = 0.34 nm).
- **Hydrophilic Backbone:** The sugar-phosphate backbone is highly negatively charged and faces outward towards the aqueous environment.
- **Hydrophobic Core:** The bases are stacked in the interior, perpendicular to the helix axis.  $\pi$ - $\pi$  stacking forces strongly stabilize the helix.
- **Major & Minor Grooves:** The asymmetrical attachment of base pairs to the sugar backbone creates unequal grooves, acting as binding sites for specific transcription factors.

**(iii) Schematic Diagram:**



## Part 2: Complementary Base Pairing (A–T, G–C)

(5 marks)

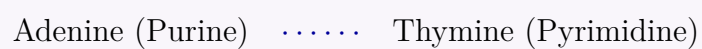
**(i) Chargaff's Rules:**

Erwin Chargaff (1950) established that in all double-stranded DNA:

$$[A] = [T] \quad \text{and} \quad [G] = [C]$$

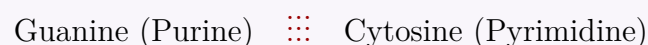
This invariant ratio strongly indicated specific pairwise binding between purines and pyrimidines.

**(ii) A–T Base Pair (2 Hydrogen Bonds):**



- Contains **2 hydrogen bonds** (one  $\text{N-H} \cdots \text{O}$ , one  $\text{N} \cdots \text{H-N}$ ).
- Forms a relatively weaker pair. A-T rich regions are easier to separate (melt) during replication.

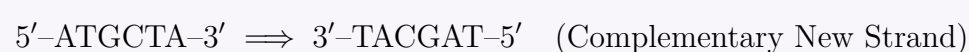
**(iii) G–C Base Pair (3 Hydrogen Bonds):**



- Contains **3 hydrogen bonds** (providing tighter electrostatic lock).
- Forms a stronger pair. High G-C content significantly increases the DNA's melting temperature ( $T_m$ ).

**(iv) Significance of Complementary Base Pairing:**

- **Uniform Width:** A purine (2 rings) always pairs with a pyrimidine (1 ring), maintaining a constant 2.0 nm helix diameter.
- **Information Redundancy:** Since A strictly binds T and G binds C, both strands carry identical genetic information.
- **Semiconservative Replication:** Each parental strand acts as a precise template to synthesize a new complementary strand:



- **Transcription Accuracy:** RNA polymerase relies on this complementarity to synthesize flawless mRNA templates for protein production.

#### Summary Comparison:

| Base Pair | Binding Type        | H-Bonds | Structural Significance                                      |
|-----------|---------------------|---------|--------------------------------------------------------------|
| A–T       | Purine – Pyrimidine | 2       | Weaker bond; origin of replication is usually A-T rich.      |
| G–C       | Purine – Pyrimidine | 3       | Stronger bond; increases thermal stability ( $T_m$ ) of DNA. |

**Conclusion:** DNA forms a right-handed double helix characterized by an external hydrophilic sugar-phosphate backbone and an internal hydrophobic core of stacked bases. The strict complementary base pairing (A-T via 2 H-bonds, G-C via 3 H-bonds) not only ensures uniform helix geometry but also provides the molecular foundation for exact genetic replication, transcription, and structural stability.

**Q3.** Explain the term “Bile Salts”. Describe the types of interactions that are involved in maintaining the tertiary structures of proteins. (15 marks) [CHEM 113 | 2020–21 (Jan 21)]

#### Answer

#### Bile Salts & Interactions Maintaining Tertiary Structure of Proteins

(15 marks)

#### Part 1: Bile Salts

(5 marks)

##### (i) Definition:

Bile salts are the **sodium and potassium salts of bile acids**, which are steroid-derived amphiphilic molecules produced in the liver from cholesterol and secreted into the small intestine via bile.

##### (ii) Structure:

A bile salt consists of:

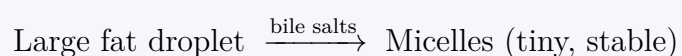
- A **hydrophobic steroid nucleus** (four fused rings: A, B, C, D) — the non-polar face.
- **Hydroxyl groups** (–OH) on one face — the polar face.
- A **carboxylate group** (–COO<sup>–</sup>) or conjugated taurine/glycine tail — highly polar.

Example: **sodium cholate** (cholic acid + Na<sup>+</sup>; three –OH groups and a –COO<sup>–</sup> Na<sup>+</sup> tail).

This dual hydrophobic–hydrophilic character makes bile salts **biological detergents (surfactants)**.

##### (iii) Function:

- **Emulsification of dietary fats:** In the small intestine, bile salts surround fat droplets with their hydrophobic faces pointing inward and hydrophilic faces outward, breaking large fat globules into tiny **micelles** (~5 nm).



- **Increase surface area** for pancreatic lipase to hydrolyze triglycerides.
- **Facilitate absorption** of fat-soluble vitamins (A, D, E, K) and fatty acids through intestinal walls.
- **Cholesterol solubilization:** Keep cholesterol soluble in bile, preventing gallstone formation.
- **Enterohepatic circulation:** ~95% of bile salts are reabsorbed in the ileum and recycled back to the liver, completing the enterohepatic cycle.

##### (iv) Bile salts as protein denaturants:

At high concentrations, bile salts can **denature proteins** by inserting their hydrophobic regions into the protein’s hydrophobic core, disrupting hydrophobic interactions and unfolding the tertiary structure. This property is exploited in laboratory protein extraction and cell lysis protocols (e.g. sodium deoxycholate used in cell lysis buffers).

#### Part 2: Interactions Maintaining Tertiary Structure of Proteins

(10 marks)

The **tertiary structure** is the complete 3-D arrangement of all atoms in a single polypeptide chain, determined by interactions between **side chains (R-groups)**. Five types of interaction are responsible:

##### (i) Hydrophobic Interactions (dominant force):

The most important driving force for protein folding. Non-polar R-groups (Leu, Val, Ile, Ala, Phe, Trp, Met) are thermodynamically unfavorable in water. Protein folding **buries** these groups in a **hydrophobic core**, away from the aqueous environment:

- Driven by **entropy gain** of water (water molecules are released from ordered hydration shells around non-polar groups when they cluster together).

- Not a true attractive force — it is the **exclusion** of non-polar residues from water.
- Accounts for ~60% of the stabilization energy of a typical globular protein.

**(ii) Hydrogen Bonding (side chain–side chain and side chain–backbone):**

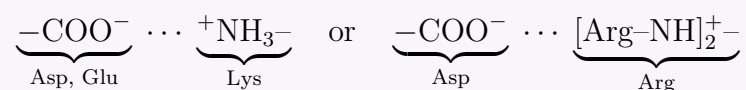
Polar R-groups containing –OH (Ser, Thr, Tyr), –NH<sub>2</sub> (Asn, Gln), –NH (His, Trp), or –COOH (Asp, Glu) can donate or accept H-bonds:



- Predominantly found on the protein **surface** where polar residues interact with water.
- Also contribute to interior stability where buried polar residues H-bond with each other.
- Each H-bond contributes ~12–38 kJ/mol.

**(iii) Electrostatic (Ionic) Interactions — Salt Bridges:**

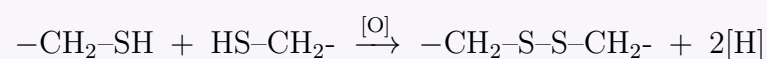
Oppositely charged R-groups attract each other:



- **Salt bridges** in the hydrophobic interior are exceptionally stable because the low dielectric constant of the interior amplifies the electrostatic interaction.
- On the surface, salt bridges are weaker due to competition with water.
- pH-dependent: disrupted at extreme pH when charged groups are protonated/deprotonated.

**(iv) Disulfide Bonds (covalent crosslinks):**

Two cysteine residues are oxidized to form a covalent **disulfide bridge**:



- The **only covalent** interaction in tertiary structure (all others are non-covalent).
- Bond energy ~250 kJ/mol — very strong.
- Abundant in extracellular proteins (insulin, keratin, antibodies) that need extra stability in harsh environments.
- Cleaved by reducing agents (DTT, β-mercaptoethanol).

**(v) van der Waals Forces:**

Weak, transient dipole–induced dipole attractions between closely packed atoms in the protein interior:

- Each interaction: ~0.4–4 kJ/mol (very weak).
- But proteins have thousands of close atom contacts ⇒ collectively significant.
- Contribute to precise packing of the hydrophobic core and shape complementarity at binding sites.

**Summary of interactions:**

| Interaction    | Type            | Strength            | R-groups involved            |
|----------------|-----------------|---------------------|------------------------------|
| Hydrophobic    | Non-covalent    | Strong (collective) | Leu, Val, Ile, Phe, Ala      |
| H-bonding      | Non-covalent    | Moderate            | Ser, Thr, Asn, Gln, Tyr, His |
| Salt bridges   | Non-covalent    | Moderate–strong     | Asp, Glu, Lys, Arg, His      |
| Disulfide bond | <b>Covalent</b> | Very strong         | Cys–Cys only                 |
| van der Waals  | Non-covalent    | Weak (cumulative)   | All R-groups                 |

**Conclusion:** Bile salts are amphiphilic steroid-based biological detergents that emulsify dietary fats for digestion and absorption, and can denature proteins at high concentrations. Tertiary protein structure is maintained by five interactions: **hydrophobic interactions** (dominant), **H-bonds**, **salt bridges**, **disulfide bonds** (only covalent), and **van der Waals forces** — together defining the unique 3-D fold and biological function of each protein.

**Q4.** How does hydrogen bonding and hydrophobic interaction influence the structure of biomolecules? **(10 marks)**

[CHEM 113 | 2020–21]



## Answer

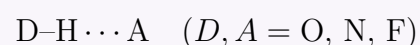
## Hydrogen Bonding and Hydrophobic Interactions in Biomolecule Structure

(10 marks)

## Part 1: Hydrogen Bonding in Biomolecules

(5 marks)

A **hydrogen bond** forms between a hydrogen attached to an electronegative atom (N–H, O–H) as donor and another electronegative atom (N, O, F) as acceptor:



Bond energy:  $\sim 12\text{--}38$  kJ/mol.

## (i) Role in Protein Structure:

- **$\alpha$ -Helix:** Backbone N–H  $\cdots$  O=C H-bonds between residues  $i$  and  $i + 4$  coil the chain into a helix. All backbone amide groups participate.
- **$\beta$ -Sheet:** Inter-strand N–H  $\cdots$  O=C H-bonds hold adjacent strands together in a sheet.
- **Tertiary structure:** Side-chain H-bonds (Ser, Thr, Asn, Gln, Tyr) stabilize the folded conformation.
- **Enzyme active sites:** Precise H-bond networks position substrates for catalysis and determine specificity.

## (ii) Role in Nucleic Acid (DNA/RNA) Structure:

- **Watson–Crick base pairing:** A–T: **2 H-bonds**; G–C: **3 H-bonds**. These H-bonds hold the two DNA strands together in the double helix.
- G–C pairs (3 H-bonds) make DNA more thermally stable than A–T-rich regions.
- H-bonds also stabilize RNA secondary structures (stem-loops, pseudoknots).

## (iii) Role in Carbohydrate Structure:

- Multiple –OH groups in sugars form extensive H-bond networks with water  $\Rightarrow$  high water solubility.
- In cellulose, inter- and intra-chain H-bonds between –OH groups of adjacent glucose units create rigid, crystalline microfibrils of extraordinary tensile strength.

## Part 2: Hydrophobic Interactions in Biomolecules

(5 marks)

**Hydrophobic interaction** is the tendency of non-polar groups to **cluster together** in aqueous solution to minimize their exposure to water. It is driven by **entropy** — water molecules gain freedom when released from ordered hydration shells around non-polar groups:

$$\Delta G = \Delta H - T\Delta S \quad \Rightarrow \quad \Delta S_{\text{water}} > 0 \quad \Rightarrow \quad \Delta G < 0 \quad \Rightarrow \quad \text{spontaneous clustering}$$

## (i) Role in Protein Structure:

- **Primary driving force for folding:** Non-polar R-groups (Leu, Val, Ile, Phe, Ala) are buried in the protein's hydrophobic core; polar and charged groups face the aqueous exterior.
- Determines the overall globular shape of proteins.
- Loss of this force (by detergents like SDS or organic solvents) leads to denaturation.

## (ii) Role in Lipid Bilayer (Cell Membrane) Structure:

- Phospholipids are amphiphilic: polar head (H-bonding with water) and two non-polar fatty acid tails.
- In water, hydrophobic interaction drives the fatty acid tails to cluster together, spontaneously forming a **lipid bilayer** — the fundamental structure of cell membranes.
- The bilayer is **self-sealing** and **self-assembling** due to hydrophobic effects.

## (iii) Role in Nucleic Acid Structure:

- The planar aromatic bases in DNA stack on top of each other — **base stacking** — via hydrophobic and van der Waals interactions.
- Base stacking contributes significantly to DNA stability (arguably as much as H-bonding).
- Keeps the hydrophobic bases shielded in the helix interior, with the hydrophilic sugar-phosphate backbone on the outside.

## (iv) Role in Micelle and Liposome Formation:

Bile salts, soap molecules, and detergents self-assemble into **micelles** with hydrophobic tails in the interior and polar heads facing water — purely driven by hydrophobic interaction. Liposomes (hollow bilayer vesicles) form similarly.

| Biomolecule   | H-bonding role                                    | Hydrophobic role                                  |
|---------------|---------------------------------------------------|---------------------------------------------------|
| Proteins      | $\alpha$ -helix, $\beta$ -sheet, tertiary folding | Hydrophobic core formation (dominant fold driver) |
| DNA/RNA       | Watson–Crick base pairing (A-T, G-C)              | Base stacking in helix interior                   |
| Carbohydrates | Water solubility, cellulose rigidity              | Minimal (mostly hydrophilic)                      |
| Lipids        | Polar head interactions with water                | Bilayer/micelle self-assembly                     |

**Conclusion:** H-bonding and hydrophobic interactions are the two most critical forces shaping biomolecular structure. H-bonds provide specificity and directionality (base pairing in DNA,  $\alpha$ -helix in proteins, cellulose rigidity). Hydrophobic interactions provide the thermodynamic driving force for protein folding, DNA base stacking, and lipid self-assembly into membranes and micelles. Together, they define the form and function of all biological macromolecules.

**Q5.** Describe the forces that work efficiently to hold proteins in their most stable three-dimensional structures. Briefly explain the different ways in which protein structures could be unfolded. **(15 marks)** [CHEM 113 | 2019–20]

Answer

Forces Stabilizing 3-D Protein Structure & Protein Unfolding (Denaturation) **(15 marks)**

Part 1: Forces Holding Proteins in Their Most Stable 3-D Structure **(8 marks)**

A protein’s stable 3-D structure (native conformation) is the result of minimizing free energy through a combination of non-covalent and covalent interactions among its residues.

**(i) Hydrophobic Interactions:**  
The **dominant driving force** for protein folding. Non-polar R-groups (Leu, Val, Ile, Phe) are sequestered in the protein interior (hydrophobic core), away from water. The **entropy gain** of bulk water (released from ordered hydration cages around non-polar groups) is the thermodynamic driving force:

$$\Delta G_{\text{folding}} < 0 \quad \text{primarily because} \quad \Delta S_{\text{water}} > 0$$

Contributes ~4–8 kJ/mol per buried non-polar residue.

**(ii) Hydrogen Bonds:**  
**Backbone H-bonds** stabilize secondary structure ( $\alpha$ -helix,  $\beta$ -sheet) via  $\text{N-H} \cdots \text{O}=\text{C}$ . **Side-chain H-bonds** further stabilize tertiary structure between polar R-groups (Ser, Thr, Asn, Tyr). Contribute ~12–38 kJ/mol each; collectively essential.

**(iii) Electrostatic Interactions (Salt Bridges):**  
Charged R-groups ( $\text{Asp}^-/\text{Glu}^-$  and  $\text{Lys}^+/\text{Arg}^+/\text{His}^+$ ) form ion pairs. Interior salt bridges are amplified by the low dielectric constant of the non-polar core. Disrupted by pH changes.

**(iv) Disulfide Bonds:**  
Covalent S–S bonds between Cys residues. Provide extra kinetic stability (energy barrier to unfolding). Critical in extracellular proteins (insulin, lysozyme, antibodies). Reduced by DTT,  $\beta$ -ME.

**(v) van der Waals Forces:**  
Packing interactions between atoms ~3–4 Å apart in the hydrophobic core. Individually <4 kJ/mol but thousands of such contacts sum to significant stabilization.

**(vi) Metal coordination (in metalloproteins):**  
Some proteins use **metal ions** ( $\text{Zn}^{2+}$ ,  $\text{Fe}^{2+}$ ,  $\text{Ca}^{2+}$ ) coordinated by specific R-groups (His, Cys, Asp, Glu) to stabilize structure. Example: zinc finger proteins, calcium-binding proteins.

**Overall thermodynamic balance:**

$$\Delta G_{\text{stability}} = \Delta H_{\text{interactions}} - T\Delta S_{\text{chain}} \approx -20 \text{ to } -65 \text{ kJ/mol}$$

(Surprisingly small — proteins are only marginally stable!)

Part 2: Ways in Which Protein Structures Can Be Unfolded (Denaturation) **(7 marks)**

**Denaturation** is the disruption of the native 3-D structure (secondary, tertiary, quaternary) without breaking the primary structure (peptide bonds). The protein loses its biological activity.

**(i) Heat (Thermal denaturation):**  
Increasing temperature provides kinetic energy to overcome the non-covalent stabilizing interactions. The protein **unfolds** as H-bonds, salt bridges, and hydrophobic interactions are disrupted.

- Most proteins denature at 60–80°C.
- Heat-stable (thermophilic) proteins may withstand >100°C due to extra disulfide bonds and salt bridges.
- Example: egg white albumin coagulates on cooking.

**(ii) Extreme pH:**

Acids and bases alter the protonation state of ionizable R-groups (His, Asp, Glu, Lys, Arg), disrupting salt bridges and H-bonds:

- Low pH: protonates carboxylate groups ( $\text{COO}^- \rightarrow \text{COOH}$ ), destroying negative charges.
- High pH: deprotonates ammonium groups ( $\text{NH}_3^+ \rightarrow \text{NH}_2$ ), destroying positive charges.
- Ionic interactions collapse  $\Rightarrow$  protein unfolds.

**(iii) Chemical Denaturants (urea, guanidinium chloride):**

High concentrations (6–8 M) of urea or guanidinium Cl (GdmCl):

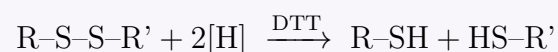
- Form extensive H-bonds with the protein backbone, competing with intrachain H-bonds.
- Disrupt the hydrophobic core by making the aqueous environment more compatible with non-polar groups.
- GdmCl additionally disrupts electrostatic interactions as a chaotropic salt.

**(iv) Detergents (SDS, bile salts):**

Sodium dodecyl sulfate (SDS) and bile salts insert their **hydrophobic tails** into the protein's hydrophobic core, disrupting hydrophobic interactions and unfolding the protein. The detergent coat keeps the denatured chain soluble.

**(v) Reducing Agents (DTT,  $\beta$ -mercaptoethanol):**

Cleave disulfide bonds by reduction:



For proteins stabilized primarily by disulfide bonds (e.g. ribonuclease), this alone can cause significant unfolding.

**(vi) Organic solvents (ethanol, acetone):**

Disrupt hydrophobic interactions and H-bonding by altering the dielectric constant of the solvent. Also compete with water in solvating polar groups.

**(vii) Heavy metal ions ( $\text{Pb}^{2+}$ ,  $\text{Hg}^{2+}$ ,  $\text{Ag}^+$ ):**

React with  $-\text{SH}$ ,  $-\text{COOH}$ ,  $-\text{NH}_2$  groups, forming inappropriate metal-ligand bonds that distort the protein structure.

**Summary of denaturation methods:**

| Denaturing Agent  | Force Disrupted                            |
|-------------------|--------------------------------------------|
| Heat              | All non-covalent forces                    |
| Extreme pH        | Salt bridges, H-bonds                      |
| Urea / GdmCl      | H-bonds, hydrophobic interactions          |
| SDS / bile salts  | Hydrophobic interactions                   |
| DTT / $\beta$ -ME | Disulfide bonds (covalent)                 |
| Organic solvents  | Hydrophobic interactions, H-bonds          |
| Heavy metals      | Covalent coordination, $-\text{SH}$ groups |

**Conclusion:** Proteins are stabilized by hydrophobic interactions (dominant), H-bonds, salt bridges, disulfide bonds, and van der Waals forces. The overall stability is surprisingly modest ( $\Delta G \approx -20$  to  $-65$  kJ/mol), making proteins susceptible to denaturation by heat, extreme pH, chaotropic agents (urea, GdmCl), detergents (SDS), reducing agents (DTT), organic solvents, and heavy metals — each targeting specific stabilizing interactions.

**Q6.** Draw the mechanisms of the acid hydrolysis of methyl- $\beta$ -D-glucopyranoside. (5 marks)

[CHEM 113 | 2017–18]

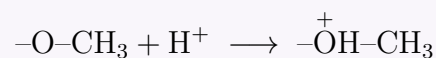
**Answer****Acid Hydrolysis of Methyl- $\beta$ -D-Glucopyranoside: Mechanism**

(5 marks)

The acid hydrolysis of methyl- $\beta$ -D-glucopyranoside converts the glycoside (acetal) back to the hemiacetal ( $\beta/\alpha$ -D-glucopyranose) plus methanol. The mechanism follows **specific acid catalysis ( $\text{S}_\text{N}1$ )**:

**Step 1 — Protonation of the glycosidic oxygen:**

The lone pair on the **ring oxygen** or the exocyclic **glycosidic oxygen** (O-1) is protonated by  $H^+$  (from  $HCl$ ,  $H_2SO_4$ , etc.):



**Step 2 — Departure of methanol (oxocarbenium ion formation):**

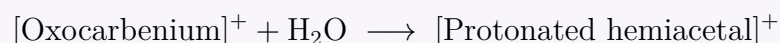


Loss of methanol generates a **cyclic oxocarbenium (carbocation) intermediate** stabilized by resonance with the ring oxygen:



The positive charge is delocalized over C-1 and the ring O. This intermediate is **planar** at C-1.

**Step 3 — Attack of water:**



Water attacks C-1 from either face of the planar oxocarbenium ion:

- **Attack from  $\alpha$ -face**  $\Rightarrow$   $\alpha$ -D-glucopyranose
- **Attack from  $\beta$ -face**  $\Rightarrow$   $\beta$ -D-glucopyranose

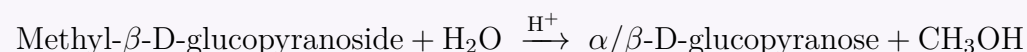
Both anomers are formed in the equilibrium ratio ( $\sim 36\%$   $\alpha$ ,  $64\%$   $\beta$  in water).

**Step 4 — Deprotonation:**



Loss of a proton from the newly formed  $-OH$  at C-1 gives the free sugar. The catalyst  $H^+$  is regenerated.

**Overall:**



**Conclusion:** Acid hydrolysis of methyl- $\beta$ -D-glucopyranoside proceeds via: (1) protonation of the glycosidic O, (2) departure of  $CH_3OH$  forming a planar oxocarbenium ion, (3) water attack on both faces to give both anomers, and (4) deprotonation. The mechanism is  $S_N1$  via a stabilized cyclic oxocarbenium intermediate.

**Q7.** Draw the chemical structure of  $\beta$ -D-(+)-glucopyranose in chair conformation. Discuss the significance of  $\beta$ , D and (+). **(10 marks)**  
[CHEM 113 | 2017–18]

### Answer

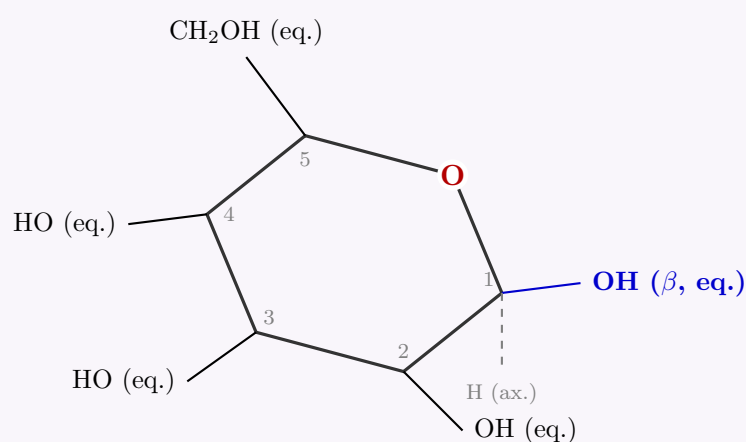
**$\beta$ -D-(+)-Glucopyranose in Chair Conformation; Significance of  $\beta$ , D and (+)**

**(10 marks)**

#### Part 1: Chair Conformation of $\beta$ -D-(+)-Glucopyranose

**(5 marks)**

$\beta$ -D-Glucopyranose adopts the  ${}^4C_1$  (C4 up, C1 down) chair conformation in which **all five substituents** (four  $-OH$  groups and the  $-CH_2OH$  group) are in the **equatorial position** — making it the most stable pyranose ring:



**${}^4C_1$  chair: all substituents are equatorial**

**Key features of  $\beta$ -D-glucopyranose chair:**

- The C-1  $-OH$  is **equatorial** ( $\beta$  configuration) — trans to the C2- $-OH$ .

- All four –OH groups (C2, C3, C4, C1) and the –CH<sub>2</sub>OH are equatorial.
- Equatorial positions minimize 1,3-diaxial steric interactions ⇒ **most stable** glucose anomer.
- This explains why β-D-glucopyranose (~64%) predominates over α (~36%) at equilibrium in water.

## Part 2: Significance of β, D and (+)

(5 marks)

### (i) Significance of D (absolute configuration):

The **D/L designation** refers to the absolute configuration of the **highest-numbered chiral carbon** (C-5 in glucose), based on its relationship to D-glyceraldehyde:

- **D-glucose:** The –OH group at C-5 is on the **right** in the Fischer projection. Same spatial arrangement as D-glyceraldehyde at the reference carbon.
- **L-glucose:** The –OH at C-5 is on the **left** — the mirror image of D-glucose.
- D and L are **enantiomers** (non-superimposable mirror images).
- Nearly all naturally occurring sugars are **D-sugars**; biological enzymes are stereospecific and only process D-sugars.
- D/L is **not** the same as (+)/(-) — they refer to different properties.

### (ii) Significance of (+) (optical rotation):

The **(+) or d-designation** refers to the direction of rotation of plane-polarized light as measured by a polarimeter:

- **(+) or d-glucose:** Rotates plane-polarized light **clockwise** (dextrorotatory).  $[\alpha]_D^{25} = +52.7^\circ$  (equilibrium mixture).
- **(-) or l:** Rotates light **anticlockwise** (levorotatory).
- (+)/(-) is **determined experimentally** and cannot be predicted from D/L alone.
- Example: D-fructose is **D-(-)**-fructose — D configuration but levorotatory.

### (iii) Significance of β (anomeric configuration):

The **α/β designation** refers to the configuration at the **anomeric carbon (C-1)** — the new chiral centre created upon ring closure:

- **α-glucose:** C-1 –OH is **axial** (trans to C-2 –OH); on the same side as the ring oxygen in Haworth projection (below the ring plane for D-sugars).
- **β-glucose:** C-1 –OH is **equatorial** (cis to C-2 –OH); on the opposite side from the ring oxygen (above the ring plane for D-sugars).
- β is more stable for D-glucose due to all equatorial substituents.
- α and β are **anomers** — diastereomers differing only at C-1.
- They interconvert in solution via **mutarotation**.

### Summary:

| Symbol | Property                | Significance                                                       |
|--------|-------------------------|--------------------------------------------------------------------|
| D      | Absolute config. at C-5 | –OH at C-5 on the right in Fischer; same as D-glyceraldehyde       |
| (+)    | Optical rotation        | Rotates plane-polarized light clockwise; experimentally determined |
| β      | Anomeric config. at C-1 | –OH at C-1 equatorial (trans to C-2 –OH); more stable anomer       |

**Conclusion:** β-D-(+)-Glucopyranose adopts the <sup>4</sup>C<sub>1</sub> chair conformation with **all substituents equatorial** — the most stable hexose configuration. **D** denotes the absolute configuration at C-5 (same as D-glyceraldehyde); **(+)** indicates dextrorotatory optical activity ( $[\alpha]_D = +52.7^\circ$ ); **β** specifies the equatorial C-1 –OH (anomeric configuration), making β-D-glucose predominate (~64%) at equilibrium.

**Q8.** Define carbohydrates. Give one example of monosaccharides, disaccharides and polysaccharides. Draw their structural formulas. (10 marks) [CHEM 113 | 2016–17]



## Answer

## Definition of Carbohydrates; Monosaccharide, Disaccharide &amp; Polysaccharide with Structural Formulas (10 marks)

## Part 1: Definition of Carbohydrates

(2 marks)

**Carbohydrates** (saccharides) are organic compounds composed of carbon, hydrogen, and oxygen in the general formula  $(CH_2O)_n$  or  $C_n(H_2O)_n$ . More precisely, they are **polyhydroxy aldehydes or ketones** (and their derivatives and polymers).

General formula:  $C_nH_{2n}O_n$  ( $n \geq 3$ )

- If the carbonyl is at C-1  $\Rightarrow$  **aldose** (aldehyde sugar).
- If the carbonyl is at C-2  $\Rightarrow$  **ketose** (ketone sugar).

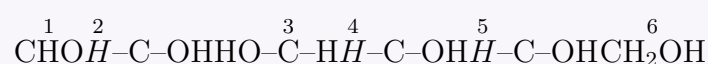
Classified by size: **mono-**, **oligo-**, and **polysaccharides**.

Part 2: Monosaccharide — D-Glucose ( $C_6H_{12}O_6$ )

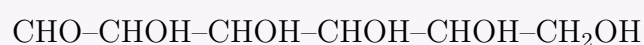
(3 marks)

A **monosaccharide** is the simplest carbohydrate unit — a single polyhydroxy aldehyde or ketone that cannot be hydrolyzed further. Example: **D-glucose** (dextrose, blood sugar).

**Open-chain (Fischer) form:**



Written linearly:



(C2, C4, C5 –OH on right; C3 –OH on left in Fischer)

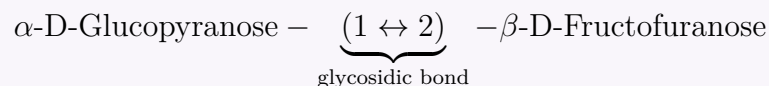
**Cyclic (Haworth)  $\beta$ -form:** Six-membered ring ( $\beta$ -D-glucopyranose) with all –OH equatorial in chair conformation.

Part 3: Disaccharide — Sucrose ( $C_{12}H_{22}O_{11}$ )

(2 marks)

A **disaccharide** consists of two monosaccharide units joined by a **glycosidic bond** with loss of water. Example: **sucrose** (table sugar) = glucose + fructose, linked by an  $\alpha(1\rightarrow2)\beta$  glycosidic bond.

Structure of sucrose:



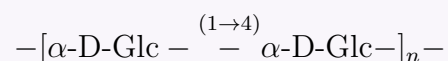
- The glycosidic bond involves the anomeric carbons of **both** sugars  $\Rightarrow$  **non-reducing sugar** (no free anomeric OH).
- Hydrolysis by sucrase: sucrose  $\rightarrow$  glucose + fructose (**invert sugar**,  $[\alpha]$  changes from  $+66.5^\circ$  to  $-22^\circ$ ).

## Part 4: Polysaccharide — Starch

(3 marks)

A **polysaccharide** is a polymer of many monosaccharide units ( $n = 10^2\text{--}10^6$ ) joined by glycosidic bonds. Example: **starch** (amylose component).

**Amylose:** Linear polymer of  $\alpha$ -D-glucose linked by  $\alpha(1\rightarrow4)$  glycosidic bonds:



- Each glucose unit: chair conformation, C1 anomeric carbon linked to C4 of next unit.
- Amylose forms left-handed helices (6 glucose/turn).
- Amylopectin: branched, with  $\alpha(1\rightarrow6)$  branch points every 24–30 glucose units.
- Energy storage polymer in plants.
- Hydrolyzed by amylase to maltose, then glucose.

**Summary table:**

| Class          | Example          | Formula              | Linkage                      |
|----------------|------------------|----------------------|------------------------------|
| Monosaccharide | D-Glucose        | $C_6H_{12}O_6$       | None                         |
| Disaccharide   | Sucrose          | $C_{12}H_{22}O_{11}$ | $\alpha(1\rightarrow2)\beta$ |
| Polysaccharide | Starch (amylose) | $(C_6H_{10}O_5)_n$   | $\alpha(1\rightarrow4)$      |

**Conclusion:** Carbohydrates are polyhydroxy aldehydes/ ketones with formula  $(CH_2O)_n$ . Monosaccharides (glucose,  $C_6H_{12}O_6$ ) are the building blocks; disaccharides (sucrose) consist of two monosaccharides joined by a glycosidic bond; polysaccharides (starch) are large polymers of monosaccharide units serving structural and energy-storage functions.



Q9. Draw the Fischer projection formula of D-glucose and L-glucose. (5 marks)

[CHEM 113 | 2016–17]

### Answer

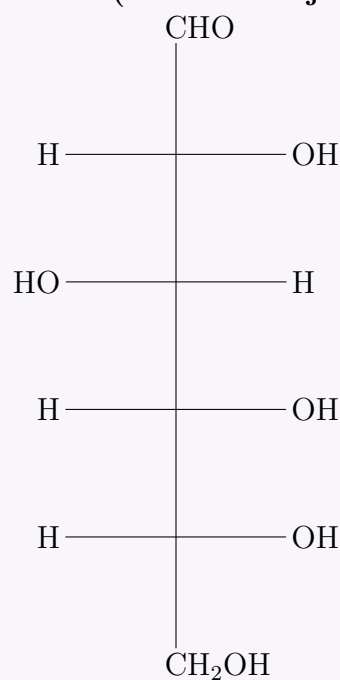
#### Fischer Projection of D-Glucose and L-Glucose

(5 marks)

In a **Fischer projection**:

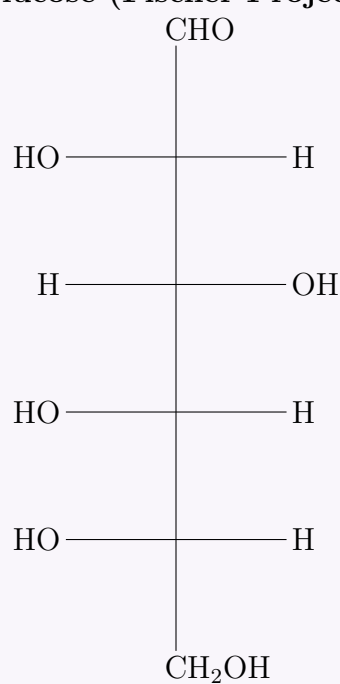
- The carbon chain is written **vertically** with C-1 (aldehyde) at the top.
- Horizontal bonds** project **toward** the viewer; **vertical bonds** project **away**.
- The **D/L designation** is determined by the –OH at C-5: **right = D**, **left = L**.

#### D-Glucose (Fischer Projection)



**L-Glucose:** The mirror image of D-glucose — every –OH is on the **opposite side**:

#### L-Glucose (Fischer Projection)



**Key differences:**

| Property           | D-Glucose                 | L-Glucose |
|--------------------|---------------------------|-----------|
| C-5 –OH position   | Right                     | Left      |
| Optical rotation   | (+)52.7°                  | (–)52.7°  |
| Natural occurrence | Yes (abundant)            | No (rare) |
| Relationship       | Mirror image (enantiomer) |           |

**Conclusion:** D-Glucose has –OH at C-5 on the **right** in the Fischer projection; L-glucose has it on the **left**. They are enantiomers with equal and opposite optical rotations ( $\pm 52.7^\circ$ ). Only D-glucose occurs naturally and is metabolized by biological systems.

**Q10.** Write down the mechanism of the formation of methyl glycoside from  $\beta$ -D-glucopyranose. Explain which anomeric product of methyl glycoside is more stable than the other. (10 marks) [CHEM 113 | 2016–17]

### Answer

#### Mechanism of Methyl Glycoside Formation from $\beta$ -D-Glucopyranose; Anomeric Stability

(10 marks)

#### Part 1: Mechanism of Methyl Glycoside Formation

(6 marks)

When  $\beta$ -D-glucopyranose is treated with **dry methanol** ( $\text{CH}_3\text{OH}$ ) and an **acid catalyst** ( $\text{HCl}$ ,  $\text{H}_2\text{SO}_4$ ), the anomeric  $-\text{OH}$  at C-1 reacts with methanol to form a **methyl glycoside** (an acetal). The mechanism proceeds via the same **oxocarbenium ion intermediate**:

**Step 1 — Protonation of C-1  $-\text{OH}$ :**



The anomeric  $-\text{OH}$  is protonated by the acid catalyst.

**Step 2 — Loss of water; oxocarbenium ion formation:**



Water leaves, generating the **planar cyclic oxocarbenium ion** stabilized by the ring oxygen:



**Step 3 — Nucleophilic attack by methanol on C-1:**

$\text{CH}_3\text{OH}$  attacks C-1 of the planar oxocarbenium ion from **both faces**:

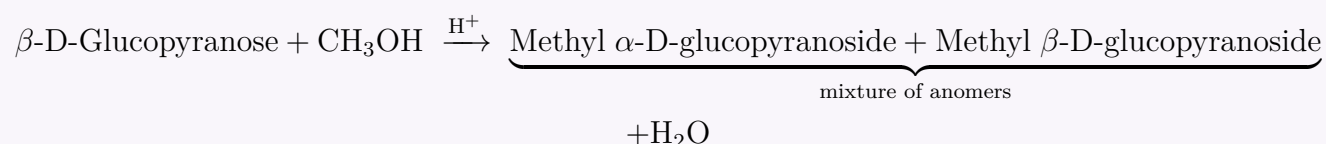
- **Attack from  $\alpha$ -face** (same side as ring O)  $\Rightarrow$  **methyl  $\alpha$ -D-glucopyranoside** (C-1  $-\text{OCH}_3$  axial)
- **Attack from  $\beta$ -face** (opposite side from ring O)  $\Rightarrow$  **methyl  $\beta$ -D-glucopyranoside** (C-1  $-\text{OCH}_3$  equatorial)

**Step 4 — Deprotonation:**



Loss of  $\text{H}^+$  from  $-\text{OH}_2^+$  gives the neutral methyl glycoside.

**Overall:**



#### Part 2: Which Anomeric Product is More Stable?

(4 marks)

The two products differ only at C-1:

- **Methyl  $\beta$ -D-glucopyranoside:**  $-\text{OCH}_3$  at C-1 is **equatorial**.
- **Methyl  $\alpha$ -D-glucopyranoside:**  $-\text{OCH}_3$  at C-1 is **axial**.

**Steric argument (standard):**

In the  ${}^4C_1$  chair, an equatorial substituent experiences less 1,3-diaxial steric strain than an axial substituent. Therefore, **methyl  $\beta$ -D-glucopyranoside** (equatorial  $-\text{OCH}_3$ ) should be more stable by steric argument.

**The Anomeric Effect (electronic argument):**

However, for electronegative substituents ( $-\text{OR}$ ,  $-\text{OH}$ ,  $-\text{F}$ ) at the anomeric carbon, the **axial** orientation is actually **preferred** — this is the **anomeric effect**.

- In the axial position, the lone pair of the ring O can overlap with the  $\sigma^*$  orbital of the C-1-OR bond (back-lobe interaction / hyperconjugation) —  **$\text{n} \rightarrow \sigma^*$  stabilization**.
- This electronic stabilization *favours the axial* (i.e.  $\alpha$ ) anomer for electronegative groups.
- In methyl glucoside: anomeric effect stabilizes **methyl  $\alpha$ -D-glucopyranoside**.
- The anomeric effect is stronger in **non-polar solvents** (less shielding of the dipole interaction).

**Net result:**

| Anomer                     | –OCH <sub>3</sub> | Stability                                                               |
|----------------------------|-------------------|-------------------------------------------------------------------------|
| Methyl α-D-glucopyranoside | Axial             | Stabilized by <b>anomeric effect</b> (preferred in non-polar media)     |
| Methyl β-D-glucopyranoside | Equatorial        | Stabilized by <b>steric argument</b> (preferred in polar/aqueous media) |

In practice, methyl α-D-glucopyranoside is often the **thermodynamic product** because the anomeric effect outweighs the steric penalty for the axial –OCH<sub>3</sub>. The α : β ratio in methanol under acid catalysis is ~66:34 at equilibrium, confirming α is more stable.

**Conclusion:** Methyl glycoside formation proceeds via protonation of C-1 –OH, loss of water to form the oxocarbenium ion, then nucleophilic attack by CH<sub>3</sub>OH from both faces giving α and β anomers. While steric effects favor the β-anomer (equatorial –OCH<sub>3</sub>), the **anomeric effect** (n→σ\* stabilization) makes the **α-anomer thermodynamically more stable**, with α : β ≈ 66 : 34 at equilibrium.

Q11. What is mutarotation? Why does glucose undergo mutarotation? (5 marks)

[CHEM 113 | 2016–17]

### Answer

#### Mutarotation: Definition and Why Glucose Undergoes It

(5 marks)

##### (i) Definition:

**Mutarotation** is the spontaneous change in optical rotation of a freshly prepared solution of a reducing sugar until an equilibrium value is reached. It occurs because the α and β anomers of the sugar interconvert in solution until equilibrium is established.

##### (ii) Optical rotation values for D-glucose:

| Form              | $[\alpha]_D^{25}$ | At equilibrium         |
|-------------------|-------------------|------------------------|
| α-D-Glucopyranose | +112.2°           | +52.7° (~36% α, 64% β) |
| β-D-Glucopyranose | +18.7°            |                        |

##### (iii) Why glucose undergoes mutarotation:

Glucose undergoes mutarotation because it can exist as both a **cyclic hemiacetal** (pyranose ring) and a transient **open-chain aldehyde**:

1. The α or β pyranose ring **opens reversibly** at C-1 to give the open-chain aldehyde (acyclic form, < 0.02% at equilibrium).
2. The open-chain aldehyde **recycles** by attack of the C-5 –OH on the C-1 carbonyl.
3. Ring closure can occur from either face of the carbonyl, regenerating either α (C-1 –OH axial) or β (C-1 –OH equatorial).



##### (iv) Why only reducing sugars mutarotate:

Only sugars with a **free anomeric –OH** (reducing sugars) can open the ring to give the open-chain aldehyde. **Non-reducing sugars** (e.g. sucrose, methyl glycosides) have the anomeric carbon blocked by a glycosidic bond and **cannot mutarotate**.

**Conclusion:** Mutarotation is the change in  $[\alpha]_D$  until equilibrium, caused by reversible ring opening of each anomer to the open-chain aldehyde followed by reclosure to give both anomers. Glucose mutarotates because its C-1 –OH (free anomeric –OH) allows ring opening. Equilibrium gives  $[\alpha]_D = +52.7^\circ$  (~36% α + 64% β).

Q12. Prove that glucose is a reducing sugar. (5 marks)

[CHEM 113 | 2016–17]

### Answer

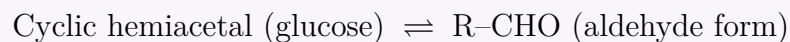
#### Proof that Glucose is a Reducing Sugar

(5 marks)

A **reducing sugar** is one that can **reduce** an oxidizing agent (e.g. Cu<sup>2+</sup>, Ag<sup>+</sup>) in alkaline solution. This property requires a **free anomeric –OH** (hemiacetal group) that can open to give a free aldehyde, which acts as the reducing agent.

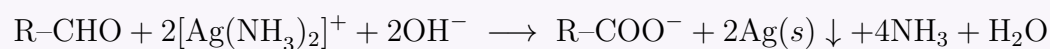
##### (i) Structural basis:

Glucose has a **free anomeric –OH at C-1** (hemiacetal). In solution, the ring opens to give the **open-chain aldehyde**:



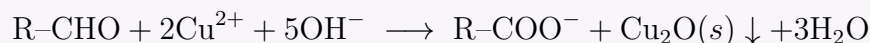
The aldehyde form is readily oxidized (< 0.02% at equilibrium, but continuously regenerated).

**(ii) Tollens' Test (Silver Mirror):**



Glucose reduces silver(I) to metallic silver (silver mirror on glass). **Positive result  $\Rightarrow$  reducing sugar.**

**(iii) Fehling's / Benedict's Test:**



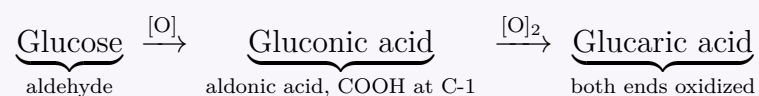
Glucose reduces  $\text{Cu}^{2+}$  (deep blue) to  $\text{Cu}_2\text{O}$  (**brick-red precipitate**). **Positive result  $\Rightarrow$  reducing sugar.**

**(iv) Why non-reducing sugars fail:**

Sucrose has **no free anomeric –OH** (both C-1 of glucose and C-2 of fructose are involved in the glycosidic bond). The ring cannot open to give an aldehyde  $\Rightarrow$  **cannot reduce**  $\text{Cu}^{2+}$  or  $\text{Ag}^+$   $\Rightarrow$  negative Fehling's/Tollens' tests.

**(v) Oxidation product:**

Controlled oxidation of glucose produces:



**Conclusion:** Glucose is a reducing sugar because its free anomeric –OH (hemiacetal) allows ring opening to give the reactive aldehyde form, which readily reduces  $\text{Ag}^+$  (Tollens': silver mirror) and  $\text{Cu}^{2+}$  (Fehling's/Benedict's: brick-red precipitate). Sucrose is non-reducing because it lacks a free anomeric –OH.

**Q13.** Draw the primary, secondary, tertiary and quaternary structures of proteins. **(10 marks)**

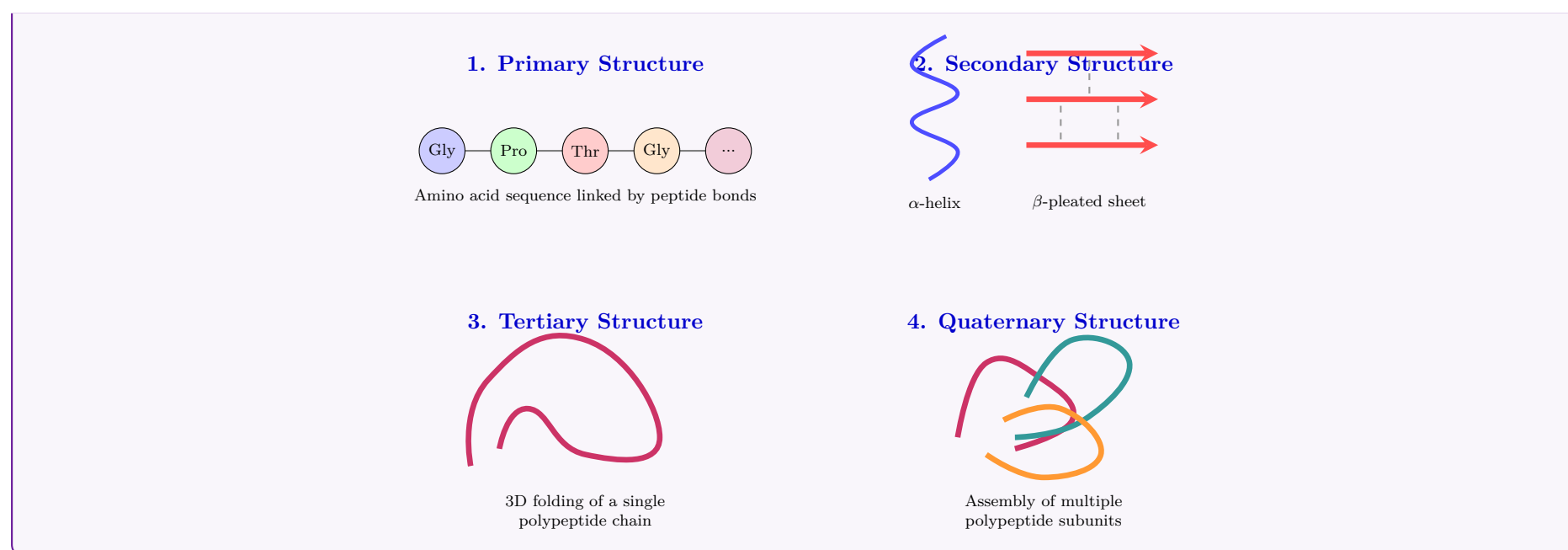
[CHEM 113 | 2016–17]

### Answer

#### Four Levels of Protein Structure

Proteins have a complex hierarchy of structures that determine their final three-dimensional shape and biological function.

- 1. Primary Structure (1°):** The linear sequence of amino acids in a polypeptide chain. It is determined by genetic coding and held together by strong covalent **peptide bonds**.
- 2. Secondary Structure (2°):** The local, regular folding of the polypeptide backbone. It is stabilized predominantly by **hydrogen bonds** between the carboxyl oxygen and the amino hydrogen of the peptide backbone. The two most common arrangements are:
  - $\alpha$ -helix:** A right-handed coil/spiral.
  - $\beta$ -pleated sheet:** Laterally aligned strands forming a sheet-like structure.
- 3. Tertiary Structure (3°):** The complete, three-dimensional arrangement (folding) of a single polypeptide chain. It is stabilized by interactions between the amino acid side chains (R-groups), including **disulfide bridges, hydrophobic interactions, ionic bonds, and hydrogen bonds**.
- 4. Quaternary Structure (4°):** The spatial arrangement and association of **two or more** polypeptide chains (called subunits) to form a single, fully functional protein complex (e.g., hemoglobin). It is stabilized by the same non-covalent interactions as the tertiary structure.



**Q14.** What sequence of bases on one strand of DNA is complementary to the sequence TATGCAT on another strand? **(3 marks)** [CHEM 113 | 2015–16]

### Answer

#### Complementary DNA Sequence to TATGCAT

**(3 marks)**

In DNA double helix, the two strands are **antiparallel** and **complementary** following Watson–Crick base pairing:

A pairs with T (2 H-bonds)      G pairs with C (3 H-bonds)

**Given strand (5'→3'):**

5'—T—A—T—G—C—A—T—3'

**Complementary strand (written 3'→5'):**

3'—A—T—A—C—G—T—A—5'

**Complementary strand (rewritten 5'→3'):**

5'-ATACGTA-3'

**Verification (antiparallel alignment):**

|    |   |   |   |   |   |   |   |    |
|----|---|---|---|---|---|---|---|----|
| 5' | T | A | T | G | C | A | T | 3' |
|    |   |   |   |   |   |   |   |    |
| 3' | A | T | A | C | G | T | A | 5' |

Each base pair: T–A, A–T, T–A, G–C, C–G, A–T, T–A ✓

**Conclusion:** The complementary sequence is **5'-ATACGTA-3'** (antiparallel to the given 5'-TATGCAT-3' strand).

**Q15.** “Mutarotation occurs by a reversible ring-opening of each anomer to the open-chain aldehyde, followed by reclosure” — Explain with a suitable example. **(5 marks)** [CHEM 113 | 2015–16]

### Answer

#### Mutarotation: Reversible Ring Opening and Reclosure with Example

**(5 marks)**

**Statement:** “Mutarotation occurs by a reversible ring-opening of each anomer to the open-chain aldehyde, followed by reclosure.”

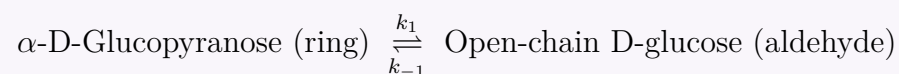
**Example: D-Glucose**

**Step 1 — Pure  $\alpha$ -D-glucopyranose dissolves:**

When pure  $\alpha$ -D-glucopyranose ( $[\alpha]_D = +112.2^\circ$ ) is dissolved in water, the solution's optical rotation immediately begins to **decrease** toward  $+52.7^\circ$ .

**Step 2 — Ring opening ( $\alpha \rightarrow$  open chain):**

The C-1 hemiacetal bond in the  $\alpha$ -anomer breaks via **acid/base catalysis by water**, regenerating the open-chain aldehyde:



The open chain is present at  $< 0.02\%$  at equilibrium (too low to detect directly, but continuously regenerated).

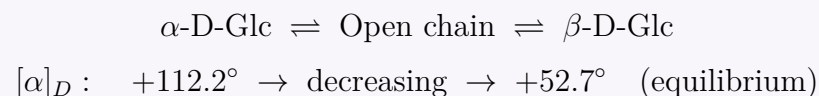
**Step 3 — Ring reclosure:**

The C-5 –OH attacks the C-1 aldehyde carbonyl **from either face**:



- **Attack from below** (same face as C-5 –O)  $\Rightarrow$  C-1 –OH **axial**  $\Rightarrow$   $\alpha$ -anomer.
- **Attack from above** (opposite face)  $\Rightarrow$  C-1 –OH **equatorial**  $\Rightarrow$   $\beta$ -anomer.

**Step 4 — Equilibrium reached:**



Similarly, if pure  $\beta$ -D-glucopyranose ( $[\alpha]_D = +18.7^\circ$ ) is dissolved, the rotation **increases** to  $+52.7^\circ$  as  $\alpha$ -anomer is generated.

**Equilibrium composition:**

$$\sim 36\% \alpha + \sim 64\% \beta + \sim 0.02\% \text{ open chain} = 100\%$$

$$[\alpha]_D^{\text{calc}} = 0.36(112.2) + 0.64(18.7) = 40.4 + 12.0 = 52.4 \approx +52.7^\circ \quad \checkmark$$

**Conclusion:** Mutarotation of D-glucose proceeds via reversible ring opening of each anomer ( $\alpha$  or  $\beta$ ) to the transient open-chain aldehyde, followed by ring reclosure from either face. This interconverts  $\alpha$  ( $[\alpha]_D = +112.2^\circ$ ) and  $\beta$  ( $+18.7^\circ$ ) until the equilibrium value of  $+52.7^\circ$  is reached (36%  $\alpha$ , 64%  $\beta$ ).

**Q16.** What do you mean by rRNA, tRNA and mRNA? What are the differences between DNA and RNA? **(9 marks)** [CHEM 113 | 2015–16]

**Answer**

**rRNA, tRNA, mRNA; Differences Between DNA and RNA**

**(9 marks)**

**Part 1: Types of RNA**

**(4 marks)**

RNA (ribonucleic acid) is a single-stranded polynucleotide that plays various roles in protein synthesis. Three major functional types:

**(i) mRNA (Messenger RNA):**

- Carries the **genetic code** transcribed from DNA to the ribosome.
- Contains codons (triplets of bases, e.g. AUG = start/Met) that specify amino acid sequence.
- Most abundant RNA in terms of variety ( $\sim 4\%$  of total cell RNA in bacteria).
- In eukaryotes: processed (5'-cap, poly-A tail, splicing of introns) before export from nucleus.
- **Short-lived** — turned over rapidly to regulate gene expression.

**(ii) tRNA (Transfer RNA):**

- Adaptor molecule that **decodes mRNA codons** and carries the correct amino acid to the ribosome.
- $\sim 73\text{--}93$  nucleotides; folds into a **cloverleaf** (2-D) or **L-shaped** (3-D) structure via intramolecular H-bonds.
- Key features: **anticodon loop** (3 bases complementary to the mRNA codon) at one end; **3'-CCA-OH** amino acid attachment site at the other.
- $\sim 15\%$  of total cell RNA.
- One tRNA for each amino acid (charged by aminoacyl-tRNA synthetase).

**(iii) rRNA (Ribosomal RNA):**

- Most abundant RNA ( $\sim 80\%$  of total cell RNA).
- Structural and catalytic component of **ribosomes** — the molecular machines of translation.



- Prokaryote ribosome (70S): 23S rRNA + 5S rRNA (50S large subunit); 16S rRNA (30S small subunit).
- Eukaryote ribosome (80S): 28S, 5.8S, 5S rRNA (60S); 18S rRNA (40S).
- The **peptidyl transferase activity** (forming peptide bonds) is catalyzed by rRNA (**ribozyme**) in the 23S/28S rRNA.

Part 2: Differences Between DNA and RNA

(5 marks)

| Property     | DNA                                      | RNA                                           |
|--------------|------------------------------------------|-----------------------------------------------|
| Full name    | Deoxyribonucleic acid                    | Ribonucleic acid                              |
| Sugar        | <b>2'-Deoxyribose</b> (no –OH at C-2')   | <b>Ribose</b> (–OH at C-2')                   |
| Bases        | A, T, G, C                               | A, <b>U</b> (uracil), G, C (T replaced by U)  |
| Strands      | <b>Double-stranded</b> (usually)         | <b>Single-stranded</b> (usually)              |
| Helix form   | B-DNA: right-handed double helix         | A-form helix (in dsRNA regions)               |
| Location     | Nucleus (+ mitochondria, chloroplasts)   | Nucleus, cytoplasm, ribosomes                 |
| Stability    | Very stable (no 2'-OH)                   | Less stable (2'-OH promotes hydrolysis)       |
| Function     | Permanent storage of genetic information | Temporary; gene expression (mRNA, tRNA, rRNA) |
| Amount/cell  | Constant (diploid cells)                 | Variable (depends on gene activity)           |
| Base pairing | A–T (2 H-bonds), G–C (3 H-bonds)         | A–U (2 H-bonds), G–C (3 H-bonds)              |

**Conclusion:** mRNA carries genetic information from DNA to ribosomes; tRNA decodes codons and delivers amino acids; rRNA forms the structural and catalytic core of ribosomes. DNA and RNA differ in sugar (deoxy vs ribo), base (T vs U), strandedness (double vs single), stability (DNA more stable due to no 2'-OH), location, and function (information storage vs gene expression).

**Q17.** Why are amino acids amphiprotic? Find out what species are present in a 1.00 M solution of alanine at pH = 9.00. The  $pK_{a1}$  and  $pK_{a2}$  values of alanine are 2.34 and 9.69, respectively. (10 marks) [CHEM 113 | 2015–16]

Answer

Why Amino Acids Are Amphiprotic; Species in 1.00 M Alanine at pH 9.00

(10 marks)

Part 1: Why Amino Acids Are Amphiprotic

(4 marks)

**Amphiprotic** species can act as **both an acid (proton donor) and a base (proton acceptor)** depending on the pH of the solution.

**(i) Structure of amino acids:**

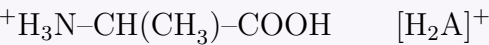
Every amino acid has:

- An **acidic group** (–COOH,  $pK_{a1} \sim 2$ ): can donate  $H^+$ .
- A **basic group** (–NH<sub>2</sub>,  $pK_{a2} \sim 9\text{--}10$ ): can accept  $H^+$ .

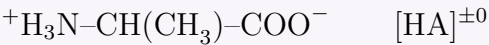
**(ii) Three protonation states:**

At different pH values, an amino acid (e.g. alanine) exists in three forms:

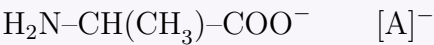
**Low pH** (fully protonated, cationic form):



**Intermediate pH** (zwitterion, isoelectric form):



**High pH** (fully deprotonated, anionic form):



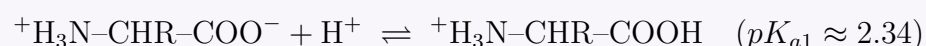
**(iii) Acid–base reactions:**

The zwitterion (HA) is **amphiprotic** — it acts as:

- An **acid** (donates  $H^+$  from –NH<sub>3</sub><sup>+</sup>):



- A base (accepts  $\text{H}^+$  on  $-\text{COO}^-$ ):



(iv) Isoelectric point (pI):

$$pI = \frac{pK_{a1} + pK_{a2}}{2} = \frac{2.34 + 9.69}{2} = 6.02$$

At  $\text{pH} = pI$ , the zwitterion is the dominant species and the molecule carries zero net charge.

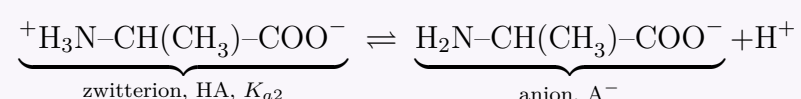
## Part 2: Species Present in 1.00 M Alanine at pH 9.00

(6 marks)

**Given:**  $pK_{a1} = 2.34$  ( $-\text{COOH}$ ),  $pK_{a2} = 9.69$  ( $-\text{NH}_3^+$ ),  $\text{pH} = 9.00$ .

**Step 1 — Identify the relevant equilibrium:**

Since  $\text{pH} = 9.00$  is between  $pK_{a1} = 2.34$  and  $pK_{a2} = 9.69$ , the relevant acid–base equilibrium is:



The  $-\text{COOH}$  group is **fully deprotonated** at  $\text{pH} 9.00$  ( $\text{pH} \gg pK_{a1} = 2.34$ ). Only the second equilibrium matters.

**Step 2 — Apply Henderson–Hasselbalch equation:**

$$\text{pH} = pK_{a2} + \log \frac{[\text{A}^-]}{[\text{HA}^\pm]}$$

$$9.00 = 9.69 + \log \frac{[\text{A}^-]}{[\text{HA}^\pm]}$$

$$\log \frac{[\text{A}^-]}{[\text{HA}^\pm]} = 9.00 - 9.69 = -0.69$$

$$\frac{[\text{A}^-]}{[\text{HA}^\pm]} = 10^{-0.69} = 0.204$$

**Step 3 — Calculate concentrations:**

Let  $[\text{HA}^\pm] = x$  and  $[\text{A}^-] = 0.204x$ . Total concentration = 1.00 M (ignoring the fully protonated form  $[\text{H}_2\text{A}]^+$  which is negligible at  $\text{pH} 9.00$ ):

$$x + 0.204x = 1.00 \quad \Rightarrow \quad 1.204x = 1.00 \quad \Rightarrow \quad x = 0.831 \text{ M}$$

$$[\text{HA}^\pm] = [\text{zwitterion}] = 0.831 \text{ M}$$

$$[\text{A}^-] = [\text{anion}] = 0.204 \times 0.831 = 0.169 \text{ M}$$

**Step 4 — Check the cationic form  $[\text{H}_2\text{A}]^+$ :**

At  $\text{pH} 9.00 \gg pK_{a1} = 2.34$ :

$$\frac{[\text{HA}^\pm]}{[\text{H}_2\text{A}]^+} = 10^{9.00-2.34} = 10^{6.66} \approx 4.6 \times 10^6$$

$[\text{H}_2\text{A}]^+ \approx \frac{0.831}{4.6 \times 10^6} \approx 1.8 \times 10^{-7} \text{ M}$  — completely negligible.

**Result:**

| Species                                                    | Form                            | Concentration                         |
|------------------------------------------------------------|---------------------------------|---------------------------------------|
| $^+\text{H}_3\text{N}-\text{CH}(\text{CH}_3)-\text{COOH}$  | Cation $[\text{H}_2\text{A}]^+$ | $\sim 10^{-7} \text{ M}$ (negligible) |
| $^+\text{H}_3\text{N}-\text{CH}(\text{CH}_3)-\text{COO}^-$ | Zwitterion $[\text{HA}]^\pm$    | <b>0.831 M</b> (dominant)             |
| $\text{H}_2\text{N}-\text{CH}(\text{CH}_3)-\text{COO}^-$   | Anion $[\text{A}]^-$            | <b>0.169 M</b> (significant)          |

**Physical interpretation:**

At  $\text{pH} = 9.00 < pK_{a2} = 9.69$ , the solution is more acidic than the second  $pK_a$ , so the protonated form (zwitterion,  $\text{HA}^\pm$ ) dominates over the anion ( $\text{A}^-$ ). The ratio is about 5:1 (zwitterion:anion).

**Conclusion:** Amino acids are amphiprotic because they contain both an acidic ( $-\text{COOH}$ ) and a basic ( $-\text{NH}_2$ ) group, allowing them to donate or accept protons. At  $\text{pH} 9.00$ , a 1.00 M alanine solution contains predominantly the **zwitterion** ( $^+\text{H}_3\text{N}-\text{CH}(\text{CH}_3)-\text{COO}^-$ , 0.831 M) and a significant amount of the **anion** ( $\text{H}_2\text{N}-\text{CH}(\text{CH}_3)-\text{COO}^-$ , 0.169 M), with the cationic form negligible ( $\sim 10^{-7} \text{ M}$ ).

BUET · CHEM 113

# Chemistry

Section S12

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Computational Chemistry

## S12 — Computational Chemistry

**Q1.** What are the salient features of molecular mechanics in computational chemistry? Under what conditions or scenarios would you choose molecular mechanics as your preferred computational method? (12 marks) [CHEM 113 | 2023–24]

## Answer

## Salient Features of Molecular Mechanics &amp; When to Choose It

(12 marks)

## Part 1: Salient Features of Molecular Mechanics (MM)

(7 marks)

**Molecular Mechanics (MM)** treats molecules as collections of **classical balls (atoms) connected by springs (bonds)**, using Newtonian mechanics rather than quantum mechanics. Electrons are not explicitly considered.

## (i) Force Field Based:

MM uses a **force field** — a set of mathematical functions and parameters that describe the potential energy of the system as a sum of contributions:

$$E_{\text{total}} = E_{\text{bond}} + E_{\text{angle}} + E_{\text{torsion}} + E_{\text{vdW}} + E_{\text{electrostatic}}$$

- $E_{\text{bond}}$ : harmonic potential for bond stretching ( $k(r - r_0)^2$ )
- $E_{\text{angle}}$ : harmonic potential for bond angle bending
- $E_{\text{torsion}}$ : periodic potential for dihedral angle rotation
- $E_{\text{vdW}}$ : Lennard-Jones potential for non-bonded interactions
- $E_{\text{electrostatic}}$ : Coulombic interactions between partial charges

## (ii) No Electrons — Classical Treatment:

Electrons are not treated explicitly. Atomic partial charges are assigned empirically. This makes MM fundamentally incapable of describing bond breaking/forming or electronic effects (no quantum mechanics).

## (iii) Parameterized Force Fields:

All parameters ( $k$ ,  $r_0$ , partial charges, vdW radii) are derived from **experimental data** or high-level QM calculations. Common force fields:

- **AMBER**: Proteins and nucleic acids.
- **CHARMM**: Biomolecules, membrane systems.
- **OPLS**: Organic liquids and proteins.
- **UFF**: Universal, applies to all elements.

## (iv) Computationally Very Cheap:

Since electrons are ignored, MM scales as  $O(N^2)$  or even  $O(N)$  with cutoffs, where  $N$  is the number of atoms. Systems of **millions of atoms** can be handled in reasonable time.

## (v) Geometry Optimization and Conformational Analysis:

MM excels at finding energy-minimum geometries and mapping out conformational energy surfaces (e.g. protein folding, drug binding poses).

## (vi) Molecular Dynamics (MD) Compatible:

MM force fields are the foundation of **molecular dynamics simulations**, where Newton's equations of motion are integrated over time to study dynamic behavior.

## (vii) Transferability:

MM parameters are transferable across similar chemical environments — a C–H bond parameter in one molecule applies to C–H bonds in other molecules.

## Part 2: When to Choose Molecular Mechanics

(5 marks)

MM is the preferred computational method under the following conditions:

## (i) Very large systems:

When the system contains  $> 10^3$  atoms (proteins, DNA, polymers, lipid bilayers, nanoparticles), quantum mechanical methods are computationally prohibitive. MM is the **only practical choice** for systems with  $10^3$ – $10^7$  atoms.

## (ii) Conformational sampling and geometry optimization:

When the goal is to explore the conformational space of a molecule (e.g. find the lowest-energy conformation of a drug molecule or protein side chain), MM is fast enough to sample thousands of conformations.

## (iii) Molecular dynamics simulations:

For studying the dynamic behavior of biomolecules over nanosecond–microsecond timescales (protein folding, membrane permeation, drug unbinding), MM-based MD is the standard approach.

**(iv) Molecular docking and virtual screening:**

In drug discovery, MM is used to rapidly evaluate binding poses of millions of drug candidates against a protein target — impossible with QM methods.

**(v) No bond breaking or forming involved:**

When the chemical process does not involve bond formation/ breaking or electronic rearrangements, MM gives reliable results. Examples: protein–ligand binding, crystal packing, polymer chain dynamics.

**(vi) When speed is more important than accuracy:**

For initial screening, preliminary geometry estimates, or pre-processing before higher-level QM calculations, MM provides **fast and reasonably accurate results**.

**Summary — Choose MM when:**

| Scenario                 | Why MM is suitable              |
|--------------------------|---------------------------------|
| System > 1000 atoms      | QM too expensive                |
| Protein/DNA structure    | Force fields well-parameterized |
| MD simulation (ns–μs)    | Only feasible method            |
| Drug binding pose        | Fast scoring needed             |
| No bond breaking/forming | Classical treatment valid       |
| Conformational search    | Fast energy evaluation          |

**Conclusion:** Molecular mechanics treats atoms as classical balls-and-springs, computing total energy from parameterized force fields without explicit electrons. Its extreme computational efficiency makes it uniquely suited for large biological systems, conformational analysis, molecular dynamics, and drug discovery workflows where quantum mechanical accuracy is not required.

**Q2.** Draw a flow diagram illustrating the key stages of the modern drug discovery process. Explain how molecular docking is applied within this process. **(13 marks)** [CHEM 113 | 2023–24]

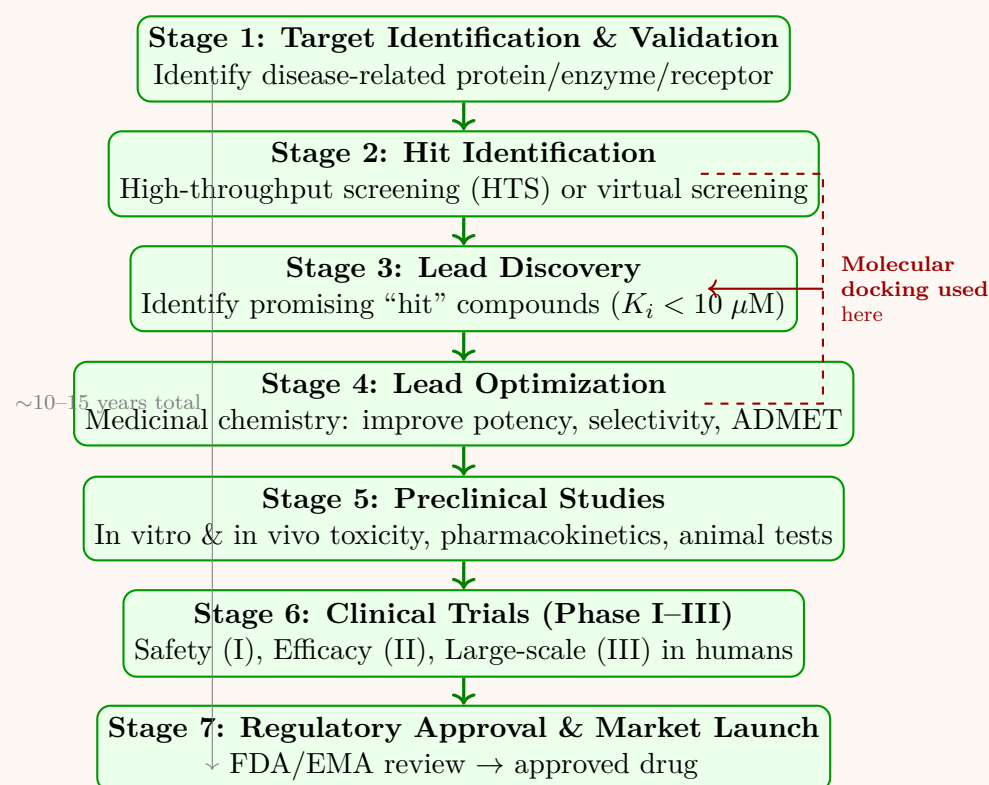
**Answer**

**Flow Diagram of Drug Discovery Process & Application of Molecular Docking**

**(13 marks)**

**Part 1: Flow Diagram of Modern Drug Discovery Process**

**(7 marks)**



**Part 2: How Molecular Docking is Applied in Drug Discovery**

**(6 marks)**

**(i) Definition:**

**Molecular docking** is a computational method that predicts the **preferred orientation and binding affinity** of a small molecule (ligand/drug candidate) within the **binding site (active site)** of a target **macromolecule** (receptor/enzyme/protein).



**(ii) Key components of molecular docking:**

- **Receptor structure:** 3-D structure from X-ray crystallography, NMR, or cryo-EM (from PDB — Protein Data Bank).
- **Ligand library:** Millions of candidate drug molecules (virtual compound library).
- **Search algorithm:** Explores all possible orientations, conformations, and positions of the ligand within the binding pocket. Examples: genetic algorithms (AutoDock), shape complementarity (Glide), Monte Carlo sampling.
- **Scoring function:** Estimates the binding affinity  $\Delta G_{\text{bind}}$  for each pose:

$$\Delta G_{\text{bind}} = \Delta G_{\text{vdW}} + \Delta G_{\text{H-bond}} + \Delta G_{\text{electrostatic}} + \Delta G_{\text{desolvation}}$$

Lower (more negative) score = stronger binding.

(iii) **Applications within the drug discovery pipeline:**

- **Stage 2 — Virtual Screening:** Instead of physically synthesizing and testing millions of compounds (HTS), docking screens a **virtual library** of  $10^5$ – $10^9$  compounds computationally in days, selecting the top  $\sim 100$ – $1000$  candidates for experimental testing. Reduces cost by  $\sim 90\%$ .
- **Stage 3 — Hit Identification:** Docking identifies the **binding pose** of hits — which functional groups interact with which residues of the active site (H-bonds, hydrophobic contacts, electrostatic interactions).
- **Stage 4 — Lead Optimization:** Structure-based drug design uses docking to guide medicinal chemistry modifications:
  - Add groups to fill empty pockets.
  - Remove groups causing steric clashes.
  - Introduce H-bond donors/acceptors to match receptor residues.
  - Predict how structural modifications change binding affinity before synthesis.
- **ADMET prediction:** Some docking protocols also consider binding to off-target proteins (selectivity) and metabolic enzymes (CYP450) to predict toxicity and drug interactions.

(iv) **Advantages of molecular docking:**

- Screens millions of compounds in days vs years experimentally.
- Provides atomic-level insight into binding mode.
- Reduces cost and animal testing.
- Guides rational drug design.

(v) **Limitations:**

- Scoring functions are approximate ( $\Delta G_{\text{pred}} \neq \Delta G_{\text{exp}}$ ).
- Protein flexibility not always treated adequately (“induced fit” issue).
- Cannot replace experimental validation.

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**Conclusion:** Modern drug discovery proceeds through target identification, hit finding, lead optimization, and clinical trials over  $\sim 10$ – $15$  years. Molecular docking is a central computational tool applied at the hit identification and lead optimization stages, screening millions of compounds by predicting protein–ligand binding poses and affinities, dramatically accelerating and reducing the cost of drug discovery.

**Q3.** Write down the salient features of Density Functional Theory. Discuss the advantages and disadvantages of the Molecular Mechanics method. **(12 marks)** [CHEM 113 | 2022–23]

**Answer**

**Salient Features of DFT; Advantages & Disadvantages of Molecular Mechanics**

**(12 marks)**

**Part 1: Salient Features of Density Functional Theory (DFT)**

**(6 marks)**

**Density Functional Theory** is a quantum mechanical method that replaces the many-electron wavefunction  $\Psi(r_1, r_2, \dots, r_N)$  with the simpler **electron density**  $\rho(r)$  as the fundamental variable.

**(i) Hohenberg–Kohn Theorems (1964) — Foundation:**

- **Theorem 1:** The ground-state energy of any electronic system is a **unique functional** of the electron density  $\rho(r)$ :  $E = E[\rho(r)]$ .



- **Theorem 2:** The true ground-state density  $\rho_0(r)$  minimizes the energy functional (variational principle).

**(ii) Kohn–Sham Equations (1965) — Practical DFT:**

The real interacting many-electron problem is mapped onto a fictitious system of non-interacting electrons moving in an effective potential:

$$\left[ -\frac{\hbar^2}{2m} \nabla^2 + V_{\text{ext}} + V_H[\rho] + V_{xc}[\rho] \right] \psi_i = \varepsilon_i \psi_i$$

where:

- $V_{\text{ext}}$ : nuclear–electron attraction
- $V_H$ : Hartree (Coulomb) potential
- $V_{xc}[\rho]$ : **exchange-correlation functional** — contains all many-body effects

**(iii) Key features:**

- **Includes electron correlation** (unlike Hartree–Fock) through the  $V_{xc}$  functional.
- **Computationally efficient:** Scales as  $O(N^3)$  vs  $O(N^5)$  for MP2 and  $O(N^7)$  for CCSD(T) — very favorable.
- **Accuracy:** For geometry, vibrational frequencies, and reaction energies, DFT (B3LYP, M06-2X,  $\omega$ B97X-D) gives near-chemical accuracy ( $\sim 1$ – $5$  kcal/mol error).
- **Wide applicability:** Molecules, solids, surfaces, nanoparticles, transition metals, excited states (TD-DFT).
- **Popular functionals:** B3LYP (most cited), PBE, M06-2X, CAM-B3LYP.
- **Limitation:** The exact  $V_{xc}$  is unknown; all functionals are approximations. DFT has difficulty with dispersion (van der Waals) — corrected by DFT-D3 or DFT-D4 corrections.
- **No systematic improvement:** Unlike wavefunction methods, there is no ladder of systematic improvement toward exact solution.

**Part 2: Advantages of Molecular Mechanics**

**(3 marks)**

- **Extremely fast:** Can handle  $10^6$ – $10^7$  atoms; MM runs are orders of magnitude faster than any QM method.
- **Large systems:** Only practical method for proteins, DNA, membranes, polymers at full atomistic resolution.
- **Molecular dynamics:** Enables ns– $\mu$ s simulations of biomolecular dynamics.
- **Geometry optimization:** Rapidly finds local energy minima; excellent for conformational search.
- **Simple implementation:** Easy to use; well-parameterized force fields available.
- **Reasonably accurate:** For geometries and conformational energies of well-parameterized systems, MM gives results within 1–5 kcal/mol.
- **Transferability:** Parameters work across similar chemical environments.

**Part 3: Disadvantages of Molecular Mechanics**

**(3 marks)**

- **No electrons:** Cannot describe bond breaking/ forming, chemical reactions, transition states, or electronic spectra.
- **Force field dependent:** Results depend heavily on the quality of the force field; poor parameters give poor results.
- **No electronic properties:** Cannot calculate charge distributions, polarizabilities, NMR chemical shifts, UV-vis spectra.
- **No quantum effects:** Tunneling, zero-point energy, and other quantum phenomena are ignored.
- **Parameterization required:** Force fields must be developed/validated for each new atom type; exotic elements or novel functional groups may lack parameters.
- **Classical approximation:** H-bonding and electrostatics are treated very approximately (fixed charges, no polarization in most force fields).
- **Limited accuracy for absolute energies:** Cannot reliably predict absolute binding free energies or reaction enthalpies without extensive calibration.

**Conclusion:** DFT is the dominant quantum mechanical method in computational chemistry, offering a balance of accuracy and efficiency through the electron density functional framework — including electron correlation at  $O(N^3)$  cost. Molecular mechanics is extremely fast and scalable to millions of atoms, making it ideal for biomolecular simulations, but cannot describe electronic properties or chemical reactions.

**Q4.** “The accuracy of a calculation is dependent on both the model and the type of basis set applied to it” — Explain. **(10 marks)**  
[CHEM 113 | 2021–22; 2015–16]

### Answer

#### Accuracy of Computation: Dependence on Model and Basis Set

**(10 marks)**

#### Part 1: The Two Pillars of Computational Accuracy

**(2 marks)**

In quantum chemistry, any calculation involves two independent choices that together determine the accuracy:

- **The model (level of theory):** The quantum mechanical method used to treat electron–electron interactions (HF, DFT, MP2, CCSD, CCSD(T)...).
- **The basis set:** The mathematical functions used to represent atomic orbitals and hence the molecular wavefunction.

Neither alone is sufficient — both must be appropriate for the problem at hand.

#### Part 2: Effect of the Model (Level of Theory)

**(4 marks)**

The **model** determines how electron–electron repulsion and correlation are treated:

**(i) Hartree–Fock (HF):** Each electron moves in the average field of all other electrons. **No electron correlation** is included. Good geometries but poor reaction energies and binding. Computationally  $O(N^4)$ .

**(ii) Density Functional Theory (DFT):** Includes electron correlation approximately through the exchange–correlation functional  $V_{xc}[\rho]$ . **Good balance of accuracy and cost** ( $O(N^3)$ ). B3LYP, M06-2X widely used.

**(iii) Post-HF Methods:**

- **MP2:** 2nd-order Møller–Plesset perturbation theory. Better than HF/DFT for weak interactions.  $O(N^5)$ .
- **CCSD:** Coupled cluster with singles and doubles. Very accurate.  $O(N^6)$ .
- **CCSD(T):** Gold standard of quantum chemistry.  $O(N^7)$ . Benchmark accuracy (<0.5 kcal/mol error).

**Hierarchy of accuracy:**

$$\text{HF} < \text{DFT} \approx \text{MP2} < \text{CCSD} < \text{CCSD(T)}$$

$$(\text{cost:}) \quad O(N^4) \ll O(N^3) \ll O(N^5) \ll O(N^6) \ll O(N^7)$$

#### Part 3: Effect of the Basis Set

**(4 marks)**

A **basis set** is a set of mathematical functions (Gaussian-type orbitals, GTOs) used to approximate atomic orbitals. The **more functions** used, the closer the approximation to the true wavefunction.

**(i) Minimal basis sets (STO-3G):** Only the minimum number of functions to describe each occupied orbital. Very fast but very inaccurate. Used only for qualitative trends or very large systems.

**(ii) Split-valence basis sets (3-21G, 6-31G):** Valence orbitals described by two sets of functions (inner and outer). More flexible than minimal. Common first choice for medium-sized molecules.

**(iii) Polarization functions (\* or p):** Add higher angular momentum functions (*d* on heavy atoms, *p* on H) to describe distortions in the electron density due to bonding. Essential for accurate geometries and reaction energies.

$$6\text{-}31\text{G}^* \equiv 6\text{-}31\text{G}(\text{d}) \quad (\text{adds d functions to heavy atoms})$$

**(iv) Diffuse functions (+ or aug):** Add functions with large radii to describe electrons far from the nucleus. **Essential for:** anions, excited states, hydrogen bonding, dispersion interactions.

$$6\text{-}31+\text{G}(\text{d},\text{p}), \quad \text{aug-cc-pVDZ}$$

**(v) Correlation-consistent basis sets (cc-pVXZ):** Dunning basis sets (cc-pVDZ, cc-pVTZ, cc-pVQZ) designed for use with correlated methods (MP2, CCSD). Allow **extrapolation to the complete basis set (CBS) limit**.

**Convergence with basis set size:**

$$\text{STO-3G} \ll 6\text{-}31\text{G} < 6\text{-}31\text{G}^* < 6\text{-}311+\text{G}(2\text{d},2\text{p}) < \text{cc-pVTZ} \rightarrow \text{CBS}$$

**Why both model AND basis set matter:**

If a high-level method (CCSD) is used with a small basis set (STO-3G), the result is poor because the wavefunction cannot be accurately represented. Conversely, a large basis set with HF still gives no electron correlation. Only the combination of an appropriate model **and** an appropriate basis set yields reliable results:

| Model     | Basis Set | Result                                    |
|-----------|-----------|-------------------------------------------|
| CCSD(T)   | STO-3G    | Poor (basis set limits accuracy)          |
| HF        | cc-pVQZ   | Poor (no correlation despite large basis) |
| DFT/B3LYP | 6-31G*    | Good for most organic chemistry           |
| CCSD(T)   | cc-pVTZ   | Excellent ( $\sim$ chemical accuracy)     |

**Conclusion:** Computational accuracy requires both the right model (level of electron correlation treatment) and the right basis set (quality of orbital representation). A high-level model with a small basis set, or a large basis set with a poor model, both give inaccurate results. The gold standard CCSD(T)/cc-pVTZ combines excellent electron correlation with a large basis set, approaching chemical accuracy ( $<1$  kcal/mol) — but at significant computational cost ( $O(N^7)$ ).

**Q5.** What kinds of questions can computational chemistry answer? How are the different types of models used in computational chemistry? What advantages does computational chemistry have over “wet chemistry”? (20 marks) [CHEM 113 | 2020–21 (Jan 21)]

### Answer

**Questions Computational Chemistry Answers; Models Used; Advantages over Wet Chemistry** (20 marks)

#### Part 1: Questions Computational Chemistry Can Answer (6 marks)

- (i) **Molecular geometry and structure:** What is the equilibrium geometry of a molecule? What are the bond lengths, bond angles, and dihedral angles? Example: Predicting the geometry of a transition-metal catalyst.
- (ii) **Electronic structure and properties:** What is the electron density distribution? What are the atomic charges, dipole moments, polarizabilities, and ionization energies? Where are the HOMO and LUMO located?
- (iii) **Energetics and thermochemistry:** What is the heat of formation, reaction enthalpy, or bond dissociation energy? Which reaction pathway is thermodynamically favorable?
- (iv) **Reaction mechanisms and transition states:** What is the transition state structure? What is the activation energy? Which mechanism (SN1, SN2, E1, E2) is preferred?
- (v) **Spectroscopic properties:** What are the IR/Raman frequencies, NMR chemical shifts, UV-vis absorption wavelengths, and CD spectra? These can be directly compared to experiment.
- (vi) **Intermolecular interactions:** How strongly does a drug bind to its receptor? What is the hydrogen bond network in a protein? How do solvation effects alter reaction rates?
- (vii) **Materials properties:** What is the band gap of a semiconductor? What is the bulk modulus of a crystal? How does a polymer chain behave under mechanical stress?
- (viii) **Drug design:** Which molecular modifications will improve binding affinity? Which compound from a library of  $10^6$  molecules is most likely to inhibit a given protein?

#### Part 2: Different Types of Models in Computational Chemistry (8 marks)

##### (i) Molecular Mechanics (MM):

- Classical, no electrons, force field-based.
- Systems:  $10^3$ – $10^7$  atoms.
- Uses: Conformational analysis, MD simulations, protein structure, drug docking.
- Cost: Very low ( $O(N^2)$  or lower).

##### (ii) Semi-empirical Methods (AM1, PM3, PM7):

- Quantum mechanical but with parameterized integrals (neglect of certain electron repulsion integrals).
- Systems:  $10^2$ – $10^3$  atoms.
- Uses: Organic molecule properties, drug-like molecules, quick geometry optimization.
- Cost: Low ( $O(N^3)$ ).

##### (iii) Ab Initio Methods (HF, MP2, CCSD):

- Pure quantum mechanics, no empirical parameters.
- Solve the Schrödinger equation from first principles.

- Systems:  $\leq 100$  atoms (HF/DFT),  $\leq 30$  atoms (CCSD(T)).
- Uses: Accurate thermochemistry, reaction mechanisms, spectroscopy.
- Cost:  $O(N^4)$ – $O(N^7)$  depending on method.

**(iv) Density Functional Theory (DFT):**

- Quantum mechanical, uses electron density  $\rho(r)$ .
- Includes correlation via exchange-correlation functional.
- Systems:  $\leq 500$  atoms (molecules), larger for periodic systems.
- Uses: Geometry, reaction energies, spectroscopy, materials, catalysis.
- Cost:  $O(N^3)$  — best accuracy/cost ratio.

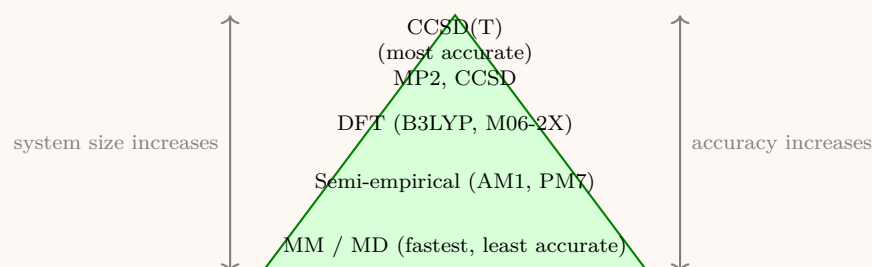
**(v) QM/MM (Hybrid Methods):**

- Treats the chemically active region (active site) with QM; the rest with MM.
- Combines QM accuracy with MM speed for biomolecular systems.
- Uses: Enzyme catalysis, drug binding, photoreactions in proteins.

**(vi) Molecular Dynamics (MD) and Monte Carlo (MC):**

- Statistical mechanical simulations using MM or QM energy functions.
- Sample conformational space; study dynamics over time.
- Uses: Protein folding, solvation, phase transitions.

**Accuracy vs cost pyramid:**



**Part 3: Advantages of Computational Chemistry over Wet Chemistry**

**(6 marks)**

- (i) Cost and speed:** Computational calculations cost a fraction of experimental synthesis and testing. Screening  $10^6$  virtual compounds computationally takes days; synthesizing them would take decades and billions of dollars.
- (ii) Safety:** Extremely toxic, explosive, radioactive, or unstable compounds can be studied computationally without any physical synthesis or exposure. No hazardous waste is generated.
- (iii) Access to inaccessible information:** Transition states, reaction intermediates, and excited states exist for femtoseconds to picoseconds — too short to characterize experimentally. Computation provides their full geometry, energy, and electronic structure.
- (iv) Systematic variation:** Substituent effects, isotope effects, solvent effects, and temperature dependence can be systematically studied by changing one variable at a time — difficult in the laboratory without confounding factors.
- (v) Complement to experiment:** When experimental data is ambiguous or conflicting, computational results can discriminate between proposed structures or mechanisms. Computational IR/NMR spectra confirm experimental assignments.
- (vi) Prediction before synthesis:** Properties of yet-to-be-synthesized compounds (new drugs, new materials, new catalysts) can be predicted, guiding synthetic effort toward the most promising candidates.
- (vii) No resource limitations:** Multiple reactions, solvents, temperatures, and pressures can be studied simultaneously in parallel on high-performance computers. No glassware, reagents, or laboratory time needed.
- (viii) Visualization:** 3-D molecular models, electron density maps, orbital visualizations, and potential energy surfaces provide intuitive insight into chemical bonding and reactivity that is not directly accessible experimentally.

| Criterion            | Computational       | Wet Chemistry           |
|----------------------|---------------------|-------------------------|
| Cost per compound    | Very low            | High                    |
| Speed of screening   | Days ( $10^6$ /day) | Slow                    |
| Safety               | No risk             | Hazardous materials     |
| Transition states    | Fully characterized | Not directly observable |
| Systematic variation | Trivial             | Difficult               |
| Waste generated      | None                | Chemical waste          |
| Physical realization | Not required        | Requires synthesis      |



**Conclusion:** Computational chemistry answers questions about molecular geometry, energetics, mechanisms, spectroscopy, and material properties. Methods range from classical MM (large systems, low accuracy) to CCSD(T) (small systems, highest accuracy), with DFT offering the best cost-accuracy balance. Advantages over wet chemistry include lower cost, safety, access to transient species, systematic variation, and the ability to predict properties before synthesis.

**Q6.** What computable properties are studied by computational chemistry? Discuss the methods most commonly used in computational chemistry in terms of theories and cost (time) effectiveness. (10 marks) [CHEM 113 | 2020–21]

### Answer

#### Computable Properties in Computational Chemistry; Methods, Theories and Cost Effectiveness (10 marks)

##### Part 1: Computable Properties (4 marks)

Computational chemistry can calculate a wide range of molecular properties:

- (i) **Structural properties:** Bond lengths, bond angles, dihedral angles, molecular volume, surface area, van der Waals radii.
- (ii) **Energetic properties:** Heat of formation ( $\Delta H_f$ ), reaction enthalpy ( $\Delta H_{\text{rxn}}$ ), bond dissociation energy (BDE), activation energy ( $E_a$ ), lattice energy, solvation energy.
- (iii) **Electronic properties:** Atomic partial charges, dipole moment, polarizability, ionization energy, electron affinity, HOMO/LUMO energies and shapes, electrostatic potential maps.
- (iv) **Spectroscopic properties:**
  - IR and Raman frequencies and intensities
  - NMR chemical shifts and coupling constants
  - UV-vis absorption wavelengths and oscillator strengths (TD-DFT)
  - CD (circular dichroism) spectra
  - EPR/ESR parameters
- (v) **Thermodynamic properties:** Entropy ( $S$ ), heat capacity ( $C_p$ ), Gibbs free energy ( $\Delta G$ ), equilibrium constants ( $K_{eq}$ ), partition functions.
- (vi) **Dynamic properties (from MD):** Diffusion coefficients, viscosity, thermal conductivity, protein conformational flexibility, membrane permeability.
- (vii) **Intermolecular properties:** Hydrogen bond strengths, van der Waals interactions,  $\pi$ - $\pi$  stacking energies, binding free energies, protein–ligand  $K_d$ .

##### Part 2: Methods — Theories and Cost Effectiveness (6 marks)

| Method                    | Theory basis          | Scaling  | Max size     | Best use            |
|---------------------------|-----------------------|----------|--------------|---------------------|
| MM (AMBER, CHARMM)        | Classical, Newton     | $O(N^2)$ | $10^7$ atoms | Proteins, MD        |
| Semi-empirical (PM7, AM1) | QM with fitted params | $O(N^3)$ | $10^3$ atoms | Organic molecules   |
| HF                        | QM, no correlation    | $O(N^4)$ | 300 atoms    | Baseline QM         |
| DFT (B3LYP, M06)          | QM + $V_{xc}[\rho]$   | $O(N^3)$ | 500 atoms    | Best cost/accuracy  |
| MP2                       | Perturbation theory   | $O(N^5)$ | 100 atoms    | Weak interactions   |
| CCSD                      | Coupled cluster       | $O(N^6)$ | 50 atoms     | Accurate properties |
| CCSD(T)                   | Gold standard         | $O(N^7)$ | 30 atoms     | Benchmark           |

**Cost effectiveness ranking:**

$$\underbrace{\text{MM}}_{\text{fastest}} \gg \text{Semi-emp.} > \text{DFT} > \text{HF} > \text{MP2} > \text{CCSD} > \underbrace{\text{CCSD(T)}}_{\text{most accurate}}$$

**Practical choice:**

- **Large biomolecules:** MM/MD
- **Drug-like molecules, routine properties:** DFT/B3LYP/6-31G\*

- **Accurate reaction energies:** DFT/M06-2X or MP2 with large basis
- **Benchmark quality:** CCSD(T)/cc-pVTZ

**Conclusion:** Computational chemistry computes structural, energetic, electronic, spectroscopic, thermodynamic, and dynamic properties. Methods range from classical MM (cheapest, largest systems) to CCSD(T) (most accurate, smallest systems), with DFT offering the best balance of accuracy and computational cost for most chemical applications.

Q7. What precautions should one follow in computational chemistry? (5 marks)

[CHEM 113 | 2020–21]

#### Answer

#### Precautions in Computational Chemistry

(5 marks)

##### (i) Choose an appropriate level of theory:

The method must be suited to the problem. Using MM for a reaction that involves bond breaking gives completely wrong results. Using HF (no correlation) for systems where electron correlation is critical (transition metals, radical reactions, dispersion-dominated complexes) gives poor energetics. Always ask: “Does this method describe the physics of my problem?”

##### (ii) Use an adequate basis set:

A basis set that is too small (STO-3G) introduces **basis set error** and gives unreliable results even with a high-level method. For charged species or anions, always include diffuse functions (+). For geometry, at minimum use 6-31G\*. Perform basis set convergence checks.

##### (iii) Verify geometry optimization convergence:

Always confirm that the geometry optimization has **truly converged** to a stationary point (forces  $< 10^{-4}$  a.u., energy change  $< 10^{-6}$  a.u.). Perform a **frequency calculation** to verify: all real frequencies = true minimum; one imaginary frequency = transition state.

##### (iv) Validate against experiment:

Never trust computational results without some experimental validation. Compare computed bond lengths, IR frequencies, or reaction energies to known experimental values for similar systems. If agreement is poor, the method or basis set needs revision.

##### (v) Beware of local vs global minima:

Geometry optimization finds the nearest **local** minimum, not necessarily the global minimum. For flexible molecules (drugs, polymers), **conformational searching** must be performed (MM or MD) before higher-level QM refinement to ensure the true lowest-energy structure is found.

##### (vi) Check for spin state and charge:

Always specify the correct **multiplicity** (spin state) and **charge**. For transition metal complexes, multiple spin states may be close in energy — all must be checked. Wrong charge or spin state gives completely meaningless results.

##### (vii) Use benchmark reference data:

For a new class of compounds, test several methods against **high-level reference data** (CCSD(T) or experimental) before using the method for predictions. Do not assume a method that works for organic molecules is valid for transition metals or excited states.

##### (viii) Solvation effects:

Gas-phase calculations may not represent solution-phase behavior. For reactions in polar solvents, always include solvation models (PCM, SMD, explicit solvent MD) to account for solvent stabilization of polar species and ions.

**Conclusion:** Key precautions in computational chemistry include choosing an appropriate method and basis set, verifying optimization convergence by frequency analysis, validating against experiment, performing global conformational searches, correctly specifying spin state and charge, and including solvation when relevant. Computational results are only as reliable as the assumptions built into them.

Q8. Discuss the importance of computational technology in drug delivery. (15 marks)

[CHEM 113 | 2019–20]

#### Answer

#### Importance of Computational Technology in Drug Delivery

(15 marks)

##### Part 1: Overview

(2 marks)

**Drug delivery** refers to formulating and administering a therapeutic agent to achieve the desired pharmacological effect at the right location, at the right time, and in the right amount. **Computational technology** plays an increasingly central role by:



- Predicting how drug molecules interact with biological barriers and carrier systems.
- Designing nanoparticle and liposomal delivery vehicles.
- Optimizing pharmacokinetics (ADMET) computationally.
- Simulating controlled release profiles.

## Part 2: Specific Applications

(10 marks)

### (i) Molecular Docking and Binding Site Prediction:

Computational docking predicts **where and how** a drug molecule binds to its target protein. This identifies:

- The binding pose (orientation in the pocket).
- Key interactions (H-bonds, hydrophobic contacts).
- Predicted binding affinity ( $K_d$ ,  $\Delta G_{\text{bind}}$ ).

Software: AutoDock Vina, Glide, GOLD.

### (ii) ADMET Prediction (Absorption, Distribution, Metabolism, Excretion, Toxicity):

Computational models predict whether a drug candidate will:

- Cross intestinal membranes (oral absorption) — predicted from logP, MW, H-bond count (Lipinski's Rule of Five).
- Cross the blood-brain barrier (CNS drugs).
- Be metabolized by liver enzymes (CYP450 binding predicted by docking).
- Cause cardiac toxicity (hERG channel binding).
- Be excreted renally or hepatically.

Tools: SwissADME, pkCSM, ADMETLab. This eliminates toxic or poorly bioavailable candidates **before synthesis**.

### (iii) Nanoparticle Design for Drug Delivery:

Computational MD simulations model:

- **Liposome formation:** How phospholipids self-assemble; drug encapsulation efficiency.
- **Polymer nanoparticles (PLGA):** How polymer chain length and composition affect drug loading and release kinetics.
- **Dendrimers:** How drug molecules bind to dendrimer branches; optimal generation for loading.
- **Targeted delivery:** How antibody or ligand functionalized nanoparticles bind to tumor receptors (e.g. folate receptor targeting in cancer).

### (iv) Blood–Brain Barrier (BBB) Permeation Modeling:

Computational models (QSAR, MD simulations with explicit BBB membrane models) predict which CNS drug candidates can cross the BBB — essential for designing drugs for neurological disorders (Alzheimer's, Parkinson's).

### (v) Controlled Release Modeling:

Mathematical/computational models (diffusion equations, coarse-grained MD) simulate:

- Drug release kinetics from hydrogels, microspheres, patches.
- pH-triggered, temperature-triggered, or enzyme-triggered release from smart delivery systems.
- Optimizing polymer matrix composition for desired release profile.

### (vi) Pharmacophore Modeling and Drug Design:

A **pharmacophore** is the set of structural features essential for biological activity. Computational pharmacophore modeling identifies the minimum required features (H-bond donors/acceptors, aromatic rings, hydrophobic groups) and screens databases for compounds matching the pharmacophore.

### (vii) Quantitative Structure–Activity Relationship (QSAR):

QSAR models mathematically relate molecular descriptors (logP, MW, surface area, electronic properties) to biological activity:

$$\text{Activity} = f(\log P, \text{MW}, \text{pK}_a, \text{TPSA}, \dots)$$

Machine learning QSAR models can predict activity, toxicity, and solubility for millions of untested compounds.

### (viii) Formulation Optimization:

Molecular simulations predict:

- Solubility enhancement by cyclodextrins or co-solvents.
- Stability in different excipient matrices.
- Optimal solid-state form (polymorph, salt, cocrystal) for maximum bioavailability.

**Part 3: Impact and Future Directions**
**(3 marks)**

| Impact Area           | Computational Contribution                                                        |
|-----------------------|-----------------------------------------------------------------------------------|
| Drug discovery cost   | Reduces cost from ~\$2.6B to lower through virtual screening and AD-MET filtering |
| Time to market        | Accelerates hit-to-lead optimization by 2–5 years                                 |
| Personalized medicine | Models patient-specific drug metabolism (genotype-based)                          |
| Nanomedicine          | Designs targeted nanocarriers at atomic detail                                    |
| COVID-19 example      | Remdesivir identified by docking into SARS-CoV-2 RNA polymerase within weeks      |

**AI/ML integration:** Deep learning models (AlphaFold2 for protein structure, generative AI for de novo drug design) are revolutionizing computational drug delivery by predicting protein structures and generating novel drug candidates with desired properties.

**Conclusion:** Computational technology is indispensable in modern drug delivery, enabling molecular docking for binding prediction, ADMET screening for safety, MD simulations for nanoparticle design, QSAR models for activity prediction, and controlled release modeling. These tools collectively reduce cost, accelerate discovery, improve safety prediction, and enable rational design of delivery systems at the molecular level.

**Q9.** How can computational chemistry be utilized to address real-world problems in chemistry? Briefly explain with specific examples. **(15 marks)** [CHEM 113 | 2019–20]

**Answer**
**Computational Chemistry for Real-World Problems: Specific Examples**
**(15 marks)**
**Part 1: Overview**
**(1 marks)**

Computational chemistry bridges the gap between fundamental molecular science and practical real-world challenges by providing **atomic-level insight** into chemical processes that are difficult or impossible to study experimentally. Key application areas include medicine, materials, energy, and environmental science.

**Part 2: Specific Real-World Applications**
**(14 marks)**
**(i) Drug Discovery and Design:**

**Problem:** Finding effective inhibitors for disease targets (HIV protease, cancer kinases, SARS-CoV-2 proteins).

**Computational approach:** Virtual screening (docking) of  $10^6+$  compounds against the target crystal structure, followed by MD simulations to confirm binding stability.

**Real example:** Ritonavir and indinavir (HIV protease inhibitors) were designed using structure-based computational docking. During COVID-19, docking studies identified remdesivir, nirmatrelvir, and other antivirals within weeks.

**(ii) Catalysis Design:**

**Problem:** Designing more efficient, selective, and sustainable catalysts for industrial chemical processes.

**Computational approach:** DFT calculations compute reaction energy profiles (reactants, transition states, products) for different catalyst candidates. The Sabatier principle (optimal binding energy) guides catalyst selection without synthesizing every possibility.

**Real example:** Computational screening of transition metal alloys for the hydrogen evolution reaction (HER) led to the discovery of NiMo and CoP as inexpensive alternatives to platinum for green hydrogen production.

**(iii) Materials Design (Solar Cells & Batteries):**

**Problem:** Designing new organic photovoltaic (OPV) materials and battery electrode materials.

**Computational approach:**

- TD-DFT calculates optical absorption spectra of new organic dyes for dye-sensitized solar cells.
- DFT + periodic boundary conditions computes band gaps of perovskite materials for photovoltaics.
- MD simulations model  $\text{Li}^+/\text{Na}^+$  diffusion in battery electrode materials ( $\text{LiFePO}_4$ , graphene).

**Real example:** High-throughput DFT screening of  $10^4$  perovskite compositions ( $\text{ABX}_3$ ) identified  $\text{CsSnI}_3$  as a lead-free perovskite solar cell material.

**(iv) Environmental Chemistry:**

**Problem:** Understanding atmospheric chemistry, ozone depletion, and pollutant degradation.

**Computational approach:** QM calculations compute reaction rate constants and mechanisms for atmospheric reactions ( $\text{OH}$  radical + VOCs, CFCs + ozone). MD simulations model adsorption of pollutants onto soil and water surfaces.

**Real example:** The mechanism of stratospheric ozone depletion by CFCs was elucidated computationally (QM calculations of  $\text{Cl} + \text{O}_3$  reaction pathways) before experimental verification — contributing to the Montreal Protocol.

**(v) Protein Structure Prediction:**

**Problem:** Determining protein 3-D structure from amino acid sequence (crucial for drug design).

**Computational approach:** AlphaFold2 (deep learning + evolutionary data) predicts protein structures with near-experimental accuracy without X-ray crystallography or cryo-EM.

**Real example:** AlphaFold2 (DeepMind, 2021) predicted structures for >200 million proteins, deposited in the AlphaFold Protein Structure Database — revolutionizing structural biology and drug discovery.

**(vi) Polymer and Biomaterial Design:**

**Problem:** Designing biodegradable polymers for medical implants and drug delivery.

**Computational approach:** MD simulations model polymer chain flexibility, crystallinity, and degradation kinetics. DFT computes hydrolysis activation energies for polyester bonds (PLA, PLGA) at different pH.

**Real example:** Computational design of PLGA copolymer compositions optimized degradation rate for sustained drug release over 30 days in bone regeneration scaffolds.

**(vii) Semiconductor and Nanoelectronics:**

**Problem:** Designing new semiconductor materials for next-generation electronics.

**Computational approach:** DFT + non-equilibrium Green's function (NEGF) calculates electron transport through molecular junctions and graphene nanoribbons.

**Real example:** Computational prediction of the band gap of bilayer graphene under an electric field guided the development of graphene-based transistors.

| Field             | Method used       | Real example          |
|-------------------|-------------------|-----------------------|
| Drug design       | Docking, MD       | Ritonavir, Remdesivir |
| Catalysis         | DFT               | NiMo for HER          |
| Solar cells       | TD-DFT            | Perovskite screening  |
| Environment       | QM kinetics       | CFC ozone mechanism   |
| Protein structure | AI/ML (AlphaFold) | 200M structures       |
| Biomaterials      | MD, DFT           | PLGA drug release     |
| Electronics       | DFT + NEGF        | Graphene transistors  |

**Conclusion:** Computational chemistry addresses real-world problems across drug discovery (docking, virtual screening), catalysis design (DFT energy profiles), renewable energy (solar/battery materials), environmental chemistry (atmospheric reaction mechanisms), protein structure (AlphaFold2), biomaterials (MD degradation modeling), and nanoelectronics (DFT transport). These examples demonstrate that computational chemistry has become an indispensable partner to experiment in virtually every branch of applied chemistry.

**Q10.** What do you mean by basis sets in computational chemistry? “The accuracy of a calculation is dependent on both the model and the type of basis set applied to it” — Explain. (6 marks) [CHEM 113 | 2015–16]

**Answer**

**Basis Sets in Computational Chemistry; Accuracy Dependence on Model and Basis Set**

(6 marks)

**Part 1: What Are Basis Sets?**

(3 marks)

A **basis set** is a collection of mathematical functions (usually **Gaussian-type orbitals, GTOs**) used to approximate the molecular wavefunction. Each atomic orbital in a molecule is expressed as a **linear combination** of these basis functions:

$$\phi_{\text{MO}} = \sum_i c_i \chi_i$$

where  $\chi_i$  are the basis functions and  $c_i$  are coefficients determined by the variational principle.

**Types of basis sets:**

- **Minimal (STO-3G):** One function per orbital. Fastest but least accurate.
- **Split-valence (3-21G, 6-31G):** Two sets of functions for valence orbitals. Inner electrons contracted, outer electrons flexible.
- **Polarized (6-31G\*, 6-31G\*\*):** Adds *d*-functions on heavy atoms (\*) and *p*-functions on H (\*\*). Essential for geometry and reaction energies.
- **Diffuse (6-31+G, aug-cc-pVDZ):** Adds large-radius functions. Essential for anions, excited states, H-bonding.
- **Correlation-consistent (cc-pVDZ/TZ/QZ):** Systematically improvable; designed for correlated methods; allow CBS extrapolation.

**Part 2: Accuracy Depends on Both Model and Basis Set****(3 marks)**

**The model** determines how electron correlation is treated (HF, DFT, MP2, CCSD...). **The basis set** determines how accurately the orbitals are represented.

Both must be appropriate:

- **Good model + small basis:** High-level electron correlation cannot be expressed in a limited basis  $\Rightarrow$  result is poor. Example: CCSD(T)/STO-3G gives worse bond lengths than DFT/cc-pVTZ.
- **Poor model + large basis:** Even an exact basis cannot compensate for missing electron correlation in HF. Example: HF/cc-pVQZ still misses dispersion and correlation energy.
- **Matched model + basis:** DFT/6-31G\* for organic molecules, CCSD(T)/cc-pVTZ for benchmark thermochemistry — both are well-matched and give reliable results.

$$\text{Error}_{\text{total}} = \underbrace{\text{Error}_{\text{model}}}_{\text{incomplete correlation}} + \underbrace{\text{Error}_{\text{basis}}}_{\text{incomplete orbital expansion}}$$

Both contributions must be minimized for accurate results. The concept of the **complete basis set (CBS) limit** is the energy obtained with a perfect (infinite) basis set for a given model — only CCSD(T)/CBS represents true chemical accuracy.

**Conclusion:** Basis sets are mathematical functions representing atomic orbitals, ranging from minimal (STO-3G) to large correlation-consistent (cc-pVQZ) sets. Accuracy requires both a suitable model (electron correlation treatment) and an adequate basis set (orbital representation) — neither alone is sufficient, and the total error is the sum of model error and basis set error.

**Q11.** Discuss the advantages and disadvantages of the Semi-empirical method of computation. **(5 marks)**

[CHEM 113 | 2015–16]

**Answer****Advantages and Disadvantages of Semi-empirical Methods****(5 marks)**

**Semi-empirical methods** (AM1, PM3, PM6, PM7, MNDO, INDO) are quantum mechanical methods that **approximate** certain computationally expensive integrals in the HF equations using **empirical parameters** fitted to experimental data, dramatically reducing computational cost.

**Advantages:**

- **Much faster than ab initio/DFT:** Computationally  $O(N^3)$  with small prefactor; can handle 500–2000 atoms in minutes. Drug-sized molecules ( $\sim 50$ –200 atoms) in seconds.
- **Quantum mechanical treatment:** Unlike MM, semi-empirical methods explicitly include electrons and can describe bond breaking/forming, electronic excitations, and charge distributions.
- **Geometry optimization:** Gives reasonable molecular geometries for organic molecules. PM7 gives bond lengths within  $\sim 0.02$  Å of experiment for organic compounds.
- **Heat of formation:** AM1 and PM3 are parameterized to reproduce heats of formation of organic compounds, giving useful thermochemical data.
- **Initial estimate:** Excellent for generating starting geometries before DFT optimization, or for quick screening of large numbers of molecules.
- **Large biomolecule QM treatment:** The linear-scaling MOZYME (PM7) can handle  $>10,000$  atoms quantum mechanically.

**Disadvantages:**

- **Parameter dependent:** Accuracy depends entirely on the quality of empirical parameters; poorly parameterized atom types give unreliable results.
- **Limited element coverage:** Parameters available for common elements (C, H, N, O, S, P, halogens) but many transition metals, lanthanides, and uncommon elements are poorly parameterized.
- **Lower accuracy than DFT for:** Transition states, hydrogen bonds, aromatic systems, dispersion interactions, and excited states.
- **No systematic improvement:** Unlike ab initio methods, there is no clear path to improve semi-empirical results other than re-parameterization.
- **Fails for unusual bonding:** Hypervalent species, highly strained molecules, and metal-organic compounds often give poor results.



**Conclusion:** Semi-empirical methods offer a valuable middle ground — faster than DFT/ab initio (handling 500–2000 atoms) yet quantum mechanical in nature, allowing description of bond breaking and electronic properties. Their main limitations are parameter dependence, limited element coverage, and lower accuracy than DFT for sensitive properties.

**Q12.** Write down the salient features of the Ab-initio method. When would you choose this method for computation? (7 marks) [CHEM 113 | 2015–16]

### Answer

#### Salient Features of Ab Initio Method; When to Choose It

(7 marks)

#### Part 1: Salient Features of Ab Initio Methods

(4 marks)

**Ab initio** (Latin: “from the beginning”) methods solve the Schrödinger equation using only **fundamental physical constants** (electron mass, charge, Planck’s constant, speed of light) and **no empirical parameters**:

$$\hat{H}\Psi = E\Psi$$

**(i) No empirical parameters:**

All integrals are computed exactly (to numerical precision) from quantum mechanics. No fitting to experimental data. Results are **entirely from first principles**.

**(ii) Systematic improvability:**

There is a clear hierarchy of methods that converges toward the exact solution:



Adding more electron correlation and larger basis sets **systematically improves** accuracy toward the complete basis set (CBS) full configuration interaction (FCI) limit.

**(iii) Hartree–Fock (HF) — simplest ab initio:**

Each electron moves in the mean field of all others. No electron correlation. Computationally  $O(N^4)$ . Foundation for all higher ab initio methods.

**(iv) Post-HF methods — including electron correlation:**

- **MP2:** 2nd-order perturbation theory. Captures ~80–90% of correlation energy.  $O(N^5)$ . Good for weak interactions and geometries.
- **CCSD:** Coupled cluster with singles and doubles. Very accurate.  $O(N^6)$ .
- **CCSD(T):** CCSD with perturbative triples. **Gold standard.** <1 kcal/mol error for most properties.  $O(N^7)$ .
- **CASSCF/CASPT2:** For multi-reference systems (bond breaking, excited states, biradicals).

**(v) Known and controllable errors:**

Errors arise from: (a) incomplete basis set (basis set error) and (b) incomplete correlation treatment (truncation error). Both can be systematically reduced by using larger basis sets and higher-level methods. Error analysis is transparent and well-understood.

**(vi) Benchmark quality:**

CCSD(T)/CBS results are used as **benchmarks** against which DFT functionals, semi-empirical methods, and force fields are validated.

#### Part 2: When to Choose Ab Initio Methods

(3 marks)

**(i) When high accuracy is essential:**

Thermochemical data (heats of formation, bond dissociation energies, reaction barriers) to within  $\pm 1$  kcal/mol require CCSD(T)/cc-pVTZ or higher. DFT cannot reliably achieve this.

**(ii) When validating DFT functionals:**

Before using a new DFT functional for a class of compounds, CCSD(T) calculations on representative small molecules establish whether the functional is reliable.

**(iii) For weak non-covalent interactions:**

Dispersion-dominated complexes ( $\pi$ – $\pi$  stacking, van der Waals complexes), hydrogen bonded systems, and halogen bonds require MP2 or CCSD(T) for accurate interaction energies (DFT without dispersion correction fails).

**(iv) For multi-reference systems:**

When a single determinant is insufficient (bond breaking, transition metal excited states, biradicals), CASSCF/CASPT2 or MRCI ab initio methods are required.

**(v) For spectroscopic accuracy:**

Rotational constants, vibrational frequencies, and NMR chemical shifts that must match experimental spectra to high precision require CCSD(T) or MP2 quality calculations.

**(vi) For small molecules:**

When the system has <30–50 atoms and high accuracy is needed (atmospheric reaction mechanisms, interstellar chemistry, astrochemistry), CCSD(T) is the method of choice.

**Summary:**

| Scenario                   | Method          | Cost        |
|----------------------------|-----------------|-------------|
| Quick geometry, <100 atoms | HF/6-31G*       | $O(N^4)$    |
| Routine energetics         | MP2/cc-pVDZ     | $O(N^5)$    |
| Accurate thermochemistry   | CCSD(T)/cc-pVTZ | $O(N^7)$    |
| Multi-reference problems   | CASSCF/CASPT2   | High        |
| Exact (benchmark)          | FCI/CBS         | Impractical |

**Conclusion:** Ab initio methods solve the Schrödinger equation from first principles with no empirical parameters, offering systematic improvability toward exact results. CCSD(T) is the gold standard for small molecules (<30 atoms), providing benchmark accuracy (<1 kcal/mol). Choose ab initio methods when high accuracy is essential, when validating other methods, for weak interactions, multi-reference systems, and high-precision spectroscopy.



BUET · CHEM 113

# Chemistry

Section S13

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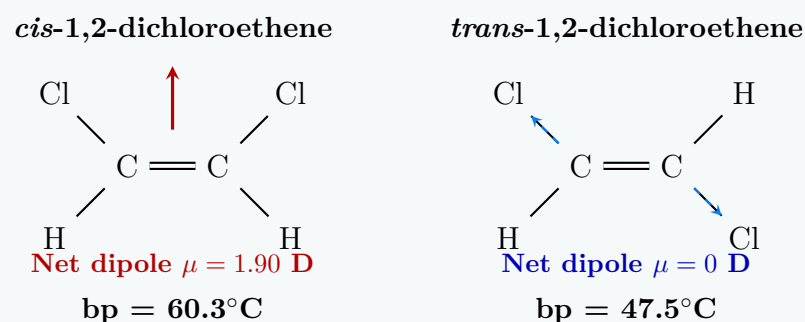
Miscellaneous

## S13 — Miscellaneous

Q1. Explain why *cis*-1,2-dichloroethene has a higher boiling point compared to *trans*-1,2-dichloroethene. (5 marks) [CHEM 113 | 2023–24]

## Answer

## Cis vs Trans-1,2-Dichloroethene — Boiling Points:



## Explanation:

Both isomers have the same molecular formula ( $\text{C}_2\text{H}_2\text{Cl}_2$ ) and molecular mass (96.94 g/mol), so London dispersion forces are essentially equal. The difference in boiling point arises from **dipole–dipole interactions**:

- In *cis*-1,2-dichloroethene, both Cl atoms are on the **same side** of the double bond. The two C–Cl bond dipoles point in the **same general direction** and **reinforce** each other, giving a significant **net molecular dipole moment** ( $\mu = 1.90 \text{ D}$ ).
- In *trans*-1,2-dichloroethene, the Cl atoms are on **opposite sides**. The two C–Cl bond dipoles are **equal and opposite**, cancelling completely to give a **zero net dipole moment** ( $\mu = 0 \text{ D}$ ).

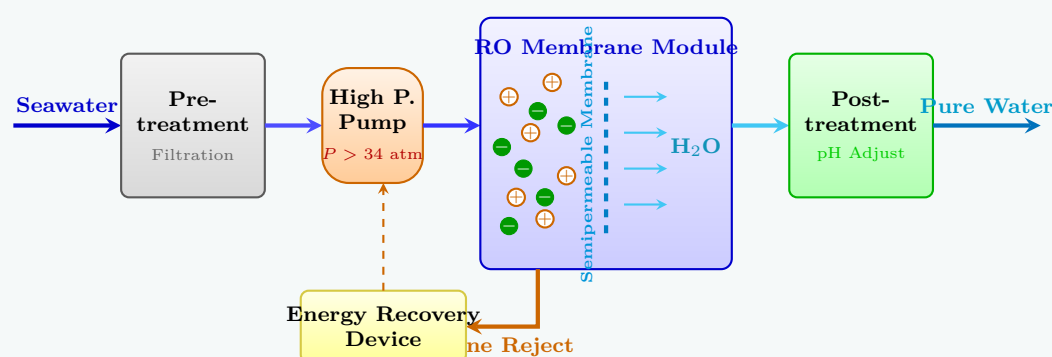
Because the *cis* isomer is **polar**, molecules attract each other via **dipole–dipole forces** in addition to London dispersion forces. More energy is required to overcome these additional intermolecular attractions, resulting in a **higher boiling point** (60.3°C vs 47.5°C for *trans*).

**Conclusion:** The *cis* isomer has a higher boiling point because its non-zero dipole moment creates additional dipole–dipole intermolecular attractions absent in the non-polar *trans* isomer.

Q2. Draw a schematic diagram of the water desalination process and explain. (10 marks) [CHEM 113 | 2020–21 (Jan 21)]

## Answer

## Water Desalination by Reverse Osmosis:



## Explanation of Each Stage:

- Pre-treatment:** Seawater passes through coarse filters (sand, activated carbon) to remove suspended solids, algae, and bacteria. Antiscalants are added to prevent membrane fouling by  $\text{CaCO}_3$  and  $\text{MgSO}_4$  deposits.
- High-pressure pump:** Pressure of 50–80 atm is applied — well above the osmotic pressure of seawater ( $\sim 27 \text{ atm}$ ). This forces water molecules through the membrane **against** the natural osmotic gradient (**reverse osmosis**).
- RO Membrane Module:** A semipermeable membrane (thin-film composite polyamide) allows only water molecules to pass. Salt ions ( $\text{Na}^+$ ,  $\text{Cl}^-$ ,  $\text{Mg}^{2+}$ ,  $\text{SO}_4^{2-}$ ) and larger molecules are **rejected** (> 99% rejection rate).

$$\pi_{\min} = iMRT = 2 \times 1.13 \times 0.0821 \times 298 = 55.3 \text{ atm}$$

- Brine disposal:** Concentrated salt reject (brine,  $\sim 2\times$  seawater salinity) is discharged carefully to minimise environmental impact. An **energy recovery device** captures pressure energy from the brine to reduce overall energy consumption.
- Post-treatment:** Permeate water is slightly acidic and mineral-deficient. pH adjustment (lime addition), remineralisation ( $\text{CaCO}_3$  bed), and final disinfection (UV/chlorination) produce safe drinking water.

**Conclusion:** Reverse osmosis desalination applies pressure exceeding osmotic pressure to force seawater through a semipermeable membrane, producing pure water with > 99% salt removal. Modern RO plants achieve specific energy consumption of 2–4 kWh/m<sup>3</sup> with energy recovery systems.

**Q3.** Sea water contains dissolved salts at a total concentration of about 1.13 mol/L. Draw a desalination diagram for obtaining drinking water from sea water. (6 marks) [CHEM 113 | 2016–17]

**Q4.** What is soap? Write down the mechanism of saponification of triacylglycerols. (7 marks) [CHEM 113 | 2016–17]

### Answer

#### What is Soap?

A **soap** is the sodium or potassium salt of a **long-chain fatty acid**, produced by alkaline hydrolysis (saponification) of fats or oils. General structure:



where R is a long hydrocarbon chain ( $\text{C}_{11}$ – $\text{C}_{17}$ ). The molecule has a **hydrophilic head** ( $-\text{COO}^-\text{Na}^+$ ) and a **hydrophobic tail** (alkyl chain).

#### Mechanism of Saponification of Triacylglycerols:

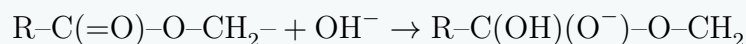
A **triacylglycerol** (triglyceride) consists of glycerol esterified with three fatty acids.

#### Overall reaction:



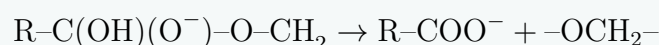
#### Step-by-step mechanism:

(a) **Nucleophilic attack:** Hydroxide ion ( $\text{OH}^-$ ) acts as a nucleophile and attacks the electrophilic carbonyl carbon of one ester linkage:



(b) **Tetrahedral intermediate:** A tetrahedral alkoxide intermediate forms at the carbonyl carbon.

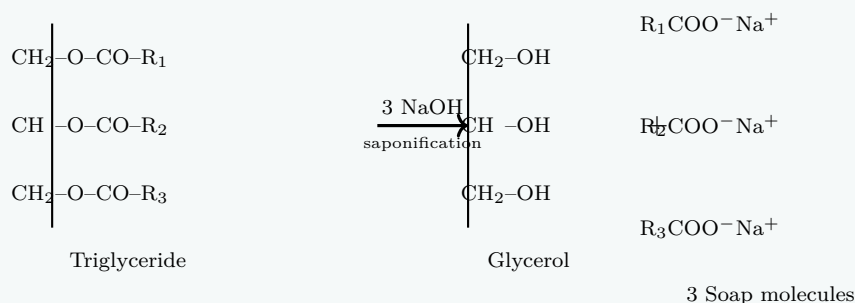
(c) **Elimination of alkoxide:** The C–O bond to glycerol breaks — the glyceroxide ( $-\text{O-CH}_2-$ ) leaves as an alkoxide:



(d) **Proton transfer:** The carboxylate  $\text{R-COO}^-$  is stabilised by resonance. The alkoxide picks up a proton from solvent to regenerate  $-\text{OH}$  on glycerol.

(e) **Repeat  $\times 3$ :** Steps 1–4 repeat for the second and third ester linkages, releasing:

- 3 soap molecules ( $\text{R}_1\text{COO}^-\text{Na}^+$ ,  $\text{R}_2\text{COO}^-\text{Na}^+$ ,  $\text{R}_3\text{COO}^-\text{Na}^+$ )
- 1 glycerol molecule



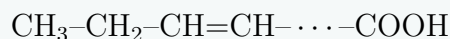
**Conclusion:** Saponification is a base-catalysed nucleophilic acyl substitution reaction.  $\text{OH}^-$  attacks each ester carbonyl, forming a tetrahedral intermediate, then collapsing to release carboxylate soap and glycerol.

**Q5.** What is omega-3 fatty acid? Why is consumption of omega-3 fatty acid containing oils good for our health? (7 marks) [CHEM 113 | 2016–17]

### Answer

#### Omega-3 Fatty Acids:

**Omega-3 fatty acids** are **polyunsaturated fatty acids** (PUFAs) containing multiple  $\text{C}=\text{C}$  double bonds, with the first double bond located at the **third carbon** from the methyl ( $\omega$ ) end of the chain:



The three most important omega-3 fatty acids are:

- **ALA** (alpha-linolenic acid, 18:3 $\omega$ 3): found in flaxseed, chia, walnuts
- **EPA** (eicosapentaenoic acid, 20:5 $\omega$ 3): found in oily fish (salmon, mackerel, sardines)
- **DHA** (docosahexaenoic acid, 22:6 $\omega$ 3): found in oily fish, algae

All double bonds are **cis** configuration, causing the chain to **bend**, preventing tight packing and keeping membranes fluid.

#### Why Omega-3 Consumption is Beneficial:

(a) **Cardiovascular health:**

- EPA and DHA reduce **triglyceride levels** in blood by 20–50%
  - Decrease platelet aggregation, reducing risk of **blood clots and stroke**
  - Lower blood pressure by increasing nitric oxide (NO) production, relaxing blood vessels
  - Raise HDL (“good”) cholesterol
- (b) **Anti-inflammatory effects:** Omega-3s are precursors to **eicosanoids** (prostaglandins, thromboxanes, leukotrienes) that are **less inflammatory** than those derived from omega-6 fatty acids. They reduce production of inflammatory cytokines (IL-6, TNF- $\alpha$ ), benefiting arthritis and inflammatory bowel disease.
- (c) **Brain and neurological function:** DHA is a major structural component of brain cell membranes and the retina (~40% of brain PUFAs). Adequate DHA supports:
- Cognitive development in infants
  - Memory and learning
  - Reduced risk of Alzheimer’s disease
- (d) **Membrane fluidity:** The kinked cis double bonds prevent tight packing of membrane phospholipids, maintaining **optimal membrane fluidity** for cell signalling, receptor function, and ion channel activity.
- (e) **Mental health:** EPA and DHA are associated with reduced rates of depression, anxiety, and bipolar disorder.
- (f) **Cancer prevention:** Some evidence suggests omega-3s reduce risk of breast, colon, and prostate cancer by reducing chronic inflammation and inhibiting tumour cell growth.

**Conclusion:** Omega-3 fatty acids are essential PUFAs with the first double bond at the  $\omega$ -3 position. They provide significant health benefits through anti-inflammatory eicosanoid production, cardiovascular protection, brain development support, and membrane fluidity maintenance.

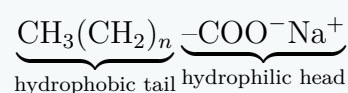
Q6. What is soap? Show the cleaning mechanism of soaps. (8 marks)

[CHEM 113 | 2015–16]

### Answer

#### What is Soap?

A **soap** is the sodium or potassium salt of a long-chain fatty acid ( $C_{12}$ – $C_{18}$ ):

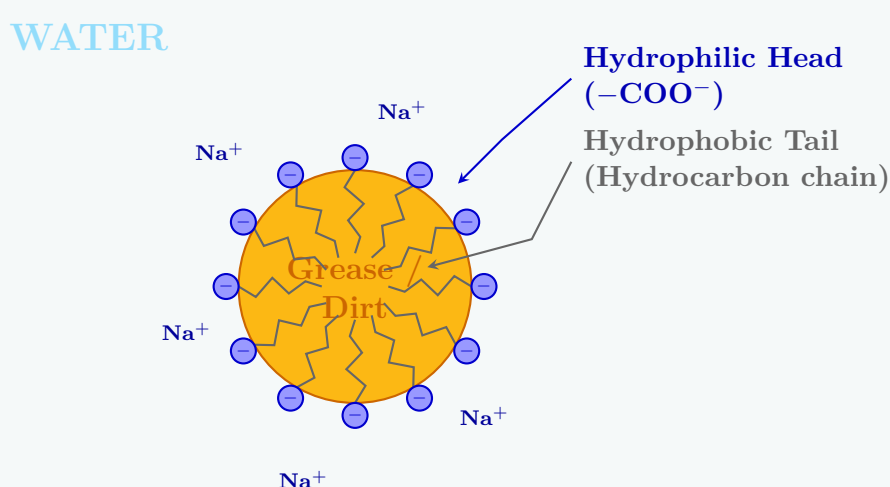


It is an **amphiphilic** molecule — having both a water-loving (ionic head) and water-fearing (hydrocarbon tail) part.

#### Cleaning Mechanism of Soap:

##### Step 1 & 2 — Orientation and Micelle formation:

Soap molecules orient at the surface of a grease droplet. The hydrophobic tails dissolve in the grease, while the hydrophilic heads remain in the surrounding water, forming a spherical structure called a **micelle**:



##### Step 3 — Emulsification:

The negatively charged outer surface of micelles causes them to **repel each other**, preventing coalescence and keeping the grease stably **emulsified** (dispersed) in water.

##### Step 4 — Rinsing:

The grease-filled micelles are carried away with the wash water, removing the dirt.

#### Summary of Cleaning Action:

| Step | Process                                                |
|------|--------------------------------------------------------|
| 1    | Soap dissolves in water and orients at grease surface. |
| 2    | Tails penetrate grease; heads remain in water.         |
| 3    | Micelle forms enclosing grease in its interior.        |
| 4    | Negative micelle surface prevents re-deposition.       |
| 5    | Grease-filled micelles rinsed away with water.         |

**Conclusion:** Soap cleans by forming micelles that **emulsify** grease — the hydrophobic tails dissolve in grease while the hydrophilic heads maintain water solubility, lifting grease off surfaces into stable dispersed micelles that are rinsed away.

**Q7.** List four factors that can shift the position of an equilibrium. Which one can alter the value of the equilibrium constant? Why? (8 marks) [CHEM 113 | 2014–15]

**Q8.** The pH of a bicarbonate-carbonic acid buffer is 8.00. Calculate the ratio of the concentration of carbonic acid ( $\text{H}_2\text{CO}_3$ ) to that of the bicarbonate ion ( $\text{HCO}_3^-$ ).  $K_{a1} = 4.2 \times 10^{-7}$ . (5 marks) [CHEM 113 | 2014–15]

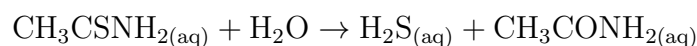
**Q9.** Define reaction quotient. How does it differ from equilibrium constant? What are the uses of reaction quotient? (8 marks) [CHEM 113 | 2014–15]

**Q10.** The equilibrium constant  $K_c$  of the reaction  $\text{N}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{NO}(\text{g})$  equals  $4.6 \times 10^{-31}$  at  $25^\circ\text{C}$ . (i) What does the magnitude of  $K_c$  tell you about the reaction? (ii) If you assume that the concentrations of  $\text{N}_2$  and  $\text{O}_2$  are 1.0 M, find the concentration of NO. (9 marks) [CHEM 113 | 2014–15]

**Q11.** At  $128^\circ\text{C}$  the equilibrium constant ( $K_c$ ) for the reaction  $\text{Br}_2(\text{g}) \rightleftharpoons 2\text{Br}(\text{g})$  is  $1.1 \times 10^{-3}$ . If the initial concentrations are  $[\text{Br}_2] = 6.3 \times 10^{-2}$  M and  $[\text{Br}] = 1.2 \times 10^{-2}$  M, calculate the concentrations of these species at equilibrium. (10 marks) [CHEM 113 | 2014–15]

**Q12.** What are the four variables or factors that can affect the rate of reaction? Which of the factor(s) affect the magnitude of the rate constant? Which factor(s) do not affect the magnitude of the rate constant? Why? (8 marks) [CHEM 113 | 2014–15]

**Q13.** The reaction of thioacetamide with water is shown by the equation below:



The rate of reaction is given by the rate law:  $\text{Rate} = k[\text{H}_3\text{O}^+][\text{CH}_3\text{CSNH}_2]$ . Consider 1 L of solution that is 0.20 M in  $\text{CH}_3\text{CSNH}_2$  and 0.15 M in  $\text{HCl}$  at  $25^\circ\text{C}$ . (i) For each of the changes listed below, state whether the rate of reaction increases, decreases, or remains the same. Why? (A) A 4-g sample of  $\text{NaOH}$  is added to the solution. (B) 500 mL of water is added to the solution. (ii) For each of the changes listed below, state whether the magnitude of the rate constant  $k$  will increase, decrease or remain the same. (A) A catalyst is added to the solution. (B) The reaction is carried out at  $10^\circ\text{C}$ . (8 marks) [CHEM 113 | 2014–15]

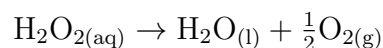
**Q14.** The following values of the rate constant were obtained for the decomposition of nitrogen dioxide at various temperatures:

| Temperature ( $^\circ\text{C}$ ) | $k$ (L/mol·s) |
|----------------------------------|---------------|
| 320                              | 0.527         |
| 330                              | 0.776         |
| 340                              | 1.121         |
| 350                              | 1.607         |

Using the Arrhenius equation, determine the activation energy. (10 marks)

[CHEM 113 | 2014–15]

**Q15.** The decomposition of hydrogen peroxide is a first-order reaction:



The half life of the reaction is 17.0 minutes. (i) What is the rate constant of the reaction? (ii) If you had a bottle of  $\text{H}_2\text{O}_2$ , how long would it take for 86.0% to decompose? (iii) If you started the reaction with  $[\text{H}_2\text{O}_2] = 0.100$  M, what would be the  $\text{H}_2\text{O}_2$  concentration after 15.0 minutes? (9 marks) [CHEM 113 | 2014–15]

**Q16.** Define (i) rate of reaction, (ii) rate of conversion, and (iii) rate constant. (8–9 marks)

[CHEM 113 | 2013–14; 2010–11]

**Q17.** Define Hard and Soft Acids and Bases (HSAB). Discuss the classification of HSAB taking three examples from each class. (10 marks) [CHEM 113 | 2013–14]

### Answer

**Hard and Soft Acids and Bases (HSAB) — Pearson's Concept (1963):**

**Hard species:** small, high charge density, low polarisability. They hold their electrons tightly and interact via **ionic/electrostatic**

forces.

**Soft species:** large, low charge density, high polarisability. Electrons are loosely held; interactions are **covalent/polarisation-based**.

**Classification:**

| Class              | Acids (Lewis)                                                                                                                                                      | Bases (Lewis)                                                                                                                                                                  |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Hard</b>        | H <sup>+</sup> , Li <sup>+</sup> , Na <sup>+</sup> , K <sup>+</sup> , Mg <sup>2+</sup> , Ca <sup>2+</sup> , Al <sup>3+</sup> , Cr <sup>3+</sup> , Fe <sup>3+</sup> | F <sup>−</sup> , OH <sup>−</sup> , H <sub>2</sub> O, NH <sub>3</sub> , ROH, RNH <sub>2</sub> , CO <sub>3</sub> <sup>2−</sup> , SO <sub>4</sub> <sup>2−</sup> , Cl <sup>−</sup> |
| <b>Border-line</b> | AlCl <sub>3</sub> , Fe <sup>2+</sup> , Co <sup>2+</sup> , Ni <sup>2+</sup> , Cu <sup>2+</sup> , Zn <sup>2+</sup> , Pb <sup>2+</sup>                                | C <sub>6</sub> H <sub>5</sub> NH <sub>2</sub> , N <sub>3</sub> <sup>−</sup> , NO <sub>2</sub> <sup>−</sup> , Br <sup>−</sup>                                                   |
| <b>Soft</b>        | SO <sub>2</sub> , Cu <sup>+</sup> , Ag <sup>+</sup> , Au <sup>+</sup> , Hg <sup>2+</sup> , Pd <sup>2+</sup> , Pt <sup>2+</sup> , I <sub>2</sub> , carbenes         | I <sup>−</sup> , S <sup>2−</sup> , RS <sup>−</sup> , CN <sup>−</sup> , CO, alkenes, R <sub>2</sub> S, H <sup>−</sup>                                                           |

**HSAB Principle:**

Hard acids prefer hard bases;      Soft acids prefer soft bases

**Three Examples from Each Class:**

**Hard–Hard interactions:**

(a) **Al<sup>3+</sup> + F<sup>−</sup>:** Both hard — AlF<sub>3</sub> is very stable; Al<sup>3+</sup> prefers F<sup>−</sup> over I<sup>−</sup> ( $K$  for AlF<sub>3</sub>  $\gg$  AlI<sub>3</sub>)

(b) **Mg<sup>2+</sup> + OH<sup>−</sup>:** Both hard — Mg(OH)<sub>2</sub> is sparingly soluble but stable; Mg<sup>2+</sup> found as MgCO<sub>3</sub>/MgSO<sub>4</sub> in nature

(c) **H<sup>+</sup> + H<sub>2</sub>O:** Hard acid + hard base — protonation of water is highly favoured; H<sub>3</sub>O<sup>+</sup> is the dominant protonated species in aqueous solution

**Soft–Soft interactions:**

(a) **Ag<sup>+</sup> + I<sup>−</sup>:** Both soft — AgI is extremely insoluble ( $K_{sp} = 8.5 \times 10^{-17}$ ) compared to AgF ( $K_{sp} = 200$ ); silver found as AgI in nature

(b) **Hg<sup>2+</sup> + S<sup>2−</sup>:** Both soft — HgS (cinnabar) is one of the most insoluble compounds ( $K_{sp} \sim 10^{-54}$ ); Hg<sup>2+</sup> has strong affinity for sulfur ligands

(c) **Pt<sup>2+</sup> + CN<sup>−</sup>:** Both soft — [Pt(CN)<sub>4</sub>]<sup>2−</sup> is exceptionally stable; CN<sup>−</sup> is a strong ligand for soft metals (basis of gold cyanide leaching: Au<sup>+</sup> + 2CN<sup>−</sup>)

**Conclusion:** HSAB theory classifies Lewis acids and bases as hard or soft based on polarisability and charge density. The guiding principle “hard likes hard, soft likes soft” predicts the stability of complexes, solubility products, and reaction outcomes across inorganic and biochemistry.

**Q18.** Discuss the principle of application of HSAB in the following fields: (i) Medicinal Chemistry, (ii) Geochemical classification of the elements and their occurrence in nature as minerals, (iii) Predicting favorable equilibria, (iv) Toxicology. **(20 marks)** [CHEM 113 | 2013–14]

**Answer**

**Applications of HSAB Principle:**

**(i) Medicinal Chemistry:**

HSAB guides the design of **metallo drugs** and understanding of drug–receptor interactions:

- **Cisplatin** [Pt(NH<sub>3</sub>)<sub>2</sub>Cl<sub>2</sub>]: Pt<sup>2+</sup> is a **soft acid**. It preferentially binds to **soft bases** — specifically the N7 position of guanine in DNA (soft N-donor after Cl<sup>−</sup> displacement). This forms **intrastrand crosslinks** that distort DNA and prevent replication in cancer cells. Hard ligands like OH<sup>−</sup> and SO<sub>4</sub><sup>2−</sup> are not preferred.
- **Gold compounds** (auranofin, Au<sup>+</sup>): Soft Au<sup>+</sup> targets **soft thiol groups** (–SH, soft base) on cysteine residues of thioredoxin reductase — the mechanism of anti-rheumatic and anticancer activity.
- **Chelation therapy:** EDTA (hard donor O,N) is used to remove **hard toxic metals** like Pb<sup>2+</sup>, while dimercaprol (–SH, soft donor) removes **soft toxic metals** like Hg<sup>2+</sup> and As<sup>3+</sup> (both soft acids).
- **Antidotes:** British Anti-Lewisite (BAL, dithiol) chelates soft metal arsenic As<sup>3+</sup>; EDTA chelates hard metal Pb<sup>2+</sup> — HSAB dictates which antidote to use.

**(ii) Geochemical Classification (Goldschmidt):**

HSAB mirrors Goldschmidt’s geochemical classification:

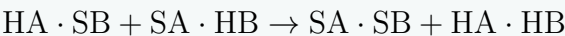


| HSAB       | Gold-schmidt | Minerals formed                                                                                                                |
|------------|--------------|--------------------------------------------------------------------------------------------------------------------------------|
| Hard acids | Lithophile   | Found as oxides, carbonates, silicates, sulfates (ionic minerals): $\text{Al}_2\text{O}_3$ , $\text{MgCO}_3$ , $\text{CaSO}_4$ |
| Soft acids | Chalcophile  | Found as sulfides (covalent minerals): $\text{HgS}$ , $\text{PbS}$ , $\text{CuS}$ , $\text{ZnS}$ , $\text{Ag}_2\text{S}$       |
| Very soft  | Siderophile  | Found as free metals in iron cores: $\text{Au}$ , $\text{Pt}$ , $\text{Ir}$                                                    |

**Rationale:** Hard metal cations ( $\text{Na}^+$ ,  $\text{Mg}^{2+}$ ,  $\text{Al}^{3+}$ ) prefer hard bases ( $\text{O}^{2-}$ ,  $\text{CO}_3^{2-}$ ) in the earth’s crust. Soft metals ( $\text{Hg}^{2+}$ ,  $\text{Pb}^{2+}$ ,  $\text{Cu}^+$ ) prefer soft  $\text{S}^{2-}$  — occurring as sulfide ores.

**(iii) Predicting Favourable Equilibria:**

The HSAB principle predicts which direction a metathesis reaction will proceed:



(HA = hard acid, HB = hard base, SA = soft acid, SB = soft base)

**Examples:**

- **$\text{AgF} + \text{NaI} \rightarrow \text{AgI} + \text{NaF}$ :** Soft  $\text{Ag}^+$  moves to soft  $\text{I}^-$ ; hard  $\text{Na}^+$  moves to hard  $\text{F}^-$  — strongly favoured ( $K \gg 1$ )
- **$\text{BF}_3 + (\text{CH}_3)_3\text{N} \rightarrow (\text{CH}_3)_3\text{N-BF}_3$ :** Hard  $\text{BF}_3$  (hard acid) prefers N-donor (borderline base): stable adduct
- **Displacement of CO by  $\text{CN}^-$  from  $\text{Ni(CO)}_4$ :** Soft  $\text{CN}^-$  displaces soft CO from soft  $\text{Ni}^0$  — both reactions involve soft–soft, so thermodynamic differences are small

General rule: **reactions that create harder hard–hard and softer soft–soft pairs are favoured.**

**(iv) Toxicology:**

HSAB explains the mechanism of toxicity of heavy metals and predicts their biological targets:

- **Mercury ( $\text{Hg}^{2+}$ , soft acid):** Binds strongly to **soft sulfur bases** in biological systems — cysteine –SH groups in enzymes (glutathione, thioredoxin), inhibiting them.  $\text{Hg}^{2+}$  crosses the blood–brain barrier as  $\text{Hg}(\text{CH}_3)_2$  (methylmercury, very soft), causing neurological damage (Minamata disease).
- **Lead ( $\text{Pb}^{2+}$ , borderline soft):** Replaces  $\text{Ca}^{2+}$  (hard, similar size) in bones and  $\text{Ca}^{2+}$ -binding enzymes, disrupting calcium signalling. Also binds to –SH groups, inhibiting  $\delta$ -aminolevulinate dehydratase (ALAD), blocking haem synthesis — causing anaemia.
- **Arsenic ( $\text{As}^{3+}$ , soft acid):** Binds vicinal –SH groups (dithiols) in **pyruvate dehydrogenase** and lipoic acid, disrupting energy metabolism (Krebs cycle). Antidote: BAL (2,3-dimercaptopropanol) — two –SH groups chelate soft  $\text{As}^{3+}$ .
- **Cadmium ( $\text{Cd}^{2+}$ , soft acid):** Displaces  $\text{Zn}^{2+}$  from zinc-finger proteins and enzymes ( $\text{Zn}^{2+}$  is borderline/soft); binds –SH and –NH bases, disrupting DNA repair enzymes and causing renal toxicity.
- **Cyanide ( $\text{CN}^-$ , soft base):** Binds  **$\text{Fe}^{3+}$**  in cytochrome c oxidase (though  $\text{Fe}^{3+}$  is borderline, the iron–CN bond is highly stable due to  $\pi$ -backbonding with soft  $\text{CN}^-$ ), blocking mitochondrial electron transport chain and ATP synthesis.

| Toxin            | HSAB       | Target          | Effect                          |
|------------------|------------|-----------------|---------------------------------|
| $\text{Hg}^{2+}$ | Soft acid  | –SH (soft)      | Enzyme inhibition, neurological |
| $\text{Pb}^{2+}$ | Borderline | –SH, Ca sites   | Haem synthesis blocked          |
| $\text{As}^{3+}$ | Soft acid  | dithiols (soft) | Energy metabolism disruption    |
| $\text{Cd}^{2+}$ | Soft acid  | Zn enzymes      | DNA repair inhibition           |
| $\text{CN}^-$    | Soft base  | Fe in CcO       | ATP synthesis blocked           |

**Conclusion:** The HSAB principle has wide-ranging applications — from designing metallodrugs that exploit soft metal–soft ligand affinity, to explaining mineral distribution in the earth’s crust, to predicting thermodynamically favourable reactions, to understanding heavy metal toxicity through selective binding to biological soft-base sites (thiols, amines, DNA).

- Q19.** Explain why: (i) Picric acid is as strong as nitric acid though picric acid is an organic acid. (ii) Gold and copper are colored but silver is not, though they are elements of the same group IB. (iii) The coordination number of  $\text{Co}^{3+}$  is six but that of  $\text{Zn}^{2+}$  is four, though they are ions of 'd' block elements. (iv) Potassium is more reactive than copper though they have  $4s^1$  electron in their outer shell orbital. (v) Transition metals generally form colored compounds. **(20 marks)** [CHEM 113 | 2013–14]

### Answer

#### (i) Picric acid is as strong as nitric acid:

Picric acid (2,4,6-trinitrophenol) is an organic phenol, yet has  $K_a \approx 0.42$  — comparable to  $\text{HNO}_3$  ( $K_a \approx 24$ , strong acid).

**Reason:** The phenol  $-\text{OH}$  in picric acid loses a proton to give the **picrate anion** (2,4,6-trinitrophenoxide). Three strongly **electron-withdrawing nitro groups** ( $-\text{NO}_2$ ) at ortho and para positions:

- Withdraw electron density from the ring via **resonance** and **induction**, greatly stabilising the negative charge on  $\text{O}^-$  in the picrate anion
- The negative charge is delocalised over the ring *and* into all three  $-\text{NO}_2$  groups via resonance:  

$$\text{ArO}^- \longleftrightarrow \text{resonance structures with charge on NO}_2$$
- This exceptional **stabilisation of the conjugate base** shifts the equilibrium strongly toward dissociation — making picric acid unusually strong for an organic acid.

#### (ii) Gold and copper are coloured but silver is not:

All three are group IB metals with configuration: Cu:  $[\text{Ar}]3d^{10}4s^1$ , Ag:  $[\text{Kr}]4d^{10}5s^1$ , Au:  $[\text{Xe}]4f^{14}5d^{10}6s^1$ .

**Colour arises when visible light promotes an electron from a filled  $d$ -band to the empty  $s/p$ -band** (interband transition).

The energy gap determines which wavelengths are absorbed:

- Copper (Cu):** The  $d \rightarrow s$  band gap corresponds to **blue/green light** absorption — reflected colour is **orange-red**.
- Gold (Au):** Due to **relativistic effects** — electrons in the 6s orbital travel at relativistic speeds, causing orbital contraction and **lowering the 6s band energy**. This narrows the  $d \rightarrow s$  gap to correspond to **blue light** absorption — reflected colour is **yellow/gold**.
- Silver (Ag):** The  $d \rightarrow s$  gap is larger, corresponding to **UV absorption** — all visible wavelengths are reflected equally, giving silver its **colourless/white metallic** appearance. Relativistic effects are much smaller for Ag (5s electrons are less relativistic than Au 6s).

#### (iii) Coordination number of $\text{Co}^{3+}$ is 6 but $\text{Zn}^{2+}$ is 4:

- $\text{Co}^{3+}$ :** configuration  $[\text{Ar}]3d^6$  (low-spin with strong field ligands). It has **three empty orbitals** available for bonding:  $d_{z^2}$ ,  $d_{x^2-y^2}$  and combinations with 4s, 4p give 6 hybrid orbitals ( $d^2sp^3$  or  $sp^3d^2$ ).  $\text{Co}^{3+}$  forms **kinetically inert** octahedral complexes (large CFSE with  $d^6$  low-spin) — CN = 6.
- $\text{Zn}^{2+}$ :** configuration  $[\text{Ar}]3d^{10}$  — **completely filled  $d$  orbitals**. No  $d$  orbitals available for  $d^2sp^3$  hybridisation. Only  $sp^3$  hybridisation is possible, giving **tetrahedral geometry** — CN = 4. There is also no CFSE (no unpaired  $d$  electrons to stabilise octahedral geometry), so tetrahedral is preferred.

#### (iv) Potassium is more reactive than copper despite both having $4s^1$ :

Both K ( $[\text{Ar}]4s^1$ ) and Cu ( $[\text{Ar}]3d^{10}4s^1$ ) have one 4s electron, but their reactivities differ enormously.

**Reason:**

- Ionisation energy:** K has  $\text{IE}_1 = 419 \text{ kJ/mol}$  vs Cu =  $745 \text{ kJ/mol}$ . In Cu, the  $3d^{10}$  electrons **poorly shield** the 4s electron from the nuclear charge ( $Z = 29$  vs  $Z = 19$  for K). The effective nuclear charge ( $Z_{\text{eff}}$ ) felt by the 4s electron is much higher in Cu.
- Shielding:** In K, the 4s electron is shielded by 18 core electrons ( $Z_{\text{eff}} \approx 2.2$ ). In Cu, 3d electrons are in the same shell and shield poorly, giving  $Z_{\text{eff}} \approx 4.4$  for the 4s electron.
- Consequence:** Cu's 4s electron is held much more tightly — Cu is much less reactive, does not react with water, and has a positive reduction potential ( $E^\circ = +0.34 \text{ V}$ ), while K reacts violently with water ( $E^\circ = -2.93 \text{ V}$ ).

#### (v) Transition metals generally form coloured compounds:

Transition metals have **partially filled  $d$  orbitals**. When ligands surround the metal ion, the five degenerate  $d$  orbitals split into two groups of different energy (**crystal field splitting,  $\Delta$** ):

$$\text{octahedral: } \underbrace{t_{2g}}_{\text{lower}} \quad \underbrace{e_g}_{\text{upper}} \quad \Delta_o = E(e_g) - E(t_{2g})$$

- Electrons can be promoted from lower ( $t_{2g}$ ) to upper ( $e_g$ ) levels by absorbing **visible light** photons:

$$\Delta E = h\nu = \frac{hc}{\lambda} \quad \lambda \in 400\text{--}700 \text{ nm}$$

- The **complementary colour** is observed (colour of unabsorbed wavelengths)
- $\Delta_o$  depends on the metal, its oxidation state, and the ligand field strength — this is why different complexes show different colours
- Example:**  $[\text{Ti}(\text{H}_2\text{O})_6]^{3+}$  ( $d^1$ ) absorbs at 493 nm (green) — appears **purple**;  $[\text{Cu}(\text{H}_2\text{O})_6]^{2+}$  ( $d^9$ ) absorbs red — appears **blue**

Compounds with **completely filled** ( $d^{10}$ , e.g.  $\text{Zn}^{2+}$ ) or **empty  $d$  orbitals** ( $d^0$ , e.g.  $\text{Ti}^{4+}$ ) cannot undergo  $d-d$  transitions and are therefore **colourless** — confirming the rule.

- Q20.** Define buffer solution. How does buffer solution work when small quantities of strong acid or base are added to it? **(16 marks)**  
[CHEM 113 | 2013–14]
- Q21.** Derive an expression for the rate of a first-order reaction which is opposed by another first-order reaction. How can you determine the equilibrium constant of such a reaction? **(17 marks)**  
[CHEM 113 | 2013–14]
- Q22.** In studying the decomposition of ozone  $2\text{O}_3(\text{g}) \rightleftharpoons 3\text{O}_2(\text{g})$  in a two liter (2 L) reaction vessel, it is found that  $\frac{d[\text{O}_3]}{dt} = -1.5 \times 10^{-2} \text{ mol}\cdot\text{L}^{-1}\cdot\text{s}^{-1}$ . (i) What is the rate of reaction? (ii) What is the rate of conversion? (iii) What is the value of  $\frac{d[\text{O}_2]}{dt}$ ? **(9 marks)** [CHEM 113 | 2013–14]
- Q23.** Explain why: (i) Bond in  $\text{BeCl}_2$  is covalent but bond in  $\text{CaCl}_2$  is ionic though Be and Ca are in the same group of the periodic table. (ii) Picric acid is as strong as nitric acid though picric acid is an organic acid. (iii) Gold and copper are colored but silver is not, though they are elements of the same group IB. (iv) The coordination number of  $\text{Co}^{3+}$  is six but that of  $\text{Zn}^{2+}$  is four, though they are ions of 'd' block elements. (v) Potassium is more reactive than copper though they have  $4s^1$  electron in their outer shell orbital. (vi) Transition metals generally form colored compounds. **(24 marks)**  
[CHEM 113 | 2013–14]
- Q24.** Explain how a buffer solution resists the change in its pH value upon the addition of small quantities of strong acid or base. **(8–11 marks)**  
[CHEM 113 | 2012–13; 2011–12; 2010–11]
- Q25.** Define buffer solution and buffer capacity. **(6 marks)**  
[CHEM 113 | 2011–12; 2012–13]
- Q26.** What are Hard and Soft Acids and Bases (HSAB)? Explain the HSAB principle with examples. Mention the applications of hard and soft acids and bases concept. **(13 marks)**  
[CHEM 113 | 2012–13]
- Q27.** Discuss different classes of Lewis acids and bases. Cite two examples in each class. **(10 marks)**  
[CHEM 113 | 2012–13]
- Q28.** How will you determine the strength of acids and bases? Illustrate your answer with examples. **(12 marks)** [CHEM 113 | 2012–13]
- Q29.** Outline the significant features in the properties of the following pairs of molecules: (i)  $\text{H}_2\text{O}$  and  $\text{H}_2\text{S}$ , (ii)  $\text{CO}_2$  and  $\text{SiO}_2$ , (iii)  $\text{AgF}$  and  $\text{AgI}$ . **(10 marks)**  
[CHEM 113 | 2012–13]
- Q30.** What are the differences between order and molecularity of a reaction? **(5 marks)**  
[CHEM 113 | 2012–13]
- Q31.** Show how you can determine the order of a reaction by the half-life method. **(6 marks)**  
[CHEM 113 | 2012–13]
- Q32.** Derive an expression for the rate constant of a first-order reaction. **(9 marks)**  
[CHEM 113 | 2012–13]
- Q33.** From the following data for the decomposition of  $\text{N}_2\text{O}_5$  at  $45^\circ\text{C}$ , show that the reaction is of the first order.
- |                                   |   |      |      |       |       |          |
|-----------------------------------|---|------|------|-------|-------|----------|
| Time (min)                        | 0 | 10   | 15   | 20    | 25    | $\infty$ |
| Vol. of $\text{O}_2$ evolved (mL) | 0 | 6.30 | 8.95 | 11.40 | 13.50 | 34.75    |
- (12 marks)**  
[CHEM 113 | 2012–13]
- Q34.** Prove that buffer solution shows maximum buffer capacity when  $\text{pH} = \text{pK}$ . **(14 marks)**  
[CHEM 113 | 2011–12]
- Q35.** Define order of reaction. Show how you can determine the order of reaction by (i) half-life method and (ii) differential method. **(11 marks)**  
[CHEM 113 | 2011–12]
- Q36.** Derive an expression for the rate constant of a second order reaction. What are the characteristic features of a second order reaction? **(16 marks)**  
[CHEM 113 | 2011–12]
- Q37.** The decomposition of a gas follows second order kinetics. It takes 40 minutes for 40% of the gas to be decomposed when the initial concentration is  $4 \times 10^{-2} \text{ mol}\cdot\text{L}^{-1}$ . Calculate the specific reaction rate. **(8 marks)**  
[CHEM 113 | 2011–12]
- Q38.** Establish the condition for maximum buffer capacity. **(12 marks)**  
[CHEM 113 | 2011–12]
- Q39.** Explain what is meant by the diagonal relationship. List two pairs of elements that show this relationship. **(8 marks)** [CHEM 113 | 2011–12]
- Q40.** What is buffer solution? How is it prepared? **(6 marks)**  
[CHEM 101 | 2010–11]
- Q41.** Prove that buffer solution shows maximum buffer capacity when  $\text{pH} = \text{pK}$ . **(14 marks)**  
[CHEM 101 | 2010–11]
- Q42.** Calculate the pH of a buffer solution consisting of  $\text{CH}_3\text{COOH}$  and 1.0 M  $\text{CH}_3\text{COONa}$  solutions. What will be the pH after the addition of 0.10 mol of  $\text{HCl}$  to one litre of this buffer solution? **(7 marks)**  
[CHEM 101 | 2010–11]
- Q43.** Discuss with example the principle of Hard and Soft Acids and Bases (HSAB) and also discuss one of its applications. **(8 marks)**  
[CHEM 101 | 2010–11]

- Q44.** Calculate the normality of 95% (W/W) sulphuric acid solution of specific gravity (sp. gr.) 1.95 and also calculate the amount of water required per litre for this acid to convert its sp. gr. to 1.40. **(10 marks)** [CHEM 101 | 2010–11]
- Q45.** The reaction  $\text{H}_2 + \text{Br}_2 = 2\text{HBr}$  is carried out in a 0.250 L reaction vessel. The change in the amount of  $\text{Br}_2$  in 0.01 s is  $-0.001$  mol. (i) What is the rate of conversion? (ii) What is the rate of reaction? (iii) What are the values of  $d[\text{H}_2]/dt$ ,  $d[\text{Br}_2]/dt$  and  $d[\text{HBr}]/dt$ ? **(9 marks)** [CHEM 101 | 2010–11]
- Q46.** Define rate law. Write the rate law for the following reaction:  $a\text{A} + b\text{B} \rightarrow n\text{N} + m\text{M}$ . **(10 marks)** [CHEM 101 | 2009–10]
- Q47.** What is a reversible reaction? Derive the rate expression for a first-order reaction opposed by another first-order reaction. **(15 marks)** [CHEM 101 | 2009–10]
- Q48.** The reaction  $\text{SO}_2\text{Cl}_2 \rightarrow \text{SO}_2 + \text{Cl}_2$  is a first-order reaction with the rate constant  $2.2 \times 10^{-5} \text{ s}^{-1}$  at  $302^\circ\text{C}$ . What percentage of  $\text{SO}_2\text{Cl}_2$  will get decomposed in 90 minutes when the reaction is carried out at  $302^\circ\text{C}$ ? **(10 marks)** [CHEM 101 | 2009–10]
- Q49.** What is buffer solution? Classify buffer solution and show how they operate. **(9 marks)** [CHEM 101 | 2009–10]
- Q50.** Explain the terms ‘rate’ and ‘rate constant’ of reactions. **(8 marks)** [CHEM 101 | 2008–09]
- Q51.** Derive an expression for the rate constant of a second-order reaction involving one reactant only. What are the characteristic features of a second-order reaction? **(18 marks)** [CHEM 101 | 2008–09]
- Q52.** What are the modern concepts of acids and bases? Which one is the most acceptable concept in your opinion? Give reasons. **(12 marks)** [CHEM 101 | 2007–08]
- Q53.** Discuss five important bases for determining the relative acidity of different acids. Cite an example in each case. **(10 marks)** [CHEM 101 | 2007–08]
- Q54.** What are buffer solutions? How can they be classified? Show how a buffer solution functions. **(10 marks)** [CHEM 101 | 2007–08]
- Q55.** Derive the rate law for a first-order reaction being opposed by another first-order reaction. **(8 marks)** [CHEM 101 | 2007–08]
- Q56.** Derive and explain the Arrhenius equation. Show its application. **(16 marks)** [CHEM 101 | 2007–08]
- Q57.** For the decomposition of an acid, the value of the rate constant ( $k$ ) at  $0^\circ\text{C}$  is  $2.46 \times 10^{-5}$  and at  $30^\circ\text{C}$  is  $1.63 \times 10^{-5}$ . Calculate the energy of activation ( $E_a$ ) of the process and comment on the result. **(11 marks)** [CHEM 101 | 2007–08]
- Q58.** What do you mean by the following terms used in chemical kinetics: (i) Order and Molecularity; (ii) Rate and Rate constant. **(10 marks)** [CHEM 101 | 2006–07]
- Q59.** Derive the integrated rate equation for the second order reaction  $2\text{A} \rightarrow \text{P}$ . Show how  $t_{1/2}$  for such a reaction is dependent on the initial concentration. **(11 marks)** [CHEM 101 | 2006–07]
- Q60.** Describe two methods for the determination of the order of a reaction. **(6 marks)** [CHEM 101 | 2006–07]
- Q61.** The half-life period  $t_{1/2}$  for a first-order reaction is 69.3 min at  $27^\circ\text{C}$  and 34.7 min at  $37^\circ\text{C}$ . Find out the energy of activation ( $E_a$ ) for the reaction. **(3 marks)** [CHEM 101 | 2006–07]